Technical Reference Guide

ST Engineering

iDirect Velocity™

Release 4.1.0

Revision 1

September 15, 2025

iDIRECT

© 2025 ST Engineering iDirect, Inc. dba iDirect and/or its affiliates. All rights reserved.

All rights reserved. Reproduction in whole or in part without permission is prohibited. Information contained herein is subject to change without notice. The specifications and information regarding the products in this document are subject to change without notice. While every effort has been made to ensure the accuracy of the statements, information and recommendations in this document, they are provided without warranty of any kind, express, or implied. Users must take full responsibility for their application of any products. Trademarks, brand names and products mentioned in this document are the property of their respective owners. All such references are used strictly in an editorial fashion with no intent to convey any affiliation with the name or the product's rightful owner.

ST Engineering iDirect is a global leader in satellite communications (satcom) providing technology and solutions that enable its customers to expand their business, differentiate their services and optimize their satcom networks. Through the merger with Newtec, a recognized industry pioneer, the combined business unites over 35 years of innovation focused on solving satellite's most critical economic and technology challenges, and expands a shared commitment to shaping the future of how the world connects. The product portfolio, branded under the names iDirect and Newtec, represents the highest standards in performance, efficiency and reliability, making it possible for its customers to deliver the best satcom connectivity experience anywhere in the world. ST Engineering iDirect is the world's largest TDMA enterprise VSAT manufacturer and is the leader in key industries including broadcast, mobility and military/government.

Company Web site: http://www.idirect.net | Main Phone: 703.648.8000

TAC Contact Information: Phone: 703.648.8151 | Email: tac@idirect.net | Web site: http://support.idirect.net

iDirect Government created in 2007, is a wholly owned subsidiary of ST Engineering iDirect and was formed to better serve the U.S. government and defense communities. In 2019, iDirect Government acquired Glowlink Communications Technology, Inc.

Company Web site: http://www.idirectgov.com | Main Phone: 703.648.8118

TAC Contact Information: Phone: 703.648.8111 | Email: tac@idirectgov.com | Web site: http://-partnerportal.idirectgov.com/

Document Name: TRG_Velocity_4.1.0_Rev1_T0001665_15-Sep-2025.pdf

Document Number: T0001665

Revision History

The following table shows all revisions for this document. To determine if this is the latest revision, check the Technical Assistance Center (TAC) Web site.

Revision	Date	Updates
1	September 15, 2025	Initial release document for Velocity Release 4.1.0.

Contents

Revision History iii

Contents v

List of Figures xxii

List of Tables xxx

About xxxi

- Purpose xxxi
- Audience xxxi
- Getting Help xxxii
  - iDirect Contact Information xxxii
  - iDirect Government Contact Information xxxii
- Related Documents xxxii

Velocity Technical Reference Guide—An Introduction 1

- Velocity System Overview 2
- Velocity System Operations Overview 3
- Satellite Link Operations Overview 4
- Global Bandwidth Management Overview 5
- Redundancy and the Velocity Network Overview 7
- Pulse NMS Operations Overview 8
- Satellite Terminal Operations Overview 8
- Velocity Features Chapters 10

1 Velocity System Overview 11

- 1.1 About the Velocity Network 12

1.2 Velocity Space Segment 13
1.3 Velocity Ground Segment 15
1.3.1 SAS and CDC/RF Gateway Architectures 16
1.3.1.1 The SAS Site Architecture 17
1.3.1.2 CDC and RF Gateway Site Architecture 18
1.3.1.3 SAS, CDC, and RF Gateway Components 19
1.3.2 The Network Operating Center Site 22
1.3.3 The Extended Access Portal (EAP) Site 22
1.3.4 The Satellite Terminal 23
1.4 Velocity Network Architectures 24
1.4.1 Velocity IP Routing Domains 25
1.5 Satellite Business Service Models 26
1.5.1 Basic Satellite Business Model Terminology 26
1.5.2 The Vertically Integrated Service Model 28
1.5.3 The Managed Service Model 29
1.5.4 The Co-Located Service Model 30
1.6 Velocity System Characteristics 31
2 Velocity System Operations 33
2.1 About Maps and Regulatory Areas 34
2.1.1 Figure of Merit Maps 35
2.1.1.1 FOM Threshold for Smaller Terminals 35
2.1.2 Transmit Regulatory Offset Maps 39
2.1.3 Low Resolution Beam Maps for Acquisition 39
2.1.4 Service Area Maps 41
2.1.4.1 Service Area Groups 41
2.1.4.2 Geoscope Regions Based on Service Areas 43
2.1.5 Regulatory Areas 44
2.2 About Satellite Virtual Networks 45

2.2.1 Velocity SVN Representation in Pulse 46
2.2.2 Velocity SVN Types 47
2.3 Velocity Layer 3 (IP) Network Architecture 48
2.3.1 Velocity Layer 3 Routing Operations 49
2.3.2 Velocity IP Address Assignment 49
2.3.2.1 Data SVN — IP Address Assignment 50
2.3.2.2 Admin SVN — IP Address Assignment 51
2.3.2.3 Tunnel SVN — IP Address Assignment 51
2.3.3 Velocity IP Multicast 52
2.4 The Velocity Backbone Network 53
2.5 Velocity Subscriber Networks and Internet 54
2.6 The Velocity SAS Network 54
2.6.1 Velocity SAS Network — Layer 1 Physical Architecture 54
2.6.1.1 Shared Chassis Operation 55
2.6.2 Velocity SAS Network — Layer 2 Architecture (VLANs) 56
2.6.3 Velocity SAS Network — Layer 3 Network Architecture 57
2.7 The Velocity CDC Network 58
2.7.1 Layer 2 CDC Network 59
2.8 The Velocity RF Gateway Network 60
2.9 The CDC and RF Gateway WAN Link 61
2.10 Multiple Satellites per CDC 62
2.10.1 Deployment Architecture 63
2.11 The Velocity Over the Air Network 66
2.12 The Velocity Terminal Network 68
2.13 The Velocity NOC Network 70
2.14 The Monitoring and Control Network 71
2.15 Velocity Data Traffic Protocols 71
2.15.1 IPv4 Packet Forwarding 71

2.15.2 IPv6 Packet Forwarding ... 71
2.15.3 Layer 2 Over Satellite (L2oS) ... 72
2.15.4 Multicast ... 72
2.15.5 Transmission Control Protocol (TCP) ... 72
2.15.5.1 TCP Acceleration ... 72
2.15.5.2 TCP Connection Acceleration ... 73
2.15.5.3 TCP Header Compression ... 73
2.15.5.4 HTTP Acceleration ... 73
2.15.5.5 Generic Routing Encapsulation (GRE) ... 73
2.15.6 User Datagram Protocol (UDP) ... 74
2.16 Velocity Routing and Control Protocols ... 74
2.16.1 Border Gateway Protocol (BGP) ... 74
2.16.2 IGMP ... 75
2.16.3 Virtual Routing and Forwarding (VRF)-Lite ... 76
2.17 Velocity Satellite Router Protocols ... 76
2.17.1 DHCP ... 76
2.17.2 Network Address/Port Address Translation (NAT/PAT) ... 77
2.17.3 Domain Name System (DNS) ... 77
2.17.4 Session Initiation Protocol/Application Layer Gateway ... 78
2.17.5 RADIUS ... 79
2.17.5.1 Accounting ... 80
2.17.5.2 RADIUS Session Disconnect ... 80
2.17.5.3 RADIUS Authentication ... 80
3 Satellite Link Operations ... 81
3.1 Network Acquisition ... 82
3.1.1 Acquisition Signaling Carriers (ASC) ... 82
3.1.1.1 The ASC in a Satellite Network ... 83
3.1.1.2 Dedicated ASC ... 84

3.1.1.3 Dedicated ASC with Fan-Out ... 85
3.1.1.4 Shared Data Carrier/ASC ... 86
3.1.2 Acquisition Methods ... 87
3.1.2.1 Acquisition Bursts ... 88
3.2 DVB-S2 Overview ... 88
3.2.1 DVB-S2 Baseband Modulation and Coding ... 89
3.2.2 DVB-S2 Physical Layer Framing ... 90
3.2.2.1 DVB-S2 Roll-Off Factor ... 91
3.2.2.2 Guard Band Reduction for Improved Throughput ... 91
3.2.2.3 Group Delay and Downstream Performance ... 92
3.2.2.4 DVB-S2 Adjacent Channel Interference ... 92
3.3 Downstream Carrier Operation with DVB-S2 ... 94
3.3.1 ILC-T Multi-Carrier DVB-S2 Support ... 95
3.3.2 ACM Operation ... 96
3.3.3 DVB-S2 Fade Control ... 97
3.4 About the Upstream TDMA Waveform ... 99
3.4.1 TDMA Burst Formats ... 99
3.4.2 Spread Spectrum ... 99
3.5 Upstream Carrier Operation with Adaptive TDMA ... 100
3.5.1 Inbound Channel Configuration Messaging ... 100
3.5.1.1 System Information Tables ... 100
3.5.1.2 Network Clock Recovery Timestamp ... 101
3.5.1.3 Burst Time Plan ... 101
3.5.2 Inroute Group Composition ... 102
3.5.2.1 Differentiating C/N0 and C/N ... 103
3.5.2.2 Adaptivity Modes ... 104
3.5.2.3 Fade Margin and Measurement Interval Calculations ... 106
3.5.2.4 C/N Dynamic Range for Individual Carriers ... 107

3.5.2.5 C/N0 Dynamic Range for Inroute Group 108
3.5.2.6 Maximum Power and Control 108
3.5.2.7 Frame-Filling Efficiency 109
3.5.3 Adaptive TDMA Configuration and Constraints 109
3.5.4 Upstream TDMA LBA Loss at Specific Symbol Rates 111
3.6 About the Uplink Control Process 113
3.6.1 TDMA Initial Transmit Power 113
3.6.2 UCP Operation 115
3.6.2.1 UCP Correction Processing 116
3.6.2.2 UCP Symbol Timing 116
3.6.2.3 UCP Frequency Tracking 116
3.6.2.4 UCP Power Control and Fade Detection 117
4 Global Bandwidth Management 119
4.1 About Global Bandwidth Management 120
4.2 Velocity Service Providers 120
4.3 Velocity Network QoS Service Modes 122
4.4 About Velocity Service Plans 123
4.4.1 GSP and Subscriber Service Plan Properties 123
4.4.2 Quality of Service Basics 130
4.4.2.1 Throughput 130
4.4.2.2 Latency 131
4.4.2.3 Jitter 131
4.4.2.4 Packet Loss 131
4.4.3 Bandwidth Allocation 132
4.4.3.1 Allocation Algorithms 132
4.4.3.2 Allocation Fairness Algorithms 134
4.5 Service Plan Definition 137
4.5.1 Subscriber Service Plans 137

4.5.1.1 Unicast Subscriber Service Plans 137
4.5.1.2 Multicast Subscriber Service Plans 140
4.5.1.3 Subscriber Service Plan Limits per Terminal 141
4.5.2 GBWM Tree Structure 142
4.5.3 Group Service Plans (GSPs) 143
4.5.3.1 GSP Configuration and QoS Service Mode Options 144
4.5.3.2 Unicast Group Service Plans 144
4.5.3.3 Geographic Scope in GSPs 145
4.5.3.4 Multicast Group Service Plans (M-GSPs) 147
4.5.4 Features Common to GSP and Subscriber Service Plans 148
4.5.4.1 Geoscope 148
4.5.4.2 Child GSP/SSPP Geoscope Restrictions 148
4.5.4.3 Service Scheduling 148
4.5.5 Outbound Service Plan Parameters 149
4.5.5.1 Outbound (DVB-S2) QoS 149
4.5.5.2 Single Beam Restrictions and GSP MODCOD Scaling 153
4.5.5.3 IP Minimum Efficiency 157
4.5.6 Inbound Service Plan Parameters 158
4.5.6.1 Inbound (Adaptive TDMA) QoS 158
4.5.6.2 Nominal MODCOD (for GSPs only) 159
4.5.6.3 Inbound Fairness Base Bitrate (for SSPCs only) 159
4.5.7 NMS Service Plans 159
4.6 GBWM Operations 160
4.6.1 Global QoS Enforcement (GQE) Tree 160
4.6.2 Bandwidth Allocation Within an iNet 162
4.6.3 Global Bandwidth Allocation 165
4.6.3.1 Bandwidth Allocation for Management Traffic 167
4.6.3.2 GQE Enhancements 167

4.6.4 Allocation Fairness 168
4.6.4.1 Allocation Fairness Relative to Bandwidth (default) 168
4.6.4.2 Allocation Fairness Relative to CIR 169
4.6.4.3 Allocation Fairness Relative to CIR and MIR 170
4.6.4.4 Allocation Fairness Relative to MODCOD 170
4.6.5 Packet Classification 173
4.6.6 Packet Scheduling 173
4.6.6.1 Priority Queues 175
4.6.6.2 Class-Based Weighted Fair Queues 175
4.6.6.3 Best-Effort Queues 175
4.6.7 FAP Service Enforcement 175
4.6.8 Operating in Symbol-based Mode 176
4.6.9 Operating in IP-based Mode 176
4.6.10 Switching Between QoS Service Modes 177
4.6.11 IP Efficiency Operation 177
4.6.12 IP-Based QoS and Efficiency Stats 178
4.6.12.1 IP-Based QoS Stats 179
4.6.12.2 IP-Based Efficiency Stats 180
5 Redundancy and the Velocity System 181
5.1 Automatic Rain Diversity Switchover 182
5.1.1 SAS Configuration 182
5.1.1.1 Network Setup 182
5.1.1.2 Physical Layer 183
5.1.1.3 Service Implications and Timing 185
5.1.1.4 Backbone Network Traffic Requirements 185
5.1.2 CDC/RF Gateway Configuration 186
5.1.2.1 Network Setup 186
5.1.2.2 Physical Layer 187

5.1.2.3 Service Implications and Timing 189
5.1.2.4 Backbone Network Traffic Requirements 190
5.2 SAS Site Redundancy 191
5.2.1 SAS Redundancy Overview 191
5.2.2 Understanding SAS Switchover 193
5.2.2.1 SAS Switchover — Service Implications and Timing 194
5.2.2.2 SAS Switchover — Backbone Network Traffic Requirements 194
5.3 CDC and RF Gateway Redundancy 196
5.3.1 CDC Site Redundancy and Switchover 196
5.3.2 RF Gateway Site Redundancy and Switchover 198
5.3.3 RF Gateway Switchover Across HEO Satellites 199
5.4 NOC Site Redundancy 201
5.4.1 NOC Redundancy Overview 201
5.4.2 Understanding NOC Switchover 203
5.4.2.1 NOC Switchover — Service Implications and Timing 204
5.4.2.2 NOC Switchover — Backbone Network Traffic Requirements 204
5.5 NMS Redundancy and Failover 205
5.5.1 NMS Failure 205
5.5.2 Service Failure 205
5.5.3 Server Failure 205
5.6 Component Redundancy 207
5.6.1 Line Card Redundancy 207
5.6.1.1 IF Domain 208
5.6.1.2 Line Cards in an IF Domain 209
5.6.1.3 Line Card Switchover Conditions 210
5.6.1.4 Line Card Switchover Precedence 210
5.6.1.5 Failed Line Card Replacement 211
5.6.1.6 Line Card Switchover Scenarios 212

5.6.2 PP Redundancy 217
5.6.2.1 PP Server Redundancy 217
5.6.2.2 Process and Data Resiliency 218
5.6.3 Global Bandwidth Management Redundancy 218
5.6.3.1 GBWM Process and Data Resiliency 218
5.6.3.2 GBWM Server Redundancy 218
5.7 Network Resiliency 219
5.7.1 SAS Network Resiliency 219
5.7.2 NOC and EAP Network Resiliency 220
6 Pulse NMS Operations 221
6.1 About Pulse NMS 222
6.1.1 The NOC NMS 222
6.1.2 The NMS External Access Portal 222
6.1.3 The SAS NMS 223
6.2 Pulse NMS Features and Operations 223
6.2.1 NMS Browser Support 224
6.2.2 User Sessions 224
6.2.3 NMS Web UI for Configuration and Control 224
6.2.4 Kafka Streaming 225
6.2.5 User Groups and Roles 226
6.3 Virtualization/NMS Cluster 227
6.3.1 Virtualized Modular Architecture 227
6.3.2 Robust System Security 228
6.4 About SAS NMS Processes 229
6.4.1 Authentication Manager 230
6.4.2 Configuration Manager 230
6.4.3 Discrete Data Engine 231
6.4.4 Stats Manager 231

6.4.5 Update Manager 232
6.4.6 Web Core 232
6.4.7 Latency Measurement Manager 232
6.4.8 RT Command 232
6.5 About NOC NMS Processes 233
6.5.1 Certificate Authority (CA) Foundry 233
6.5.2 Thresholding Engine 233
6.6 External Access Portal NMS 234
6.7 Network Monitoring and Reporting 235
6.7.1 Types of Report Requests 235
6.7.2 Types of NMS Monitoring and Reporting 236
6.8 NMS Troubleshooting Tools 237
6.8.1 Probe Commands 237
6.8.2 Engineering Debug Console 237
7 Satellite Terminal Operations 239
7.1 About Satellite Terminals 240
7.2 Terminal Physical Components 241
7.2.1 Satellite Router 241
7.2.2 Built-In Ethernet Switch 241
7.2.3 Antenna 241
7.2.4 Antenna Control Unit 242
7.2.5 LNB 242
7.2.6 BUC 242
7.2.7 Terminal Type 243
7.2.8 RF Terminal Types 243
7.3 Terminal Software Components 244
7.3.1 Board Support Software 244
7.3.2 Over-the-Air Signaling/Protocols 244

7.3.2.1 Out-of-Band Signaling 244
7.3.2.2 RF Monitoring and Link Maintenance 245
7.3.2.3 Over-The-Air Data Framing and Processing 245
7.3.3 IP Application Support 246
7.3.3.1 SVN and IP Support 246
7.3.3.2 Domain Name System 247
7.3.3.3 Caching and HTTP Acceleration 247
7.3.3.4 Session Initiation Protocol Application Layer Gateway 247
7.3.3.5 Dynamic Host Configuration Protocol 248
7.3.3.6 Generic Routing Encapsulation 248
7.3.3.7 QoS Configuration 248
7.3.3.8 Terminal Routing 249
7.3.3.9 Multicast 249
7.3.4 Terminal Software and Configuration Updates 249
7.3.4.1 Software Updates 249
7.3.4.2 Configuration Updates 250
7.4 Satellite Router Interfaces 251
7.4.1 Web User Interface 251
7.4.2 Statistics Collection and Error Reporting 251
7.4.3 Switch Management 252
7.4.4 OpenAMIP 252
7.4.5 OpenBMIP™ 252
7.5 Terminal Installation and Commissioning 253
7.5.1 Installation and Commissioning Process 253
7.5.1.1 Terminals with OpenAMIP Support 254
7.5.1.2 Terminals with OpenBMIP Support 254
7.5.1.3 One-Touch Calibration 255
7.5.1.4 One-Touch Calibration-Multi-Band Smart BUC Support 255

7.6 Satellite Terminal Activation 257
7.7 Satellite Router Configuration 258
7.7.1 Static Configuration 258
7.7.2 Dynamic Configuration 258
7.8 Link Encryption Terminal Licensing 259
7.9 Terminal Network Acquisition 260
7.9.1 Network Acquisition Sequence 260
7.9.2 Authentication and Admission Control Sequence 262
7.9.3 Terminal Fails to Enter Network 264
7.10 Satellite Terminal Operational States 265
7.10.1 Satellite Router Functions 267
7.10.2 Normal Operation — Satellite Terminal Passes Traffic 268
7.11 Terminal Beam Switching 271
7.11.1 Mechanics of Terminal Switching 271
7.11.2 Terminal Make-Before-Break Beam Switch 273
7.11.3 Terminal Break-Before-Make Beam Switch 275
7.11.4 Terminal Satellite Switch 277
7.11.5 Using the Beam Switch API 279
7.12 Terminal Congestion Management 280
7.12.1 SAS-to-NOC Congestion Metrics Communication 281
7.12.2 NOC-to-SAS Consolidated iNet CM Communication 282
7.12.3 NOC/SAS Communications 283
7.12.4 Determining Congestion Metric (CM) Values 283
7.12.5 Configurable Congestion Parameters 285
7.12.6 Per-iNet Congestion Metrics Calculations 287
7.12.6.1 Configurable Congestion Weight Values 288
7.12.6.2 Configurable Upstream and Downstream Congestion 288
7.12.7 Congestion Metric and Decision Metric Calculations 289

7.12.8 Beam Switching Decision Logic 290
7.12.8.1 Determining the Decision Metric 290
7.12.8.2 Staggered Terminal Switching 291
7.12.9 Congestion Switching Scenarios 292
7.13 Terminal Global Positioning System Validation 294
7.14 Highly-Elliptical Orbit Support in Velocity 295
7.14.1 HEO Acquisition and Physical Layer Operations Including GSC 295
7.14.1.1 Acquisition Algorithm 296
7.14.2 Dynamic iNet Creation/Deletion 297
7.14.3 HEO Satellite and Beam Switching 299
7.14.4 Dynamic Beam Changes 299
7.15 Enhanced Inclined Orbit Support in Velocity 301
7.16 Centroid-Based Initial Beam Selection 303
8 Satellite Terminal Features 305
8.1 Terminal VLAN Mapping 306
8.2 Terminal Component Mismatch 307
8.3 Bonded Modems 308
8.3.1 Master Mode 308
8.3.2 Subordinate Mode 308
8.4 Discreet Mode Terminals 310
8.4.1 Beam Switch Decision 310
8.4.2 iNet Load Balancing 311
8.4.3 Skew Control 312
8.4.4 Service Areas 312
8.5 Mitigating Skew Impairment in Mobile Terminals 312
8.5.1 Non-Circular Aperture Antennas 312
8.5.2 PSD Limits Per Beam, Terminal, and Regulatory Area 314
8.5.2.1 Feature Description 314

8.5.2.2 Configuration Considerations ... 316
8.5.3 Hybrid Skew Margin Support ... 317
8.5.4 Real Skew ... 318
8.5.4.1 Real Skew with Flat Plate Antennas ... 318
8.6 Fast SNR Reporting ... 320
8.7 External Beam Select Mode ... 321
8.8 Commissioning Enhancements (API) ... 322
8.7.1 Prerequisites ... 322
8.7.1.1 First-Time Commissioning of Terminal ... 322
8.7.1.2 Previously-Commissioned Terminal ... 322
8.7.2 Commissioning Steps ... 322
9 Fair Access Policy ... 325
9.1 About Fair Access Policy ... 326
9.2 FAP Volume Allowance ... 326
9.2.1 FAP with Multicast Traffic ... 326
9.2.2 FAP Allowance Period ... 327
9.2.3 FAP Restoration Rules ... 327
9.2.4 FAP Rollover ... 328
9.2.5 FAP Top-UP Account ... 328
9.2.6 FAP Overage ... 330
9.2.7 FAP HTTP Redirect ... 332
9.3 FAP Feature Interaction ... 333
9.3.1 Regular Consumption (Fixed or Rolling Updates) ... 333
10 Multicast Overlay ... 335
10.1 About Multicast Overlay ... 336
10.1.1 One-Way iNet ... 336
10.1.2 Multicast Traffic on Terminal Secondary Receiver ... 336
10.1.3 Persistent Multicast ... 337

10.1.3.1 Beam/iNet Switching 337
10.2 Multicast Overlay Use Cases 338
10.2.1 Persistent Multicast via 1-way Overlay 338
10.2.2 Persistent Multicast Receive-Only Mode 339
10.2.3 Switching Between Overlay Beams 341
10.2.4 Unicast on Overlay Beams 342
10.2.5 Multicast Overlay Antenna and Terminal Requirements 344
10.2.6 Multicast Overlay Limitations and Constraints 344
11 Statistics Reports and SLA Management 347
11.1 About SLA Management and Statistics Reporting 348
11.2 CIR/MIR Fulfillment Statistics 348
11.2.1 Delayed Fulfillment Based on CIR/MIR Demand 350
11.3 CIR/MIR Fulfillment Parameters 350
11.4 CIR/MIR Fulfillment Ratio Statistics 352
11.4.1 CIR/MIR Fulfillment Ratio Implementation 352
11.4.1.1 Configuring the Fulfillment Window Size 353
11.4.1.2 Enable/Disable Fulfillment Ratio Statistics 354
11.4.1.3 Deriving CIR/MIR Fulfillment Ratio Statistics 354
11.4.2 CIR/MIR Performance Metrics Compared 355
11.4.3 Fulfillment Ratio Stats Calculation and Reporting 356
11.4.3.4 Fulfillment Ratio Calculation at Frame Level 356
11.4.3.5 Example Calculations at Frame Level 357
11.4.3.6 Fulfillment Ratio Reporting at SAS Level 358
11.4.3.7 Fulfillment Ratio Reporting at NOC Level 358
11.5 CIR/MIR Statistics Calculation Exclusions 359
12 Layer 2 Over Satellite 361
12.1 About Layer 2 Over Satellite (L2oS) 362
12.1.1 L2oS Benefits 363

12.2 L2oS Reference Model 364
12.3 SVN and Service Delimiting Tags 366
12.4 Layer 2 Services -SVN 367
12.4.1 L2 SVN for IP VPN 368
12.4.2 Layer 2 Network 368
12.5 L2oS Service Mode 369
12.6 Layer 2/Layer 3 Hybrid Mode 370
12.7 TCP Acceleration 370
12.8 Advanced Header Compression 370
12.9 MAC Address Learning 373
12.9.1 Persistent MAC FIB Table 373
12.10 L2oS Failover and Resiliency 373
12.11 Forwarding Rules 374
12.11.1 Flooding Avoidance 375
12.11.2 Ethernet Control Protocols 375
12.12 MEF Carrier Ethernet 2.0 Services 375
Appendix A: Acronyms & Abbreviations 379
Appendix B: Glossary 395
Appendix C: Debugging Terminal Events 414
Overview 414
Supported Remotes 414
Events Report 415
Using Event Viewer 415
Using Terminal Web UI Dashboard 416
Events Snapshot 418
Event Snapshot Example: 420

List of Figures

Figure 1-1. The Velocity Ground System Solution *12*

Figure 1-2. Beams, Channels, Carriers, and iNets *13*

Figure 1-3. Velocity System—SAS and CDC/RF Gateway Ground Segments *15*

Figure 1-4. Satellite Access Station Architecture *17*

Figure 1-5. Central Data Center and Radio Frequency Gateway Architecture *18*

Figure 1-6. Example Pulse EAP Deployment *22*

Figure 1-7. Satellite Terminal in Velocity System *23*

Figure 1-8. Velocity System *24*

Figure 1-9. Velocity Routing Domains *25*

Figure 1-10. a) Satellite Operator & Service Providers b) Satellite Operator as Service Provider *27*

Figure 1-11. Vertically Integrated Service Model *28*

Figure 1-12. Managed Service Model — Full or Partial *29*

Figure 1-13. Co-Located Model with Virtual Network Operator (VNO) *30*

Figure 2-1. Example Figure of Merit Map *35*

Figure 2-2. Basic FOM Contour Operation Overview *36*

Figure 2-3. FOM Contour Optimization Configuration Flow *38*

Figure 2-4. Example Transmit Regulatory Offset Map *39*

Figure 2-5. Comparison of Low-Resolution Map and Actual Beams *40*

Figure 2-6. Example Service Areas *41*

Figure 2-7. GSP Geoscope with Single Region/Single SA & Multiple Regions/Multiple SAs *42*

Figure 2-8. Two Service Area Groups for the Same Geographic Area *42*

Figure 2-9. Overlapping Service Areas in Two Different SA Groups *43*

Figure 2-10. Satellite Virtual Network *45*

Figure 2-11. Velocity SVN Representation in Pulse *46*

Figure 2-12. Velocity IP Network Architecture for an SVN *48*

Figure 2-13. IP Multicast in the Velocity System *52*

Figure 2-14. A Sample SAS Network Architecture *55*
Figure 2-15. Layer 2 Architecture *56*
Figure 2-16. SAS Switch Trunk and Access Ports *56*
Figure 2-17. CDC Site Network *58*
Figure 2-18. Velocity L2 CDC Network *59*
Figure 2-19. RF Gateway Site Network *60*
Figure 2-20. CDC and RF Gateway Wide Area Network (WAN) Link *61*
Figure 2-21. Legacy Single Satellite/CDC Architecture *62*
Figure 2-22. Three Satellites Managed by a Single CDC *63*
Figure 2-23. Single Chassis Supporting Two Satellites *63*
Figure 2-24. Three Satellites Managed by a Single CDC *64*
Figure 2-25. Multiple RF Gateways *64*
Figure 2-26. Single Chassis Supporting Multiple Satellites *65*
Figure 2-27. OTA Network Topology *66*
Figure 2-28. Example Terminal Network *68*
Figure 2-29. Simple Example of a Terminal Network *69*
Figure 2-30. Complex Example of a Terminal Network *69*
Figure 2-31. NOC Network Architecture *70*
Figure 2-32. RADIUS *79*
Figure 3-1. Contained Beam *83*
Figure 3-2. Dedicated DVB-S2 ASC *84*
Figure 3-3. DVB-S2 with Fan-out *85*
Figure 3-4. Shared DVB-S2 Carrier/ASC *86*
Figure 3-5. DVB-S2 Short Frame Protocol Structure *89*
Figure 3-6. Comparison of Constant Coding and Adaptive Coding Modes *90*
Figure 3-7. Physical Layer Frames *91*
Figure 3-8. Adjacent Carrier Interference Example *93*
Figure 3-9. Downstream DVB-S2 Carrier *94*

Figure 3-10. Feedback Loop ... ***97***

Figure 3-11. DVB-S2 Fade Control ... ***97***

Figure 3-12. TDMA Burst Format with Distributed Pilots ... ***99***

Figure 3-13. Heterogeneous Inroute Groups vs. Homogeneous Inroute Groups ... ***102***

Figure 3-14. A-TDMA Adaptivity Modes ... ***104***

Figure 3-15. Short-Term Adaptivity ... ***104***

Figure 3-16. Medium-Term Adaptivity ... ***106***

Figure 3-17. Optimal C/N Detection Range ... ***114***

Figure 3-18. Skewed C/N Detection Range — Initial Transmit Power Too High ... ***115***

Figure 3-19. Skewed C/N Detection Range — Initial Transmit Power Too Low ... ***115***

Figure 4-1. a) Satellite Operator and Service Providers b) Satellite Operator as Service Provider ... ***121***

Figure 4-2. Example of a Simple GBWM Tree ... ***124***

Figure 4-3. CIR/MIR Demands and CIR/MIR Distribution ... ***126***

Figure 4-4. GSP Modes - Service Provider Legacy vs. New Velocity 3.5 Modes ... ***128***

Figure 4-5. Example Allocation Scenario ... ***133***

Figure 4-6. SSPC with Multiple Parent GSPs ... ***139***

Figure 4-7. Example GBWM Tree Structure ... ***143***

Figure 4-8. GSP Region Geoscope Types ... ***146***

Figure 4-9. Multicast Traffic in Velocity ... ***147***

Figure 4-10. Scheduling, recurrent modification to CIR during business hours on first 5 days of month ... ***149***

Figure 4-11. Bandwidth (in symbols, normalized) vs. Information Rate ... ***152***

Figure 4-12. CIR Based on Base MODCOD and Bandwidth Utilization ... ***153***

Figure 4-13. Example Procedure for Calculating Nominal MODCOD for each Beam ... ***156***

Figure 4-14. Procedure for Calculating Average MODCOD for each Beam ... ***157***

Figure 4-15. Outbound iNet QoS Enforcement Tree ... ***161***

Figure 4-16. Multicast GSP Tree ... ***162***

Figure 4-17. iNet QoS Enforcer Tree ... ***163***

Figure 4-18. Global QoS Enforcer ... ***165***

Figure 4-19. Dynamic Bandwidth Management ... ***166***

Figure 4-20. Management Traffic Partition Node in iNet QoS Enforcement Tree ... ***167***

Figure 4-21. Example of Allocation Fairness Relative to Bandwidth ... ***168***

Figure 4-22. Example of Allocation Fairness Relative to Bandwidth with child SSPCs ... ***169***

Figure 4-23. Example of Allocation Fairness Relative to CIR ... ***170***

Figure 4-24. Example of Allocation Fairness Relative to CIR and MIR ... ***170***

Figure 4-25. Example of Allocation Fairness Relative to Base MODCOD ... ***171***

Figure 4-26. Example of Allocation Fairness Relative to Operating MODCOD ... ***172***

Figure 4-27. iDirect Packet Scheduling Algorithm ... ***174***

Figure 5-1. Rain Diversity Network Architecture - SAS Configuration ... ***183***

Figure 5-2. Before Rain Diversity Switchover - SAS Configuration ... ***184***

Figure 5-3. After Rain Diversity Switchover - SAS Configuration ... ***185***

Figure 5-4. Rain Diversity Network Architecture - RF Gateway Configuration ... ***187***

Figure 5-5. Before Rain Diversity Switchover - RF Gateway Configuration ... ***188***

Figure 5-6. After Rain Diversity Switchover - RF Gateway Configuration ... ***189***

Figure 5-7. Redundant SASs ... ***191***

Figure 5-8. Data Path when Primary SAS is Active ... ***192***

Figure 5-9. Velocity Network after SAS 1 Switchover ... ***193***

Figure 5-10. Data Path after a SAS Switchover ... ***194***

Figure 5-11. Before CDC Switchover ... ***197***

Figure 5-12. After CDC Switchover ... ***198***

Figure 5-13. RF Gateway Switchover Across HEO Satellites ... ***200***

Figure 5-14. Redundant NOCs ... ***201***

Figure 5-15. Velocity Network after a NOC Switchover ... ***202***

Figure 5-16. IF Domains ... ***208***

Figure 5-17. Line Cards in a Network ... ***210***

Figure 5-18. Available Standby Line Card Scenario — before failure ... ***212***

Figure 5-19. Available Standby Line Card Scenario — after failure ***213***

Figure 5-20. Failover based on iNet Priority — before failure ***214***

Figure 5-21. Failover based on iNet Priority — after failure on Low Priority iNet ***215***

Figure 5-22. Failover based on iNet Priority - after failure in High Priority iNet ***216***

Figure 5-23. Switchover Process from the Active Site to the Standby (Redundant) Site ***217***

Figure 5-24. A Sample SAS Network Architecture ***219***

Figure 6-1. Pulse NMS Distributed Components ***222***

Figure 6-2. Pulse NMS Operations ***223***

Figure 6-3. Virtualized Modular Architecture ***227***

Figure 6-4. SAS NMS Processes ***229***

Figure 6-5. NOC NMS Processes ***234***

Figure 7-1. Satellite Terminals in Velocity System ***240***

Figure 7-2. Terminal Interactions during Network Acquisition ***261***

Figure 7-3. Network Authorization and Admission Control Sequence ***263***

Figure 7-4. Terminal Actions ***266***

Figure 7-5. Satellite Router Uplink and Downlink Configuration Information and Data Flow ***267***

Figure 7-6. Operational Sequence for Traffic from Satellite Terminal to Internet ***268***

Figure 7-7. Operational Sequence for Traffic from Internet to Satellite Terminal ***269***

Figure 7-8. Beam Switching, before switch ***272***

Figure 7-9. Beam Switching, during switch, Make-before-Break ***273***

Figure 7-10. Beam Switching, single beam (after switch), Make-before-Break ***274***

Figure 7-11. Beam Switching, multiple beams (during switch), Break-before-Make ***275***

Figure 7-12. Beam Switching, single beam (after switch), Break-before-Make ***276***

Figure 7-13. Satellite Switching, before switch ***277***

Figure 7-14. Satellite Switching, after switch ***278***

Figure 7-15. SAS-to-NOC iNet CM Values Communication ***281***

Figure 7-16. NOC-to-SAS iNet Consolidated CM Values Communication ***282***

Figure 7-17. SAS-to-NOC iNet CM Values Communication ***283***

Figure 7-18. Congestion Metric Examples ***284***

Figure 7-19. Congestion Metric Value Calculation ***287***

Figure 7-20. Multiple SASs with Different Satellites ***292***

Figure 7-21. Multiple SASs with Different Satellites ***292***

Figure 7-22. Multiple SASs with Different Satellites ***293***

Figure 7-23. HEO Satellites ***295***

Figure 7-24. SSP Contour Map ***297***

Figure 7-25. Dynamic iNet Creation/Deletion ***298***

Figure 7-26. Beam Capacity ***298***

Figure 7-27. Dynamic Beam Changes ***299***

Figure 7-28. Beam Selection ***303***

Figure 8-1. VLAN ID Mapping ***306***

Figure 8-2. Discreet Mode beam selection logic illustration ***311***

Figure 8-3. Low-Profile Antenna a) Favorable Alignment b) Unfavorable Alignment ***313***

Figure 8-4. Example a) Standard Skew Table b) Pessimized Skew Table. ***315***

Figure 8-5. Limit determination using combined angular and PSD margins ***317***

Figure 8-6. Example a) Standard Skew Table b) Pessimized Skew Table. ***319***

Figure 9-1. Simple Rollover Example ***328***

Figure 9-2. Simple Top-Up Example – Fixed Allowance ***329***

Figure 9-3. Simple Top-Up Example – Rolling Allowance ***329***

Figure 9-4. Simple Overage Example – Fixed Allowance ***331***

Figure 9-5. Simple Overage Example – Rolling Allowance ***331***

Figure 9-6. FAP Flowchart ***333***

Figure 10-1. IP Multicast in the Velocity System ***337***

Figure 10-2. Multicast Overlay Aero Services ***338***

Figure 10-3. Persistent Multicast Receive-Only Mode ***340***

Figure 10-4. Switching Between Overlay Beams ***341***

Figure 10-5. Unicast on Overlay Beams ***342***

Figure 11-1. Delayed Fulfillment for CIR ... ***349***

Figure 11-2. Delayed Fulfillment for MIR ... ***349***

Figure 11-3. CIR Delayed Fulfillment Algorithm ... ***349***

Figure 11-4. Algorithm for Calculating the CIR (MIR) fulfillment Ratio ... ***354***

Figure 12-1. Velocity Transparent Layer 2 Emulation ... ***362***

Figure 12-2. L2oS Reference Model ... ***364***

Figure 12-3. L2oS Reference Model Applied to iDirect ... ***365***

Figure 12-4. SDT Mode = VLAN ... ***366***

Figure 12-5. Velocity L2oS Broadcast domain ... ***367***

Figure 12-6. L3 SVN ... ***368***

Figure 12-7. L2 SVN ... ***368***

Figure 12-8. Velocity L2 Network ... ***369***

Figure 12-9. L2oS Forwarding Rules ... ***374***

Figure 12-10. MEF E-Line Overview ... ***376***

Figure 12-11. MEF Ethernet Virtual Connection ... ***377***

Figure C-1. Sample evtview Table ... ***415***

Figure C-2. evtview Commands ... ***416***

Figure C-3. Terminal Events Displayed on the Terminal WUI ... ***417***

Figure C-4. Event Snapshot ... ***418***

Figure C-5. Event Verbose Snapshot ... ***419***

Figure C-6. On Startup Before the Search Cycle Begins ... ***420***

Figure C-7. Search Cycle Initiated and the Remote Tunes to the Signaling Carrier ... ***421***

Figure C-8. Remote Tries to Download the OTA Beam Map from the Signaling Carrier ... ***422***

Figure C-9. Waiting for ACU to Lock to the Satellite ... ***423***

Figure C-10. Waiting for Downstream Lock ... ***424***

Figure C-11. Waiting to Receive Valid SI Tables ... ***425***

Figure C-12. Waiting for a Valid Upstream Carrier to Transmit ... ***426***

Figure C-13. Waiting to Unmute the Transmitter ... ***427***

Figure C-14. Waiting to Send an Acquisition Burst to the Hub ***428***

Figure C-15. Waiting for the Hub to Acknowledge the Acquisition Burst ***429***

Figure C-16. Waiting for Allocated Slots ***430***

Figure C-17. Waiting to Receive UCP Updates ***431***

Figure C-18. Waiting to Receive BTP Updates ***432***

Figure C-19. Waiting for Authentication to Complete ***433***

Figure C-20. Remote Waiting for Second UCP Update to Ensure the Remote is in-network .....

Figure C-21. Remote is in-network and Stable .....

List of Tables

Table 2-1. Globally-Defined Layer 2 VLAN/SVN Assignments ***59***

Table 3-1. Increasing Information Rate by Reducing Guard Band ***92***

Table 3-2. Affected Symbol Rate Ranges for ULC-R MCD-TDMA at all MODCODs (Msps) ***111***

Table 3-3. Affected Symbol Rate Ranges ILC-R MCD-TDMA at BPSK 1/2, BPSK 2/3 and QPSK 1/2 (Msps) ***112***

Table 3-4. Additional Affected Symbol Rate Ranges for ILC-R MCD-TDMA at BPSK 1/2 (Msps) ***112***

Table 4-1. Inbound (IB)/Outbound (OB) Valid Allocation Fairness Methods ***135***

Table 4-2. Service Plan Limits ***141***

Table 4-3. MODCOD Scaling Factor Decision Logic ***154***

Table 4-4. GSP/SSPC QoS Stats for IP-Based Service Mode ***179***

Table 4-5. GSP/SSPC/iNet/Terminal Efficiency Stats for IP-Based Service Mode ***180***

Table 7-1. BUC Band Configuration Characteristics ***256***

Table 7-2. Configurable Congestion Parameters ***285***

Table 7-3. Operator Configurable US/DS Congestion Weight (CW) Parameters ***288***

Table 7-4. Sample Upstream/Downstream Congestion Calculation ***288***

Table 7-5. Congestion Metric and Decision Metric Calculations ***289***

Table 10-1. Sequential Use of Rx1 and Rx2 as Flight Path is Traversed ***339***

Table 11-1. CIR/MIR Fulfillment Statistics ***348***

Table 11-2. CIR/MIR Fulfillment Ratio Statistics ***352***

Table 12-1. SDT Support Matrix ***366***

Table 12-2. IPv4 Header Compression ***371***

Table 12-3. IPv4 Header Compression with IP Extensions ***372***

Table 12-4. MEF Service Summary ***376***

About

Purpose

The ***iDirect Technical Reference Guide*** provides detailed descriptions of iDirect Velocity system technology and major features as implemented in Velocity Release 4.0.

Audience

This document is intended for use by iDirect Network Operators, Network Architects and Engineers, and any other personnel that needs to understand iDirect Velocity and iDirect Pulse technology concepts and implementations.

Getting Help

The iDirect Technical Assistance Center (TAC) and the iDirect Government Technical Assistance Center (TAC) are available to provide assistance 24 hours a day, 365 days a year. Software user guides, FAQs, installation procedures, and other documents that support iDirect and iDirect Government products are available on the respective TAC Web site:

iDirect Contact Information

- Web site at http://support.idirect.net
- Telephone: 703.648.8151
- E-mail: tac@idirect.net

iDirect Government Contact Information

- Web site at http://partnerportal.idirectgov.com
- Telephone: 703.648.8111
- Email: tac@idirectgov.com

iDirect and iDirect Government produce documentation that are technically accurate, easy to use, and helpful to our customers. Please assist us in improving this document by sending feedback comments to:

- iDirect: techpubs@idirect.net
- iDirect Government: techpubs@idirectgov.com

For sales or product purchasing information contact iDirect Corporate Sales using the following:

- Telephone: 703.648.8000
- E-mail: sales@idirect.net

Related Documents

The following iDirect documents, available on the Service Now customer portal, contain information relevant to installing and using iDirect satellite network software and equipment:

- *iDirect Velocity™ Terminal Web UI User Guide*
- *iDirect Pulse® Network Management System User Guide*
- *Configuring Velocity™ Networks Using Pulse*
- *iDirect Velocity™ Software Release Notes*
- *iDirect Pulse® NMS Software Release Notes*
- Technical Note—Auto-Commissioning Using Integrasys SatMotion
- Technical Note—Installing iGateway in a Velocity Network

- *Installation, Support, and Maintenance Guide for Satellite Routers*

Velocity Technical Reference Guide—An Introduction

This chapter presents a high-level overview of the Velocity Technical Reference Guide document. This overview is presented as brief introductions to the chapters and topics covered in this guide. Each of these brief overviews is covered in full in its own chapter in the remainder of this Technical Reference Guide.

The following overviews are included in this chapter:

Velocity System Overview 2
Velocity System Operations Overview 3
Satellite Link Operations Overview 4
Global Bandwidth Management Overview 5
Redundancy and the Velocity Network Overview 7
Pulse NMS Operations Overview 8
Satellite Terminal Operations Overview 8
Velocity Features Chapters 10

Velocity System Overview

The Velocity System Overview chapter provides a high level summary overview of the iDirect Velocity™ System and the iDirect Pulse® Network Management System (NMS). Each topic of this overview chapter presents a high level introduction to topics that will be presented in other chapters.

Topics presented in the ***Velocity System Overview*** chapter include:

- **Network Overview** — presents a brief high-level tour of the Velocity System.
- **Space Segment** — defines key terminology of the Velocity space segment - for example beam, channel, carrier, iNet, inroute, and inroute group — which collectively support worldwide coverage through dynamic bandwidth reallocation to all parts of the globe, using DVB-S2 (Digital Video Broadcasting - Satellite - Second Generation) downstream and A-TDMA (Adaptive Time Division Multiple Access) upstream transmission.
- **Ground Segment** — broadly describes the Velocity segment component systems, their associated equipment and functions, and the physical location of each. The ground segment includes SAS, CDC, RF Gateway, NOC, and EAP sites, as well as distributed satellite terminals.
- **Velocity Network Architecture** — briefly introduces the Velocity IP network, which comprises a terrestrial backbone network, backbone network peering sites/POPs, SAS, NOC, and satellite terminals. In addition, the three logically organized routing domains (SAS routing, OTA routing, and terminal routing) are each briefly described.
- **Service Models** — introduces how the Velocity System encompasses the capabilities and the flexibility to support a variety of business models and the services they can provide, by delivering high-throughput satellite systems.
- **Major System Characteristics** — outlines some of the many strong Velocity system characteristics and functionality, for example — automatic load balancing, Make-before-Break beam switching, real-time monitoring and management, rain diversity support, service model fidelity, scalable architecture, standards-based interoperability, robust hardware and software security.

For information and full coverage of these topics, see Chapter 1 , Velocity System Overview.

Velocity System Operations Overview

The Velocity System Operations chapter introduces and describes those key operations that are fundamental to the operation of the Velocity network system. Gaining an understanding of these system operations, their system-wide implications and requirements, is key to implementing and leveraging the rich features of the Velocity system.

Topics presented in the ***Velocity System Operations*** chapter include:

- **Maps and Regulatory Areas** — reviews the various map file types and how the information they contain is used in the Velocity system.
- **Satellite Virtual Networks (SVNs)** — reviews how SVNs allow remote terminals to send and receive data over satellite as if directly connected to the customer's private network.
- **Overall Layer 3 (IP) Network Architecture** — reviews the Velocity SVN types, IP architecture, support for IPv4/IPv6 addressing and the BGP implementation.
- **Backbone Network** — reviews the possible global/regional backbone network that provides connectivity between Velocity SASs, NOCs and multiple POPs, peering sites, or other locations that support high-bandwidth connectivity to customer networks.
- **Subscriber Networks and the Internet** — briefly describes how the physical backbone network connects to and provides interfaces to each subscriber's private terrestrial network.
- **SAS Network** — reviews the SAS network, which provides IP connectivity to satellite routers within the footprint of the associated satellites; and examines the SAS network architecture for layer 1 through layer 3.
- **Over the Air Network** — introduces the RF satellite OTA network (described further in Chapter 3) and examines how the OTA network can also be viewed as a per-satellite star topology of IP subnets or LANs.
- **Terminal Network** — reviews how Velocity supports a variety of terminal networks deployed at the remote location — a network as simple as a few hosts directly connected to the satellite router, or a complex collection of VLANs and subnets.
- **NOC Network** — reviews how the NOC site and connections between the SAS and the NOC, are expected to meet high-availability requirements. Also introduced is the EAP portal, which is open to external users and typically deployed in a network DMZ.
- **Monitor and Control (M & C) Network**— introduces the M & C network that carries internal Velocity traffic (all administrative traffic between the Pulse NMS, SAS components, and satellite routers) across the backbone network and the OTA network in an SVN referred to as the Management or “Admin” SVN.
- **Data Traffic Routing and Control Protocols** — review routing and data protocols supported by Velocity, as well as those protocols supported by satellite routers.
- **Network QoS Service Modes** — introduces the Velocity system concept QoS Service mode concept, available to Network Operators. The QoS Service mode determines the type of service the network provides in the downstream direction and impacts the manner in which global bandwidth management is handled.

For information and full coverage of these topics, see Chapter 2 , Velocity System Operations.

Satellite Link Operations Overview

The Satellite Link Operations chapter describes how the interface between Velocity Satellite Access Station (SAS) sites and satellite terminals is managed. Satellite link operations include those processes and timing mechanisms by which terminals acquire into the network — as well as the various modulation, encoding, and error correction techniques applied to both upstream and downstream transmissions.

Topics presented in the ***Satellite Link Operations*** chapter include:

- **Network Acquisition** — describes how satellite terminals use signaling carriers to join Velocity TDMA networks. Satellite beam information, including beam footprint data, and link parameters such as outbound carrier frequency, symbol rate, and polarization, are broadcast to terminals using a signaling carrier.
- **DVB-S2 Overview** — introduces the DVB-S2 open standard - an extension to the widely-used DVB-S standards set for satellite digital video broadcasting, along with implementation enhancements.
- **Downstream Carrier Operation with DVB-S2** — describes how Velocity downstream carrier operation uses DVB-S2 with ***Adaptive Coding and Modulation (ACM)***. ACM enables each terminal to operate at its most efficient coding and modulation at any moment in time, depending on the location within the satellite contour, the antenna size, and atmospheric conditions.
- **Upstream TDMA Waveform** — describes the ***Time Division Multiple Access (TDMA)*** waveform, used on the Velocity inbound channel. The line card demodulator enables the configuration of multiple heterogeneous channels. These channels share a dedicated receiver that is responsible for timing, frequency and phase corrections, using state-of-the-art signal processing.
- **Upstream Carrier Operation with A-TDMA** — describes how the Velocity ***Adaptive-TDMA*** (A-TDMA) implementation allows terminals to dynamically adapt their SAS transmissions, in real time, to the optimal symbol rate and MODCOD based on the prevailing conditions.
- **Uplink Control Process** — introduces the iDirect ***Uplink Control Process*** (UCP), which executes at the SAS to bring terminals into the network and to maintain the correct frequency, power, and timing during uplink operation.

For information and full coverage of these topics, see Chapter 3 , Satellite Link Operations.

Global Bandwidth Management Overview

The Global Bandwidth Management (GBWM) chapter describes how this primary Velocity™ subsystem manages bandwidth distribution and allocation, as well as enforces global bandwidth usage across the entire network.

Topics presented in the ***Global Bandwidth Management*** chapter include:

- **Introduction** — starting with the GBWM tree structure, this topic illustrates how a total pool of bandwidth capacity is partitioned into smaller partitions for distribution and allocation to various market segments, business partners, and subscribers. This introduction is covered in the following topics:
  - **Services** – describes how service plans in Velocity, represented by GSPs and the child SSPPs/SSPCs, form the GBWM tree structure. Services that are created by service providers and provided to subscribers, represent the foundation of the satellite operator and service provider business models.
  - **Bandwidth Allocation** – further describes how available bandwidth is allocated according to the GBWM tree, derived Global QoS Enforcement (GQE) trees, the configuration of GSP and SSPC tree nodes, and current network conditions. How from this structure, demand originates and flows up the tree from subscribers, and finally, how available capacity is allocated down the tree.
- **Service Plan Definition** — introduces how the Velocity GBWM supports the construction of very sophisticated service and bandwidth allocation models — how network operators can create network subgroups with various levels of service on the same outbound carrier or inroute group; how bandwidth is subdivided among customers or service providers; and finally how bandwidth can be over-subscribed, so bandwidth unused during a given period, does not have to go unused, but can be redirected. This section is covered in the following topics:
  - **Subscriber Service Plans** – examines basic tools and configurable options in Velocity, used by service providers to develop unicast/multicast SSPPs — to provide services directly to subscriber terminals.
  - **GBWM Tree Structure** – examines how the tree structure formed by unicast and multicast GSPs and their child SSPPs/SSPCs, represent the partitioned design of the total bandwidth pool and the competition for that bandwidth.
  - **Group Service Plans (GSPs)** – describes unicast GSPs and multicast GSPs, the role they play in creating a pool of bandwidth for inbound or outbound traffic, and the main configuration parameter sets of each plan type.
  - **Service Features Common to GSP and SSPP** – describes plan features such as Start/End Date schedule, Geographic scope of coverage, Recurrence Patterns, and Fair Access Policy (FAP), which can be configured for both GSPs and SSPPs.
  - **Service Plan Satellite Link-Related Parameters** – examines the main service plan parameters, as well as operational characteristics and configuration components of the DVB-S2 outbound links and the Adaptive-TDMA inbound links.
  - **NMS Service Plans** – provides a brief overview of the NMS requirement of having bandwidth allocated on each beam for handling of management traffic.

- **GBWM Operations** — introduces the QoS Enforcement Engine components, including the global QoS (GQoS), SAS QoS, and iNet QoS, and examines closer how the Velocity GBWM subsystem allocates available bandwidth according to the GQE tree, which reflects the configuration of GSP and SSPP nodes in the GBWM tree and bandwidth demand originated by terminal SSPCs. This section is covered in the following topics:
  - **Global QoS Enforcement (GQE) Tree** – describes the development of the GQE tree and how its governing rules support bandwidth allocation methods across groups and subgroups, to determine allocation when demand exceeds availability.
  - **Bandwidth Allocation within an iNet** – examines how the iNet QoS Enforcer builds its iNet GQE tree from the GBWM tree and allocates bandwidth on a per iNet basis. The topic also describes iNet QoS demand and allocation concepts such as ***outbound demand***, ***inbound demand***, ***outbound allocation*** and ***inbound allocation***.
  - **Global Bandwidth Allocation** – examines how the Global QoS Enforcer (GQE) component of the GBWM works in conjunction with the iNet QoS Enforcer and the distributed NOC and SAS GQE components to manage global QoS enforcement for the entire system.
  - **Allocation Fairness** – examines a number of allocation fairness methods that enable service providers to control how bandwidth is allocated by a GSP when bandwidth demand from child nodes exceed the bandwidth allocated to the GSP.
  - **Operating in IP-based/Symbol-based Mode** – briefly describes the Velocity system operating in either the IP-based or Symbol-based QoS Service mode option; and switching between the modes.

Redundancy and the Velocity Network Overview

The Redundancy and the Velocity Network chapter describes the various operational modes, component and subsystem redundancy schemes, and associated failover methods supported by the Velocity network. These flexible operational modes, which operate between primary and backup sites, support high-availability and continuous Velocity operation despite interruptions due to inclement weather, hardware failures, or manual backup switchovers.

Topics presented in the ***Redundancy and the Velocity Network*** chapter include:

- **Rain Diversity Configuration** — describes the Rain Diversity configuration, which allows network designs that minimize the impact of weather events on over-the-air performance.
- **SAS Site Redundancy** — examines SAS redundancy and outlines the switchover from the primary SAS to a backup SAS.
- **CDC Site Redundancy** — examines CDC site redundancy and outlines the switchover from the primary CDC to a backup CDC site.
- **RF Gateway Site Redundancy** — examines RF Gateway redundancy and outlines the switchover that operators can perform to externally control rain diversity, perform maintenance on an idle RF Gateway, and protect against catastrophic failures. This switchover can be performed using real-time commands from the NMS UI and API.
- **NOC Site Redundancy** — examines NOC redundancy and switchover, which ensures the system continuity that provides service even upon losing the primary NOC.
- **NMS Site Redundancy** — examines how NMS redundancy and failover methods ensure that a failure of any NMS element does not cause an NMS service outage or degradation of Velocity system performance.
- **Component Redundancy** — describes the processes and system design features that ensure that the failure of any individual component does not cause service outage or degradation of system performance.
- **Network Resiliency** — describes how the SAS, NOC, and EAP network architectures are designed to provide high-availability.

For information and full coverage of these topics, see Chapter 5 , Redundancy and the Velocity System.

Pulse NMS Operations Overview

The Pulse NMS Operations chapter offers a broad description of the iDirect Pulse® Network Management System (NMS). Pulse facilitates the configuration of all system components, the control of system operations, and the reporting of network events and alarms. Normal operation of the NMS is centralized at the NOC. However, the NMS also supports distributed operations including SAS-level operations when the NOC is not accessible or when the SAS requires maintenance and troubleshooting. The EAP Portal also extends the NMS support.

Topics presented in the ***Pulse NMS Operations*** chapter include:

- **Virtualization/NMS Cluster** — provides a quick overview of how iDirect Pulse employs virtualization technologies to minimize the time and cost of deployment, and supports improved system management for IT staff.
- **SAS NMS Processes** — describes each of the server processes that comprise the Pulse SAS NMS. Each of these processes performs a specific set of back-end functions vital to the coordination and management of the overall operation of the NMS.
- **NOC NMS Processes** — describes each of the server processes that comprise the Pulse NOC NMS. The NOC processes are identical to those operating in the SAS NMS, however, the Latency Measurement Manager operating in the SAS NMS, is not in the NOC NMS.
- **External Access Portal (EAP) NMS** — describes the Pulse EAP system that resides outside of the primary NMS system, but supports controlled access to the primary NMS via the EAP NMS Proxy. Each EAP instance is responsible for providing limited NMS functionality to users and systems that are external to the service provider, including secondary service providers and subscribers.
- **NMS Redundancy and Failover** — describes how the NMS redundancy and failover implementation ensures that failure of any NMS element does not cause an NMS service outage or degradation of Velocity system performance - for example, using the underlying virtualization platform for reliable storage and virtual machine operations.
- **Network Monitoring and Reporting** — introduces NMS monitoring and reporting capabilities, which provide network operators with access to detailed real-time and historical alarms and events reports, as well as operational and statistical performance information about the Velocity network.
- **NMS Troubleshooting Tools** — briefly introduces the NMS ***probe commands***, which support real-time interaction with the operating systems of satellite terminals, line cards, and server elements; and the engineering ***debug console***, which provides direct access to line cards, terminals, NMS and PP servers, using a shell window.

For information and full coverage of these topics, see Chapter 6 , Pulse NMS Operations.

Satellite Terminal Operations Overview

The Satellite Terminal Operations chapter describes the Velocity network terminals, which provide the interface between the customer LAN — for example, shipboard, land-based, or Aero-based LAN — and the space segment of a Velocity network. Terminal operations are one side of the Over-the-Air (OTA) interface for communication with the Velocity SAS.

Topics presented in the ***Satellite Terminal Operations*** chapter include:

- **Functional Overview** — provides a quick overview of how terminals enter the Velocity network and the primary function they perform.
- **Terminal Components** — describes the terminal and the components it consists of in its simplest form; and additional equipment, which may vary for fixed and mobile units.
- **Satellite Router Software** — provides a brief overview of satellite router software components — such as board support software, OTA software components, and IP application support — and the functions for which they are responsible.
- **Satellite Router Interfaces** — provides a brief overview of the satellite router interfaces and routing protocols.
- **Satellite Terminal Commissioning** — provides an overview of the installation and commissioning sequence, and how the process ensures the configuration and setup will not cause interference or disruption of service after the terminal acquires the network.
- **Satellite Terminal Activation** — describes how a configured terminal is unable to enter the network until activated — and that activation is performed on the NMS by an operator setting the terminal to "active," prior to terminal commissioning.
- **Satellite Router Configuration** — introduces and describes how a terminal can operate in Velocity, based on either its ***static configuration*** or its ***dynamic configuration***.
- **Satellite Terminal Network Acquisition** — describes the events of the terminal acquisition process — a process by which a terminal starts in an initial, unconnected state, and then enters into a state where it can transfer user data.
- **Satellite Terminal Component Mismatch** — identifies the sequence of events that occur when a terminal's key value string, identifying its key components, does not match the equivalent string at the SAS.
- **Satellite Terminal Operational States** — describes the four operational states of a satellite terminal, including ***satellite search***, ***locked to acquisition signaling carrier (ASC)***, ***acquisition***, and ***data communication***.
- **Satellite Terminal Beam Switching** — extends the discussion of satellite/beam switching of Chapter 2, to include the switching mechanics to a new beam or satellite.
- **Satellite Terminal GPS Validation** — describes how GPS information is provided to terminals, either manually through user input or through an external device.

For full coverage of these topics, see to Chapter 7 , Satellite Terminal Operations. Also see Chapter 8 , Satellite Terminal Features.

Velocity Features Chapters

The *Velocity Technical Reference Guide (TRG)*, for the most part, describes the various components and arrangements of a Velocity network, and its inherent network operations.

Velocity operations, described in the TRG, have been grouped as overall *System Operations*, *Satellite Link Operations*, *Bandwidth Management Operations*, *Redundancy Operations*, *NMS Operations*, and finally, *Satellite Terminal Operations*.

The remaining chapters of the TRG, are dedicated to describing specific Velocity features. These chapters are listed below:

Satellite Terminal Features

Fair Access Policy

Multicast Overlay

Statistics Reports and SLA Management

Layer 2 Over Satellite

1 Velocity System Overview

This chapter presents a high-level overview of the iDirect Velocity™ System and iDirect Pulse® Network Management System, by way of brief introductions to some of the main architectural and operational elements of Velocity and Pulse systems.

The following sections are included in this chapter:

1.1 About the Velocity Network 12
1.2 Velocity Space Segment 13
1.3 Velocity Ground Segment 15
1.4 Velocity Network Architectures 24
1.5 Satellite Business Service Models 26
1.6 Velocity System Characteristics 31

1.1 About the Velocity Network

The iDirect Velocity™ network system delivers high-speed mobile broadband services on a global basis using satellite-based access technology operating in the Ku- and Ka-bands. Velocity networks are also able to support spectral inversion on both the hub-side and remote-side, to allow these components to operate with C-band networks. These services are the basis of the iDirect ground system solution, which supports mobile broadband communications for deep-sea vessels, in-flight connectivity for aircraft; and streaming high-resolution video, voice, and data applications.

The Velocity System is complemented by complete management and configuration software services enabled by the iDirect Pulse® Network Management System.

In its simplest form this iDirect solution is represented in the following illustration. The Velocity System, at the center of the diagram, consists of the following:

- Satellite Access Station (SAS)
- Central Data Center (CDC)
- RF Gateway
- Network Operations Center (NOC)
- External Access Portal (EAP)
- Satellite Terminals

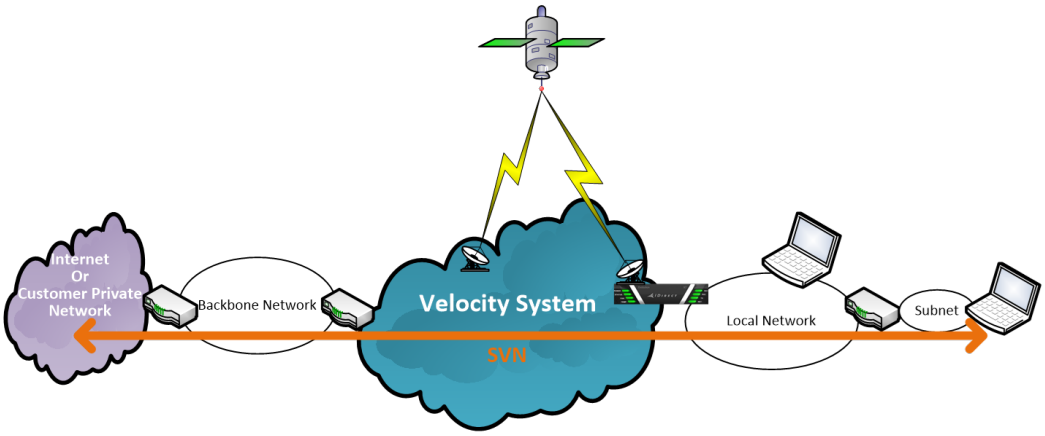


Figure 1-1. The Velocity Ground System Solution

The connection from the Internet or private network to the user is defined as a Satellite Virtual Network, or SVN. SVNs implement service provider (SP), Value Added Reseller (VAR), Virtual Network Operator (VNO), distribution partner (DP) and subscriber business models by extending a private network across the shared satellite network, allowing customer devices to send and receive data as if they were directly connected to the customer's private network. An SVN is essentially a Virtual Private Network (VPN).

1.2 Velocity Space Segment

The Velocity space segment is designed for worldwide coverage and supports dynamic bandwidth allocation to different parts of the earth. This segment, as illustrated in the following diagram, uses downstream [DVB-S2] and upstream [A-TDMA] satellite transmissions.

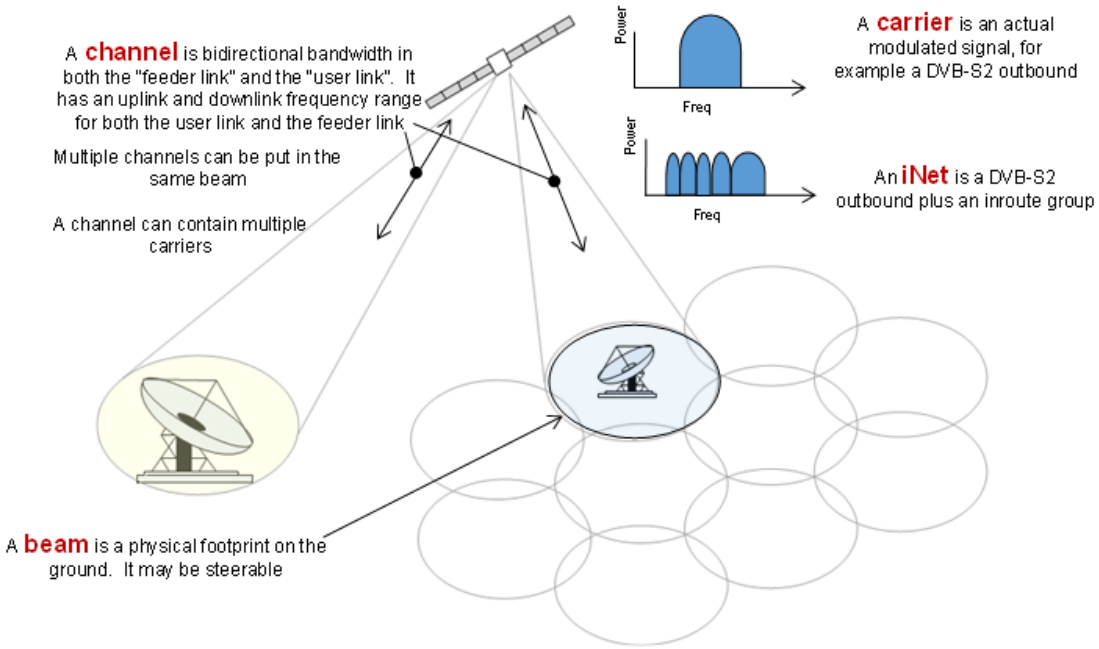


Figure 1-2. Beams, Channels, Carriers, and iNets

Velocity Space Segment Components

The following terms describe components of the space segment and how they relate to the Velocity Over-the-Air (OTA) system.

- **Beam** — Each satellite provides a collection of beams that are characterized by their footprint on the earth's surface. Depending on the satellite, the beam may have the same or different footprints for the uplink (toward the satellite) and downlink (toward the ground). A terminal must be within a beam to enter the satellite network. At any time a given beam may have zero or more associated channels. If a beam is assigned zero channels, it is unusable by terminals and is sometimes referred to as a ***dark beam***.
- **Channel** — A satellite also provides channels and associates those channels with beams. A channel is bidirectional bandwidth in both the feeder link and the user link. It has an uplink and downlink frequency range for both. A channel is either associated with exactly one beam, or it is unused. A specific channel contains uplink and downlink carriers that form one or more iNets. The bandwidth of a channel is the sum of the bandwidth of the iNets it contains. Channels in different beams may use the same frequencies. Where beam footprints overlap, the assignment of channels to beams must

be such that no location on the Earth's surface is illuminated by two channels of the same frequency or polarization.

- **Carrier** — A channel consists of a set of carriers. A carrier is a single modulated RF signal or waveform carrying encoded data in one direction. It is referred to as an inbound or outbound carrier with respect to the antenna sending the signal (outbound) or receiving it (inbound). Carrier types supported by Velocity include DVB-S2, and TDMA.

  If a carrier is used only to broadcast (signal) beam map information to terminals, it is referred to as a ***signaling carrier***. If it is used to provide actual user data service, it is referred to as a ***service carrier***. Note that Velocity also broadcasts beam map information on service carriers.

  The signaling carrier is used to broadcast the satellite network beam configurations to the terminals located within the network. For each beam, this contains:

  - A low resolution beam map
  - An outbound carrier frequency, symbol rate, and polarization

  The broadcast needs to be received by all terminals within the beam, regardless of the signal strength at their particular location and the sensitivity of the terminal's antenna. Thus the lowest available MODCOD (modulation and coding), QPSK 1/4, is the default for signaling carriers.

- **Feeder and User Links** — The group of uplink and downlink carriers between the Satellite Access Station (SAS) and the satellite is sometimes referred to as a ***feeder link***. The group of uplink and downlink carriers between the user terminal and the satellite is likewise referred to as a ***user link***.

- **iNet** — A channel contains one or more ***iNets***. From the perspective of the SAS/Teleport, an iNet consists of an outbound DVB-S2 carrier of up to 45 Msps, together with a set of TDMA inbound carriers. Transmit (Tx) and receive (Rx) line cards are assigned to an iNet, and each card supports exactly one iNet.

- **Inroute/Inroute Group** — A satellite terminal transmits a burst during a timeslot on an ***inroute***. The collection of carriers (inroutes) available to the terminal are an ***inroute group***. All inroutes in an inroute group are assigned consistent transmission attributes so that a terminal may transmit on various inroutes without changing anything other than transmission frequency.

For additional information, see Chapter 3 , Satellite Link Operations.

1.3 Velocity Ground Segment

The Velocity ***ground segment*** refers to the terrestrial components involved in providing satellite services. In a Velocity system, the term ***site*** refers to a collection of equipment and processes installed at a designated location for a specific purpose. As the term implies, a site is usually a physical location that is different from other sites, but this is not always the case.

A user can configure any one of the following site types, where each site can be designated as a primary or backup site:

- SAS
- NOC
- EAP
- CDC
- RF Gateway

Although Pulse allows the type of site to be configured, changing the type of site requires the installation and configuration of additional or different hardware elements.

A ***teleport*** is a facility that serves one or more satellites. In the context of the Velocity network, a teleport is a physical location at which one or more SAS or RF Gateway sites are installed. CDC, NOC and EAP sites are usually installed at a data center, but may also be collocated in a teleport.

As illustrated below, the Velocity system supports full backup elements, which may be added to increase availability and reliability. See Chapter 5 , Redundancy and the Velocity System.

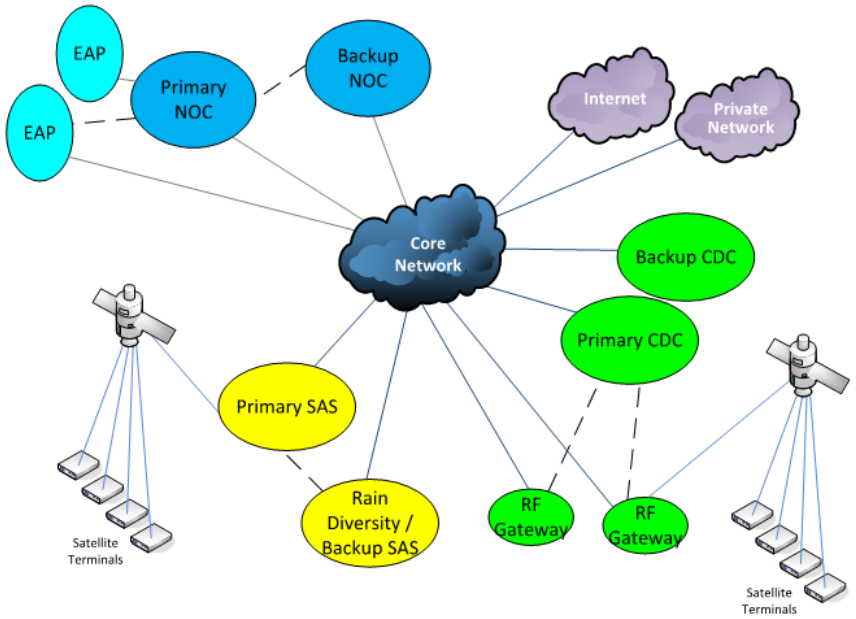


Figure 1-3. Velocity System—SAS and CDC/RF Gateway Ground Segments

1.3.1 SAS and CDC/RF Gateway Architectures

Two ground segment network options are available for deployment in the Velocity network—the traditional Satellite Access Station (SAS) architecture, and the Central Data Center (CDC) and Radio Frequency (RF) Gateway architecture. Both ground segments provide communications between the Service Provider Data communications Network (DCN) and Satellite Terminals.

The SAS site, sometimes referred to as a hub, is generally based on an NMS cluster, PP cluster, hub chassis and line cards, all co-located at the same site and generally sufficient to provide service over a single feeder link for one satellite.

The CDC/RF Gateway implementation is based on geographically distributed components in which the CDC, which contains an NMS and PP cluster, is geographically separated from the RF equipment. It offers a scalable alternative to the SAS, where the total bandwidth capacity of spot beams outstrip the feeder-link capacity of any single location. Packet Processing (PP) functions, in the CDC, are consolidated at one data center instead of within the footprint of the satellite's feeder-link. In this arrangement, multiple geographically separated RF gateways can support different feeder-links. In addition to supporting the same features and functions as the SAS, the CDC and RF Gateway architecture manages terrestrial delay and jitter, as well as packet loss and reordering between the RF Gateways and Data Centers.

A summary of CDC and RF Gateway Architecture supported capabilities are listed below:

- Supports integration of new components, including ILC/iHub and iGW, which accommodate newer and larger satellites and networks with higher throughput
- Configuration, Monitoring and Debugging of CDC/RF Gateway components, including ILC, iHub, and iGW objects are supported by the NMS UI and API
- All components support redundancy and switchover with no single-points of failure
- CDC Site switchover performed using manual steps from both iGW/PP and NMS consoles
- CDC Site switchover performed using manual steps from the NMS UI and API
- Active/Standby mode of CDC and components is obtainable from NMS UI and API
- RF Gateway Redundancy supports external control over rain diversity and maintenance on an idle RF Gateway
- RF Gateway switchover performed using real-time commands from the NMS UI and API
- CDC and RF Gateway architecture supports association of beam sets with RF Gateways to align Velocity and Satellite payload configurations

1.3.1.1 The SAS Site Architecture

The term *Satellite Access Station (SAS)* site refers to a site that has an operation mode of SAS. Multiple SAS sites can exist at the same physical location—in other words, in a teleport.

A SAS site supports a single satellite within a satellite network. This type of site provides the network interface between the IP network of the backbone network and the radio frequency (RF) links to a satellite. The space segment configuration can be scaled up by adding beams and channels to an existing satellite. Likewise, the supporting SAS can be scaled up, using additional line cards and/or possibly additional servers.

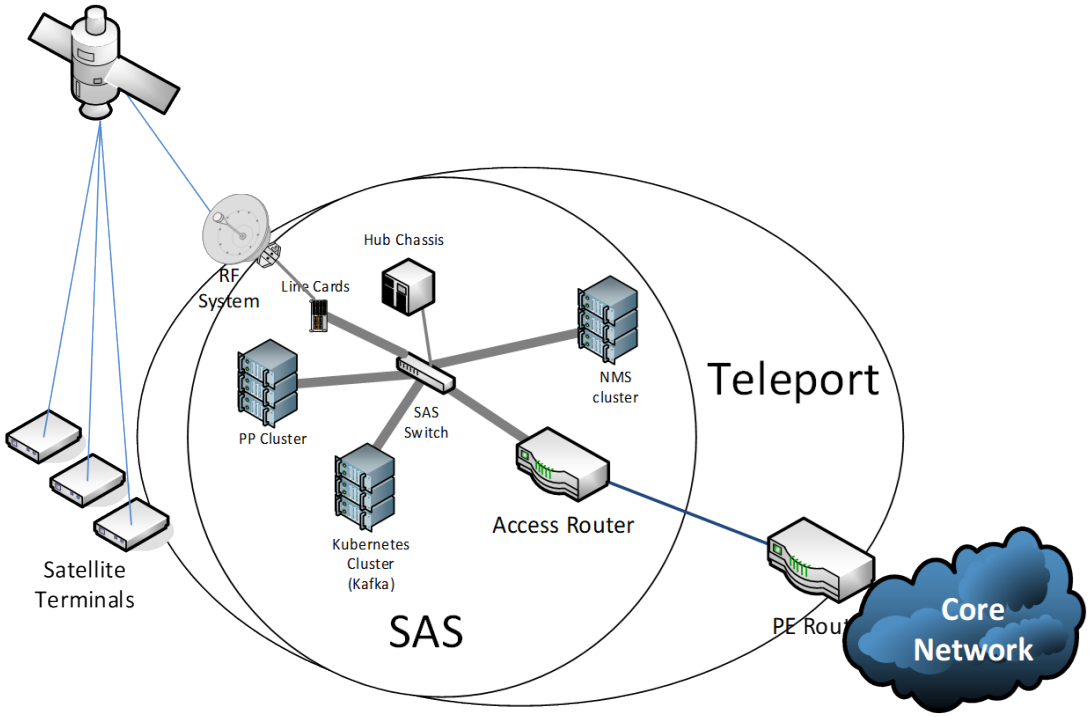


Figure 1-4. Satellite Access Station Architecture

1.3.1.2 CDC and RF Gateway Site Architecture

Using the CDC architecture, Service Providers can consolidate Packet Processing (PP) functions at a data center, typically within a metropolitan location, instead of within the footprint of a satellite feeder-link. The RF gateway functions are maintained at one or more geographically remote locations. This arrangement places the hub and line cards at a site referred to as an “RF Gateway” site, and the PP and management functions at a “CDC site". As seen in the diagram, from one PP function in a central data center, this architecture supports satellites having multiple feeder-links.

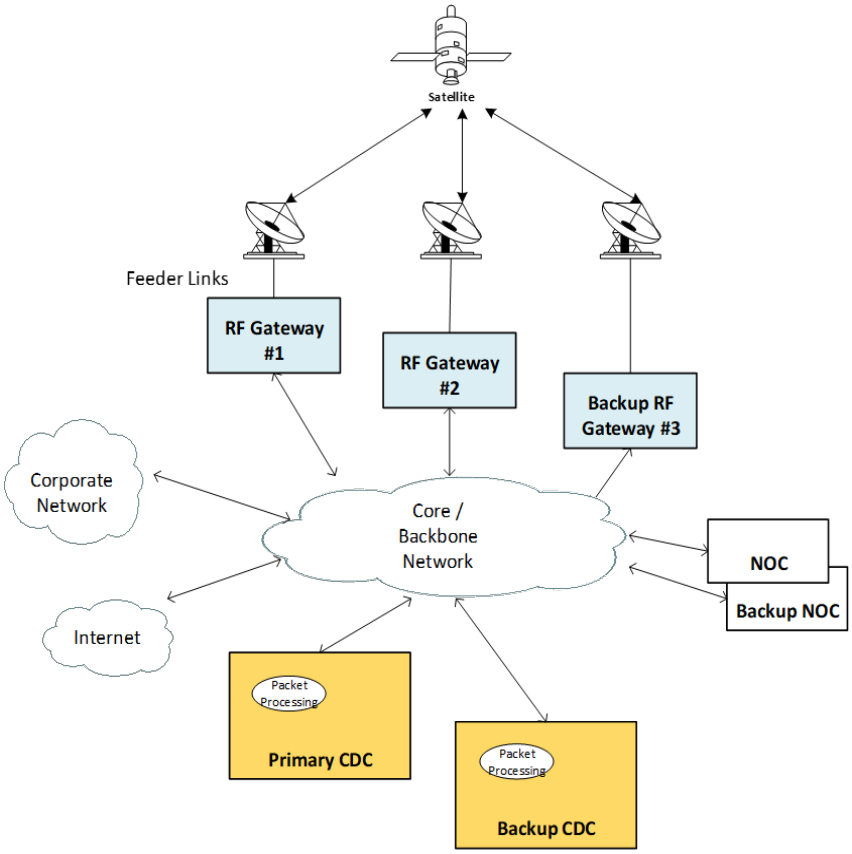


Figure 1-5. Central Data Center and Radio Frequency Gateway Architecture

Since a major characteristic of the CDC architecture is that the RF Gateway is located away from the Protocol Processors, across a WAN, the Velocity system also introduces mitigation that enables the system to tolerate the effects of latency, jitter, packet loss, and packet reordering for both control and data traffic exchanged by the PP and line cards. The same features and functions available in the traditional SAS are available, while accommodating terrestrial delay, jitter, packet loss and reordering between RF Gateways and the Data Center.

The CDC and RF Gateway architecture also supports Velocity system high availability. Site redundancy and switchover, with no single point of failure, are provided for both the RF Gateways and the CDC itself. Rain diversity switchover is accomplished through an RF

Gateway site switchover that is triggered either through manual operator action or automatically using the NMS or NMS API.

Finally, to support increase of scale and efficiency, this architecture integrates the Intelligent Gateway (iGW) which hosts the PP functions, and the Intelligent Hub and Intelligent Line Cards (iHub and ILCs). Further, downstream packet encapsulation for ILCs is performed on PP/iGW, rather than on the line card.



NOTE: The CDC switch and router are expected to support jumbo frames and an MTU of 9000 without duplication. Furthermore, all components are expected to be connected to the CDC switch using 10GigE ports, must support jumbo frames and with the exception of the WAN link, are expected to support an MTU of 9000 without fragmentation.

1.3.1.3 SAS, CDC, and RF Gateway Components

With the exception of the hub chassis, all SAS components support exactly one satellite. A physical hub chassis can be shared by multiple SASs; the Pulse operator assigns a subset of the chassis' slots to the Pulse representation of the chassis within its representation of the SAS. Furthermore, multiple SASs may also support the same satellite; the Pulse operator creates separate representations of the satellite for each SAS and assigns the appropriate subset of the beams, channels to the satellite representations.

SAS, CDC, and RF Gateway components listed (applicable components depend on site):

SAS Site

- **15152 Hub Chassis**—Provide housing and bus structure for RF transmit/receive line cards.
- **Line Cards**—Provide intermediate-frequency (IF) modulation and demodulation for all outbound and inbound OTA traffic.
- **SAS NMS Cluster** — An element of the Pulse NMS, responsible for event monitoring, alarming, and statistics collection for the SAS components and terminals they support.
- **SAS Switch** — Provides local area network (LAN) connectivity for components in the SAS.
- **Protocol Processor (PP) Cluster**—Performs data stream processing, packet routing and forwarding, terminal control and monitoring, load balancing and automatic redistribution of terminal load across PP servers, and other network operations. The PP is essentially an IP router. A PP cluster also resides at the NOC to handle global bandwidth management.
- **Access Router** — Provides IP networking between the components of the SAS network and the terrestrial backbone network.
- **RF System** — Provides the radio frequency interface to and from satellites, includes a large earth station antenna, low noise amplifiers (LNAs), high power amplifiers (HPAs), and associated equipment.
- **iHub**-Provides housing and a high speed bus structure with 10GbE uplink support for the intelligent RF transmit and receive line cards (ILC-T /ILC-R). A compact 3 RU Chassis, the

iHub supports redundancy, hot-swappable modules, control, timing, power, and cooling for up to 12 line cards.

- **iGateway-**The iGW is a virtualized platform that hosts the Velocity Protocol Processor cluster. Each virtual PP node operates like a bare-metal PP server. The iDirect virtual server is a Protocol Processor (PP) server that is installed, executed, and hosted as a guest virtual machine on the designated server machine. Physically speaking, the iGW platform is a number of sleds in one or more FX2 chassis. Each sled hosts a number of virtual machines, each of which is a node in the PP cluster.

CDC Site

- **iGateway (iGW)**—The iGW is a virtualized platform that hosts the Velocity Protocol Processor cluster. Each virtual PP node operates like a bare-metal PP server. The iDirect virtual server is a Protocol Processor (PP) server that is installed, executed, and hosted as a guest virtual machine on the designated server machine. Physically speaking, the iGW platform is a number of sleds in one or more FX2 chassis. Each sled hosts a number of virtual machines, each of which is a node in the PP cluster.
- **SAS NMS Cluster** — An element of the Pulse NMS, responsible for event monitoring, alarming, and statistics collection for the SAS components and terminals they support.
- **SAS Switch** — Provides local area network (LAN) connectivity for components in the SAS.
- **Protocol Processor (PP) Cluster**—Performs data stream processing, packet routing and forwarding, terminal control and monitoring, load balancing and automatic redistribution of terminal load across PP servers, and other network operations. The PP is essentially an IP router. A PP cluster also resides at the NOC to handle global bandwidth management.
- **Access Router** — Provides IP networking between the components of the SAS network and the terrestrial backbone network.
- **iGateway-**The iGW is a virtualized platform that hosts the Velocity Protocol Processor cluster. Each virtual PP node operates like a bare-metal PP server. The iDirect virtual server is a Protocol Processor (PP) server that is installed, executed, and hosted as a guest virtual machine on the designated server machine. Physically speaking, the iGW platform is a number of sleds in one or more FX2 chassis. Each sled hosts a number of virtual machines, each of which is a node in the PP cluster.

RF Gateway Site

- **15152 Hub Chassis**—Provide housing and bus structure for RF transmit/receive line cards.
- **Line Cards**—Provide intermediate-frequency (IF) modulation and demodulation for all outbound and inbound OTA traffic.
- **RF System** — Provides the radio frequency interface to and from satellites, includes a large earth station antenna, low noise amplifiers (LNAs), high power amplifiers (HPAs), and associated equipment.
- **Intelligent Hub (iHub) Chassis**—Provides housing and a high speed bus structure with 10GbE uplink support for the intelligent RF transmit and receive line cards (ILC-T /ILC-

R). A compact 3 RU Chassis, the iHub supports redundancy, hot-swappable modules, control, timing, power, and cooling for up to 12 line cards.

- **Intelligent Line Cards (ILC)**—With higher capacity and much smaller footprint than ULCs, ILCs provide the intermediate-frequency (IF) modulation and demodulation for all outbound and inbound OTA traffic.

  The ILC-T supports up to four DVB-S2 carriers per ILC-T line card. In this arrangement, all four carriers must be configured the same, although each has a separate power control loop to properly manage the frequency-dependent cable loss. One ILC-T can also be shared across different beams, and even in such cases, all of carriers across the beams must be configured the same.

1.3.2 The Network Operating Center Site

The Velocity ***Network Operations Center (NOC)*** site is, as the name implies, an operations site. In terms of Velocity system resources, it contains the NOC Network Management System (NMS), NOC Kubernetes/Kafka cluster, and the Velocity Global Bandwith Management system.

The NOC Pulse NMS cluster, responsible for system configuration, control, monitoring, alarming, and reporting, also provides an interface to back-office systems. Pulse manages network upgrades and longer-term storage of statistics, alarms, and historical reports.

The GBWM cluster enforces QoS globally for the satellite network by adjusting satellite bandwidth allocation according to aggregate demand from all SASs and configured committed and maximum information rates (CIR and MIR) on group service plans (GSPs). It also performs FAP usage and control, and service scheduling at the GSP level. For more details refer to Chapter 4 , Global Bandwidth Management, and Chapter 9 , Fair Access Policy.

The components located in a NOC site are:

- NOC Pulse NMS Cluster
- Kubernetes/Kafka Cluster
- GBWM (Global Bandwidth Management) Cluster

1.3.3 The Extended Access Portal (EAP) Site

The EAP site contains the External Access Portal. The External Access Portal is responsible for providing secure portals for service providers and subscribers. If desired, an EAP may be installed in the same physical location as the NOC. In that case, the EAP is typically open to networks not controlled by the service provider and the NOC must be protected and segregated from intrusion via those networks. For example, the EAP could be confined behind a firewall in a networking DMZ, as shown in the diagram.

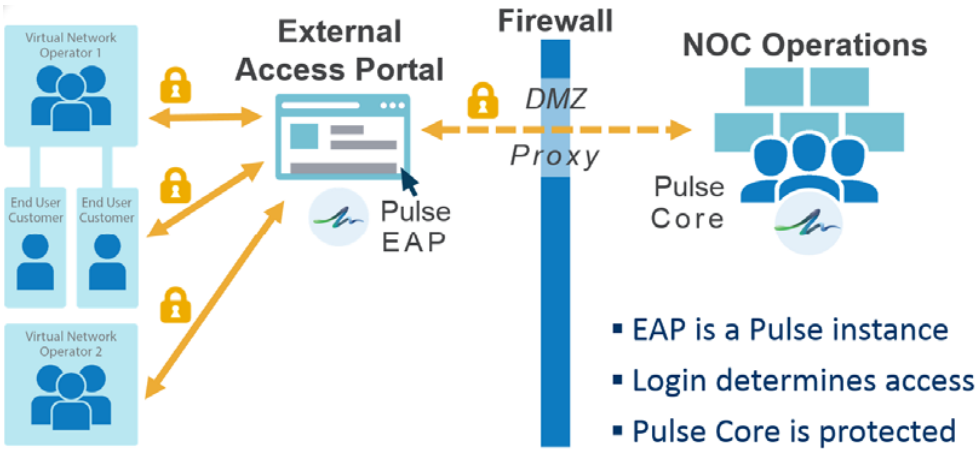


Figure 1-6. Example Pulse EAP Deployment

1.3.4 The Satellite Terminal

The satellite terminal is located at the user side of the satellite communications link. It communicates through the Over the Air (OTA) interface with the SAS associated with the satellite providing the beam in which it is physically located.

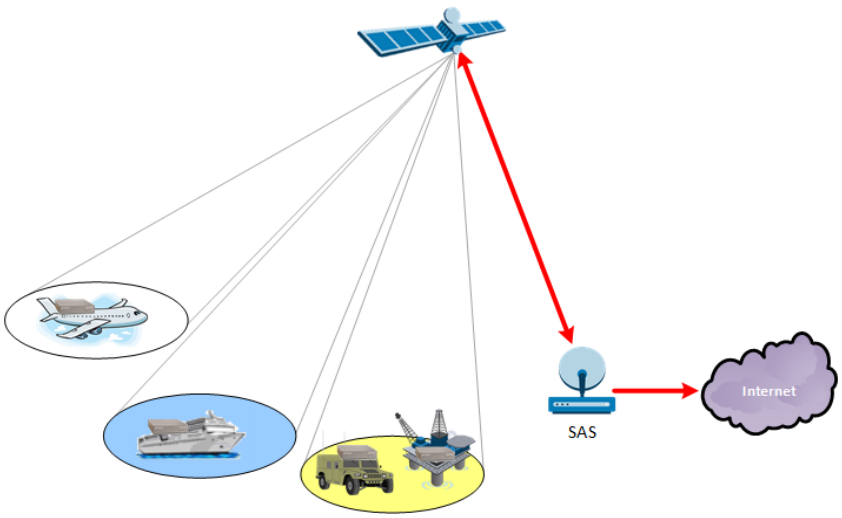


Figure 1-7. Satellite Terminal in Velocity System

Terminal Components

The equipment associated with a terminal, may vary based on whether the terminal is deployed as fixed or mobile. In its simplest form, the terminal is comprised of the following:

- **Satellite Router** (also called modem) — The satellite router is the processing and modem component of the terminal. It provides connectivity to the SAS at the satellite link, virtual local area network (VLAN), and IP layers in order to forward IP packets, enabling applications such as voice-over-IP (VoIP), video, and data. The satellite router also directs the operation of other terminal components, including the ACU, BUC, and LNB.
- **BUC (Block Up-Converter)** — The BUC is used for uplink satellite transmission and amplification, converting a band from a lower frequency to a higher frequency.
- **LNB (Low Noise Block Converter)** — A combination low noise amplifier and down-converter built into one device attached to the feed, the LNB is used for satellite downlink signal reception by converting a higher frequency band to a lower frequency.
- **Antenna** — The equipment that transmits and receives electromagnetic waves, and usually referred to as the satellite dish. The antenna contains a feed assembly that gathers received high frequency signals and routes them to the LNB. The feed assembly includes an orthomode transducer that combines the Tx and Rx RF signals, which allows simultaneous reception and transmission of signals through the same feed and reflector. The feed assembly may also contain a servo motor that rotates the probe to allow for reception and transmission of both horizontally and vertically polarized signals.

- **Antenna Control Unit (ACU)** — The antenna control unit connects to the satellite router over OpenAMIP. It provides geo-location information to the satellite router, points the antenna at the satellite, and performs related functions such as providing status and, in some terminals, muting the transmitter when off-pointed or blocked.
- **Built-In Ethernet Switch** — Some satellite routers contain an Ethernet switch, which provides Ethernet interfaces that can be configured either as access or trunk ports. An access port carries the traffic of a single VLAN, whereas a trunk carries traffic for multiple VLANs.

1.4 Velocity Network Architectures

The Velocity network is an IP network that comprises a terrestrial backbone network, backbone network peering sites/POPs (points of presence), Satellite Access Station sites, a Network Operating Center site (NOC), and fixed or mobile Satellite Terminals. The terrestrial backbone network has interfaces to the Internet, to customers' private networks, and to network control equipment.

The following diagram shows the architectural components of the Velocity network, in which the Velocity IP network is logically organized into three routing domains.

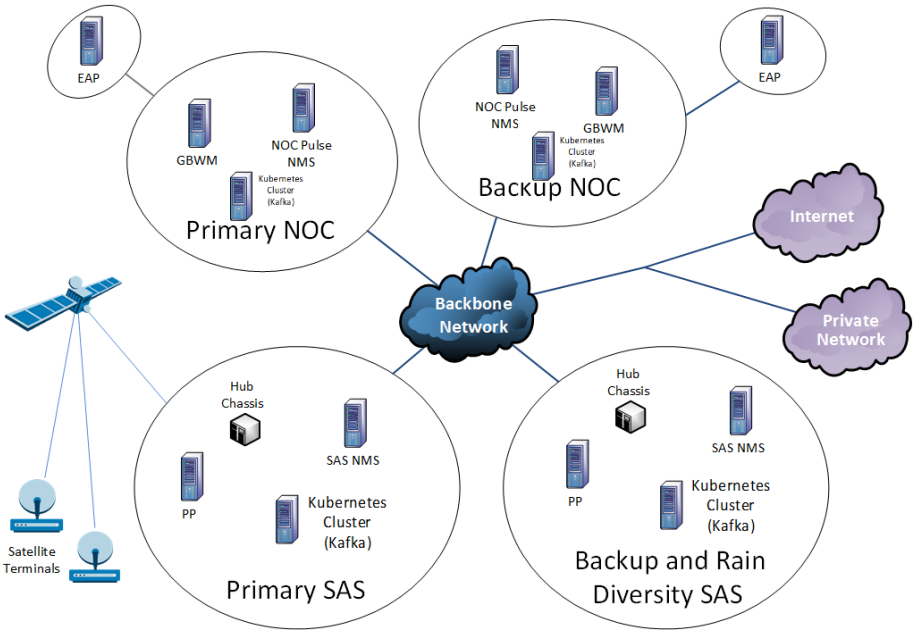


Figure 1-8. Velocity System

1.4.1 Velocity IP Routing Domains

The Velocity IP routing domains, illustrated in the following diagram, each are configured for each SVN and may simultaneously support IPv4 addressing or IPv6 addressing, or both.

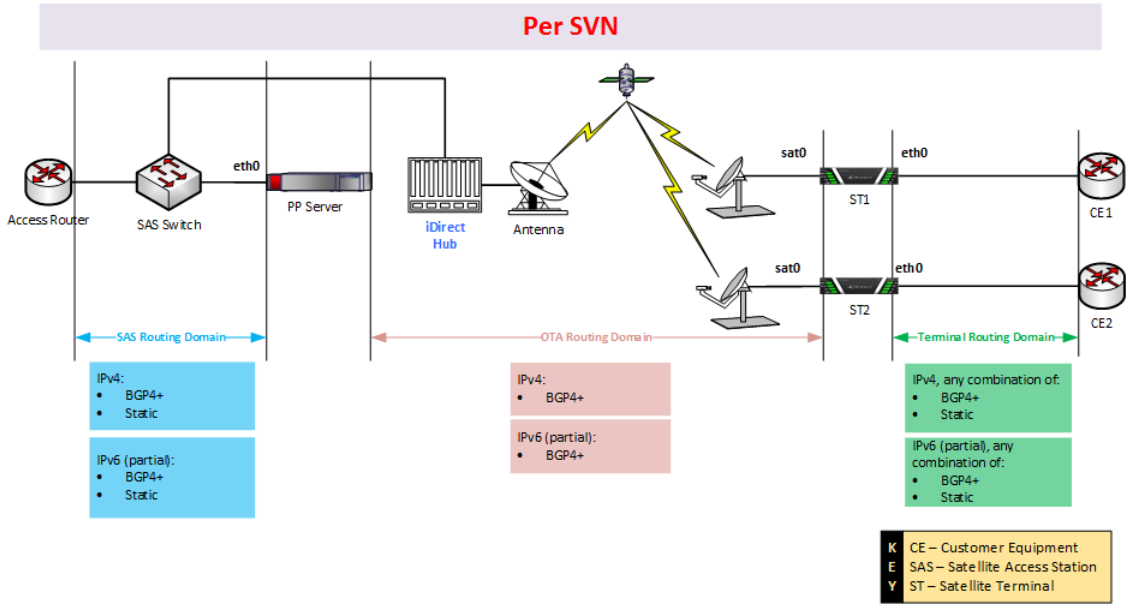


Figure 1-9. Velocity Routing Domains

The SAS or CDC routing domain of an SVN includes the IP interfaces of all PP servers and the IP interface of SAS access routers in the SAS network segment. The routing protocols available in this domain are iBGP (internal Border Gateway Protocol) and eBGP (external Border Gateway Protocol). Static IP routes may also be configured. Administrative distances are used in route selection decisions, and may be configured for iBGP, eBGP, static and connected routes.

The OTA routing domain of an SVN includes all of the virtual IP interfaces automatically created by the PP to represent the terminals it is currently serving and that are configured to support that SVN. This domain also includes the satellite interfaces (sat0) configured on the satellite terminals. In this routing domain, either eBGP or iBGP may be selected, or no routing protocol if all routes to equipment in the terminal routing domain are statically configured.

The terminal routing domain of an SVN is the satellite terminal LAN, and includes the IP interface (eth0) configured on the terminal, as well as the IP interfaces of all the devices (i.e., hosts and routers) in the terminal LAN. Static routes may be used in this domain, as well as iBGP or eBGP. Administrative distances are used for route selection decisions.

DHCP (Dynamic Host Configuration Protocol) may be used to dynamically assign IP addresses to the equipment on the terminal LAN, and RADIUS (remote authentication dial-in user service) may be used to assign the IP address of the terminal's interface to the LAN (i.e., eth0). If RADIUS is used, DHCP server must also be enabled.

For additional information, refer to Chapter 2 , Velocity System Operations.

1.5 Satellite Business Service Models

Availability of high-throughput satellite (HTS) systems is changing the way capacity is being brought to market. Instead of traditional wide-beams, HTS satellites encompass multiple spot beams with varying sizes, throughputs and functional capabilities. The Velocity Network and Pulse NMS Systems provide flexible support for implementing a blend of traditional as well as new business models to provide a variety of market solutions.

1.5.1 Basic Satellite Business Model Terminology

Before examining some of the business model options, a few useful terms are defined here:

- **Satellite Operator** — A satellite owner/operator who offers satellite bandwidth in MHz or Mbps (Megabits per second) to other service providers. In general, the satellite operator sells bandwidth to service providers such as virtual network operators (VNOs).
- **Satellite Operator/Service Provider** — A satellite operator who also provides services directly to subscribers, as described in Section 1.5.2, The Vertically Integrated Service Model. The satellite operator may simultaneously provide subscriber services and bandwidth to other service providers.
- **Service Provider** — A company who provides services over satellite links to terminals. Such a service may involve simple network connectivity, or a higher-level service like disaster recovery. Some examples of service provider offerings are:
  - Multicast services to end customers
  - Network connectivity [Mbps] to end customers
  - Disaster recovery services
  - Voice-over-IP (VoIP) services to end customers

  Service Providers who are not also satellite operators may also be referred to as ***Virtual Network Operators (VNOs)***. Specifically, a VNO is a company that purchases global or regional bandwidth capacity from a satellite operator or another VNO for the purpose of partitioning and reselling that capacity to subscribers. In general, a VNO has the ability to create and sell/resell service plans to subscribers or other VNOs. As described in Section 1.5.3, The Managed Service Model and Section 1.5.4, The Co-Located Service Model, Velocity supports a wide range of potential service models for service providers.

  VNOs are sometimes referred to as **Value-Added Reseller (VARs)**. A VNO who is limited to purchasing subscriber service plans on a wholesale basis and reselling them to individual subscribers may be referred to as a **Dealer**. In general, a dealer can only manage subscribers.
- **Subscriber** — An end user who receives service through a satellite terminal.
- **Subscriber Organization** — An organization owning many satellite terminals. Examples of subscriber organizations might include ocean-going shipping companies, airlines, companies that manage oil rigs, emergency relief organizations, infrastructure distribution networks, cellular network providers, and store chains. Some examples of subscriber organization services include:

- Connectivity to corporate network
- Internet access
- Business continuity/Disaster recovery
- SCADA (Supervisory Control and Data Acquisition)
- Voice and/or video provided by the organization

The following diagram illustrate possible relationships between these organizations. Different access levels to the system may be given to members of these organizations. This access management is handled through the Pulse NMS via user assigned permissions. For details on defining user roles, refer to ***Configuring Velocity Networks Using Pulse***.

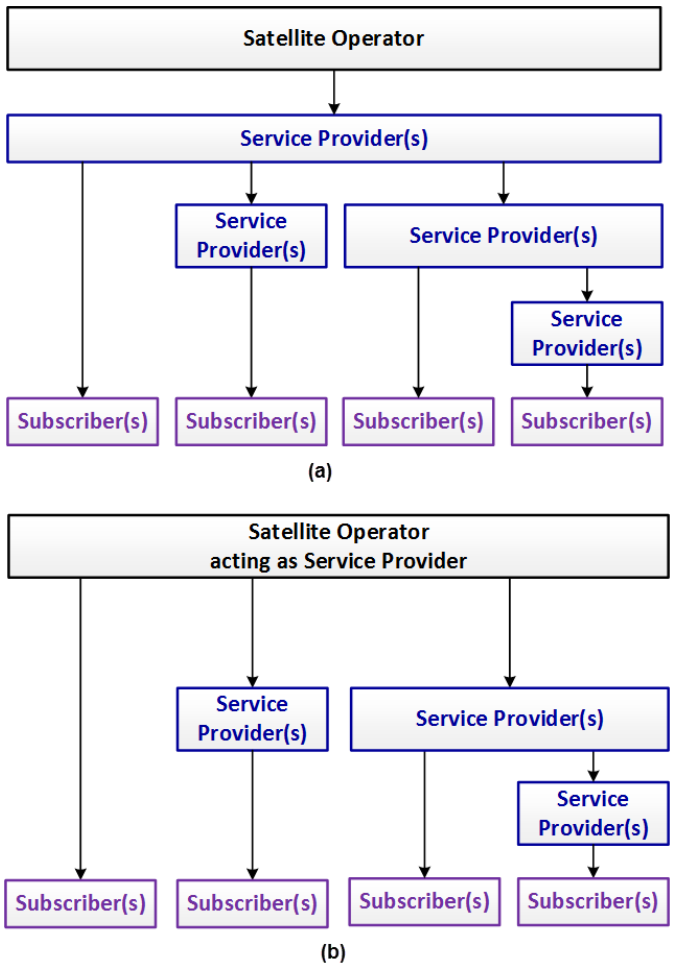


Figure 1-10. a) Satellite Operator & Service Providers b) Satellite Operator as Service Provider

	NOTE: Although three levels of service providers are shown in the previous illustrations, the tree of service providers may be any depth.

1.5.2 The Vertically Integrated Service Model

In the ***vertically integrated service model***, the satellite operator/service provider owns and operates the entire value chain — the satellites, SAS (hub infrastructure), network operations, service provision and customer relationships.

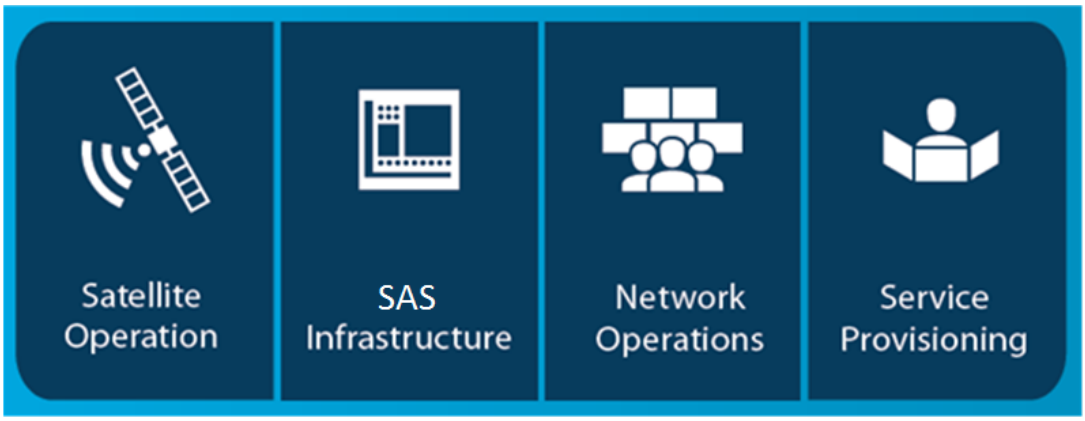


Figure 1-11. Vertically Integrated Service Model

The vertically integrated service model offers the following benefits:

- ***Benefits to the satellite operator/service provider***
  - Centralized control of service delivery, operational efficiency, economics of scale,
  - Reduced intermediary impact and network contention controls,
  - Improved usage visibility, networking segmentation, and service-specific views for end customers.

1.5.3 The Managed Service Model

In the ***managed service model***, a satellite operator owns and manages the majority of the satellite infrastructure, yet service providers have a certain amount of control over their network. This model can be a ***partially-managed model***, in which the service provider also manages a bandwidth pool; or a ***fully-managed model***, in which the service provider simply sells pre-defined services and manages customer relationships. The service provider may be referred to as a Virtual Network Operator (VNO).

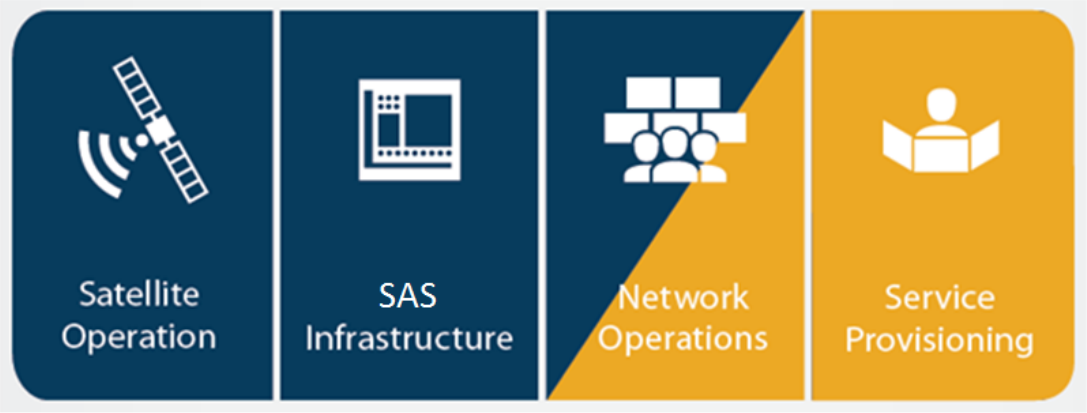


Figure 1-12. Managed Service Model — Full or Partial

The managed service model offers the following benefits:

- ***Benefits to the satellite operator***
  - Decreased initial and ongoing subscriber-related expenses, as service providers take on service provisioning and management tasks
  - Lower operating expenses with service providers taking on the more network operations tasks delegated by the satellite operator (managed service models)
  - Network versioning and control of ground systems infrastructure, end-to-end control of service profiles, and network reachability and availability
- ***Benefits to the service provider***
  - Decreases costs and time to market, rationalizing hardware-intensive HTS deployments (both models)
  - Fully-managed model minimizes capital expenditure requirements and accelerates time to market
  - Partially-managed model offers greater network control and bandwidth management

1.5.4 The Co-Located Service Model

In the ***co-located service model***, a satellite operator leases bandwidth and SAS infrastructure to a service provider. The service provider, also referred to as the VNO, establishes HTS service and has full control of its own network and end users.

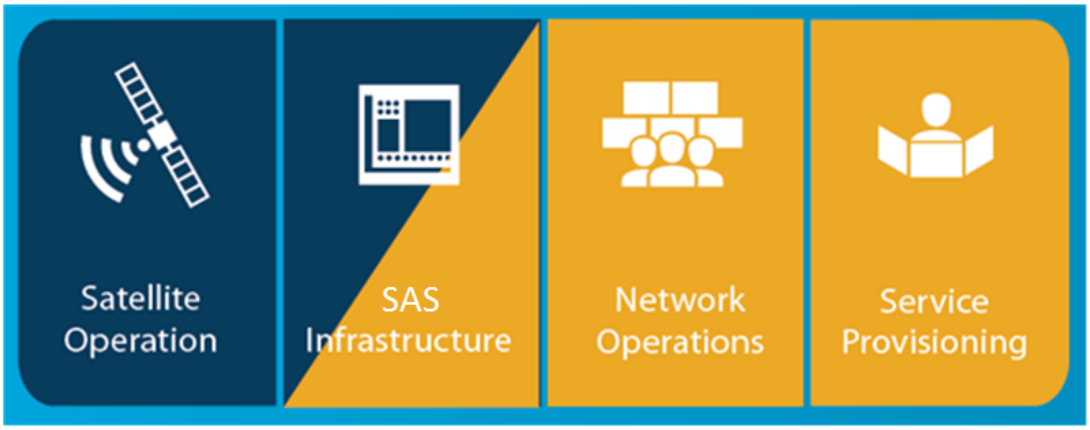


Figure 1-13. Co-Located Model with Virtual Network Operator (VNO)

The co-located service model offers the following benefits:

- ***Benefits to the Satellite Operator***
  - Lower operating expenses as service providers take on full network operations
- ***Benefits to the Service Provider***
  - Greater flexibility to tailor service levels in response to customer demand

1.6 Velocity System Characteristics

The iDirect Velocity™ system supports the following characteristics and functionality:

- **Hardware**
  - Hub Chassis/Line Cards Optimized for Large Scale HTS Networks
- **Extensibility and Scalability**
  - Scalable Architecture
  - Upgradeable Firmware/Software Components
- **Resiliency**
  - High system Availability
  - No Single Point of SAS Failure
  - Minimal-Impact Upgrades
  - Rain Diversity Support (Reduction in availability impact due to weather events)
- **Management**
  - A highly-scalable, web-based, API driven Network Management System (NMS)
  - Complete real-time monitoring and management
  - Full-system Configuration
- **Bandwidth Management**
  - Global Bandwidth Management
  - Automatic Load Balancing among overlapping beams and channels within a satellite
  - Service Areas/Independent Service Areas
  - Fair Access Policy (FAP)
  - MODCOD Scaling
- **Beam Switching**
  - Fast beam switching, with beams served by one or more gateways
  - Make-Before-Break with 2nd receiver
  - Beam switch between dissimilar beams
  - Break-Before-Make Beam Switch
  - Invitation-Based Re-Acquisition (As needed)
  - Doppler Compensation
  - Skew Angle Support
- **Performance, including packet performance**

- **Service Model Fidelity**
  - A-TDMA (Adaptive Time Division Multiple Access) technology
  - Bandwidth efficiency
  - DVB-S2/ACM (Digital Video Broadcasting - Satellite - Second Generation/Adaptive Coding Modulation)
  - Enforcement of group and subscriber service plans
  - Well-defined QoS (Quality of Service) semantics under terminal fade conditions
- **Standards-Based Interoperability**
  - End-to-end BGP (Border Gateway Protocol) Routing
  - IPv6 Addressing
  - IPv4/IPv6 Dual Stack
  - VLAN Capabilities
  - Tagging for Multi-Protocol Label Switching (MPLS)
- **Robust Hardware and Software Security**
  - Token-Based Terminal Authentication
  - X.509 SAS Authentication Certificates
  - AES 256 Over-the-Air Encryption
  - Extensible Public Key Infrastructure (PKI) Based Security Architecture
  - TLS and IPSec Secure-All Inter-Component Messaging

2 Velocity System Operations

This chapter introduces and describes those operations that either affect iDirect Velocity networks on a system wide basis or that have system wide implications or requirements.

The following sections are included in this chapter:

2.1 About Maps and Regulatory Areas 34
2.2 About Satellite Virtual Networks 45
2.3 Velocity Layer 3 (IP) Network Architecture 48
2.4 The Velocity Backbone Network 53
2.5 Velocity Subscriber Networks and Internet 54
2.6 The Velocity SAS Network 54
2.7 The Velocity CDC Network 58
2.8 The Velocity RF Gateway Network 60
2.9 The CDC and RF Gateway WAN Link 61
2.10 Multiple Satellites per CDC 62
2.11 The Velocity Over the Air Network 66
2.12 The Velocity Terminal Network 68
2.13 The Velocity NOC Network 70
2.14 The Monitoring and Control Network 71
2.15 Velocity Data Traffic Protocols 71
2.16 Velocity Routing and Control Protocols 74
2.17 Velocity Satellite Router Protocols 76

2.1 About Maps and Regulatory Areas

All information about where a satellite terminal may access the satellite network around the globe, and how to codify the information and use it in the system is contained within a collection of various map files.

There are five different map types, which are configurable in the iDirect system to help define geo-location dependent behavior. A satellite terminal gets into the network using the Low Resolution Acquisition map. The terminal then reports its location to the PP, which makes decisions based on the Figure of Merit (FOM) map, Transmit Regulatory Offset (Tx RO) map, and regulatory areas (RAs). Service Area (SA) maps are also used by the PP to determine the services that are provided at the terminal's location.

All geographic information on the system is contained in following five items.

- Figure of Merit Maps
- Transmit Regulatory Offset Maps
- Low Resolution Acquisition Maps
- Service Area Maps
- Regulatory Areas



NOTE: All map files, including FOM and Tx RO maps, live entirely on the PP. A high-resolution map is not needed nor maintained/updated on individual terminals.

2.1.1 Figure of Merit Maps

Figure of Merit (FOM) maps are imported into the Velocity™ system through the NMS and saved on the Protocol Processor. The FOM maps define a figure of merit for each beam and for each geo-location. This FOM lets the SAS balance congestion vs. operation in overlapping areas.

The FOM map is specified as a set of closed latitude/longitude contours, with a figure of merit for each contour. The figure of merit is always a positive number or zero. A larger figure of merit indicates a less desirable location. The FOM is scaled to kilobits per second (Kbps); this helps with load balancing. For more information on using Figure of Merit maps, see Terminal Beam Switching on page 271.

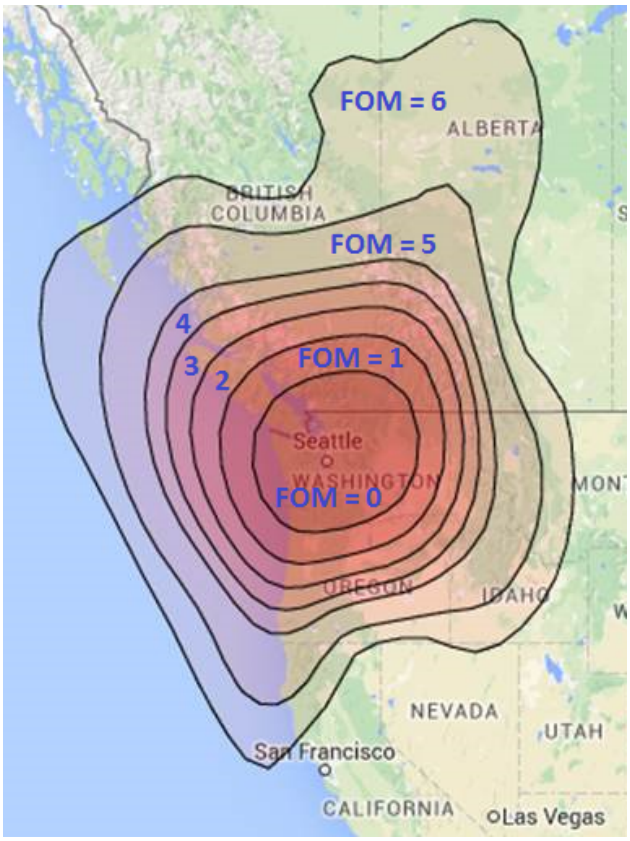


Figure 2-1. Example Figure of Merit Map

2.1.1.1 FOM Threshold for Smaller Terminals

The FOM Threshold for Smaller Terminals Feature enables a Velocity operator to define which Figure of Merit (FOM) Contours more closely match the capabilities of constrained smaller terminal types. The operator can define per channel, per RF Terminal Type (RFTT), FOM Threshold Values that result in the reduction of the outermost FOM Contours of a beam to

match the capabilities of the terminal using the given RFTT. As a result, the terminal will not switch to a beam in which it is not yet capable of receiving/transmitting.

The operator can configure the optimum outermost contour for a given terminal type (from a group of concentric contours defined by the operator for a given beam). For example, if the FOM threshold is set to 5,000 for a particular terminal type, the PP will use a FOM contour with a FOM of value less than or equal to 5,000 as the beam edge for that terminal type.

The FOM Threshold for Smaller Terminals Feature helps match beam performance with the terminal's capabilities, avoiding connection/switching issues.

2.1.1.1.1 Basic FOM Contour Operation

The Velocity platform supports a single FOM map for all terminal types, assuming that all terminals can operate within the edge of a beam, even at the very edge. This approach can be problematic for smaller terminals, as they may struggle to maintain service at the beam's edge.

Different terminals have different transmit and receive capabilities. These capabilities determine locations within a beam where the terminal can close the link with hub and can access the network. Satellite beams are usually stronger toward the center of the beam. The performance of a terminal is the best in the innermost contour and degrades towards the outer contours. The outermost contour for a terminal represents the boundary of where the terminal can access and operate in a given beam.

This is shown in Figure 2-2 below. The red and blue beams with FOM contours are represented by circles. High-end terminals can function within both red and blue FOM contours, while lower-end terminals can only operate within the inner FOM contours (represented by solid lines). If a lower-end terminal is at position x2 and the system tries to switch it to the blue beam, it may lose connection due to its limitations. Similarly, at position x1, the system may incorrectly assume the terminal can operate within the red beam’s contour, leading to a loss of connection.

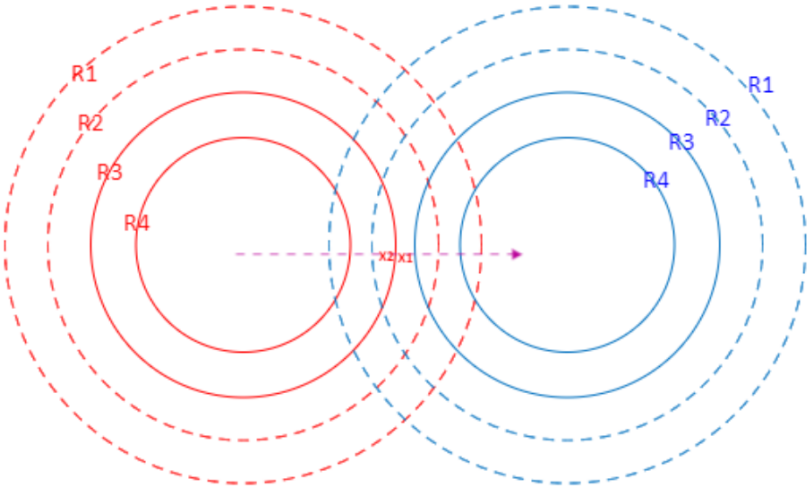


Figure 2-2. Basic FOM Contour Operation Overview

2.1.1.1.2 FOM Contour Optimization Configuration Flow

FOM Contours are usually based on link budget calculations that the operator should perform for terminals they are planning to deploy. For example, an operator may have a range of terminals with reflector sizes ranging from 2.4m to 60 cm. If they do link budget calculations, the 2.4 m will be the least constrained and can be used to determine the outermost Contour for the service. Smaller antenna sizes will be more constrained and will have their own outermost FOM Contour boundary, which is expected to be an inner contour with respect to that of the most unconstrained. The smallest terminals, which are most constrained, will be able to operate to yet another inner contour.

The steps below give an overview of the process to follow to customize FOM Contours for different RFTTs.

1. Operator prepares FOM Contours for all beams and all satellites using the highest capability terminal type in the system. These represent the outer limits of performance. Note that the Maximum FOM contour that the Operator can prepare is 25400 as the outermost contour. In Figure 2-3 below, The maximum value that contour R1 can represent is 25400.
2. Operator determines the best FOM Contour each type of terminal can use. Smaller or less capable terminals will use contours closer to the center of the beam.
3. Operator groups terminals that have the same RF performance in the same category and assigns them the same RFTT.
4. Operator configures the FOM Threshold Value for every RFTT in every beam (channel for more granular control per iNet).
5. Operator loads the FOM maps through the NMS.
6. The PP will get the FOM maps and will apply the appropriate FOM back-off by eliminating the outermost contour that cannot be used by the terminal.
7. The terminals will now only connect to beams they can handle, according to the set FOM Contours.

This process helps ensure that each terminal is matched with the appropriate beam based on its performance capabilities. The examples below show how this feature can be used to customize the FOM Contours for two (2) different RF Terminal Types.

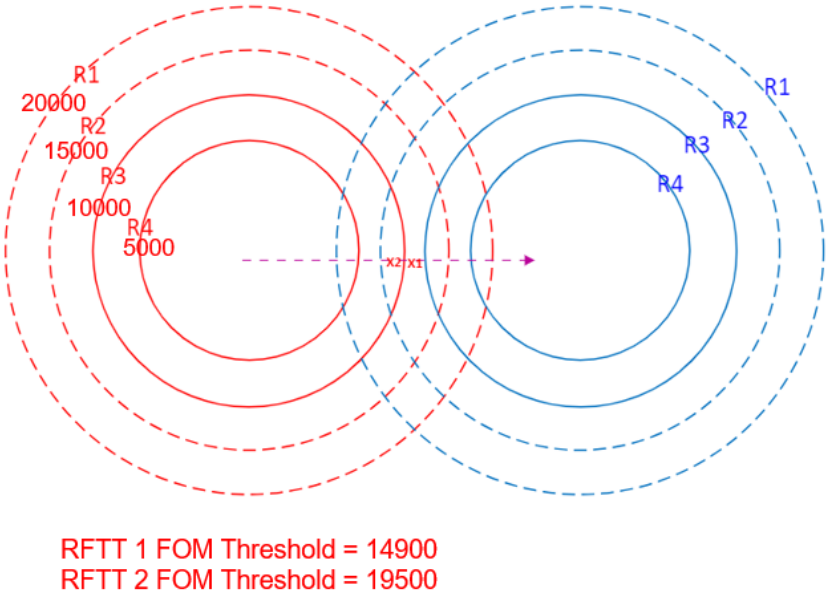


Figure 2-3. FOM Contour Optimization Configuration Flow

Example 1 - Operator wants to exclude two of the outermost contours in a beam for RFTT 1.

- Exclude the outermost two contours which have values of 20,000 and 15,000 respectively (e.g., contours R1 and R2).
- Operator sets the FOM Threshold Value to 14,900. Both 20,000 and 15,000 are greater than 14,900, so both outer contours will be excluded for this RFTT.

Example 2 - Operator wants to exclude the outermost contours in a beam for RFTT 2.

- Operator sets the FOM Threshold Value for that RFTT in that iNet to 19,500. This would exclude only the outermost FOM Contour for that RFTT (e.g., R1).

	NOTE: By default, the FOM Threshold Value is disabled for an RFTT (the maximum FOM Threshold Value is applied to the RFTT). It is only enabled for an RFTT when a user configures a Threshold Value for the RFTT. See the 4.0 Configuring Velocity Using Pulse User Guide for more information on configuring FOM Threshold Values.

2.1.2 Transmit Regulatory Offset Maps

The Transmit Regulatory Offset (Tx RO) maps are used when the satellite terminal enters the network for the first time. These maps are used to adjust the maximum usable C/N of a terminal, which typically varies with the beam G/T footprint. This allows a terminal to operate at the regulatory limit for transmit power spectral density (PSD), which increases the overall system efficiency. The Tx RO maps are imported into the Velocity™ system through the NMS and saved on the PP.

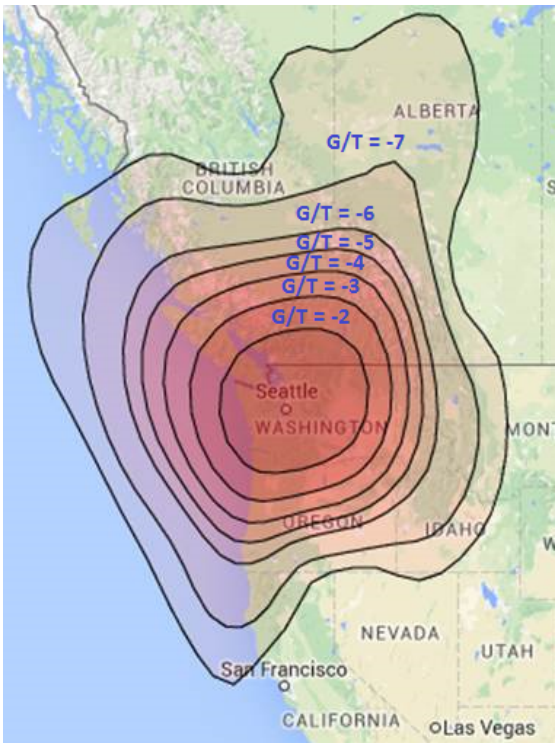


Figure 2-4. Example Transmit Regulatory Offset Map

2.1.3 Low Resolution Beam Maps for Acquisition

A satellite terminal requires descriptions of the footprints of the beams in which it can operate. The Low Resolution Beam maps define, for each beam, the area on the earth where terminals are permitted to establish a network connection. A low-resolution map for a beam consists of one or more contours containing a small set of latitude/longitude points (typically < 20), or a center point and radius if the beam is circular. This often is the outermost contour of the FOM map.

A beam's low-resolution map does not need to include all points of the FOM contour, just enough that the terminal can determine whether it is located within the beam's footprint:

- For circular beams — a beam center and radius for circular beams.
- For non-circular beams — a polygon of latitude/longitude points.

The points are stored in the Pulse® database, which provides it to the PP in an options file. When a terminal wants to join the network, it will attempt to use a beam only when it is

inside the defined contours. The terminal only uses its low-resolution beam contours to select a beam to join the satellite network. These latitude/longitude points are either entered into Pulse® by the operator or loaded from a txt file.

The solid lines of the image shows three beam footprints on the earth, while the dashed lines show the contours for the low-resolution acquisition maps. The linked latitude/longitude points are indicated by small stars. Note that the satellite operator may choose to place the low-resolution contours inside the beam footprint, as shown below, which maximizes the likelihood a terminal can enter the network on its first try. Alternatively, the satellite operator may design the contours outside the beam footprint, so that terminals will attempt to enter on a weak signal.

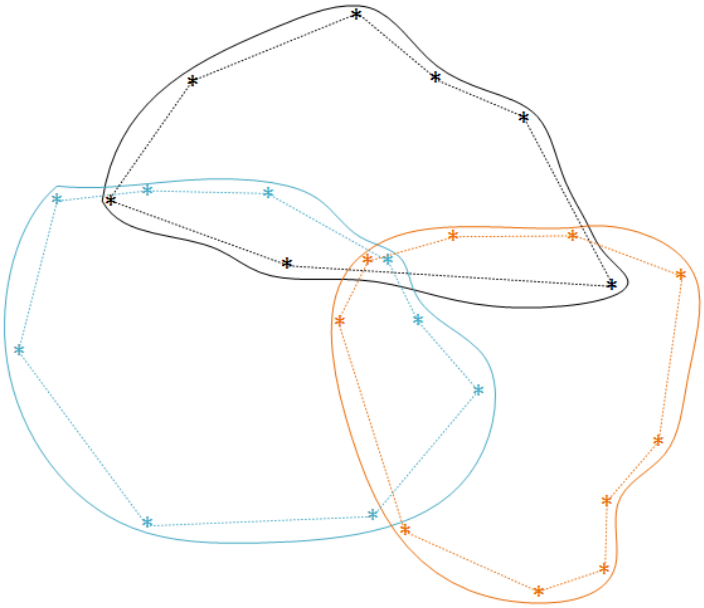


Figure 2-5. Comparison of Low-Resolution Map and Actual Beams

When the terminal is powered up, it uses its locally-stored configuration files to identify a signaling carrier it can lock to. Once it locks to a signaling carrier, it receives the beam low-resolution maps for the satellite network on the carrier. It can then use its current location to identify the beam footprint or footprints in which it is located and attempts to join the network in one of those beams.

For information on how the maps are transmitted to the terminal and kept up to date on the terminal, see Acquisition Signaling Carriers (ASC) on page 82.

2.1.4 Service Area Maps

Service areas (SAs) define geographic areas, which are not related to a satellite or beam footprint. Velocity allows service areas to be used to define:

- geographic areas in which service is provided (service geoscope)
- areas where regulatory restrictions are enforced or service is barred (regulatory area)
- areas that serve both service area geoscopes and regulatory areas

Individual service areas are defined using one or more contours in a gxt file. A group of related service areas are imported through the NMS into the Velocity system as an SA map. The service areas in an SA map need not be contiguous; and if a service area is defined using multiple contours, the contours need not be contiguous.



Figure 2-6. Example Service Areas

2.1.4.1 Service Area Groups

A Service Area Group SA Group), as its name suggests, is a group of service areas used to create Geoscope regions of service areas or Regulatory Areas. Each SA Group, when defined, references an SA map. A given SA map may be referenced by one or more SA Groups. Within an SA Group, SA coverage may not overlap. Each SA has an identifier 16-bit unsigned code known as the service area ID. Valid SA ID values range from 0 to 2^{15}-1 and are unique within an SA Group, but not across SA Groups.

The service provider may use a ***private SA Group*** for one subscriber, or may define a ***public SA Group*** for multiple subscribers. If a single SA Group is used for all subscribers, it is referred to as the ***global SA Group***.

Multiple regions, as illustrated in the image, can be specified using the same SA Group with the same set of SAs in the SA Group; or multiple regions specified based on different SA Groups.

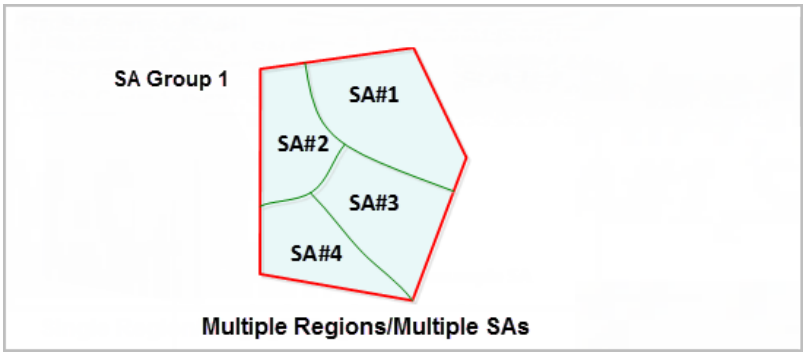


Figure 2-7. GSP Geoscope with Single Region/Single SA & Multiple Regions/Multiple SAs

In the above diagram, four Geoscope Regions or Service Areas could be defined so that

- Region 1: contains SA #1
- Region 2: contains SA #1 and SA #2
- Region 3: contains SA #1, SA #2, and SA #3
- Region 4: contains SA #1, SA #2,SA #3, and SA #4
- Or, the region can contain any subset of the SA Group

The Velocity system may contain multiple SA Groups that cover the same geographic area. This arrangement, known as i***ndependent service areas***, allows the satellite operator to support two different service providers who offer services in independently-defined service regions. In the following image, for example, the same geographic area is represented by two Service Area Groups built from different map files. On the left, the geographic area is represented by a single SA, and on the right by four SAs.

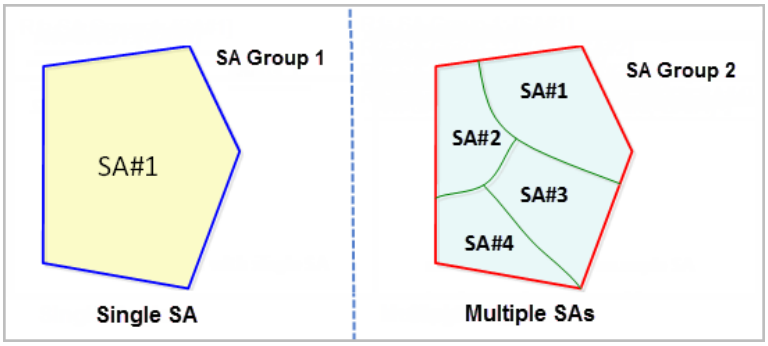


Figure 2-8. Two Service Area Groups for the Same Geographic Area

Independent Service Areas also allow a satellite operator to support different service providers who offer services in independently-defined and unrelated service areas , which may happen to overlap, as illustrated in the following image. In this illustration, Service Provider A (in red) operates in the service areas labeled A1 and A2. Service Provider B (in blue) operates in service areas B1 and B2. As shown, however, service areas within one SA Group, may not overlap.

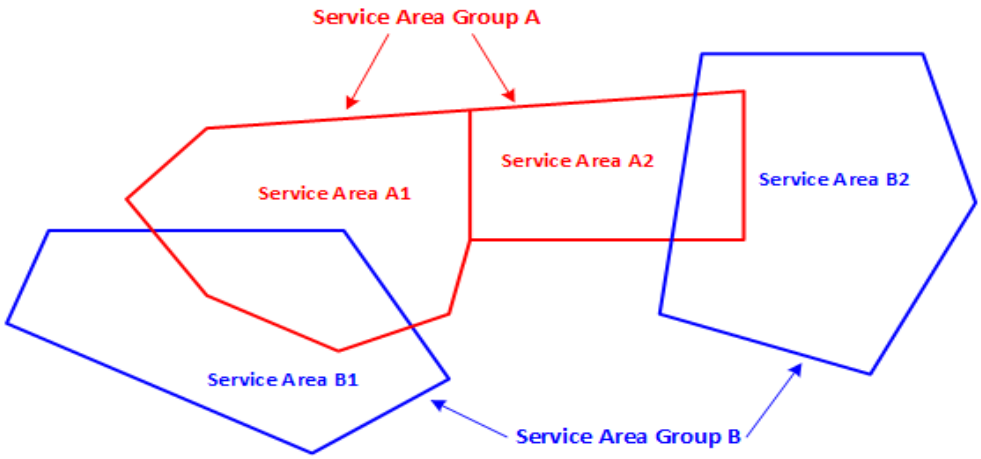


Figure 2-9. Overlapping Service Areas in Two Different SA Groups

A satellite router can be associated with exactly one service geoscope SA Group and one regulatory area SA Group. Thus, a satellite router will be located in one and only one service geoscope SA at a time, even if it lies in multiple beam footprints. Similarly, the satellite router will be located in one and only one RA SA at a time. When a satellite router reports its geo-location, the PP looks up the SA for the satellite router in the appropriate geoscope and RA SA Groups and makes all decisions based on that lookup.

SAs and SA Groups are also used to restrict the operation of satellite routers to conform to regulatory restrictions as described in Regulatory Areas on page 44.

2.1.4.2 Geoscope Regions Based on Service Areas

Geoscope Regions are used by a service provider to guarantee a defined level of service within a specific geographic area. Specifically, they are used in the following examples:

- A subscriber service plan specifying a global committed information rate/maximum information rate (CIR/MIR) may specify a different CIR/MIR in a high demand SA.
- Different billing parameters may be applied based on service area ID
- An area may be limited to receive only
- Service areas can be used for barring service

In Velocity, a ***Region Geoscope*** defines a geographic area in which a Satellite Operator/Service Provider can sell services. A region geoscope may be specified as a group of beams, or a group of service areas.

The following list of constraints are applicable when defining service polygons in regional geoscopes:

- A region geoscope/RA can contain a single service area or multiple service areas
- A region geoscope/RA must contain at least one region
- A region geoscope/RA can contain multiple regions only within the same SA Group
- A service area can be used in multiple geoscopes/RAs

For more information on using Service Polygons for providing services, see Geographic Scope in Unicast Subscriber Service Plans on page 138 and Geographic Scope in GSPs on page 145.

2.1.5 Regulatory Areas

Regulatory areas (RAs) are used to enforce transmit restrictions, such as no-transmit areas, maximum EIRP and power, as well as frequency ranges. An RA is defined as a union of service areas defined within one SA Group.

Typically, all satellite routers will be configured to use a single set of RAs defined by the satellite operator who is responsible for providing RA compliance. However, the Velocity system also allows more than one set of RAs to be defined, each in a different SA Group. This allows, for example, a satellite operator to delegate regulatory compliance to their service provider customers, or to support service providers who are subject to different regulatory compliance rules or boundaries.

2.2 About Satellite Virtual Networks

Satellite virtual networks (SVNs) allow the satellite operator to implement service provider and subscriber business models by extending a private network across the shared satellite network so that the subscriber's remote devices can send and receive data as though they were directly connected to their private network.

An SVN is analogous to a terrestrial Virtual Private Network (VPN). Both provide connectivity across a common network infrastructure while segregating the traffic of each private network, and providing each with a separate and independent IP address space. This method is called "virtual routing" in an iDirect network. Both the PP and the satellite router maintain logically separate virtual routing tables in a manner similar to router vendors' Virtual Routing and Forwarding (VRF) features. The combination of iDirect virtual routing and VRF allows the same IP address to be used by multiple SVNs.

One possible scenario of how an end-user's SVN could be implemented by a satellite operator, is shown here.

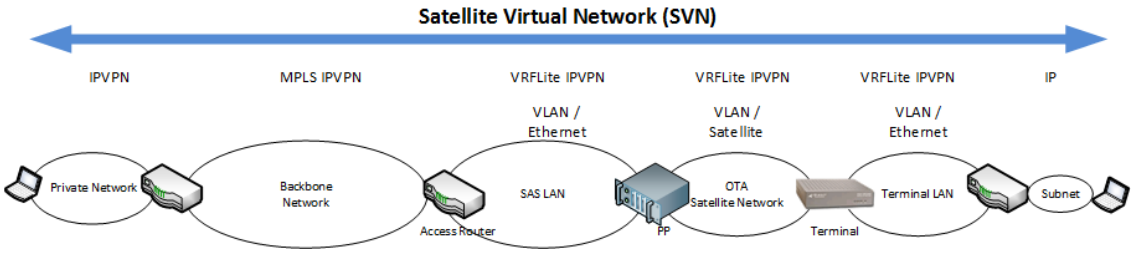


Figure 2-10. Satellite Virtual Network

This SVN extends from the subscriber's private network on the left, through an MPLS (Multi-Protocol Label Switching) backbone network via an MPLS pseudo wire (PW), into and through the Satellite Access Station (SAS) including the PP, and onto and through the satellite router. At the access router the MPLS PW is terminated and mapped to a unique 802.1q-compliant VLAN tag that has the same value as the SVN's ID. This tag is maintained through the SAS and over the air (OTA) routing domains to the satellite router. The service provider may choose to use the same VLAN tag value on the terminal LAN, or to configure the satellite router to map it to an arbitrary local VLAN tag value.

Unicast traffic may be carried upstream or downstream within an SVN — for example, from a subscriber's terrestrial network to equipment on the terminal LAN or vice versa. Multicast traffic may only be carried downstream within an SVN — for example, to equipment on the terminal LAN.

Multiple SVNs may be present on a single satellite router. The GQoS service plans (SSPCs) associated with the satellite router may apply to a single SVN, or to multiple SVNs on the same satellite router.

2.2.1 Velocity SVN Representation in Pulse

In Velocity, each SVN has logical elements within the backbone network, within the SAS, and within and possibly beyond the satellite routers. The Pulse NMS representation of the example SVN, shown here, is represented as the logical elements that would be configured in Velocity.

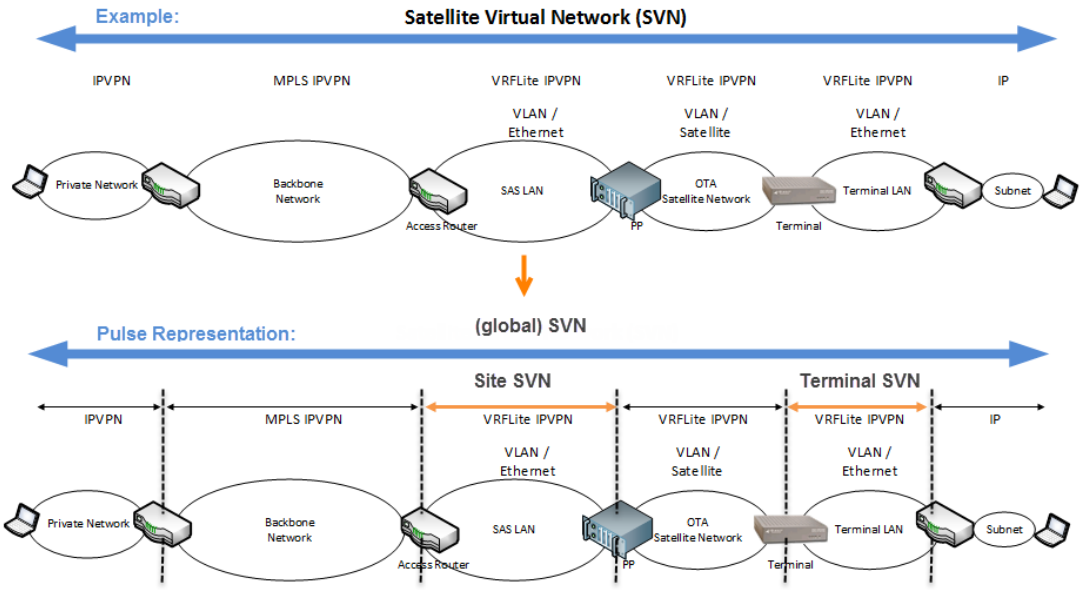


Figure 2-11. Velocity SVN Representation in Pulse

These logical elements, in Velocity, and as configured in the NMS are as follows:

- **SVN** — An ***SVN***, sometimes called "Global SVN, in Pulse, is at the top of the Velocity SVN structure and connects a subscriber's equipment in one location to another location through a Velocity network — hence global.
- **Site SVN** — A ***Site SVN*** represents the specific segment of an SVN within a site — for example within a SAS site, which extends across the DCN to another site. All global SVNs are implemented in all SASs to allow satellite routers to move freely from one satellite to another. One Site SVN, associated with a specific global SVN, is created for each site.
- **Terminal SVN** — A Terminal SVN represents the specific segment of an SVN located in the terminal network behind a satellite router. Within a given terminal, up to 12 terminal SVNs may be configured.

2.2.2 Velocity SVN Types

In Velocity, each SVN represents a separate routing domain, and IP addresses may be reused across multiple domains. Traffic flow between VLANs is not allowed inside a Velocity system.

The Velocity network uses the following SVN types:

- **Data SVN** — Transports user traffic from a satellite router to its final destination in a subscriber's private network, and vice versa. It extends from the terminal network, through the satellite router, a SAS, the backbone network, and finally to an interface point to the subscriber's private network. Each SVN is identified by an SVN ID that is defined by the satellite operator, and has its own IP address space, which may be managed by the service provider. Also referred to as a ***Subscriber or Customer SVN***.
- **Admin SVN** — Transports internal Velocity administrative traffic, between equipment within a SAS, between SASs, between the SAS and the NOC, and finally between the NOC or SAS NMS and satellite routers. This SVN must be assigned **SVN ID 1**. The management SVN IP address space is managed by the satellite operator who assigns management IP addresses to all network equipment.
- **Tunnel SVN** — Transports user traffic between Velocity PP processes and line cards located in the same SAS or in the backup/diversity SAS. The SVN ID of the Tunnel SVN is configured by the satellite operator (Tunnel default SVN ID = 4048).

2.3 Velocity Layer 3 (IP) Network Architecture

The Velocity system supports MP-BGP (Multi-Protocol — Border Gateway Protocol) with the IPv4 and IPv6 address families. Thus, BGP (Border Gateway Protocol) advertises both IPv4 and IPv6 IP addresses.

The Velocity system is quite flexible, allowing satellite operators, service providers, and subscribers to design IP network architectures tailored to their particular needs. Specifically, they design their Autonomous Systems (AS), determine where to use iBGP (interior Border Gateway Protocol) versus eBGP (external Border Gateway Protocol), determine where to use route reflectors, and decide what BGP features to enable.

For a description of the BGP features available in the Velocity satellite router and PP servers, see Border Gateway Protocol (BGP) on page 74.

An overview of the Velocity IP Network architecture for an SVN is illustrated in the diagram.

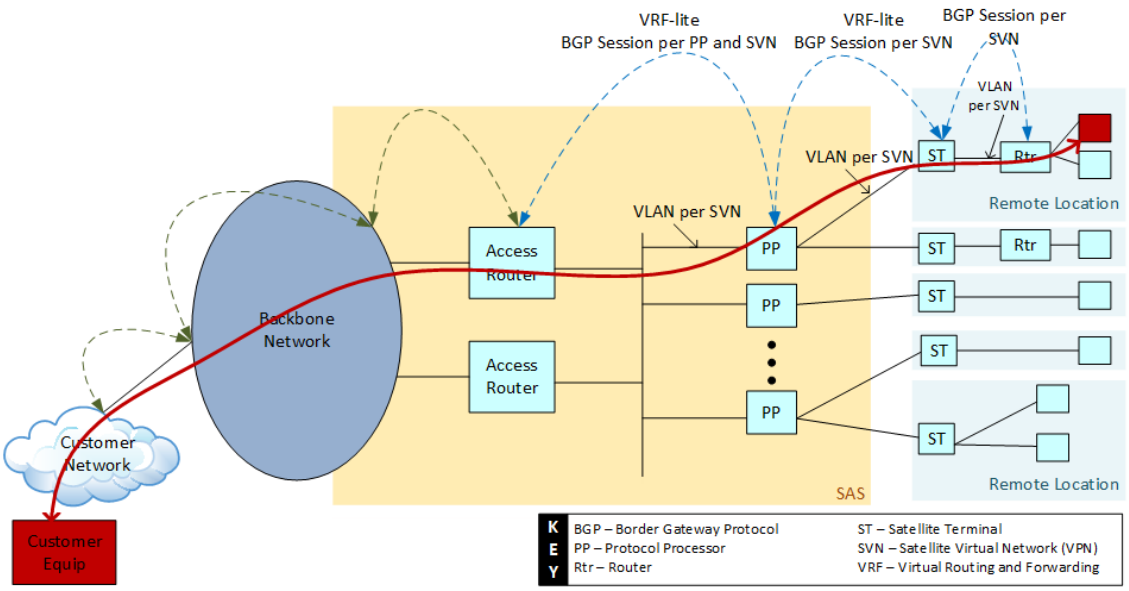


Figure 2-12. Velocity IP Network Architecture for an SVN

NOTE: Although BGP is configured for each SVN, the AS must be the same for all SVNs configured on a given piece of equipment. Furthermore, to reduce memory usage and processing load, service providers and subscribers are advised to use the maximum prefix and route filtering features to control the number of BGP prefixes advertised to and received from BGP peers.

2.3.1 Velocity Layer 3 Routing Operations

Layer 3 routing configuration and operations, in Velocity, are outlined in the following points:

- When initially configuring the SAS, the service provider configures the BGP peers on the access routers and PPs , as well as an OTA BGP Peer template that defines parameters common to all terminal BGP sessions on all SVNs. Once this configuration is done, the access router and PP establish BGP sessions, begin exchanging route advertisements, and the PP is ready to peer with terminals.
- During terminal configuration, the service provider or subscriber configures a similar OTA BGP template for each SVN defined on that terminal.
- When a satellite router acquires into the satellite network, the PP provides it with detailed configuration information via Dynamic File Options Exchange (DFOE). This information includes the BGP OTA Peer template and the identity of the PP BGP peer within each SVN configured on the satellite router, as well as all options keys other than those required for the satellite router to acquire into the satellite network and those needed for it to function in the IP network.
- Once the BGP peering relationship is established, the PP and satellite router begins exchanging route advertisements. The satellite router advertises the route table to directly-connected subnets, as well as any subnets it learns via BGP sessions with routers on the terminal LAN. The PP then advertises these subnets to the access router, which then advertises them to the backbone network, and hence into the subscriber's private network. IP packets from equipment on the subscriber's private network can then be routed to equipment on the terminal network.
- Similarly, the PP has already learned routes to the subscriber's private network from the access router and advertises them to the satellite router. IP packets from equipment on the terminal network can then be routed to equipment on the terminal network.

The PP also will not deliver traffic from one SVN to another. If desired, the satellite operator may implement gateways between SVNs within the backbone network.

2.3.2 Velocity IP Address Assignment

For IP routing and forwarding to work correctly, the equipment participating in each SVN must have IP addresses assigned to their interfaces. The type and number of IP addresses assigned to these interfaces depends first on whether the SVN supports IPv4, IPv6, or both.

- If the SVN supports IPv4, the interfaces must be configured with an IPv4 address
- If the SVN supports IPv6, the interfaces must be configured with IPv6 addresses
- If the SVN supports IPv4 and IPv6, the interfaces must have both IPv4 and IPv6 addresses

The following address assignment guidelines can be applied:

- The satellite operator or service provider decides how to manage an SVN's IP address space and allocate IP addresses to the equipment in the SVN.
- The IP address space of each SVN is completely separate and independent from the IP address space used by the other SVNs in the Velocity system.

- The same IP address can be used within multiple SVNs.
- Velocity is agnostic to the IP addressing scheme implemented by the satellite operator or their customer service providers.
- The operator responsible for an SVN should implement best practices for their particular environment. For example, IP subnets should be allocated to routing domains in a way that allows for efficient route aggregation.
- The responsible operator must also decide whether (or when) to use public IP addresses vs. private IP addresses.
- Although a physical SAS network uses redundant ports to provide a highly available network, the iDirect equipment uses port bonding to create a single “bonded interface” out of redundant ports, in which only one IP address is required.
- The IP addresses that must be assigned depends on the Velocity SVN type.

2.3.2.1 Data SVN — IP Address Assignment

The Data SVN transports user traffic from a satellite router to its destination and vice versa. It extends from the terminal network, through the satellite router and SAS to the backbone network. PP servers, access routers, and satellite routers participate in subscriber (Data) SVNs.

The following guidelines apply to IP address assignment in Velocity Data SVNs:

- Each Data SVN has its own independent IP address space, which may be allocated and managed by the subscriber or by the service provider. The service provider also has the option to manage IP addresses in the SAS and OTA routing domains, and to delegate IP address assignment, to their subscribers, in the terminal routing domain .
- Unless otherwise stated below, the IP interface address assignments must be unique within the global Data SVN — that is, across all PP servers, access routers, and satellite routers in all locations and sites of the satellite network.
- Protocol Processor (PP) servers have an IP interface (ETH0), in each Data SVN on the SAS network, to which the operator assigns an IP address. Sometimes this interface is referred to with a name that includes the SVN ID, for example, “eth0.5” for SVN 5. PP servers also have an IP interface in the OTA routing domain, but that IP interface is not explicitly configured and does not have an IP address.
- Access Routers, like PP servers, have an IP interface with an operator assigned IP address, in every Data SVN.
- Satellite routers have two IP interfaces, as described below, for each subscriber SVN configured on the satellite router. Each interface is assigned an IP address.
  - The satellite router IP interface in the OTA routing domain is referred to as “sat0”, and the IP address assigned to this interface must be unique within the SVN.
  - The satellite router IP interface in the terminal routing domain is referred to as “eth0” or “eth0.5” (for SVN 5), and an IP address is assigned to it. The IP address assigned to the eth0 interface is either operator configured or dynamically assigned by a RADIUS server. See RADIUS on page 79.

If the satellite router's eth0 address is routable, it must be unique within the global subscriber SVN. However, there are network designs in which this IP address can be un-routable, and therefore not unique. For example, if there is only a router directly connected to the satellite router's eth0 interface and BGP is running between the satellite router and that router, the operator may choose to configure BGP to not advertise a route to the IP subnet associated with the terminal routing domain — that is to say, the IP addresses of the satellite router and router interfaces in it.

2.3.2.2 Admin SVN — IP Address Assignment

The Velocity Admin SVN transports internal administrative traffic between Velocity equipment wherever it is located: within a SAS, between SASs, between the SAS and the NOC, and between the NOC or SAS and satellite routers.

The following guidelines apply to IP address assignment in the Velocity Admin SVN:

- The Admin SVN is owned by the satellite operator, and therefore the SVN IP address space is managed by the satellite operator, who assigns management IP addresses to all network equipment.
- All Velocity equipment participates in the Admin SVN — NMS servers, iGW/PP servers, GBWM servers, line cards, chassis, and satellite routers.
- The access router is in the Admin SVN to allow traffic to flow across the backbone network to the NOC and between primary and diversity SASs.
- The equipment IP address assignments in the Admin SVN is the same address as used in NMS to configure and identify the components. Thus, the "Management IP address" configured for the component must also be the IP address configured in the SVN.

2.3.2.3 Tunnel SVN — IP Address Assignment

The Velocity Tunnel SVN transports user traffic between PP processes and line cards. If the SAS is one of a pair of primary and backup/diversity SASs, the traffic may also be sent across the backbone network between the PP servers in the primary SAS and line cards in the backup/diversity SAS. Finally, in those networks that employ the CDC/RF Gateway architecture, the Tunnel SVN transports traffic across the WAN between the CDC and the RF Gateway.

The following guidelines apply to IP address assignment in the Velocity Tunnel SVN:

- PP servers and line cards participate in the Tunnel SVN.
- The access router also participates in the Tunnel SVN, if the SAS is one of a pair of primary and backup/diversity SASs.
- As with Data SVNs, each PP has a Tunnel SVN "ETH0" IP interface and requires an IP address assignment. Each line card also has an IP interface, and requires an IP address.
- When the CDC+RF GW architecture is being deployed, the access router participates in the Tunnel SVN.

2.3.3 Velocity IP Multicast

The Velocity system supports downstream IPv4 multicasting within any SVN. Any equipment on the terminal network may subscribe to multicast streams in its SVNs.

The Velocity system currently supports the following types of multicast:

- **Dynamic Multicast** — The satellite router is an IGMP (Internet Group Management Protocol) proxy and implements IGMPv3 and IGMPv2. For dynamic multicast, the satellite router software accepts IGMPv2 or IGMPv3 join and leave messages from hosts on the terminal network, and requests the multicast streams on behalf of the hosts from the SAS site using IGMPv3. The PP also is an IGMPv3 proxy, and similarly requests the multicast streams from the upstream multicast source.
- **Static Multicast** — For static multicast streams, a fixed set of multicast IP addresses are configured to receive the streams in the satellite router configuration. After it enters the iNet, the satellite router generates IGMPv3 requests to the SAS site for those statically configured multicast IP addresses on behalf of the hosts on the terminal network. Note that Evolution's use of the term “static multicast” is the same as persistent multicast in Velocity.
- **Persistent Multicast** — Persistent Multicast represents a multicast method, in Velocity, in which the protocol processor sends the multicast stream on the configured iNets, without waiting for an IGMP join. When a Persistent Multicast stream is configured in Velocity, the PP will send an IGMP request to the multicast source for that stream, transmitting the request without waiting for a join, on the iNets identified in that configuration.

The diagram below illustrates the IGMP proxies and the IP multicast protocols involved in implementing multicast in a Velocity network.

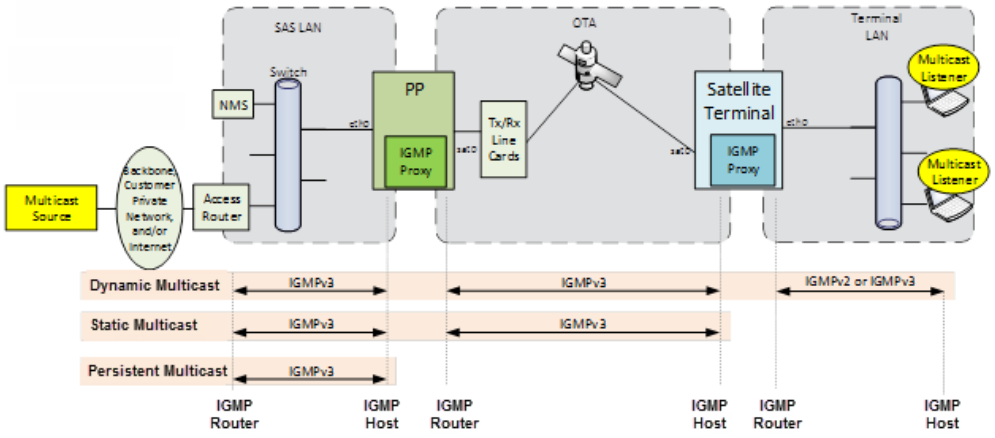


Figure 2-13. IP Multicast in the Velocity System

The following are some key concepts in IP multicast using Velocity:

- Multicast is only sent within an SVN, just as unicast packets are only sent within an SVN.
- If the same multicast stream is requested on more than one SVN, a copy of that stream is sent on each of those SVNs.

- Only satellite routers that have requested a multicast stream will put it on their terminal network (their VLAN).
- Enabling IGMP snooping on the SAS switch avoids unnecessary load on the PP servers and links in the SAS network. The SAS switch will replicate and send multicast packets only on interfaces on which it has observed an incoming IGMP join.

In the OTA Routing Domain, each multicast packet is identified by an HDLC address as it is sent from the PP to the satellite router. There are 128 multicast HDLC addresses available in each beam. However, this does not limit the number of multicast streams in a beam to 128. The multicast HDLC address is assigned to the Multicast Subscriber Service Plan Profile (SSPP), which will typically cover a range of multicast IP streams.

Implementing multicast for streaming applications is advantageous using broadcast satellites, but can be challenging in spot beam satellite networks. Velocity addresses these challenges by sending a multicast stream only on iNets containing a satellite router that has requested that stream.

The multicast stream follows a terminal as it switches beams: the PP begins pro-actively sending the stream to the iNet to which the terminal is moving, before the terminal actually makes the switch. See Multicast Subscriber Service Plans on page 140.

2.4 The Velocity Backbone Network

The backbone network is a global or regional network that provides connectivity between SASs, NOCs, CDCs, RF Gateways, and multiple POPs, peering sites, or other locations that support high bandwidth connectivity to subscriber networks. The backbone network typically consists of IP routers interconnected via high-speed terrestrial links as required to support the expected traffic. The backbone network may also consist of leased lines or IP VPNs procured from a terrestrial network operator. Regardless, the backbone network's design is determined by the satellite operator, who ensures the backbone network provides sufficient capacity and resiliency. No iDirect-supplied equipment is included in the backbone network.

Backbone networks are divided logically into core routing functionality and edge routing functionality. Typically, Velocity sites (SASs, NOCs, CDCs, RF Gateways, and EAPs) connect to the backbone network via a redundant pair of edge routers and standard IP protocols. Only standard protocols and interfaces are required on the edge routers.

Most backbone networks are MPLS networks that provide VPN functionality using MPLS IP VPN, and interface to peer networks using BGP. The number of SVNs the backbone network architecture supports is a function of the edge and core routing capacity and the backbone logical network design.

2.5 Velocity Subscriber Networks and Internet

The physical backbone network also provides interfaces to each subscriber's private terrestrial network. Each of these networks is a peer network connected to the backbone network via a physical interface and protocol agreed to between the satellite operator and the subscriber. A subscriber's network will typically have multiple interfaces to the backbone network for redundancy, and peer to the backbone network using BGP. No iDirect equipment is installed in the subscriber's private terrestrial network. Similarly, the backbone network typically connects to the Internet via multiple BGP peering arrangements.

2.6 The Velocity SAS Network

The SAS provides IP connectivity to the satellite routers within the footprint of the satellite associated with the SAS. The SAS is required to meet high-availability requirements and therefore has been architected with redundancy that provides no single point of failure.

The PP components function as IP routers and send and receive standards-compliant IP packets and dynamic routing updates to the backbone network, and hence to subscriber terrestrial networks. Similarly, the PP exchanges IP packets and routing updates on the OTA interface with satellite routers located in the satellite's footprint. The SAS access routers are BGP peers of the PPs in every SVN. Likewise, satellite routers function as IP routers and are BGP peers of the PPs.

The line cards interface with the RF subsystem over an L-Band interface for the transmission and reception of all RF signals.

All SAS components function as IP hosts for management and administrative traffic.

2.6.1 Velocity SAS Network — Layer 1 Physical Architecture

The SAS LAN is architected with a high performance switching fabric. The SAS LAN relies on redundant SAS switches and access routers as well as key features of that networking equipment to provide a highly available service. These features include Virtual Port Channel (VPC), Link Aggregation Control Protocol (LACP), Port Channel, etc. For a more complete description, including redundancy and automatic failover for Velocity components, see SAS Site Redundancy on page 191 and its subsections.

A sample SAS network architecture, using Cisco Nexus switches and ASR routers is shown here. This diagram shows two router pairs typically physically located within the SAS: the backbone network PEs (provider edge routers) and the access routers. Two router pairs provide a clear demarcation between the core network and the SAS LAN.

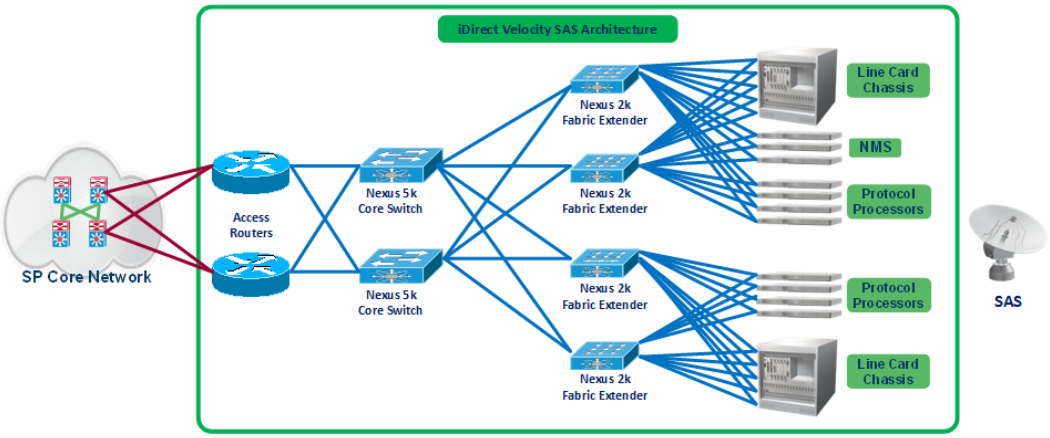


Figure 2-14. A Sample SAS Network Architecture

2.6.1.1 Shared Chassis Operation

In a Velocity network, a single hub chassis (not iHub chassis) can be configured to be shared by multiple SASs. This means the network is able to support additional satellites without the cost and effort of installing an additional chassis for every satellite. This is especially important when only a few beams on the additional satellite will be used by the Velocity network. The satellite operator will need additional line cards, a PP cluster and a SAS NMS dedicated to the additional satellite.

This operation is based on having multiple PP clusters share the chassis. Specific chassis slots are cabled together into an IF domain associated with each satellite and indirectly assigned to each PP cluster through configuration. Each PP cluster then determines what slots it can use from the configuration.

The chassis supports multiple simultaneous sessions to different clients, which allows it to simultaneously process commands from multiple PP clusters. Control and monitoring of the chassis as a whole, such as switching slots on/off and checking power supply status, is performed by every PP cluster as if it were the only PP cluster communicating with the chassis. Since each PP cluster limits its control to its assigned chassis slots, multiple PP clusters are able to share a chassis without any other coordination between the PP clusters.

2.6.2 Velocity SAS Network — Layer 2 Architecture (VLANs)

Within the SAS, each SVN is implemented as an IP subnet within a VLAN. The VLAN ID of the Admin SVN is 1, and the VLAN ID of the Tunnel SVN can be defined by the operator. The following diagram shows the Layer 2 topology of the SAS.

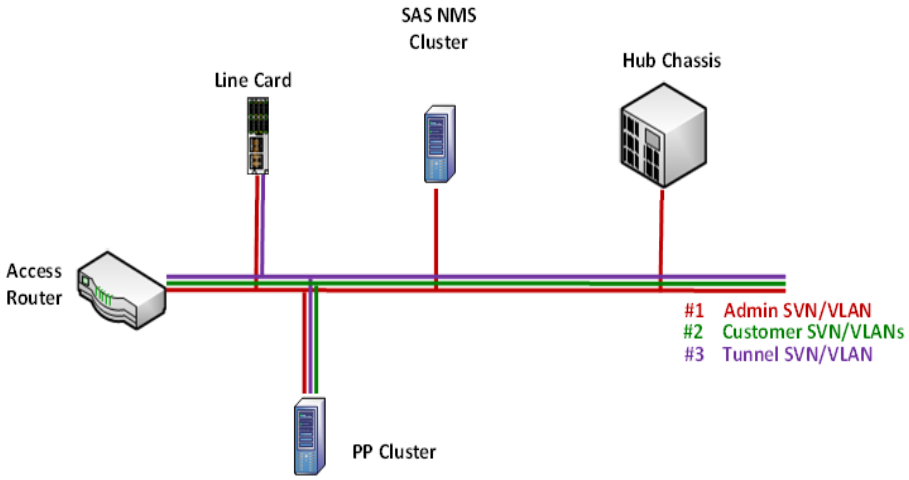


Figure 2-15. Layer 2 Architecture

On a switch, all ports are classified as trunk or access. Access ports carry untagged traffic, and all traffic is assumed to be in the same VLAN. Trunk ports support multiple VLANs and can carry both VLAN (802.1q) traffic and untagged traffic. The following diagram shows the trunk and access ports on the SAS switch.

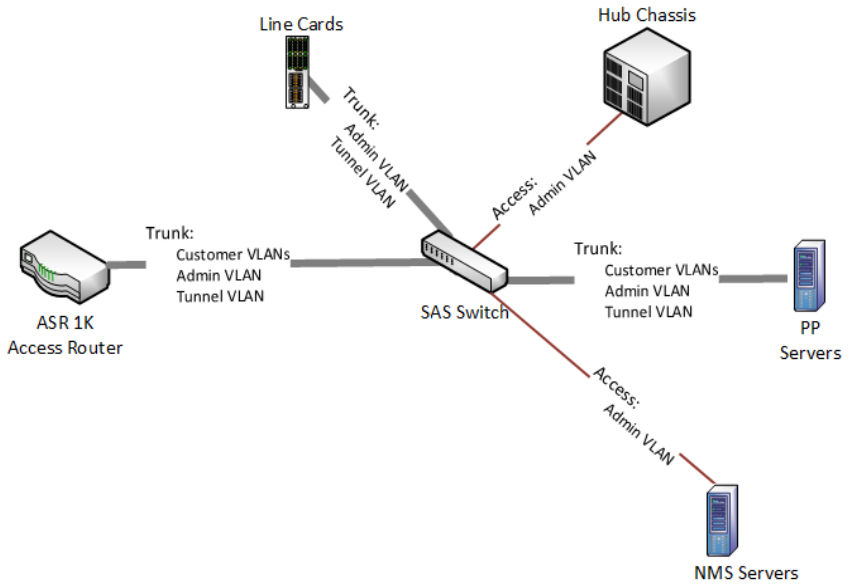


Figure 2-16. SAS Switch Trunk and Access Ports

All subscriber SVNs/VLANs and the Tunnel SVN/VLAN are 802.1q VLAN-tagged, and the VLAN tag has the same value as the SVN ID. Although the satellite operator is given the choice whether to use VLAN tags on the Admin SVN/VLAN, the operator must choose to enable Admin VLAN tagging; failing to do so causes a heavy CPU load on the PPs.

2.6.3 Velocity SAS Network — Layer 3 Network Architecture

In the Velocity system, BGP must be enabled in the SAS network domain in each SVN. Enabling BGP enables the access router to learn routes to the IP subnets on the satellite routers, and to in turn advertise them into the backbone network and the subscriber's private network.

The following configuration guidelines are recommended:

- Each SAS be configured as a separate BGP Autonomous System (AS)
- iBGP be used within the SAS
- The access routers be configured as route reflectors to avoid configuring a full mesh of BGP peering relationships between all PPs
- The access router use BGP outbound filters to minimize routing table sizes in PP server
- The access router not store the Internet routing table; instead, it should store backbone routes only
- IP addresses be chosen to facilitate route aggregation

2.7 The Velocity CDC Network

As shown in the diagram, the *CDC site* contains three iDirect components—the Intelligent Gateway (iGW), Pulse SAS NMS, and a Kafka Kubernettes Cluster.

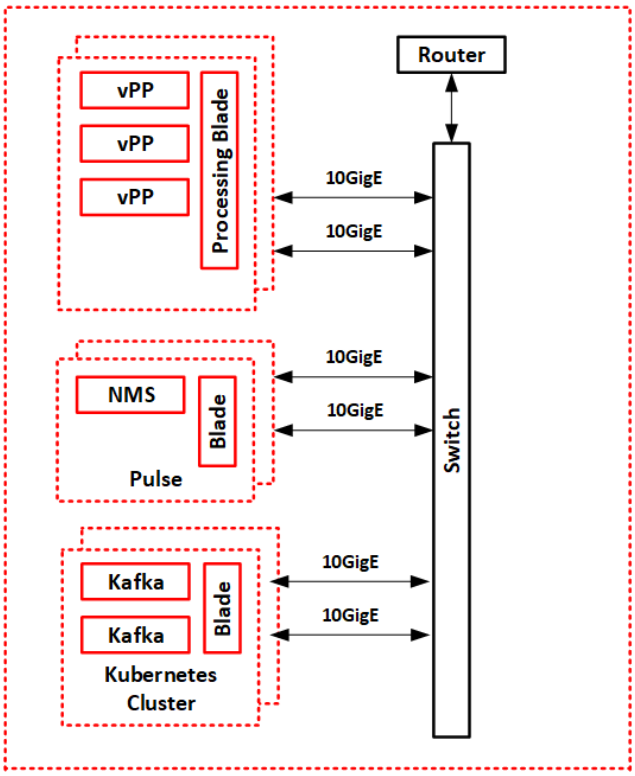


Figure 2-17. CDC Site Network

The iGW is a virtualized platform that hosts the Velocity PP cluster. In physical terms, the iGW platform is a number of sleds in one or more FX2 chassis. Each sled hosts a number of virtual machines (VMs), each of which is a node in the PP cluster. For additional details on the iGW platform, refer to the document *Installing iGateway in a Velocity Network*.

The iGW PP cluster operates in the same manner as with a bare-metal PP server cluster. As in the case of the bare-metal PP cluster, the number of nodes/VMs is determined by the satellite bandwidth and number of supported terminals. Extended PP software manages the timing synchronization across RF Gateways to minimize the effects of the WAN on control and customer traffic.

The NMS in the CDC site is the same NMS as deployed in the traditional SAS, as is the Kafka cluster. The CDC network is also similar to that of the traditional SAS.

2.7.1 Layer 2 CDC Network

Finally, globally-defined layer 2 VLAN/SVN assignments, shown below, are implemented in all SAS and CDC sites. At layer 3, the BGP routing protocol is used for advertising routes between PPs and the access router as in the traditional SAS architecture.

Table 2-1. Globally-Defined Layer 2 VLAN/SVN Assignments

VLAN/SVN Purpose	VLAN ID
Admin	1
Tunnel	Range: 2-4094; default: 4048
User Traffic	Range: 2-4094
Forced Routing I2R (iDirect to Regulatory Authority	Range: 2-4094
Forced Routing R2I (Regulatory Authority to iDirect)	Range: 2-4094

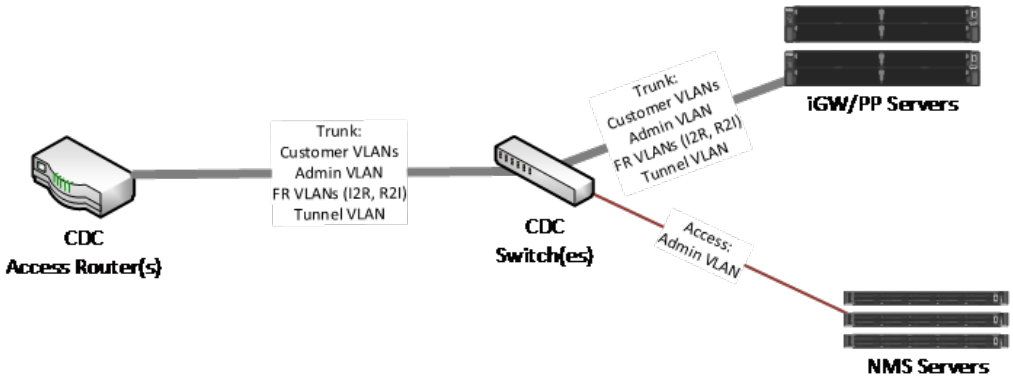


Figure 2-18. Velocity L2 CDC Network

2.8 The Velocity RF Gateway Network

The RF Gateway site contains the RF equipment necessary to support a single feeder-link—including a satellite antenna, up and down converters, RF high-power amplifiers, a third-party Carrier Measurement System (CMS), if applicable, and iDirect IF modulation and demodulation equipment.

In its initial availability, the RF Gateway supports user traffic using the Intelligent Hub (iHub) and Intelligent Line Cards (ILC). As with the CDC site, the RF Gateway network design and networking equipment, including switches and routers, are generally provided by the Satellite Network Operator.

The line cards in the RF Gateway Site provide IF modulation and demodulation for traffic sent and received over the satellite link. The iHub and ILC equipment provide high density and performance to support high throughput satellites.

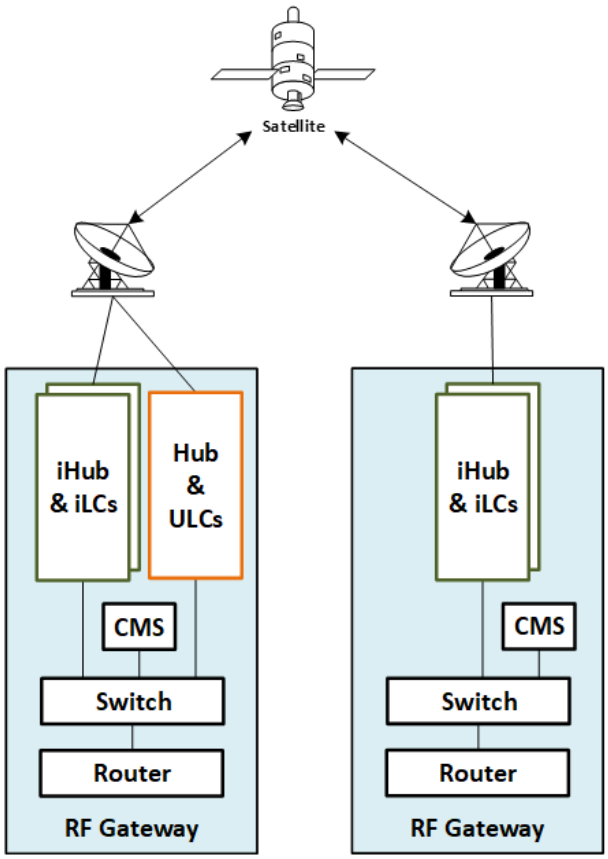


Figure 2-19. RF Gateway Site Network

In a typical RF Gateway network design, iHub components are expected to be connected to a switch using 4 10GigE and 2 1GE ports (ILC traffic flows through iHub ports). The ULCs continue to use 2 1G ports each, and the 15152 hub chassis 1 1GE port. Ultimately, the Satellite

Operator determines the exact design of RF Gateway site network, including the choice of switch and router manufacturer and model.

2.9 The CDC and RF Gateway WAN Link

A defining characteristic of the CDC architecture is that the RF Gateway and Protocol Processors are not co-located, but instead are at separate locations that are connected over a Wide Area Network (WAN) link. All communication between the CDC and RF Gateway components is based on standard IP, carried over an IP/MPLS WAN, or other networking technology such as MetroEthernet.

The WAN link will likely introduce the effects of latency, jitter, packet loss and/or packet reordering to both control and data traffic between the PP and line cards. Although Velocity is defended, as much as possible, against these effects, the SLA for the Network Operator's WAN must minimize these factors to avoid degraded service. Both Tunnel and Admin SVN traffic must also be prioritized across the WAN by policies applied at the access or PE routers.

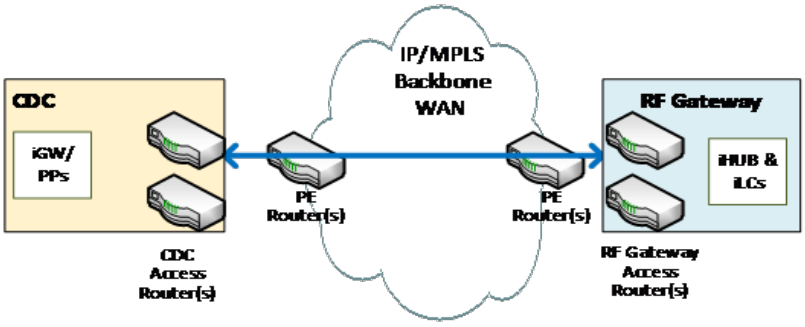


Figure 2-20. CDC and RF Gateway Wide Area Network (WAN) Link

On the CDC and RF Gateway WAN link, Velocity is designed to sustain the following:

- One way latency up to 75 ms
- Up to 2 ms jitter (+/-1 ms, packet-to-packet)

The control plane will continue to function given the following:

- Up to 0.01% packet drop
- Up to 0.01% packet reordering

Other impairments, such as packet duplication, while remotely possible, are not considered.

In many data flows through the system, no change is needed since the Velocity already accounts for these effects, which remain much higher on the satellite link than on the terrestrial links. The Velocity system also prioritizes control flows over data flows. For some packet flows, however, timers will be extended and for others packet buffering will be used. See the following subsections related to impact on control and end-user data flows.

Since BB frames will be transmitted over the WAN link between the CDC and RF Gateway, the WAN must support packets larger than 1500 bytes without fragmentation. The WAN must also support jumbo frames.

2.10 Multiple Satellites per CDC

A single Central Data Center (CDC) can support multiple satellites. This removes the need to set up a separate SAS NMS and PP Cluster for each satellite connected to a teleport; saving cost on CAPEX and ongoing OPEX by removing the need for redundant equipment that might not be fully utilized for each satellite.

This improves the Velocity software for situations where multiple satellites are landing at the same teleport, or different teleports (RF Gateways) connected to a common CDC, and allows for the chassis (15152 or iHub) to be shared with multiple satellites, as well as consolidation of CDCs.

NOTE: The number of terminals that can be supported by a CDC is limited by the SAS NMS implementation and the size of the PP cluster.

The figure below shows the legacy implementation required for supporting multiple satellites with the CDC architecture. The diagram shows a single CDC site managing two RF Gateways for Sat 0, and another CDC Site managing one RF Gateway and Sat 30. For each satellite, an iGW/PP cluster and NMS instance is required.

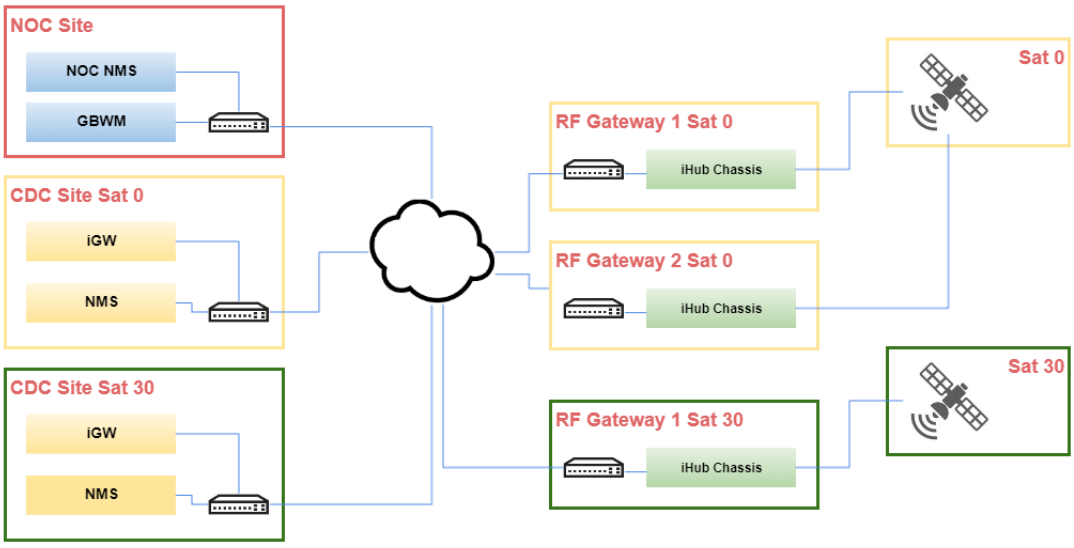


Figure 2-21. Legacy Single Satellite/CDC Architecture

However, with the Multiple Satellites per CDC feature, a single CDC can manage multiple satellites without the need for additional iGW/PP Clusters, NMS instances, and Kafka clusters.

The figure below shows a case of three satellites managed by a CDC. There is no imposed limit on the number of additional satellites that can be added to the CDC, or the number of additional RF Gateways that can be added to support them. The limitation is in the required resources in PP processing and the terminal scale restriction of the particular SAS NMS implementation.

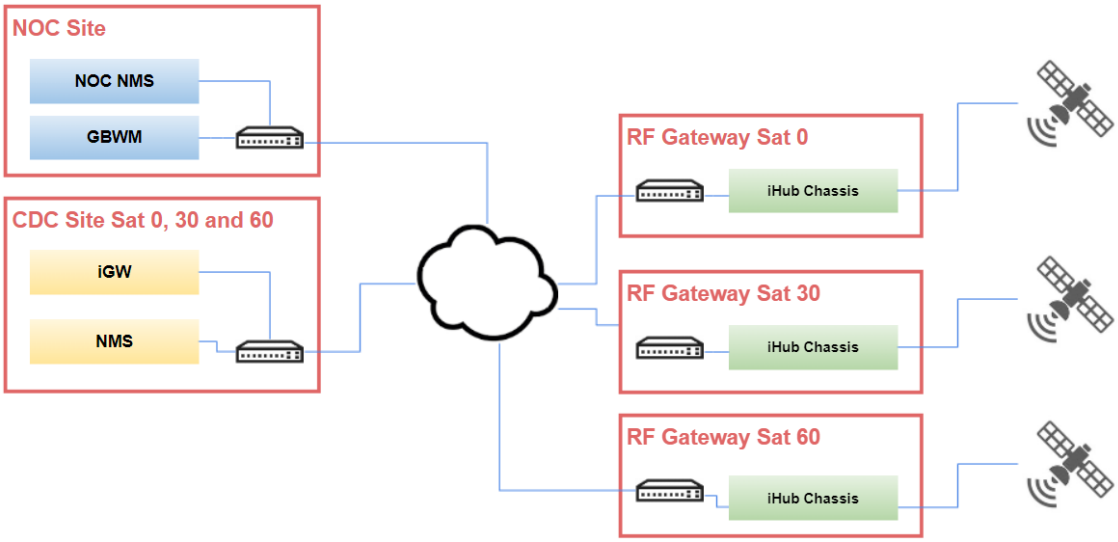


Figure 2-22. Three Satellites Managed by a Single CDC

The figure also shows multiple RF Gateways communicating with the same satellite, which requires the use of a Carrier Measurement System (CMS) for ranging measurements for each RF Gateway communicating with that satellite. An alternative approach, introduced in Velocity 4.0, is to use Two-Line Element Sets (TLE) as described in the RF Gateway Redundancy Feature.

A chassis (15152 or Hub) can be shared with the satellites associated to the CDC site. The figure below shows a single chassis supporting two satellites.

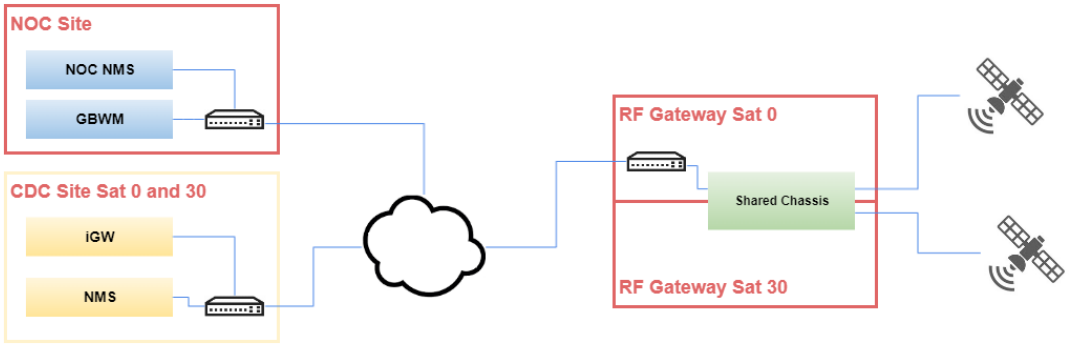


Figure 2-23. Single Chassis Supporting Two Satellites

2.10.1 Deployment Architecture

Multiple satellites can be managed by a single CDC through multiple RF Gateways. The figure below shows a CDC working with three (3) satellites, but there is no limit on the number of satellites that can be supported.

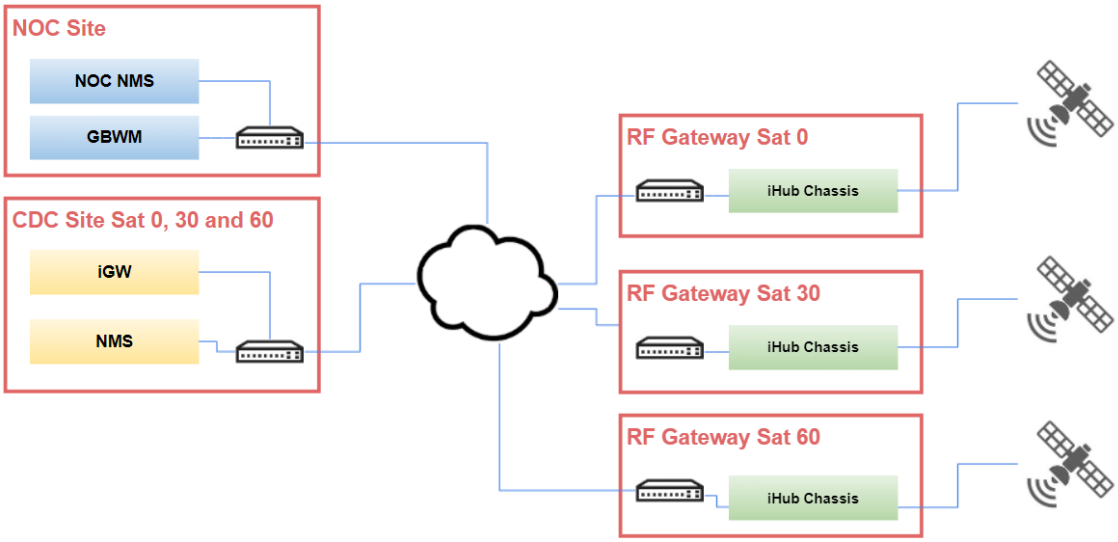


Figure 2-24. Three Satellites Managed by a Single CDC

If multiple RF Gateways are used for a satellite, a CMS is required. (An alternative approach, using Two-Line Element Sets (TLE) was introduced in the RF Gateway Redundancy Feature in Velocity 4.0.) The CMS provides measurement information that is used by the PP to ensure the downstream NCR timestamps agree at the satellite. This is what allows Seamless Beam Switch of a remote terminal between different beams on the same satellite landing at different RF Gateways. In this configuration, PP LCC will not activate iNets until timing information is available.

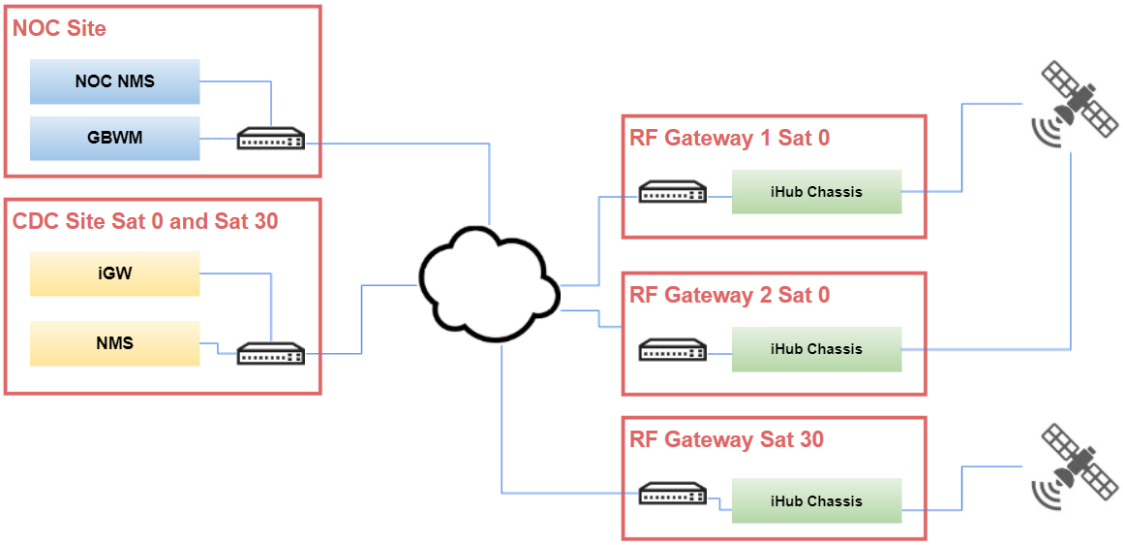


Figure 2-25. Multiple RF Gateways

This feature also allows for a chassis to support multiple satellites by assigning each chassis slot to a specific RF Gateway.

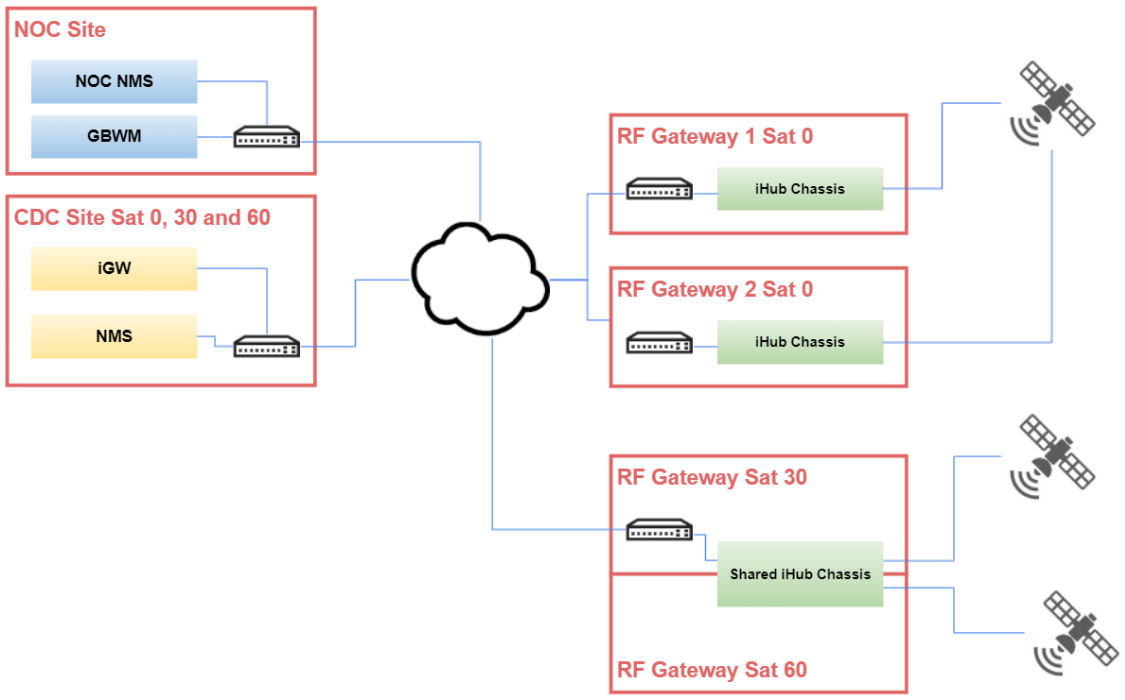


Figure 2-26. Single Chassis Supporting Multiple Satellites

2.11 The Velocity Over the Air Network

At one level, an Over the Air (OTA) network is the RF satellite network, comprised of satellite beams and channels. This is described in Chapter 3 , Satellite Link Operations.

The OTA network can also be viewed as per-satellite star topology IP subnets or LANs, though the frames and packets on this network are designed specifically for the satellite environment, to minimize overhead and compensate for satellite communications attributes like latency.

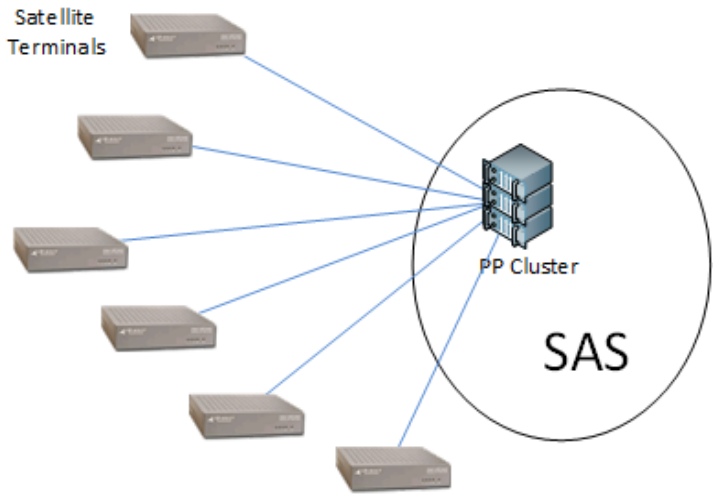


Figure 2-27. OTA Network Topology

When thought of as a LAN, the OTA network links are LAN trunks on which all traffic is 802.1q tagged, and each SVN is a VLAN.

When the OTA network is considered to be an IP network, each SVN is an IP subnet containing an interface on each PP server and an interface on each satellite router (sat0). Further, the PP and satellite router are BGP peers exchanging route advertisements to allow packets to be routed between the terminal network domain and the SAS network domain. Either iBGP or eBGP may be configured, depending on the overall IP network design.

To minimize unnecessary traffic on the satellite network, the operator should:

- Disable BGP Keep Alive messages by setting the BGP keep alive and hold timers to 0
- Limit the number of route updates sent to satellite routers through the use of route filters and route aggregation
- Configure the PPs as route reflectors for the satellite routers if iBGP is configured in the OTA Routing Domain

	NOTE: Next hop self must always be enabled in the OTA Routing Domain.

If the satellite router should be re-assigned to a new PP because the satellite router reacquired into the satellite network or the PP went off line, the satellite router and new PP will automatically establish a new BGP peering session. The satellite router and previous PP will tear down the old BGP peering relationship.

2.12 The Velocity Terminal Network

The Velocity system is designed to support any terminal network deployed at the remote location. The terminal network may be as simple as a few hosts directly connected to the satellite router, or a complex collection of VLANs and subnets.

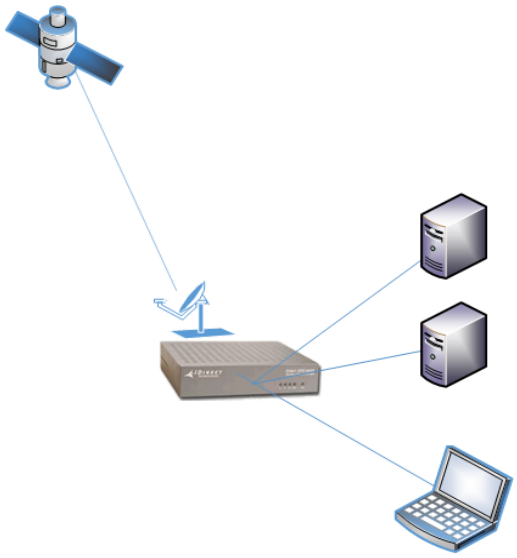


Figure 2-28. Example Terminal Network

The number of ports available on a satellite terminal depends on its composition. In its simplest form factor, the terminal has a single Ethernet port to connect to the terminal network. The terminal may also contain an internal switch, providing multiple Ethernet ports.

On the terminal switch, ports may be trunk or access. An access port carries untagged traffic, i.e. all traffic is mapped to the same SVN. Trunk ports support multiple VLANs and all frames have VLAN (802.1q) tags. Each VLAN tag is mapped to an SVN. The satellite router supports up to twelve SVNs.

The Velocity satellite terminal is often paired with an external router, as illustrated in the following image. In this topology, the router is a BGP peer of the satellite router and advertises the IP subnets behind it. The topology and routing protocols of the networks behind the router are transparent to the satellite router.

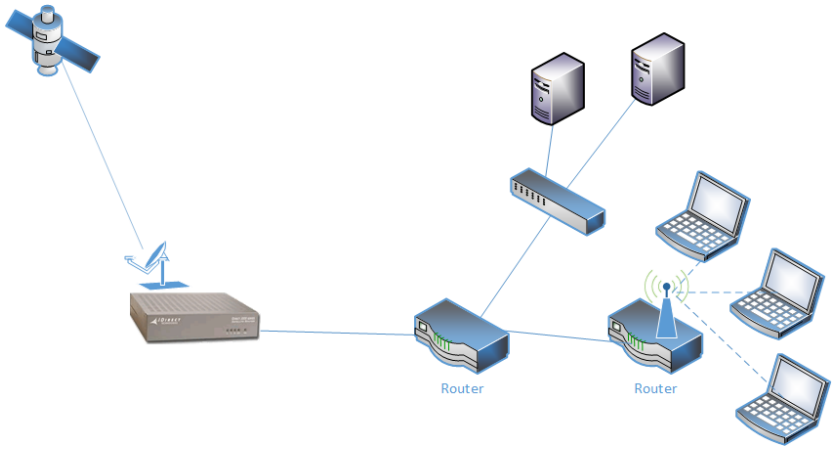


Figure 2-29. Simple Example of a Terminal Network

Even more complex topologies are possible, with multiple routers or switches connected to the Velocity satellite terminal. The only restriction is that all equipment directly connected to the terminal within a VLAN must be in the same IP subnet.

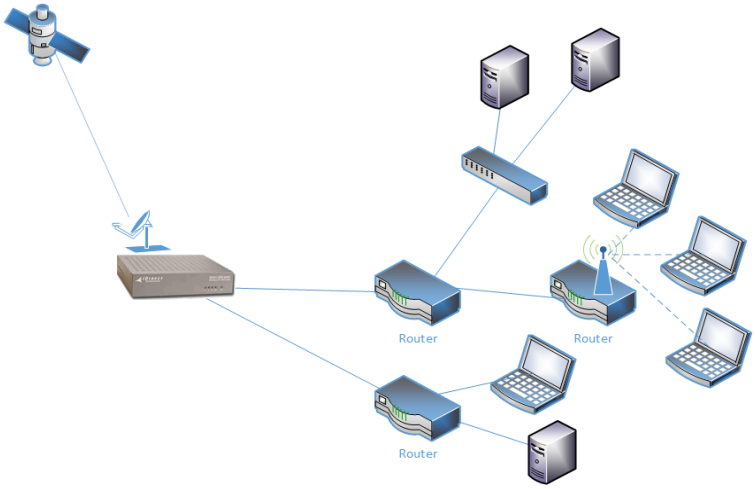


Figure 2-30. Complex Example of a Terminal Network

The Velocity terminal may be used to switch traffic on the terminal network, even when the satellite network is unavailable or the satellite link is disabled administratively. Similarly, the terminal will be active in the terminal routing domain — i.e., receiving and sending BGP messages — when the satellite network is unavailable or the satellite link is disabled.



NOTE: Also see Satellite Terminal Features on page 305, for additional information on Velocity Satellite Terminals — for example VLAN Mapping.

2.13 The Velocity NOC Network

Often, the NOC is co-located with other management systems and back office systems. Thus, the NOC network architecture is typically designed by the satellite operator and conforms to their existing practices and standards. Some concepts, however, are universal.

- The NOC is expected to meet high-availability requirements and, therefore, the NOC network contains redundant links, and redundant routing and switching equipment.
- The connection between the SASs and the NOC is likewise expected to meet high-availability requirements, thus, multiple connections to the backbone network are typically provided. Generally, these connections are to redundant edge routers.
- The EAP is a portal open to external users, and as such requires network security. It typically is deployed in a network DMZ, with firewalls between it and the external network(s), as well as a firewall between it and the NOC NMS.

These concepts are illustrated in the following diagram.

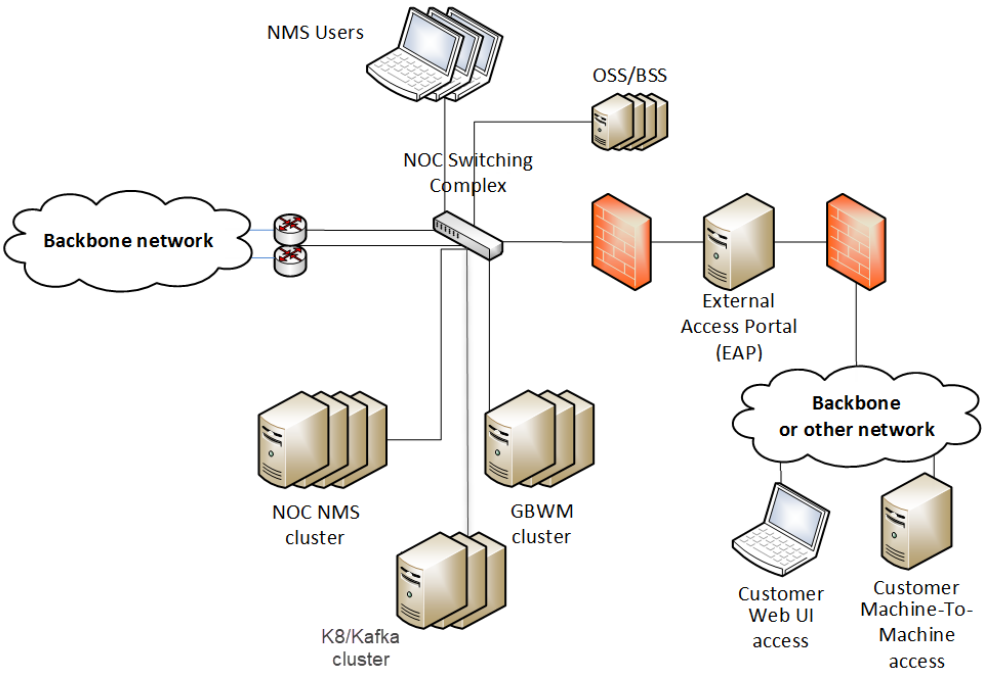


Figure 2-31. NOC Network Architecture

2.14 The Monitoring and Control Network

Internal Velocity traffic is carried across the backbone network and the OTA network in an SVN with ID 1, referred to as the Admin SVN. This includes all administrative traffic between the Pulse NMS, SAS components, and satellite routers.

Other equipment in the SAS and NOC sites (such as switches, routers, and firewalls) are typically managed using a separate logical or physical monitor and control (M&C) network. Depending on the operator's philosophy, the M&C network may also include dial-in modem connections on terminal servers in each POP and SAS to provide last-resort M&C connectivity. This M&C network may also be used to manage backbone network components.

A port on each server in Velocity clusters is designated as a debug LAN, or Dell Remote Access Controller (DRAC) port. Optionally, the satellite operator may connect this port to the M&C network to provide last-resort connectivity to debug, shut down, or recover a failed server that has become unreachable.

2.15 Velocity Data Traffic Protocols

This section describes the protocols used for handling Velocity traffic.

2.15.1 IPv4 Packet Forwarding

IPv4 packets are forwarded through the system based on the forwarding table built by the underlying routing protocols. The IPv4 implementation is compliant to all applicable standards.

An individual routing and forwarding domain exists for each SVN in the system. Incoming IPv4 packets are checked for IP header checksum validity and are discarded if the checksum fails. As the IPv4 packets are forwarded either through the satellite router or the hub-side satellite terminal proxy, the TTL (time to live) is decremented. If the TTL decrements to zero an ICMP TTL Expired message is sent back to the source address of this packet. When the TTL is decremented, the IP header checksum is updated.

By default, the DSCP (differentiated services code point) and ECN (Explicit Congestion Notification) bits are not used. However, the DSCP bits can be used in the QoS engine to prioritize packets according to their DSCP values. The system can also rewrite the DSCP bit values if this option has been configured.

Velocity does not reassemble fragmented packets; the fragment bits are ignored. Packet fragmentation is not recommended, because fragmented packets cannot be accelerated. (For the importance of being able to accelerate packets, see Transmission Control Protocol (TCP) on page 72.

2.15.2 IPv6 Packet Forwarding

The Velocity network is capable of seamlessly and simultaneously carrying traffic using both IPv4 and IPv6. Unicast IPv6 is supported with IPv4 in all Velocity datapath components in dual-stack mode in conformance with RFC 4213, though most Performance Enhancing Proxies (PEPs) do not yet support IPv6. To use IPv6, all third-party routers and switches must also support IPv6. IPv6 multicast and the Multicast Listener Discovery (MLD), however, are not

supported by the Velocity system, and thus multicast must use IPv4. The Admin SVN is also required to use IPv4.

If an SVN supports IPv6, all PP servers must be assigned an IPv6 address for that SVN.

2.15.3 Layer 2 Over Satellite (L2oS)

In Velocity, Layer 2 over Satellite (L2oS) enables the satellite link to emulate an Ethernet connection. When an iDirect network is configured for L2oS, Ethernet frames, rather than IP packets, are transported across the satellite link. See Chapter 2 , Velocity System Operations.

2.15.4 Multicast

The Velocity system supports downstream IPv4 multicast operation within any SVN. On these SVNs, the PP is an IGMPv3 proxy; and the satellite router is an IGMPv3/IGMPv4 proxy.

Velocity implements all messages defined in RFC 3376 for IGMPv2 and IGMPv3, namely Membership Query, Version 3 Membership Report, Version 2 Membership Report and Version 2 Leave Group.

There is one multicast controller process per iNet (for additional details on iNets, beams, and channels, see Velocity Space Segment on page 13. This process supports multicast independently in each SVN. When a satellite router subscribes or unsubscribes to a multicast stream, the satellite terminal proxy process on the PP server informs the multicast controller in its iNet, and the multicast controller subscribes to the stream and forwards it on the downstream. The satellite terminal proxy process also notifies the multicast controller when the satellite router switches iNets — for example, when it switches beams.

2.15.5 Transmission Control Protocol (TCP)

In satellite networks long round trip delays cause low TCP throughput. Velocity implements acceleration and compression algorithms to accelerate TCP traffic across the satellite links. Each of these features can be enabled/disabled separately for each SVN on a satellite router, and are configurable.

2.15.5.1 TCP Acceleration

The protocol layer that implements TCP acceleration in the Velocity system is a Performance Enhancing Proxy (PEP) for TCP traffic. The TCP PEP is a symmetric protocol layer implemented inside the datapath modules at the SAS and also in the remote. TCP acceleration is configurable per-SVN-per-satellite router.

TCP acceleration divides every accelerated TCP session into three links: two terrestrial links (between the communicating devices and the Velocity network), and the over the air link (between Velocity endpoints). The protocol is independent of the direction of traffic flow. On the terrestrial links, all TCP packets are sent according to the standard RFCs (Requests for Comment).

When a TCP packet arrives on the LAN (at the satellite router from a local host, or at the SAS) the TCP PEP responds immediately with locally-generated acknowledgments, so that the host on the LAN side thinks the round-trip delay is significantly lower than the actual value. The

TCP PEP handles the reliable transmission of the packet and holds the data packets until they are acknowledged by the peer end point. The size of the buffers needed to hold the unacknowledged packets is configurable and the default value is a function of configured bandwidth for the link, OTA delay, and the amount of memory available.

The TCP PEP is responsible for TCP session management to the host on the LAN side and is responsible for retransmission and handling of out of sequence packets on the LAN side. The TCP PEP handles over the air retransmission and out of sequence packets that may occur. The TCP PEP supports slow start and TCP congestion control.



NOTE: Current implementation of TCP/UDP related PEPs (acceleration and compression) is applicable to IPv4 only.

2.15.5.2 TCP Connection Acceleration

TCP can optionally be configured for accelerated connection establishment. In standard TCP, a three-way hand shake must be completed before the client can begin sending data. The TCP PEP can speed up this process by responding to the connection request immediately, in parallel with sending a connection request over the air. If the actual connection fails, the TCP PEP sends a TCP reset to the local host. Connection acceleration is performed for TCP connection requests received by both the satellite router and the PP.

2.15.5.3 TCP Header Compression

The TCP Acceleration layer compresses and decompresses the header for TCP traffic sent over the satellite, following the OTA compressed header message format. The compression context must first be established by exchanging several packets with a full header OTA message format, either IPv4 or IPv6.

2.15.5.4 HTTP Acceleration

In Velocity, HTTP requests and responses are typically transported via TCP. Thus, the time to load web pages can be greatly decreased by enabling the TCP PEP features of data acceleration (spoofing), connection acceleration, and compression.

In addition to web page acceleration, by enabling the TCP PEP features, DNS (Domain Name System) caching also speeds up web access. The DNS cache works to speed up lookups of objects in a web page without going over the air for every lookup. For additional details on DNS, see Velocity Data Traffic Protocols on page 71.

2.15.5.5 Generic Routing Encapsulation (GRE)

The Velocity system may be configured to accelerate TCP traffic in GRE tunnels that pass through the system. These tunnels do not originate or terminate on Velocity equipment, but the existence of those tunnels must be configured in the Velocity system for TCP acceleration to occur for the traffic within them.

2.15.6 User Datagram Protocol (UDP)

Header compression PEP is used to reduce the bandwidth usage of UDP and RTP (Real-Time Protocol) packets. UDP packets have 28 bytes of header overhead in every packet transmitted over IPv4 networks. VoIP (Voice over Internet Protocol) packets in RTP can be compressed even more as they have an additional 12-byte header.

cRTP significantly reduces bandwidth usage over the air. UDP header compression follows the specification in RFC 3095/3545, ***Compressing IP/UDP/RTP Headers for Low-Speed Serial Links***. This RFC defines both cRTP (header compression for RTP packets) and UDP header compression (for other UDP packets).

Header compression is performed only on packets which have a length below a certain threshold. Above that threshold, bandwidth savings with compression is not significant.

Compressed Real-Time Protocol (cRTP)—The satellite operator can enable cRTP to significantly reduce the bandwidth requirements of VoIP flows. cRTP is implemented using standard header compression techniques. It allows for better use of real-time bandwidth, especially for RTP-based applications that use large numbers of small packets, since the 40-byte IP/UDP/RTP header often accounts for a significant fraction of the total packet length. iDirect has implemented a standard header compression scheme including heuristic-based RTP detection with negative cache support for misidentified UDP streams. For example, G729 voice RTP results in less than 12 Kbps (uncompressed is 24 Kbps).

2.16 Velocity Routing and Control Protocols

This section presents the protocols used in Velocity for routing and control.

2.16.1 Border Gateway Protocol (BGP)

The Velocity system supports MP-BGP with the IPv4 and IPv6 address families. The IPv4 BGP implementation advertises both IPv4 and IPv6 IP addresses, allowing the system to support both IPv4 and IPv6. This implementation is based on RFC 4271 and RFC 1771.



NOTE: The vpnv4, vpnv6, and l2vpn address families are not supported.

Features supported by the Velocity BGP implementation include:

- Connected and static route redistribution
- Route reflection (iBGP)
- Next hop self (iBGP)
- Peer groups
- Outbound route filters (ORF) — address, community, and extended-community based
- Passive mode
- Maximum prefixes learned from a BGP peer

- Conditional advertisement
- Route aggregation
- MD5 authentication

The Velocity BGP implements the following BGP attributes:

- Origin
- AS_Pat
- Next_Hop
- Local_Pref
- MED (Multi-exit discriminator)
- Community and extended community
- Weight (Cisco-defined)

The following optional and discretionary BGP attributes are not supported by the Velocity BGP implementation:

- Atomic-aggregate
- Originator ID
- Cluster list
- Aggregator

The standard BGP timers are configurable, namely keep-alive timer, connection retry timer, and hold timer.

BGP can be used to build complex policy-based architectures. BGP provides a list of attributes through which policies can be set. The Velocity system implements the following policies:

- Set the LOCAL_PREF if a community is present
- Set a weight based on the AS path
- Filter out all IPv4 address prefixes of length greater than 32
- Filter out all IPv6 address prefixes of length greater than 128
- Remove route targets from a route
- Set a specific community value based on the AS path
- Prepend the AS path
- Match and replace an AS number in the AS path

2.16.2 IGMP

See Multicast on page 72.

2.16.3 Virtual Routing and Forwarding (VRF)-Lite

The Velocity system supports VRF-Lite. VRF-lite is a particular way of using the VRF technology. A VRF is a standalone routing table with its own set of interfaces, its own CEF instance and its own rules about populating and sharing its contents. Only interfaces associated with a particular VRF can communicate with each other (provided that by normal routing rules, a packet entering one of this set of interfaces has its destination reachable via another interface in this set, and this destination is properly recorded in the VRF).

Interfaces in different VRFs in general cannot talk to each other. There are, however, some specific exceptions. VRFs are to routers what VLANs are to switches—they both allow you to create multiple virtual devices on top of the physical device. With VLANs, you virtualize a single switch into multiple virtual switches. With VRFs, you virtualize a single router into multiple virtual routers. VRFs, used as described here, is considered VRF-Lite.

In a way similar to switches and VLANs, you group a set of interfaces into a VRF and thereby limit their mutual visibility to them alone, isolating them from all other interfaces on the router.



NOTE: The Velocity system employs what is referred to as VRF-Lite. It is considered a full VRF when another technology such as MPLS or LISP is added to communicate with other routers having similar VRFs while allowing to carry all traffic via a single interface and being able to tell the packets apart.

2.17 Velocity Satellite Router Protocols

In addition to the Velocity Data Traffic Protocols on page 71 and the Velocity Routing and Control Protocols on page 74, the satellite router also supports the protocols described in this section. These protocols are on the terminal LAN, as described in The Velocity Terminal Network on page 68.

2.17.1 DHCP

The Dynamic Host Configuration Protocol (DHCP) is a configuration protocol for hosts on IP networks. DHCP eliminates manual configuration and allocates IP addresses dynamically to clients from an address pool managed by DHCP.

The satellite router may be configured to work as a DHCP server or a DHCP relay. The configuration is per-terminal and per-SVN, thus for each SVN on a given satellite router one of the following is true:

- DHCP is disabled.
- DHCP is in server mode, and the client address space and the other values contained in the DHCP response are configured.
- DHCP is in relay mode, and the IP addresses of the DNS server to which DHCP requests are relayed are configured.

The service provider or subscriber specifies one of these as part of the satellite router configuration.

In server mode, the satellite router responds to a client's DHCP request with:

- A host IP address dynamically allocated from a pool of IP addresses
- The IP addresses of the primary and secondary DNS servers the client should use
- The IP address of the default gateway the client should use
- The broadcast IP address the client should use
- The configuration, including host IP address, which can be valid for a specified duration

In relay mode:

- The satellite router forwards the clients DHCP request to a configured DHCP server.
- The satellite router receives a response from the server and forwards it to the client.
- The contents of the response are determined by the configured DHCP server.

2.17.2 Network Address/Port Address Translation (NAT/PAT)

The satellite router implements network address translation (NAT) and port address translation (PAT) for IPv4 networks. Port forwarding rules may also be configured on the satellite router. The public address that is used for translation is configured on the satellite router.

NAT modifies the IP addresses and port numbers in the IP packet header, so nodes that reside on a private network can share a public IP address for communication with the outside world. It is frequently used for TCP, UDP and ICMP packets. However, NAT inherently breaks protocols in which IP addresses are carried in the body of the packet. SIP (Session Initiation Protocol) is one of these protocols. Velocity provides a SIP ALG (Application Layer Gateway) to overcome this limitation.

NAT can be enabled and configured independently for each SVN instance on the satellite router. The public IP address is the satellite router satellite (sat0) interface address within that SVN. The satellite router supports NAT and PAT, and port forwarding.

2.17.3 Domain Name System (DNS)

The Domain Name System (DNS) is a distributed database used by IP applications to map between host names and IP addresses. It provides the protocol that allows clients and servers to communicate with each other to resolve domain names to IP addresses and vice versa.

The satellite router's DNS server is not a zone authority (that is, it does not maintain a database for a site or zone). The DNS server allows hosts on the remote network to resolve domain names to IP addresses without having to go over the satellite link by acting as a caching proxy.

If the DNS component is enabled and if hosts on the remote network are configured to look to the remote satellite terminal as their primary DNS server, then the DNS component acts as a

recursive caching name server to hosts on the remote network. There can be one DNS server per VLAN. Each DNS server maintains its own independent cache.

The DNS component functions as follows:

1. Clients on the remote network issue DNS queries to the DNS component through UDP port 53 (the well-known DNS port) of the remote satellite terminal.
2. The DNS receives the query and checks if recursion is desired. If recursion is not desired, the DNS replies with the addresses of the primary and secondary name servers.
3. The DNS searches its cache for a matching response.
4. When a matching response is found, the DNS replies with the cached response.
5. When a matching response is not found, the DNS sends the query to its primary DNS server.
6. When a response is received within the time specified by the forward timeout parameter the DNS performs the following:
   a. Adds the response to its cache
   b. Forwards the response to the client
7. When a response is not received within the time specified by the forward timeout parameter, the DNS performs the following:
   a. Forwards the query to the secondary name server
   b. Resets the timestamp associated with the forwarded query
8. When a response is received from the secondary name server within the time specified by the forward timeout parameter, the DNS performs the items listed in Step 6 above.
9. When a response is not received from the secondary name server within the time specified by the forward timeout parameter, the client's DNS query times out.

2.17.4 Session Initiation Protocol/Application Layer Gateway

Session Initiation Protocol/Application Layer Gateway (SIP ALG) packets can make use of UDP or TCP as the transport protocol, but the use of UDP is more common. Thus, they are transported through the Velocity system and are treated in the same way as other TCP/UDP packets. In particular, header compression may be enabled.

However, when NAT is enabled at the satellite router, the SIP protocol is unable to establish connections, because SIP connection negotiation includes the RTP connection information including IP address within the SDP. iDirect has implemented a SIP ALG to solve this problem.

The SIP packets are snooped by the SIP ALG. When the SIP message contains IP addresses that are subject to NAT, a port forwarding rule is dynamically created in the NAT layer and the SIP proxy modifies the contents of the address field with the address provided by the NAT layer for this port forwarding rule. When the call is terminated, the port forwarding rule is removed from the NAT table.

2.17.5 RADIUS

The Velocity system provides RADIUS clients for:

- Authentication of a terminal SVN by the owner of the SVN
- Dynamic address assignment to satellite router SVN IP (eth0) interfaces
- Reporting traffic usage, i.e., input and output octets

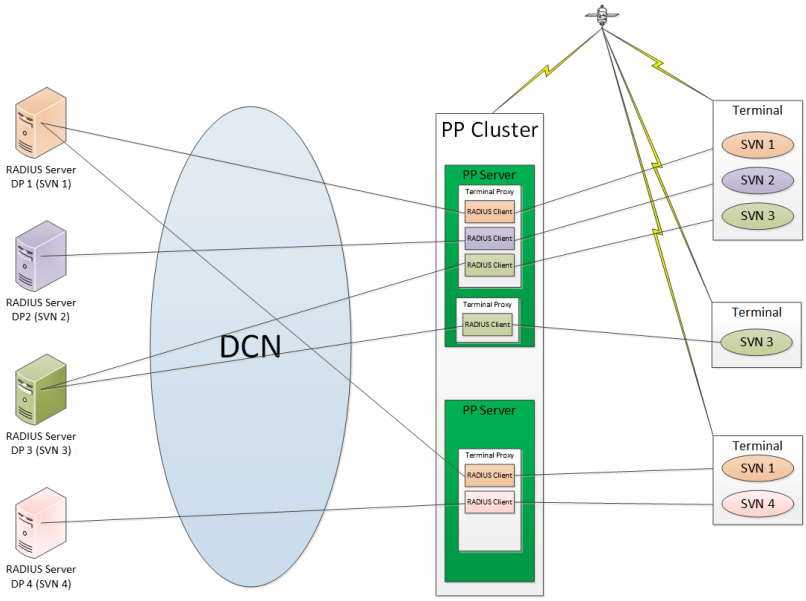


Figure 2-32. RADIUS

The optional use of RADIUS is configured on a per-terminal, per-SVN basis. When a terminal SVN is so configured, the PP server acts as a RADIUS client on behalf of the satellite router for that SVN. This is done independently within each SVN the satellite router is configured to use. Note that if RADIUS is used to dynamically assign an SVN IP address and subnet to the eth0 interface, DHCP must be enabled so that the satellite router will dynamically assign IP addresses to the hosts on the terminal network.

The Velocity RADIUS implementation requires that all three functions are performed for an SVN for which RADIUS is enabled.

The RADIUS server functionality is the responsibility of the administrator of the SVN and is not provided by iDirect. The administrator must also assign an IP address within the SVN to each RADIUS server.

The Velocity RADIUS implementation complies with the applicable portions of the following standards:

- ***RFC 2865 — Remote Authentication Dial In User Service (RADIUS)***
- ***RFC 2866 — RADIUS Accounting***

- ***RFC 5176 — Dynamic Authorization Extensions to Remote Authentication Dial In User Service (RADIUS)***
- ***RFC 3162 — RADIUS and IPv6***
- ***RFC 2869 — RADIUS Extensions***
- ***RFC 2548 — Microsoft Vendor-Specific RADIUS Attributes***
- ***3GPP TS 29.061*** — 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; ***Inter-working between the Public Land Mobile Network (PLMN) supporting packet based services and Packet Data Networks (PDN)***, Release 6, version 6.7.0, December 2005
- ***3GPP TS 24.008*** — 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; ***Mobile radio interface Layer 3 specification***; ***Core network protocols***; ***Stage 3***, Release 11. Version 11.4.0, September 2012

2.17.5.1 Accounting

RADIUS session accounting is done by the PP server, which sends updates for any authenticated SVN to the appropriate RADIUS accounting server. These updates typically report traffic usage for the session, which includes both input and output octets. The IP addresses may also be included in these updates. The intervals between two consecutive updates is configurable, but may not be less than 30 minutes.

2.17.5.2 RADIUS Session Disconnect

A RADIUS session can be disconnected by a RADIUS server. Upon receiving the session disconnect request from the RADIUS server, the PP responds, with the appropriate reply depending upon whether the request is valid or not; meanwhile, it sends a message to the satellite router to disable the SVN.

When a satellite router travels beyond the boundary of a SAS or goes offline, the PP determines that the satellite router is offline and terminates the RADIUS sessions for its SVNs with the RADIUS server.

2.17.5.3 RADIUS Authentication

Included in the DFOE message is the IP address for the satellite router's interface to the satellite terminal network. The operator may configure that IP address using the Pulse NMS, or be assigned an IP address by a RADIUS server when the terminal comes into the network.

3 Satellite Link Operations

Satellite link operations include those processes and timing mechanisms by which satellite terminals acquire into the network — the various modulation, and error correction techniques applied to both upstream and downstream carriers.

This chapter introduces and describes those uplink and downlink operations that facilitate communication between the Velocity SAS and satellite terminals.

The following sections are included in this chapter:

3.1 Network Acquisition 82
3.2 DVB-S2 Overview 88
3.3 Downstream Carrier Operation with DVB-S2 94
3.4 About the Upstream TDMA Waveform 99
3.5 Upstream Carrier Operation with Adaptive TDMA 100
3.6 About the Uplink Control Process 113

3.1 Network Acquisition

This section describes how satellite terminals join iDirect TDMA networks. In a Velocity™ network, satellite beam configuration data is broadcast to terminals within the network using a ***signaling carrier***. In Velocity, this signaling carrier is called an *acquisition signaling carrier*. This data includes beam footprint data contained in a low-resolution beam map, and link parameters such as outbound carrier frequency, symbol rate, and polarization.

Terminals use this information to join the network. Once a terminal is able to receive downstream information, the inbound acquisition process requires the satellite terminal to transmit one or more acquisition bursts on designated inroutes to allow traffic flow in the upstream direction and keep the satellite terminal in network using the Uplink Control process (UCP). The protocol processor at the SAS controls the acquisition process.

3.1.1 Acquisition Signaling Carriers (ASC)

The signaling carrier is a way to inform terminals of the current satellite network configuration — more specifically, beam footprint and or/and parameters such as outbound carrier frequency, symbol rate and/or polarization. The ***acquisition signaling carrier*** is a Velocity signaling carrier in which the SAS broadcasts the satellite configuration on a single satellite uplink, which the satellite in turn replicates on a number of downlinks in different beams. This signaling carrier has no return channel.

Terminals require this information in order to join (or come into) the satellite network. This information exchange is one-way (downstream only), and is necessary to allow satellite network operators to reconfigure their satellite. If a change in downstream carrier configuration, in terms of frequency or symbol rate, is not properly communicated to a terminal, it is likely to become stranded.

There are several reasons that can lead to a change in carriers, including:

- The satellite was reconfigured (either with a channelizer, or steerable beams, or otherwise reconfigurable beams).
- The satellite supports power sharing between beams and the operator wants to adjust the outbound symbol rates to use the power in a way that best suits the business need.
- The customer originally licensed a limited amount of bandwidth support from iDirect, and is increasing that because the network is growing.
- Frequency plan changes to mitigate satellite channel interference

These changes may occur daily, or multiple times per day as the satellite operator moves capacity to follow demand. To ensure the terminals in-network can adjust to changes in the satellite configuration, the Velocity™ system periodically broadcasts the current satellite configuration on all its service channels. Thus, any terminal currently in-network can take the appropriate actions to remain in the satellite network.

A powered off terminal will not receive the satellite configuration update, and will not be able to enter the network when it is powered on. Therefore, terminals are configured to know of one or more signaling carriers. A terminal can lock onto a signaling carrier to receive the broadcast satellite configuration.

3.1.1.1 The ASC in a Satellite Network

For all areas in the satellite network where service is required, at least one beam with an ASC must be visible to every terminal. This means that if one beam lies entirely within another beam, the fully-contained beam may have, but is not required to have, an ASC of its own. For example, as illustrated in the diagram, Beam 2 lies fully within Beam 1. Therefore, Beam 2 would not need its own ASC, as it would have access to Beam 1's ASC. However, since Beam 3 does not lie entirely within Beam 1, Beam 3 would need its own ASC.

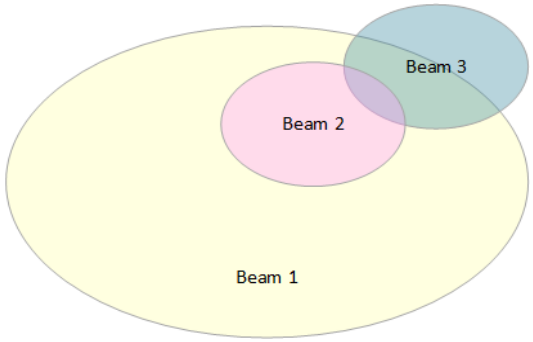


Figure 3-1. Contained Beam

In Velocity, three ASC variations are supported, whereby the SAS broadcasts the satellite configuration:

3.1.1.2 Dedicated ASC

As implied, the dedicated ASC is used for signaling only. Having a single uplink and downlink, this ASC option is based on a 1-to-1 relationship involving a single DVB-S2 (digital video broadcasting over satellite, 2nd generation) uplink and a single downlink per beam with a minimum of 1 Msps associated symbol rate.

In those business models where a satellite operator sells blocks of satellite MHz (megahertz) bandwidth to other providers, service channels are assigned to each provider. An ASC assigned to each provider is suited to this model.

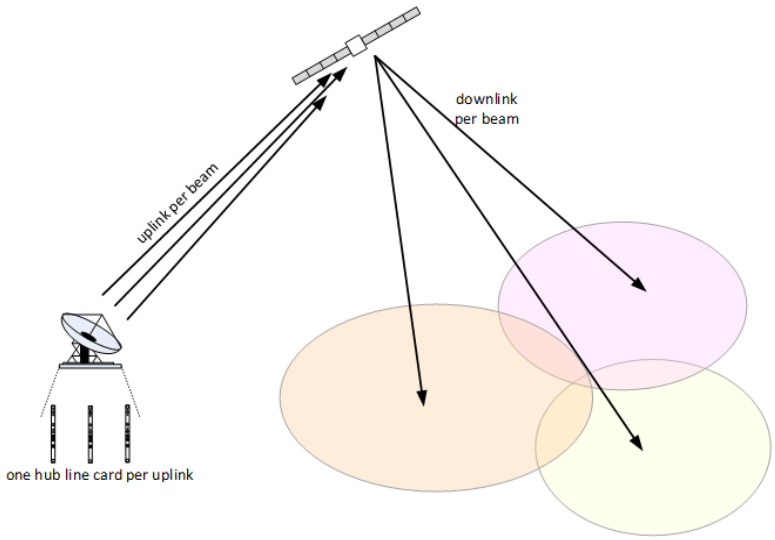


Figure 3-2. Dedicated DVB-S2 ASC

3.1.1.3 Dedicated ASC with Fan-Out

This ASC option is dedicated to signaling only, and supports a single uplink that is replicated on multiple downlinks in different beams. Each downlink uses the same symbol rate (minimum of 1 Msps) and modulation and coding (MODCOD) parameters, but could have different frequency and polarization parameters, depending on the specific beam on which the downlink is transmitted.

This configuration is possible in high throughput satellites that are equipped with an onboard digital channelizer.

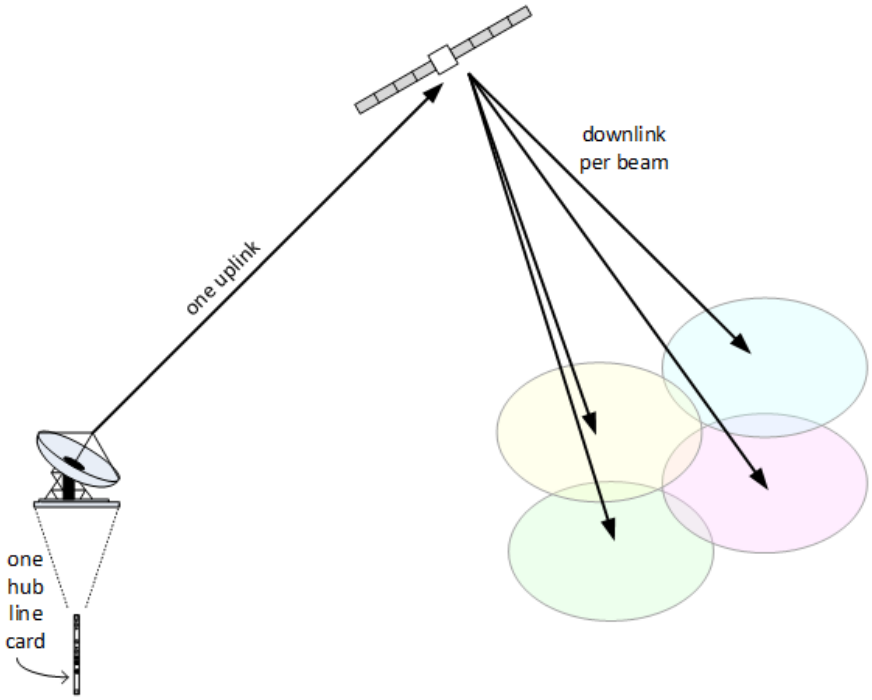


Figure 3-3. DVB-S2 with Fan-out

3.1.1.4 Shared Data Carrier/ASC

This ASC option offers a cost-effective signaling carrier solution that provides the ASC function using standard service carriers, as opposed to using one or more dedicated ASCs. In this case, the satellite terminal signaling carrier search function uses defined service carriers that have been designated as signaling carriers as well.

In this configuration, terminals receive the beam map information only from the broadcasts on the designated service carriers. The shared Data Carrier/ASC is suited to situations in which the satellite network configuration is relatively static and does not typically require reconfiguration.

When using the data carrier in the ASC function, the beam map traffic is sent at the highest priority, ahead of all data traffic, and is not applied against customer bandwidth allocation.

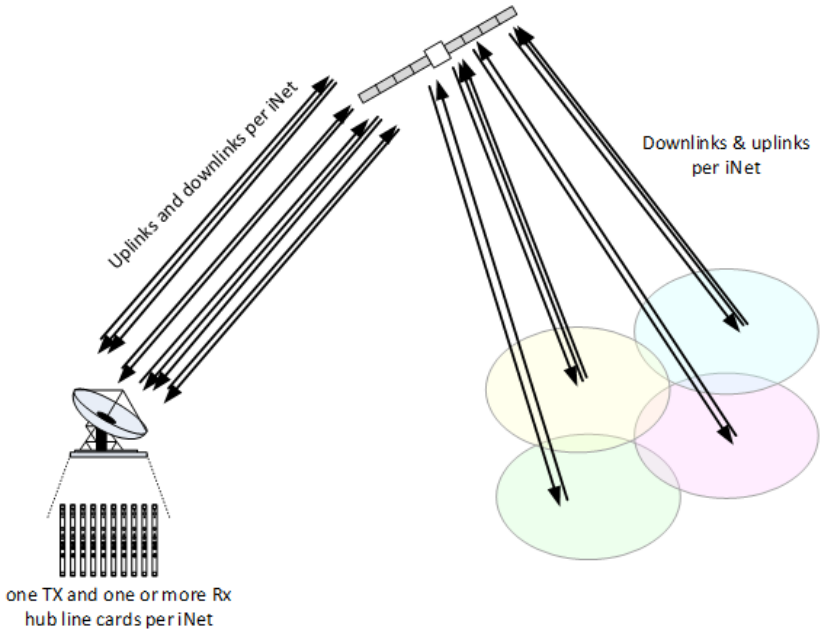


Figure 3-4. Shared DVB-S2 Carrier/ASC

3.1.2 Acquisition Methods

The protocol processor broadcasts time-plan messages on the iNet downstream carrier that allocates upstream traffic slots to the satellite terminals in the network, and available acquisition slots to satellite terminals that are not in the network. There are two types of acquisitions — "acquisition by invitation" and "acquisition by contention".

Acquisition by invitation is typically used, by terminals, during a recovery process when the SAS has knowledge of which terminals may be in a given beam. ***Acquisition by contention*** is used by terminals for the typical or normal acquisition operation.

Information pertinent to the acquisition process, and included in the broadcast signaling, includes the following:

- The symbol rate and detection threshold of the reference carrier
- The initial antenna flange transmit power at the reference symbol rate per terminal type
- The maximum operational ***C/N*** per terminal type
- Symbol rates and detection thresholds for carriers with actual acquisition slots, whether used in contention or invitation mode

If the satellite terminal does not receive a response to the acquisition transmission, it is assumed to have been lost (or suffered a collision in a contention-based arrangement). The terminal will then transmit another acquisition burst. This will be transmitted in the next invitation assignment received, or in a contention slot selected following a random interval chosen from a range that grows exponentially with each failed attempt.

The satellite terminal thus handles transitions between invitation-based and contention-based operation transparently; no re-configuration is required. A satellite terminal that may not have succeeded during the invitation phase, will automatically continue its log on procedure using contention slots. In both cases, the satellite terminal will ensure the maximum allowed power spectral density is not exceeded.

The physical-layer acquisition is considered complete when the acquisition burst has been received and the first correction message has been sent. Following this, the satellite terminal is authenticated and accepted into the iNet, given its QoS allocations, etc., and is allowed to start sending traffic and capacity requests. These aspects are outside the physical layer acquisition process.

3.1.2.1 Acquisition Bursts

Velocity supports two types of satellite terminal acquisition: superburst acquisition and spread spectrum acquisition.

Superburst acquisition greatly improves the time and bandwidth required by satellite terminals to join the network. ***Spread spectrum acquisition*** supports high-speed AERO terminals with a new acquisition burst waveform that may be received at a much lower ***C/N*** than superburst, while being similarly tolerant to Doppler. These bursts are arranged with the following properties in mind:

- Bursts can be received at a very low ***C/N***, accommodating rain fade — which is unknown at the time of logon — as well as conservative estimates of the initial transmitted power that may be used to avoid violating regulatory limits.
- The performance of the acquisition bursts is primarily defined by the ability of the scheme to overcome uncertainties in terms of frequency, timing, and signal level that inevitably exist when the satellite terminal has been out of the network for any length of time. It is for these reasons that the acquisition slots have sufficiently large guard times to accommodate initial timing uncertainty.
- The superburst receiver has a frequency tolerance of ±7.5% of the symbol rate.
- The spread-spectrum acquisition burst receiver has a tolerance of ±3% of the chip rate.
- A satellite terminal may need to perform frequency sweeping that involves transmission of a small number of additional acquisition bursts. Sweep parameters are configurable in the network management system (NMS).
- More acquisition attempts may be required in the case of high-speed mobile satellite terminals due to the high Doppler shift. To increase the probability of successful attempts and provide additional robustness for the spread-spectrum acquisition, Doppler correction is employed as a means of reducing frequency uncertainty.

  This correction can ultimately serve to allow lower symbol rates, in particular for the high-speed profile. An initial Doppler estimate is applied to both the carrier frequency and timing drift based on the downstream frequency component and the satellite terminal's position, heading and speed values.

3.2 DVB-S2 Overview

Digital Video Broadcasting (DVB) represents a set of open standards for satellite digital broadcasting. DVB-S2 (Digital Video Broadcasting over Satellite, 2nd generation) is an extension to the widely-used DVB-S (Digital Video Broadcasting over Satellite) standard.

The following are supported by DVB-S2:

- A powerful forward error correction (FEC) system with concatenation of BCH (Bose-Chaudhuri-Hocquenghem) with low-density parity check (LDPC) inner coding
- Higher order modulation schemes with 8PSK (phase shift keying) and16APSK (amplitude and phase shift keying) in addition to QPSK (quadrature phase shift keying)

- Dynamic variation of the transmission parameters on a frame-by-frame basis with the introduction of adaptive coding and modulation (ACM)

These improvements lead to greater efficiencies and flexibility in use of available bandwidth.

3.2.1 DVB-S2 Baseband Modulation and Coding

A BBFRAME (Baseband Frame) is the basic unit of the DVB-S2 protocol. DVB-S2 supports two frame sizes: short and long. Each frame type is defined in the DVB-S2 standard in terms of the number of coded bits: short frames contain 16200 coded bits; long frames contain 64800 coded bits. The Velocity system supports short frames, as illustrated in the diagram.

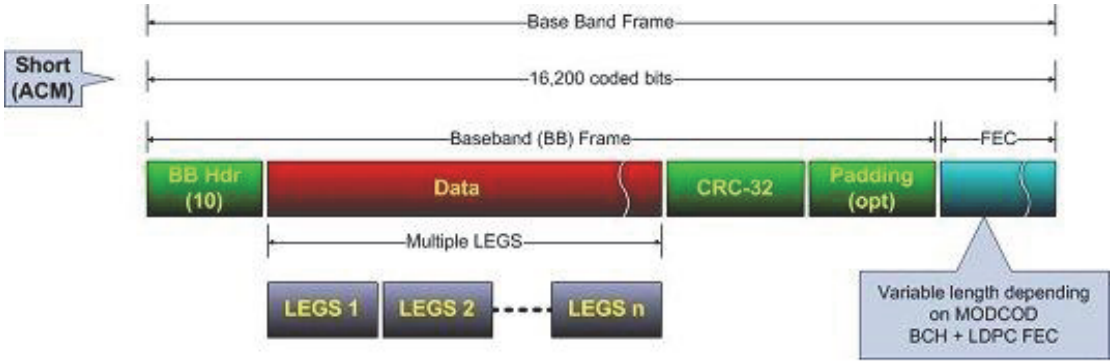


Figure 3-5. DVB-S2 Short Frame Protocol Structure

iDirect uses a proprietary data encapsulation protocol called LEGS (Lightweight Encapsulation for Generic Streams) for efficient encapsulation of downstream data between the DVB-S2 line cards and satellite terminals using generic streams. The LEGS protocol maximizes the efficiency of data packing into BBFRAMES on the downstream. For example, if a timeplan only takes up 80% of a BBFRAME, the LEGS protocol allows the transmit (Tx) line card to include a portion of another packet that is ready for transmission in the same frame. This results in maximizing the use of the downstream bandwidth.

MODCOD refers to the combinations of modulation types and error coding schemes supported by the DVB-S2 standard. The higher the modulation, the greater the number of bits per symbol (or bits per Hz). The modulation types specified by the DVB-S2 standard and supported by the iDirect system are documented in the ***iDirect Link Budget Analysis Guide***.

Coding refers to the error-correction coding schemes available. Low-density parity-check (LDPC) and Bose-Chaudhuri-Hocquenghem (BCH) codes are used in DVB-S2. Effective rates are 1/4 through 9/10. The 9/10 coding rates are not supported for short BBFRAMEs.

The DVB-S2 standard does not support every combination of modulation and coding. In general, the lower the MODCOD, the more robust the error correction, and the lower the spectral efficiency. The higher the MODCOD, the less robust the error correction, and the greater the spectral efficiency.

DVB-S2 defines three methods of applying modulation and coding to a data stream:

- ACM (adaptive coding and modulation) specifies that every BBFRAME can be transmitted on a different MODCOD. Terminals receiving an ACM carrier cannot anticipate the

MODCOD of the next BBFRAME. A DVB-S2 demodulator must be designed to handle dynamic MODCOD variation. iDirect DVB-S2 networks support ACM on the downstream carrier with all modulations up to 16APSK. An iDirect DVB-S2 network uses short DVB-S2 BBFRAMES for ACM.

- CCM (constant coding and modulation) specifies that every BBFRAME is transmitted at the same MODCOD. It is possible to simulate a CCM outbound carrier by selecting ACM and setting the terminal's maximum and minimum MODCODs to the same value.
- VCM (variable coding and modulation) specifies that MODCODs are assigned according to service type. As with ACM, the resulting downstream contains BBFRAMEs transmitted at different MODCODs. iDirect does not support VCM on the downstream carrier.

The iDirect CCM Mode and ACM Mode are compared in the following diagram.

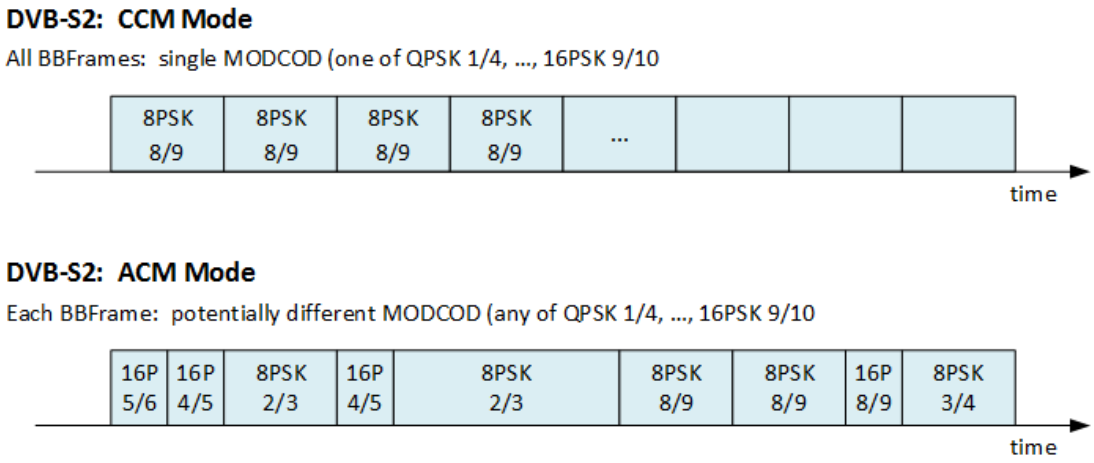


Figure 3-6. Comparison of Constant Coding and Adaptive Coding Modes

The list of DVB-S2 MODCODs and modes of operation are documented in the ***iDirect Link Budget Analysis Guide***.

3.2.2 DVB-S2 Physical Layer Framing

An iDirect DVB-S2 downstream contains a continuous stream of Physical Layer Frames (PLFRAMEs). The PLHEADER indicates the type of modulation and error correction coding used on the subsequent data. It also indicates the data format and frame length. An example of physical layer frames is illustrated in the diagram.

The PLHEADER always uses π/2 BPSK modulation. The overhead associated with pilot symbols injected into the data stream is 2.4%.

The symbol rate remains fixed on the DVB-S2 downstream. Variation in throughput is realized through the variation of MODCODs in ACM mode. For example, the throughput of the DVB-S2 carrier is approximately 143 Mbps when using a symbol rate of 45 Msps at the highest MODCOD, 16APSK 8/9, with a 5% roll off factor.

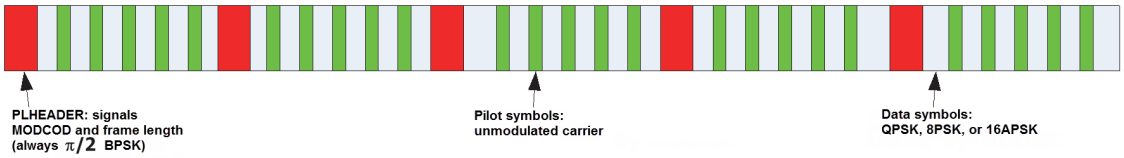


Figure 3-7. Physical Layer Frames

3.2.2.1 DVB-S2 Roll-Off Factor

Digital communication systems employ waveform pulse shaping using identical matched filters at the transmit and receive sides to limit the occupied bandwidth required by a carrier and to maximize the received signal-to-noise ratio (SNR). The matched filters used for linear modulation signals belong to a class of Nyquist filters having Raised-Cosine filter shapes with different carrier roll-off factors.

The roll-off factor of a digital filter defines how much more bandwidth the filter occupies than that of an ideal brick wall filter, which defines the theoretical minimum occupied bandwidth. This dictates the guard band requirement for the carrier. The DVB-S2 standard specifies roll-off factors of 20%, 25%, and 35%.

iDirect has reduced the DVB-S2 roll-off factor (and, therefore, the required carrier guard band) to as low as 5% of the carrier symbol rate. With a 5% roll-off factor, the transmitted spectrum more closely resembles the ideal brick wall spectrum with occupied bandwidth at 1.05 times the carrier symbol rate. Not all DVB-S2 carriers support a 5% guard band. Specifically, DVB-S2 carriers below 10 Msps require a guard band of at least 10%. In addition, guard bands of 10% or 5% require a minimum MODCOD of QPSK-1/2.

3.2.2.2 Guard Band Reduction for Improved Throughput

A lower bandwidth requirement between carriers makes it possible to fit a higher bit rate carrier into the same occupied bandwidth or to fit additional carriers into the same overall bandwidth. The example in this section shows how a lower guard band requirement can allow a satellite operator to increase the information rate of a DVB-S2 carrier without changing the occupied bandwidth and allotted power.



NOTE: Increasing the information rate of existing carriers may affect the link budget of the satellite network. This is especially true for upstream carriers since uplink power control (UPC) will automatically increase the transmit power of the satellite terminals. Consult with the link budget provider prior to adjusting carrier configurations.

3.2.2 shows the increase in information rate that can be achieved on a DVB-S2 downstream carrier by decreasing the guard band while holding the occupied bandwidth constant. The calculations were performed using the 8PSK-8/9 MODCOD. In the example, reducing the guard band from 20% to 5% of the symbol rate results in a 14.29% increase in information rate.

Table 3-1. Increasing Information Rate by Reducing Guard Band

Guard Band	Occupied Bandwidth (kHz)	Symbol Rate (ksym/s)	Information Rate (kbps)
20%	20000	16666.667	41729
15%	20000	17391.304	43543
10%	20000	18181.818	45522
5%	20000	19047.619	47690

3.2.2.3 Group Delay and Downstream Performance

DVB-S2 waveform performance is affected on terminals when operated under channel conditions characterized by high transponder group delay. Such atypical conditions are encountered in single-carrier-per-transponder applications with certain transponder model types and to the extent symbol rate occupies the transponder bandwidth depending on the configured roll-off factor. When the symbol rate exceeds 75% of transponder bandwidth (18/27/40.5 Msps for 24/36/54 MHz transponders), customers are encouraged to check the cascaded IMUX (input multiplexer) and OMUX (output multiplexer) filter group delay of the transponder, usually available from the satellite operator. These plots or specifications capture the peak-to-peak ripple (in nanoseconds) of the group delay at different frequency offsets from the transponder center.

The demodulator handles the group delay levels with robust equalizers usually up to 1.5 symbol periods across symbol rate bandwidth with negligible SNR degradation. Group delay exceeding 1.5 symbols periods (i.e. approximately 43.8 ns peak-topeak based on the 34.285 Msps with 5% ROF carrier on a 36 MHz transponder) will result in noticeable SNR degradation and negatively affect link performance.

A group delay exceeding these limits usually requires symbol rate/roll-off factor reduction to keep away from the group delay edges for satisfying identified limits.

3.2.2.4 DVB-S2 Adjacent Channel Interference

The channel spacing between carriers is generally dictated by the roll-off factors of the carriers, although the satellite operator may mandate spacing higher than that allowed by the carrier roll-off. The occupied bandwidth of the carrier, with symbol rate Fsym and roll-off factor a, is given as (1 + a)*Fsym.

Figure 3-8, shows three carriers spaced by their occupied bandwidth constraints with the planned DVB-S2 carrier (F_{wanted}) in the center. The difference in power spectral density between the wanted carrier and the adjacent carriers is denoted as the adjacent-channel interference (ACI) level in dBc (decibel relative to carrier).

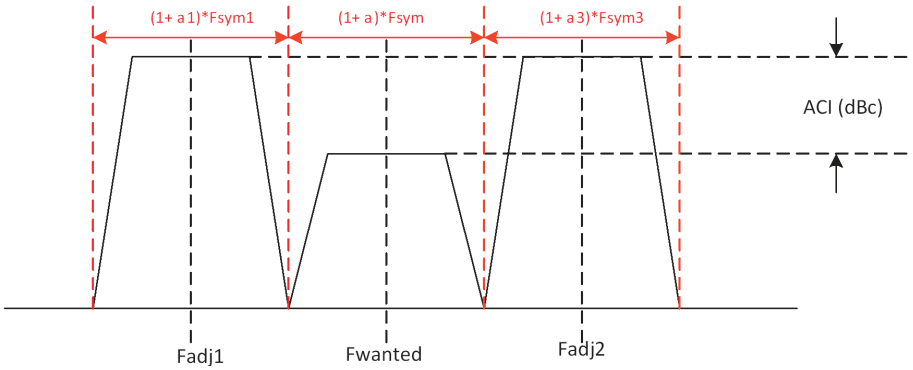


Figure 3-8. Adjacent Carrier Interference Example

ACI Performance for X7/CX700/CX751/CX753/CX780/900/9350/950mp/980

For DVB-S2 roll-off factors of 10, 15 and 20%, the BER degradation of the wanted carrier does not exceed 0.25 dB from LBA guide across all supported MODCODs for ACI levels of +7 dBc (each) on either side independent of the symbol rate, roll-off and waveform modulation type of the adjacent carriers. However, for 5% roll-off factor, the robustness to ACI levels is degraded slightly.

Based on carrier arrangements robustness to ACI levels for 5% roll-off factor vary from +4 dBc to +7 dBc on either side. The worst case typically occurs at the highest operating MODCOD under the following conditions:

- The wanted carrier is less than 15 Msps and the adjacent carriers have an identical symbol rates shaped with 5% roll-off factor
- With strong, narrow-band adjacent carriers that occupy 5% to 10% of symbol rate of the wanted carrier on either side

In most cases, the DVB-S2 downstream carrier operates at the highest power spectral density in the transponder, maximizing the contracted equal power equal bandwidth (EPEBW) limit. This makes ACI levels exceeding +4 dBc unlikely. In addition, adjacent carriers shaped at 5% (bullet 1) or narrow-band carriers stronger than the DVB-S2 downstream carrier (bullet 2) that are likely to violate the EPEBW mask are unusual.

Network designers can realize benefit of 5% roll-off by planning around identified constraints.

ACI Performance for iQ-Desktop/iQ-200 Series Remotes

For all DVB-S2 roll-off factors from 5 to 20%, the BER degradation of the wanted carrier does not exceed 0.25 dB from LBA guide across all supported MODCODs for ACI levels of up to +7 dBc. There is a transient period of 30 seconds to 4 minutes for the wanted carrier to recover to within 0.25 dB loss following turn-on of adjacent channels.

ACI Performance for iQ-Desktop/iQ-800

For all DVB-S2 roll-off factors from 5 to 20%, BER degradation of the wanted carrier does not exceed 0.25 dB from LBA guide across all supported MODCODs for ACI levels of up to +7 dBc.

3.3 Downstream Carrier Operation with DVB-S2

The Velocity downstream carrier operation uses DVB-S2 with ACM. ACM enables each satellite terminal to operate at its most efficient coding and modulation scheme at any moment in time, depending on the location within the satellite contour, the antenna size, and the atmospheric conditions. The ACM algorithm evaluates current channel conditions through a return channel to determine the ideal modulation and FEC rate for each individual DVB-S2 frame and makes adjustments in real time.

The downstream carrier provides communication from the SAS to the satellite terminals or in the case of the CDC/RF GW architecture, from the RF Gateway to the satellite terminals. For each iNet, only one downstream carrier is present. Using time-division multiplexing (TDM), it is possible to allocate traffic for both individual (unicast) and multiple (multicast) satellite terminals in a single shared carrier.

The protocol processors at the SAS or at the CDC in the case of the CDC/RF Gateway architecture, use the Group Quality of Service (GQoS) configuration to determine which satellite terminals receive which TDM blocks. The information transmitted on the downstream carrier is broadcast to all satellite terminals, however, only the intended recipients receive it. This method is achieved by marking each IP packet, sent over the air, with the unique HDLC (high-level data link control) address of the targeted satellite terminal. The modems ignore packets that does not have its address.

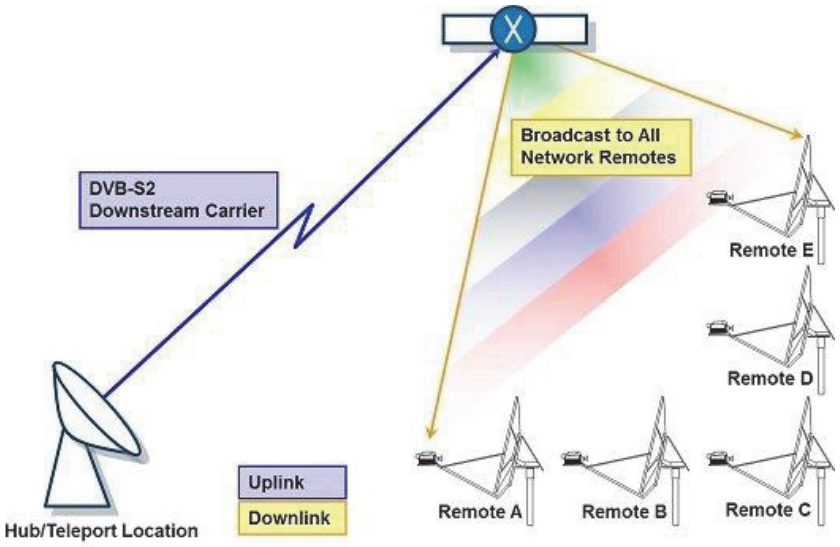


Figure 3-9. Downstream DVB-S2 Carrier

3.3.1 ILC-T Multi-Carrier DVB-S2 Support

The ILC-T multi-carrier feature supports up to four DVB-S2 carriers on an ILC-T line card. In this arrangement, all four carriers must be configured the same, based on the iNet Profile. Each ILC-T has a separate power control loop to properly manage the frequency-dependent cable loss and other factors at the RF Gateway site. The ILC-T can also be shared across beams, but in such cases, all carriers in the associated beams must be configured the same.

The Pulse NMS configuration work flow allows adding beams, channels, iNets, Inroute Groups, iNet Profiles, and Inroute Group Profiles. The user can combine all related workflow pages and use the Line Card Group field on the iNet to define a group of iNets that will share the same line card.

With the ILC-T multi-carrier feature, the number of required transmit line cards and the associated chassis, racks, floor space, power and climate control, and intra-site network equipment, all is reduced by as much as a factor of four.

With this feature, the Pulse NMS supports the following:

- Adding Beams, Channels, and iNets
- Deleting Channels and iNets
- Performing Line Card Failover
- Pulse NMS Beam-Related Elements Page (Channels, iNets, Inroute Groups, iNet Profiles and Inroute Group Profiles)

Some ILC-T Multi-Carrier Characteristics, Limitations and Constraints:

- Maximum 4 carriers per ILC-T
- Maximum 45Msps per DVB-S2 carrier
- An ILC-T is limited to a single IF Domain
- Carrier configurations must be non-overlapping
- Supports a maximum of up to 10 iNets per Beam
- The ILC-T is useable (can be shared) across multiple beams
- Reduces number of ILC-T line cards required to fill transponders
- Carriers in same Line Card Group must use same symbol rate across all beams
- Carriers on an ILC-T must be configured with the same iNet Profile (Carrier Spacing, Power, Modulation (ACM/CCM), DVB-S2 MODCOD range, and symbol rate)
- Line Card Center Frequency must be maintained, within +/- 250 MHz of the current center frequency, when adding new carriers
- Initial power configured equally for all downstream carriers based on iNet Profile

3.3.2 ACM Operation

iDirect supports ACM configuration only on the DVB-S2 downstream. With ACM configuration, there is no constraint on the downstream to operate at a fixed MODCOD. Instead, the modulation and coding of the downstream varies within a configurable range of MODCODs.



NOTE: An ACM downstream can be configured to simulate a CCM downstream by constraining the MODCOD range to a single MODCOD.

ACM allows satellite terminals operating in better signal conditions to receive data on higher MODCODs by varying the MODCOD of data to match the targeted satellite terminal's current receive capabilities.

Not all data is sent to each satellite terminal at its best MODCOD. The PP/iGW transmits important system information (for example, time-plan messages), as well as broadcast traffic, at the minimum MODCOD configured for the outbound carrier. This allows all satellite terminals in the network, even those operating at the worst MODCOD, to reliably receive this information. Multicast is transmitted at a MODCOD configured in the applicable multicast subscriber service plan profile (SSPP). See Multicast Subscriber Service Plans on page 140.

The protocol processor determines the maximum MODCOD for all data sent to the DVB-S2 line card for transmission over the outbound carrier. However, the line card does not necessarily respect these MODCOD assignments. In the case of the ULC, this process is managed as described in the following paragraph by the line card itself; in the case of the ILC, this process is managed by the "Rook" process.

In the interest of downstream efficiency, some data scheduled for a high MODCOD may be transmitted at a lower one as an alternative to padding bytes into a BBFRAME. When a BBFRAME is assembled for transmission, all available data for the chosen MODCOD is first packed into the frame. If space is left in the BBFRAME, and no data left for transmission at that MODCOD, the remainder of the frame is packed with data for higher MODCODs. This method takes advantage of the fact that a satellite terminal can demodulate any MODCOD in the range between the carrier's minimum MODCOD and the current maximum MODCOD of the terminal.

The maximum MODCOD of a satellite terminal is based on the latest signal-to-noise ratio (SNR) reported by the terminal to the protocol processor. Each terminal continually measures its downstream SNR and reports the current value to the protocol processor. When the protocol processor assigns data to an individual satellite terminal, it uses the last reported SNR value to determine the highest MODCOD on which that satellite terminal can receive data without exceeding a specified bit error rate (BER). The protocol processor includes this information when sending outbound data to the line card. The line card then adjusts the MODCOD of the BBFRAMES to the targeted satellite terminals accordingly. This feedback loop, whereby the PP/iGW adjusts the MODCODs of the transmissions to the satellite terminals, is illustrated in the diagram. The SNR (or ***C/N***) thresholds per MODCOD and the spectral efficiencies of iDirect terminals are documented in the ***iDirect Link Budget Analysis Guide***.

For additional details on MODCODs for DVB-S2/ACM, see Outbound (DVB-S2) QoS on page 149.

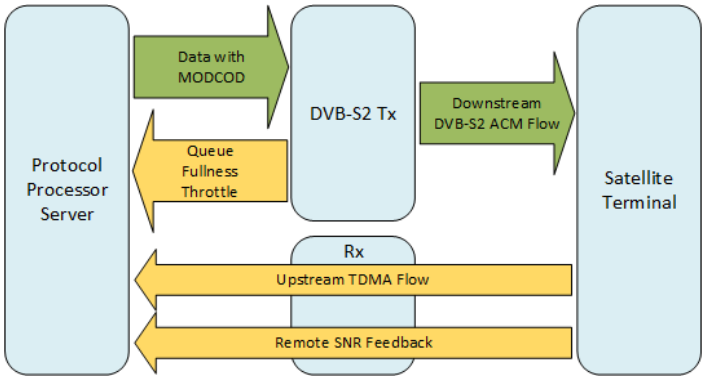


Figure 3-10. Feedback Loop

3.3.3 DVB-S2 Fade Control

During steady state conditions, DVB-S2 satellite terminals report their current SNR measurements to the SAS based on the following:

- Sends an ACM message once every 5 seconds in Steady State
- Sends an ACM message once every 1 second in Fast Fade

The SAS monitors the SNR measurements of each satellite terminal. Using the network settings for the fast fade threshold and the fade slope threshold, the SAS determines whether or not a satellite terminal has entered a fast fade state. If either the fast fade threshold or the fade slope threshold is exceeded, the satellite terminal places itself into the fast fade state.

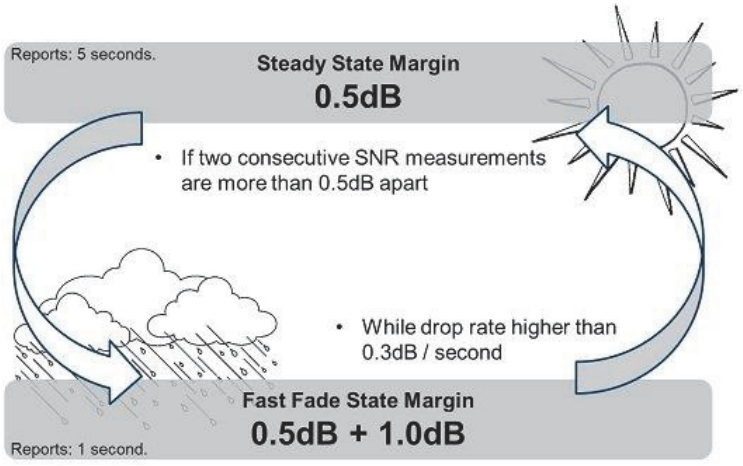


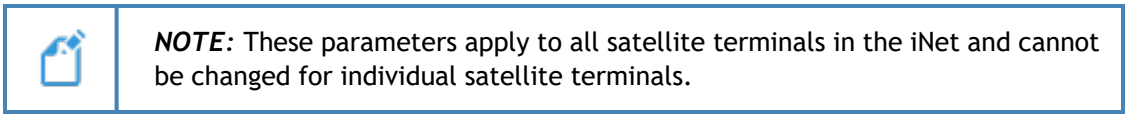
Figure 3-11. DVB-S2 Fade Control

When in the fast fade state, the satellite terminal tries to report its current SNR to the SAS every second. Upon reception of a fast fade message, the SAS increases the operational margin of the satellite terminal by an additional 1.0 dB over the normal operational margin (the steady state default margin is 0.5 dB), which means a total of 1.5 dB is added to the required SNR threshold to select the operating MODCOD for that satellite terminal.

As a result, lower MODCODs are selected to ensure no data loss and keep the satellite terminal in network during the rain fade. The system can manage a fade slope of 0.3 dB/s with default steady state and fast fade margins. Fade slopes outside this range or sudden changes in SNR (for example , blockage) can lead to data loss or satellite terminals falling out of network.

The reporting time of the ACM status message is not configurable, but the operational steady state margin and fast fade margin are configurable. The parameters are described below:

- **Steady State Margin** (default = 0.5 dB) — The margin added to the SNR thresholds measured at hardware qualification to arrive at the operational SNR threshold during steady state operation.
- **Fast Fade Margin** (default = 1 dB) — The additional margin added to the SNR thresholds measured at hardware qualification to arrive at the operational threshold during a fast fade condition. During a fade, this margin is added to the Steady State Margin.
- **Fast Fade Threshold** (default = 0.5 dB) — The drop in receive signal strength between two consecutive SNR measurements by a satellite terminal that causes the satellite terminal to enter a fast fade state. If, during steady state operation, a satellite terminal reports an SNR drop greater than or equal to the Fast Fade Threshold, the SAS considers the satellite terminal to be in the fast fade state.
- **Fade Slope Threshold** (default = 0.3 dB per second) — The rate of drop in receive signal strength by a satellite terminal that causes the satellite terminal to enter a fast fade state. If, during steady state operation, a satellite terminal's SNR drops at a rate greater than or equal to the Fade Slope Threshold, the SAS considers the satellite terminal to be in a fast fade state.



NOTE: These parameters apply to all satellite terminals in the iNet and cannot be changed for individual satellite terminals.

3.4 About the Upstream TDMA Waveform

This section describes the TDMA (Time Division Multiple Access) waveform used on the inbound channel. The demodulator on the line card enables the configuration of multiple heterogeneous channels that share a dedicated receiver responsible for timing, frequency, and phase corrections using state-of-the-art signal processing algorithms.

3.4.1 TDMA Burst Formats

The proprietary upstream iDirect waveform consists of short blocks of known symbols (pilots) distributed evenly throughout the burst as shown in the diagram.

The payload of each burst consists of a fixed 170-byte block coded using a 2D 16-State FEC code at rates published in the ***iDirect Velocity Link Budget Analysis Guide***. The pilot structure is optimized to provide a frequency tolerance of up to 1.5% of the symbol rate across all modulation and FEC rate combinations. The ***C/N*** dynamic range and granularity permits the use of adaptive coding and modulation on the inbound channel by configuring heterogeneous inroute groups with multiple carriers.



Figure 3-12. TDMA Burst Format with Distributed Pilots

3.4.2 Spread Spectrum

Spread spectrum is employed in iDirect networks to minimize ***adjacent satellite interference*** (ASI). ASI can occur in applications such as Comms-on-the-Move (COTM) because the small antennas (typically sub-meter) used on mobile vehicles have a small aperture size, large beam width, and high pointing error. Enabling spread spectrum reduces the spectral density of the transmission to where it is low enough to avoid interfering with adjacent satellites.

Although this feature may be useful for other applications, such as overpowering interference or hiding carriers from hostile jammers, customers should consult with iDirect sales engineering to ensure that a specific application is appropriate for this technology.

iDirect spread spectrum is an extension of BPSK modulation. The signal is spread over wider bandwidth according to a spreading factor (SF) selected for the carrier in the NMS user interface. For a TDMA upstream carrier, the satellite operator can select from one of four options: No Spreading, 2, 4, or 8.

Each symbol in a spread burst is called a “chip” with an associated chip rate that incorporates the spreading factor into the symbol rate. For example, selecting a spread factor of 4 means the chip rate is four times the symbol rate, hence occupying a larger bandwidth and resulting in lower spectral efficiencies compared to non-spread BPSK waveforms. The pilot structure in spread spectrum TDMA is optimized to provide a frequency tolerance of up to 1.5% of the symbol rate.

3.5 Upstream Carrier Operation with Adaptive TDMA

Adaptive TDMA (A-TDMA) allows satellite terminals to dynamically adapt their transmissions to the SAS in real time to the optimal symbol rate and MODCOD for the prevailing conditions. This process reduces the amount of rain margin designed into the upstream link, significantly improving the satellite terminals' clear-sky throughput when compared to non-adaptive TDMA networks. Reducing the system resources set aside for worst-case conditions makes these resources available for data transmission.

Each satellite terminal transmits to the SAS on a shared upstream channel and dynamic time slots determined by the protocol processor. Based on QoS considerations and the link conditions of each satellite terminal, the protocol processor determines the amount of time and the frequency for each burst transmitted by the satellite terminal and provides a synchronized Burst Time Plan (BTP) to all satellite terminals.

3.5.1 Inbound Channel Configuration Messaging

A-TDMA provides the ability to optimize the upstream physical layer characteristics in real time to adjust for dynamic link conditions. The SAS conveys the dynamically changing inroute information to the satellite terminals using:

- System Information (SI) Tables
- Network Clock Recovery (NCR) Timestamp
- Burst Time Plan

3.5.1.1 System Information Tables

The SI tables describe the inroute configuration contained in each inroute (carrier) group. These tables are based on the DVB-RCS standard and have been modified to support the iDirect system. There are five tables that include:

- **Timeslot Composition Table (TCT)** — Defines the properties of generic time slots and the bursts they contain, such as the timeslot type, MODCOD, payload size, guard intervals and symbol rate.
- **Frame Composition Table (FCT)** — Defines center frequency, frame duration, and carrier construction based on different types of timeslots for each carrier.
- **Superframe Composition Table (SCT)** — Defines the collection of carriers that make up the inroute group. This is done by referencing entries in the next-level signaling element (the FCT). The signaling format allows any combination of carriers in a group.
- **Satellite/SAS Position Table (SPT)** — Describes information about the positions of the satellites and SASs used for outbound and inbound links.
- **Broadcast Terminal Information Message Table** (TIM-B) — Transmits static or quasi-static information about the downstream and upstream links (for example, configuration parameters).

The SI tables are periodically broadcast on the DVB-S2 outbound, with the re-broadcast interval nominally sets to every 5 seconds.

3.5.1.2 Network Clock Recovery Timestamp

The iDirect system uses a variant of the Network Clock Reference (NCR) scheme as described in the DVB-RCS standard. It is modified to work with LEGS encapsulation instead of MPEG Transport Streams. A high-accuracy or precision timing source, such as a global positioning system (GPS) receiver, is commonly used as a timing reference for the NCR module inside the PP stack. All Tx line cards, receive (Rx) line cards, stand-by line cards in a SAS, and all satellite terminals from all networks that belong to the same SAS are synchronized with the NCR module clock. Furthermore, dynamic MODCOD switching between inroutes in iDirect's A-TDMA system requires precise synchronization between Rx line cards and satellite terminals.

In the iDirect system, NCR packets are always transmitted on the lowest MODCOD supported by the network. The message is transmitted 32 times per second to indicate the current NCR count at the SAS. The NCR timestamp messages define a precise Rx NCR time at the terminal, referenced to the incoming L-band signal. However, in order to line up the transmit times of different satellite terminals, each satellite terminal must have a unique Tx NCR time. Events transmitted from different terminals at the Tx NCR time for each satellite terminal should arrive at the satellite simultaneously. All transmissions are done relative to the Tx NCR time. In particular, the NCR time in the SCT is the Tx NCR time. In addition, the UCP messages are corrections applied to the Tx NCR time. Typically, the Tx NCR time is managed by keeping track of an offset to the Rx NCR time.

3.5.1.3 Burst Time Plan

The Burst Time Plan (BTP) performs the following functions:

- Assigns time slots to satellite terminals
- Assigns ID numbers to satellite terminals
- Sends UCP corrections to satellite terminals
- Sends the return channel link encryption initialization vector seed

3.5.2 Inroute Group Composition

The core element of the iDirect A-TDMA system is a heterogeneous inroute group made up of TDMA carriers that support different transmission rates and provide different levels of protection against adverse channel effects such as rain fade. A heterogeneous inroute group, in relation to a homogeneous inroute group, is illustrated in the diagram.

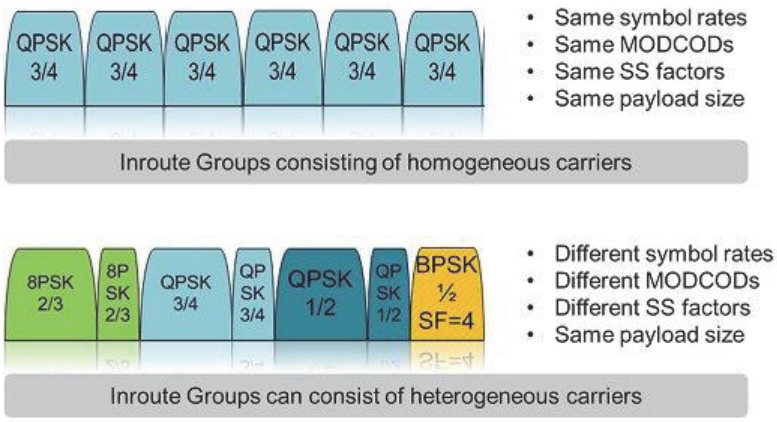


Figure 3-13. Heterogeneous Inroute Groups vs. Homogeneous Inroute Groups

The protocol processor assigns time slots on upstream carriers to individual satellite terminals according to their current demand and capability, as determined by the channel state. Inroute group configuration automatically adapts over time to maximize system efficiency.

- **Inroute Group (IG)** — A fixed portion of space segment bandwidth and power partitioned into TDMA carriers with different properties.
- **Inroute Group Composition (IGC)** — A set of carrier definitions assigned to an inroute group.

The IGC assigned to an inroute group determines how the bandwidth in the inroute group is partitioned in real time. An inroute group can include carriers with different symbol rates and MODCODs. Note, however, that the symbol rates and center frequencies of the individual carriers remain the same in all IGCs. The IGC currently assigned to the inroute group defines the MODCODs of the various carriers in the group.

The MODCODs of the upstream carriers do not vary frame by frame. The protocol processor assigns TDMA slots to individual satellite terminals on the upstream carriers based on the upstream link conditions of the individual satellite terminals. In the short term, satellite terminals adapt to changing conditions by frequency hopping among carriers with different properties. Over time, the protocol processor may also adjust the configuration of the inroute group by selecting different IGCs as network conditions change. Whenever a different IGC is assigned to the inroute group, the MODCODs of the upstream carriers change to match the configuration of the newly-applied IGC. Thus, the system automatically adapts in two ways: frame-by-frame through satellite terminal frequency hopping, and longer term by selecting the optimal IGC for the prevailing conditions.

3.5.2.1 Differentiating C/N_0 and C/N

With the introduction of A-TDMA, most upstream TDMA signal quality are now expressed in terms of C/N_0 (carrier-power to noise-spectral-density ratio). In previous (non-adaptive) releases, the signal quality was expressed in terms of ***C/N*** (carrier-power to noise-power ratio). This section explains why this metric was changed and the relationship between ***C/N*** and C/N_0.

C/N depends on the bandwidth of the upstream carrier. Using ***C/N*** as the signal quality metric makes it difficult to view inroute groups of upstream carriers with different symbol rates in a unified way. Since A-TDMA allows carriers in the same inroute group to have different symbol rates, it is necessary to use a measure such as C/N_0 that is independent of the signal characteristics.

C/N is always qualified by a corresponding measurement bandwidth. The received carrier power (***C***) has a certain value, given the various system parameters and link conditions. However, the additive noise is defined in terms of its spectral density (N_0). It has a much larger bandwidth than the signal. The total noise power (***N***) in a receiver is the product of N_0 and the measurement bandwidth (***B***). In iDirect documentation that uses ***C/N***, such as the ***iDirect Link Budget Analysis Guide***, ***B*** is assumed to equal the symbol rate of the carrier.

An adaptive system constantly evaluates and compares the received ***C/N*** on one carrier with what can be achieved on another carrier. For example, if a line card is receiving a satellite terminal at a ***C/N*** of 11 dB on a carrier with a symbol rate of 1 Msps, what can be achieved on a 2 Msps carrier? In this example, the corresponding ***C/N*** on the 2 Msps carrier is 8 dB.

Although the carrier power (***C***) is the same for both carriers, doubling the bandwidth doubles the noise power. Therefore, the ***C/N*** is halved (lowered by 3 dB). Rather than making comparisons with arbitrary references (and risking mistakes and confusion), the iDirect system uses C/N_0 as the common measure.

C/N_0 can be thought of as the ***C/N*** achievable in a bandwidth of 1 Hz. The fact that this is a stand-alone measure that does not need to be qualified by any signal properties is the primary reason the iDirect system now uses C/N_0 extensively to measure and characterize upstream performance.

C/N_0 is equal to ***C/N*** + 10log10(Symbol Rate) and is expressed in dBHz.

For this example: $C/N_0 = 11 + 10\log_{10}(1x106) = 8 + 10\log_{10}(2x106) = 71$ dBHz

Demodulation thresholds can also be expressed in terms of C/N_0. The carrier MODCOD corresponds to a certain threshold ***C/N***. The ***C/N*** threshold combined with the symbol rate is used during operation to compute the carrier threshold C/N_0.

A satellite terminal is allowed to transmit on a given upstream carrier when the C/N_0 that can be achieved is greater than or equal to the ***C/N*** threshold - C/N_0 of the carrier. The C/N_0 is therefore a direct and unified measure of the ability to use any of the available carriers in the inroute group. This is another useful property of C/N_0.

3.5.2.2 Adaptivity Modes

On a real-time basis, the channel characteristics of each upstream carrier are always static. Frame by frame, the protocol processor allocates TDMA slots on the upstream carriers based on the upstream link conditions of each satellite terminal. This method requires that the protocol processor implement adaptivity modes based on three distinct time scales, as illustrated in the following diagram.

The adaptivity modes are:

- **Short-Term** — Satellite terminals use frequency hopping to find the best carriers.
- **Medium-Term** — Carriers change their working MODCOD to adapt to satellite terminal levels.
- **Long-Term** — An operator defines inroute group compositions using statistical data.

These modes are described in the following sections.

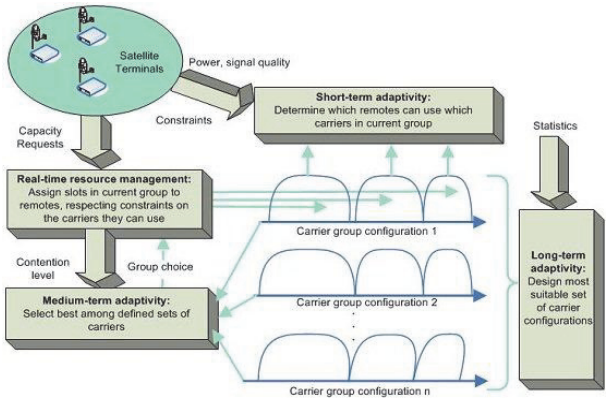


Figure 3-14. A-TDMA Adaptivity Modes

Short-Term Adaptivity

Short-term adaptivity operates in real time and considers only the current IGC. The Upstream Control process monitors the ***C/N*** of each satellite terminal's upstream carrier, as measured at the SAS, to determine the satellite terminal's nominal carrier. A nominal carrier is the upstream carrier with the highest threshold C/N_0 that the satellite terminal can currently sustain and validly use.

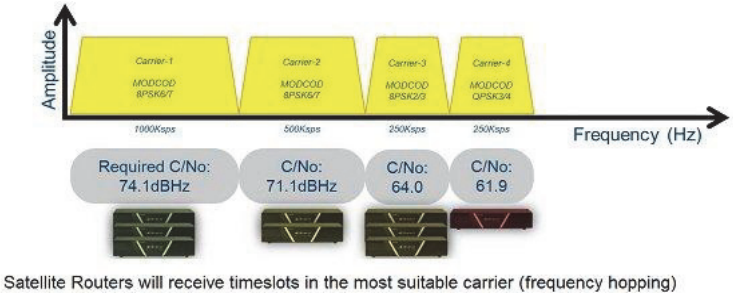


Figure 3-15. Short-Term Adaptivity

The protocol processor uses a bandwidth allocation algorithm to assign each satellite terminal slot with parameters (such as symbol rate and MODCOD) that are estimated to be within its capabilities for the current link conditions. The algorithm attempts to allocate slots on the satellite terminal's nominal carrier and on any carriers having a C/N_0 lower than the nominal carrier. However, this is not always possible due to the overall demand for upstream traffic slots in the inroute group.

The bandwidth allocation algorithm schedules bursts in TDMA frames for each satellite terminal in accordance with the rules and priorities defined by the Group QoS settings for the inroute group. However, for A-TDMA, the algorithm must also account for the dynamic nature of the total capacity available for each satellite terminal, because the subset of carriers on which a terminal is able to transmit can change according to real-time network conditions.

As an example of short-term adaptivity, consider a satellite terminal experiencing a rain fade. When the satellite terminal enters the fade, the ***C/N*** of the satellite terminal's bursts, as measured at the SAS, drops. This drop limits the set of upstream carriers on which the satellite terminal can successfully transmit. This causes the UCP to change the satellite terminal's nominal carrier to a more protected carrier.

When allocating new bandwidth to the satellite terminal during the fade, the bandwidth allocation algorithm only considers available slots on the more limited subset of carriers. Because the satellite terminal's nominal carrier is set to a more protected carrier with lower throughput, the satellite terminal is able to stay in the network at the expense of bandwidth efficiency. When the fade passes, the satellite terminal's nominal carrier is changed back to the more efficient carrier.

Medium-Term Adaptivity

Medium-term adaptivity is the automated process of selecting the best IGC for current network conditions. The frequency at which the selection algorithm executes is configurable through the NMS. By default, the IGC selection algorithm executes once per minute.

The goal of the IGC selection algorithm is to determine the IGC that best matches the current overall state of the system. At each invocation the algorithm carries out a trial scheduling of a single frame for each defined IGC, using demand levels that are computed statistically over the preceding interval. The algorithm takes into account the total number of slots assigned (capacity) as well as the level of bandwidth contention, which represents the number of slots that could not be allocated due to lack of capacity. If the optimal IGC selected by the algorithm is different from the current IGC, then the carrier configuration for the inroute group is changed to the new IGC.

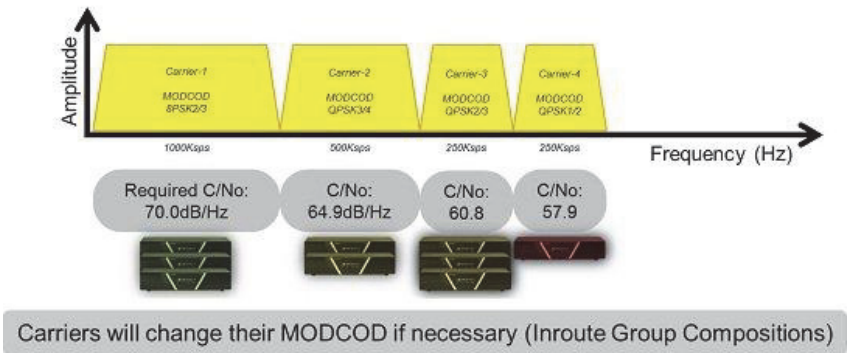


Figure 3-16. Medium-Term Adaptivity

When there are events affecting many satellite terminals, IGC selection is a compromise between total capacity and the accepted level of contention on the most protected carriers.

User-configurable parameters for the inroute group, as described below, help to guide the IGC selection process.

These parameters include:

- **IGC Selection Interval** — The frequency (in seconds) at which the IGC selection algorithm is executed.
- **Allowed Dropout Fraction** — The fraction of active satellite terminals allowed to drop out of the network. These satellite terminals are unable to stay in the network on the most protected carrier and are logged out by the UCP.

  The selection algorithm attempts to select an IGC that does not allow the number of satellite terminals that drop out of the network to exceed the Allowed Dropout Fraction. If the Allowed Dropout Fraction is exceeded for all IGCs, the default IGC is selected for the inroute group.
- **Maximum Allowed Downlink Fade** — The maximum downlink (hub-side) fade (in dBs) allowed for the current IGC before a new IGC is selected. This parameter is only used if periodic hub side fade is measured by a teleport RF subsystem or through a beacon receiver configured in the network.
- **Default IGC** — The IGC selected if the dropout rate is exceeded for all IGCs defined for the inroute group or if the downlink fade exceeds the maximum allowed downlink fade.



NOTE: A severe downlink fade is observed at the SAS as the simultaneous fading of all satellite terminals. iDirect recommends including a very robust IGC among those defined for the inroute group to allow the system to operate during a severe downlink fade, rather than collapsing entirely or risk driving the user uplinks to saturation by UCP.

Velocity also supports rain diversity switchover, which avoids weather impact through the use of a second SAS site. See Automatic Rain Diversity Switchover on page 182.

Long-Term Adaptivity

Long-term adaptivity is a manual process that uses system feedback to refine the IGC configurations. Using the NMS user interface, satellite operators can view A-TDMA operational statistics. These statistics include IGC usage, the distribution of signal quality, and the number of satellite terminals logged off due to network conditions. The system planner uses these statistics to review and revise the IGC definitions and system parameters discussed in the previous sections.

3.5.2.3 Fade Margin and Measurement Interval Calculations

This section provides some general guidance on how suitable values for the signal quality measurement frequency and fade margins can be established. The default values provided are

expected to be adequate to get the system running in most situations; any further optimization is left to the satellite operator as part of the long-term adaptivity.

Terminals are expected to transmit a traffic burst at least once every measurement interval, which is a user configurable value recorded under “Measurement Spacing”. The fade depth and fade slope are normally assessed after each such interval using a sliding record of a pre-configured number of recent measurements. Following a non-zero power control command or change of nominal carrier (including IGC change), the measurement history is reset. In clear weather, changes are relatively rare in comparison to scenarios involving significant fade events. The total system reaction time accounts for processing time and a round-trip delay.

The M_1 margin is intended to cover measurement uncertainties as well as allow for the system reaction time for fade detection; a value of 1.0 dB has been found to be a suitable default for handling 0.1 dB/s fade slope with detection uncertainties. In the presence of time-varying fade, this parameter is closely related to the spacing between measurements. Hence, an additional trade-off can be accessed, between margin and overhead. Dedicated measurement “probe” bursts are only used by terminals that are typically idle. Actual traffic bursts are also used for the measurements, so there is no measurement overhead for an active terminal. Therefore, for systems with active terminals, a “low margin” setting will be the preference. The “low-overhead” setting is intended for systems that serve as back-up to other communications means, or other situations where a large population of terminals is normally idling, so it is desirable to keep the overhead low.

The M_2 system hysteresis margin is intended to prevent spurious switching of the nominal carrier. It should be set higher than the power control step size supported by the terminals. Initial experience suggests that this margin also caters effectively for a number of additional minor uncertainties; a value of 0.6 dB has been found to be a suitable default for a variety of situations. In extreme cases, terminals may be forced to log out because the system cannot keep up with the rapid carrier changes. If the M2 is set too high, capacity will be wasted because the carrier usage (time slot assignments) will be more conservative than necessary.

The M_3 margin is applied to acquisition bursts for the purpose of selecting an initial nominal carrier: its value is related to the ***C/N*** estimation accuracy for this signal. The M3 margin is also used to accommodate initial uncertainties in EIRP (Effective Isotropic Radiated Power) settings, BUC (block up-converter) gain variation with frequency. A fixed value of 2 dB is suggested based on initial experience, design analyses of the RF components, and the relevant estimation algorithms.

3.5.2.4 C/N Dynamic Range for Individual Carriers

For any given carrier in an inroute group, the MODCODs used in the various compositions should be chosen such that the range of variation of the threshold ***C/N*** does not exceed approximately 5 dB. This restriction is associated with the finite dynamic range inherent in any digital demodulator design. If the constraint is violated, the sudden, large variation in signal level happening at IGC changes may result in temporarily increased rates of lost or bursts with error. These episodes typically last for a few seconds, until the demodulator automatic gain control is able to adjust. A higher range of threshold variation can be used if these temporary episodes of increased errors are acceptable.

The 5 dB range is more than sufficient to vary a carrier from BPSK r=1/2 to QPSK r=2/3 or QPSK r=3/4 to 8PSK r=4/5, for example.

3.5.2.5 C/N_0 Dynamic Range for Inroute Group

In order to make the best use of the available capacity, individual terminals may be assigned time slots on any carrier in the inroute group for which it is able to close the link budget. Disadvantaged terminals have priority for the time slots on the more protected carriers, but when there is no specific demand for these slots, they may be assigned to any terminals. The Free Slot Assignment (FSA) feature ensures all slots do indeed get assigned, to the extent possible.

The transmit power required on any carrier is proportional to the carrier demodulation threshold expressed as $\boldsymbol{C/N_0}$; that is, the ***C/N*** threshold plus allowance for the symbol rate. Therefore, within a very short span of time, the terminal may be required to transmit on carriers with power requirements that differ by as much as the overall span of $\boldsymbol{C/N_0}$ threshold values in the inroute group. This includes the various compositions, since they can change instantaneously on any frame boundary.

All terminals have a finite dynamic range of transmitter power/EIRP. At one end, this is limited by the maximum power of the BUC. At the other end, it is limited by the level-control attenuators and by stability considerations. The total dynamic range of an iDirect modem's IF output power is large; however, not all of it is available to accommodate signal differences. In particular, substantial allowances must be made in order to accommodate differences in BUC gain and IF cable losses.

For all iDirect modems, the dynamic range available for real-time adjustment of burst power is 15 dB. Hence, the inroute group should be arranged such that the difference between the lowest and highest $\boldsymbol{C/N_0}$ threshold of the carriers that any terminal needs to access does not exceed it. This 15 dB constraint, which applies across all compositions, is not very limiting if taken into consideration when designing the inroute group and its compositions.

A 15 dB fade is a very rare event, even in extreme locations. As an example, the city of Recife in Brazil (the 0.01% rainfall rate is 69 mm/hr, old climatic zone “N”) has an elevation of 20° to a satellite, 15 dB still corresponds to 99.9% availability at Ku-band and 99.3% at Ka-band. For a location with slightly more moderate but still substantial rainfall (Montreal, 38 mm/hr, old zone “K”), the figures are 99.99% at Ku-band and 99.85% at Ka-band.

3.5.2.6 Maximum Power and Control

For A-TDMA to work well in the network, it is important that maximum transmit power is set accurately. To maintain the transmit power value close to the ODU (outdoor unit) limit, it is recommended to perform 1 dB compression tests corresponding to beam center frequency on ODUs that do not support Modem-BUC calibration. Having accurate maximum power settings can be challenging for beam switching satellite terminals, as the variation in gain flatness of a BUC can impact the maximum power used in UPC for a given terminal in different beams and requires maximum power determined per beam. For ODUs that support Modem-BUC calibration the maximum power at L-band is predetermined across RF band prior to network entry and over the air 1dB compression testing is not necessary across any of the beams.

Power control transitions typically occur during rain fades and changes in inroute compositions. CRC are observed when a remote drastically changes its power level. For all remote types, power accuracy is not always 100% when moving from one power level to

another. As they try to self-correct through the uplink power control, there are chances of CRCs happening during the short transition period.

3.5.2.7 Frame-Filling Efficiency

The multi-frequency TDMA frame employed in the iDirect inroute provides a very flexible transmission capacity resource for a population of terminals — even when those terminals present the system with very disparate and highly dynamic capacity demands. In ideal conditions, the frame can be filled 100% — that is, all time slots on all carriers can be used to transport user data. For frame filling efficiency, tests with real-world arrangements, in terms of carrier rates and numbers of terminals, typically provide filling efficiencies above 98% on all carriers. This section discusses some of the real-world constraints that force the filling efficiency to slightly lower than 100% in some cases, as well as some constraints that determine the actual spectral efficiency.

Time slots in the iDirect system can have very different durations. The main constraint currently imposed is that all slots have the same payload size (170 bytes). The choice of symbol rates, FEC rates, and modulation schemes on different carriers impacts the number of symbols in a burst. Time slot boundaries typically do not align between carriers in the inroute group. Since the terminals can only transmit on one carrier at a time, assigning a slot on one carrier will generally prevent it from using more than one time slot on all other carriers. This phenomenon, known as ***shadowing***, is a process that assigns specific slots to specific terminals in an attempt to minimize any issues. The problem is effectively eliminated when the number of terminals is far greater than the number of carriers. A slot that cannot be assigned to one terminal, due to shadowing or for other reasons, can be assigned to another terminal with higher probability. This, when combined with a sophisticated algorithm, allows the iDirect system to achieve the 98+% figure noted above.

It is possible to create artificial situations where the task of filling the frame completely becomes impossible or impractical. Either with a very small number of terminals or a small number of carriers and, in particular, if the time slot durations vary widely between carriers, the achievable frame filling may be reduced substantially due to excessive shadowing. This is not a major issue in most practical situations.

Generally, frame filling works well for simple inroute groups with only two types of carriers. However, as the inroute groups become more complex with multiple carriers, frame filling efficiencies are lowered to the 80% and 90% range. Additionally, depending on the GQoS rules configured for the inroute group, a distortion of allocation for each satellite terminal may occur. This issue is more apparent for complex A-TDMA use cases that require multiple types of carriers. For these complex A-TDMA use cases, additional trade-offs are required with some over provisioning of bandwidth for disadvantaged carriers.

3.5.3 Adaptive TDMA Configuration and Constraints

All upstream carriers configured in Velocity are adaptive; and frequency plan (signal bandwidth and channel center frequencies) is fixed. The MODCOD of an upstream carrier, which can change from one IGC to another, is configured after the carrier is added to the IGC.

The following constraints apply to inroute groups and TDMA upstream carrier configuration:

- All configured upstream carriers are adaptive.

- Velocity supports 1 inroute group per iNet.
- The Payload Block Size is 170-bytes for all upstream carriers configured in the NMS, therefore the same for all carriers defined for any inroute group and in all IGCs
- Velocity supports the configuration of up to eight IGCs per inroute group. iDirect recommends configuring no more than three IGCs — an IGC for Clear Sky, Rain Fade, and Hub-Side fade conditions. The granularity provided by more than three IGCs may add to the complexity of operation.
- At least one IGC must be configured in each inroute group.
- For each carrier, the center frequency and symbol rate must be the same for all IGCs.
- A different MODCOD can be selected for each carrier in each IGC.
- The adaptive carrier can be configured to have Superburst, SS-ACQ bursts for acquisition slot (based on traffic slot MODCOD and spreading factor) or have no acquisition slot.



NOTE: With Velocity 1.6, the maximum return channel symbol rate on upstream carriers has be increased from to 5.875 Msps to 7.5 Msps. This increase allows for a greater than 25% increase in maximum burst rate from any single terminal. In addition, with spread spectrum, a maximum chip rate of 7.5 Mcps is supported.

3.5.4 Upstream TDMA LBA Loss at Specific Symbol Rates

Due to the multichannel demodulator (MCD) design, both ULC-R and ILC-R receive line cards may degrade from 0.5 dB to several dBs in received C/N under specific very narrow symbol-rate ranges. For the ULC-R, the magnitude of the affected symbol-rate ranges are from 2 ksps at the lowest symbol-rate end to 20 ksps at the highest symbol-rate end. The full lists of these affected ranges for ULC-R are provided in Table 3-2.

Spread spectrum BPSK mode for ULC-R is affected as well in which case the ranges provided in the Table 3-2 are inferred as chip rate.



NOTE: Supported chip rates are from 1 to 7.5 Mcps.

For ILC-R, the affected symbol-rate ranges are narrower, from 2ksps to 4ksps in magnitude, and far fewer. Unlike the ULC-R, only the lowest 3 MODCODs are affected. Details of these affected symbol-rate ranges are provided after the ULC-R coverage below.

All Velocity releases are affected by this issue. iDirect recommends avoiding these symbol rates when configuring inroute groups.



NOTE: A script is available to detect any existing upstream carriers that fall in the affected ranges for ULC-R. Contact TAC for assistance in running the script.

Table 3-2. Affected Symbol Rate Ranges for ULC-R MCD-TDMA at all MODCODs (Msps)

Minimum - Maximum Symrate (Msps)	Minimum - Maximum Symrate (Msps)
0.173 - 0.175	1.387 - 1.391
0.194 - 0.196	1.561 - 1.565
0.207 - 0.209	1.665 - 1.669
0.216 - 0.218	1.734 - 1.738
0.222 - 0.224	1.784 - 1.788
0.346 - 0.348	2.773 - 2.783
0.390 - 0.392	3.120 - 3.130
0.416 - 0.418	3.328 - 3.338
0.433 - 0.435	3.467 - 3.477
0.445 - 0.447	3.566 - 3.576
0.693 - 0.695	5.551 - 5.561
0.780 - 0.782	6.245 - 6.255
0.832 - 0.834	6.662 - 6.672

Table 3-2. Affected Symbol Rate Ranges for ULC-R MCD-TDMA at all MODCODs (Msps) (Cont.)

Minimum - Maximum Symrate (Msps)	Minimum - Maximum Symrate (Msps)
0.867 - 0.869	6.939 - 6.949
0.892 - 0.894	7.138 - 7.148

The impact on ILC-R is limited to the lowest 3 MODCODs: BPSK 1/2, BPSK 2/3 and QPSK 1/2. The lowest MODCOD, BPSK ½, is further affected by 5 additional symbol rate ranges.



NOTE: Spread spectrum BPSK MODCODs are **not** affected for the ILC-R.

Table 3-3. Affected Symbol Rate Ranges ILC-R MCD-TDMA at BPSK 1/2, BPSK 2/3 and QPSK 1/2 (Msps)

Minimum - Maximum Symrate (Msps)
0.206 - 0.210
0.415 - 0.419
0.831 - 0.835
1.665 - 1.669
3.331 - 3.335
6.665 - 6.669

Table 3-4. Additional Affected Symbol Rate Ranges for ILC-R MCD-TDMA at BPSK 1/2 (Msps)

Minimum - Maximum Symrate (Msps)
0.249 - 0.251
0.499 - 0.501
0.999 - 1.001
1.999 - 2.001
3.999 - 4.001

The ranges listed in Table 3-4 are only slightly impacted with the LBA loss being no more than 0.4 dB. As these symbol rates fall in the commonly used range, but only one MODCOD is affected, network operators can choose one of 3 mitigation options:

a. Avoid these symbol rate ranges.

b. Avoid the BPSK 1/2 MODCOD.

c. Add 0.5 dB additional margin to the M1 parameter.

3.6 About the Uplink Control Process

The iDirect Uplink Control process (UCP) executes at the SAS to bring satellite terminals into the network and to maintain the correct frequency, power, and timing during uplink operation. Then, the UCP continuously monitors each satellite terminal's upstream performance and instructs the satellite terminal to adjust its transmission as required to remain within acceptable limits. This section discusses both phases of TDMA uplink control.

By default, the system allows for up to 1.7 ms of uncertainty in start-of-frame (SOF) timing. The 1.7 ms timing uncertainty consists of ±0.1 degree of satellite station keeping uncertainty and approximately 50 miles of satellite terminal position uncertainty.

This variation in symbol timing is accounted for during satellite terminal acquisition by providing a larger guard band in the TDMA frame for acquisition slots than for traffic slots. A larger guard band in the acquisition slot prevents the acquiring satellite terminal from bursting into the traffic slots allocated to other satellite terminals.

Frequency offset is the difference between the nominal frequency at which the satellite terminal transmits and the frequency at which the SAS demodulator receives the upstream carrier. Frequency offset occurs even after the satellite terminal's clock is synchronized to the SAS clock. The SAS down converter equipment and the ***Doppler Effect*** due to satellite and satellite terminal motion are major contributors to frequency offset.

A satellite terminal attempts to join an iDirect TDMA network by sending an acquisition burst to the SAS in an upstream carrier acquisition slot, in contention or invitation mode. To join the network, the satellite terminal must transmit the acquisition burst at a power level that allows the burst to be successfully demodulated by the hub line card (HLC) receiving the upstream carrier. Uplink control selects the initial nominal upstream carrier for the satellite terminal based on the measured ***C/N*** of the acquisition burst.

This initial selection is based on the acquisition margin configured in the NMS user interface on the M3 Margin tab of the Inroute Group Profile configuration. The acquisition margin is separate from the normal margins, due to the larger uncertainty of the ***C/N*** estimation on the acquisition burst and to allow for other system uncertainties, such as BUC aging and frequency response variations. After the satellite terminal has joined the network, the UCP at the SAS takes over to keep the satellite terminal in the network at the correct power.

3.6.1 TDMA Initial Transmit Power

TDMA initial transmit power is the power level at which the satellite terminal transmits acquisition bursts. The initial transmit power is determined during remote commissioning. After the satellite terminal is commissioned, the value contained in the satellite terminal options file is used whenever the satellite terminal re-joins the network.

This section describes why it is important to correctly configure the TDMA initial transmit power for all satellite terminals. Additional information is contained in the following iDirect documentation:

- The ***Terminal Web User Interface User Guide for iDirect Satellite Routers*** contains the procedure for determining the correct TDMA initial power and reference carrier parameters for a satellite terminal.

- Terminal Network Acquisition on page 260 describes the process by which a satellite terminal joins an iDirect network.
- About the Uplink Control Process on page 113 describes uplink power control in iDirect networks.

Each upstream carrier in an inroute group can have a different MODCOD, symbol rate, and spreading factor. Specific values for these upstream carrier parameters, which are collectively called the satellite terminal ***reference carrier***, are entered along with the TDMA initial transmit power. The value configured for the TDMA initial transmit power is defined in relation to the configured reference carrier parameters.

A satellite terminal may acquire into the network on any upstream carrier in the inroute group. When sending an acquisition burst on an upstream carrier with characteristics that differ from the configured reference carrier, the satellite terminal uses its reference carrier parameters to adjust the initial transmit power so the acquisition burst is received by the HLC within the correct ***C/N*** range for that carrier.

In a burst mode demodulator, the gain must be set at some nominal point prior to the arrival of a data burst so that the burst is correctly detected and demodulated. Since a single HLC receives bursts from many different remote modems, it constantly calculates the optimal gain point by taking into account the average levels of all bursts arriving at that HLC.

If all the bursts arrive at similar ***C/N*** levels, the average ***C/N*** is close to the optimal value for all bursts. However, if many bursts arrive at varying ***C/N*** levels, the higher-level and lower-level bursts can skew the average such that the average is no longer optimal.

The nominal ***C/N*** range is 2 dB. The actual range at which bursts can be optimally detected is approximately 8 dB, centered at the nominal gain point. As shown in the following diagram, under ideal circumstances, the average ***C/N*** of all satellite terminals on the upstream carrier is equal to the center of the UCP adjustment range. Notice that the optimal detection range extends below the threshold ***C/N*** of the upstream carrier.

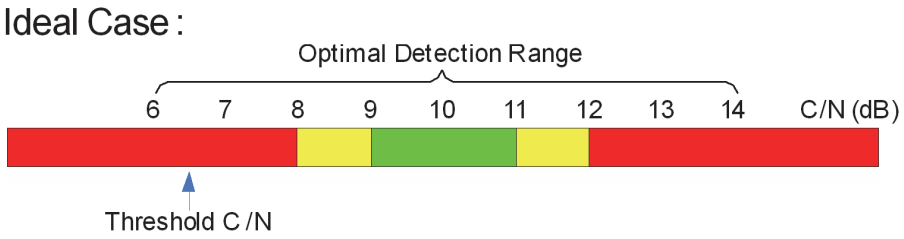


Figure 3-17. Optimal *C/N* Detection Range

If the TDMA transmit initial power of a satellite terminal is not set correctly, network performance can be negatively affected. Post-acquisition, the center point of the 8 dB wide detection range is set at the ***C/N*** value at the time of acquisition.

Satellite terminals joining the network with initial transmit power set too high skew the average ***C/N*** to be above the center of the UCP adjustment range, as shown in the following diagram. Since UCP updates may occur as infrequently as every 20 seconds, it can take a minute or more for satellite terminals with too much initial power to decrease their power to be within the nominal range. During this period the optimal detection range does not include

the threshold ***C/N*** and the performance of satellite terminals experiencing rain fade may degrade. Those satellite terminals could drop out of the network because their bursts no longer fall within the optimal detection range.

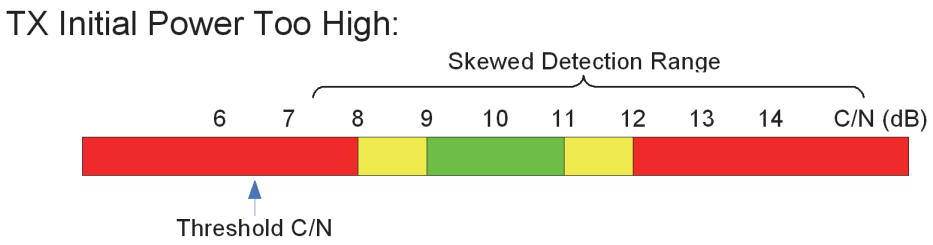


Figure 3-18. Skewed *C/N* Detection Range — Initial Transmit Power Too High

Satellite terminals joining the network with initial transmit power set too low skew the average ***C/N*** to be below the center of the UCP adjustment range, as shown in the following diagram. This could cause satellite terminals coming into the network at the higher end (for example, 14 dB) to experience some distortion in the demodulation process. Additionally, a satellite terminal acquiring at below the ***C/N*** threshold may experience a large number of CRC errors after joining the network until its power is sufficiently increased.

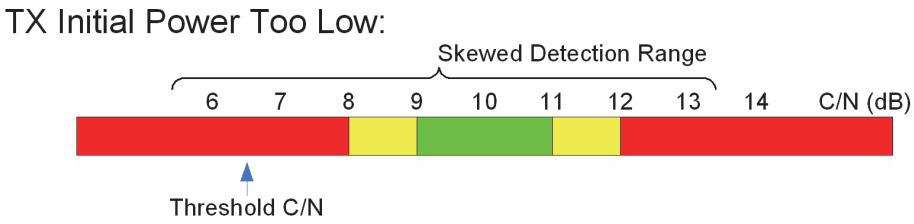


Figure 3-19. Skewed *C/N* Detection Range — Initial Transmit Power Too Low

As in the case where the initial power was set too high, since UCP updates may occur as infrequently as every 20 seconds, it may take a minute or more for carriers with initial power set too low to increase their power to within the nominal range. During this time, terminals operating under clear sky conditions could drop out of the network because the bursts no longer fall within the optimal detection range. Satellite terminals trying to acquire with a ***C/N*** value of greater than 13 dB may not acquire the network. Bursts can still be detected below the threshold ***C/N*** but the probability of detection and demodulation is reduced.

3.6.2 UCP Operation

After the satellite terminal has acquired the network, the UCP continuously corrects the frequency, symbol timing and power while the satellite terminal is in the network. For satellite terminals in adaptive inroute groups, the UCP process is also responsible for switching satellite terminals to new nominal upstream carriers as required.

A satellite terminal's nominal carrier is the upstream carrier with the highest threshold C/N_0 the satellite terminal can sustain and is allowed to use. All power control and other real-time corrections are related to the nominal carrier.

3.6.2.1 UCP Correction Processing

The SAS periodically sends UCP correction information to the satellite terminal. By default, the SAS sends this information every 20 seconds to stationary satellite terminals. For high-speed mobile satellite terminals, UCP correction information should be sent to the satellite terminal more often (for example, Aero terminals every 2 seconds).

In order for the UCP algorithm to function correctly, the terminal must periodically transmit bursts on the TDMA upstream carrier even when idle. Stationary satellite terminals send a minimum of 1 burst every 2 seconds by default. For high-speed mobile applications, some satellite terminal model types require a user to send a minimum of 1 burst per second.

An idle satellite terminal automatically increases the frequency at which it transmits bursts to the SAS when a fade condition is detected. For a satellite terminal in an adaptive inroute group, this allows the SAS to monitor a fast fade and move the satellite terminal to a more protected carrier before the SAS can no longer detect the satellite terminal's bursts.

3.6.2.2 UCP Symbol Timing

Satellite terminals transmitting on the same TDMA carrier must time their transmissions so that all bursts arrive at the SAS modem burst demodulator in sequence and without collisions. To achieve correct burst timing synchronization, each satellite terminal uses its own geographic coordinates, the geographic coordinates of the SAS, and the longitude of the satellite to calculate a Frame Start Delay (FSD). The different FSDs of the satellite terminals compensate for the variation in transmission delay between the individual satellite terminals and the SAS. The satellite terminal adds the FSD to the frame start time received from the SAS to derive the satellite terminal's TDMA frame start time.

During network operation, the symbol timing changes as the satellite moves within the station-keeping box. Symbol timing can also change due to timing Doppler shifts in mobile satellite terminals. The UCP at the SAS adjusts the satellite terminal's symbol offset by averaging timing drifts every UCP period and sending corrections to the satellite terminal.

The guard band between bursts for a TDMA upstream carrier is automatically set during IGC carrier configuration in the NMS user interface. The configured guard band depends on the lowest symbol rate in the IGC and accounts for timing Doppler shifts in addition to system timing uncertainties.

3.6.2.3 UCP Frequency Tracking

Frequency drifts are caused by clock drifts in various components of the system. These drifts are primarily due to temperature variations and the age and quality of the oscillators. The protocol processor tracks frequency offsets over every UCP period and sends corrections to the satellite terminals. The satellite terminals then adjust their transmit frequencies accordingly.

For mobile satellite terminals, Doppler shifts also contribute to frequency drift. These additional frequency drifts in mobile satellite terminals are primarily caused by variation in the satellite terminals' speed. In high-speed COTM applications, the protocol processor uses a predictive algorithm to correct for frequency drifts.

3.6.2.4 UCP Power Control and Fade Detection

The demodulator at the SAS functions optimally when the received power spectral density is similar for all satellite terminals. Therefore, the goal of the UCP is to adjust the individual transmit powers of the satellite terminals so that all satellite terminals are received at the same ***C/N*** as measured at the SAS.

For a given terminal, the ***C/N*** may vary due to fade conditions or as mobile terminals move across the satellite footprint, requiring the UCP to adjust the satellite terminal's transmit power to keep the ***C/N*** within range. To normalize the ***C/N*** across satellite terminals, the protocol processor averages the ***C/N*** deviations detected by the Rx line cards for each satellite terminal. If the deviations are within the acceptable range of the target ***C/N***, no corrections are sent to the satellite terminal. If the deviations are outside the acceptable range, the SAS sends corrections to the satellite terminal until the ***C/N*** is within range.

The power control algorithm has been designed to accommodate the heterogeneous nature of the upstream carriers in adaptive inroute groups. The goal of the power control algorithm is for each satellite terminal to be received at the target ***C/N*** on the best carrier the satellite terminal can receive at any given time. Therefore, the power control algorithm is responsible not only for adjusting the satellite terminal's power on the current nominal carrier, but also for selecting a new nominal carrier when required.

The target ***C/N*** is calculated based on the ***C/N*** thresholds for the inroutes from the iDirect Link Budget Analysis Guide and the following margins configured in the NMS, the details of which are described in Fade Margin and Measurement Interval Calculations on page 106.

- **Fade Slope Margin (M1)** — allows for an incremental fade, which can occur during the reaction time of the power control algorithm, as well as the uncertainty in the C/N_0 estimations.
- **Hysteresis Margin (M2)** — is added to the fade slope margin to prevent unnecessarily frequent switching between carriers.

The system adds the sum of these two margins to the ***C/N*** thresholds to determine the target ***C/Ns*** for each carrier in the inroute group. The target values and margins are adjusted automatically by the power control algorithm as necessary, based on the MODCOD and symbol rate of each carrier. Highlights of the UCP power control algorithm include:

- If a satellite terminal's ***C/N*** falls below the target ***C/N***, the power control algorithm increases the satellite terminal's transmit power (if possible) to bring the receive signal back up to the target level. If the satellite terminal's power cannot be increased on the nominal carrier and a more protected carrier is available, the satellite terminal's nominal carrier is changed to the more robust carrier. This keeps the satellite terminal in the network at the expense of diminished throughput.
- If the satellite terminal is consistently below the threshold defined by the target ***C/N*** minus the hysteresis margin on the most protected carrier in the inroute group, the satellite terminal is logged out of the network. A satellite terminal that has been logged out must re-acquire the network before it can continue to transmit user traffic.
- If there is additional power headroom available over the satellite terminal's current ***C/N*** plus the hysteresis margin, the power control algorithm looks for a more efficient carrier with a higher MODCOD at which the satellite terminal can maintain the target ***C/N***. If

such a carrier is found, the satellite terminal's nominal carrier is changed to the higher-MODCOD carrier.

- If the satellite terminal cannot switch to a better carrier, the power control algorithm decreases the power as necessary to return the satellite terminal's receive signal to the target ***C/N***.
- If a satellite terminal fade is detected, the update interval is increased based on the severity of the fade. Increasing the update interval may require the system to allocate additional upstream slots to the satellite terminal. For idle satellite terminals, the configured MIR is ignored and more slots are allocated if the UCP requires additional burst measurements.
- The power control algorithm reacts faster when the SAS determines that a terminal has entered an upstream fade. When monitoring the satellite terminal's receive signal, the algorithm periodically samples the receive ***C/N***. The algorithm determines a fade slope for the satellite terminal based on the difference in receive ***C/N*** between successive samples. The algorithm is designed to keep the fade slope linear by increasing the measurement interval during a fade. The algorithm adjusts the measurement interval depending upon the fade slope, such that it is shorter when the fade varies rapidly.

4 Global Bandwidth Management

This chapter describes key aspects of Global Bandwidth Management (GBWM) such as bandwidth allocation and distribution, contention, and service plan enforcement.

The following sections are included in this chapter:

4.1 About Global Bandwidth Management 120
4.2 Velocity Service Providers 120
4.3 Velocity Network QoS Service Modes 122
4.4 About Velocity Service Plans 123
4.5 Service Plan Definition 137
4.6 GBWM Operations 160



NOTE: Also see Satellite Terminal Features on page 305, for additional information on Velocity Satellite Terminals.

4.1 About Global Bandwidth Management

Global Bandwidth Management (GBWM) is the Velocity™ subsystem that manages global bandwidth capacity across the entire satellite network — a constellation of beams potentially from multiple spot beam and broad beam satellites. All beams on all satellites defined in the global satellite network contribute bandwidth to a single pool that is managed by GBWM, giving service providers the flexibility to create service plans for a single beam or across multiple beams on multiple satellites.

This Velocity functionality supports advanced service providers with a high degree of flexibility in subdividing satellite bandwidth and creating groups of satellite terminals with various levels of service plans tailored to the characteristics of supported user applications.



NOTE: Many service providers will be familiar with the existing iDirect bandwidth management and Group Quality of Service (GQoS) of the Evolution platform. In many ways, Velocity's GBWM can be viewed as an extension of the existing GQoS architecture - from satellite networks that operated on single transponders, to satellite networks spanning multiple beams and satellites.

Criteria used for packet classification and the iNet demand and allocation tree structure are essentially the same in Evolution GQoS and Velocity GBWM. However, GBWM introduces a number of new concepts and features, such as SAS and system-wide QoS enforcement, Fair Access Policies (FAP), and geographic scope. Knowledge of Evolution GQoS can be useful, but is only the core of a broader set of features available in Velocity GBWM.

The remaining topics of this section describe some of the over-arching components of Velocity networks, including:

- Service Providers
- Network QoS Service Modes
- Service Plans
- Quality of Service Basics

4.2 Velocity Service Providers

As noted in Chapter 1 , Velocity System Overview, a satellite operator owns and operates the satellite network and defines business rules with respect to allocation and distribution of available bandwidth to their service providers. Service providers create and provide services to subscribers - the end consumer of the provided services - and/or provide bandwidth to other service providers. In some cases, a satellite operator may also decide to offer services directly to a subscriber, thereby acting as both a satellite operator and a service provider. These possible relationships are illustrated below. Note that while only three levels of service providers are shown, the tree of service providers may be any depth.

These services are the foundation of the satellite operator and service provider business models. Velocity in general, and GBWM in particular, is designed to provide both with a wide variety of options to meet the unique needs of both satellite operators and service providers.

Satellite bandwidth is not statically allocated - instead it is allocated only when demand is actually present. This approach allows service providers to provide best effort services. It also allows service providers to support CIR-based service level agreements and, based on statistical analysis, oversubscribe with respect to CIR.

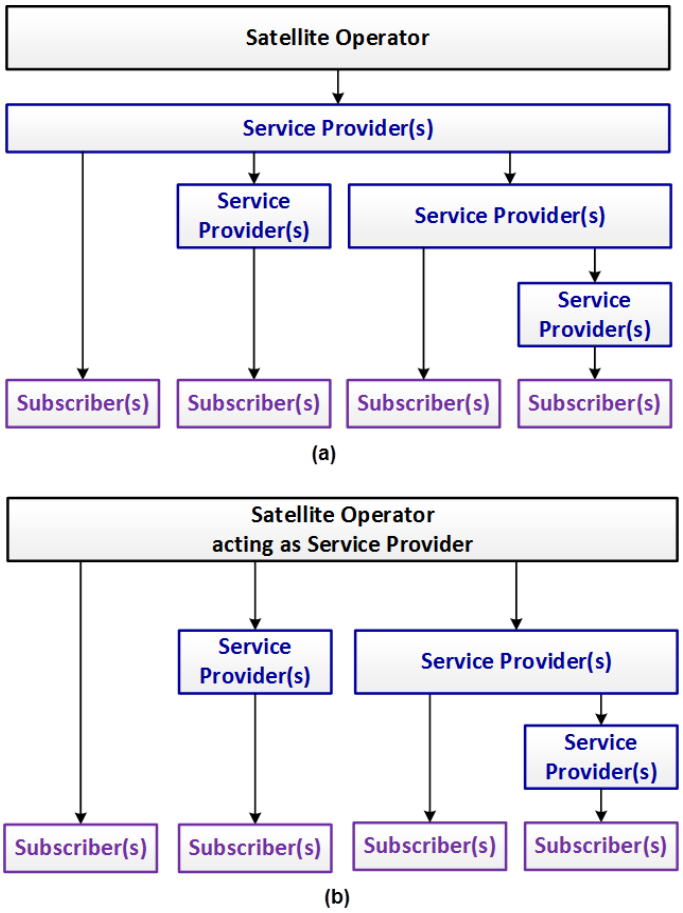


Figure 4-1. a) Satellite Operator and Service Providers b) Satellite Operator as Service Provider

4.3 Velocity Network QoS Service Modes

In Velocity systems, Network Operators are able to select between two QoS Service Mode options for system operation. These options, which include the Symbol-based mode of operation and the new IP-based mode, determine the type of service the network will provide in the downstream direction (hub to remotes), and impacts the manner in which global bandwidth management is handled.

The ***Symbol-based mode***, the QoS operation of all iDirect Evolution networks, and previous Velocity networks, supports data rates in terms of Mbps at a nominal MODCOD or the equivalent of Symbols/s or MHz bandwidth.

With the ***IP-based mode***, Service Providers (SPs) can provide Layer-3/IP outbound services, with CIR/MIR specified, strictly in Mbps. This option allows Service Providers to both purchase and further distribute bandwidth in Mbps, and to monitor QoS stats against a given region, based on the IP data rate without overhead.

The QoS service mode option is selected in Pulse at the Network level. See GSP Configuration and QoS Service Mode Options on page 144.

4.4 About Velocity Service Plans

Services are represented in the Velocity system by subscriber service plans (SSPPs/SSPCs) and group service plans (GSPs). A ***subscriber service plan profile (SSPP)*** is a generic service plan offered by a service provider to subscribers. A ***subscriber service plan component (SSPC)*** is a terminal-specific service, based on an instance of a specific SSPP, and implemented on a subscriber's terminal. A satellite terminal typically has multiple SSPCs, each with its own independent service level.

A ***group service plan (GSP)*** creates a pool of bandwidth that the terminal-specific services will draw from. A GSP created by a service provider, may be used by the service provider to implement their business model, or be provided to another service provider.

For further information on SSPPs, SSPCs and GSPs, see Subscriber Service Plans on page 137.

The relationships between services and the partitioning of bandwidth between them are represented by the configured GBWM tree. Each service plan — whether a GSP or an SSPP/SSPC — is represented as a node in that tree. The GBWM subsystem uses the GBWM tree to create, at runtime, GQE (Global QoS Enforcement) trees at the SAS, NOC, and iNet levels, and allocates available satellite bandwidth to fulfill the demand of subscribers in real-time according to those GQE trees, their service plan parameters, variations in available bandwidth, contention, and congestion.



NOTE: The GQE tree is often referred to as the GBWM tree, with the meaning derived from the context of the reference. In this document, the two trees are named explicitly.

GBWM's allocation algorithms take into account several GSP and subscriber service plan configuration properties. Refer to Bandwidth Allocation on page 132 for details of the bandwidth allocation algorithms used when demand exceeds the available bandwidth.

4.4.1 GSP and Subscriber Service Plan Properties

GSPs and Subscriber Service Plans include the following:

- **Priority** — Priority defines the strict order in which bandwidth is allocated among competing nodes. In general, bandwidth is allocated to higher-priority nodes before being allocated to lower-priority nodes.
- **Cost** — Cost is used to divide and allocate bandwidth among competing nodes that have the same priority.
- **Maximum Information Rate (MIR)** — The maximum information rate (MIR) specifies the maximum amount of bandwidth that may be allocated to a GBWM node, regardless of the amount of bandwidth requested.
- **Committed Information Rate (CIR)** — The committed information rate (CIR) specifies an amount of requested bandwidth given preference during allocation. Generally speaking, bandwidth is allocated to fulfill demand up to a child nodes' CIR prior to meeting any additional demand, subject to the child nodes' MIR.

- **IP Data Rate** — The IP data rate refers to user data transferred over the satellite link and does not include DVB-S2 or TDMA overhead. However, it does include SVN (satellite virtual network) and link encryption headers, which are classified as user data.
- **Information Rate** — The information rate also includes the link layer header overhead and Cyclic Redundancy Check (CRC).

  The system internal information rate is fixed; thus a higher IP data rate on the satellite link is achieved by enabling Velocity compression features, which end users will perceive as higher throughput. Since TCP acceleration prevents some TCP overhead, like local acknowledgments from being transmitted over the air, they too are not included in the IP data rate.

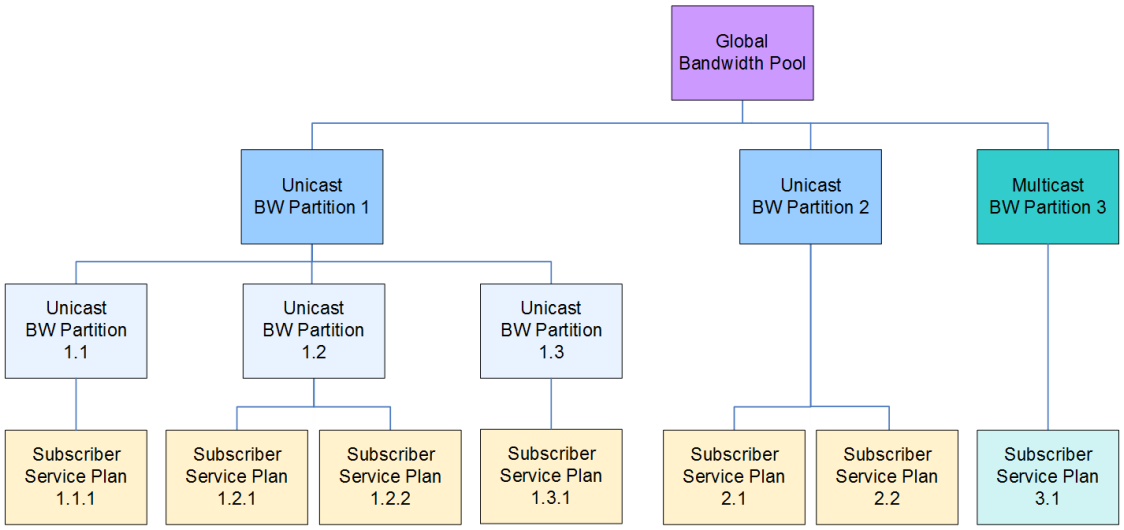


Figure 4-2. Example of a Simple GBWM Tree

NOTE: CIR and MIR are information rates, and differ from their IP data rates. In IP-based mode, configured CIR/MIR values are based on IP bit rates without overhead; in symbol-based mode, information rates include overhead as described.

- **Service Mode** — GSP Service modes allow configuration of mixed modes of operation in the GSP tree, which give the satellite operator and the service providers maximum flexibility in configuring services in a way that meets their service objectives and SLAs. There are four (4) different Service Modes for GSPs:
  - **End User Strict** — CIR distribution is limited strictly to CIR demand (originating from lower nodes). MIR distribution is limited to MIR demand.
  - **End User Flexible** — CIR distribution does not include MIR demand; however MIR distribution can include CIR demand.

    The End User modes are geared toward allocation that takes into account the breakout of the aggregated demands from the SSPCs into aggregated CIR and MIR

demand separately. In Service Provider mode, the allocation takes into account the total demand regardless of whether it originated from SSPC CIR or MIR demand.

- **Service Provider** — CIR distribution is not limited strictly to CIR demand. MIR demand can be partially or fully fulfilled if there is sufficient CIR capacity left after the CIR round.
- **Service Provider Legacy** — This mode gives the operator the ability to configure the GSP in a way so that its behavior is similar to the behavior before an upgrade. It is the default configuration after an upgrade.

  Two mode types (Service Provider and End User) are not mutually exclusive. Depending on the operator requirements, some GSPs may want to operate in an End User mode while others may want to operate in Service Provider mode.

  Figure 4-3summarizes the contribution of CIR and MIR demands to CIR and MIR distribution.

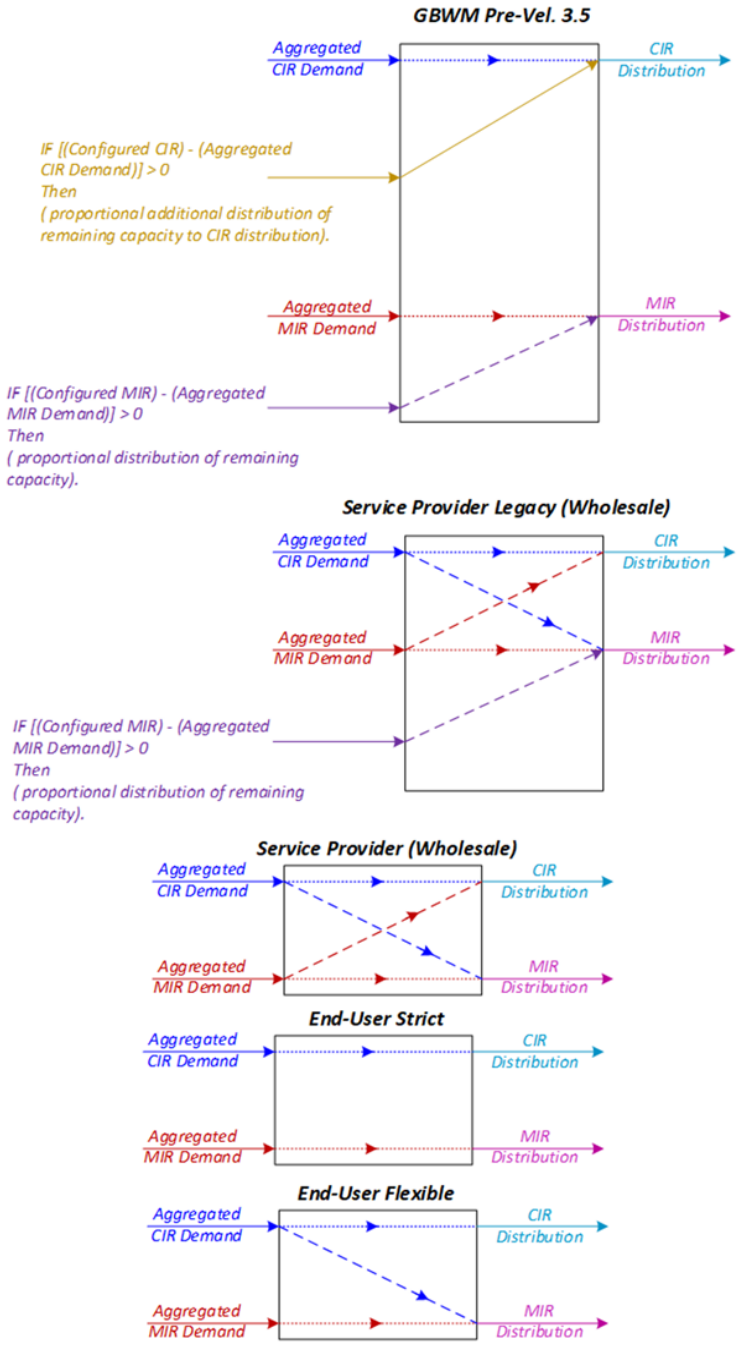


Figure 4-3. CIR/MIR Demands and CIR/MIR Distribution

Service Provider, End User Strict, and End User Flexible Modes

The Service Provider mode and the two End User modes (Strict and Flexible) in Velocity 3.5 have a tighter control on bandwidth allocation. GSP capacity is distributed strictly based on demand from the SAS/CDC and the iNets. There is a 5% extra allocation (from the GSP configured capacity) to accommodate real-time traffic

fluctuations. It should be noted that if the GSP is fully or oversubscribed (aggregated demand is greater than configured capacity), there will be no extra capacity to allocate. If GSP is heavily subscribed, extra allocation will be done until the configured capacity of the GSP is reached.

Real-time traffic fluctuations can result from a sudden increase in demand from terminals using a specific GSP in one or more iNets. This happens when actual demand for existing terminals suddenly increases by a large percentage value or when new terminal(s) enter a sparsely populated iNet(s) using that GSP.

If the traffic fluctuation (increase) is more than 5%, there will be a delay of up to 10 seconds to catch up with the new demand by the GQE process. During this time, there will be less distribution than actual demand. TCP traffic flows usually adjust automatically to this fluctuation. However, for UDP traffic flows, some traffic losses are likely to be encountered.

For GSPs that have on average 10 or more active terminals per iNet, the impact is expected to be relatively negligible.

It is worth noting here that the operator has the option to configure more than the pre-set 5% extra allocation mentioned above. This can be done using an EDK at the GSP level (configured independently per GSP). The detail of that configuration is as follows:

This EDK goes in GSP_DOWNSTREAM_OPT and GSP_UPSTREAM_OPT:

```
"QOS": {
"gsp_allocation_scale_factor": 10
}
```

The default value is 5, the allowed range is 0 - 100. In the above example, it is configured for 10%.

Service Provider Legacy Mode

The Service Provider Legacy mode works as in pre-3.5 PP software. After meeting all of the demand, the leftover GSP capacity is proportionally distributed. Any increase in demand on specific iNets due to new terminals joining the iNets or changes in existing terminals, is immediately served without waiting for GQE demand-distribution delay, and expected to have very minimal impact on any type of traffic streams.

Guidance in Selection of GSP Service Modes (Legacy vs. New Modes)

The flowchart below can be used to decide between Service Provider Legacy GSP Mode and the new modes available in Velocity 3.5.

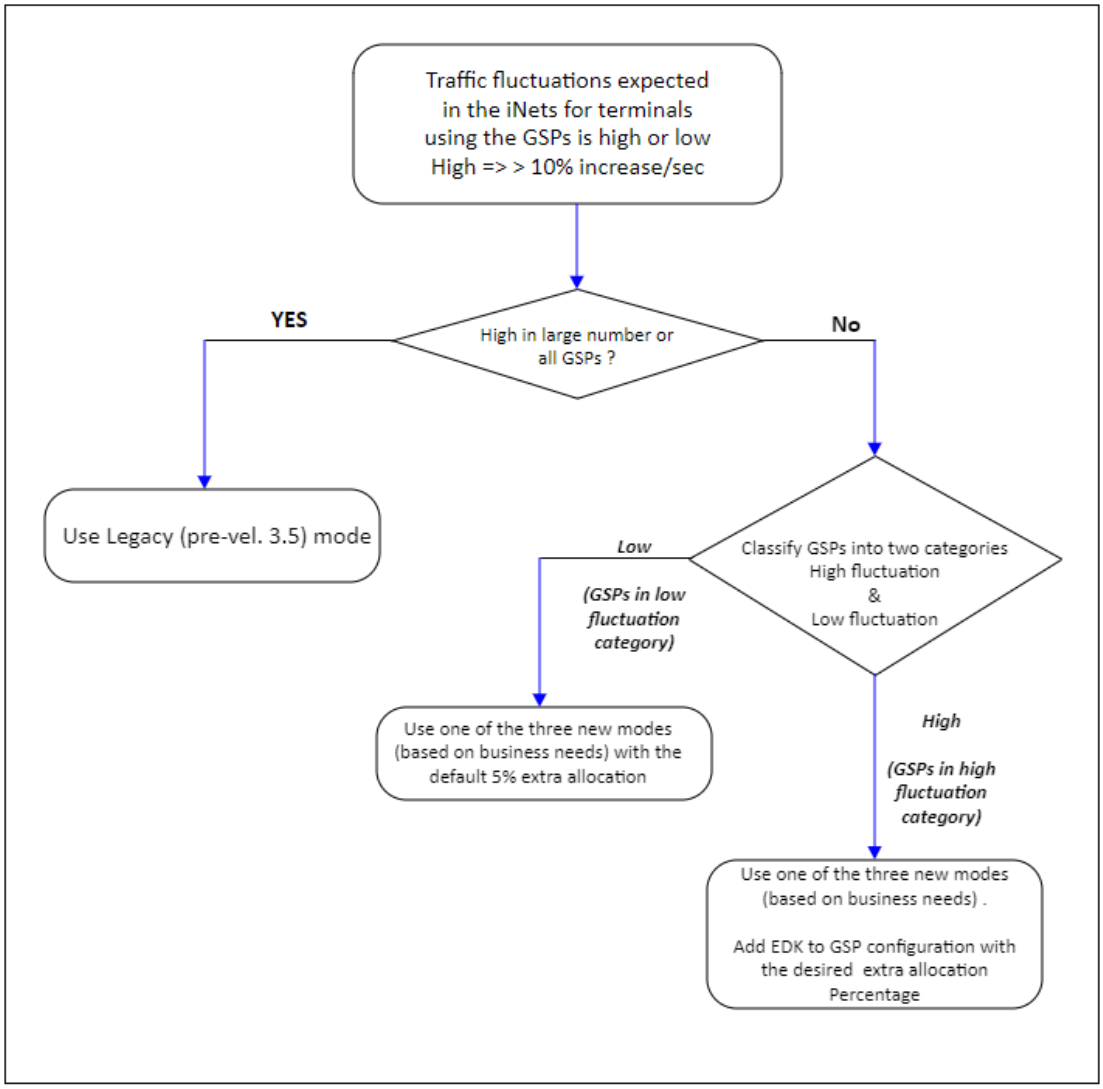


Figure 4-4. GSP Modes - Service Provider Legacy vs. New Velocity 3.5 Modes

Feature Comparison of All GSP Service Modes

The table below provides a summary of Service Mode attributes for all GSP Service Modes.

	Pre-3.5 GBWM Behavior Not supported in 3.5 *	Service Provider Legacy Avail in 3.5	Service Provider Avail in 3.5	End User Strict Avail in 3.5	End User Flexible Avail in 3.5
Contribution of CIR/MIR demand to CIR/MIR distribution	CIR capacity is used for CIR demand and MIR capacity is used for MIR demand. (independent -ly).	CIR capacity is used for both CIR & MIR demand, MIR capacity is used for unmet CIR and MIR demand.	CIR capacity is used for both CIR & MIR demand, MIR capacity is used for unmet CIR and MIR demand.	CIR capacity is used for CIR demand and MIR capacity is used for MIR demand.	CIR capacity is used for CIR demand, MIR capacity is used for unmet CIR and MIR demand.
What do we do with unused GSP CIR and MIR Capacity	CIR and MIR leftovers are proportionally given as CIR and MIR distribution.	CIR and MIR leftovers are proportionally given as MIR distribution.	CIR and MIR leftovers are discarded.	CIR and MIR leftovers and unmet demand are discarded.	CIR and MIR leftovers are discarded.
CIR & MIR Beam Limits	May violate both CIR & MIR beam limits.	May violate both CIR & MIR beam limits.	May violate CIR beam limits but MIR limit is good & no violation.	No CIR & MIR beam limit violation.	No CIR & MIR beam limit violation.
Behavior at the iNet level with different fairness configurations (iNet Congested)	In all fairness modes, gets advantage of more BW allocation compared to other GSPs due to higher CIR & MIR distribution.	In all fairness modes gets advantage of more BW allocation compared to other GSPs due to higher CIR & MIR distribution.	In default & CIR fairness modes, gets advantage of more BW allocation compared to other GSPs due to higher CIR distribution.	In all fairness modes, gets fair BW allocation compared to other GSPs.	In all fairness modes, gets fair BW allocation compared to other GSPs.
BW allocation response time to initial traffic demand request	Takes 5 – 10 seconds.	Takes 5 – 10 seconds.	Takes 5 – 10 seconds.	Takes 5 – 10 seconds.	Takes 5 – 10 seconds.

	Pre-3.5 GBWM Behavior Not supported in 3.5 *	Service Provider Legacy Avail in 3.5	Service Provider Avail in 3.5	End User Strict Avail in 3.5	End User Flexible Avail in 3.5
BW allocation response time to any changes in traffic demand request (GSP Congested)	Takes 5 – 10 seconds on congested cases and 1 – 2 seconds on uncongested cases.	Takes 5 – 10 seconds on congested cases and 1 – 2 seconds on uncongested cases.	Takes 5 – 10 seconds on both congested and uncongested cases.	Takes 5 – 10 seconds on both congested and uncongested cases.	Takes 5 – 10 seconds on both congested and uncongested cases.
Remember last allocation to handle bursty traffic conditions (GSP & iNet is not congested)	Takes 0 – 1 seconds. Will remember and use if needed the last NON-ZERO allocation. (NON-ZERO allocation logic redesign required).	Takes 0 – 1 seconds. Will remember and use if needed the last NON-ZERO allocation. (NON-ZERO allocation logic redesign required).	Takes up to 5 – 10 seconds. Will remember and use if needed the last NON-ZERO allocation. (NON-ZERO allocation logic redesign required).	Takes up to 5 – 10 seconds. Will remember and use if needed the last NON-ZERO allocation. (NON-ZERO allocation logic redesign required).	Takes up to 5 – 10 seconds. Will remember and use if needed the last NON-ZERO allocation. (NON-ZERO allocation logic redesign required).

4.4.2 Quality of Service Basics

Management of bandwidth and quality of service (QoS) is not unique to satellite networks. However, in satellite networks, the bandwidth (MHz) provided by each of the beams on each of the satellites is a precious and expensive resource — one that is carefully managed by satellite operators and service providers. They will take care to deploy ground equipment that is link efficient (generates the most Mbps per MHz) and provides the most flexibility in creating service plans for a variety of applications.

When discussing QoS, several interrelated measures are considered: throughput, latency, jitter, and packet loss. This section describes these QoS characteristics in general terms, without specific regard to a Velocity satellite network.

4.4.2.1 Throughput

Throughput is the rate of data use by an entity such as an end user application. Many applications function properly only with a minimum throughput, which depends on proper

classification and prioritization of packets, and proper management of available bandwidth. For example, if a Voice-Over IP application requires 16 Kbps and a remote is only allocated 10 Kbps, the application fails regardless of priority, as available bandwidth is not enough.

4.4.2.2 Latency

In general, latency measures the amount of time between events. When the term "latency" is used in communications networks, it refers to the amount of time between the transmission of a packet from its source and the receipt of that packet at the destination. In general, latency accounts for the total delay between events and it includes transit time, queuing, and processing delays.

Application latency is typically a concern for transaction-based applications, such as credit card verification systems, which require a rapid completion time per transaction; and interactive applications, like VoIP calls. For such applications, it is important that latency-sensitive traffic be expedited through the system at the expense of less important background traffic. This is especially true in bandwidth-limited conditions where a remote may have a limited number of TDMA (time division multiple access) slots on which to transmit its packets.

4.4.2.3 Jitter

Jitter refers to the variation in arrival time between packets for a specific traffic flow. Time sensitive applications such as VoIP typically require limited latency and jitter to perform properly. A network with high jitter may not satisfy the requirements of such time-sensitive applications, even if their latency requirements are met.

There are many sources of jitter in networks, two of which are significant in a satellite network. While jitter plays a role in both downstream and upstream directions, a satellite network with TDMA inbounds, such as Velocity, tends to introduce more jitter in the upstream direction than in the downstream. This characteristic is due to the discrete nature of TDMA, which only allows a remote to transmit data in assigned slots. Jitter is introduced when the inter-slot times assigned to a particular remote do not match the desired play out rate.

Another source of jitter is additional traffic transmitted between (or in front of) successive packets in the real-time stream. When a large packet needs to be transmitted in front of a real-time packet, jitter is introduced because the remote must wait longer than desired before transmitting the next real-time packet.

Consider an application like VoIP: the transmitting application spaces each packet at a known fixed interval (for example, every 20 ms). However, in the packet-switched network that this application's traffic traverses, there is no guarantee the packets will arrive at their destination with the same 20 ms interval rate. To compensate for this uneven interval, the receiving equipment stores incoming packets in a jitter buffer and attempts to play out the arriving packets at the interval at which they were transmitted. To accomplish this method, additional latency is introduced by buffering packets for a certain amount of time before forwarding them at the fixed interval.

4.4.2.4 Packet Loss

Packet loss measures the number of packets transmitted by a source, but not received by the destination. The most common cause of packet loss is network congestion. Congestion occurs

when the volume of traffic exceeds the available bandwidth, causing packets to fill queues internal to network devices at a rate faster than those packets can be transmitted from the device. When this condition exists, devices drop packets to keep network conditions stable.

Applications built on a TCP (Transmission Control Protocol) transport interpret the absence of these packets (and the absence of their related ACKs) as congestion and they invoke standard TCP slow-start and congestion avoidance techniques. With real-time applications, such as VoIP or streaming video, it is often impossible to gracefully recover these lost packets for lack of enough time to retransmit lost packets. Packet loss may affect the application in adverse ways. For example, parts of words in a voice call may be missing or there may be an echo, or video images may break up or become block-like (pixilation effects).

4.4.3 Bandwidth Allocation

Over-subscription of satellite bandwidth, as well as network conditions such as rain fade, cause situations in which the available bandwidth is not sufficient to satisfy all subscribers' demand at a given point in time.

The Velocity GBWM subsystem allocates available bandwidth according to the configured GBWM tree, configuration of the GSPs and SSPCs in that tree, and current satellite network conditions. GBWM uses a demand and allocation algorithm, whereby demand originates from SSPCs on individual satellite terminals and flows up the Global QoS Enforcement (GQE) tree to the satellite network level, and capacity is allocated down the tree.

The GBWM subsystem uses the configured GBWM tree to create iNet-specific GQE trees, and allocates available satellite capacity to fulfill the demand of subscribers in real-time according to those GQE trees, their service plan parameters, variations in available bandwidth, contention, and congestion. Each GSP and SSPC has properties that control bandwidth allocation and bandwidth requests. These properties are configured specifically at each node in the tree. How these properties are configured determines how available bandwidth is distributed among groups of competing nodes in the tree.

SSPCs, resident on the satellite terminal, are the 'leaves' of the tree, and do not have any nodes below them. SSPCs originate demand requests. GSPs cannot originate any demand, and only ever aggregate demand from the nodes below them, either SSPC or other GSP nodes.



NOTE: The GQE tree is often also referred to as the GBWM tree, with the meaning derived from the context of the reference. In this document, the two trees are named explicitly.

The bandwidth allocation function of the GBWM subsystem consists of a Global QoS Enforcer, a per satellite SAS QoS Enforcer, per iNets/per satellite QoS Enforcers, and the GQE trees. For more details on these portions of the GBWM subsystem, see GBWM Operations on page 160.

4.4.3.1 Allocation Algorithms

Beginning at the highest (root) node in the GBWM tree, available bandwidth is allocated by a GSP node to all nodes immediately below it, based on satellite operator or service provider-defined properties (refer to Velocity Service Providers on page 120) and the current demand

as indicated by those nodes. Child nodes compete for the bandwidth allocated to their parent GSP.

> ***NOTE:*** Bandwidth allocation algorithms only apply fairness options when there is contention; when there is no contention, all nodes below the GSP receive their demand.

A GSP node may be configured to allocate bandwidth using “Absolute” (strict priority) mode, or using “Per Allocation” mode. Both modes perform multiple bandwidth allocation rounds; they differ in which nodes are included in a given round, and how much of the nodes' demand is satisfied in that round.

In absolute mode, the allocation rounds are organized by child node priority from highest priority node to lowest priority node until the available bandwidth is exhausted. Within a priority round, there are two sub-rounds — one for demand that conforms to the child node's configured CIR, and another for its remaining demand, subject to its configured MIR.

If multiple child nodes are configured with the same priority, an allocation fairness algorithm determines bandwidth allocation. Refer to Allocation Fairness Algorithms on page 134.

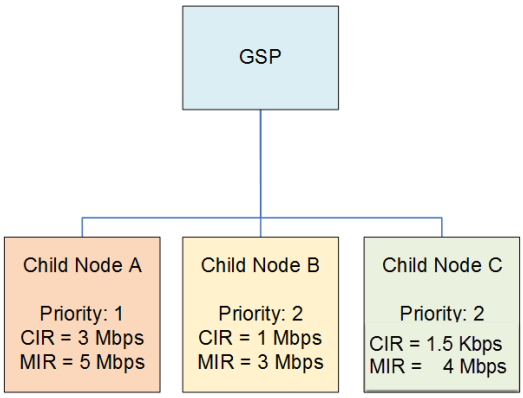


Figure 4-5. Example Allocation Scenario

In the scenario shown above, the allocation order, in absolute mode, would be as follows:

In Priority 1 round:

1. Node A - receives bandwidth up to CIR demand of 3 Mbps
2. Node A - receives bandwidth of demand-delta (MIR Demand - CIR Allocated) in MIR round

In Priority 2 round:

3. Nodes B and C- receive bandwidth up to their CIR demand, 1 Mbps and 1.5 Kbps respectively in CIR round
4. Nodes B and C - receive bandwidth for their demand-delta (MIR demand - CIR allocated) in MIR round

In ***Per-Allocation Round*** mode, there are two allocation rounds — the first round attempts to fulfill demand up to CIR for all child nodes, and the second allocates remaining bandwidth for

"best effort" — that is, MIR). Within each round, bandwidth is allocated according to the child nodes' configured priority and cost, similar to absolute mode. In the same scenario, but using Per-Allocation mode, the bandwidth allocation order would be as follows:

CIR round:

1. Priority 1 sub-round
   a. Node A - receives demand up to 3 Mbps CIR
2. Priority 2 sub-round
   a. Nodes B and C - receive bandwidth up to their CIR demand of 1 Mbps and 1.5 Kbps respectively

MIR round:

1. Priority 1 sub-round
   a. Node A - receives bandwidth of demand-delta (MIR Demand - CIR Allocated) in MIR round
2. Priority 2 sub-round
   a. Nodes B and C - receive bandwidth for their demand-delta (MIR demand - CIR allocated) in MIR round

The GBWM bandwidth allocation algorithm allocates iNet excess capacity in upstream (after meeting all the demand iNet) equally to all active terminals in the iNet. It is possible, therefore, for a satellite terminal to receive more bandwidth than the configured maximum. However, it strictly enforces the downstream allocation based real-time demand and no excess capacity distribution on iNet downstream direction.

4.4.3.2 Allocation Fairness Algorithms

GBWM provides the service provider with the ability to specify how equally-deserving child service plans share a bandwidth allocation from their parent GSP, when the child service plans demand exceeds the GSP received allocation—in other words, there is bandwidth contention. Allocation fairness is the method of determining how this allocation is shared among GSP child service plans.



NOTE: 1) Allocation fairness only applies when there is contention between service plans configured with the same priority.
2) The concept of inbound fairness is not related to the allocation fairness methods described here. Inbound fairness is comparable to the configuration of base MODCOD on the outbound. For additional details, see Inbound Fairness Base Bitrate (for SSPCs only) on page 159.

The GBWM system provides a choice of several ***allocation fairness*** algorithms. These options support various business models. The allocation fairness algorithms are:

- **Allocation Fairness relative to Bandwidth (default)** — favors child nodes with low CIRs. Typically used when the CIR/MIR of the child nodes and their RF terminal types do not vary considerably, and beams within the GSP are also consistent.

- **Allocation Fairness relative to CIR** — favors child nodes with larger CIRs. Preferred when the service provider's commitments to customers include penalties (or credits) determined by the size of the CIR commitment violation.
- **Allocation Fairness relative to CIR and MIR** - favors child nodes with greater CIRs (in CIR round) and greater MIR (in MIR round). Fairness relative to CIR is done in the CIR round and relative to MIR in the MIR rounds of allocation.
- **Allocation Fairness relative to base MODCOD** — preferred for the GSP parent of SSPCs, when the terminals are fixed. See Table (QoS Mode).
- **Allocation Fairness relative to operating MODCOD** — preferred for GSP parent of SSPCs, when the terminals are mobile. See Table (QoS Mode).

See Velocity Network QoS Service Modes on page 122.

Table 4-1. Inbound (IB)/Outbound (OB) Valid Allocation Fairness Methods

Allocation Fairness Method	OB	IB	Useable In	QoS Mode
relative to bandwidth (default)	•	•	all GSPs	Symbol-based/IP-based
relative to CIR	•	•	all GSPs	Symbol-based/IP-based
relative to CIR and MIR	•	•	all GSPs	Symbol-based/IP-based
relative to MODCOD (base or operating)	•	**n/a**	lowest level GSPs	Symbol-based only

These fairness algorithms may be combined on the same service plan, with some restrictions. The algorithms ***Relative to Bandwidth*** and ***Relative to CIR*** or ***Relative to CIR and MIR***are mutually exclusive. Likewise, the two algorithms relative to MODCODs are also mutually exclusive. Only those GSPs whose child nodes are SSPPs may be configured to use ***Relative to Base MODCOD*** or ***Relative to Operating MODCOD***.

Thus, GSPs that have other GSPs as children may be configured to use either:

- Relative to Bandwidth
- Relative to CIR
- Relative to CIR and MIR.

And GSPs that have SSPPs as children may be configured to use one of the following:

- Relative to Base MODCOD
- Relative to Operating MODCOD
- Relative to Bandwidth
- Relative to Bandwidth + Relative to Base MODCOD
- Relative to Bandwidth + Relative to Operating MODCOD

- Relative to CIR
- Relative to CIR + Relative to Base MODCOD
- Relative to CIR + Relative to Operating MODCOD
- Relative to CIR and MIR
- Relative to CIR and MIR + Relative to Base MODCOD
- Relative to CIR and MIR + Relative to Operating MODCOD

For additional details on allocation fairness methods, see Allocation Fairness on page 168.



NOTE: Beginning In Velocity 3.5 the operator can configure a fairness mode that applies to the top level GSPs. In previous versions, fairness at this level was not configurable and the only available mode was fairness relative to bandwidth.

4.5 Service Plan Definition

GBWM enables the construction of very sophisticated and complex service and bandwidth allocation models. It allows service providers to create network subgroups with various levels of service on the same outbound carrier or inroute group. It allows bandwidth to be subdivided among customers or service providers. It also allows that bandwidth to be over-subscribed, so that if one customer is not using all of their bandwidth at a given point in time, that excess bandwidth can be used by another customer. For a description of QoS properties to consider when defining services on Velocity, see GBWM Operations on page 160.

Details on configuring service plans, are provided in the document ***Configuring Velocity Networks Using Pulse***.

4.5.1 Subscriber Service Plans

Subscriber service plans are defined by the service provider to represent services offered directly to subscriber terminals. A subscriber service plan profile (SSPP) is a generic service plan offered by a service provider to multiple subscribers. In general, an SSPP defines:

- Bandwidth (CIR and MIR in Mbps)
- Geographic scope
- Traffic classification rules and filters
- Fair access policy (optional)

The service provider may also specify which terminal types an SSPP may be used with.

Once an SSPP is purchased by a subscriber, it is represented on the subscriber's terminal as a subscriber service plan component (SSPC), which contains the SSPP parameters and subscriber or terminal-specific parameters, such as: service start and end time/date scheduling.

In GBWM Enforcement (see GBWM Operations on page 160), SSPCs are the demand-generating objects in the GQE Tree, and an SSPC reports its demand to its parent GSP.

4.5.1.1 Unicast Subscriber Service Plans

A unicast subscriber service plan applies to bidirectional traffic sent to and from one subscriber terminal. The service plan (SSPP and/or SSPC) is configured with the following features:

- Bandwidth (CIR and MIR in Mbps)
- Geographic scope
- Start and end time/date scheduling
- Recurring modifications
- Fair Access Policy (FAP)
- Multi-parent GSP

Details about these features can be found in the immediately following topics or in Features Common to GSP and Subscriber Service Plans on page 148.

Geographic Scope in Unicast Subscriber Service Plans

The geographic scope for a subscriber service plan is defined by one or more regions, where each region can be either a combination of one or more beams from one or more satellites, or a set of service areas. Each region must be defined with the CIR/MIR, Priority, and Cost, when the satellite terminal operates in the specified region. A subscriber service plan may have multiple geographic components, each associated with a geographic scope and GBWM parameters.

geoscope region of type service area (SA) consists of a set of predefined service areas. The service areas (SA) in a geoscope region must be defined within a single service area group (SA Group). SA Groups allow the satellite operator or service provider to define different service areas for a customer or group of customers. For additional information on SAs and SA Groups, see Service Area Maps on page 41.

SSPP Geoscope Rules and Guidelines

SSPP geoscope is restricted to regional service-area based geoscope that is a subset of the parent GSP geoscope, when the parent geoscope is based on regions. The exception to this rule, is when the geoscope is configured of beam-based regions and when configuring NMS/Global SSPPs. In such cases, these restrictions do not apply and the SSPP can be configured with either a global geoscope or a regional geoscopes with service area-based regions or beam-based regions.

Since an SSPP may have a single parent or, in some cases, multiple parents, these parents all will influence the possible geoscopes that can be configured for the SSPP.

The following rules and guidelines, which apply when configuring SSPP geoscopes, are based on the geoscope configured in the parent GSPs.

- If any one of the GSP parents has a Global geoscope, then the SSPP geoscope may be configured as Global or Regional.
- If all GSP parents have beam-based geoscopes, then the SSPP geoscope may be configured as Global or Regional.
- If GSP parents have a geoscope mix — some are global geoscopes and some are regional beam-based geoscopes, then the SSPP geoscope may be Global or Regional with a subset of the parent GSP regions.
- If GSP parents have a geoscope mix — some service area-based and some beam-based, then SSPP geoscopes must be regional, where each region is restricted to a subset of the parent GSP.

Multi-parent GSP

Mobile satellite terminals may not always be situated in the geographic scope defined for a parent GSP. The Velocity platform allows for a subscriber service plan to be configured with multiple possible GSP parents. The list of GSP parents is prioritized, and an SSPC is linked to only one of the GSP parents at any given time.

> ***NOTE:*** 'Multi-parent GSP' is a characteristic of **unicast** subscriber service plans only.

For example, support for multi-parent GSPs enables the following situation: a service provider wishes to offer a global service, but wants to provide subscribers with higher throughput when they are located in a regional coverage area and lower throughput when they are outside that area. As shown in the diagram below, the service provider can define a regional GSP (GSP1) and a global GSP (GSP2), and configure the subscriber service plan (SSPC1) to have both GSPs as parents. The subscriber service plan would draw bandwidth from only one of its possible GSP parents at a time - the regional GSP when it is located in the regional coverage area, and from the global GSP when outside the regional coverage area.

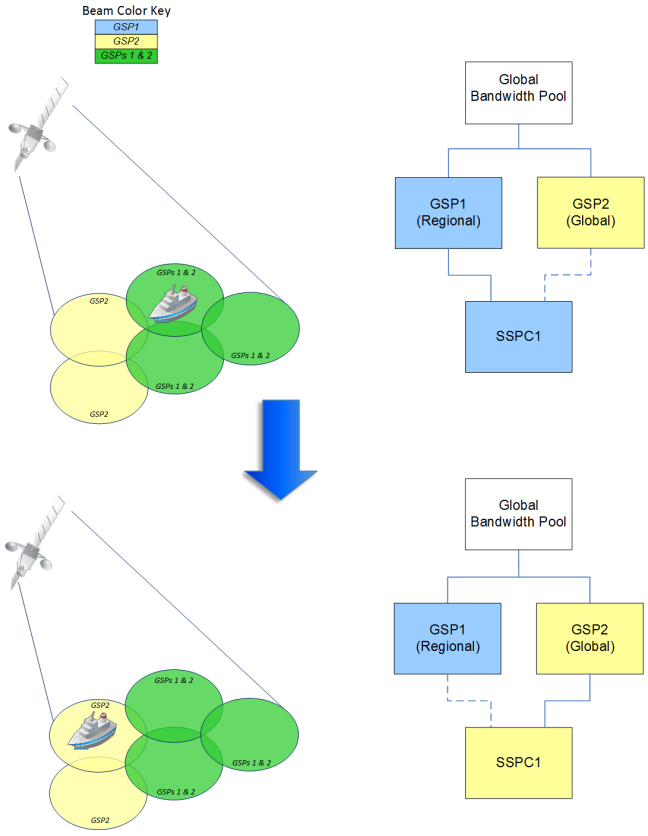


Figure 4-6. SSPC with Multiple Parent GSPs

4.5.1.2 Multicast Subscriber Service Plans

Multicast traffic differs from unicast traffic in that a single multicast stream transmitted on a beam may be received by any terminal in the beam's footprint. Thus, the bandwidth on a beam or iNet required for transmission is the same regardless of the number of recipients.

A multicast subscriber service plan (M-SSPP / M-SSPC) defines a multicast service that may consist of one, several, or unlimited multicast streams. The GBWM options are similar to that of unicast traffic - geographic scope, packet filtering, and classification rules — however the following are also true:

- All multicast traffic travels downstream to terminals.
- The MODCOD used to transmit multicast packets is defined in the M-SSPP.
- An M-SSPP can only be configured to have one parent M-GSP.
- FAP is not available for M-SSPPs; it can only be configured at the GSP level.
- Recurring service plan modifications are not supported.

The multicast streams that the M-SSPP enables are specified in its application service level classification rules. The classification rules may specify one multicast stream (i.e., multicast IP address), multiple multicast streams, or even match all multicast streams. If the same stream is specified in two service plans on a terminal, the one with higher priority is applied.

4.5.1.3 Subscriber Service Plan Limits per Terminal

While there is no explicit limit on the number of SSPPs that can be created in the system, an explicit limit is placed on the number of unicast SSPCs, applications (VRs) and service levels that can be applied to a terminal. Each terminal can support up to 13 active unicast SSPCs plus one for NMS management unicast SSPC (per direction).

If the user decides to create one or more separate unicast SSPC for IGMP (multicast control) unicast traffic, multicast counts against the terminal limits. However, the operator is not required to create a separate unicast SSPC, they could instead cover IGMP traffic in the NMS unicast SSPC, or in one of their other unicast customer SSPC(s).

Furthermore, a unicast SSPC is not restricted to handling a certain type of traffic, but may be configured to handle multiple types of traffic — they may be configured with multiple service levels (for additional information, see Packet Classification on page 173). While multiple service levels can be configured per unicast SSPC, and there is a maximum 13 active unicast SSPCs per terminal, there is also a maximum of 31 service levels across all unicast SSPCs per terminal. The network administrator can configure as many deactivated unicast SSPCs as needed and the VRs and service levels defined within the deactivated unicast SSPCs do not count against the limits defined in the table below.

Table 4-2. Service Plan Limits

Unicast SSPC Limits Per Direction	
Maximum SSPCs per terminal (carrying only unicast traffic)	13 Active SSPCs; +1 NMS mgmt. SSPC
SSPC Limits per terminal *per direction*	
Minimum Service Levels per SSPC	1 service level
Minimum applications (VRs) per SSPC	**1**
Maximum applications (VRs) per terminal	13 + NMS application
Maximum Service Levels per terminal	31 service levels including service levels defined in the NMS mgmt. SSPC

4.5.2 GBWM Tree Structure

All beams on all satellites defined in the global network contribute to a single bandwidth pool managed by GBWM. The GBWM tree defines how the service provider subdivides that bandwidth pool into smaller bandwidth pools to control how individual subscribers compete for the finite satellite bandwidth.

Services are represented in the Velocity system by subscriber service plans (SSPPs / SSPCs) and GSPs. The relationships between these services and partitioning of bandwidth among them are represented by nodes in the GBWM tree. GSPs can be parents of other GSPs, many times over, down to the leaf-nodes of the tree, which are SSPCs. This hierarchy and the rules configured at each node determine how bandwidth is distributed by GQE. For additional information on Global QoS Enforcement, see GBWM Operations on page 160.

When designing the GBWM tree, it is useful to think of the activity as designing pools of bandwidth and competition for that bandwidth. A GSP represents a pool of bandwidth. At runtime, its child nodes - whether other GSPs or SSPCs - compete with each other for the bandwidth available to the GSP. The bandwidth available to the GSP varies over time to reflect the result of competition with its sibling nodes and environmental conditions such as rain fade. Please refer to Outbound Service Plan Parameters on page 149 and Inbound Service Plan Parameters on page 158.

Velocity allows bandwidth to be oversubscribed in any GSP and at multiple levels of the tree. A service provider's knowledge of their customers' usage patterns and statistical analysis, typically guide the degree of oversubscription and where it should be used.

The GSP bandwidth pools often reflect one or more of the following:

- Market segments
- Service offerings
- Customer or traffic priority
- Service providers — for example, VNOs)
- Geographic scope — for example, regions or beams
- Terminals of similar antenna capabilities (antenna size, transmit EIRP, receive G/T)

Multiple GSPs with different geographic scopes and/or priorities can be purchased by a given service provider. These purchases result in the service provider owning multiple GSP nodes in the GBWM tree.

The following diagram shows an example GBWM tree structure in which a satellite operator is providing services to two service providers, one of which is providing services to its customer service providers. The satellite operator is also providing multicast services directly to subscribers.

- The satellite network consists of ten beams on two satellites; this is the scope of the root of the tree.
- Service Provider #1 is providing unicast services in the rightmost three beams. This is the scope of GSP 1.

- Service Provider #1 has subdivided its bandwidth pool into three smaller pools. Each of these GSPs has a scope that is one or more beams defined in the scope of GSP 1.
- Service Provider #2 is providing unicast services in the leftmost four beams.
- The satellite operator is providing multicast service in all beams. This is the scope of GSP 3. The subscriber service plan has the same scope.
- The geoscope of each SSPP is the same as its parent GSP.

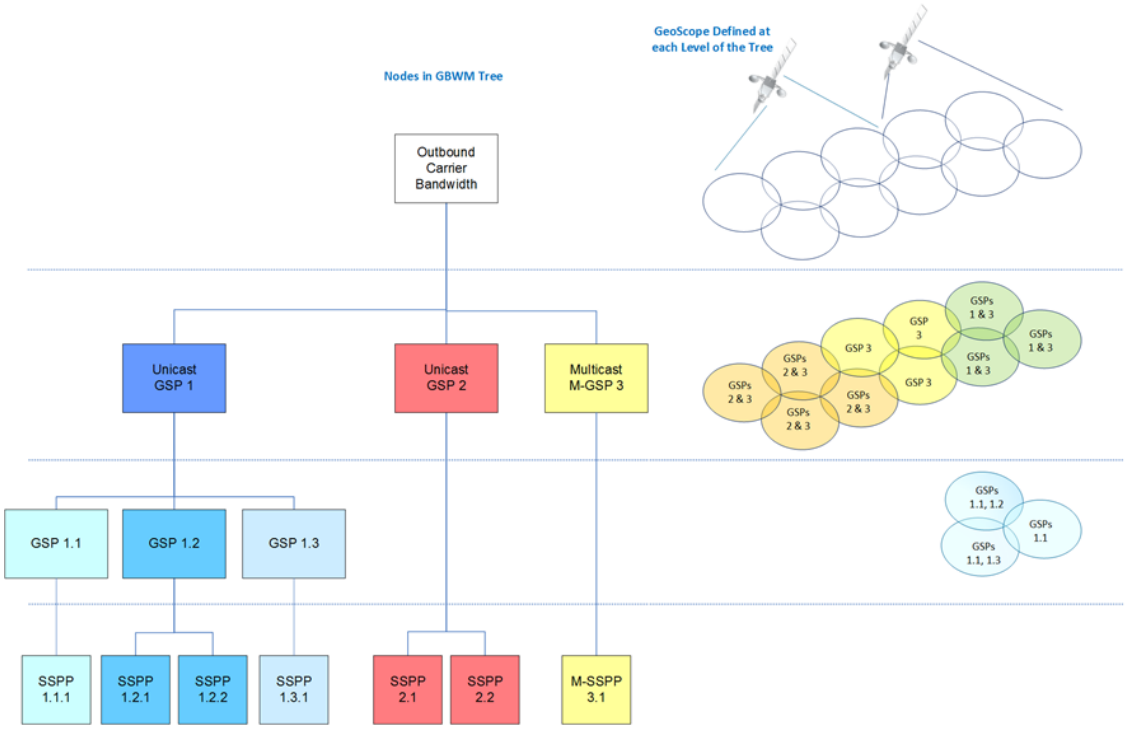


Figure 4-7. Example GBWM Tree Structure

NOTE: Only a single level of multicast GSPs (M-GSP) is supported — beneath which are multicast SSPCs (M-SSPCs). Each M-SSPC applies to one or more multicast streams.

4.5.3 Group Service Plans (GSPs)

A group service plan creates a pool of bandwidth capacity as described in GBWM Tree Structure on page 142. Separate GSPs are required for outbound and inbound bandwidth.

In the forward, or outbound, direction the bandwidth is specified in terms of Mbps (Megabits per second), at a nominal MODCOD that the system translates directly to Msps (Mega-symbols per second). In the inbound direction it is specified in Mbps. This is true only in the Symbol-based mode.

Each GSP represents a single node in the GBWM tree structure. Each GSP may have different GBWM parameters, geographic scopes, priorities, and schedules. The GBWM parameters

include: nominal MODCOD (only applicable in the outbound direction), priority, cost, CIR, and MIR and the allocation fairness algorithm used during contention.

In addition to defining a bandwidth purchase, each GSP may also include an associated Fair Access Policy (FAP) by which it is governed. A FAP may be configured with multiple rules and options. For additional details on FAP, see Fair Access Policy on page 325.

Service providers may optionally specify the terminal types and SVNs that may be used with the GSP. This restriction is enforced on any child GSPs or SSPCs created below the GSP parent.



NOTE: There is no explicit limit on the number of GSPs that can be created.

4.5.3.1 GSP Configuration and QoS Service Mode Options

Before creating and configuring Group Service Plans, Velocity network operators should have specified the QoS Service Mode in which the system will operate — the Symbol-based mode of operation or the IP-based mode of operation. These options, which are mutually exclusive, are selected at the Velocity Network level.

If the Symbol-based mode is specified, Group Service Plan configuration supports data rates in terms of Mbps at a nominal MODCOD, which is equivalent to Symbols/s or MHz. Bandwidth allocation algorithms are based on symbols. As such, the system translates the configured MIR/CIR (Mbps) to symbols using the configured nominal MODCOD, and thereby scales the demand accordingly. QoS Stats are reported in bytes and contain the L2 overhead.

If the IP-Based option is specified, Satellite Operators and Service Providers create and configure Group Service Plans (GSPs) with CIR/MIR specified as a Mbps data rate and thereby offer Mbps as a Service. Service Providers can provision and manage bandwidth at the GSP/SSPC level at a real IP rate, with MIR enforcement at the GSP level. Bandwidth demand, allocation, and usage statistics for both GSPs and SSPCs are monitored at the real IP rate for unicast traffic.

In the IP-based mode, system bandwidth allocation algorithms are all natively performed in Mbps. In general, the Mbps mode is more accurate than the symbols mode - both in bandwidth allocation and statistics measurements, for both GSPs and SSPCs. Also, in Mbps mode, the GBWM management is in pure IP bytes (with no OTA overhead for unicast traffic), which ultimately results in stats that are more accurate. Furthermore, MIR is strictly enforced for downstream traffic for GSPs and SSPCs, and for upstream traffic for GSPs. The GBWM bandwidth allocation algorithm does not strictly enforce MIR for upstream traffic at the SSPC level of the GQE Tree when the iNet contains excess bandwidth.

See Operating in Symbol-based Mode on page 176 and Operating in IP-based Mode on page 176.

4.5.3.2 Unicast Group Service Plans

A unicast GSP applies to traffic sent to or from a number of subscriber terminals. A unicast GSP may be configured with the following features:

- Geographic Scope

- QoS (Quality of Service) Parameters
- Start and End Date/Time Schedule
- Recurring Service Plan Modifications
- Fair Access Policy (See Fair Access Policy on page 325.)

Details about these features can be found in the following topics or in Features Common to GSP and Subscriber Service Plans on page 148.

4.5.3.3 Geographic Scope in GSPs

A GSP may have a geographic scope that is global (the entire satellite network). Alternatively, a regional geographic scope may be configured as a specific region. A regional geographic scope is defined as a collection of one or more beams or service areas. Beams forming a region may or may not be adjacent, and may be provided by the same or different satellites.

In previous Velocity releases, GSP regional geoscopes were based on regions that only consisted of beams and satellites, which did not allow operators to specify custom geographic areas where they intended to sell a service. Network operators are now able to specify a GSP regional geoscope based on service areas in which the GSP is used to provide bandwidth. Although a regional geoscope may contain one or more beam-based or service area based regions, the service areas in a region must be a subset of the same SA Group. See Geoscope Regions Based on Service Areas on page 43.

The following are some key aspects of GSP region geoscopes:

- Satellite Operators/Service Providers can sell unicast GSPs with CIR/MIR and downstream efficiency threshold specified in Mbps for specific geographic areas
- Supports GSP/SSPP geoscopes configuration based on single region with SA
- Supports GSP/SSPP geoscopes configuration based on multiple regions with multiple SAs
- IP-based multicast traffic supported using geoscope regions based on beams

4.5.3.3.1 GSP Regional Geoscope Rules and Guidelines

Although the GSP supports both Global and Regional geoscope configuration, this discussion relates to a set of rules and guidelines that apply when configuring regional GSP geoscopes. These rules and guidelines, illustrated in the following diagram, apply to beam-based regional geoscopes and to SA-based regional geoscopes, for both parent GSPs and child GSPs, but do not apply to GSP global geoscopes.

In the Pulse UI, these geoscope rules are enforced by enabling or disabling the Global scope option, where applicable.

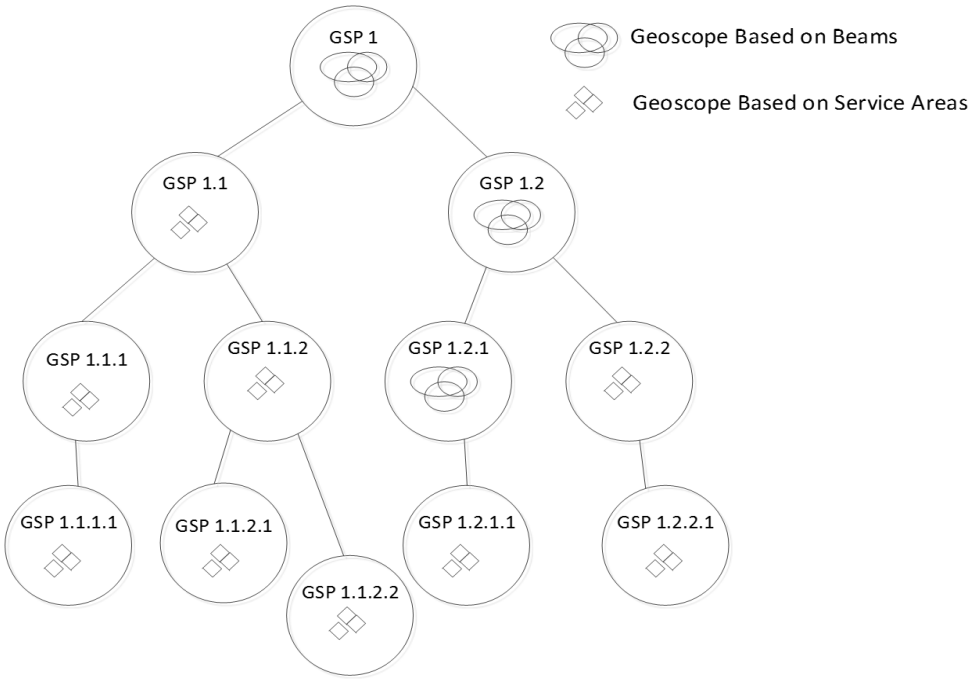


Figure 4-8. GSP Region Geoscope Types

The following guidelines apply to parent and child GSP Regional geoscopes:

- The top level of a GSP tree must be configured using a beam-based or global geoscope.
- Child GSPs may specify one or more regions based on beams only.
- Child GSPs may specify one or more regions based on service areas only.
- Child GSPs whose parent GSP has a service area-based geoscope must also specify a region geoscopes based on service areas.
- Child GSPs whose parent has a service area-based geoscope must specify regions that contains a subset of the service areas of the parent GSP. See **Example 1**.
- A GSP with a beam-based regional geoscope has no restrictions and supports child GSPs with geoscopes as follows:
  - Child GSPs may have global or regional geoscope
  - All children have geoscopes that are based on service areas
  - All children have geoscopes that are based on beams or global
  - Some children have service area based geoscopes and others beam-based geoscopes

The following example illustrates that a child GSP inherits restrictions from parent geoscope:

Example 1: ***Child GSP geoscope must be a subset of SAs used in parent GSP geoscope***

If GSP-1 has a regional geoscope of **R1**[**SAG1**:SA1,SA2,SA3] and **R2**[**SAG2**:SA1,SA2], then a **GSP-1.1** geoscope must use any of the following combinations: **"R1** & **R2"**, **"R1"** or **"R2"**.

4.5.3.4 Multicast Group Service Plans (M-GSPs)

Multicast service plans are only supported on the outbound, and can only be transmitted to a terminal using a bandwidth defined by a parent multicast GSP. Any multicast traffic received from the backbone terrestrial network, at the SAS, under a non-multicast GSP is discarded.

The configuration of a multicast GSP applies to every iNet of every beam in the geoscope. This includes CIR, MIR, priority and cost. The CIR and MIR configured for an M-GSP must account for transmission of a multicast stream on each iNet in which there is a listening terminal. For details of multicast operations in Velocity, refer to Velocity IP Multicast on page 52.

The following diagram shows two multicast streams sent under two GSPs that have different regional geoscopes.

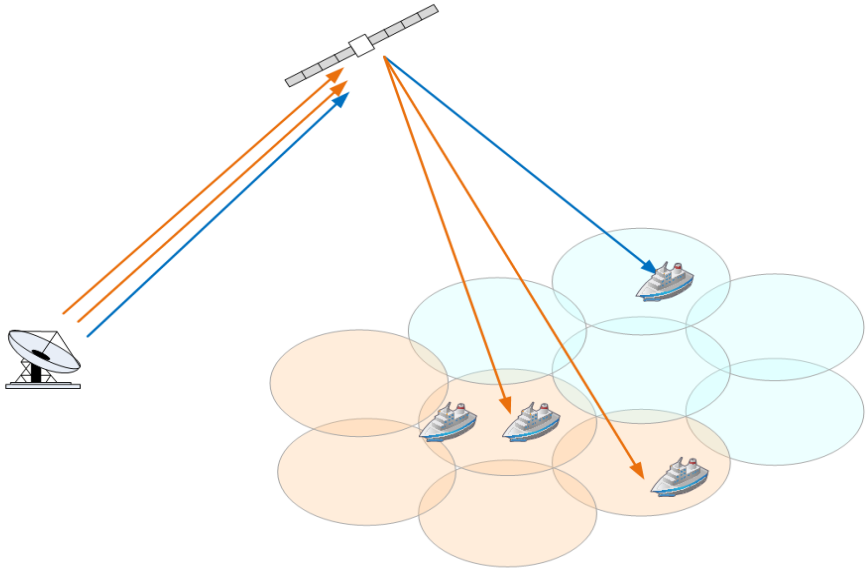


Figure 4-9. Multicast Traffic in Velocity

Multicast GSPs have:

- A schedule, to provide start and end dates
- A geoscope, defined in beams
- A QoS configuration (CIR/MIR/priority and cost configurations). This configuration is replicated internally in every iNet of every beam in the geoscope.
- Fair Access Policy (See Fair Access Policy on page 325.)

Multicast GSPs support the same features as unicast GSPs. The GBWM Tree may contain only a single level of multicast GSPs (M-GSP).

4.5.4 Features Common to GSP and Subscriber Service Plans

When it comes to the configuration of GSPs and Subscriber Service Plans in the NMS, there are several aspects of the configuration that are common to both the GSP configuration and the Subscriber Service Plan Components (SSPC) configuration.

4.5.4.1 Geoscope

Geoscope is a characteristic of service plans that describes where a service plan can be used, and by extension that bandwidth in the satellite network can be used to satisfy their demand. For each GSP, satellite operators and service providers can indicate it is supported globally, or is applicable to a subset of the beams in the satellite network. Additionally, each GSP can be configured with maximum throughput limits per beam, limiting the maximum throughput delivered to an individual service plan in specific beams.

For SSPCs, service providers can specify the beams and/or service areas where the service plan can be used. See Geoscope Regions Based on Service Areas on page 43, for additional details on geographic scope and service areas.

4.5.4.2 Child GSP/SSPP Geoscope Restrictions

If a GSP or SSPP has a parent GSP that has a regional geoscope composed of service area-based or beam-based regions, the GSP/SSPP children are inherently restricted to configure regions that are consistent with the regions configured in the parent GSPs.

4.5.4.3 Service Scheduling

Service scheduling provides a service provider with the means to configure the start and end times of a service plan (GSP or subscriber service plan), or to configure a temporary modification in the characteristics of the service plan such as CIR/MIR, in both the GSP and the SSPC, as well as the priority and/or cost in the SSPC. Unicast and multicast service plans both support service scheduling in the same way.

Start/End Times

Service scheduling supports time-of-day GSP/SSPCs, where different service plans may be used at certain times during the day. For example, a group may have different service plans for business hours, after hours, and weekends. Service scheduling also supports the scheduled creation of a service plan, where a customer requests a temporary GSP or SSPC for a duration specified by start/end dates and times.

When a GSP that is parent to other GSPs or SSPCs reaches its end date, all of its child nodes will also be effectively ended, regardless of their own start/end date configurations. The exception is an SSPC configured with multiple parent GSPs. In that case, the ending of one GSP will simply cause the SSPC to attempt to draw bandwidth from another eligible GSP.

Notifications (alarms) are sent to the NMS when the GSP/SSPC end date and time is approaching to indicate service termination is near. The service provider is able to configure the lead time between the delivery of the service termination warning and the actual termination of services.

Recurring Service Plan Modification

Recurring service plan modification, also known as recurrence, describes a repeating change to the configuration of a GSP or SSPC. The schedule can be specified as:

- Daily, during a specific time of the day
- Weekly, on specific days of the week
- Monthly, on specific dates of the month

The recurrence features differ for GSPs and SSPPs/SSPCs.

- For a GSP, recurrence can be configured for the entire geoscope of the GSP, and cannot be applied to a subset of the beams within a GSP's geoscope. Only CIR and MIR can be modified as part of a GSP's recurrence.
- For a subscriber service plan, each geoscope region may be configured with a separate CIR, MIR, priority, and cost. Recurrence can modify CIR, MIR, priority and cost for each region covered by an SSPP.

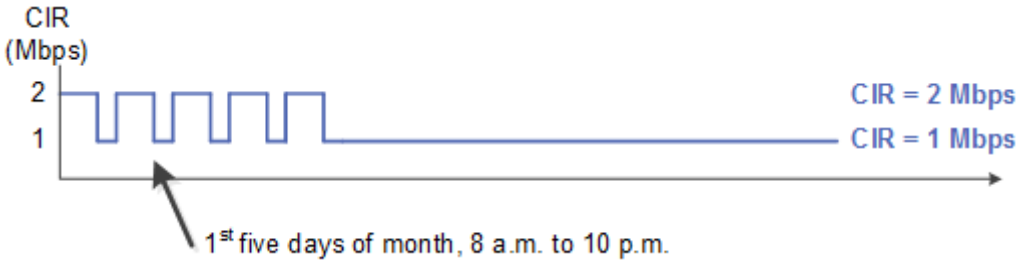


Figure 4-10. Scheduling, recurrent modification to CIR during business hours on first 5 days of month

Interaction with FAP

Service scheduling actually changes the base CIR and MIR (Kbps) of the GSP or SSPC for a temporary period. This method is different from FAP throttling, which is implemented as a percentage reduction in the base CIR and MIR, corresponding to the desired throttling. Modifications resulting from service scheduling can be applied at the same time as FAP throttling.



NOTE: See Fair Access Policy on page 325.

4.5.5 Outbound Service Plan Parameters

This section describes GSP and subscriber service plan (SSPP/SSPC) parameters specific to the DVB-S2 satellite outbound link.

4.5.5.1 Outbound (DVB-S2) QoS

The modulation and coding (MODCOD) at which a satellite terminal is operating in DVB-S2/ACM (adaptive coding and modulation) determines the amount of satellite bandwidth consumed to provide a given information rate.

The following terms are used when discussing MODCODs in the context of GQoS:

- ***Maximum MODCOD*** — The network's maximum MODCOD is the highest (or most efficient) MODCOD supported by the Velocity network. The Maximum MODCOD on the DVB-S2 outbound is 16APSK 8/9. Although the maximum MODCOD value is a global parameter that applies to all iNets, iNets can have different configured Maximum MODCODS.
- ***Minimum MODCOD*** — The network's minimum MODCOD is the lowest (least spectrally efficient) MODCOD supported by the Velocity network. The Minimum MODCOD on the DVB-S2 outbound is QPSK 1/4. Although the minimum MODCOD value is a global value that applies to all iNets, iNets can have different configured Minimum MODCODS.
- ***Base MODCOD*** — The lowest MODCOD at which an application's CIR/MIR is honored in DVB-S2/ACM. The satellite terminal can still operate in rain fade all the way down to the network's minimum MODCOD. See following detailed description.
- ***Nominal MODCOD (for GSPs)*** — The nominal MODCOD is the estimated average operating MODCOD for all satellite terminals in a group. Nominal MODCOD is only available in the downstream direction.
- ***Operating MODCOD*** — The operating MODCOD refers to the satellite terminal's real-time operating point. Under normal operation, a satellite terminal's operating MODCOD is expected to be above the base MODCOD.
- ***Minimum Efficiency*** — The ratio of IP bits per symbol — above which the GSP is expected to operate. If the current operating efficiency of the GSP falls below the configured efficiency, then the aggregate CIR/MIR of the GSP is scaled down on the downstream. This parameter applies only when operating in the IP-based mode. See IP Efficiency Operation on page 177.



NOTE: See the following sections for detailed descriptions of ***Base MODCOD***, ***Operating MODCOD***, and ***Nominal MODCOD***.

The **Nominal MODCOD** parameter applies only when operating in Symbol-based mode.

Scaled Information Rate

While service plans are configured with a CIR and an MIR in Mbps, the amount of satellite bandwidth (in Msps) consumed will depend on the operating MODCOD of the terminal. Delivering 1 Mbps to a terminal operating at a low and inefficient MODCOD, say QPSK 1/4, will require more symbols (measured in Msps) than delivering the same 1 Mbps to a terminal operating at the more efficient 16PSK 8/9 MODCOD.

To accommodate this requirement, the number of bytes being requested (demand) in the 100-ms allocation window is converted to a Scaled Information Rate (SIR) with a unit of scaled bytes. Scaled bytes take the MODCOD into account, so as to give a more true measure of the amount of satellite bandwidth (Msps) that is actually used over the air.

Nominal MODCOD

A GSP is an abstract service plan, not tied to the real-world performance of a single satellite terminal. When configuring a GSP, while operating in the Symbol-based mode, the service

provider must define a nominal MODCOD.

- ***Nominal MODCOD (for GSPs)*** — Nominal MODCOD is the estimated average operating MODCOD for all satellite terminals in a group. It is used by a service provider to estimate the capacity required by a group of satellite terminals in the equivalent of satellite bandwidth rather than information rate. The nominal MODCOD is a GSP parameter that enables the primary service provider to sell capacity based on a contractual MODCOD, and is only applicable when operating in Symbol-based mode.

  For example, if a service provider purchases 10 Mbps based on a nominal MODCOD of QPSK 7/8 and most satellite terminals are in a hurricane area operating between QPSK 1/2 and QPSK 1/4, a significant amount of additional satellite bandwidth would be required to achieve the 10 Mbps information rate. In a situation like this, the service plan does not obligate the primary service provider to meet the 10 Mbps because the nominal MODCOD specified in the GSP plan is QPSK 7/8. Note that when the average operating MODCOD for the group is above the nominal MODCOD, the group will get a higher CIR/MIR while the consumed bandwidth remains the same.



NOTE: Nominal MODCOD is only available in the downstream direction, and when the QoS Service Mode options is set for Symbol-based.

For additional details on how to determine the nominal MODCOD for a GSP, see Calculation of the Nominal MODCOD for an Outbound GSP on page 154.

Base MODCOD and Operating MODCOD (for SSPCs only)

An SSPC exists on a specific satellite terminal that will experience changing RF signal conditions over time. Adaptive coding and modulation (ACM) allows satellite terminals to operate at their optimal MODCOD, based on their real-time signal-to-noise ratio (SNR). The SNR is dependent on the satellite terminal antenna size, the carrier downlink EIRP, geographic location within the beam coverage, and rain fade conditions. Those real-time RF conditions will determine the operating MODCOD for traffic received by that satellite terminal. Operating MODCOD is not a configured parameter.

- ***Operating MODCOD*** — The operating MODCOD refers to the satellite terminal's real-time operating point. Under normal operation, a satellite terminal's operating MODCOD is expected to be above the base MODCOD.

The amount of bandwidth (in MHz or Msps) required to deliver the CIR configured for the SSPC will vary depending upon the signal conditions at the satellite terminal. Satellite terminals operating in clear-sky conditions will require fewer MHz to deliver the CIR in the service plan. However, if that satellite terminal experiences fade, ACM will use more robust and less efficient MODCODs to transmit data to the satellite terminal, requiring more MHz or Msps to deliver the same CIR Mbps required by the service plan.

The diagram illustrates that the bandwidth needed to deliver a fixed information rate increases significantly when terminals operate at lower SNR, and therefore, lower MODCODs.

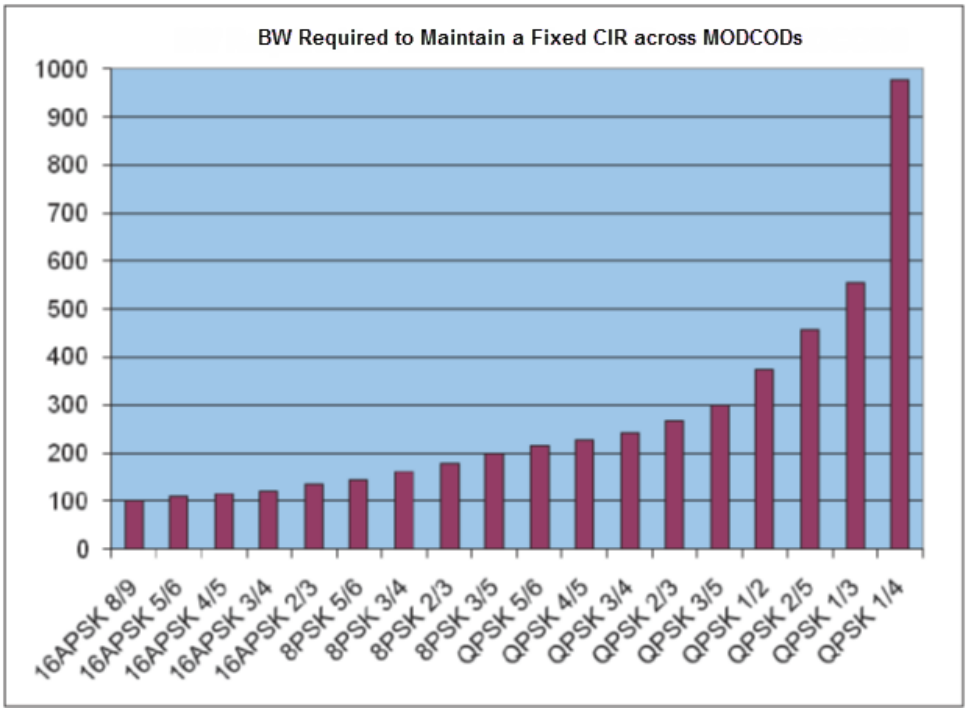


Figure 4-11. Bandwidth (in symbols, normalized) vs. Information Rate

To prevent the situation where a satellite terminal goes into deep fade, where the system would consume a large number of Msps in attempting to satisfy this satellite terminal's CIR, the concept of a base MODCOD is introduced.

- ***Base MODCOD*** — The base MODCOD is the lowest MODCOD at which an application's (SSPC) CIR/MIR is honored in DVB-S2/ACM. The satellite terminal can still operate in rain fade all the way down to the beam's minimum MODCOD. In a mobile environment, the base MODCOD is typically selected based on the expected operating point at the beam edge with reasonable rain fade. It allows service providers to specify a base MODCOD in the service plan and offer lower pricing, if desired, for customers using larger antennas or customers with no requirements for maintaining CIR/MIR in deep rain fade. When a satellite terminal is in deep rain fade operating below its base MODCOD, its CIR/MIR is scaled down.

On the ACM outbound, the system allows the user to configure a base MODCOD for each satellite terminal SSPC. The CIR/MIR configured for the SSPC is honored as long as the satellite terminal is operating above or at the base MODCOD. When the satellite terminal operates above its base MODCOD, the SSPC's CIR/MIR is not increased to a higher value. However, if the satellite terminal operates below the base MODCOD, the CIR/MIR is scaled down based on the bandwidth efficiency of the operating MODCOD. The allocated bandwidth remains fixed at the base MODCOD bandwidth, but the CIR/MIR is gradually degraded because the SSPC continues to use the same amount of satellite bandwidth that was allocated at the base MODCOD. The comparable concept on the inbound is that of inbound fairness, which limits CIR commitments to an SSPC if its satellite terminal is operating below a defined operating point. For information on inbound fairness, see Inbound Fairness Base Bitrate (for SSPCs only) on page 159.

The following diagram illustrates the scaling effects of CIR/MIR when a terminal operates below its base MODCOD.

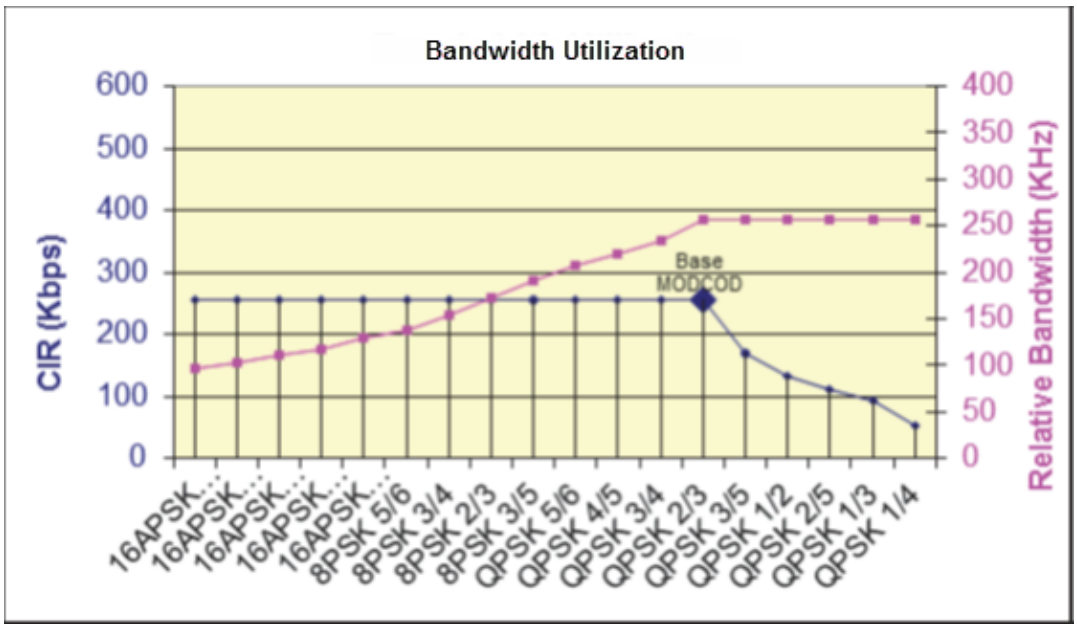


Figure 4-12. CIR Based on Base MODCOD and Bandwidth Utilization

4.5.5.2 Single Beam Restrictions and GSP MODCOD Scaling

Single beam restrictions and GSP MODCOD scaling are characteristics of GSPs that both give service providers a more granular control of the amount of bandwidth a GSP uses in each of the beams from which it is allowed to draw bandwidth.

Single Beam Restrictions

The single beam restrictions parameter defines the maximum CIR/MIR allowed for a GSP in a single beam. This restriction is used to avoid a situation where too many satellite terminals with high demand from one GSP are in a single beam. The system allows the configuration of a default maximum CIR/MIR for a GSP, which applies to all beams. The system allows exceptions for specific beams.

GSP MODCOD Scaling

As described in Outbound (DVB-S2) QoS on page 149, the GSP configuration includes a nominal MODCOD that is used to scale the CIR/MIR commitment in Mbps to scaled bytes. (For additional information on scaled bytes, see Scaled Information Rate on page 150.) However, not all beams offer the same signal performance, and for a set of beams with widely varying signal performance a single nominal MODCOD field is not appropriate.

CIR/MIR configurations in the outbound direction for GSPs, using GSP MODCOD scaling, allow service providers to offer the same quality of service across satellite beams with different capabilities, while managing the configuration for all of the beams in an efficient and consistent manner. GSP MODCOD scaling applies an appropriate scaling factor to each beam in order to normalize the performance between stronger and weaker beams. With this feature

the CIR/MIR for any given beam is scaled either up or down based on the ratio of the spectral efficiencies of the nominal MODCOD and the average MODCOD in the beam.

GSP MODCOD scaling is applicable only when the system is in the Symbol-based QoS mode.

Key characteristics of GSP MODCOD scaling include:

- Satellite terminals are supported uniformly when moving among non-identical beams.
- Scaling factors in outbound GSP geoscope are configured manually.
- Scaling factors for multiple outbound GSPs are configured using the GSP Service Plan profile.
- Multiple scaling profiles may be configured - any profile may be applied to a GSP geoscope.
- If a default beam or beam-specific MODCOD value is not configured, the value equals $MODCOD_{nom}$.

The scaling factor of the per-beam CIR and MIR, including default, is based on the ratio of the spectral efficiencies of the nominal and average MODCODs. MODCOD spectral efficiencies used in the discussion of this topic are based on the ***Velocity Link Budget Analysis Guide***.

When the average MODCOD for a beam is equal to the nominal MODCOD the beam efficiency is 1 and, therefore, scaling is not required. If a beam has an efficiency less than the nominal MODCOD, the ratio of the efficiencies is greater than 1. For MODCOD efficiency values greater than 1, the CIR/MIR cap on symbols/sec for the beam is increased by that factor. For example, for a scaling value of 3.0 the CIR/MIR cap for the beam is increased by a factor of 3. For MODCODs that result in scaling values less than 1, the CIR/MIR cap is reduced by that factor. For example, for a value of 0.5 the CIR/MIR cap is multiplied by a factor of 0.5.

Table 4-3. MODCOD Scaling Factor Decision Logic

Entity	Ratio of Efficiencies	CIR/MIR Cap	Comments
$MODCOD_{nom} = MODCOD_{avg}$	= 1	Not Modified	Scaling Value = 1 or Not required
$MODCOD_{nom} > MODCOD_{avg}$	> 1	Increased by factor	
$MODCOD_{nom} < MODCOD_{avg}$	< 1	Decreased by factor	

Calculation of the Nominal MODCOD for an Outbound GSP

Services represented by GSPs are specified using a CIR and an MIR in Mbps. However, the capacity of the DVB-S2 outbound carriers is measured in Msps and the GQE engine uses Msps to control demand and allocation in the outbound direction. Therefore, configuring an outbound GSP requires a nominal MODCOD in addition to the CIR/MIR in Mbps. The Velocity system can then translate the CIR/MIR in Mbps into the corresponding CIR/MIR in Msps.

The objective is to select a nominal MODCOD for the GSP that, on average, represents the expected efficiency of the bandwidth consumption in all of the GSP's children nodes that demand and consume bandwidth in the outbound direction.



NOTE: iDirect does not guarantee the following process is always the optimum method for determining nominal MODCOD values. Each satellite operator or service provider must evaluate the process and make adjustments or changes to fit individual use cases and service models.

Furthermore, it is likely that the nominal MODCOD value will be adjusted periodically as more historical usage data from specific satellite networks and customers is compiled.

Procedure

This procedure is provided to aid in planning services. It is based on certain initial assumptions and the accuracy and correctness of the outcome relies on the variance between the assumptions and the real performance of the satellite terminals (the SSPCs on the satellite terminals) and their efficiency when consuming bandwidth. This procedure also assumes the availability of high quality raw data about the performance of the different types of satellite terminals being used in all of the satellite beams.

This procedure calculates the nominal and average MODCOD of a GSP used for satellite terminals that have similar performance (typically being the same RF Terminal Type). It depends upon the availability of certain raw information regarding the performance of a satellite terminal of a specific type in each of the satellite beams included in the geoscope of the GSP. In addition, an MDA (multi-destination link budget analysis) for each of the satellite beams must have been performed. The result of this MDA is a two dimensional grid. Each point on the grid represents a geographic coordinate for which the link budget analysis (LBA) has been performed and the expected ***C/N*** value is known.

- **Nominal MODCOD Calculation** — The procedure requires an initial estimate of the distribution of the CIR in Mbps across all of the satellite beams in the GSP. The procedure also requires the calculation of the average MODCOD for each satellite beam. It assumes the average MODCODs have already been calculated and the CIR required by the customer is already known. The nominal MODCOD is calculated as illustrated in the diagram, for a GSP containing ten beams.

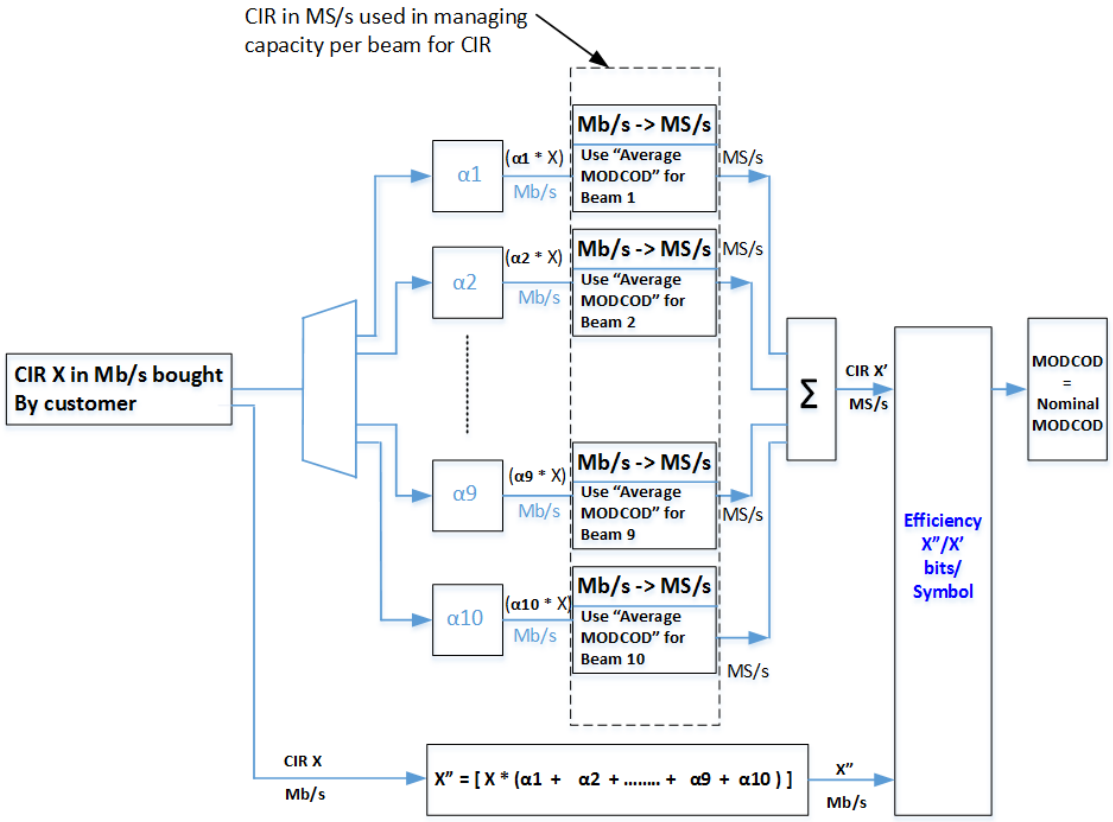


Figure 4-13. Example Procedure for Calculating Nominal MODCOD for each Beam

The initial estimate of the CIR in each beam is $\alpha_i X$. α_i is a decimal factor representing the percentage of the CIR that will be allocated to Beam i. These estimates can be changed as needed if the traffic density is known. A survey may be used to solicit information from which to create the initial estimates.

- **Average MODCOD Calculation for each Beam** — The procedure for calculating the average MODCOD for each beam is illustrated in the following diagram. The percentage availability to be used for the MDA depends upon the grade of service being offered to the service provider.

	NOTE: The average MODCOD calculation for each beam must be done before the calculation of the nominal MODCOD can be performed.

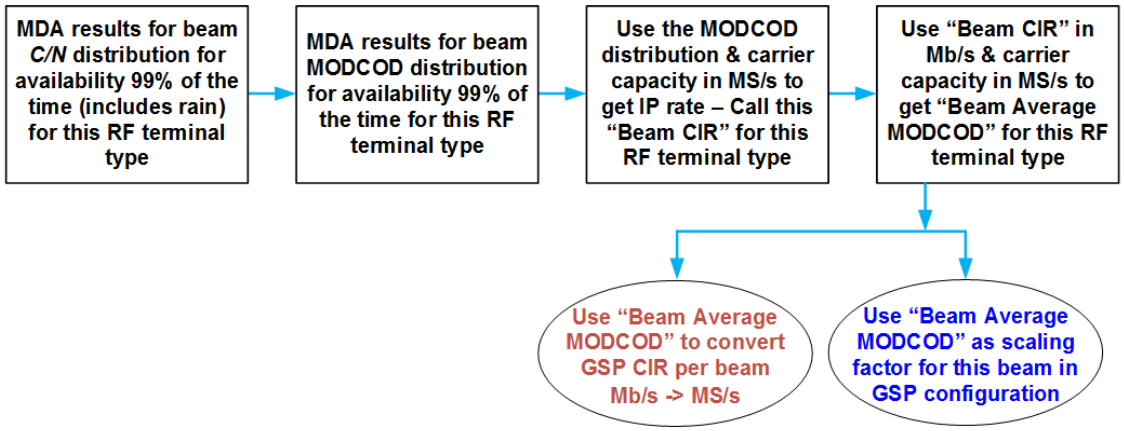


Figure 4-14. Procedure for Calculating Average MODCOD for each Beam

4.5.5.3 IP Minimum Efficiency

In the IP-based service mode, only IP bits and bytes are taken into consideration in the use and enforcement of the CIR/MIR in the downstream direction. The corresponding symbols (Msps) needed to fulfill the contracted CIR/MIR in Mbps, are not considered, and there is no cap on symbols used to satisfy a given CIR/MIR allocation. As such, without any measurement and control of the efficiency, terminals operating at low and inefficient MODCODs, exceed their fair share of the satellite bandwidth (Msps).

Configuring the IP efficiency allows Satellite Operators and Service Providers to specify, at the GSP level, a desired operating efficiency that would provide an efficiency lower limit. If the operating GSP efficiency (ratio of ***Allocated IP Bits*** to ***Distributed Bandwidth in Symbols***), reported in the last interval, is higher than the configured minimum efficiency, the GSP CIR/MIR is not scaled down or scaled up. Below this lower limit, the configured CIR/MIR on the GSP downstream is scaled down. With this down scale, the symbols used to satisfy a given CIR/MIR allocation is thereby capped, and available bandwidth is distributed fairly among GSP child nodes as long as they operate within prescribed guidelines.

Based on an assumed mix of terminal types, beam types, carrier usage, or number of terminals of each type, Satellite Operators and Service Providers can determine the efficiency (IP bits/Symbol) needed to fulfill certain operational objectives. This threshold efficiency would be based on an expected level of performance. Any operating efficiency that falls below the expected target causes the CIR/MIR to be scaled down by a scale factor equal to the measured efficiency (***M***) divided by the threshold efficiency (***E***) or ***M/E***.

f M < E then scale down the CIR/MIR as follows:

$$CIR_{scaled} = CIR_{current} * (M/E)$$

$$MIR_{scaled} = MIR_{current} * (M/E)$$

The IP-based minimum efficiency feature is characterized by the following:

- Only configurable when the QoS Service mode is set to IP-based.
- IP efficiency currently is only supported for unicast GSPs.
- Efficiency is configured only at the GSP level and in the outbound direction.

- The Network Operator determines the “minimum efficiency” value to set for a GSP.
- The valid efficiency range is 0-5, where a ‘0’ value results in no CIR/MIR scaling.
- The operator can choose to Show/Hide the efficiency control option to child GSPs.
- Ability to scale down GSP aggregate CIR/MIR if the reported operating efficiency falls below the configured minimum efficiency value (for downstream only).
- Ability to monitor each GSP efficiency stats in the NMS in bits/symbol or bits/Hz at the SSPC, iNet, Terminal, NOC and SAS levels, for downstream and upstream.

See IP Efficiency Operation on page 177.

4.5.6 Inbound Service Plan Parameters

This section describes GSP and subscriber service plan (SSPP/SSPC) parameters specific to the satellite A-TDMA inbound link.

4.5.6.1 Inbound (Adaptive TDMA) QoS

GBWM on the inbound behaves differently than on the outbound. On the outbound, there is a single DVB-S2 channel for each iNet that is shared by all satellite terminals. The MODCOD can be changed frame-by-frame. At any point in time, the aggregate bandwidth (in Mbps) delivered by the entire outbound will vary depending on the operating MODCOD at each satellite terminal.

The upstream is governed by the Adaptive TDMA (A-TDMA) algorithm. The set of inroute channels is the ***inroute group***, and the set of fixed MODCODs configured for those inroute channels is the ***inroute group composition (IGC)***. The adaptive algorithm dynamically assigns a ***nominal carrier*** to each satellite terminal based on its current C/N_0, which is the carrier with the highest C/N_0 threshold at which the satellite terminal can be assigned bursts. The satellite terminal may be assigned bursts on its nominal carrier, or any other carrier with a lower C/N_0 threshold.

For details on the Adaptive TDMA algorithm, see Differentiating C/N0 and C/N on page 103. What is important to note is that the aggregate throughput (in Mbps) delivered by a set of inbound carriers only changes when the inroute group composition (IGC) is changed.

The aggregate throughput (Mbps) of the outbound will vary according to the signal conditions at each of the individual satellite terminals it serves. However, on the inbound signal, conditions at each of the individual satellite terminals drives the selection of one of the IGC. Each IGC has a fixed throughput (Mbps) corresponding to the bandwidth, MODCOD, and length of each TDMA slot.

The satellite terminal in the inbound direction has a nominal carrier at any instance of time. The nominal carrier, one of the carriers in the currently-used IGC, is the most efficient carrier the satellite terminal can use in any given location based on a number of constraints, including the C/N_0 level received at the SAS from the satellite terminal. It has a peak IP rate that will provide an upper bound on the MIR that SSPCs on that satellite terminal can transmit at any given time. The satellite terminal is deemed to be in a fade condition if it is transmitting on a carrier less efficient than its configured nominal carrier.

4.5.6.2 Nominal MODCOD (for GSPs only)

There is no 'Nominal MODCOD defined for inbound GSPs.

4.5.6.3 Inbound Fairness Base Bitrate (for SSPCs only)

For each SSPC the service provider can decide whether or not to scale the CIR/MIR values in fade conditions (as determined by whether the satellite terminal is operating below its nominal carrier). If that scale value is set to zero, no scaling is done. If it is set to another value, that value is compared to the peak IP rate for the current nominal carrier. If the configured value is greater than the peak rate, a scaling factor equal to the ratio of the configured rate to the peak rate is applied to both the configured CIR and MIR values.

For example, if the configured rate, known as base-bit-rate, is 500 Kb/s and the peak IP rate is 300 Kb/s, then, the CIR/MIR values are scaled down by a factor of 0.6 (300/500). If the peak rate was 600 Kb/s, then no scaling down is applied, as 600/500 is greater than one. Scaling is done only at the SSPC level: no other scaling is done in the higher GQE tree nodes.

The modulation and coding of an A-TDMA carrier used by the satellite terminal determines the amount of satellite bandwidth consumed to provide a given information rate. For additional details, refer to Inbound (Adaptive TDMA) QoS on page 158.

Each subscriber service plan component (SSPC) may be configured with a ***base bitrate***. An SSPC operating on a nominal carrier with a peak data rate (in Kbps) lower than the configured base bitrate is deemed to be in fade. When in fade, the current CIR and MIR are scaled down as follows:

If (Peak Data Rate$_{\text{Nominal Carrier}}$ < base_bitrate)

$CIR_{\text{scaled-by-fade}} = CIR_{\text{current}} * (\text{Peak Data Rate}_{\text{Nominal Carrier}} / \text{base_bitrate})$

$MIR_{\text{scaled-by-fade}} = MIR_{\text{current}} * (\text{Peak Data Rate}_{\text{Nominal Carrier}} / \text{base_bitrate})$

For additional details on A-TDMA, refer to Chapter 3 , Satellite Link Operations.

4.5.7 NMS Service Plans

Velocity requires that a portion of bandwidth on each beam be allocated to network management traffic. Each terminal must thus have a subscriber service plan for administrative traffic. Typically, the NMS GSP is created at the top level of the GBWM tree and it must have the highest priority.

The NMS GSP and SSPP/SSPCs provide bandwidth for unicast management traffic such as options update, software upgrades, and control messages. If multicast user services are used, an NMS M-GSP and M-SSPP/SSPCs are required for Internet Group Management Protocol (IGMP) control traffic. Note that acquisition and broadcast traffic are accounted for separately in iNet overhead.

4.6 GBWM Operations

The Velocity GBWM subsystem allocates available satellite bandwidth according to the GQE tree, which reflects the configuration of the GSPs and SSPCs in the GBWM tree, and the requested bandwidth by terminal SSPCs. GBWM uses a demand and allocation algorithm, whereby demand originates from SSPCs on individual satellite terminals and flows up the GQE tree to the satellite network level, and capacity is allocated down the tree. The GBWM subsystem consists of:

- A global QoS Enforcer, running on the GBWM cluster in the NOC
- A SAS QoS Enforcer for each satellite, running on the PP cluster in the SAS
- An iNet QoS Enforcer for each iNet on the satellite, also running on the SAS PP cluster

An SSPC can only issue a demand and receive a bandwidth allocation within one iNet at a time. Thus, the demand/allocation of an SSPC is always handled by the local (SAS) GBWM software. The SAS GBWM software also handles demand and allocation for GSPs that have a geoscope consisting only of beams located on the satellite serviced by the SAS.

The geoscope of a GSP, however, may span multiple beams across multiple satellites, and hence across different SASs. Individual SAS QoS Enforcers report the demand/allocation from local GSPs to the central GQE. See Global Bandwidth Allocation on page 165.

4.6.1 Global QoS Enforcement (GQE) Tree

All beams on all satellites defined in the global network contribute bandwidth to a single pool that is managed by GBWM. The GBWM tree (see GBWM Tree Structure on page 142) defines the operation of the GBWM system, and is instantiated at run-time as the Global QoS Enforcement (GQE) tree. This tree actually consists of two completely separate tree structures, one for managing outbound traffic bandwidth demand and allocation and another for inbound traffic.

The GQE tree is a hierarchical construct of which containership and inheritance rules allow the iterative application of basic allocation methods across groups and child groups. QoS properties configured at each level of the GBWM tree determine how bandwidth is distributed in cases of normal demand (no over-subscription), as well as when demand exceeds availability (over-subscription).

Each QoS enforcer of the GBWM subsystem creates a QoS Enforcement Tree that reflects its scope. For example, the iNet QOS enforcer creates a tree in which the top node is a DVB-S2 outbound carrier (or the associated TDMA inbound carriers). At the bottom are the SSPCs. Between the top node and the SSPCs, there can be multiple GSP nodes that subdivide the bandwidth as described in GBWM Tree Structure on page 142.

The following diagram depicts an example outbound QoS Enforcement tree and the scope of each GSP and SSPC.

- The scope of the root of the tree is ten beams on two satellites.
- The scope of GSP 1 is the rightmost three beams. Each GSP under GSP-1 has a scope that is one or more beams within GSP-1's scope.

- The scope of GSP 2 is the leftmost four beams.
- The scope of GSP 3 is all beams in the satellite network.
- The geoscope of each SSPC is the same as its parent GSP.
- The current location of three terminals are shown, along with their SSPCs. All are within the scope of each of their SSPCs.

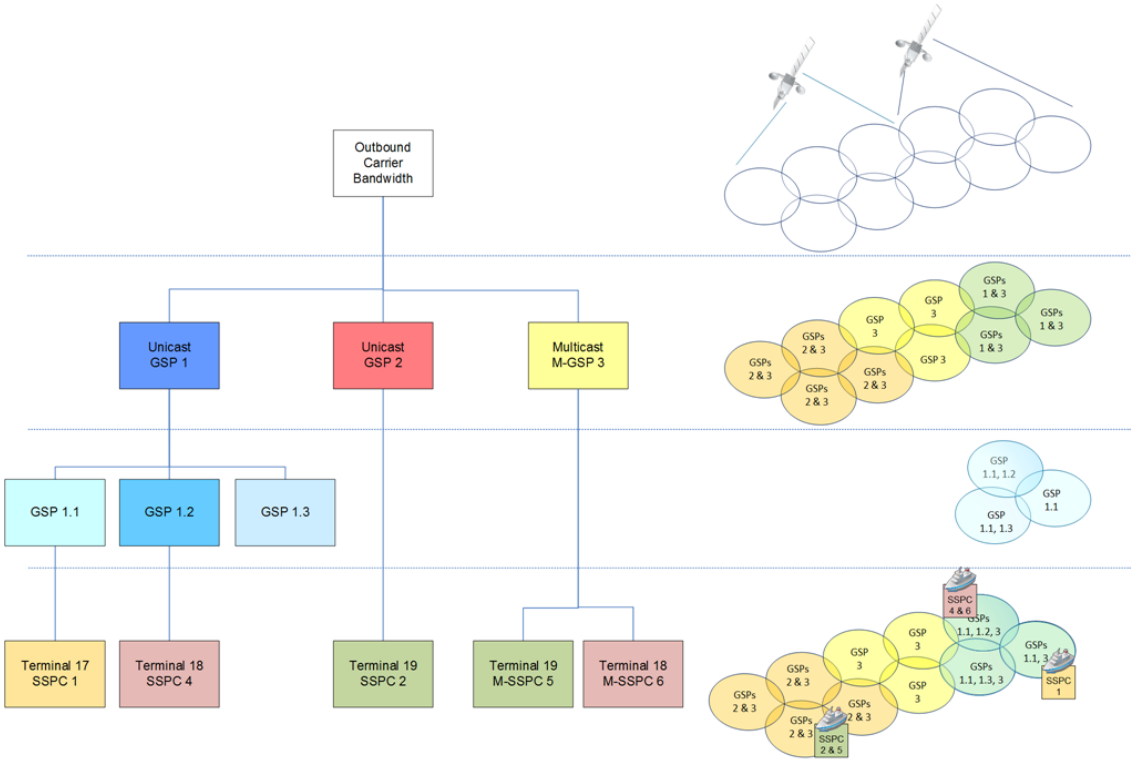


Figure 4-15. Outbound iNet QoS Enforcement Tree

An example of a multicast portion of a GQE tree is shown in the following diagram.

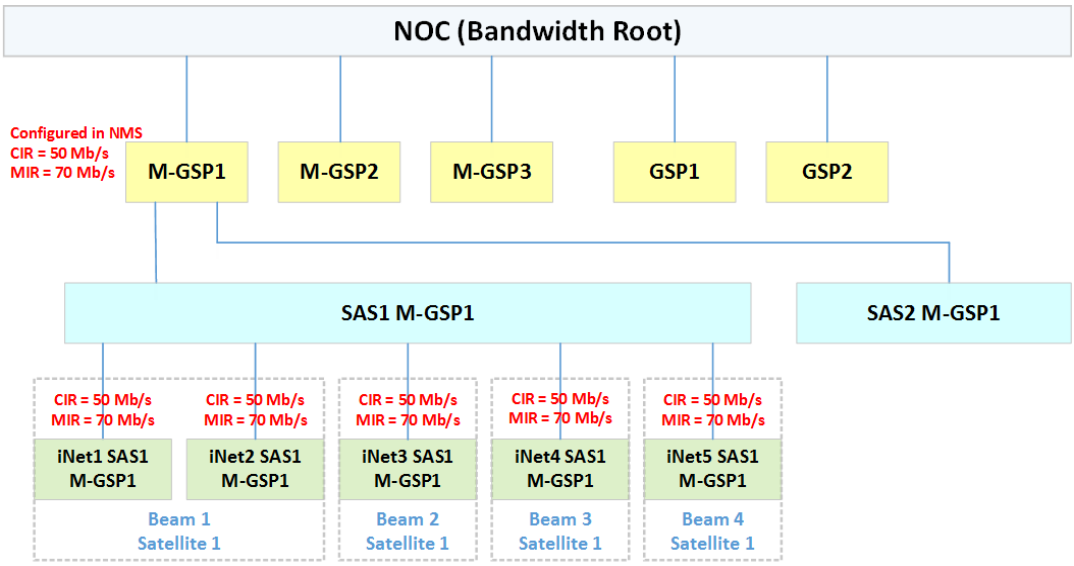


Figure 4-16. Multicast GSP Tree

At a fixed interval, demand is reported from the service plans on satellite terminals (SSPCs) and aggregated up to higher-level service plans. The fixed interval is 100 ms for the outbound, and 125 ms for the inbound. Bandwidth allocation decisions are made according to the service plans' configurations. On the outbound, the units of demand and allocation are scaled bytes (see Scaled Information Rate on page 150). On the inbound, the units of demand and allocation are TDMA slots.

4.6.2 Bandwidth Allocation Within an iNet

An iNet QoS Enforcer allocates bandwidth on a per iNet basis. It builds its iNet GQE tree from the GBWM tree. For GSP nodes, some configuration parameters are static and inherited from the configured GSP, such as priority, cost, and fairness options. However, the value of the CIR/MIR for the GSP in the iNet are dynamic and are controlled by the SAS QoS Enforcer. For additional bandwidth allocation information, see Global Bandwidth Allocation on page 165.

The CIR/MIR parameters of each node are adjusted dynamically by the GQE, but bandwidth is allocated to SSPCs on satellite terminals by the iNet QoS engine. During bandwidth allocation to parent nodes, only the configuration properties of parent nodes are considered, while the demand is derived from children nodes.

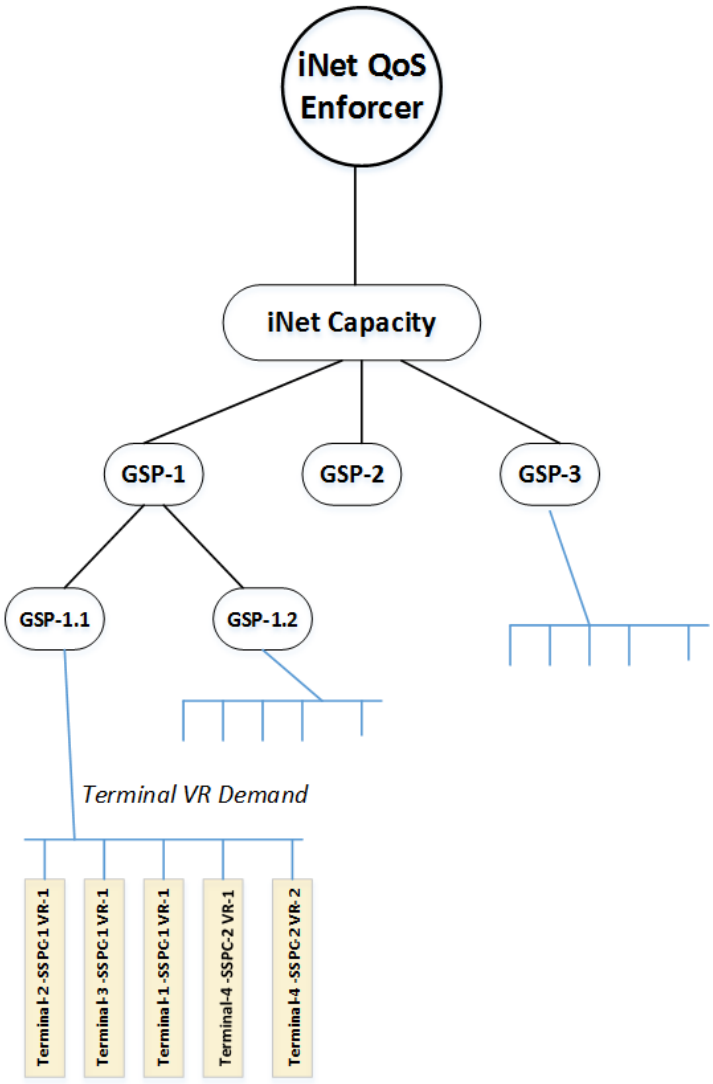


Figure 4-17. iNet QoS Enforcer Tree

The SSPC is the source of demand for the Global QoS enforcement engine. The GQE structure is a tree hierarchy, in which every node's demand is the aggregated demand of all children nodes. On the downstream (outbound), the demand is generated every 100 milliseconds. On the upstream (inbound), it is generated every 125 milliseconds.

The iNet QoS algorithm works by calculating demand, bottom up, for every node of the GQE tree. The nodes at the bottom, which are SSPCs, gather demand from service levels in the VRs. These demands are grouped according to the configured priority for the SSPCs, with higher priority groups taking precedence over lower priority groups.

Satellite bandwidth is then allocated top-down for every node of the GQE tree. The priority allocation is based on two cycles — CIR round and best-effort round. The engine offers two types of priorities — absolute and per-allocation-round. For absolute priority, nodes are grouped based on their configured priority, and the round of allocations applies to the higher priority node before moving to the next priority level. For per-allocation-round priority, nodes

are grouped based on their configured priority. However, the priority applies to the round of allocation, serving CIR by priority, then moves on to the best-effort round of allocation.

The following concepts provide more detail of the iNet QoS demand and allocation algorithm:

- **Outbound Demand** — On the DVB-S2 ACM outbound, the service level demand is generated in bytes for the next 100 ms. The demand reported by the SSPC is the maximum of rate-demand and depth-demand. The rate-demand is an estimation of how many bytes are classified in the service level queue for the next 100 ms interval. The depth-demand measures the number of bytes already in the service level queue.
- **Inbound Demand** — Each SSPC generates upstream demand based on the upstream IP packets received by the satellite terminal that match the SSPC's service levels. The demand is sent to the SAS every 125 ms to request allocation of slots in the next TDMA frame. All nodes report demand up the tree in slots. The demand is converted into the number of slots by dividing the demand, in bytes, by the configured size of the time-slot. Each node in the tree aggregates demand from its child nodes and sends it up the tree in slots.
- **Outbound Allocation** — On the ACM outbound, all node allocations are based on scaled information rate. For SSPCs, the scaled information rate is converted based on the scaling factor of the current operating MODCOD.
- **Inbound Allocation** — On the upstream, allocation is based on slots. The terminal is allocated slots, which are filled by packets waiting in the service level queues.

4.6.3 Global Bandwidth Allocation

The Global QoS Enforcer (GQE) is the portion of the GBWM system that maintains the GQE tree based on the GSPs created by service providers. Each GSP node in an iNet is dynamically assigned a CIR/MIR by the GQE, based on changes in the aggregate traffic demand from satellite terminals for that GSP node.

The GQE is distributed between the NOC and all SASs. The NOC GQE adjusts the CIR/MIR of GSP nodes in the SAS GQE tree, based on the aggregate demand from each SAS. The SAS QoS Enforcer, in turn, performs CIR/MIR adjustments for each GSP node in all of its iNets, based on the aggregate demands from each iNet.

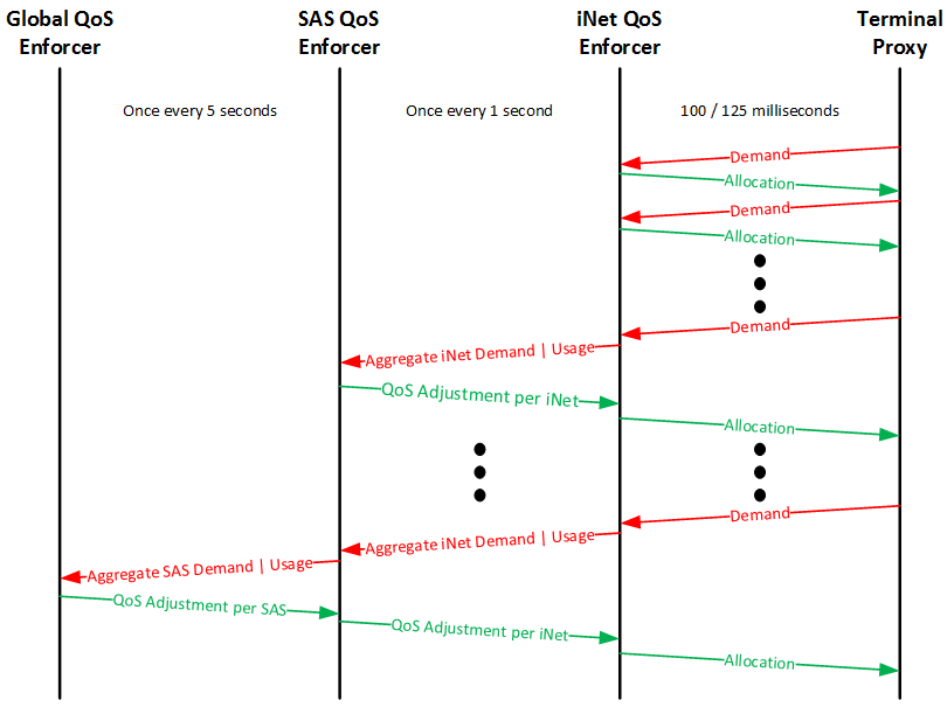


Figure 4-18. Global QoS Enforcer

The GQE dynamically adjusts CIR and MIR at the iNet level based on the global bandwidth hierarchy built from GSPs, aggregate demand, load balancing considerations, and satellite terminal distribution across the system. The CIR/MIR for the service provider is calculated dynamically by the GQE, based on traffic demand and usage from each iNet. Traffic bandwidth management is performed by the Global QoS Enforcer by maintaining a dedicated bandwidth partition node at each iNet QoS enforcement tree.

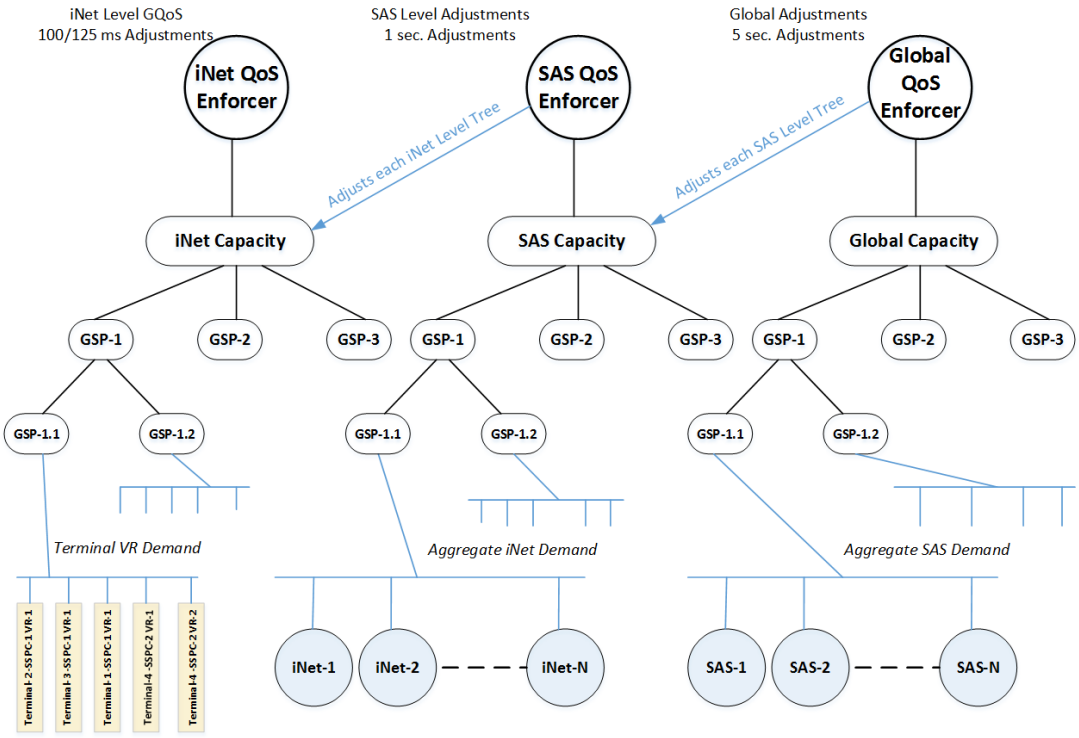


Figure 4-19. Dynamic Bandwidth Management

In case of a failure of the GQE, or when communication to it is lost, the SAS bandwidth manager sends bandwidth demands to the GQE in the backup NOC. Data is continually replicated between the primary and backup GQEs, making the backup GQE a hot standby.

If there is no backup GQE, or if there is a loss of communication between the SAS and both NOCs, the SAS continues to distribute its dynamic CIR/MIR to its iNets based on its last allocation before loss of communication with the NOC. Additionally, all satellite terminal bandwidth management functionality, including service scheduling and FAP enforcement, will also run at the SAS during the NOC outage. The GQE assumes the SAS is operating autonomously with the last allocated dynamic CIR/MIR and continues the dynamic adjustments to the other SASs.

In case of a SAS failover, as soon as the switchover is completed and the GQE starts receiving data from the backup SAS, which has now become the active SAS, global enforcement resumes at this new active SAS. If the newly active SAS is a partial SAS (i.e. not populated with line cards sufficient to provide coverage to all channels), the shift in the allocation to the SAS will happen automatically as demand from the partial SAS is reduced.

4.6.3.1 Bandwidth Allocation for Management Traffic

The diagram below shows how each iNet QoS enforcement tree dedicates a separate partition for management traffic (NMS GSP), with the highest priority at the top level. The CIR/MIR parameters for each NMS are a percentage of the iNet capacity. The CIR/MIR default is 5%.

At the satellite terminal level, each satellite terminal has an NMS SSPC that is responsible for management traffic. The NMS GSP node at each iNet collects the demand from all satellite terminals, and allocates bandwidth to all satellite terminals in its range.

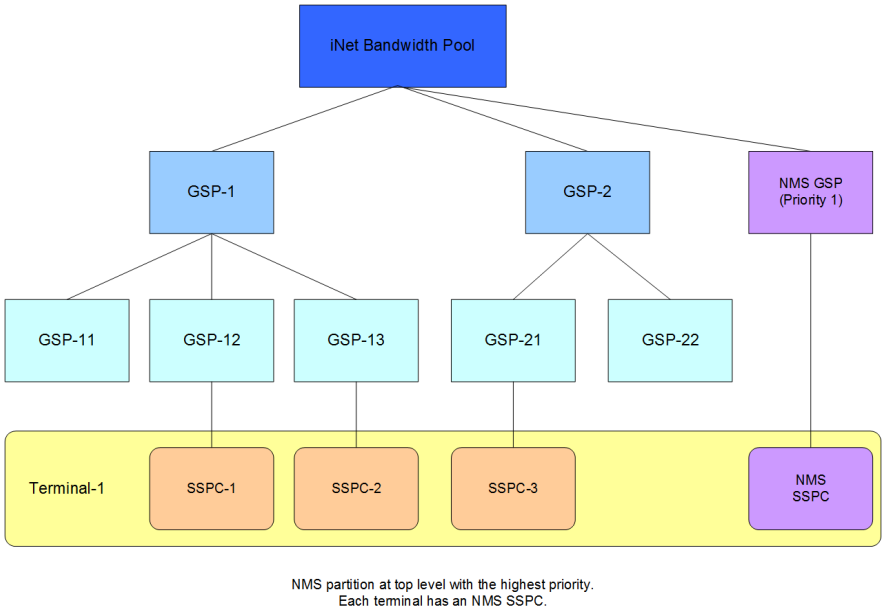


Figure 4-20. Management Traffic Partition Node in iNet QoS Enforcement Tree

4.6.3.2 GQE Enhancements

4.6.3.2.1 Beam Limits

In previous releases, the NOC GQE process had no knowledge about beam limits applied to the iNets by SAS GQE process. In cases where the GSPs were not congested, the NOC GQE over allocated bandwidth to SASs from the left over capacity. This bandwidth was wasted at SAS GQE process while distributing the bandwidth to the iNets, where beam limits were enforced.

Now, the NOC GQE process distributing "excess" bandwidth to SAS or not is controlled by the GSP Mode setting in GSP configuration. The "excess" bandwidth is distributed to SAS only when the GSP mode is set to "Service Provider Legacy" and restricted in other 3 GSP modes.

4.6.3.2.2 GQoS Bandwidth Allocation Algorithm

The GQoS bandwidth allocation algorithm was modified to use the excess iNet capacity in the downstream direction to satisfy some of the unmet demand for GSPs that are impacted by internal allocation round-trip delay. It should be noted that in the upstream direction, the free slot allocation provides some relief from the impact of delayed allocation.

4.6.4 Allocation Fairness

The GBWM system provides a number of allocation fairness algorithms that allow a service provider to control how bandwidth is allocated by a GSP when the demand from its child nodes exceeds the allocation the GSP has received — i.e. there is bandwidth contention. These options support various business models and may be combined, as described in Allocation Fairness Algorithms on page 134.

> ***NOTE:*** Allocation fairness only applies when there is contention; when there is no contention, all nodes below the GSP receive their demand, subject to their configured CIR and MIR. Allocation fairness happens at the iNet level. It is done using the dynamic CIR/MIR assigned to the GSP nodes by the GQE process.

The allocation fairness algorithms supported by Velocity are:

- **Allocation Fairness relative to Bandwidth (default)** — divides up available allocated satellite bandwidth relative to symbols.
- **Allocation Fairness relative to CIR** — allocates available satellite bandwidth proportional to the current CIR value.
- **Allocation Fairness relative to CIR and MIR** — allocates available satellite bandwidth proportional to the current CIR value in CIR round and current MIR in the MIR round.
- **Allocation Fairness relative to base MODCOD** — allocates available satellite bandwidth relative to the configured base MODCOD.
- **Allocation Fairness relative to operating MODCOD** — allocates available satellite bandwidth relative to the current operating MODCOD of the satellite terminal on which the SSPC resides.

4.6.4.1 Allocation Fairness Relative to Bandwidth (default)

The default allocation fairness method is ***allocation fairness relative to bandwidth***. Fairness according to bandwidth simply divides up available allocated satellite bandwidth relative to symbols. It may be used in both the inbound and outbound directions.

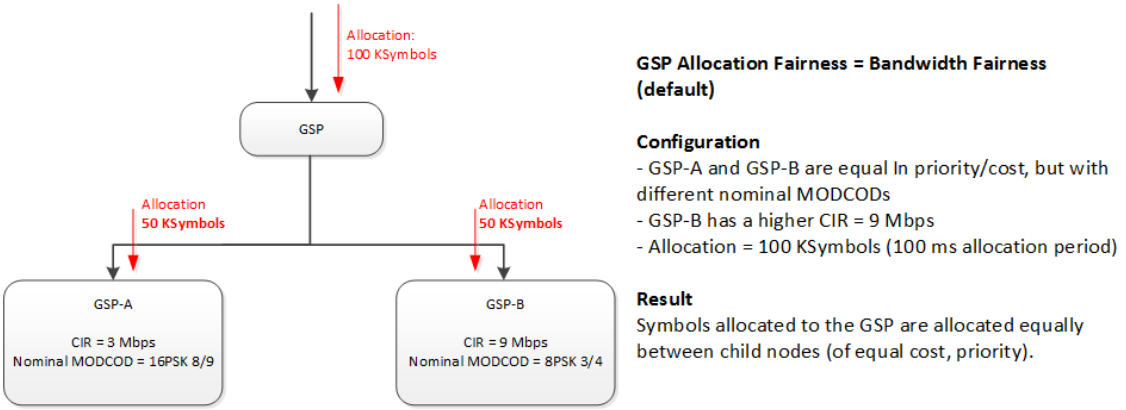


Figure 4-21. Example of Allocation Fairness Relative to Bandwidth

The previous illustration shows a simple case, with two child GSPs with equal priority and cost, but different CIR and nominal MODCOD. That GSP-B has been configured with a higher CIR and lower nominal MODCOD does not matter; any allocation available to the parent GSP is simply divided equally among the child GSPs.

The same equal distribution of bandwidth occurs if the child nodes are SSPCs.

> ***NOTE:*** Note that if there is a difference in MODCOD, fairness relative to bandwidth results in the node with the less efficient MODCOD delivering lower Kbps for this allocation period.

Consider the following example in which two SSPCs are operating at MODCODs of 8PSK3/4 and QPSK3/4, respectively. The spectral efficiency for 8PSK3/4 is approximately 2.05 and QPSK3/4 is approximately 1.37 after accounting for BBFRAME overhead.

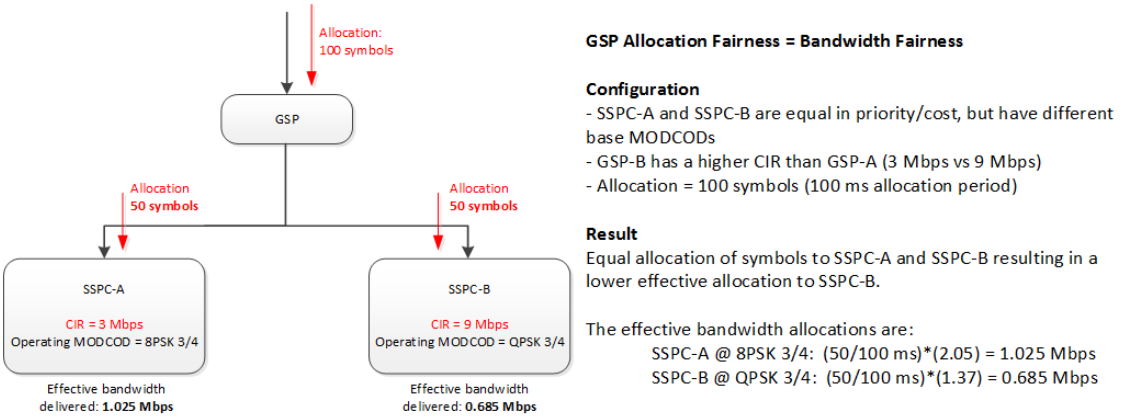


Figure 4-22. Example of Allocation Fairness Relative to Bandwidth with child SSPCs

> ***NOTE:*** Fairness relative to bandwidth is always applied, but does not preclude the additional application of other fairness methods described below.

4.6.4.2 Allocation Fairness Relative to CIR

Allocation fairness relative to CIR allows the CIR/MIR allocation to be proportional to the current CIR during periods of contention. The system computes an internal cost parameter that is used by the demand allocation algorithm to implement the proportional allocation behavior. The GSP cost parameter is still available to the user. It may be used in both the inbound and outbound directions.

For example, consider an outbound of 8 Mbps partitioned into two GSPs, for two different service providers, GSP-A with a 3 Mbps CIR and GSP-B with a 9 Mbps CIR. Assuming consistent MODCODs, allocation fairness relative to CIR, and bandwidth demand from both service providers greater than CIR, during contention the system will automatically calculate a cost parameter for each service provider proportional to the CIRs, resulting in an allocation of (8*3/(3+9)) = 2 Mbps for GSP-A and (8*9/(3+9)) = 6 Mbps for GSP-B.

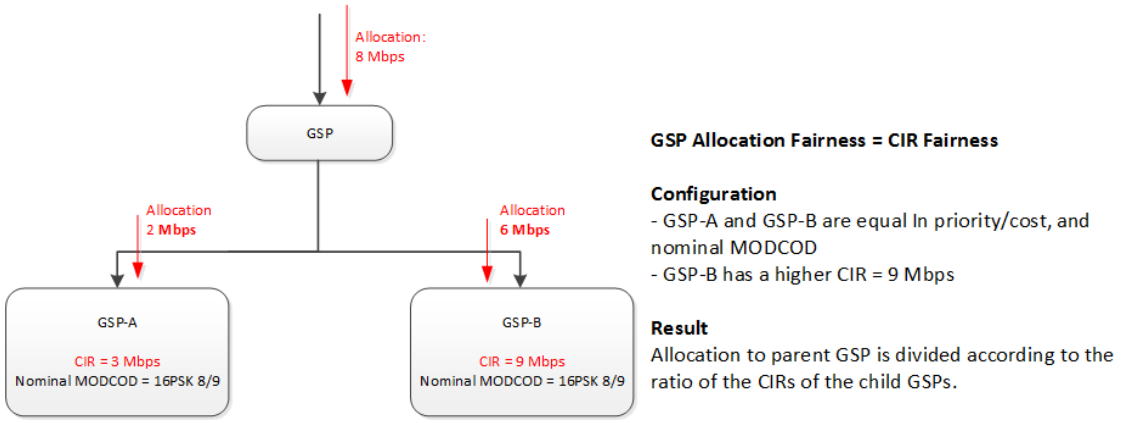


Figure 4-23. Example of Allocation Fairness Relative to CIR

> ***NOTE:*** Note that when this option is used, the best-effort round used to allocate bandwidth in excess of the CIR and up to the MIR will also be proportional to the current CIR.

4.6.4.3 Allocation Fairness Relative to CIR and MIR

Allocation fairness relative to CIR and MIR allows CIR/MIR allocation to be proportional to the current CIR value in CIR round and current MIR in the MIR round during periods of contention. The system computes an internal cost parameter that is used by the demand allocation algorithm to implement the proportional allocation behavior. The GSP cost parameter is still available to the user. It may be used in both the inbound and outbound directions.

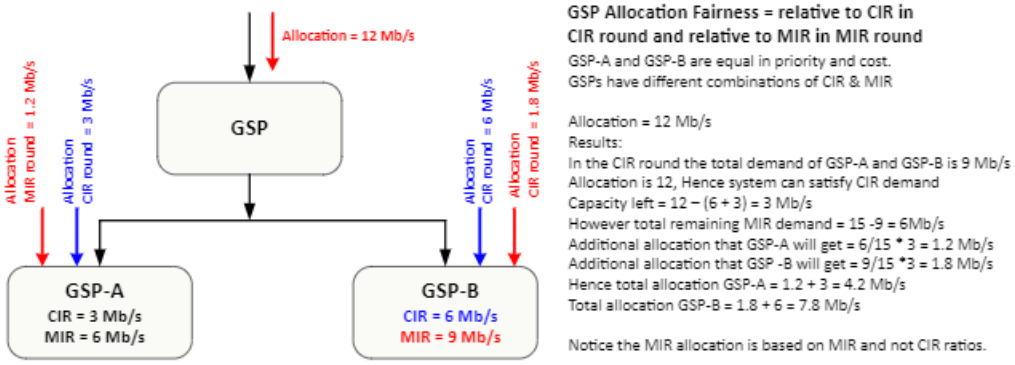


Figure 4-24. Example of Allocation Fairness Relative to CIR and MIR

4.6.4.4 Allocation Fairness Relative to MODCOD

The lowest level GSP, which serves SSPCs, may be configured to apply allocation fairness relative to base MODCOD or operating MODCOD. These options allow for a choice of fairness in allocation during contention, based on either the configured base MODCOD of child SSPCs or the operating MODCOD of the satellite terminal on which those SSPCs reside. When one of

these options is selected, the system calculates an internal cost parameter to be used by the demand allocation algorithm to implement the MODCOD fairness behavior.

The two schemes for allocation fairness relative to MODCOD are only be used in the outbound direction and for GSPs at the lowest level of the GBWM/GQE tree — for example, GSPs with SSPC child nodes.

Allocation Fairness Relative to Base MODCOD

Allocation fairness relative to base MODCOD applies an internal cost parameter to each SSPC based on the configured base MODCOD. This option is suitable for a satellite network with fixed satellite terminals where the base MODCOD is typically used as the operating point for satellite terminals with additional rain fade margin.

For example, consider two SSPCs that belong to a single GSP, where SSPC-A on Terminal A is configured with 3 Mbps CIR and has a base MODCOD of 8PSK 3/4, and SSPC-B on Terminal B is also configured with 3 Mbps CIR, but has a different base MODCOD of QPSK 3/4. (Recall that the spectral efficiency for 8PSK3/4 is approximately 2.05 and QPSK3/4 is approximately 1.37 after accounting for BBFRAME overhead.)

During contention, using the default scheme (allocation fairness relative to bandwidth), the allocation of symbols will be split equally between the two satellite terminals as described in Allocation Fairness Relative to Bandwidth (default). However, Terminal A receives 1025 Kbps while Terminal B only gets 685 Kbps due to the operating MODCOD.

However, if the allocation fairness relative to base MODCOD option is selected, allocation is based on bits/sec rather than satellite bandwidth, so both Terminals A and B will get approximately 820 Kbps.

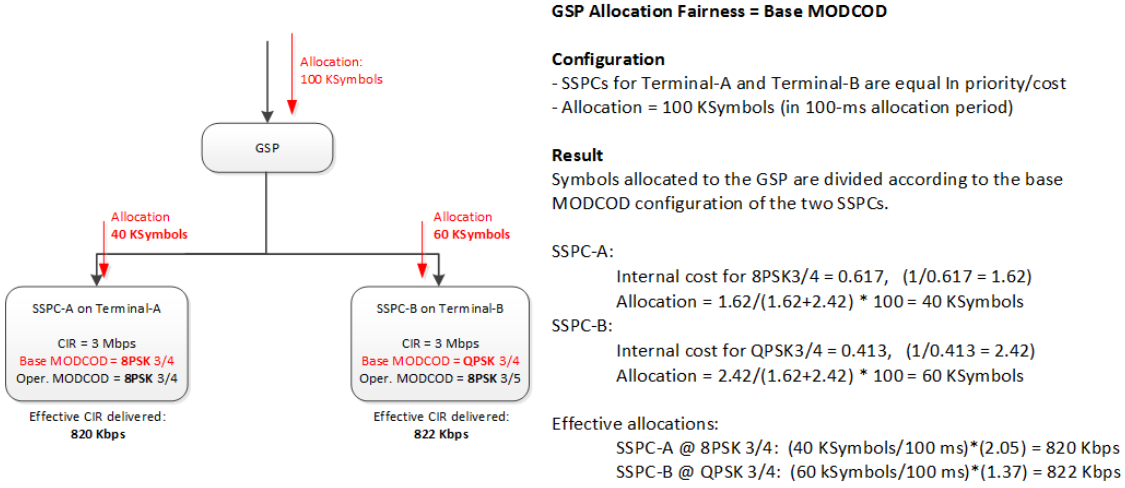


Figure 4-25. Example of Allocation Fairness Relative to Base MODCOD

> ***NOTE:*** The service provider pricing model will likely reflect the higher satellite bandwidth consumption of Terminal B relative to Terminal A.

Allocation Fairness Relative to Operating MODCOD

Allocation fairness relative to operating MODCOD applies an internal cost parameter to each SSPC based on the current operating MODCOD of the satellite terminal on which the SSPC resides. This option is best used in a mobile environment, where the base MODCOD for the satellite terminal may be the same but the operating MODCODs are different depending upon the location of the satellite terminal relative to the beam.

The system provides a mode of operation that supports bits/sec allocation in DVB-S2/ACM, which is based on the satellite terminal's operating MODCOD. This option comes into play under contention for the CIR and allocation of bursting traffic above the CIR up to the MIR. It allows all terminals in the group to be treated equally regardless of their operating MODCOD, permitting fair bits/sec allocation, as long as they are operating above their base MODCOD.

For this example, two satellite terminals with the same antenna size, same base MODCOD of QPSK 3/4, and same CIR of 1M each have an SSPC in the same GSP. One satellite terminal is in the beam center while the other is at the beam edge. Because of the difference in the SNR between the beam center and the beam edge, the two satellite terminals are operating at different MODCODs.

Terminal A's operating MODCOD is 8PSK 3/4, while Terminal B's operating MODCOD is 8PSK 3/5. (The spectral efficiency for 8PSK3/4 is approximately 2.05 and 8PSK3/5 is approximately 1.67 after accounting for BBFRAME overhead.) Given the same information rate, the satellite terminal with the less efficient MODCOD will consume more bandwidth than the satellite terminal at the beam center.

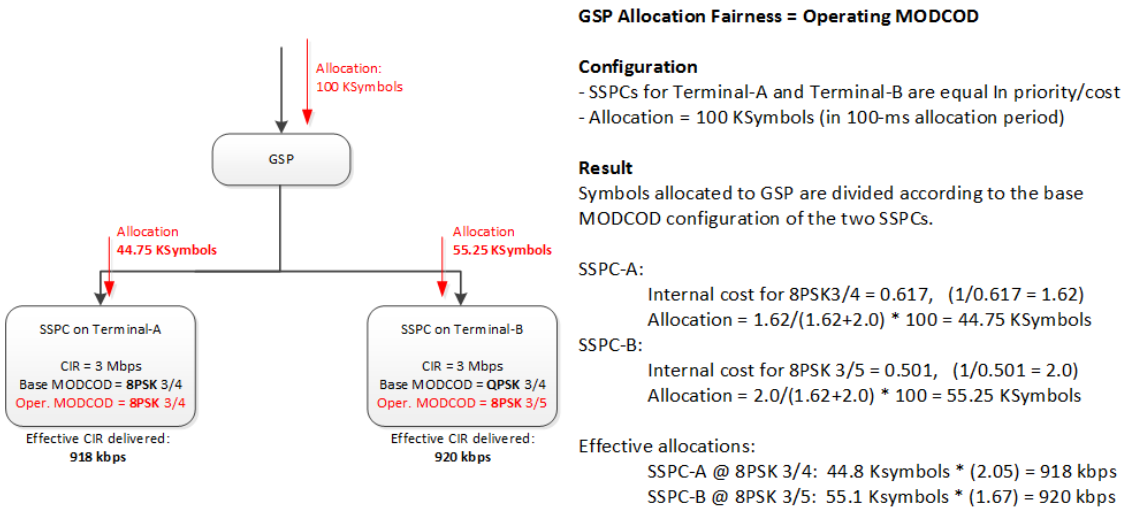


Figure 4-26. Example of Allocation Fairness Relative to Operating MODCOD

The allocation fairness relative to operating MODCOD option thus allows both satellite terminals to get equal bits/sec allocation under contention as long as they are both operating above their base MODCOD.

4.6.5 Packet Classification

Each packet entering the system, whether from the hub-side or terminal-side, must be classified and assigned into one of the configured service levels (SL) that are part of the Terminal Service Plan. A service level may represent a single application, such as VoIP traffic from a single IP address; or a broad class of applications, such as all TCP based applications.

Each service level, based on the specific application, has one or more user configured classification rules that are effectively packet-matching rules. If a presented packet matches a rule, then it is assigned to that service level. Classification rules for the SL are based on the logical combination of comparisons made between a presented packet and the following header parameters defined in the rule:

- Source IP Address
- Destination IP Address
- Source Port
- Destination Port
- VLAN ID
- Protocol
- Differentiated Services Code Point (DSCP)
- Type of Service (TOS)

4.6.6 Packet Scheduling

Packet scheduling determines the order in which packets are transmitted according to priority and classification. When a satellite terminal has sufficient bandwidth for all of its applications, packets are transmitted in the order they are received without significant delay. However, when a terminal does not have sufficient bandwidth to transmit all queued packets the terminal scheduling algorithm determines the next packet that needs to be transmitted from the set of queued packets across a number of service levels.

For each service level defined, one of the following queue types is used to determine how packets are selected for transmission: Priority Queue, Class-Based Weighted Fair Queue (CBWFQ), and Best-Effort Queue. Priority Queues are emptied before CBWFQs are serviced; CBWFQs, in turn, are emptied before Best-Effort queues are serviced.

The packet scheduling algorithm first services packets from Priority Queues in order of priority, P1 being the highest priority. The following diagram presents an overview of the iDirect packet scheduling algorithm.

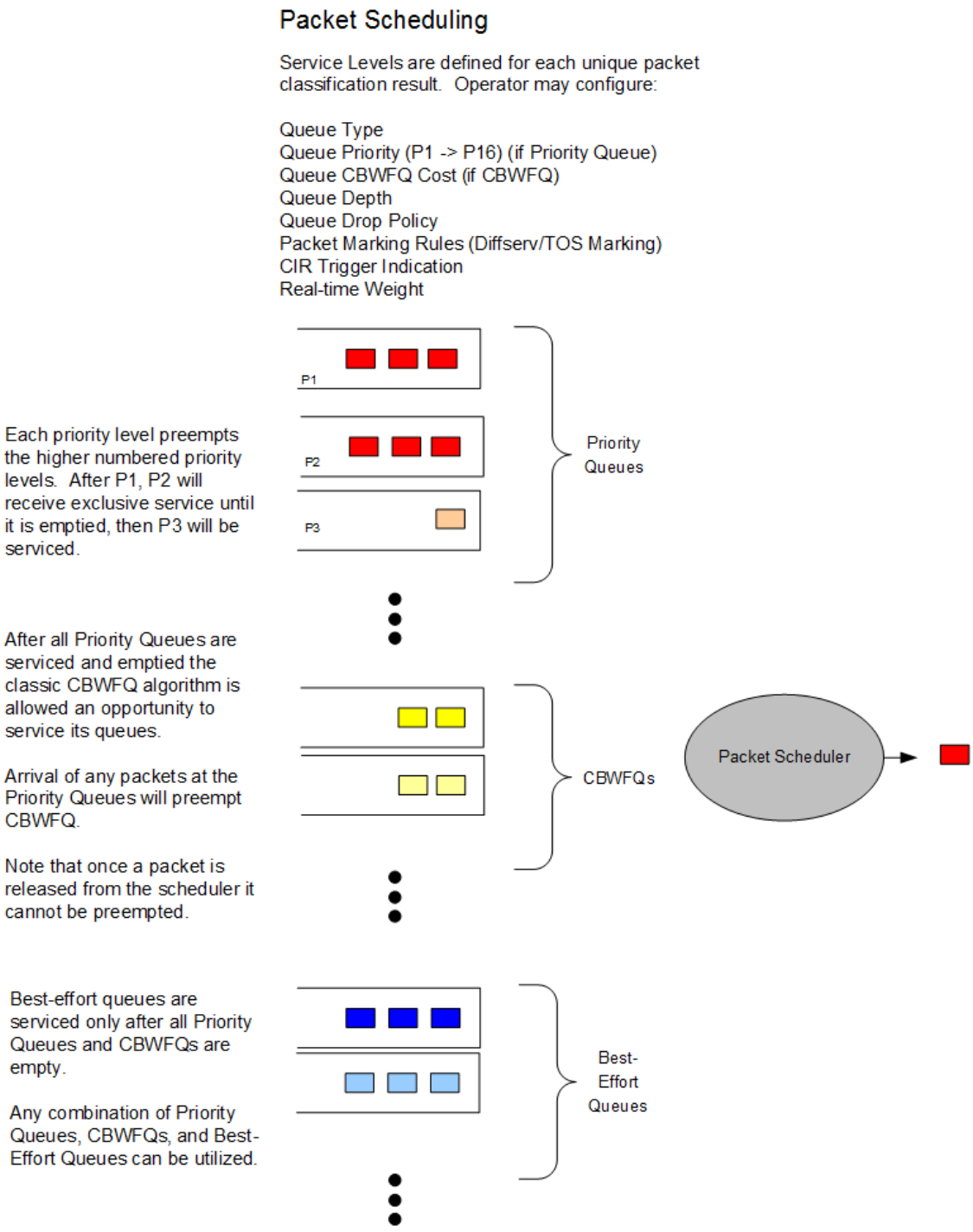


Figure 4-27. iDirect Packet Scheduling Algorithm

4.6.6.1 Priority Queues

There are sixteen levels of Priority Queues, from Level 1 (the highest) to Level 16 (the lowest). All high-priority queues are serviced before any lower-priority queues are serviced. When two or more queues are set to the same priority level, all queues of equal priority are serviced using a round-robin algorithm selection prior to selecting any packets from lower priority queues.

4.6.6.2 Class-Based Weighted Fair Queues

Packets are selected from Class-Based Weighted Fair Queues for transmission based on the service level (or ***class***) of the packet. Each service level is assigned a ***cost***. Packet cost is defined as the cost of its service level multiplied by its length. Packets with the lowest cost are transmitted first, regardless of service level.

The cost of a service level changes during operation. Each time a queue is passed over in favor of other service levels, the cost of the skipped queue is credited, which lowers the cost of the packets in that queue. Over time, all service levels have the opportunity to transmit, at least occasionally, even in the presence of higher priority traffic. Assuming there is a constant congested link with an equal amount of traffic at each service level, the total bandwidth available is divided more evenly by using transmission priority based on each service level cost.

4.6.6.3 Best-Effort Queues

Packets in Best Effort Queues do not have priority or cost. All packets in these queues are treated equally by applying a round-robin selection algorithm to the queues. Best-Effort Queues are only serviced if there are no packets waiting in Priority Queues or in CBWFQs.

4.6.7 FAP Service Enforcement

As mentioned in Fair Access Policy on page 325, FAP enforcement allows service providers to define a set of rules for throttling the throughput of GSPs and SSPCs based on the total consumption (in Mbytes) within a defined period of time. FAP enforcement for GSPs and SSPCs is largely the same, but is handled by separate software processes (Group FAP Enforcer vs. Subscriber FAP Enforcer Scheduler).

The FAP Enforcer is responsible for enforcing the FAP volume allowance defined as part of the service plan. The FAP Enforcer resides at the NOC, and receives real-time inbound/outbound traffic volumes per iNet group (for GSPs) or per SSPC. FAP enforcement verifies if any of the FAP rules have been violated; for example, if the volume usage for the allowance period has exceeded the volume allowance.

The volume usage is collected for all GSPs and SSPCs from each SAS, and is then reported to the NOC every 30 seconds. The algorithm runs periodically, every 30 seconds, or at any time when it receives an action message. The action messages are either a configuration change, a Top-Up allowance request, or an update on FAP usage. The algorithm checks for the start of a FAP update period to credit the FAP balance and applies rollover when applicable. The algorithm also checks if any FAP conditions have been reached to take action, and if restoration rules for restoring the service have been met.

4.6.8 Operating in Symbol-based Mode

After initial installation or upgrade, operators have the option of remaining in the default Symbol-base mode of operation or switching to the IP-based mode. When operating in the Symbol-based mode network operators may do the following:

- Create, Modify, and Delete GSPs
- Monitor QoS Stats in Symbols/s with L2 OTA overhead
- Switch the QoS Service Mode to IP-based

GSP configuration when using the Symbol-based option, includes the capability of configuring the Nominal MODCOD in the outbound direction, as well as ability to configure scaling factors using the Default Average MODCOD and GSP Profiles, and Average MODCOD at the beam level.

If the intent is to switch to IP-based service, then the operator must first backup the GSP configuration before switching modes. The operator will also need to determine the new CIR/MIR values to be applied after the service mode is switched to IP-based.

See Switching Between QoS Service Modes.

4.6.9 Operating in IP-based Mode

The IP-based service option is only entered after the network operator decides to switch the QoS Service Mode to IP-based and applies the change. Choosing this mode enables Service Providers to create and configure unicast GSPs with CIR/MIR specified in Mbps. In addition to this, QoS stats can be monitored in Mbps without over-the-air overhead.

Although multicast GSPs may be configured while operating in IP-based service option, operators must still configure the multicast MODCOD in the multicast SSPP, and multicast traffic still reports Stats that include over-the-air overhead. The multicast base MODCOD, specified in the multicast SSPP, is still required and used for the streams operating in IP-based service mode.

When operating in IP-based mode, operators may to do of the following:

- Create, Modify, and Delete GSPs
- Monitor QoS Stats in Mbps without OTA overhead
- Switch the QoS Service Mode to Symbol-based

See Switching Between QoS Service Modes.

4.6.10 Switching Between QoS Service Modes

QoS Service mode determines the type of service the network provides in the downstream direction (hubs to remotes). From the time the mode is switched, to either of the QoS options, operators are able to create, modify, and delete Group Service Plans (GSPs) or Subscriber Service Plan Profiles (SSPPs) as they would normally, and network operation proceeds without interruption in the newly selected service mode.

Switching from the Symbol-based mode to the IP-based mode is outlined below:

- ***Symbol-based to IP-based Mode*** — The system is operating in the Symbol-based mode, and the operator decides to switch to the IP-based mode. The following must occur:
  - Previously configured Symbol-based GSP data must be backed up by the operator.
  - Operator switches to IP-based and applies the change at the Network level.
  - Pulse GSP configuration fields/pages for the IP-based service are now available, and Pulse fields/pages associated with Symbol-based service are no longer available.
  - The QoS engine operates in Mbps and QoS stats are reported in Mbps, without over-the-air overhead.
  - Terminals in network remain stable in the network with no expected outage.
- ***IP-based to Symbol-based Mode*** — The system is operating in the IP-based mode, and the operator decides to switch to the Symbol-based mode. The following must occur:
  - Previously configured IP-based GSP data must be backed up by the operator.
  - Operator switches to Symbol-based and applies the change at the Network level.
  - Pulse GSP configuration fields/pages for the Symbol-based service are now available, and Pulse fields/pages associated with IP-based service are no longer available.
  - The QoS engine operates in Msps and QoS stats are reported in Mbps, with overhead.
  - Terminals in network remain stable in the network with no expected outage.



NOTE: After operating in one of the QoS Service modes, and having created GSPs, it is possible to switch to the opposite mode, while the system maintains backward compatibility.

If the decision is made to switch back to the previous mode, the operator must backup the configuration, prior to the switch. It will also be necessary to determine the new CIR/MIR values to be applied after the mode change.

4.6.11 IP Efficiency Operation

The minimum efficiency implementation is available only when the QoS Service mode is operating with the IP-based option, and only applies to the GSP outbound. The network operator must configure the minimum efficiency for a specific GSP, in the valid range of 0-5.

The efficiency parameter, configured in units of IP bits/Symbol, can be enabled/disabled in a parent node GSP as well as in its children nodes. Efficiency is calculated by dividing the total

IP bits (used by the GSP) by the total number of symbols (used by the GSP) during the reporting interval.

To present an example, of using minimum efficiency, the following parameters are defined:

1. the set value for efficiency is = ***E***; and the measured efficiency is ***M***.
2. the current CIR value is = $\boldsymbol{CIR}_{\text{current}}$; and the current MIR value is = $\boldsymbol{MIR}_{\text{current}}$

In a given scenario, the $\mathbf{CIR}_{current}$ and $\mathbf{MIR}_{current}$ values could be either of the following:

a. the currently configured CIR/MIR values in the GSP outbound direction
b. the recurrence CIR/MIR values set in the configuration
c. the throttled down CIR/MIR values due to a triggered FAP rule

If **E = 0**, then the CIR/MIR for the GSP will not be scaled.

If **M ≥ E**, then the CIR/MIR for the GSP is not changed (scaling up or down is not required).

If M < E then scale down the CIR/MIR as follows:

$$CIR_{scaled} = CIR_{current} * (M/E)$$

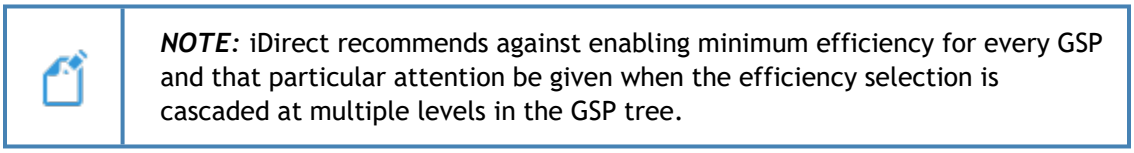
$$MIR_{scaled} = MIR_{current} * (M/E)$$

The following example illustrates how the minimum efficiency is applied in operation.

Example:

- A GSP has a geoscope that covers multiple beams
- the configured downstream CIR/MIR = 20/40 Mbps
- the configured efficiency, ***E*** = 1.8
- the last measured efficiency, ***M*** = 2.0
- since 2.0 ≥ 1.8 or M ≥ E, then the CIR/MIR is not changed

In this case the CIR/MIR would not be scaled down in the next measurement interval. At some later point during operation and at a subsequent measurement interval, the measured efficiency is 1.5. In this case the scaling factor is calculated as (1.5/1.8) = 0.8333. In the next interval the CIR/MIR values are s scaled down by a scale factor of 0.8333 or 16.67/33.33 Mbps.



NOTE: iDirect recommends against enabling minimum efficiency for every GSP and that particular attention be given when the efficiency selection is cascaded at multiple levels in the GSP tree.

4.6.12 IP-Based QoS and Efficiency Stats

In the IP-based (Mbps) mode, the Velocity system reports GSP/SSPC demand, allocation, usage, and efficiency stats in real IP rates. The IP-based QoS and Efficiency stats are based on IP bytes, excluding overhead bytes, and are suitable for reports that are based on IP Mbs.



NOTE: IP-based stats are based on IP bytes that exclude overhead bytes, and represent Mbps. The Symbol-based service mode stats, reported in symbols, have not changed, and the Stats, reported in bytes, still contain L2 overhead.

* Demand bytes/Usage bytes do not include any overhead.

4.6.12.1 IP-Based QoS Stats

In the IP-based (Mbps) mode, the Velocity system reports GSP/SSPC demand, allocation, and usage stats in real IP rates. These QoS stats are based on IP bytes, excluding overhead bytes, and are suitable for reports that are based on IP Mbs.

The available IP rate QoS stats , which can be reported at both the GSP and the SSPC level, rely on the aggregation of stats from lower levels in the QoS tree. For the SSPC, this aggregation of stats is across all SVNs used in the SSPC. For the GSP, aggregation start at the iNet level, progresses through the SAS level, and finally at the NOC level.

GSP and SSPC stats collected for the IP-Based QoS service mode are listed in the following table. Stats in symbols and associated with the previously available Symbols-based service mode have not changed. Stats in bytes used in the Symbols-based mode still contain L2 overheads. However in IP-Based mode the same stats have only IP bytes and no L2 overheads.

Stats reports can be generated from the Pulse Configurator tool, under the Monitoring menu.

Table 4-4. GSP/SSPC QoS Stats for IP-Based Service Mode

SSPC Downstream Metric	**SSPC Upstream Metric**	**Brief Description**
sspc_ds_usage	sspc_us_usage	Usage in IP bytes only *
sspc_ds_allocated_bw_bytes	sspc_us_allocated_bw_bytes	Allocated bandwidth in bytes *
sspc_ds_cir_demand_bytes	sspc_us_cir_demand_bytes	Demand in IP bytes only *
sspc_ds_mir_demand_bytes	sspc_us_mir_demand_bytes	Demand in IP bytes only *
sspc_ds_cir_current	sspc_us_cir_current	
sspc_ds_mir_current	sspc_us_mir_current	
GSP (Outbound)	**GSP (Outbound)**	**Brief Description**
gsp_ds_cir_demand_bytes	gsp_us_cir_demand_bytes	Demand in IP bytes only *
gsp_ds_mir_demand_bytes	gsp_ds_mir_demand_bytes	Demand in IP bytes only *
gsp_ds_distributed_bw_bytes	gsp_us_distributed_bw_bytes	
gsp_ds_usage	gsp_us_usage	Usage in IP bytes only *
gsp_ds_current_cir	gsp_us_current_cir	
gsp_ds_current_mir	gsp_us_current_mir	

4.6.12.2 IP-Based Efficiency Stats

In the IP-based (Mbps) mode, the Velocity system reports GSP/SSPC downstream demand, allocation, and usage stats in real IP rates. These IP rate stats rely on the aggregation of other stats from lower levels in the QoS tree. For the SSPC, this aggregation of stats is across all SVNs used in the SSPC. For the GSP, aggregation starts at the iNet level, and progresses through the SAS and NOC levels.

IP-based efficiency stats, which are collected for both the upstream and downstream, are reported for the SSPC, Terminal, iNet, and for the GSP at the NOC, SAS, and iNet levels. All stats metrics can be viewed as bits/symbol or bits/Hz.

Stats reports, which can be generated using Pulse, under the **Monitoring** menu, are available for multiple types of system users, subject to the NMS configured permissions.

Table 4-5. GSP/SSPC/iNet/Terminal Efficiency Stats for IP-Based Service Mode

SSPC Downstream Metric	**SSPC Upstream Metric**
sspc_ds_bw_efficiency	sspc_us_bw_efficiency
iNet Downstream Metric	**iNet Upstream Metric**
inet_ds_bw_efficiency	inet_us_bw_efficiency
Terminal Downstream Metric	**Terminal Upstream Metric**
ds_spectral_efficiency	us_spectral_efficiency
GSP @ iNet Downstream Metric	**GSP @ iNet Upstream Metric**
group_ds_bw_efficiency	group_us_bw_efficiency
GSP @ SAS Downstream Metric	**GSP @ SAS Upstream Metric**
gsp_sas_ds_bw_efficiency	gsp_sas_us_bw_efficiency
GSP @ NOC Downstream Metric	**GSP @ NOC Upstream Metric**
gsp_current_ds_efficiency	gsp_current_us_efficiency
gsp_configured_ds_efficiency	
gsp_current_scale_factor	

5 Redundancy and the Velocity System

The iDirect Velocity™ network supports various resiliency features, redundancy options, and switchover methods that support continuous network operation despite interruptions due to inclement weather, hardware failures, and maintenance activities. These features include primary and backup Satellite Access Station (SAS) sites, primary and backup Network Operations Center (NOC) sites, redundant data storage, redundant physical and logical links.

This chapter describes the Redundancy and Switchover features and characteristics associated with the Velocity network.

The following sections are included in this chapter:

5.1 Automatic Rain Diversity Switchover 182
5.2 SAS Site Redundancy 191
5.3 CDC and RF Gateway Redundancy 196
5.4 NOC Site Redundancy 201
5.5 NMS Redundancy and Failover 205
5.6 Component Redundancy 207
5.7 Network Resiliency 219

5.1 Automatic Rain Diversity Switchover

The Automatic Rain Diversity Switchover (RDS) configuration allows the design of networks that minimize the impact of weather events on over-the-air performance. This configuration requires a second site, or diversity site, in the same beam footprint of the feeder link of the primary site, but sufficiently far enough away to minimize the possibility of the secondary site experiencing the same weather event as the primary site. This second site also functions as a backup site. iDirect recommends careful consideration of the diversity site location when performing the rain diversity analysis.

The primary site Protocol Processor (PP) processes manage rain diversity switchover and decisions are made on an individual iNet basis.

Velocity supports Automatic RDS in both of the Velocity deployment architectures.

- SAS Automatic Rain Diversity
- CDC/RF Gateway Automatic Rain Diversity.



NOTE: Rain diversity is not available when a site is offline for maintenance.

5.1.1 SAS Configuration

The SAS Automatic Rain Diversity configuration utilizes a primary SAS site and backup SAS site to provide automatic switchover during a weather event. The configuration is described in the following sections.

5.1.1.1 Network Setup

The network architecture to support a rain diversity configuration consists of a primary SAS site and a diversity site as depicted in the following diagram.

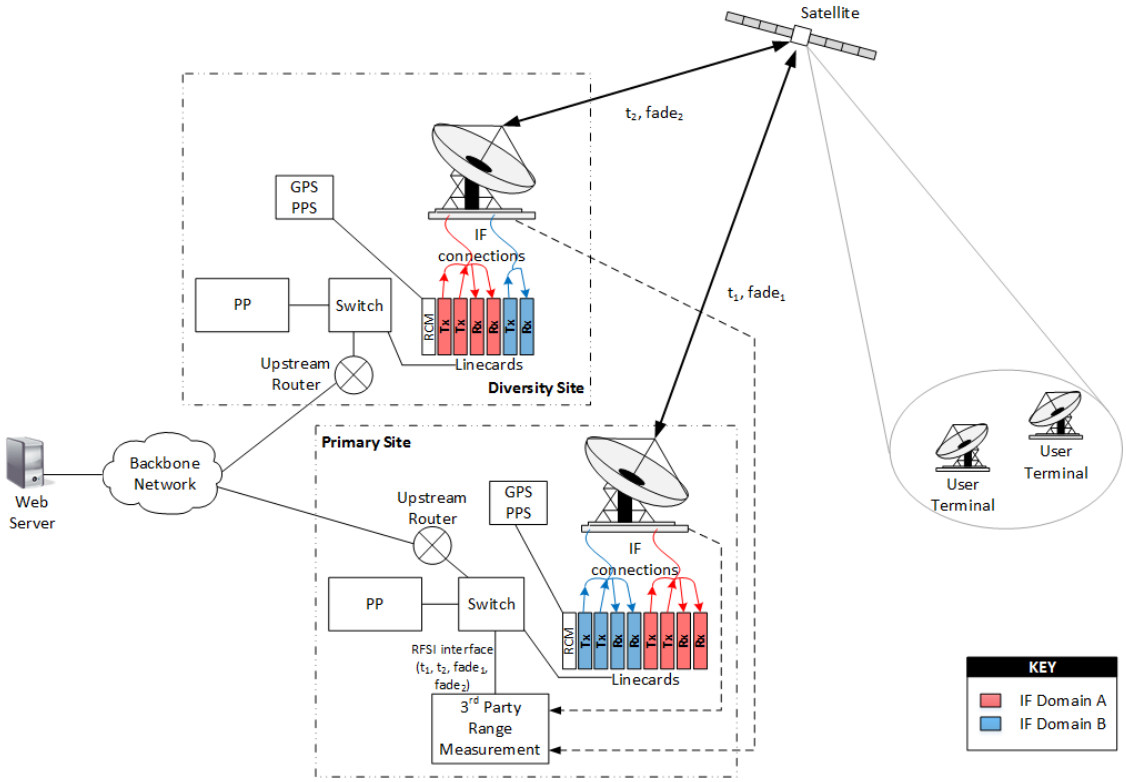


Figure 5-1. Rain Diversity Network Architecture - SAS Configuration

The diversity site can either replicate the capacity of the primary site or be a subset of the capacity of the primary site, i.e., fewer line cards and less supporting hardware. iNets are assigned to the diversity site using the same iNet priority as in the primary site.

A timing reference is required for each site and it must be capable of providing the 1PPS (packet per second) GPS (global positioning system) disciplined reference.

	NOTE: The diversity site is also a backup site.

5.1.1.2 Physical Layer

The diversity switchover is designed such that, during a switchover, active terminals in the network do not drop out of the network. During the switchover, data loss is approximately 6 seconds. To achieve this, the time it takes for the data to flow from the antennas at both sites to the satellite (t1 and t2 values in diagram) along with the fade changes for each site (fade1 and fade2 values in diagram) are measured in the Radio Frequency Subsystem (RFS) by a third-party Carrier Measurement System (CMS) and fed back to the PPs. The PPs, in turn, use this information to compensate for network timing and make switch decisions.

The Velocity network operating prior switching over to a rain diversity configuration is illustrated in the diagram.

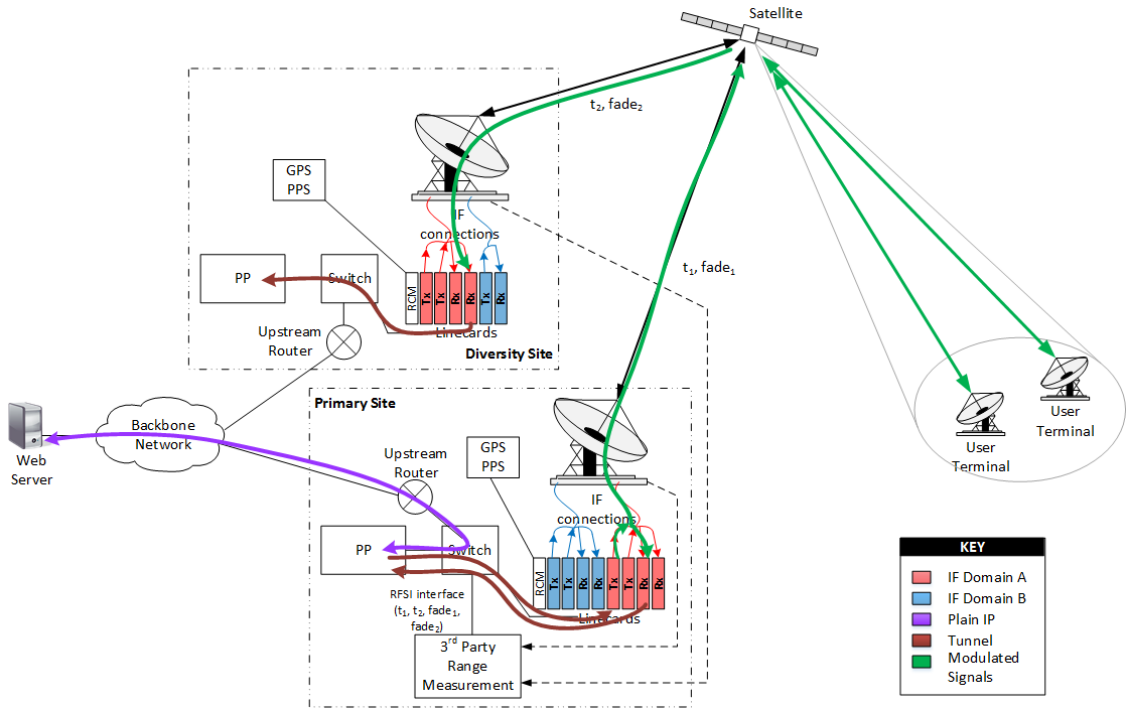


Figure 5-2. Before Rain Diversity Switchover - SAS Configuration

Under normal operation, traffic flows through the primary site, traverses the backbone network and terminates at the primary site PP. The primary site PP uses the tunnel VLAN/SVN (virtual local area network/satellite virtual network) to send data to the appropriate transmit (Tx) line cards. The data is then sent over the air to the user terminal. In the reverse direction, traffic from the terminal lands at the receive (Rx) line cards, and is sent to the primary site PP over the same tunnels.

Since the diversity site lies within the same beam footprint as the primary site, all inbound traffic to the primary site over the satellite is also received at the diversity site. This traffic is ignored by the diversity site's PP. Further, there is no outbound traffic from the diversity site as the Tx line cards in the diversity site are muted. It is also important to note that in normal operation there is no user traffic on the backbone network from the diversity site.

Upon deciding to switch to the rain diversity site, the primary site's PP mutes the Tx line cards of the affected iNets in the primary SAS and begins sending user traffic through the tunnels between the PPs in the primary SAS and the line cards in the diversity SAS over the terrestrial backbone network. The Tx line cards are unmuted and over the air traffic is established at the diversity site. Line cards are logically part of the same IF (intermediate frequency) domain.

iNets are switched between the Rain Diversity and Primary SASs based on iNet priority. This allows Rain Diversity Switchover (RDS) to be used with a backup site that does not contain a full complement of equipment. If the equipment at the diversity/backup site cannot support all user traffic switching over, iNets that remain on the nominal path may experience packet loss and other service interruption due to the rain-fade event at the primary site.

The Velocity network operating in the rain diversity configuration is shown in the diagram.

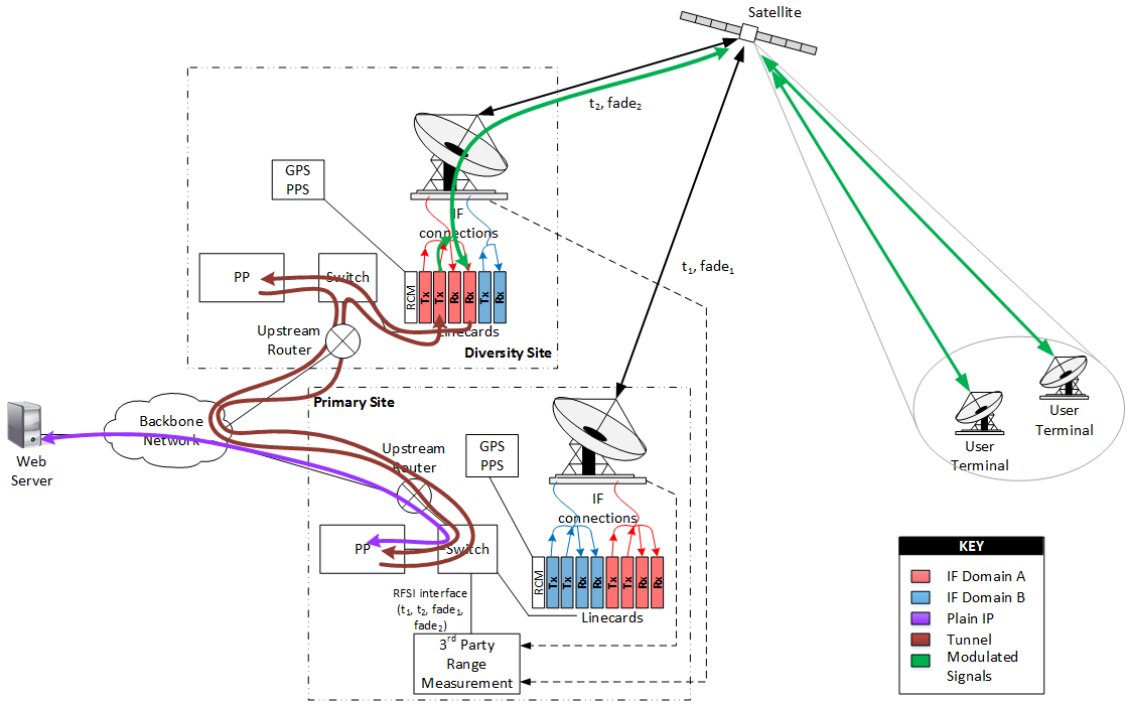


Figure 5-3. After Rain Diversity Switchover - SAS Configuration

	NOTE: To support overhead and control traffic, iDirect recommends that the link between primary and diversity SAS have at least 25% more bandwidth than the volume of user traffic crossing the link.

5.1.1.3 Service Implications and Timing

Entering and exiting a rain diversity configuration does not interrupt normal service in any way. It happens transparently and dynamically, based on rules defined in the network management system (NMS) for the protocol processor, and real-time signal quality information retrieved from the RFS.

In cases where the equipment at the rain diversity SAS cannot support all user traffic switching over, iNets that remain on the nominal path may experience packet loss and other service interruption due to the rain-fade event at the primary SAS.

5.1.1.4 Backbone Network Traffic Requirements

Under normal operating conditions, the PPs and line cards communicate across a local area network (LAN) link at Gbps (gigabits per second) speeds. In a rain diversity configuration, care must be taken to not introduce noticeable delay into the communication between these two components. iDirect requires that the terrestrial backbone latency between the primary and

diversity site should not be more than 50 msecs one-way. This latency can be translated to a distance of approximately 10,000 km between the sites.

iDirect recommends that the link across the backbone network have at least 25% more bandwidth than the volume of user traffic crossing the link.

5.1.2 CDC/RF Gateway Configuration

A CDC/RF Gateway Rain Diversity configuration consists of a CDC as well as a Primary RF Gateway Site and Backup RF Gateway Site to provide automatic switchover during a weather event. The switchover requires configuration of a Gateway Diversity Relationship (GDR) between two RF Gateways with one gateway designated as the Primary RF Gateway and one as the Diversity/Backup RF Gateway. The GDR can be configured for a manual or automatic switchover. A GDR is configured in the Pulse NMS. See the *Configuring Velocity Networks User Guide* for more information on GDRs.

5.1.2.1 Network Setup

The network architecture to support a rain diversity configuration consists of a Primary RF Gateway Site, a Diversity RF Gateway Site, and a CDC (can have an optional backup CDC) as depicted in Figure 5-4. Note the CMS (Carrier Measurement System) located in each RF Gateway. In previous CDC deployments, it was used only to measure the distance between the line cards and satellite terminals; now it will also measure signal strength, from which the PP will determine if rain fade is present.

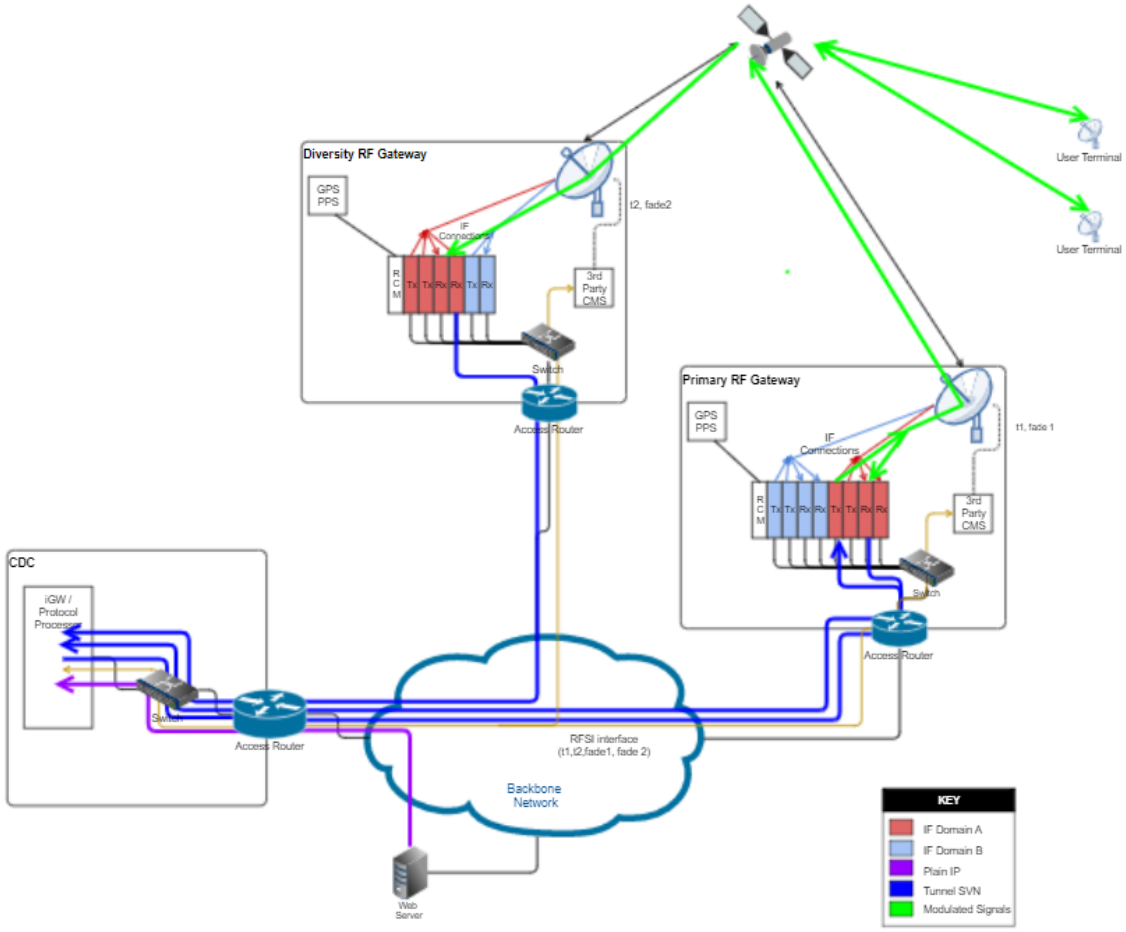


Figure 5-4. Rain Diversity Network Architecture - RF Gateway Configuration

The RF Gateway Diversity Site can either replicate the capacity of the Primary Site or be a subset of the capacity of the Primary Site, i.e., fewer line cards and less supporting hardware. iNets are assigned to the Diversity Site using the same iNet priority as in the Primary Site.

	NOTE: The Diversity Site is also a Backup Site.

5.1.2.2 Physical Layer

The diversity switchover is designed such that, during a switchover, active terminals in the network do not drop out of the network. During the switchover, data loss is approximately 6 seconds. To achieve this, the time it takes for the data to flow from the antennas at both RF Gateways to the satellite (t1 and t2 values in diagram) along with the fade changes for each site (fade1 and fade2 values in in diagram) are measured in the Radio Frequency Subsystem (RFS) by a third-party Carrier Measurement System (CMS) and fed back to the PPs. The PPs, in turn, use this information to compensate for network timing and make switch decisions.

The Velocity network operating prior switching over to a rain diversity configuration is illustrated in the diagram below.

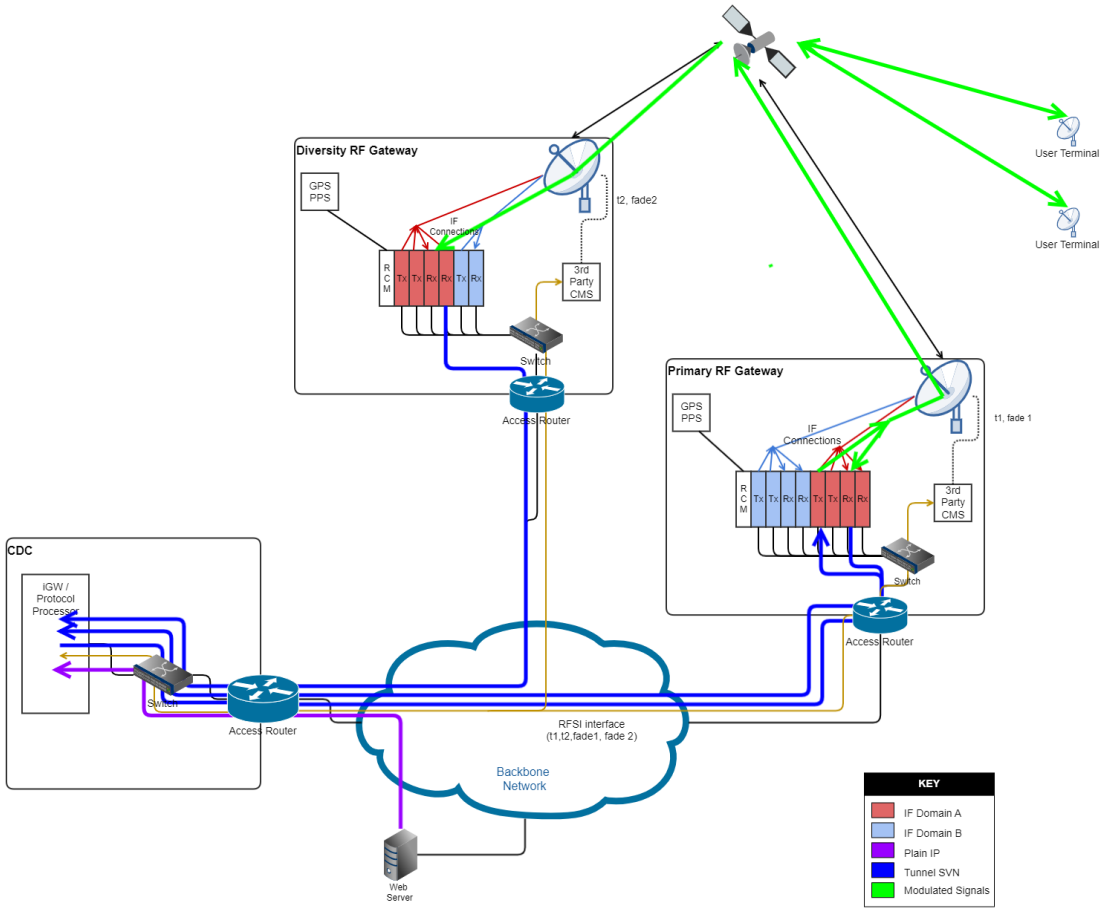


Figure 5-5. Before Rain Diversity Switchover - RF Gateway Configuration

Under normal operation, traffic traverses the backbone network and terminates at the RF Gateway Primary Site. The Primary Site uses the tunnel VLAN/SVN (virtual local area network/satellite virtual network) to send data to the appropriate transmit (Tx) line cards. The data is then sent over the air to the user terminal. In the reverse direction, traffic from the terminal lands at the receive (Rx) line cards, and is sent to the primary site PP over the same tunnels.

Since the Diversity RF Gateway lies within the same beam footprint as the Primary Site, all inbound traffic to the Primary Site over the satellite is also received at the Diversity Site.

Upon deciding to switch to the Diversity Site, the CDC's PP mutes the Tx line cards of the affected iNets in the Primary RF Gateway Site and begins sending user traffic through the tunnels between the PPs in the RF Gateway Primary Site and the line cards in the RF Gateway Diversity Site over the terrestrial backbone network. The Tx line cards are unmuted and over the air traffic is established at the Diversity site.

iNets are switched between the Diversity RF Gateway and Primary RF Gateway based on iNet priority. This allows Rain Diversity Switchover (RDS) to be used with a backup site that does

not contain a full complement of equipment. If the equipment at the diversity/backup site cannot support all user traffic switching over, iNets that remain on the nominal path may experience packet loss and other service interruption due to the rain-fade event at the primary site.

The Velocity network operating in the rain diversity configuration is shown in the diagram.

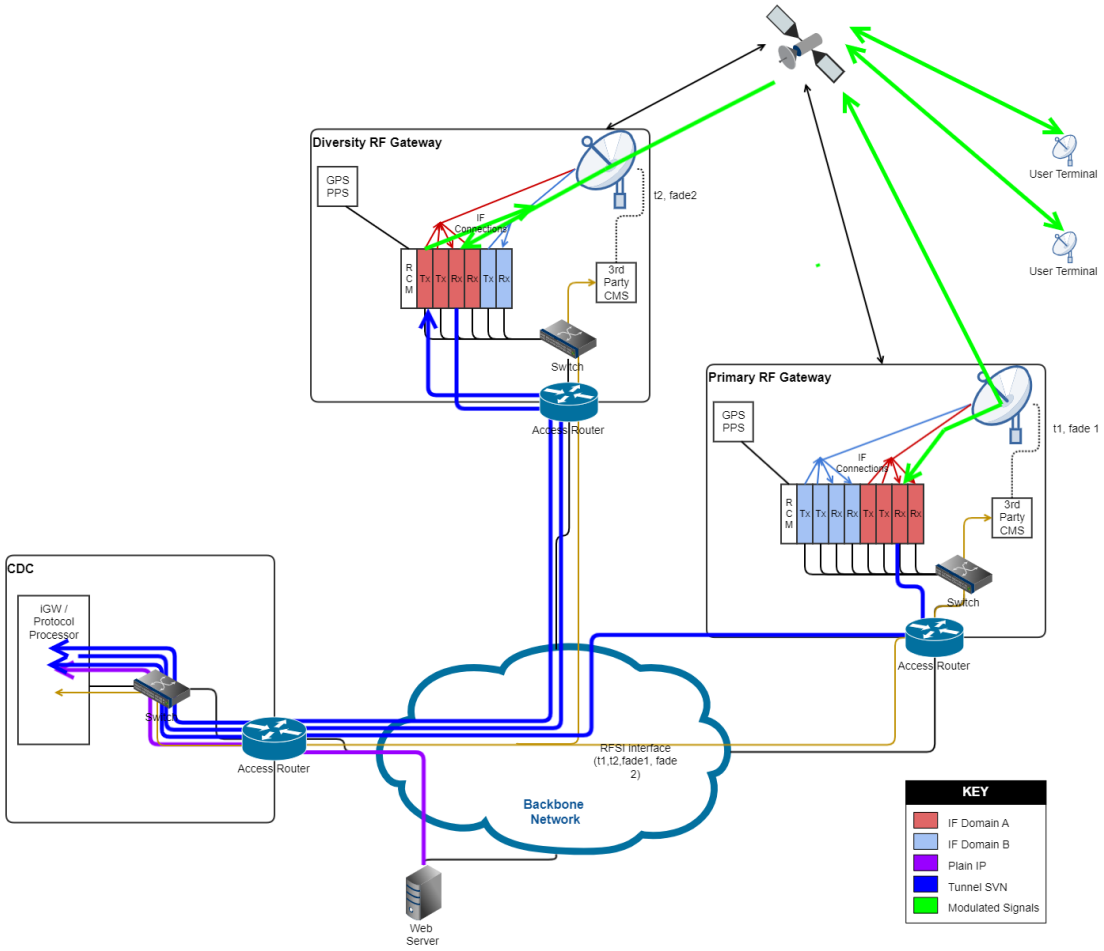


Figure 5-6. After Rain Diversity Switchover - RF Gateway Configuration

5.1.2.3 Service Implications and Timing

Entering and exiting a rain diversity configuration does not interrupt normal service in any way. It happens transparently and dynamically, based on rules defined in the network management system (NMS) for the protocol processor, and real-time signal quality information retrieved from the RFS.

In cases where the equipment at the rain diversity RF Gateway cannot support all user traffic switching over, iNets that remain on the nominal path may experience packet loss and other service interruption due to the rain-fade event at the primary RF Gateway.

5.1.2.4 Backbone Network Traffic Requirements

The WAN link bandwidth should be sufficient to handle all iNets that may be assigned to the RF Gateway.

5.2 SAS Site Redundancy

SAS site redundancy ensures that the satellite network continues to provide service even if the primary SAS is not usable due to a catastrophic natural disaster, backbone network outage, or maintenance activities such as an upgrade. Each satellite may have a backup SAS that can also function as a rain diversity site.

SAS component redundancy ensures that a failure of any component does not cause a service outage or degradation of Velocity system performance. For a description of component redundancy, see Component Redundancy on page 207.

	NOTE: The SAS backup site is also a diversity site.

5.2.1 SAS Redundancy Overview

A Velocity network may be designed to contain a pair of SASs for a satellite — for example, a primary SAS and a backup SAS. The SAS that is currently providing service for the satellite is said to be “active” and the other said to be “standby” or “redundant”.

The backup SAS may be physically identical to the primary SAS to ensure seamless system operation in the event of a switchover for an extended period, in which case the designation “primary” and “backup” is effectively arbitrary. Alternatively, the backup SAS may be sized to support less than the full capacity of the primary SAS. In that case, during a SAS switchover the iNets will be established in priority order, and lower priority iNets will become inactive.

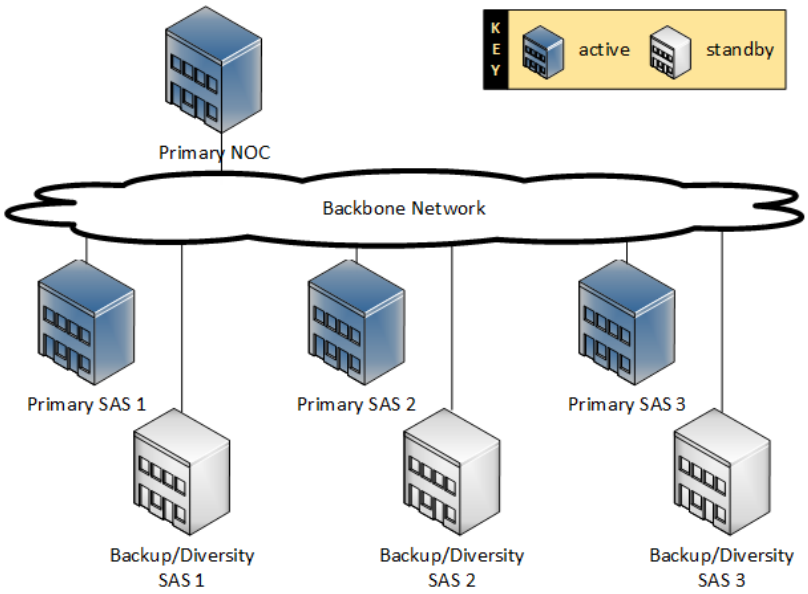


Figure 5-7. Redundant SASs

The standby (redundant) SAS is a warm standby - all components, including the access routers and SAS switch, are powered up and are reachable. The iDirect components in the standby SAS are managed by the Pulse NMS.

When the primary SAS is active, traffic flows through the active primary SAS as shown in Figure 5-8, which depicts the data path to the terminal as a thick line. Note that the standby (redundant) SAS is not active in any part of the data path.

A standby (redundant) SAS becomes the active SAS when an operator activates the Pulse NMS and PP components within it. Switchover is under operator control to reduce the risk that both sites transmit to the satellite at the same time. The manual process of switchover to a standby (redundant) SAS is generally performed from the active NOC, but can be performed from any of the following locations:

- Active NOC
- Active SAS
- Standby (redundant) NOC
- Standby (redundant) SAS

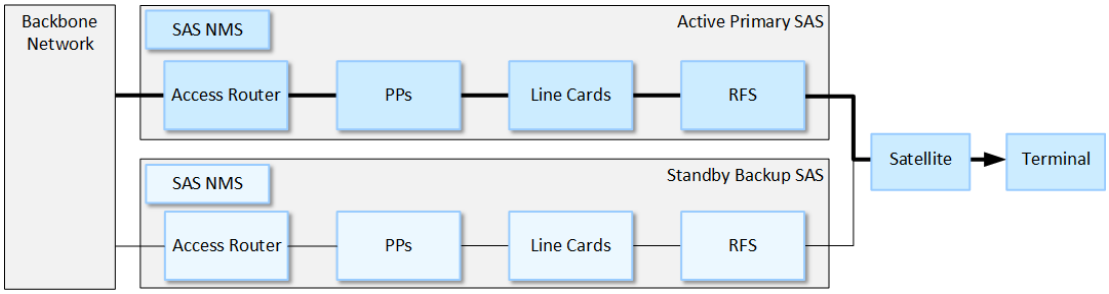


Figure 5-8. Data Path when Primary SAS is Active

NOTE: A SAS switchover operation causes a brief interruption in service to all the satellite terminals currently using the satellite supported by the primary SAS. The terminals leave the network and reacquire into the newly active SAS.

If a SAS is not reachable over the backbone network, an operator physically located at that SAS must perform the SAS switchover.

The following diagram shows the Velocity network after switchover has been performed on one SAS pair. iDirect recommends a periodically planned exercise to validate the network provider's execution of the SAS switchover process and the associated documentation.

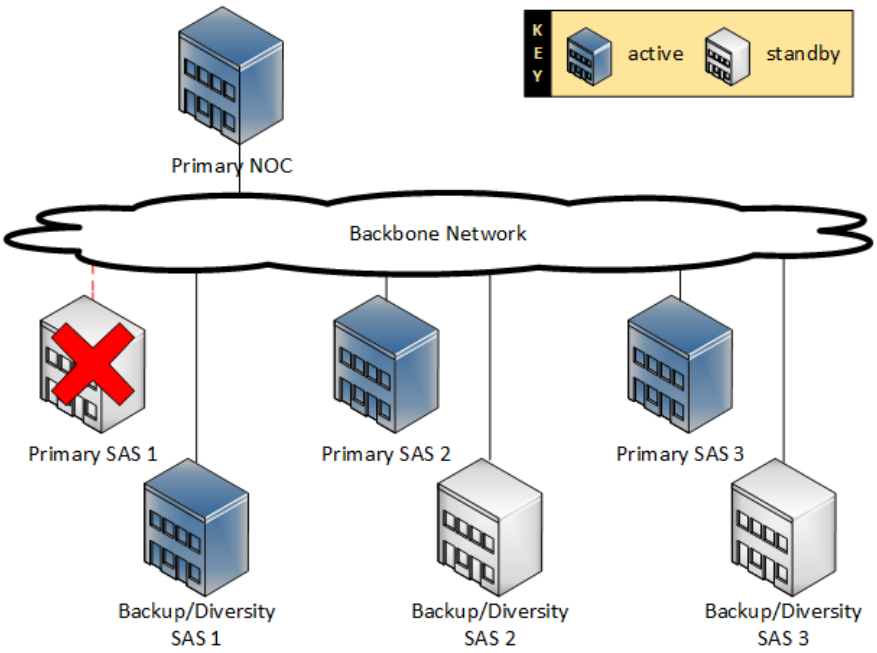


Figure 5-9. Velocity Network after SAS 1 Switchover

5.2.2 Understanding SAS Switchover

When a SAS site is designed and configured as a backup SAS site, the NOC NMS provides the entire network configuration to that site, just as it does for the primary SAS site. This includes the satellite configuration information as well as the configuration for each satellite terminal. When a SAS is operating in standby (redundant) mode, the Tx line cards do not transmit, and the PP and NMS clusters run in standby (redundant) mode.

The network operator accomplishes a SAS switchover by:

- Putting the currently active SAS into standby (redundant) mode
  - Disabling the active SAS NMS
  - Changing the mode of the SAS PP cluster from active to standby (redundant) mode
- Activating the currently standby (redundant) SAS
  - Changing the mode of the SAS NMS from standby (redundant) to active mode
  - Changing the mode of the SAS PP from standby (redundant) to active mode
- Re-establishing SAS NMS database replication
  - Putting the disabled SAS NMS in the standby (redundant) SAS in standby (redundant) mode

It is important for the formerly active SAS to have relinquished its role and no longer be transmitting before the other SAS becomes active. To ensure this, when a PP cluster receives a command to enter active mode, it first verifies that there is no other active PP cluster for the satellite by asking its paired PP cluster what mode it is operating in. If the paired PP's response is active mode or if no response is received, the command fails. This failure can

occur when connectivity between the sites via the backbone network is not available. If the operator determines through other means that the formerly active PP cluster is in fact in standby (redundant) mode or is otherwise offline, the operator may force the newly-designated SAS to become the primary.

No operator action is necessary to make the backbone network, customer private networks, or equipment on the terminal's network aware that a switchover has occurred:

- The former active SAS withdraws each satellite terminal's routes via the Border Gateway Protocol (BGP) routing protocol.
- If the former active SAS is not reachable by the backbone network, the backbone routers will deprecate the routes learned from it.
- The new active SAS learns and advertises each satellite terminal's routes via BGP.

The new primary SAS learns each satellite terminal's IP (Internet Protocol) routes and advertises them to the backbone network in exactly the same way as after a satellite switch or after a satellite terminal is newly acquired.

After the SAS switchover completes, the new standby (redundant) SAS will be available to provide fade protection via the rain diversity feature as described in Automatic Rain Diversity Switchover on page 182.

The following diagram shows the transmission path to the satellite as the thick line from the PPs through the line cards to the satellite. When the backup SAS is active, the primary SAS is in redundant mode and is not participating in any part of the data path; all of the data transmission work is performed at the active backup SAS.

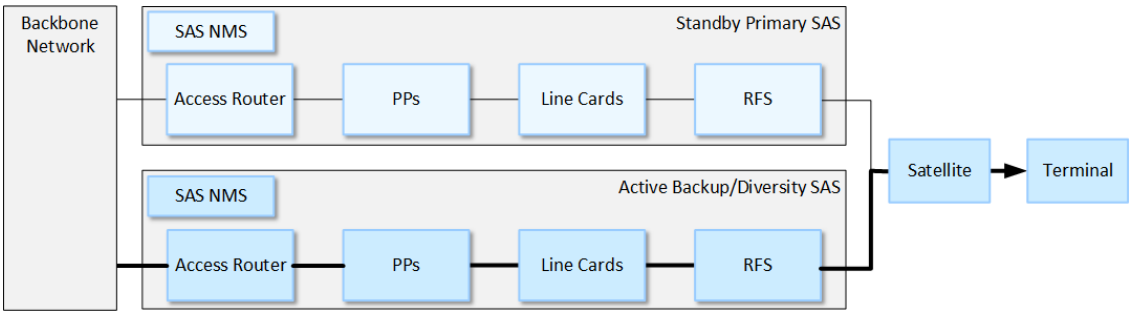


Figure 5-10. Data Path after a SAS Switchover

5.2.2.1 SAS Switchover — Service Implications and Timing

A switchover from active SAS to a standby (redundant) SAS causes a brief yet predictable interruption in service for any satellite terminals currently active on the iNets in that SAS because all processes running at the newly active SAS must be restarted or activated, and all satellite terminals need to re-acquire into the iNets. This sequence should take a few minutes once the operator has completed switching both PP cluster modes.

5.2.2.2 SAS Switchover — Backbone Network Traffic Requirements

During a SAS switchover operation, there is no spike in traffic on the backbone network. All NMS SAS processes perform regular block-level replication of stats and events data to the

other SAS.

This process adds a regular bandwidth requirement on the backbone network, which increases proportionally with the number of changes to disk blocks. The extra bandwidth requirements at the SAS are higher than those at the NOC. This is primarily due to the more granular storage of stats and events by the SAS NMS processes.

When the connection comes back up between primary and backup SAS sites, there is a spike in traffic as the two SAS NMSs re-synchronize their data. The volume and length of this additional backbone network traffic increases proportionally with the length of the outage.

5.3 CDC and RF Gateway Redundancy

CDC and RF Gateway redundancy and switchover implementations are described in the following sections.

- CDC Site Redundancy and Switchover
- RF Gateway Site Redundancy and Switchover
- RF Gateway Switchover Across HEO Satellites

5.3.1 CDC Site Redundancy and Switchover

The Velocity network supports 1:1 CDC Redundancy in which a switchover involves a manual switchover process performed on each component of the CDC site. CDC and components switchover will eventually be triggered with one action from the NMS.

An operator might trigger a CDC switchover for any of the following reasons:

- to perform maintenance
- as part of the software upgrade procedure
- a catastrophic event occurrence at the CDC
- critical errors are exhibited at one or more CDC components

The following diagram shows the CDC primary and backup, and the data flows between these components and the NOC and RF Gateway components, just prior to the CDC switchover.

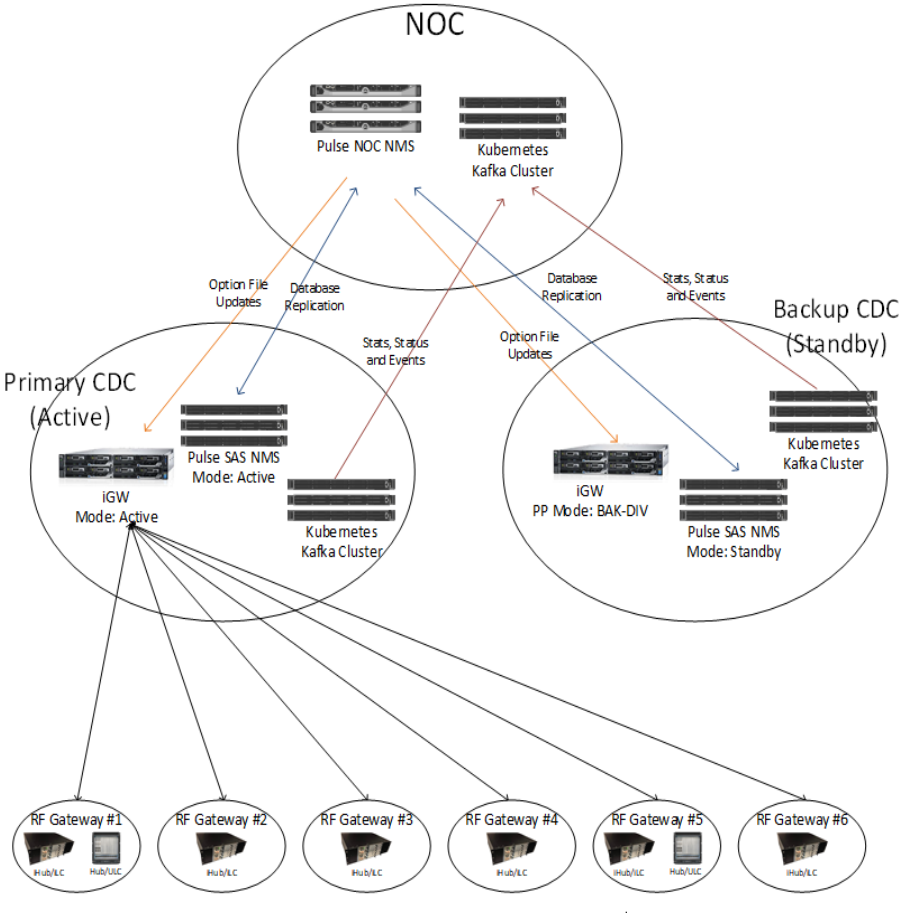


Figure 5-11. Before CDC Switchover

The following conditions can be noted just prior to a switchover:

- Primary CDC is active and the Backup CDC is standby
  - Primary and Backup refer to the configured role of a component
  - Active and Standby refer to the current operational role of a component
- All backup CDC components are healthy and not exhibiting service-affecting faults
- Pulse NMS in the primary/active CDC and Pulse NMS in the backup/standby CDC are both up, and Pulse DB replication between NOC and each CDC NMSs is operational
- Kafka clusters in the primary/active CDC and in backup/standby CDC are up
- The iGW in the primary/active CDC is active and working; Automatic Rain Diversity Switching (RDS) is disabled
- The iGW in the backup/standby CDC is in Backup/Diversity mode

The operator would perform the following to initiate manual CDC Switchover:

1. Change the mode of the iGW in the active CDC to Backup/Diversity

2. Change the mode of the NMS in the active CDC to standby
3. Change the mode of the NMS in the standby CDC to active
4. Change the mode of the iGW in the standby CDC to active
5. Enable the RF Gateways

After CDC Switchover shows the result of the CDC Switchover procedure. The Primary CDC is now in Standby, and the Backup CDC is now Active.

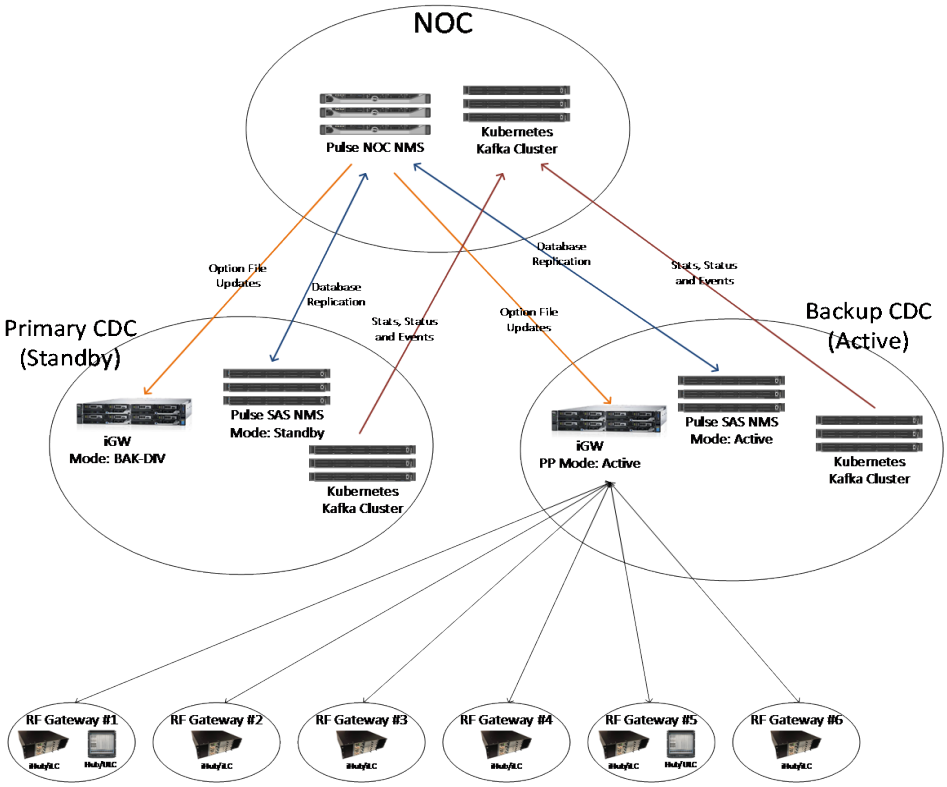


Figure 5-12. After CDC Switchover

5.3.2 RF Gateway Site Redundancy and Switchover

Velocity supports N:M redundancy for RF Gateway sites. This redundancy allows operators to externally control rain diversity, perform maintenance on an idle RF Gateway, and protect against catastrophic failures. This switchover can be performed using real-time commands from the NMS UI and API. When a rain event occurs, the feeder-link supported by one RF Gateway must be moved to an alternate RF Gateway. This move requires the satellite payload be reconfigured, making the automatic rain diversity feature inappropriate for this architecture.

The RF Gateway switchover is designed to complete while the satellite payload is being reconfigured. To ensure a fast switchover, whether the RF Gateway is currently active (supporting a set of beams), or is currently inactive (not currently supporting a set of beams), it may be modified using an RT command or using the PP console. The command to switch to active mode indicates the Beam Sets for which the RF Gateway would then be responsible.

Performing an RF Gateway Switchover involves the following:

1. **Stage Gateway Switch**: This command instructs the protocol processor to switch terminals out of any beams/channels within the specified beam sets and not to initiate any automatic switch back into those beams.
   a. Channels in a gateway are first marked as “gateway switch in progress”, permitting time for terminals to be automatically switched away if there is an unaffected beam that is available to them.
      - This command will time out after 5 minutes (by default) and terminals will be allowed to switch back into the affected beams if an RF Gateway Switchover command is not issued first and the switch command has not been explicitly canceled already. Continue with the following point.
   b. If the operator cancels the previous switch command, the configured channel states are restored for the specified beam sets.
2. **RF Gateway Switchover**: The satellite payload is reconfigured, permitting migration of these channels to a different feeder link
   a. if there are beam set assigned to the new RF Gateway other than those specified in the new beam set, will be unassigned from the gateway
   b. if the new RF Gateway has fewer active line cards then the old RF Gateway...
3. Relevant channels on the original gateway are taken down
4. These channels are then brought back up in a different RF Gateway



NOTE: If IF Domains are not identical in both the active and standby RF Gateway, a channel’s inbound or the number of iNets in the beam may need to be modified during RF Gateway switchover. If necessary these modifications may be performed via the NMS API.

5.3.3 RF Gateway Switchover Across HEO Satellites

The operator can support two (2) 'N' NGSO satellites with an N+1 gateway redundancy scheme. As the Active antenna changes, the operator controls which Velocity RF Gateway is assigned to which satellite via API calls. From Velocity's perspective, this is similar to Geosynchronous Satellites, except that the time required for the HEO RFS to make the switch is minimal.

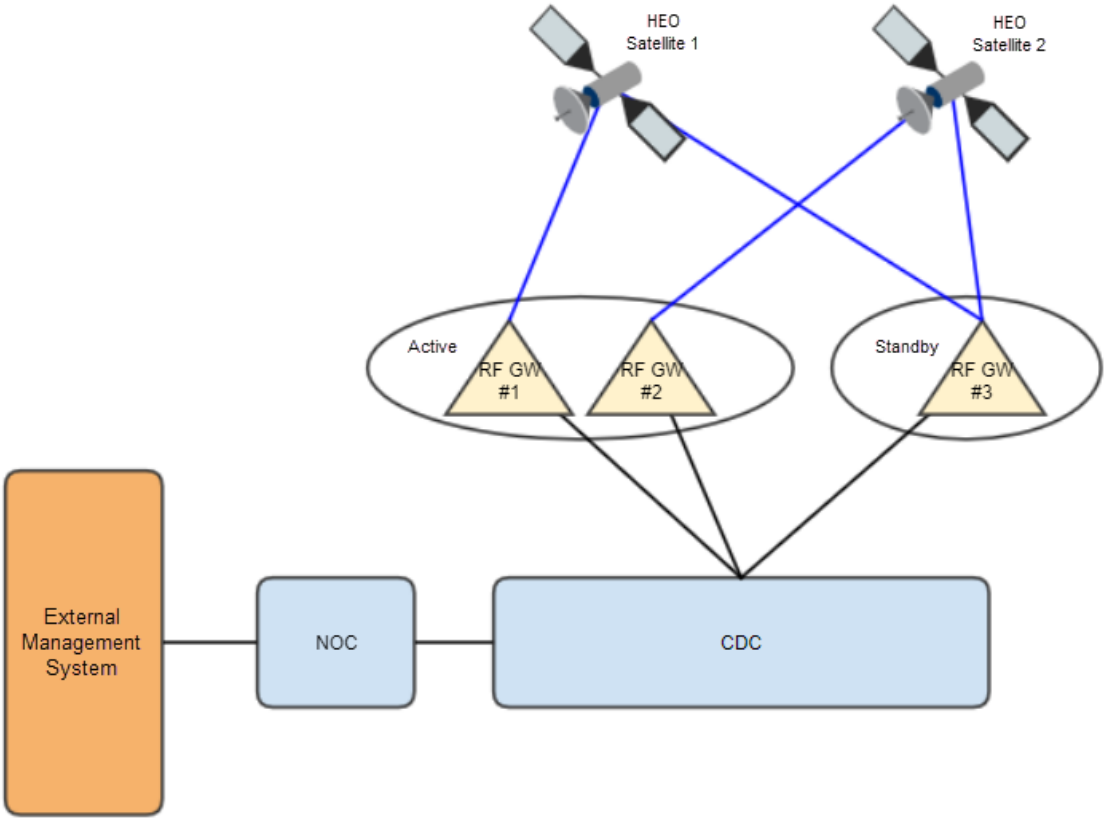


Figure 5-13. RF Gateway Switchover Across HEO Satellites

Velocity and Pulse enable an external management system or an operator to dynamically create a relationship between an Active RF Gateway and a Backup RF Gateway, which will become the Active RF Gateway's "Hot Standby". When a “Hot Standby” relationship is established, the Velocity PP configures the linecards in the “Hot Standby” RF Gateway to support the iNets in the Primary RF Gateway, and the linecards calibrate and begin processing received bursts. If the Primary RF Gateway is unable to provide service for any reason, including rain fade, an external management system or an operator can trigger an RF Gateway switchover to the “Hot Standby” RF Gateway, with minimal impact to traffic. This approach is based on the 1:1 RF Gateway Automatic Rain Diversity feature.

	NOTE: The term "Standby" means that the RF Gateway has no beam sets assigned. Its linecards are powered up, but PP_LCC has removed their beam/iNet configuration and they re not listening to incoming bursts (Rx iLC) or transmitting (Tx iLC).

5.4 NOC Site Redundancy

NOC site redundancy and switchover ensures that the satellite network continues to provide service and the network operator has visibility and can manage the satellite network even if the primary NOC is not available. This can be due to a catastrophic natural disaster, backbone network outage, or maintenance activities such as upgrade.

	NOTE: The details of this section apply to a "bare metal" Pulse install.

To ensure continuous availability of the Pulse NMS and all global bandwidth management (GBWM) features, the network operator may install a backup NOC. It contains the same Velocity components as the primary NOC — a NOC Pulse NMS and a GBWM cluster.

NOC component redundancy ensures that a failure of any component does not cause a service outage or degradation of Velocity system performance. For a description of component redundancy, see Component Redundancy on page 207.

5.4.1 NOC Redundancy Overview

The primary and backup NOCs operate in an active/standby mode. All components are powered up, reachable and managed by the active NMS. The NOC currently providing service for the satellite is said to be "active" and the other is said to be "standby" or "redundant".

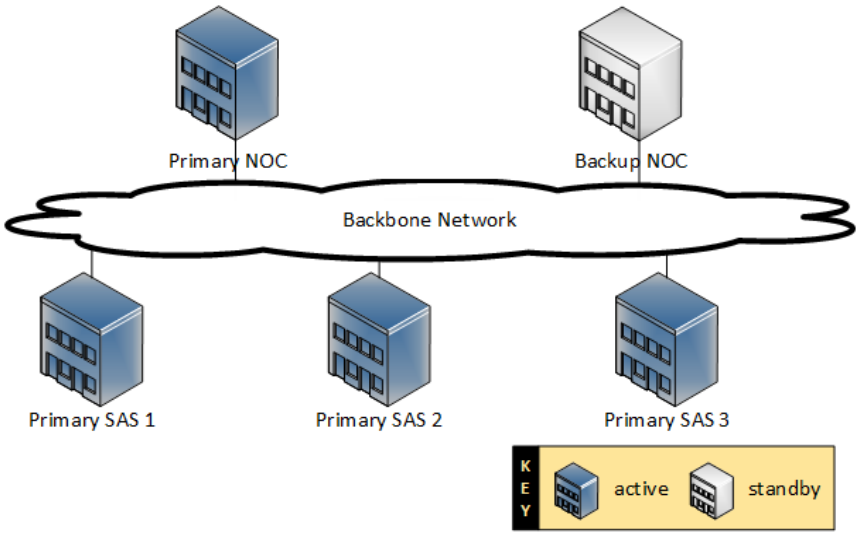


Figure 5-14. Redundant NOCs

The standby (redundant) NOC NMS is a warm standby - it is powered up, reachable and managed by the active NOC NMS.

The standby (redundant) NOC becomes the new primary (active) NOC when an operator activates the components within it. Switchover is under operator control to ensure database

integrity. The manual process of switching to a standby (redundant) NOC is generally performed from the active NOC, but can be performed from any of the following locations:

- Active NOC
- Standby (redundant) NOC
- Active SAS
- Standby (redundant) SAS

If a NOC is not reachable over the backbone network, an operator physically located at that NOC must perform the NOC switchover.

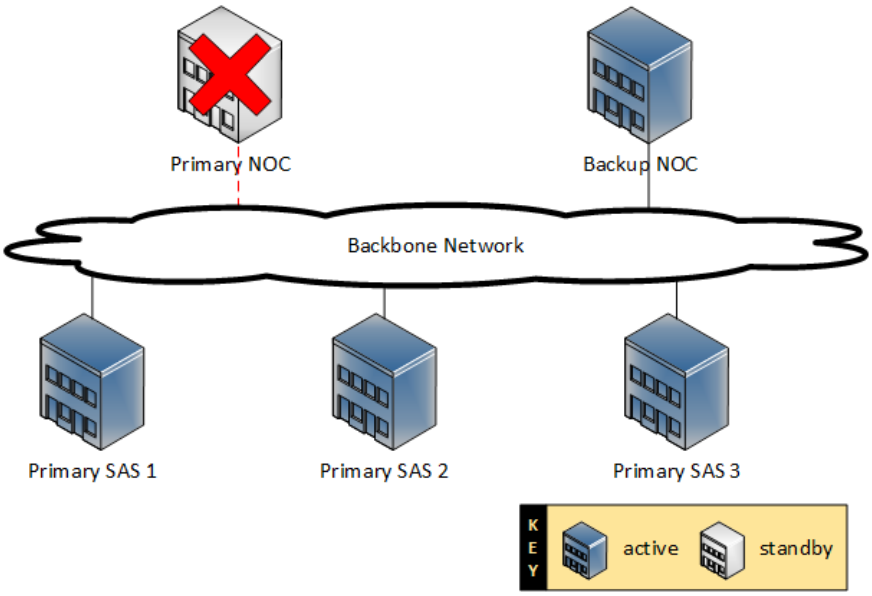


Figure 5-15. Velocity Network after a NOC Switchover

iDirect recommends a periodically planned exercise to validate the network provider's execution of the NOC switchover process and their associated documentation.

5.4.2 Understanding NOC Switchover

The standby (redundant) NOC contains the same data as the primary NOC. The active NOC NMS continuously replicates its databases to the standby (redundant) NOC NMS. It also provides the primary NOC configuration and backup NOC configuration to both sites. Thus, the only action needed to switch to backup NOC is to change the mode of the NOC components.

Prior to switchover of Primary NMS NOC to Backup NOC, the operator must perform the following:

- Switch off the Kafka Mirroring of topics from SASs to NOC K8S
- Wait until mirrored data is consumed by the NMS Processes if switch over is the planned
- Then stop the mirroring between Primary NOC K8S and Backup NOC K8S

Next step is operator failover of the NMS NOC to backup NOC cluster as described below:

- Put the active NOC into standby (redundant) mode
  - Disable the primary active NOC NMS
  - Change the mode of the primary GBWM cluster from active to redundant
- Put the standby (redundant) NOC into active mode
  - Change the mode of the backup NOC NMS from standby (redundant) to active
  - Change the mode of the backup GBWM from standby (redundant) to active
- Re-establish NMS database replication
  - Changing the mode of the disabled NOC NMS to standby (redundant)
- After the failover, restart the Mirroring from SAS to newly become Primary NOC and start the mirroring from new Primary NOC to backup NOC.



NOTE: The formerly active NOC NMS must have relinquished its role and stopped database replication before the other NOC is made active.

No action is necessary to cause the backbone network or SASs to become aware that the backup NOC is in use.

5.4.2.1 NOC Switchover — Service Implications and Timing

A NOC NMS switchover has no impact on customer traffic over the satellite because customer data and control protocols do not flow through the Pulse NMS. Configuration activities must cease during a switchover, however, and the flow of events and statistics from the SAS to the NOC are delayed. The SAS components will buffer events and statistics during the switchover.

The GBWM cluster is responsible for GBWM enforcement across SASs, and for Fair Access Policy (FAP). FAP will not be available for the duration of the NOC switchover procedure because it requires access to real-time statistics and events provided by the NMS. However, satellite network-wide GBWM enforcement will not be affected by a NOC switchover, because it does not require data from the NOC NMS. The GBWM enforcement feature functions in active-active mode and the PPs automatically send their requests to the standby (redundant) GBWM if the primary GBWM does not respond.

5.4.2.2 NOC Switchover — Backbone Network Traffic Requirements

During a NOC switchover operation, there is no spike in traffic on the backbone network. All active NOC NMS processes perform continuous regular block-level replication of stats and events data to the standby (redundant) NOC NMS. After the NOC switchover operation, the newly active NOC will receive an increase in traffic, as the SAS NMSs direct statistics, events, and GBWM queries to it.

5.5 NMS Redundancy and Failover

The NMS is responsible for all system configuration, control, monitoring, alarming, and reporting in the Velocity system. NMS redundancy and failover ensures that a failure of any NMS element does not cause an NMS service outage or degradation of system performance. In general, the NMS accomplishes this by using the underlying virtualization platform for reliable storage and virtual machine operations. The NMS then builds on this by cleanly recovering from virtual machine restarts, and monitoring processes to ensure they are restarted in case of failure.

The following sections provide information about redundancy and failover for NMS sites, servers, and services.

5.5.1 NMS Failure

An NMS site is a particular installation of the NMS that accomplishes all or part of the NMS functionality. Each NMS site is composed of a number of NMS servers that are executing software processes.

To prevent an NMS failure due to a single NMS site failure, the NMS system uses a distributed architecture with NMS stacks running both at the NOC and at each SAS.

A series of server failures or a catastrophic loss of site resources can result in a failed SAS or NOC requiring failover. However, neither type of failover is automatic. Both require manual action from an authorized user.

5.5.2 Service Failure

Each NMS service has a default home site, and an NMS process controller at each NMS site generates the proper NMS services for that NMS site. This method ensures the correct exchange of information between NMS sites. For example:

- The NOC provides configuration, historical, and alarm and incident data to the backup NOC.
- The NOC provides configuration data to the SAS.
- The SAS provides consolidated historical data from real-time data and alarm triggers to the NOC.
- The NOC provides (replicates) configuration data to backup SAS
- K8S external topics are mirrored from primary NOC K8S to backup NOC K8S.



NOTE: The NMS automatically restarts failed services.

5.5.3 Server Failure

iDirect uses virtualization technology for the NMS servers, together with process monitoring. The underlying virtualization platform provides server failure recovery. This allows the NMS to

remain available even with the following types of failures:

- Hardware failure
- Software failure
- Network failure



NOTE: Process failover due to hardware failure is automatic.

5.6 Component Redundancy

Component redundancy ensures that a failure of any component does not cause a service outage or degradation of Velocity system performance. In general, this is accomplished by:

- Redundancy for all physical servers
- Standby line cards in IF domains
- Redundant interfaces, hardware, and cabling
- Process resiliency
- Redundant copies of all system data

For details on NMS component and data redundancy, see NMS Redundancy and Failover on page 205. For details on datapath component redundancy, see the following sections.

5.6.1 Line Card Redundancy

The line card switchover process in the iDirect Velocity network is designed to avoid any network interruptions and to allow smooth network operations during hardware failures or manual switchover. Each line card is assigned to an IF domain, and line card redundancy is provided within an IF domain. An active line card supports one iNet.

Both ULCs and ILCs are assigned to IF Domains that model the physical connection between line cards and a Block Up Converter (BUC) or/and Block Down Converter (BDC). Line cards are also all dynamically assigned to iNets by the PP. Any extra ILCs, placed in the IF Domain Line, function as standby line cards.

Redundancy for the intelligent line cards (ILCs), is the same as that supported for ULCs.

5.6.1.1 IF Domain

In a Velocity network, An *IF domain* defines a specific IF frequency range that supports carrier operation on a satellite's outbound and inbound. Velocity supports what is referred to as "symmetric IF domains," and "asymmetric IF Domains."

A *symmetric IF Domain* is one in which inbound and outbound carriers are grouped into a single IF Domain, and where each IF Domain supports a combination of both Rx and Tx line cards, and is based on a one-to-one relationship with a single up-converter and down-converter.

An *asymmetric IF Domain* is one in which the configuration of both inbound or outbound carriers is possible. In this case, the NMS supports configuration of separate elements, for inbound IF Domains that contain only Rx line cards; and for outbound IF Domains that can only contain Tx line cards. Asymmetric IF Domains support all traffic flows in both CDC and SAS sites.

An IF domain can support one or more iNets. Any line cards in an IF domain that are not assigned to an iNet are standby line cards.

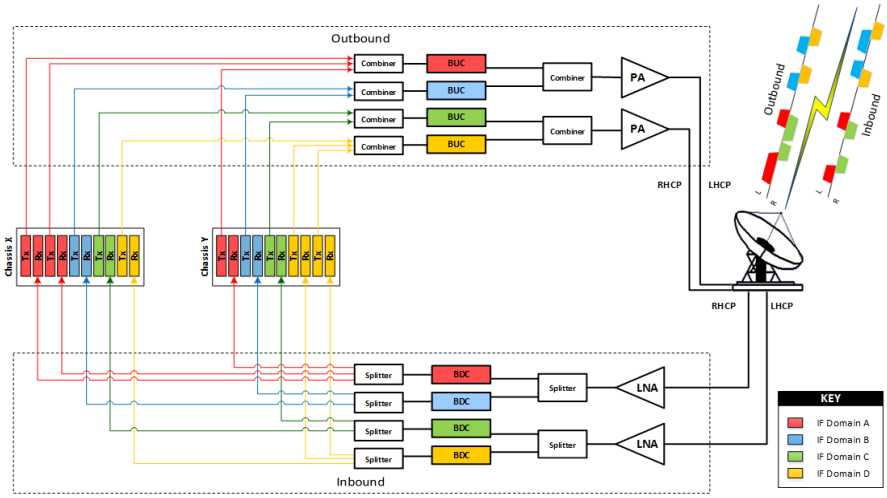


Figure 5-16. IF Domains

5.6.1.2 Line Cards in an IF Domain

All line cards configured in an iDirect Velocity network have their clocks locked together. A common pulse-per-second (pps) and a 10 MHz clock are distributed to all chassis. Additionally, a software process is involved to make sure that the network clock reference (NCR) times of all line cards are locked.

All line cards in the same IF domain can reside in any chassis and are not required to be placed next to one another in the same chassis to support an iNet.

The protocol processors (PPs) in the network are configured with the line card physical connection information. PPs know the number of available Tx and Rx line cards and their IP addresses configured in an IF domain. There is no static association between a particular iNet and line cards. When a PP gets a list of iNets that consist of one Tx and one or more Rx line cards per iNet, it starts assigning unallocated line cards to the iNets until it runs out of unallocated line cards.

The following diagram is an example of the line card distribution to IF domains in a single chassis. In this example, there are two different IF domains (A and B). IF Domain A supports one iNet with a primary Tx line card, a standby Tx line card, and two Rx line cards. IF Domain B supports two iNets, with one Tx line card and one Rx line card each. A tunnel connects the PP cluster to the active line cards. Traffic flows between the terminals and a Web server in the Internet or the subscriber's private network.

- Downstream traffic from the server flows to the PP through the upstream router, through the tunnel to the line card, and over the satellite to the user terminal.
- Upstream traffic from the terminal flows from the line card to the PP over the tunnel, and through the access router and backbone network to the server.

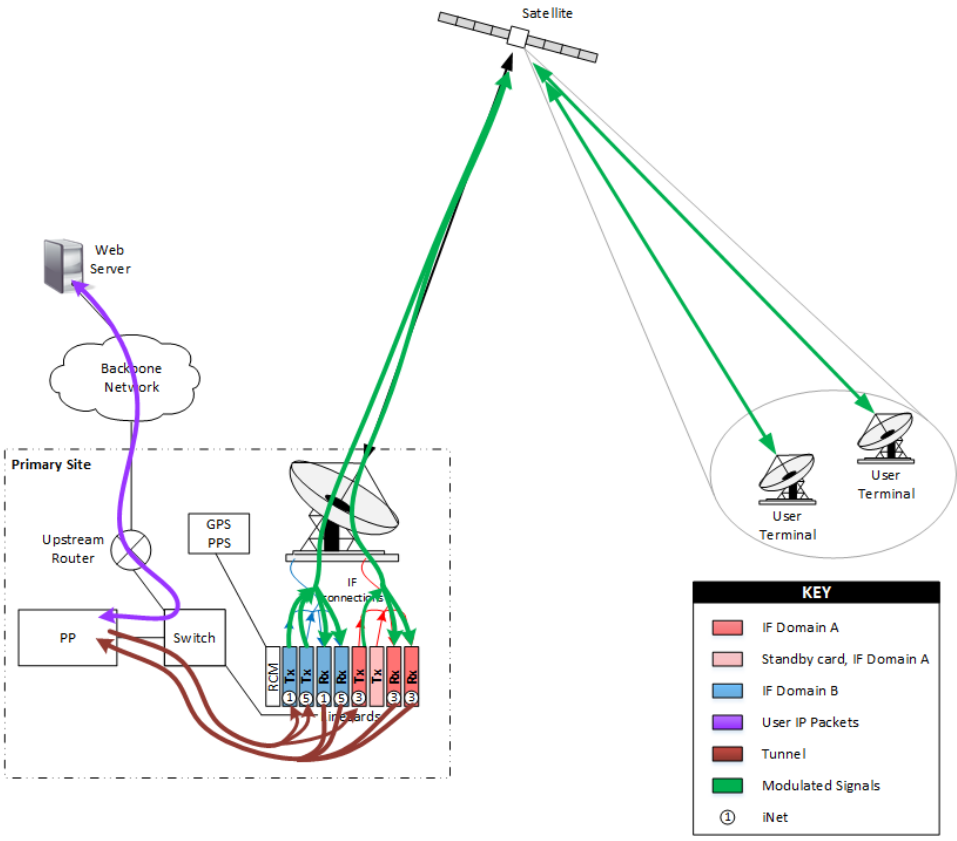


Figure 5-17. Line Cards in a Network

5.6.1.3 Line Card Switchover Conditions

Line card switchover occurs under the following conditions:

1. The Line Card Controller consistently misses heartbeats from a line card during a predefined period.
2. The Line Card Controller receives a Carrier Power Status message from the RFS Interface with an error flag set.
3. The line card self-reports a failure to the Line Card Controller.

5.6.1.4 Line Card Switchover Precedence

When an active line card fails due to a hardware failure, an event is generated, the chassis slot associated with the failed line card is powered off, the PP allocates a standby line card, and the line card fails over. No terminals should go out of the network, and minimal traffic loss would occur.

The precedence for recovering from line card failures is described below.

1. Switchover to an available standby line card in the same IF domain in the same chassis.

2. Switchover to a standby line card in the same IF domain in another chassis in the same SAS.
3. Switchover to a line card currently used by a lower-priority iNet in the same IF domain in the same SAS. The automatic line card switchover process repeats for that iNet.
4. Switchover all Tx and Rx line cards for the iNet to standby line cards in the same IF domain in the standby (redundant) SAS site.
5. Switchover all Tx and Rx line cards to line cards currently used by a lower-priority iNet in the same IF domain in the standby (redundant) SAS. The automatic line card switchover process repeats for the lower-priority iNet.
6. The iNet supported by the failed line card is lost, and terminals fall out of the network and must reacquire into a different iNet. Note that the iNet is lost, even if there is a standby line card available in a different IF domain.

5.6.1.5 Failed Line Card Replacement

To replace a failed line card, the following steps are performed:

1. When a line card failure is detected, the PP marks the failed line card as unavailable and powers it off.
2. The network operator replaces the failed line card with a working line card or analyzes the card to find and fix the issue.
3. After replacing the line card or resolving the issue with the failed line card, the network operator powers on the replaced or fixed line card.
4. Once the PP is able to communicate with the line card, the line card becomes a standby line card.

5.6.1.6 Line Card Switchover Scenarios

Details of line card switchover depend on the line card availability of an iNet within an IF Domain. Line card switchover scenarios are described in the following sections.

Switchover to Available Standby Line Card

When an active line card fails, an event is generated, the chassis slot associated with the failed line card is powered off, and the line card fails over. The terminals will not go out of the network and there would be minimal traffic loss.

In the following diagram, IF Domain A supports one iNet with an active Tx line card, and two Rx line cards. It also contains a standby Tx line card.

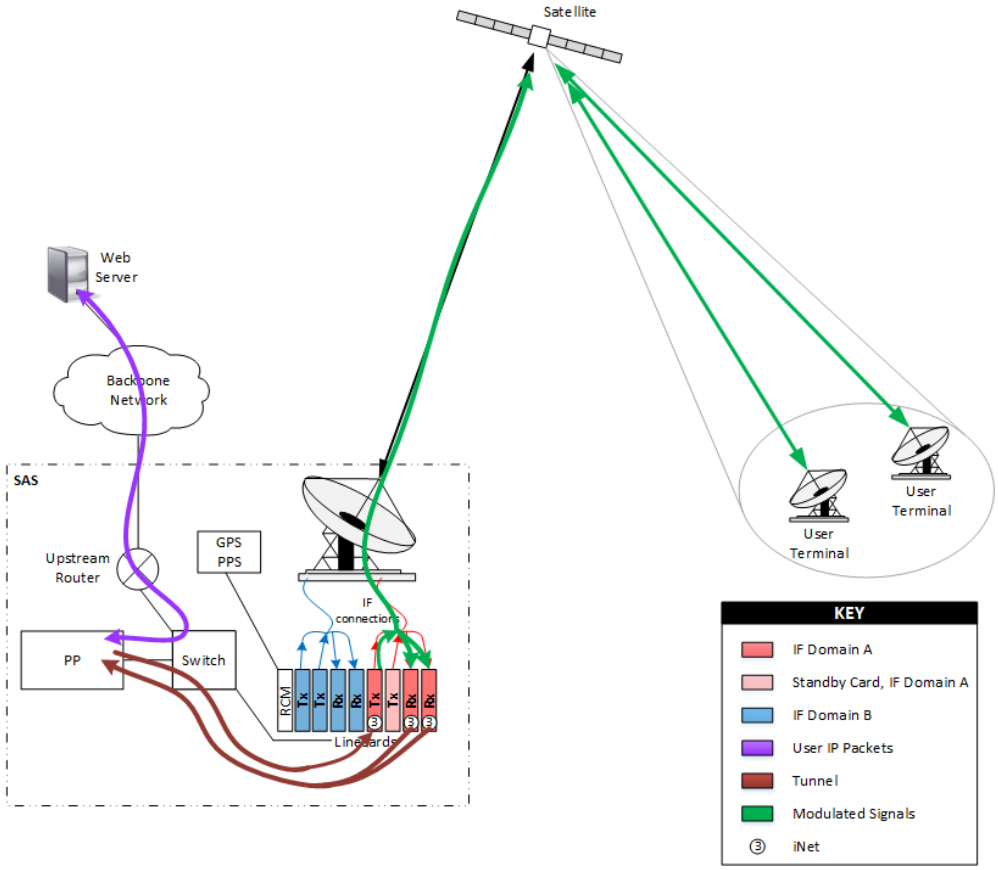


Figure 5-18. Available Standby Line Card Scenario — before failure

The active Tx line card in IF Domain A fails. Because there is an available standby line card in the IF domain, the system switches over to it.

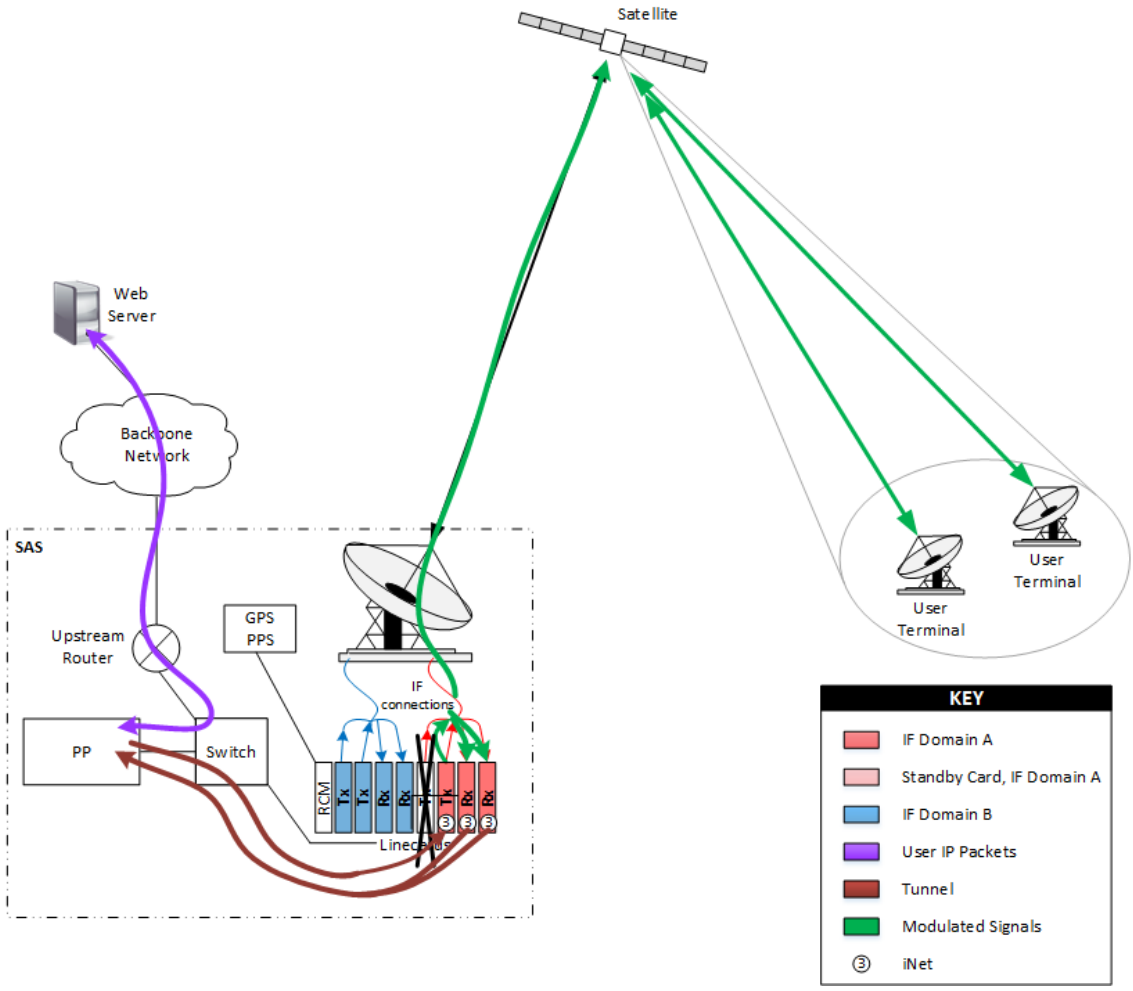


Figure 5-19. Available Standby Line Card Scenario — after failure

Switchover based on iNet IF Domain Priority

In the scenario depicted in the following diagram, there are two iNets supported by the two Tx and Rx line cards in IF Domain B. The iNets are priority of 1 and priority of 5. There are no standby cards available in this IF domain.

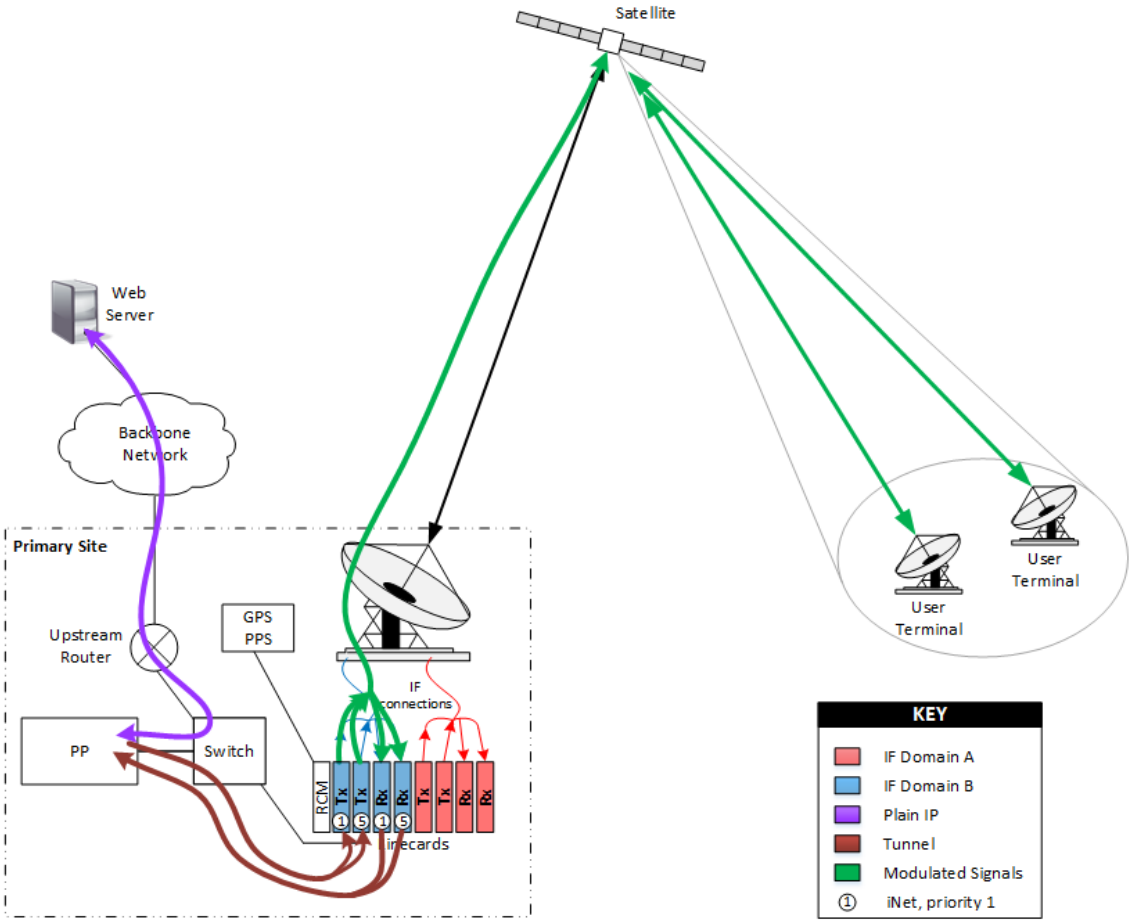


Figure 5-20. Failover based on iNet Priority — before failure

When a failure occurs, and the failed line card belongs to an iNet with a lower priority (i.e., Priority 5), and there is no standby line card available in the same IF domain, that iNet fails.

When a failure occurs, and the failed line card belongs to an iNet with a higher priority (i.e., Priority 1), and there is no standby line card available in the same IF domain, then the line card from the lowest-priority iNet is used for the switchover; thereby, causing the lowest-priority iNet to fail.

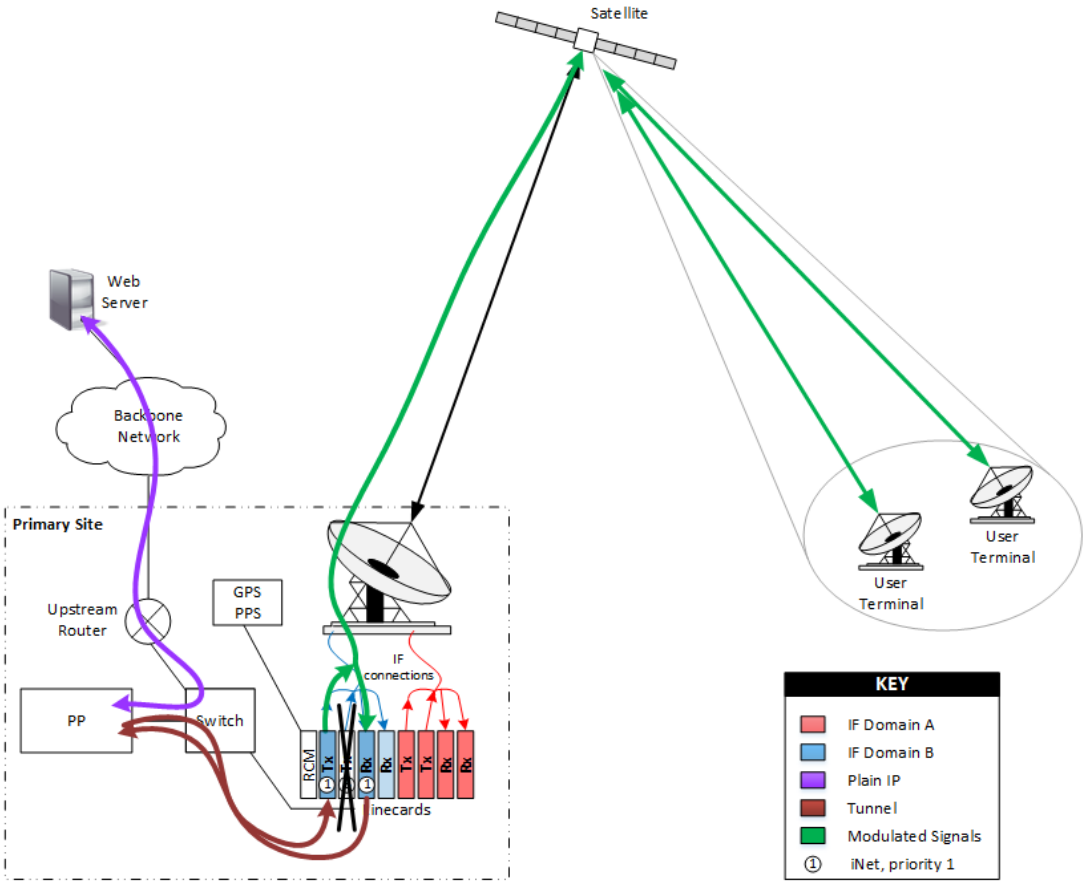


Figure 5-21. Failover based on iNet Priority — after failure on Low Priority iNet

	NOTE: The lowest-priority iNet waits for a standby line card to become available.

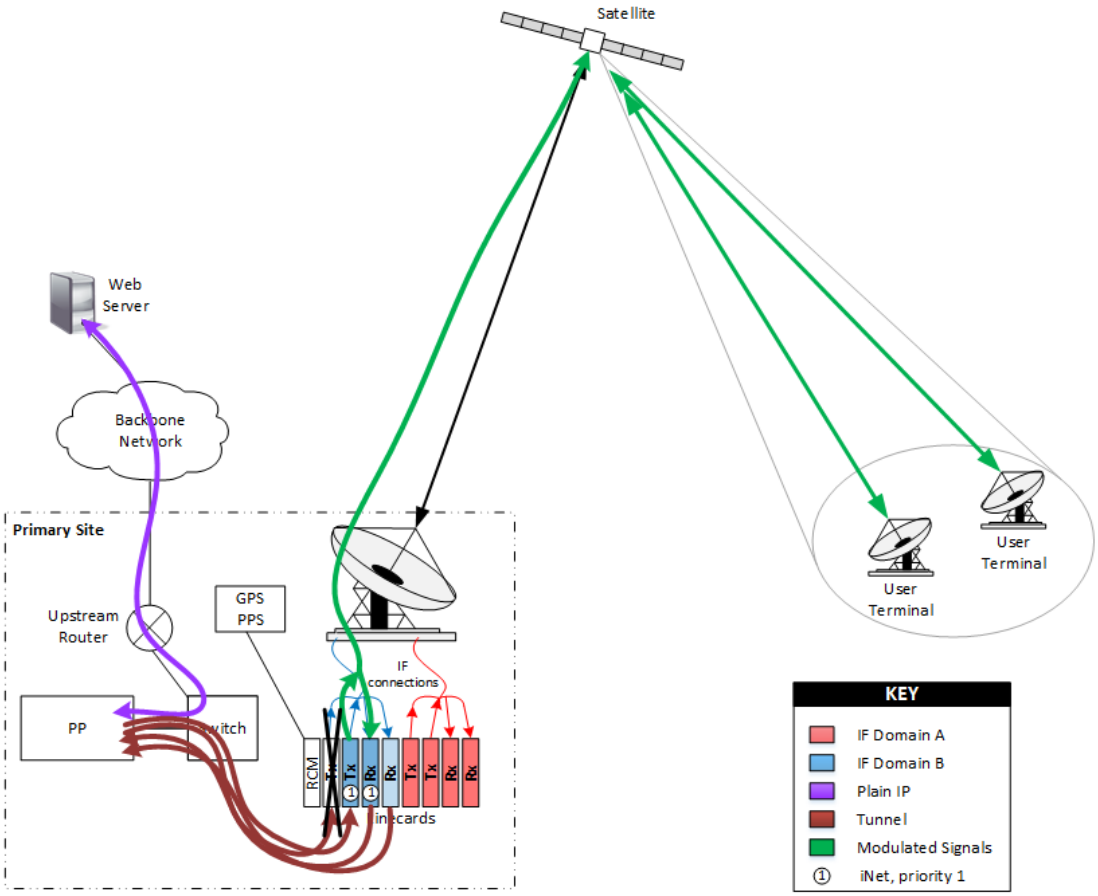


Figure 5-22. Failover based on iNet Priority - after failure in High Priority iNet

Switchover to Line Card in Standby Site

In this scenario, there are no standby line cards available in the active site. When an active line card fails, the PP will grab the first available standby line card in the same IF domain from the standby (redundant) site. For example, as illustrated in the following diagram, a line card on iNet Domain A from the active site has failed and the PP grabs the standby line card from the standby (redundant) site.

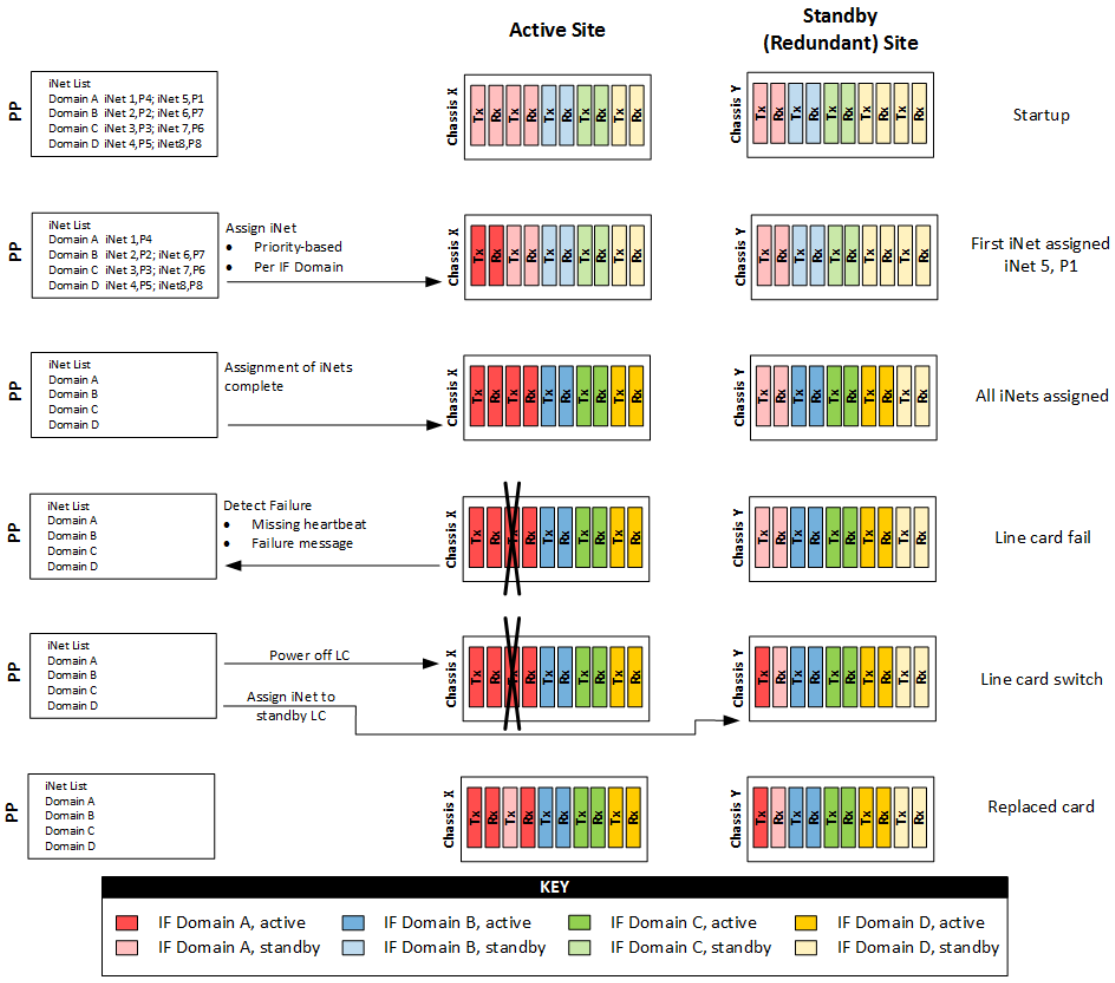


Figure 5-23. Switchover Process from the Active Site to the Standby (Redundant) Site

5.6.2 PP Redundancy

In the Velocity system, the Protocol Processor (PP) cluster is located in the SAS or CDC site, and where applicable, the SAS or CDC backup site.

5.6.2.1 PP Server Redundancy

If a PP server should fail for any reason, all terminal-specific and iNet-specific processes will restart on the remaining PP servers. When the failed PP server recovers, it re-enters the PP

cluster, and the cluster software assigns processes to it. A PP server failure does not cause any terminals to drop out of the network but may cause minimum traffic loss.

5.6.2.2 Process and Data Resiliency

If a PP process should fail for any reason, the cluster software will automatically restart it. The impact of a process failure on the rest of the PP processes and the satellite terminals depends on the purpose and design of the process:

- ***Process failure interrupts traffic*** — When these processes fail unexpectedly, traffic to and from the terminals is interrupted, but they should remain in the network and traffic should recover. The traffic impact should be minimal.
- ***Process failure has no impact on the remote or customer traffic*** — When these processes fail unexpectedly, there should be no impact on customer traffic and all terminals should remain in the network.
- ***Inter-related processes*** — The system recovers as expected when combinations of processes fail.

All data stored by PP processes is replicated across multiple servers. This data includes checkpoints made by key processes such as the terminal proxy.

There is no single point of failure for transmission of statistics and events to the NMS; there are two proxy processes for each data stream on different servers in the PP cluster for redundancy and load balancing. If the NMS is not reachable, the proxy process will buffer statistics and events, retransmitting when the NMS is again available.

5.6.3 Global Bandwidth Management Redundancy

5.6.3.1 GBWM Process and Data Resiliency

If a GBWM process fails for any reason, tit is automatically restarted by the cluster software.

Like the PP cluster, all data is replicated across the servers in the cluster, and there are multiple proxy processes for transmission of statistics and events to the NMS.

5.6.3.2 GBWM Server Redundancy

If a GBWM server should fail, its processes will restart on a remaining GBWM server. When the failed server recovers, it will re-enter the GBWM cluster, and be available to protect against another server failure.



NOTE: The GBWM cluster is located at the NOC site, and if applicable the NOC backup site.

5.7 Network Resiliency

5.7.1 SAS Network Resiliency

The physical SAS network uses redundant switches, routers, links and ports to provide a highly available network. The following diagram shows a highly available reference SAS Network Architecture using Cisco Nexus switches.

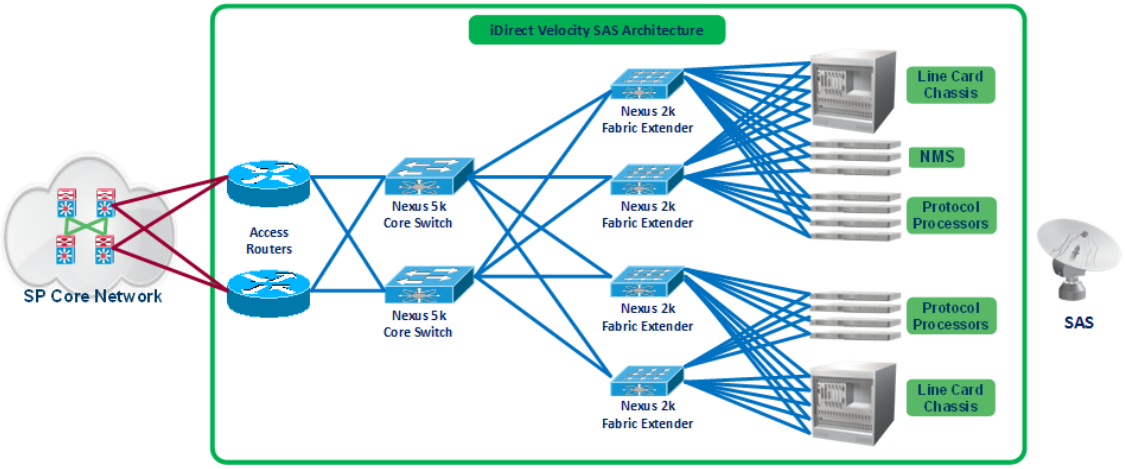


Figure 5-24. A Sample SAS Network Architecture

Each Velocity and Pulse server provides two physical ports for connection to the SAS switching fabric. These ports are bonded using Link Aggregation Control Protocol (LACP) to dynamically form a dynamic Link Aggregation Group (LAG) or with the switch as per 802.3ad. Similarly, each line card provides two physical ports for connection to the SAS switching fabric. These ports are bonded but do not support LACP; static port groups must be configured on the switch.

The SAS LAN is architected with a redundant switching fabric it uses on redundant SAS switches as well as key features of that networking equipment to provide a highly available service. These features include Virtual Port Channel (VPC), LACP, Port Channel, etc. In the example architecture, shown in the above diagram, every server and line card is dual-homed to two top-of-rack fabric extenders, which are in turn dual-homed to the core switching fabric, using the Nexus switch enhanced VPC feature.

This diagram also shows a pair of access routers. These routers are also dual-homed to the SAS switch. Typically, the SAS access routers connect to the backbone network via a redundant pair of edge routers and standard IP and/or MPLS (Multi-Protocol Label Switching) protocols. The use of these protocols and BGP between the access router and PPs ensure that data connectivity is preserved even if a router should fail.

5.7.2 NOC and EAP Network Resiliency

The NOC and EAP network architectures are designed by the network operator and usually support not only the Velocity GBWM and Pulse servers, but also other co-located management and back office systems. The NOC network is designed to meet high-availability requirements and, therefore, the NOC network contains redundant links and redundant routing and switching equipment.

The connection between the SASs and the NOC is likewise expected to meet high-availability requirements, thus, multiple connections to the backbone network are typically provided. Generally, these connections are to redundant edge routers.

6 Pulse NMS Operations

This chapter describes iDirect Pulse® Network Management System (NMS) operations for the iDirect Velocity™ network. The chapter includes the following sections:

6.1 About Pulse NMS 222
6.2 Pulse NMS Features and Operations 223
6.3 Virtualization/NMS Cluster 227
6.4 About SAS NMS Processes 229
6.5 About NOC NMS Processes 233
6.6 External Access Portal NMS 234
6.7 Network Monitoring and Reporting 235
6.8 NMS Troubleshooting Tools 237

6.1 About Pulse NMS

The iDirect Pulse NMS is designed to encompass a distributed set of components that operate on similar and geographically-separated server clusters at the Network Operations Center (NOC) and Satellite Access Station (SAS) locations, all interconnected over a high-bandwidth backbone network, as shown in the diagram.

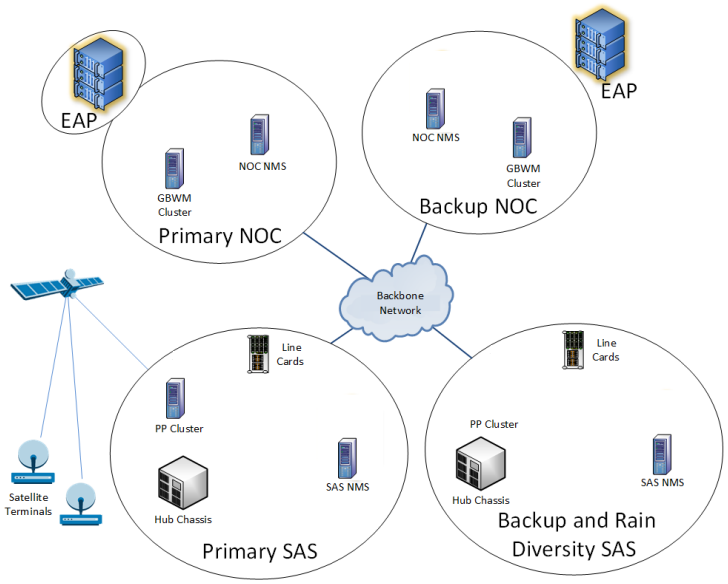


Figure 6-1. Pulse NMS Distributed Components

6.1.1 The NOC NMS

The ***NOC NMS*** is responsible for normal operational management of the entire network managed by Pulse. It is also responsible for maintaining the master copy of the configuration database for the network, monitoring events from all network elements, collecting historical stats from all SAS NMSs, collecting real-time global QoS statistics, and storing stats long-term. It is also responsible for generating thresholding events from received stats, storing stats short-term, and storing received events. In addition, the SAS NMS is responsible for latency measurements and connectivity for troubleshooting network elements.

Finally, the NOC NMS generates alarms from received events and statistics, and provides EAP user account access interfaces to the EAP NMS for control and reporting to external systems.

6.1.2 The NMS External Access Portal

An ***External Access Portal (EAP)*** to the NMS, which resides outside of the primary NOC, is responsible for providing NMS functionality for all users and systems that are external to the service provider. The Global Bandwidth Manager is located at the NOC, but it is not an integral part of the Pulse NMS. The EAP also supports limited access to NMS services for service providers and end-customers.

6.1.3 The SAS NMS

The ***SAS NMS*** is responsible for management of SAS-level operations of the SAS network and all satellite terminals within the footprint of the satellite associated with the SAS when the NOC is not accessible. The SAS NMS is responsible for the real-time collection of stats and events at the SAS and all satellite terminals associated with the SAS.

6.2 Pulse NMS Features and Operations

The Pulse NMS provides functionality for configuration, control, monitoring, alarm handling, event logging and reporting functions of the network equipment. The Pulse NMS provides a complete set of APIs and a web-based user interface for these purposes. A custom graphic user interface (GUI), or operations interfaces, to connect OSS/BSS systems can be produced using the APIs as per customer requirements.

The Pulse NMS features a fully web-based user interface for the operator, as well as a full-featured RESTful API using JSON encoding that enables machine-to-machine communication. Users accessing the system via the API and users with the same permissions accessing the system via the web-based user interface are considered equivalent and will have the same options. Pulse supports multiple users with various permission-based access.

As the main vehicle for managing and controlling iDirect networks, the Pulse NMS is the primary interface to all service provider systems, and allows network operators to configure and control iDirect networks, as well as to perform many system monitoring, reporting, and troubleshooting functions.

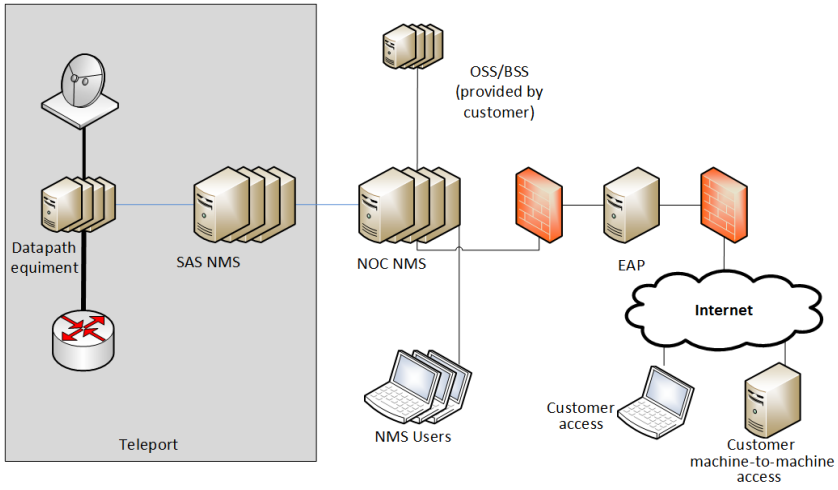


Figure 6-2. Pulse NMS Operations

The SAS is linked by a high speed data network. The NOC is linked into the SAS network for management purposes.

NMS features include:

- Monitoring and configuration of all NMS-controlled network elements
- Configuration of network behavior and resources

- CSV export of network element configuration data records
- Software updates and version management
- Secure portals with controlled access and visibility

6.2.1 NMS Browser Support

Pulse fully supports the following browser versions to establish Pulse NMS user sessions:

- Google Chrome — two most recent versions
- Mozilla Firefox — two most recent versions

6.2.2 User Sessions

After successfully entering a user name and password, users are authenticated by the system, the NMS establishes a user session, and the web interface opens to the configured landing page. The user account or user group determines the landing page to which the NMS opens.

User access privileges are based on the access privileges of a user's assigned user roles and user groups. This provides a very fine grained access control. Machine-to-machine access uses the same users and user permission system.

User sessions in which there is no activity for 30 minutes, or some other configured duration, are automatically terminated and logged out. Activity is considered any action that directly causes the browser to load contents from the server or triggers the normal asynchronous communication from the browser to the NMS server.

6.2.3 NMS Web UI for Configuration and Control

Primary access to the iDirect Velocity network is via a graphical web-based user interface. The NMS user interface supports simultaneous access by multiple users with defined access permissions and views. The web interface supports configuration and control, monitoring, and reporting capabilities for terminals and for all of the iDirect infrastructure, for example , chassis and line cards, protocol processors, and NMS servers.

The NMS web user interface employs a layered hierarchical model in which configurable NMS elements are grouped into collections of related elements. Of the many element domains, represented in the NMS, the Physical, Transport, Network, and Service domains form the primary infrastructure of the Velocity network. Together, these domains provide essential services to the Satellite Terminal domain. In turn, the Satellite Terminal domain provides IP connectivity between each remote LAN and the SAS.

The Pulse Web-based user interface is split into a number of separate areas:

- **Configuration** — Provides access to Pulse NMS pages and menus used to configure the various elements of each element domain, for the purpose of building a network. These element domains include Physical, Transport, Network, Service, and Terminal.
- **Monitoring** — Provides access to Pulse NMS pages and menus used to generate a variety of real-time reports on network events, alarms, incidents, and both status and statistics.

- **Reporting** — Provides access to Pulse NMS pages and menus used to generate a variety of real-time and historical reports on all network events, alarms, and incidents; as well as historical status and statistics.
- **Troubleshooting** — Provides access to NMS probes for NMS and PP clusters, line cards, and terminals, and provides a debug console for direct shell window access to elements.
- **NMS Management** — Provides access to NMS pages and menus used to configure and manage User Administration domain elements (such as user groups, user roles, user accounts, customers), and System Administration domain elements such a s certificates and software updates and version management.

6.2.4 Kafka Streaming

Apache Kafka® is a distributed streaming platform that supports two broad classes of applications — the first is building of real-time streaming data pipelines to reliably move data between systems or applications; and second, is building of real-time streaming applications to transform or react to streams of data. The first of these two general applications represents the Kafka implementation in Pulse 2.6.

Kafka both publishes and subscribes to streams of records, similar to a message queue; it stores streams of records, using fault-tolerant methods; and it processes streams of records as they occur. This combination of messaging, storage, and stream processing is essential to the Kafka role as a streaming platform.

Kafka is run as a cluster on one or more servers that can span multiple sites. The Kafka cluster stores streams of records in categories called topics, where each record consists of a *key*, a *value*, and a *time-stamp*. Client/server communication is accomplished with a simple, high-performance, language-agnostic TCP protocol. The Kafka implementation in Pulse operates using three core APIs:

- **Producer API**: allows an application to publish a stream of records to one or more Kafka topics — it pushes messages from the application into different topics.
- **Consumer API**: allows an application to subscribe to one or more topics and process the stream of records produced to them — it enables applications to read streams of data from topics in the Kafka cluster.
- **Connector API**: allows building and running re-usable producers or consumers that connect Kafka topics to applications or data systems.

Kafka can be enabled to manage the following:

- Enable Kafka for Status and Stats
- Enable Kafka for Events/Alarms

With Kafka enabled for Stats/Status, Events/Alarms, users can then subscribe for events/alarms/status/Real time stats/Historic stats from the respective topics as described in in detail in the document *Pulse APIs for iQ-series, 7-series, and 9-series Satellite Routers*.

6.2.5 User Groups and Roles

In an iDirect network, a ***user group*** is a collection of users with a common set of permissions and visibility. Group permissions determine which generic functions its members can perform and exactly what elements upon which these functions may be performed.

Visibility determines what group members are able to view or see in the NMS. Each user group and its members can be isolated from one another, with each having privacy from all others.

As user accounts are defined, each user is assigned membership to one or more user groups. Group membership, in turn, manages user access to the NMS. More specifically, the user group prevents users from gaining access or visibility outside of what has been allowed or assigned by the permissions assigned to the group. In essence, the user group constrains what elements a user can access.

A ***user role*** is a specific set of defined permissions that can be assigned to a user. It defines what functionality, within the NMS, can be seen or used by the user to which the role is assigned. Since roles are assigned at the creation of the user account, the user role must already exist.

The creation of a user role defines a new set of permissions. The idea is to define roles that align with the specific responsibilities of the different real-life roles in terms of working with the iDirect network. By creating multiple roles, each with specific permissions, users are allowed access and functionality based on work responsibilities.

The ***System Administrator role***, which is predefined and created when the system is installed, is permitted to create and manage network infrastructure elements, network security elements, and events and stats management profiles. This role also handles the management of network element software versions, as well as installs and upgrades.

6.3 Virtualization/NMS Cluster

iDirect Pulse utilizes virtualization technologies to minimize the time and cost of deployment and allows for improved system management for IT staff. Virtualization enables a multi-site distributed architecture that broadens flexibility and enhances resiliency.

6.3.1 Virtualized Modular Architecture

iDirect Pulse utilizes virtualization technologies to minimize the time and cost of deployment and allows for improved system management for IT staff. Virtualization enables a multi-site distributed architecture that broadens flexibility and enhances resiliency.

iDirect Pulse is configurable to support a variety of system deployments. This ranges from a single site installation of iDirect Pulse operating a small network, to complex global operations requiring management of multiple NOCs (primary and backup) and SASs where a dedicated instance of iDirect Pulse is running at each physical location.

Pulse relies on the underlying virtualization platform to provide operational resilience, and hardware redundancy. By leveraging the platform's capabilities, customers have flexibility in the level of resiliency and redundancy in their Pulse network based on their operational requirements. Should a physical host fail, the platform's high availability system, if configured, will restart the virtual machines on other physical hosts. After the virtual machines have restarted, Pulse integrates them back into the cluster and resume operations.

An example architecture for a Pulse deployment is shown in the following diagram.

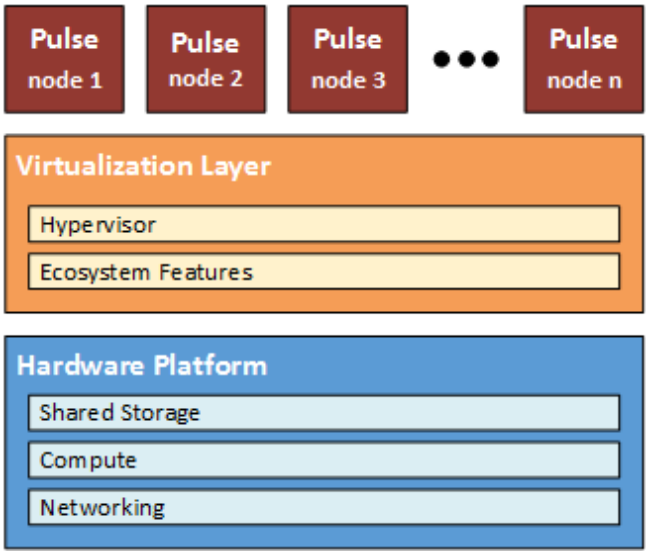


Figure 6-3. Virtualized Modular Architecture

Pulse also relies on the underlying platform to provide backups. This implementation can be integrated with a customer's existing backup strategy.

6.3.2 Robust System Security

As part of the Pulse design, all communication is encrypted using Transport Layer Security (TLS). This encryption includes external communication through the Internet, as well as internal machine-to-machine communication within iDirect Pulse itself.

By default, the system implements an internal authentication process that enforces password rules and validates users into the system through a login process.

Once a user is within the system, iDirect Pulse restricts each user's access on a "need to know" basis through a highly customizable user-groups-and-role-based mechanism. Users accessing the system via the API and users with the same permissions accessing the system via the web-based user interface are considered equivalent and will have the same options. Each user is granted visibility only to authorized network elements, and within those network elements, is only authorized for the functions they need to perform their duties. In addition, the NMS logs all user-initiated updates.

6.4 About SAS NMS Processes

The Pulse NMS is composed of a number of server processes, each of which performs a specific set of back-end functions vital to the coordination and management of the overall operation of the NMS.

The following NMS processes, illustrated in the following diagram, operate in the SAS:

- Authentication Manager
- Configuration Manager
- Discrete Data Engine
- Stats Manager
- Update Manager
- Web Core
- Latency Measurement Manager
- RT Command

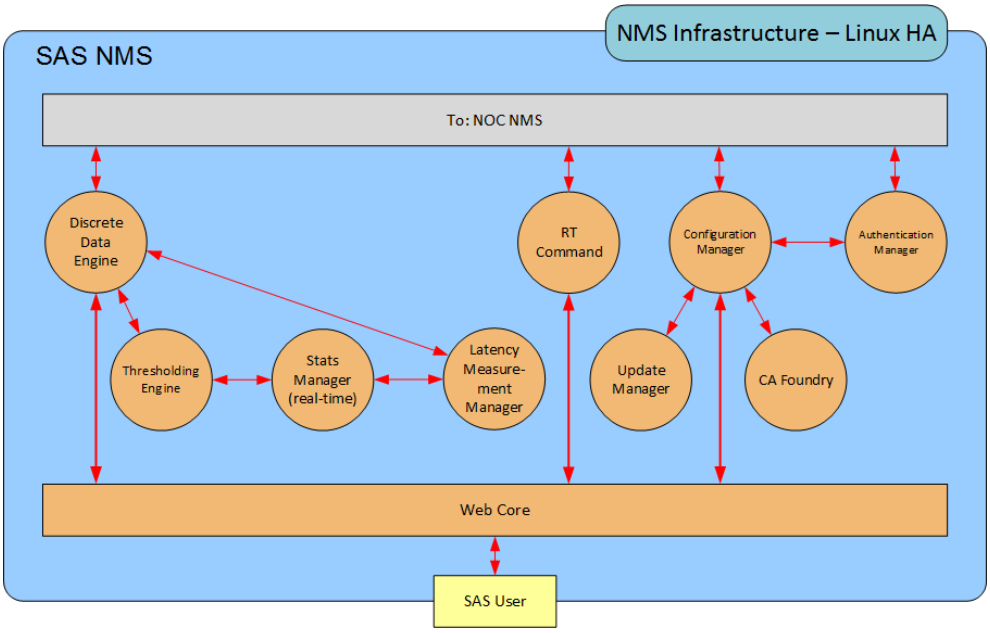


Figure 6-4. SAS NMS Processes

6.4.1 Authentication Manager

The authentication manager process serves as the authentication source in the Velocity network and is responsible for checking whether a user or process has adequate permission to perform a request. Each logged-in user or process is given a session ID that is valid until the session expires or until the user logs out. All requests initiated by this user or process are submitted with their session ID.

6.4.2 Configuration Manager

The configuration manager process is the key component of network modeling and the central storage place for all NMS configuration data. It provides an interface to all configuration data and enforces data validation and integrity. It is also responsible for generating option files for all network elements and forwarding them to the Update Manager for distribution.

The configuration manager runs in read/write mode on the NOC site only. Each SAS also has a configuration manager installed, and running in read-only mode. The SAS configuration manager stays in sync with the primary NOC using replication.

Changes to the configuration of the network are first stored as pending changes. These changes can then be applied, which pushes the changes to the network equipment via the Update Manager. Any pending changes are shown in the web-based user interface, to allow an operator to view what will change when changes are applied. Changes may also be viewed through the Impact Analysis tool, which shows which network elements will be affected by applying a change.

Incomplete or inconsistent changes may be saved in a pending state. This allows an operator to make a series of changes and save the intermediate steps. To ensure that the active configuration is always consistent, the full configuration will be checked before allowing the operator to apply the change. This step also performs validation checks to ensure the configuration being applied to the network passes business rules. A log of all changes is stored, together with the user who made the change and the time.

It is possible to “undo” any change, provided that undoing the change leaves the network in a consistent, valid state. Undo is implemented by changing values in the pending configuration back to their previous values. This allows an operator to see the effect of the undo and make any further required changes before applying the change to the network. Once a change is applied, it can be undone by making necessary changes based on the latest changes in the activity logs. Any changes will still be processed through the validation logic to ensure the network stays in a consistent state.

Every entity configurable through the web-based user interface or Web Services have one of the following configuration statuses:

- **Nominal** — There are no outstanding changes to this element, and all previous changes have been applied to the network.
- **Changes Applied** — A change has been applied to the element, and the system has already copied pending records into the active database.
- **Changes Pending** — A change has been made to this element, and the change has not yet been applied to the network.

- **Update Pending** — A change has been applied to the element, and this change is in the process of being pushed to the network.
- **Incomplete** — This object has changes that do not pass validation rules. (This state permits a user to save elements as a work in progress.)

6.4.3 Discrete Data Engine

The discrete data engine (DDE) process receives all event messages from any initiating source, including other NMS processes, SAS processes, network components, and external systems. The DDE process generates alarms and incidents based on rules and algorithms. An extensible action framework automatically invokes actions and notifications for events, alarms, or incidents.

Data path generated events are pushed from the data path to the DDE in the SAS NMS, where they are stored. The NMS state model at the SAS evaluates these events and converts them into the appropriate alarms. Alarms are pushed to the NOC NMS, where they are stored. Optionally, alarms may be grouped together into incidents for easier handling.

When the Kafka for events is switched on, the events generated at the SAS or received from the datapath are pushed to Kafka topic “event”. The event topic is mirrored to the NOC by Kafka Mirror Maker running at NOC. When kafka is On for events, events processing will not happen on SAS so no events are stored at SAS.

6.4.4 Stats Manager

The stats manager process handles querying, caching, and storage requests for statistics generated by any source. Multiple Stats Manager instances are created dependent upon the setting of the Stats Manager instantiation parameter, which is configured based on the estimated size of network elements. An instance of Stats Manager may be used for either historical statistics or for real-time statistics.

In order to reduce storage requirements, some statistics are only stored when their value changes. These are referred to as status metrics. An example of a status metric is the beam a terminal is currently operating in, which is expected to change at most once every few minutes. In other words, ***status*** refers to ***statistics*** that do not change very often, and the system is only interested in their current value and when they change.

The stats manager process is located in the NOC and at each SAS. The NOC will host two different processes, one for historical data and one for real-time global QoS statistics. Each SAS instance is dedicated to real-time and streaming requests for both statistics and status.

Real-time data is stored at the SAS at 30-second resolution. Historical data is stored at the NOC at 5-minute resolution. Stats management profiles determine how long stats and statuses are stored. The configuration of the stats management profiles and the number of elements in the network determine the storage utilization for the NMS. Typically, important stats will be stored for 10 days at the SAS and for 400 days at the NOC. Lower importance stats would be stored for 2 days at the SAS and 10 days at the NOC.

When the Kafka for Stats is switched on, the stats/status generated at the SAS or received from the datapath are pushed to Kafka topic “stats”. The stats topic is mirrored to NOC K8S

by Kafka Mirror Maker running at NOC K8S. When Kafka is ON for Stats, RT stats (of resolution 30 secs) processing will not happen on SAS so no RT stats are stored at SAS.

6.4.5 Update Manager

The update manager process delivers configuration information to network elements — for example, option files, software and firmware packages, OS images, and licenses. It queries those elements about configuration status and sends commands to those elements to apply delivered configurations as required.

6.4.6 Web Core

The web core process runs the NMS web-based user interface and the API. Its main function is to handle HTTP requests on a single IP address. It distributes the processing load across multiple web servers and implements a reverse proxy to provide a single public-facing location for the server.

6.4.7 Latency Measurement Manager

The latency measurement manager process performs periodic polling of all network elements based on configured parameters. It uses Internet Control Message Protocol (ICMP) Echo requests to measure the round trip time (RTT) in the same way the ping utility does. It also generates and stores latency statistics.

6.4.8 RT Command

The RT command process allows engineers to debug network elements by using real-time probe commands or by allowing proxy access (for example , SSH) to specific elements. The functions provided are not part of normal operation but are available for troubleshooting. Operators use a Web browser to log into the web server to define and maintain the network. The only public interface is the web core process.

6.5 About NOC NMS Processes

The following NMS processes, illustrated in the following diagram, operate in the NOC site:

- Authentication Manager
- Certificate Authority (CA) Foundry
- Configuration Manager
- Discrete Data Engine
- Stats Manager
- Thresholding Engine
- Update Manager
- Web Core
- RT Command

These processes, in addition to the Certificate Authority (CA) Foundry and the Thresholding Engine, are identical to those processes that operate in the SAS NMS. However, the Latency Measurement Manager, which operates in the SAS NMS, is not in the NOC NMS. The processes listed above, with the exception of the CA Foundry and the Thresholding Engine, are described in About SAS NMS Processes on page 229 and its subsections.

6.5.1 Certificate Authority (CA) Foundry

The CA foundry process creates, renews, and revokes all X.509 certificates created for communication among SAS network elements and maintains intermediate certificates to sign certificates for each network element.

6.5.2 Thresholding Engine

The thresholding engine process performs continuous monitoring of all real-time statistics received by its peer Stats Manager, based on centrally-configured rules. These rules specify how to compare a subset or all statistical values against thresholds and, if a threshold is exceeded, report a formatted event to the Discrete Data Engine for further processing.

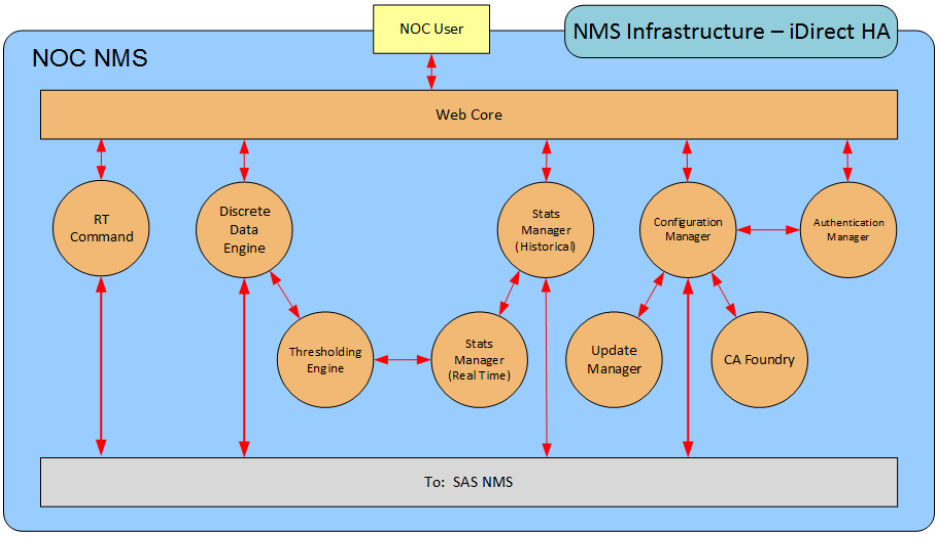


Figure 6-5. NOC NMS Processes

6.6 External Access Portal NMS

The External Access Portal (EAP) is an NMS system that resides outside of the primary NMS system, but supports access to the primary NMS via the EAP NMS Proxy. Each instance is responsible for providing NMS functionality for all users and systems that are external to the service provider, including secondary service providers and subscribers.

The EAP, although it has no direct access to a real datapath or devices, runs processes as listed below:

- Discrete Data Engine Proxy
- Stats Manager Proxy
- Update Manager Proxy
- Web Core

The EAP is a separate instantiation of the NMS that runs in a mode specifically designed to provide access to limited NMS configuration, statistics, and reporting functionality for secondary service providers, subscribers, and sales representatives. The EAP also has configuration capabilities that allow an EAP user with the correct permissions to create, modify, and delete network elements. The data source for the EAP is the NOC/backup NOC NMS and not the datapath itself. The NOC forwards data to the EAP on a regular basis (by default, every 5 minutes). As the EAP has a local data store for stats and events, it prevents large report queries from affecting the main NOC NMS. EAP user accounts are configured in the NOC.

The EAP is comprised of NMS servers that provide system configuration, control, monitoring, alarming, and reporting and the interface to all service provider systems, NMS stat storage servers, and NMS storage/compute servers.

EAP access for service providers is based on a variety of factors, such as contractual agreement, experience level, prior relationship with the service provider, number of deployed user satellite terminals, etc. Service providers will typically use the EAP NMS for one or more of the following purposes:

- Configuring group service plans to resell capacity on a wholesale basis
- Configuring subscriber service plan profiles for sale to individual subscribers or subscriber groups
- Reselling service provider-defined subscriber service plan profiles to subscribers
- Reporting on the way their group service plans have been used by user satellite terminals in the groups
- Monitoring the health and status of their satellite terminals and satellite terminals managed by their customers
- Generating reports for use by their customers
- Allowing their customers, both service providers and end customers, to access the EAP NMS for reporting and monitoring

The NMS distributes the responsibility of network operations across a set of service providers with minimal configuration and monitoring access. Part of this access includes using the NMS to apply new configurations to satellite terminals and SAS components (on behalf of terminals).

Secondary service providers cannot upgrade network elements with new software and firmware and cannot modify the configuration of any NOC or SAS infrastructure components not related to satellite terminal management.

6.7 Network Monitoring and Reporting

Pulse NMS monitoring and reporting features provide network operators with a variety of ways to access to operational and statistical performance information about the Velocity network Operators are able to access network overview and performance data as well as drill down to individual physical and logical network elements, such as line cards, satellite terminals servers, and service plans.

NMS monitoring and reporting tools support the viewing and management of network events, alarms, and incidents, as well as a variety of network Status and Statistics data, using both real-time and historical reports. Retrieved stats and operational information can be viewed in a variety of forms, for example log views, status views, and analytical views that provide visualization using pie charts, bar graphs, and line charts.

All events, alarms, and performance statistics are automatically archived according to configured management profiles that determine the sampling time and storage duration for various network statistics and performance data.

6.7.1 Types of Report Requests

Pulse NMS monitoring and reporting supports the following basic types of reporting requests:

- **Real-time Reports**—display data as it arrives at NMS; real-time requests have no end time
- **Historical Reports**—retrieve data from the historical archive based on user-specified start/end dates and times
- **Predefined Reports**—retrieve data based on variety of predefined reports such as the A-TDMA Inroute Group Composition usage, DVB-S2, and adaptive coding and modulation (ACM) Stats reports
- **Streaming Reports**—combine the historical request with the real-time request

6.7.2 Types of NMS Monitoring and Reporting

The various NMS monitoring and reporting displays present the returned data in various charts and graphs and in multi-column tabular lists. Graphical data support sorting and filtering. Tabular data can be exported directly into Excel or other CSV-compatible (comma-separated values) applications for further analysis.

The main purpose of the Log View, is to present an historical listing of network events. Generally speaking, there are events that trigger alarms and there are events that clear alarms. Events are listed in a grid format, in which each entry is listed in a single row, with default columns showing the event severity, time-stamp, element name, equipment location, and description. Which columns are displayed in a report is configurable.

The Analytical View shows events, alarms, and incidents report in a graphical format, in which event/alarm data are rendered in a correlated fashion using line, bar, and pie charts.



NOTE: Also see Statistics Reports and SLA Management on page 347.

6.8 NMS Troubleshooting Tools

Pulse NMS provides several troubleshooting features to NMS users that have permissions.

6.8.1 Probe Commands

The NMS provides the ***probe command configurator*** feature as a means of real-time interaction with the operating system of iDirect satellite terminals, line cards, and server elements. Each of these network infrastructure elements has a set of commands that allow users with permission to access and manipulate specific operations of the element.

6.8.2 Engineering Debug Console

The NMS ***Engineering debug console*** feature provides direct access to a selected element, using a shell window for debugging purposes. Network elements accessible using this tool include line cards, satellite terminals, and NMS and PP servers.

7 Satellite Terminal Operations

This chapter describes the operations of the satellite terminal. In particular, this chapter focuses on the operations of the satellite router subsystem.

The following sections are included in this chapter:

7.1 About Satellite Terminals 240
7.2 Terminal Physical Components 241
7.3 Terminal Software Components 244
7.4 Satellite Router Interfaces 251
7.5 Terminal Installation and Commissioning 253
7.6 Satellite Terminal Activation 257
7.7 Satellite Router Configuration 258
7.8 Link Encryption Terminal Licensing 259
7.9 Terminal Network Acquisition 260
7.10 Satellite Terminal Operational States 265
7.11 Terminal Beam Switching 271
7.12 Terminal Congestion Management 280
7.13 Terminal Global Positioning System Validation 294
7.14 Highly-Elliptical Orbit Support in Velocity 295
7.15 Enhanced Inclined Orbit Support in Velocity 301
7.16 Centroid-Based Initial Beam Selection 303



NOTE: Also see Chapter 8 , Satellite Terminal Features on page 305, and see .

7.1 About Satellite Terminals

The satellite terminal is located at the user side of the satellite communications link. In system operation, the satellite terminal acts as the interface between the customer local area network (LAN) and the space segment. It communicates through the over-the-air (OTA) interface with a satellite access station (SAS).

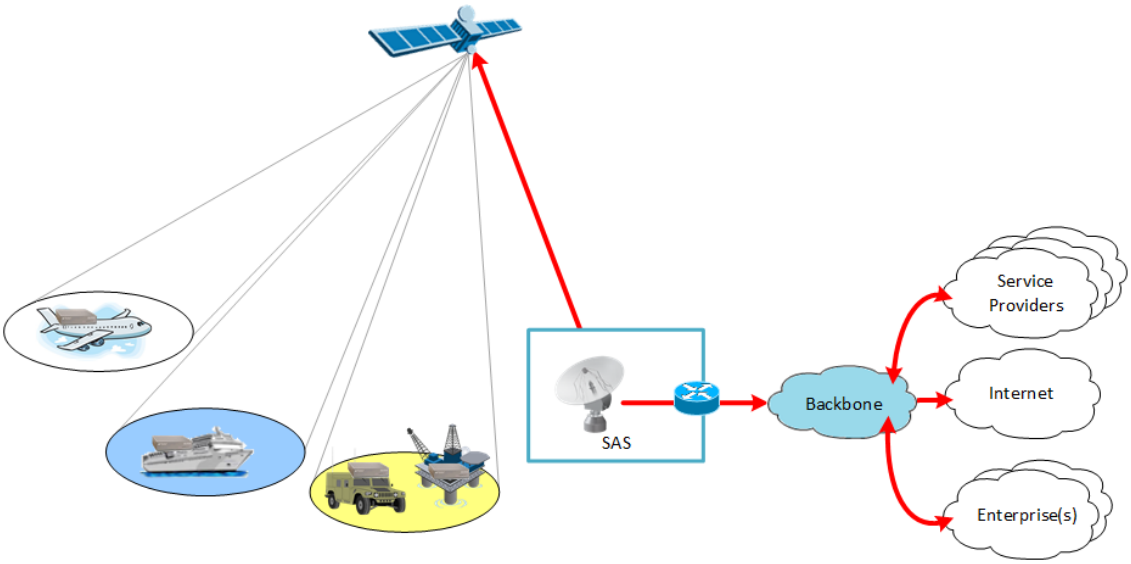


Figure 7-1. Satellite Terminals in Velocity System

Before it can enter the network, the terminal needs to obtain current information on the satellite network configuration-information such as beam footprints and outbound carrier frequencies, symbol rates, and polarization. This information is updated and continually broadcast over downstream acquisition signaling carriers. For additional details on the acquisition signaling carrier, refer to Acquisition Signaling Carriers (ASC) on page 82.

With the information it obtains from an acquisition signaling carrier, the terminal is able to lock to a downstream service carrier and acquire into the network. Once in the network, the satellite terminal is capable of providing end-to-end IP connectivity.

7.2 Terminal Physical Components

The satellite terminal is defined as all the necessary equipment between the customer's LAN connection and the satellite link interfaces. In its simplest form the satellite terminal will comprise a satellite router, BUC (block up-converter), LNB (low noise block converter), feed assembly, and antenna. This equipment may vary based on whether the satellite terminal is deployed in a fixed or mobile application.

Customer applications often require additional satellite terminal capabilities. Support for additional capabilities such as antenna controllers, multi-band LNBs and BUCs, and frequency translation are provided by the Velocity solution.

7.2.1 Satellite Router

The satellite router is the processing and modem component of a satellite terminal; it provides LAN connectivity to users and IF connectivity in the L-band to the radio frequency (RF) section of the satellite terminal. The satellite router acts as a network node in an iNet where the iNet is providing mobile satellite terminals with the capability of using different satellites, different beams on a given spacecraft, or different service channels within a given beam. The satellite router also directs the operation of internal satellite terminal components, including the antenna control unit (ACU), the BUC, and the LNB.

7.2.2 Built-In Ethernet Switch

Each terminal, depending on the type, is supported by one or more Ethernet ports built into the satellite router. Ethernet interfaces can be configured either as access ports or trunk ports. Trunks carry the traffic of multiple virtual local area networks (VLANs) over a single link and allow you to extend VLANs across the network.



NOTE: If a terminal does not contain a switch, it has only one Ethernet port/interface, for which there may be some functional limitation.

7.2.3 Antenna

Antennas using either linear or circular polarization are supported. The polarization is a property of the antenna and can be controlled through OpenAMIP (Open Antenna Modem Interface Protocol). This feature also enables the system to operate in multiple polarizations if the antenna subsystem supports more than one polarization.

Transmissions from antennas having non-circular apertures can cause interference with satellites near the intended satellite when the wider axis of the elliptical beam that these antennas project aligns with the geostationary arc. This type of aperture is common with the flat-panel phased-array antennas often used with aeronautical terminals. The extent of the interference with neighboring satellites will depend upon the alignment of the projected beam pattern with the geostationary arc, with the amount of alignment being related to a combination of the aircraft's geo-location and the local attitude of the aircraft. The satellite router can work in conjunction with the ACU to limit or stop transmissions based upon the

amount of alignment in order to keep the interference within regulatory limits. For additional details on terminal operation with this type of antenna, see Mitigating Skew Impairment in Mobile Terminals on page 312.

7.2.4 Antenna Control Unit

The antenna control unit communicates with the satellite router using the OpenAMIP protocol. The ACU provides geo-location information to the satellite router, points the antenna at the satellite, and performs related functions such as providing antenna status and (in some terminals) muting the transmitter when off-pointed or blocked. For additional details on this protocol, see OpenAMIP on page 252.

7.2.5 LNB

Single and multiple band LNBs are supported by the satellite router. Control over band selection is provided by setting the DC voltage to either 13V or 18V and by enabling or disabling a 22 KHz control tone on the receive inter-facility link (IFL). The combination of these parameters provide support of LNBs with up to four receive frequency bands. LO frequencies of multiple LNBs can be configured based on downlink frequencies.

7.2.6 BUC

Velocity systems now support multi-band BUC along with multi-band LNB devices, on the 7-Series and CX780 Remotes. LO frequencies of multiple BUCs and/or multiple LNBs can be configured based on uplink and downlink frequencies respectively. Using Pulse NMS, BUC multi-band devices can be configured with up to eight frequency bands.

At acquisition, the LNB band is selected based on the associated frequency of the selected signaling carrier in the constellation file or the selected service carrier in the OTA beam map file. Subsequently, when locked to the downstream, the BUC band is selected based on the uplink frequency communicated over-the-air to the terminal.

During acquisition or beam switching on the same satellite or between satellites, the terminal selects the appropriate band, including the associated LO and filter. When the uplink/downlink frequency is known during beam or satellite switch, the BUC/LNB band options are searched to find the first available band that can accommodate the uplink/downlink frequency. Once the appropriate band is determined, the antenna control unit is instructed to set the BUC or LNB local oscillator to the LO frequency of the band.



NOTE: Supporting multiple bands requires an ACU supporting multi-band LNBs and/or multi-band BUCs, and the ability to reconfigure LNB and BUC LO frequency based on downstream and upstream frequencies respectively. A multi-band capable ACU uses OpenAMIP messages from the satellite router to configure the LNB and BUC bands.

7.2.7 Terminal Type

A *terminal type* is a unique user-configured element in the NMS that defines a specific set of terminal components—for example, **LNB**, **BUC**, **ACU**, and **Satellite Router**. This baseline set of elements, referred to as a **Terminal Type**, represents a configuration template that can be used to create multiple terminals that will be based on this baseline set of terminal components. A terminal type does not dictate which terminal model it can be used to configure, but instead can be applied to any new terminal to which the components of the terminal type are desired and appropriate.

An important parameter that is specified when configuring a Terminal Type, is the **RF Terminal Type**. The *RF Terminal Type* is an arbitrary numerical ID (1-65535) that is unique per network, and is used to identify all terminals that share similar RF properties. The RF Terminal Type properties are actually configured on the **Terminal Type** tab of the **Add Channel** dialog or the **Add Regulatory Area** dialog.

7.2.8 RF Terminal Types

In Velocity, the RF Terminal Type is used to identify all terminals that share similar RF properties, so that the data path components can treat them in the same manner. There is no benefit to gain by defining more than one RF terminal type for terminals with the difference in max PSD or in max EIRP less than the difference in upstream carrier C/N or C/No of most inroute group compositions. The typical spacing of C/N thresholds of ATDMA MODCODs is about 1dB. Similarly, the C/No values of upstream carriers in most inroute group compositions are rarely less than 1dB apart.

Although Velocity and Pulse allow the configuration of up to 1024 RF Terminal Types, configuring a large number of RF Terminal Types incurs processing costs and overhead. Two scale aspects to consider are given below:

1. The time to load the Pulse Add Channel and Add Regulatory Area dialogs will increase proportionally with the number of configured RF Terminal Types in that channel or regulatory area. For example, an increase from 50 to 250 RF Terminal Types will cause the dialog load time to increase 5 times as well.
2. The OTA overhead to broadcast terminal information will increase as the number of RF Terminal Types increases. With up to 78 RF Terminal Types, the overhead amounts to 2.34kbps on the downstream channel. Additional 2.34 kbps broadcast overhead will accommodate another 78 RF Terminal Types and so on. For example, configuring 250 RF Terminal Types will incur 2.34 x 4 = 9.36 kbps of overhead transmission.



NOTE: Pulse 2.6.0.2 and below limit configured RF Terminal Types to 78. To use more than 78 RF Terminal Types, the user needs to upgrade to Pulse 2.7.1.2 or contact iDirect TAC for the next step. Velocity 3.2.1.x Terminal software must be installed to use RF Terminal types above 78. Terminals with earlier software version should only be configured with RF Terminal types in the range 1-77.

7.3 Terminal Software Components

The satellite router directly forwards user traffic to/from either an end host or another router. The software functional components deployed on the satellite router hardware, which are discussed in the immediately following sub topics or in subsequent sections, include:

- Board support software
- Over-the-air components
- IP application support
- Routing protocols
- External Interfaces
- Installation and Commissioning
- Terminal Configuration
- Terminal Acquisition

7.3.1 Board Support Software

The board support software comprises the following software components that serve as a hardware/software interface:

- Power-on self-tests (POSTs)
- Factory diagnostics
- Boot loaders
- Operating system
- Application software, libraries, and utilities

7.3.2 Over-the-Air Signaling/Protocols

The satellite router inter-operates with the network components at the SAS through the RF links. This over-the-air section is segmented into multiple layers, and requires software components to maintain each of these layers. The entire over-the-air section on the satellite router software comprises the following:

- Out-of-Band (OOB) Signaling
- RF Monitoring and Link Maintenance
- Over-the-Air Data Framing and Processing

7.3.2.1 Out-of-Band Signaling

The OOB signaling messages consist of all the messages necessary to monitor and control the satellite router operation within the iNet. This includes:

- Dynamic configuration exchange with the SAS
- Signaling messages to improve RF link performance using DVB-S2 ACM (adaptive coding and modulation) or A-TDMA (adaptive time division multiple access)
- Satellite terminal authentication and key management protocol
- Messages from the network management system (NMS)
- Messages to report geo-location, skew parameters, and handle beam switchovers
- Operator-initiated terminal control commands from the network operating center (NOC)

7.3.2.2 RF Monitoring and Link Maintenance

The RF monitoring and link maintenance software component of the satellite router provides link maintenance during severe or degraded weather conditions. RF monitoring and link maintenance includes:

- **Beam Switchover, Monitor and Control** — This facilitates seamless handovers across beams and satellites. The beam switch is triggered either from the SAS or locally by the satellite router based on configuration.
- **RF Receive Section, Monitor and Control** — During Make Before Break beam switching, the satellite router provides support for locking two DVB-S2 channels at the same time and, when necessary, the acquisition signaling carrier (ASC).
- **RF Transmit Section, Monitor and Control** — The satellite router software and firmware supports various modulation, error correction, and coding schemes. It also supports the ability to change all of these on the fly without data loss in order to support A-TDMA inbound.

7.3.2.3 Over-The-Air Data Framing and Processing

The satellite router supports the following:

- **Packet Classification and Scheduling**
- **Quality of Service (QoS) Enforcement**

  See Packet Classification on page 173
- **IPv4 and IPv6 Unicast Traffic Routing and Forwarding** — When forwarding packets, the satellite router validates the IP checksum and then decrements the TTL (time to live) and updates the checksum. Additionally, the satellite router implements the Address Resolution Protocol (ARP) to support inter-operating with external hosts. The satellite router interacts with the rest of the satellite terminal components using this private IP address space. See Chapter 12 , Layer 2 Over Satellite on page 361.
- **Layer 2-Over-Satellite** — The satellite router implements L2oS support to enable the satellite link to emulate Ethernet connectivity between the hub and remotes.
- **Multicast** — The satellite router implements multicast support for IPv4.

- **Network Address Translation (NAT)** — The satellite router software implements a network translation and port translation layer. The port and address translations are done on a per-synthetic virtual network (SVN) basis on the IP 5-tuple. Additionally, the NAT implementation allows for dynamic addition and deletion of ports based on the Session Initiation Protocol (SIP) traffic monitored by the SIP application layer gateway (ALG).
- **Payload and Header Compression** — Payload compression is done on a per stream basis or on a per packet basis, based on configuration.
- **Transmission Control Protocol (TCP) Acceleration** — TCP acceleration mitigates the side effects of the long-latency satellite hop by owning the end-to-end connection and splitting the sliding window protocol into three segments: two terrestrial links (between the communicating devices and the iDirect network), and the over the air link (between iDirect endpoints).

See Velocity Data Traffic Protocols on page 71

7.3.3 IP Application Support

The following features are provided by the satellite router software components in support of IP-based applications:

- SVN and IP Support
- Domain Name System
- Caching and HTTP Acceleration
- Session Initiation Protocol Application Layer Gateway (SIP ALG)
- Dynamic Host Configuration Protocol
- Generic Routing Encapsulation
- QoS Configuration
- Routing
- Multicast

For additional IP application support information, see Chapter 2 , Velocity System Operations.

7.3.3.1 SVN and IP Support

The satellite router supports up to twelve SVNs. Each SVN is identified on the remote LAN by the VLAN tag of the incoming data. Each SVN has its own independent IP address space, and thus the same IP address may be used in different SVNs — but only in different SVNs. Based on the configuration, each SVN interface is assigned either an IPv4 or an IPv6 address. However, the management SVN must be an IPv4 address. For more information, refer to About Satellite Virtual Networks on page 45.

The following information is configurable for each of the SVNs configured on the satellite router:

- SVN ID
- VLAN ID used on the remote network
- IP address of the satellite-facing satellite router interface (sat0)
- IP address and subnet of the remote network-facing satellite router interface (eth0)

IP features such as BGP, HTTP acceleration, DNS caching, DHCP, RADIUS (Remote Authentication Dial-In Service), TCP acceleration, payload compression, and SIP ALG are all enabled and configured per SVN. For more information on these various protocols, see the following sections for a brief description, or for more detailed descriptions, see Chapter 2 , Velocity System Operations on page 33.

7.3.3.2 Domain Name System

The satellite router implements a local DNS cache. The satellite router may be configured with a primary and secondary DNS server address for relaying requests for caching. Support for resolving IPv4 and IPv6 addresses is provided. An independent DNS cache is maintained on the satellite router per each SVN instance on the satellite router. For more detail, refer to Velocity Data Traffic Protocols on page 71.

7.3.3.3 Caching and HTTP Acceleration

The satellite router accelerates web traffic and caches data locally in order to avoid the long satellite delay for frequently accessed pages and objects.

TCP handshake acceleration and DNS caching speed up web access. Both should be enabled for Web traffic, whether caching is used or not. TCP handshake acceleration allows the HTTP get request to be sent before the TCP three-way handshake is completed (see Transmission Control Protocol (TCP) on page 72. DNS caching avoids sending a DNS address resolution message over the satellite link to resolve the name (URL) in the HTTP get request. This may be needed for the main page and all embedded objects. The caching software is linked to the iDirect DNS caching server.

For more details about HTTP Acceleration and caching, see Velocity Data Traffic Protocols on page 71.

7.3.3.4 Session Initiation Protocol Application Layer Gateway

NAT is commonly used to remap one IP address space into another by modifying the IP address in the IP packet headers as they transit a router such as the satellite terminal. However, protocols that embed explicit IP addresses in their payload do not work correctly with NAT.

When NAT is enabled on the satellite terminal, the satellite router supports a Session Initiation Protocol Application layer Gateway (SIP ALG). The ALG replaces the original IP addresses in the SIP packet payload with the NATed IP addresses in the IP packet header. The SIP ALG is configurable per VLAN and may be used for NAT traversal on an SVN that will carry SIP traffic. However, the operator can disable SIP ALG even if NAT is enabled.

For more details about SIP ALG, refer to Velocity Data Traffic Protocols on page 71.

7.3.3.5 Dynamic Host Configuration Protocol

The Dynamic Host Configuration Protocol (DHCP) is a network configuration protocol for hosts on IP networks. Any devices that connect to IP networks must be configured before communicating with other hosts. The most essential information is an IP address and a default route and routing prefix. DHCP eliminates manual configuration and allocates IP address dynamically to clients from an address pool managed by DHCP. Also see DHCP on page 76.

The DHCP module supports the following modes of operation:

- Server mode, where the DHCP module:
  - Responds to a client's DHCP request by dynamically allocating an IP address to a client from a pool of IP addresses



NOTE: The IP address can be valid for a specified duration.

- Relay mode, where the DHCP module:
  - Responds to a client's DHCP request by forwarding it to a configured DHCP server
  - Receives a response from the server and forwards it to a client

7.3.3.6 Generic Routing Encapsulation

Generic Routing Encapsulation (GRE) acceleration is configurable per VLAN for both IPv6 and IPv4. This feature allows TCP acceleration and compression to be applied to traffic within a GRE tunnel that passes through the iDirect system. Also see Generic Routing Encapsulation (GRE) on page 73.

7.3.3.7 QoS Configuration

The satellite router accepts and enforces the following dynamic configuration parameters from the SAS:

- Bandwidth management
- Maximum information rate (MIR)
- Committed information rate (CIR)
- Service limitations based on geographical area
- Service limitations based on date and time
- Parameters associated with specific service plan parameters for any pay-as-you-go features
- Parameters to allow for connection to a pre-configured IP address
- MIR that needs to be enforced upon service expiration
- Classification rules that allow terminal to identify traffic matching specific criteria

For additional information on QoS, see Chapter 4 , Global Bandwidth Management.

7.3.3.8 Terminal Routing

The satellite router software supports MP-BGP (Multi-Protocol - Border Gateway Protocol) for dynamic routing with both IPv4 and IPv6. While the dynamic routing information itself is exchanged through BGP, the configuration of the dynamic routing engine is set from the NMS. This includes configuration for the satellite router to inter-operate with the SAS, and any routing equipment connected to the LAN interface of the satellite router.

The satellite router software will exchange routing updates with its peer routers in each SVN on its local LAN and with its peer routers on the SAS network. Based upon the routing information received, the satellite router software consolidates the route information into a forwarding table. The forwarding engine then uses the forwarding table to forward packets either through the satellite link or through the local LAN interface. For more information on BGP, refer to Border Gateway Protocol (BGP) on page 74.

Static routing is supported by the satellite router software. When static routes are configured for all IP subnets on the terminal within an SVN, dynamic routing is not required for that SVN.

7.3.3.9 Multicast

The satellite router supports downstream multicast received from the SAS. Both dynamic and static multicast streams are supported per SVN and per terminal. The satellite router maintains separate multicast forwarding tables for each SVN. For additional details on the system's handling of multicast traffic, see Velocity IP Multicast on page 52.

7.3.4 Terminal Software and Configuration Updates

This section provides an overview of terminal software and configuration updates and processes.

7.3.4.1 Software Updates

Software updates are normally performed from the NMS, and may also be performed from the Terminal Web UI (e.g., during installation) For detailed procedures, see the *Pulse NMS User Guide* or *Velocity Terminal Web User Interface Guide*.

For resiliency, terminals have two software partitions. New software packages are downloaded, verified, and put into the partition that is not currently active. It is then installed, becomes active, and the remote restarts. If there is a problem with the new software package and the terminal is unable to come into the network, the terminal will fall back to the previous version located in the other partition.



NOTE: The fact that there are two partitions is not visible to the operator or the NMS (i.e., the operator is not able to choose which partition the software is loaded into.

The NMS periodically checks that the active software matches the version the operator has chosen, and will download the correct package if it is out of sync. This ensures that updates are only allowed if they match the version specified in the NMS. It also ensures that if someone modifies the software locally, without authorization, the NMS will override it.

7.3.4.2 Configuration Updates

The operator modifies the configuration of the terminal, services, etc. using the Pulse NMS UI and/or API. When the changes are activated, the NMS pushes the resulting config modifications in the form of one or more options files to the terminal (and the PP). An operator can also place an option file on the terminal using the Terminal WUI. This is typically used during commissioning or installation.

The NMS periodically verifies that the terminal's configuration matches what the operator has activated. If it does not, the NMS pushes the correct configuration to the terminal.

If a terminal receives an invalid remote configuration file (falcon.json) it has the ability to self-recover into the network using its previous known good configuration. When a terminal receives a configuration file and successfully comes into network using it, the terminal stores a backup copy of the configuration file. Then, if the terminal receives another new configuration file and is unable to acquire into the network using it, the terminal reverts to the backup configuration file. This protects against:

- A Missing/Corrupt Remote Configuration File: When the terminal comes into the network using the backup file, it remains in the network using that file until the operator pushes a new configuration file (falcon.json) using the NMS.
- An Invalid Remote Configuration File: This occurs when it contains a misconfiguration that prevents the terminal from coming into network, such as the LNB LO being set to a value the LNB does not support. In this case, the terminal alternates between the backup file and the new invalid configuration file based on the resiliency timeout until either a corrected configuration file is pushed by the Operator, or the LNB is modified.

If an invalid remote configuration file is sent, an event and alarm are generated to indicate to the NOC operator that the terminal had to use a backup configuration file to acquire into the network. A NOC operator can use this information to review any recent changes they have made on the terminal(s) and correct the configuration causing problem.

The alarm is cleared when the terminal successfully uses the configuration file (facon.json) to acquire into the network.



NOTE: If the configuration file is missing or is corrupted, the terminal is restarted using the old backup file. Note that this situation is rare in normal network operation.

7.4 Satellite Router Interfaces

The satellite router interfaces with the satellite terminal network components through the following software interfaces:

- Web-based User Interface
- Statistics Collection and Error Reporting
- Switch Management
- Open Antenna Modem Interface Protocol (OpenAMIP)
- Open BUC Management Interface Protocol (OpenBMIP)

7.4.1 Web User Interface

The satellite router runs a built-in web server that can serve web pages to a web browser connected through the satellite router's Ethernet interface. Current status can be viewed via status pages in this interface. This interface also provides support for administrative operations, such as a page for loading software and configuration as well as a wizard-style interface that can guide an installer through the installation and commissioning process. For details, refer to the *Terminal Web UI User Guide*.

The satellite router also has a console-based user interface accessible with root privileges either through a serial interface or through SSH (Secure Shell). This is intended for use in customer support situations under direction from iDirect, and its operation is not expected to remain constant from one software release to another.



NOTE: Underlying the Web User Interface, is a Remote Web Service API that supports machine-to-machine management of the same functions. Information related to these functions is documented in the *Remote Web Service ICD*.

7.4.2 Statistics Collection and Error Reporting

The satellite router collects statistics and reports errors as follows:

- Reports statistics and events that affect dynamic decisions on the iGW/PP to the SAS
- Reports relevant statistics and information to the web interface
- Reports relevant RF statistics to the SAS and the NMS
- Reports QoS statistics to the SAS for the SAS to implement various QoS requirements
- Reports any board statistics and events to the NMS
- Maintains a log of specific board events to aid debugging in case of in-field failures
- Reports relevant satellite terminal statistics, events, and alarms to the SAS
- Collects layer specific statistics per SVN and reports on-demand basis to SAS
- Reports the location information for non-discreet terminals to the SAS

Some satellite terminal equipment malfunctions and errors (for example, pointing error, excessive BUC power) result in the immediate shutdown of the inbound. In this case, messaging the error back to the SAS immediately is not possible. It stores these errors locally until it is again able to transmit them to the SAS or the satellite router is reset.

7.4.3 Switch Management

Before being loaded with its configuration, the satellite router's switch will operate with all ports isolated from each other and each port will be connected only to the CPU on the satellite router. This behavior provides protection from allowing temporary connectivity between ports on the switch that the user has configured to be on separate VLANs. It also allows accessibility to the satellite router's web-based user interface from any port before the switch has been configured.

Once the user's configuration has been loaded, however, the user will want the switch to act as a switch. Any ports assigned to the same VLAN will be connected so the traffic on that VLAN will be switched between those ports. This should occur for packets on that VLAN, regardless of what other VLANs have been assigned to those ports.

7.4.4 OpenAMIP

OpenAMIP is an IP-based protocol that facilitates the exchange of information between an ACU and a satellite router. When a mobile terminal passes through multiple satellite footprints, OpenAMIP allows the satellite router to command the antenna controller to move the antenna from pointing at the current satellite to pointing at the next satellite. Also, by providing the satellite router with a means to notify the antenna controller of changes to the receive frequency and polarization, it enables the use of automatic beam switching (ABS) to transfer connectivity from one satellite beam to another.

In addition, OpenAMIP enables service providers and their customers to meet government regulations by commanding the antenna to mute the signal in “no transmit” zones.

OpenAMIP commands and responses only go between the satellite router and the ACU. In order to inter-operate with the ACU, the satellite router may use OpenAMIP extensions. These include a method to query the following from the antenna controller:

- Software version
- Vendor ID
- Antenna size
- Serial number

7.4.5 OpenBMIP™

OpenBMIP™ is an optional administrative interface between the modem and BUC components of a satellite terminal. The most important function of this interface is to enable calibration of the terminal's transmitted power for regulatory compliance. This enables a terminal integrator to provide a seamless installation process, with respect to transmitter power calibration.

The BUC interface may be tailored to the application by selection of interface type (RS-422 serial or UDP over Ethernet). This standard is designed to be extensible for vendor-specific enhancements.

This interface plays a key role in setup, calibration, and installation. When the terminal is initializing, the satellite router obtains the BUC serial number and calibration data using OpenBMIP. During installation, the installer will instruct the satellite router to perform, using OpenBMIP, an auto-calibration procedure designed to determine the loss in the IFL cable across the operational frequency range. The results of this calibration procedure, along with the BUC calibration data, provides the information needed to properly control transmit power without the need for a commissioning procedure that would require over-the-air transmissions to the SAS.

Similarly, the OpenBMIP interface also plays a role during acquisition. The BUC determines the max power value, and communicates it to the terminal via BMIP, which allows the acquisition burst to be more accurate and margin to be smaller.

The OpenBMIP interface continues to be vital during normal operation, while the terminal is in network and the BUC is transmitting live data. During this time, the satellite router continually obtains the BUC current temperature, which is needed when using the BUC calibration data to accurately set transmit power.

Satellite Terminal and BUC Identification

The satellite router identifies the BUC by querying its ID and type information via OpenBMIP. This information is stored on the router. The satellite router software reads this information and sends it to the SAS where it may be used for confirmation that the components of the terminal are acceptable for operation in the network before admission of the terminal into the Velocity network.

7.5 Terminal Installation and Commissioning

After a subscriber satellite terminal is installed, it is ready for commissioning so that it can be acquired into the Velocity network. The satellite terminal type determines whether the satellite terminal is pre-calibrated at the factory or if it needs calibration during commissioning. For information on commissioning, refer to the ***Terminal Web UI User Guide***.

The commissioning process ensures the satellite terminal configuration and hardware are set up properly and will not cause interference or disruption of service after the satellite terminal acquires into the Velocity network.

7.5.1 Installation and Commissioning Process

The satellite terminal has a built-in web-based user interface (see Web User Interface on page 251), which provides users with the means to commission new sites and to monitor satellite routers from the LAN side.

The typical sequence is:

1. Installer follows antenna installation manual for physical installation of terminal and cabling.

2. Installer connects terminal with associated network equipment at site (router, switch, wireless access point, and so forth).
3. Installer uses a browser to interact with the satellite terminal's built-in Web user interface, as necessary.
4. Installer either instructs ACU to point the antenna or performs a manual pointing of the antenna. Manual pointing is facilitated by a received signal strength indication (RSSI) being available as a graphic and audible indication via the terminal's Web-based user interface. Some models of routers may send a voltage indication on the Tx IFL cable (satellite router's transmit output) that can be monitored with a voltmeter at the antenna.
5. Installer performs on-site power calibration (calculated values are stored on the satellite terminal). If the BUC supports OpenBMIP, this power calibration requires the installer to simply perform One-Touch Calibration. If the BUC does not support OpenBMIP, the installer will need to manually determine the P1dB Compression Point.
6. For satellites using linear polarization, a cross-polarization alignment can be performed to ensure proper cross-polarization isolation.
7. Satellite terminal is ready for acquisition and begins network process.

A terminal will not attempt to acquire until it has been commissioned using either the One-Touch procedure or a manual procedure. Subsequent changes to the install that would affect the results of the commissioning procedure, such as replacing the transmit IFL cable, will require that the terminal be re-commissioned before it is allowed to acquire back into the network. The terminal will not be able to enforce this after such changes have been made — so, in this case it will be up to the installer to ensure the terminal is re-commissioned.

7.5.1.1 Terminals with OpenAMIP Support

Support for OpenAMIP and support for OpenBMIP are orthogonal when it comes to their effects on the installation/commissioning process. Terminals that support OpenAMIP will auto-point their antennas, with little to no interaction required of the installer. The installer may be required to enter initial downstream parameters into the iDirect Web interface or directly into the ACU. Terminals that do not support OpenAMIP will require the installer to manually point the antenna.

7.5.1.2 Terminals with OpenBMIP Support

For terminals equipped with a BUC that supports OpenBMIP, the Velocity commissioning process does not involve any NOC operators or any tests requiring the transmission of a continuous wave (CW) to the satellite or a Pseudo random bit sequence (PRBS) pattern. The installer performs the installation of the antenna and, when activated, a One-Touch Calibration process prior to the satellite router acquiring into the Velocity network. Terminals that do not support OpenBMIP require the installer to perform a procedure to measure the P1dB compression point, which will involve the transmission of a CW to the satellite.

7.5.1.3 One-Touch Calibration

Some iDirect terminals have been designed to support a one-touch calibration process. This is a process where a satellite terminal can be commissioned with a single-click from the browser. The process is completely automated and requires no transmissions over the air to complete the calibration of the terminal's transmit power. The process requires a BUC that supports OpenBMIP, comes equipped with a built-in means to measure power, and has been calibrated at the factory and loaded with the resulting calibration data.

The calibration data provides the gain characteristics of the BUC over frequency and temperature. It does not, however, provide any information to the satellite router regarding the loss characteristics of the cable connecting it to the BUC. The cable loss characteristics are determined during One-Touch Calibration through a process where the satellite router mutes the BUC then sequences through transmissions at stepped frequencies and power levels, querying the BUC at each step for the power measurements needed to calculate the cable loss.

If the BUC is ever replaced after having performed One-Touch Calibration, the satellite router will detect the change in the serial number reported by the BUC over OpenBMIP and will automatically re-run the calibration procedure with the new BUC. If, however, the cable is replaced, an installer will need to initiate the procedure from the Web-based user interface.

7.5.1.4 One-Touch Calibration-Multi-Band Smart BUC Support

This Multi-Band Smart BUC feature is an enhancement to the One Touch Calibration (OTC) process. This enhancement enables the accurate integration of multi-band Block Up-Converters (BUCs) to Velocity 3.2 and subsequent releases.

A *Smart BUC* is one that supports the OpenBMIP interface between the BUC and remote. OTC is a process in which the effect of IF cable network losses between the remote output and the BUC input is taken into account to ensure accurate return path transmission power during installation and commissioning of a terminal. With this enhancement the OTC process, which formerly only supported single-band smart BUCs, now provides increased capability with full-band support of up to 8 BUC bands.

The upstream transmitted power from a terminal must be accurately controlled to ensure the maximum useable link budget, while ensuring that the BUC is operating linearly and that the output transmission power meets federal operating regulations.

For operation with a Multi-band BUC the terminal must be loaded with an options file (TERMINAL_OPT) that has been generated using Pulse NMS 2.6 or later, and where the BUC has been configured with multiple bands. The BUC band information, added to the TERMINAL_OPT is required to support a Multi-Band BUC.



NOTE: Initial implementation of the multiband smart BUC support is for Aviation (Aero) and Maritime applications.

BUC calibration file format 10.1 adds multiple band capability for between 1 to 8 bands. It is assumed that BUC manufacturers will use this new format for both single and multi-band BUCs, but this can be easily accommodated. Table 7-1 shows the BUC characteristics.

Table 7-1. BUC Band Configuration Characteristics

Application	Band No.	Input IF Range (MHz)	Output IF Range (MHz)	Output Band Range (MHz)
Aviation	1	950-1450	27.5 to 28 (LO=26.55 GHz)	500
	2	950-1450	28 to 29 (LO=27.05 GHz)	1000
	3	950-1450	29 to 30 (LO=28.05 GHz)	1000
Maritime/Land	1	950-1450	27.5 to 28 (LO=26.55 GHz)	500
	2	950-1450	28 to 28.5 (LO=27.05 GHz)	500
	3	950-1450	28.5 to 29 (LO=27.55 GHz)	500
	4	950-1450	29 to 29.5 (LO=28.05 GHz)	500
	5	950-1450	29.5 to 30 (LO=28.55 GHz)s	500

7.6 Satellite Terminal Activation

The provisioning system creates a record for the satellite terminal in the Velocity configuration system. The configured satellite terminal will not be able to enter the network until it is activated. Activation is performed on the NMS by an operator setting the terminal to “Activate”. Activation should be performed before the terminal is commissioned. A terminal can subsequently be deactivated at any time by an operator at the NMS.

Once activated, the terminal will be able to enter the network, but it still might not be able to transfer user traffic. In order to obtain the bandwidth needed to transfer user traffic, the terminal's subscriber service plan components (SSPCs) must be configured and must have reached their configured service start times, and must not have expired. This period of time when the service is available is typically set when the service plan is created and configured for a subscriber.

7.7 Satellite Router Configuration

The satellite router inter-operates with the other network components. Based on the network parameters and the desired operational mode, the satellite router can operate in the following configurations:

- Static configuration
- Dynamic configuration

7.7.1 Static Configuration

A ***static configuration*** consists of all the information absolutely required for the satellite router software to be able to acquire into a Velocity satellite network. It also includes the configuration of the local LAN. This information is stored on the satellite router in the form of options files that can be loaded before or during installation and commissioning of the terminal, and can be updated by loading new options files over the air from the NMS.

The satellite router starts up and acquires into an iNet with a minimal static configuration. This minimal configuration contains the following information:

- Information about the ASCs so that the satellite router can reacquire into the network even if its cached iNet information and beam map are invalid
- Configuration information needed to perform the acquisition sequence
- Information specific to the satellite terminal's commissioning

Once the satellite router has acquired into the network, this information can be updated by the NMS Update Protocol. This update will include information needed for the satellite router to perform as a switch or router for the terminal's LAN-side network. The new configuration is stored on the satellite router and will continue to be used by the terminal, even when it is out of network. If the NMS subsequently sends another update to this configuration, this new configuration will replace the existing configuration and the terminal will start using the new one, instead.

7.7.2 Dynamic Configuration

Any configuration information that is updated while the terminal remains operating in network is considered ***dynamic configuration*** information. This differs from static configuration information in that it can be updated without completely reloading the terminal's complete configuration, which would require a reset of the satellite router causing it to drop out of network during the update. This dynamic configuration information is passed from the SAS to the satellite router through an iDirect-specific over-the-air dynamic configuration protocol, allowing changes to be pushed to the terminal without losing operation of the terminal.

After satellite terminal authentication and the satellite terminal is admitted into the network, the PP provides it with detailed configuration information via Dynamic File Options Exchange (DFOE). At a high level, this includes all options keys other than those required for the satellite terminal to acquire into the satellite network and those needed for it to function in the IP network.

7.8 Link Encryption Terminal Licensing

Pulse enforces the use of Encryption Licensing on terminals deployed after the installation of Pulse Version 2.9.1.x. While terminals already operating in encrypted mode will remain in that mode after a Pulse upgrade to version 2.9.1.x, adding encryption to any terminal requires the network operator to first obtain and install a license file to Pulse.

The Pulse NMS will only accept one license file. When licenses are added or subtracted, a new license file is issued by iDirect. This license file specifies the remote types that can operate in encrypted mode and the quantity of each remote type that can operate in encrypted mode. For each network, iDirect issues a license file based on the accumulated sales records of encryption licenses for each modem type. No license file will be issued if an operator does not purchase an encryption license. In addition, a license file may also be issued based on previously-established contractual terms between an operator and iDirect.

The Network Operator is responsible for managing the export control compliance aspect of enabling encryption for any terminals. Pulse will allow an operator to add encryption to a terminal of a specific remote type as long as unused license is available for that type based on the installed license file. For example, if a license file is installed on the NMS that contains a license for (10) X7 remotes and (20) CX780 remotes, then any 10 (but no more than 10) X7s can be configured to operate in encrypted mode, and 20 (but no more than 20) CX780s can be configured to operate in encrypted mode. If encryption needs to be enabled for more terminals than are specified in the license, a new license with additional licenses must be obtained.



NOTE: Existing Velocity networks with link encryption enabled on one or more terminals via Pulse configuration (not via EDK) can be upgraded to Pulse 2.9.1.x without network disruption. Terminals operating in encrypted mode before the upgrade will continue to operate in encrypted mode after the upgrade without first installing a license file to Pulse.



NOTE: Existing Velocity networks should not have any terminals with pending EDKs before upgrade to Pulse 2.9.1.x. If there are any pending EDKs when the upgrade is performed, a warning will be shown, and the upgrade will continue. When the upgrade to Pulse 2.9.1.x is complete, any terminals with pending EDKs will be in the “Changes Pending” state.



NOTE: tion on any terminals not operating in encrypted mode before the Pulse 2.9.1.x upgrade. For more information on installing Link Encryption Licenses, see the Pulse 4.0 Network Management System User Guide.

7.9 Terminal Network Acquisition

Terminal acquisition is the process by which a terminal starts in an initial, unconnected state and goes into a state where it can be used for transferring user data. Terminal acquisition is a multi-step process that can be described by the following key events:

- Physical Layer Synchronization
- Authentication and Encryption Key Exchange
- Dynamic File Options Exchange and Initial Geo-location Report

When the terminal is initially deployed it is configured to have the information required to receive a signaling carrier.

Upon initial power up, the terminal will select, from the satellites in its constellation configuration, the satellite to which it is visible and closest. The satellite will be considered as visible only if the terminal's elevation angle is between the minimum and maximum look angle configured for the given terminal type. A relative preference can be configured for each satellite, to influence the terminal's selection when it is in situations where the closest satellite may not be the best first choice for most terminals. Once a satellite has been selected and the terminal's antenna is pointing to that satellite, the terminal will lock to a signaling carrier on that satellite and receive additional information regarding the physical layer in order to establish 2-way communication.

Once 2-way communication is established, the terminal will begin the process of authentication to establish mutual trust with the network. After mutual trust is established, the terminal will receive additional configuration information through the DFOE process. During this time the terminal will also report its geo-location. The terminal uses the information from the DFOE process and completes the acquisition process.

A terminal that has been out of network that begins the acquisition process will use information cached prior to going out of network. The use of cached information is intended to minimize the acquisition time of terminals that were previously in network. In the event a terminal is unable to complete the acquisition process based on the cached information, it will execute the acquisition process as if it was initially deployed. As long as the terminal can receive the signaling carrier it should not become stranded.

7.9.1 Network Acquisition Sequence

The satellite terminal acquires into the network when:

- The satellite terminal is provisioned, installed, and commissioned into the Velocity system.
- A channel is active in the satellite terminal's current beam.

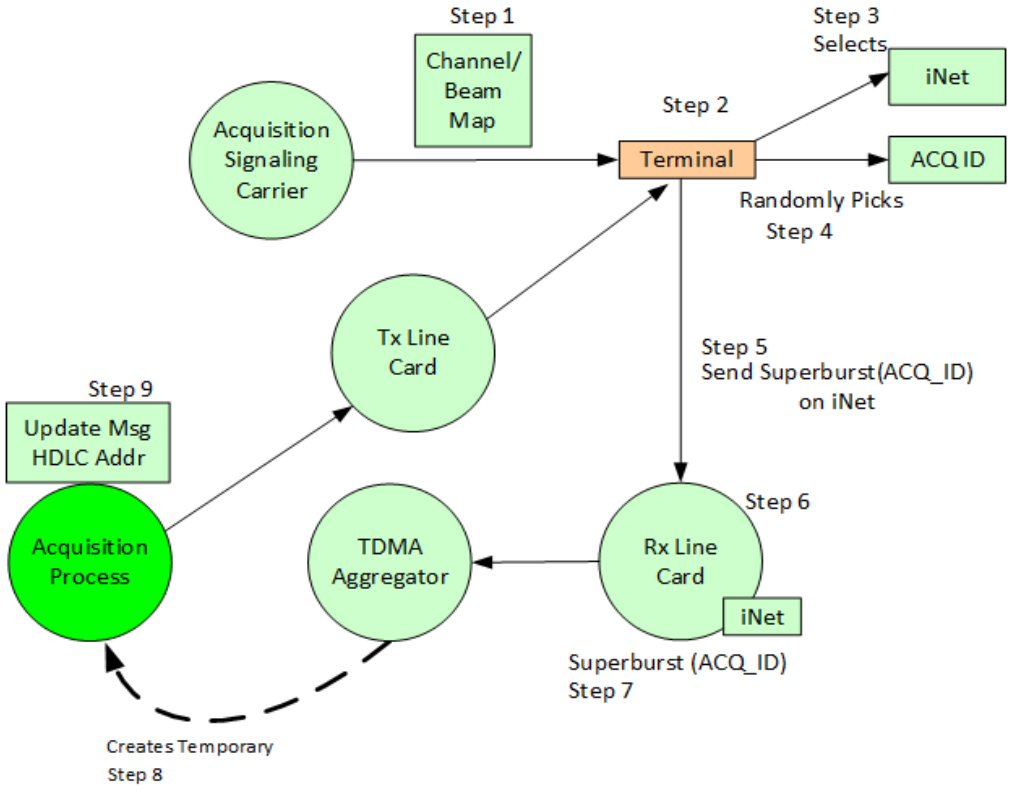


Figure 7-2. Terminal Interactions during Network Acquisition

The network acquisition sequence is:

1. The satellite terminal receives the low resolution beam map through the ASC.
2. The satellite terminal uses its geo-location, its access class, and the beam map to determine available iNets at its current location.
3. The satellite terminal selects an active iNet that it is allowed to access, based on its access class.
4. The satellite terminal randomly chooses its temporary acquisition ID.
5. The satellite terminal sends a Superburst on the selected iNet encoding its ID.
6. The Superburst is received by the iNet Rx line card.
7. The Rx line card sends the Superburst to the appropriate A-TDMA Aggregator process.
8. The A-TDMA Aggregator process forwards packet to the acquisition process for the iNet.
9. The process sends a message to the satellite terminal correcting its power, frequency, and timing, as well as assigning an HDLC address to the satellite terminal.
10. The iNet control process begins assigning A-TDMA traffic slots to the satellite terminal.

At this point, unsecured bi-directional communications have been established between the SAS and the satellite terminal.

7.9.2 Authentication and Admission Control Sequence

Once bi-directional communications have been established, authentication can be performed. Authentication, if enabled, is performed at both ends of the connection-not only does the SAS authenticate the terminal, but the terminal authenticates the SAS. Authentication of the SAS is performed by using the PKI (Public Key Infrastructure) data that has been loaded onto the terminal to validate the signature on an X.509 certificate sent from the SAS to the terminal. However, since the terminal's X.509 certificate is only self-signed and hasn't been signed by an authority by the time the terminal initially enters the network, it can't be used for authenticating the terminal.

Instead of using the terminal's certificate, the terminal is loaded with a secret that is shared with the SAS, so both the terminal and the SAS know that secret. The terminal uses cryptographic hardware to secure the secret, and this hardware supports a challenge-response approach to testing the secret without exposing it. The SAS sends a challenge, in the form of a large random number, to the terminal, and the terminal feeds that challenge into the cryptographic hardware. This hardware performs a cryptographic operation using that challenge, the shared secret, and a hardware serial number as inputs, and returns a response that is then sent back to the SAS. The same operation is performed at the SAS using the same inputs. If the response from the terminal matches that generated at the SAS, then the secret at the terminal must match the secret the SAS has on record for that terminal, and the terminal can be considered authenticated.

The shared secret must be loaded onto the terminal without exposing the secret. This is accomplished by first installing a single-use token onto the terminal, then using this token as a temporary shared secret to perform a single, first-time authentication of the terminal. Although this token is exposed to whoever installs it (usually, this is the terminal installer), its use is limited to a single authentication of a single terminal, preventing it from being used to authenticate an unauthorized terminal. Authentication using this token provides a sufficient level of trust in the terminal to then send the terminal its shared secret, over the air and encrypted. This shared secret is loaded into the cryptographic hardware on the terminal without exposing it, and it remains stored in that hardware where it can never be exposed. Subsequent authentications of the terminal are performed using this secured shared secret.

After the satellite terminal's identity is authenticated, the Velocity network is able to allow or deny a satellite terminal into the network based on configuration, network, and satellite terminal conditions. The SAS can also move the satellite terminal to a different iNet for load balancing. The following diagram illustrates the network authorization and admission control sequence that is described below.

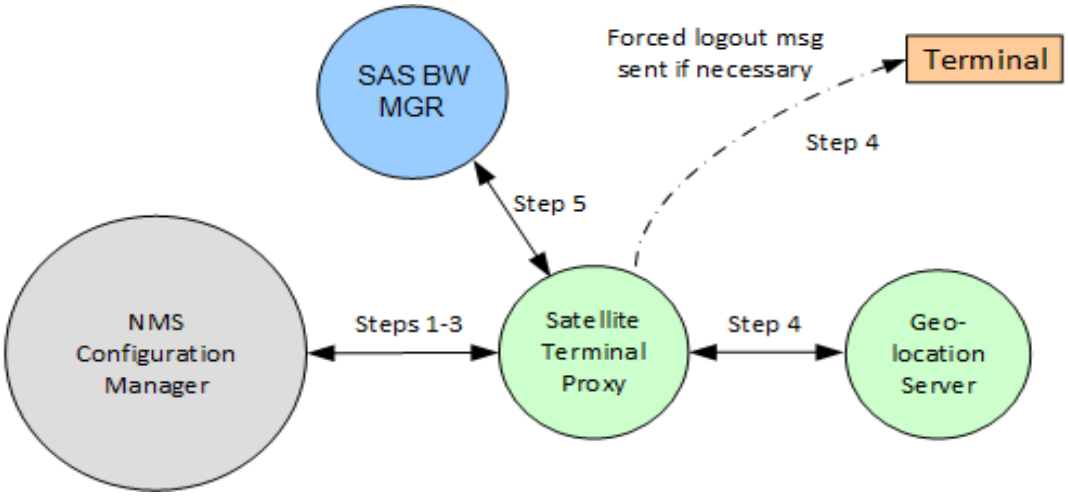


Figure 7-3. Network Authorization and Admission Control Sequence

The network authorization and admission control sequence is:

1. The Satellite Terminal Proxy process retrieves satellite terminal information based on the NMS configuration.
2. The Satellite Terminal Proxy process checks to see if the NMS has barred satellite terminal access for the particular Satellite Router ID. If so, it sends a forced logout message, and raises a warning on the NMS.
3. The Satellite Terminal Proxy process checks to see if the satellite terminal access class value is allowed into the iNet. If not, it switches the satellite terminal to another available iNet if available, or sends a forced logout message if no other iNet is available.
4. The Satellite Terminal Proxy process consults the Geo-location Server process to see if the satellite terminal is in a permit/deny geographic position. If the satellite terminal is in a deny geographic position, the Satellite Terminal Proxy sends a forced logout message to the satellite terminal with a latitude/longitude bounding box indicating where the satellite terminal is permitted to try again.
5. The normal load balancing process will move the satellite terminal to a new iNet if necessary.

7.9.3 Terminal Fails to Enter Network

A satellite terminal with beam map information can fail to acquire into an iNet for a number of reasons, such as rain fade, hardware failure, interference, or polarization or frequency mismatch.

The typical sequence is:

1. The satellite terminal locks to DVB-S2 downstream, successfully receives return channel information, and sends an acquisition burst. When no reply is received, the satellite terminal begins a random exponential backoff, involving a series of re-tries, with an increasing wait time between. The wait time increases both randomly and exponentially, to minimize the number of collisions when recovering from a complete iNet outage.
2. After a configured amount of time, the satellite terminal declares a failure to enter the network.
3. The satellite terminal begins a round robin attempt to enter all available iNets at its current geo-location, including iNets across all satellites.
4. If the satellite terminal fails on all available iNets, the satellite terminal re-tunes to the ASC to receive a new copy of the beam map and returns to Step 3.



NOTE: Once a terminal successfully acquires into an iNet by repeating step 3, it sends events to the NMS to indicate the failure to acquire on previous iNets.

7.10 Satellite Terminal Operational States

There are four satellite terminal operational states:

- **Satellite Search** — The satellite terminal is looking for an available satellite. In this state, the satellite terminal does not transmit.
- **Locked to Acquisition Signaling Carrier** — The signaling carrier is a way to inform terminals of the current satellite network configuration: more specifically - beam footprint and/or parameters such as outbound carrier frequency, symbol rate and/or polarization. However, if the satellite terminal is in a location without a suitable DVB-S2 service channel, it remains in the ASC state and, at intervals, checks whether a suitable service channel is available.
- **Acquisition** — The satellite terminal has identified a suitable DVB-S2 data carrier. It tunes to that carrier, and communicates with the SAS to authenticate and join the satellite network.
- **Data Communication** — The satellite terminal is in this state when it is logged into the network and is sending and receiving data. Because the network uses an A-TDMA scheme, transmissions are in the form of short bursts of varying time, frequency, and power level. The number of bursts depends on the transmitted data.

  In the data communication state, the satellite terminal may switch from one iNet to another. This can occur for load balancing reasons or because the satellite terminal has moved into a different beam. During this transition, the satellite terminal continues to transmit normally; however, at some point the frequency band of the transmissions shifts and the satellite terminal tracks on the new carrier.

The following diagram shows the actions a terminal takes to get into the network, and the information it uses to do so. Initial conditions are indicated by the orange oval-shaped elements in the diagram.

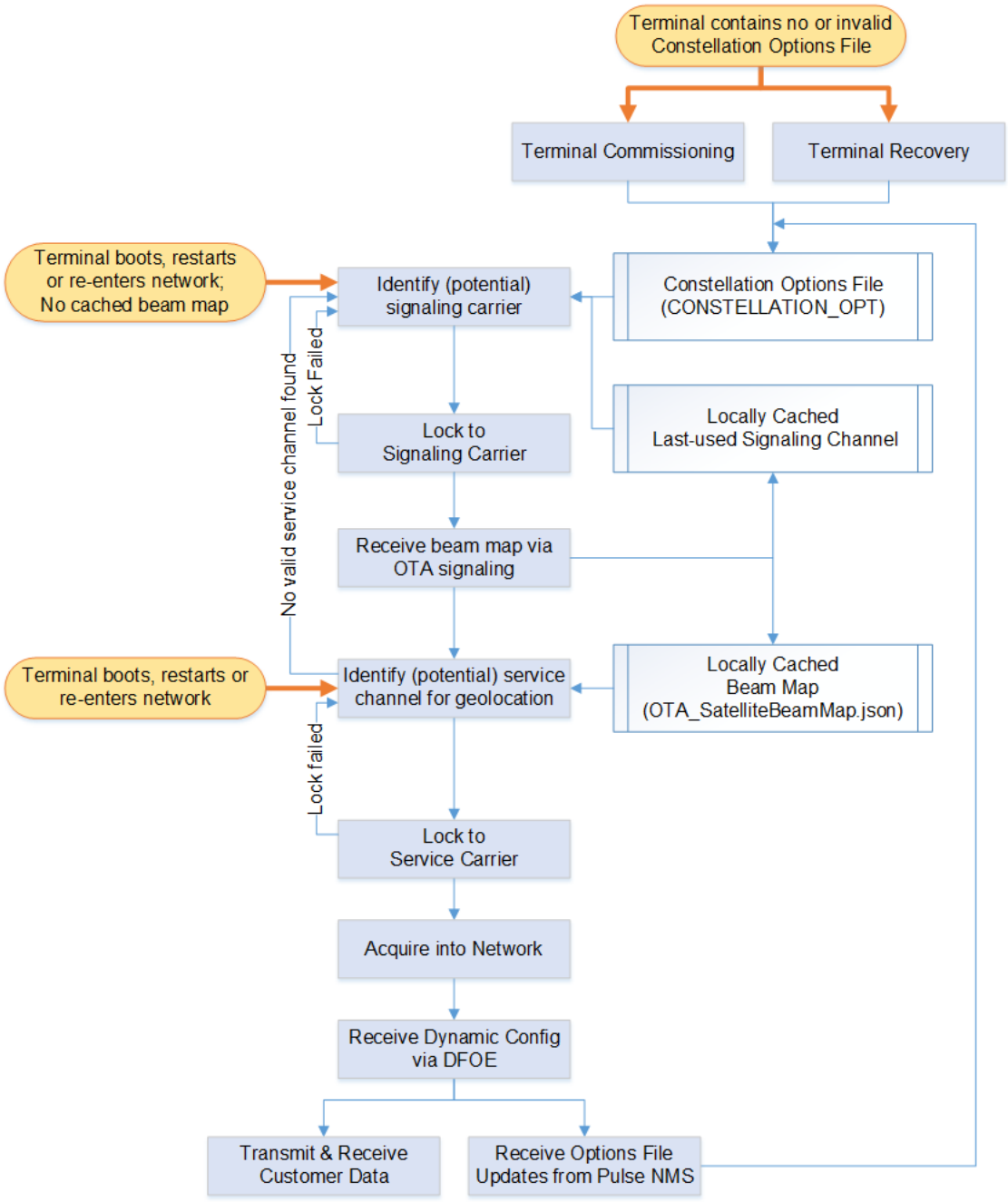


Figure 7-4. Terminal Actions

7.10.1 Satellite Router Functions

The main satellite router functions control the transfer of configuration information and data, as illustrated in the diagram.

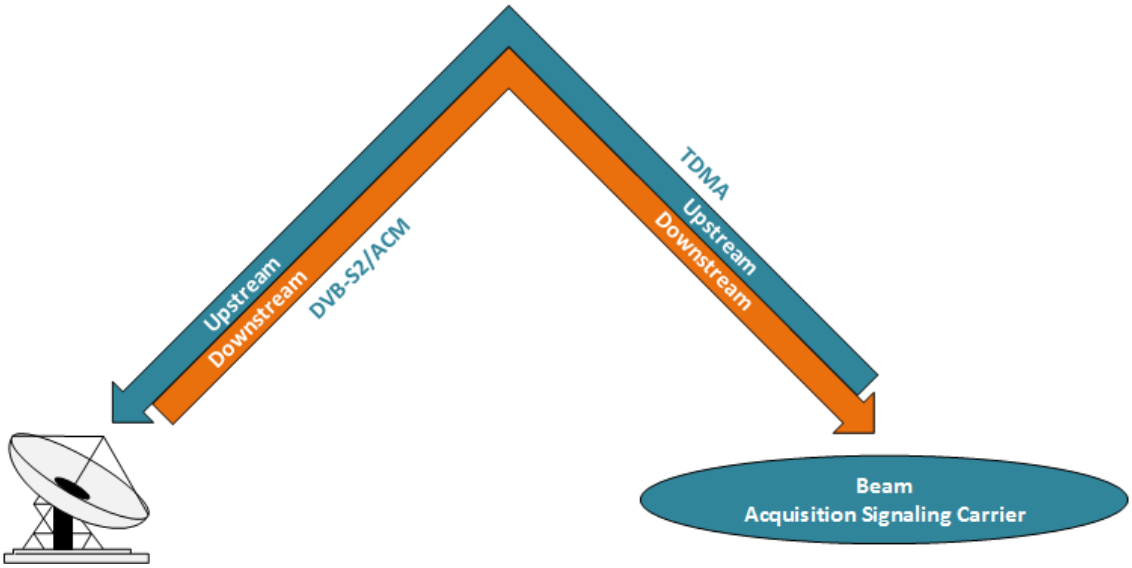


Figure 7-5. Satellite Router Uplink and Downlink Configuration Information and Data Flow

- **Acquisition Signaling Carrier Outbound (Downlink)** — The signaling carrier, irrespective of type, is used to broadcast the satellite network beam configuration to the terminals located within the network.
- **Downlink Data DVB-S2 Carrier** — After detecting the ASC, the satellite router locates a suitable DVB-S2 carrier for downlink data. The DVB-S2 carrier is > 1 Msps (typically > 20 Msps), with a minimum operational ***C/N*** of -2.0 dB (negative 2 dB). The antenna must be able to track the DVB-S2 carrier down to that signal level.

  The downlink DVB-S2 carrier encodes and modulates the data at the MODCOD (modulation and coding) that allows the highest efficiency possible given the quality of the RF link at the moment. The satellite router demodulates and decodes the downstream data. Then, satellite terminal proprietary firmware and software reassemble and decrypts the data into IP packets and send it to the Ethernet LAN port.
- **Uplink A-TDMA Carrier** — For upstream operation, the satellite router encrypts (when configured to do so), segments, and encapsulates user data from the LAN port using iDirect proprietary software and firmware. The satellite terminal hardware modulates the data for transmission to the satellite. Uplink channels use burst A-TDMA with dynamic frequency hopping across carriers of different symbol rates.

7.10.2 Normal Operation — Satellite Terminal Passes Traffic

After the satellite terminal locks onto the assigned service channel and logs into the network, it can start sending and receiving user data. For additional information on network details, see Chapter 2 , Velocity System Operations.

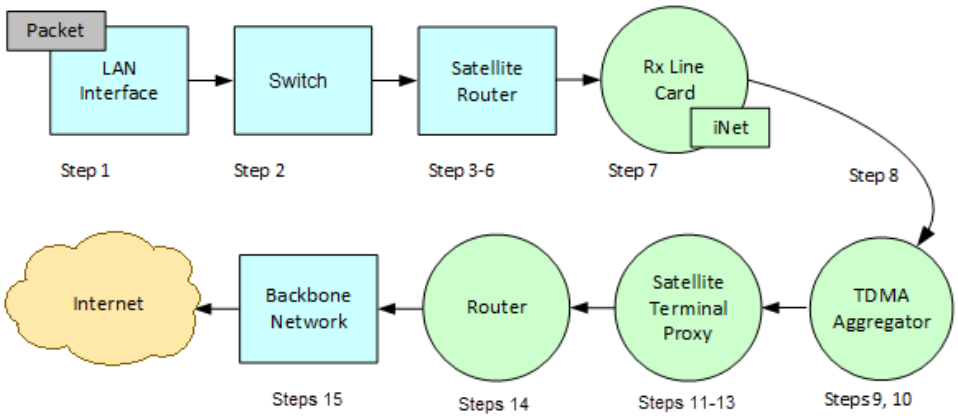


Figure 7-6. Operational Sequence for Traffic from Satellite Terminal to Internet

The normal operational sequence for a packet traveling from the satellite terminal to the Internet is:

1. Packet destined for Internet is received on a LAN interface. **Note**: The switch maybe internal or external depending on the specific terminal.
2. Packet is associated with a particular VLAN that is directly assignable to an SVN based on existing configuration. **Note**: The switch may be internal or external depending on the specific terminal.
3. Packet is passed to the LAN interface of the satellite router with appropriate VLAN tag.
4. Packet is received by the satellite router and processed according to type. TCP traffic undergoes processing pertaining to acceleration and compression associated with Performance Enhancing Proxy (PEP) operations.
5. Packet is queued into the appropriate queue according to its classification.
6. Packet is segmented and transmitted over an A-TDMA inroute as allocated bandwidth and packet scheduling dictates.
7. Packet segments are received by the iNet's associated Rx line card.
8. Packet segments (A-TDMA bursts) are transmitted from the Rx line card to the appropriate A-TDMA Aggregator process.
9. The A-TDMA Aggregator process forwards the A-TDMA bursts to the appropriate Satellite Terminal Proxy process.
10. The Satellite Terminal Proxy re-assembles the bursts into complete packets.

11. The Satellite Terminal Proxy creates standard IP packets and passes the IP packets to the IP stack. TCP traffic undergoes processing pertaining to acceleration and compression associated with PEP operations.
12. The Satellite Terminal Proxy processes the packet according to type.
13. The PP routing process forwards the packet to the access router over the SAS network in a VLAN-tagged Ethernet frame. The VLAN tag is the SVN ID.
14. Access router forwards the packet over the backbone network, using the protocols configured on the backbone, usually MPLS (Multi-Protocol Label Switching) tunnels.
15. Packet exits the backbone network and traverses the Internet.

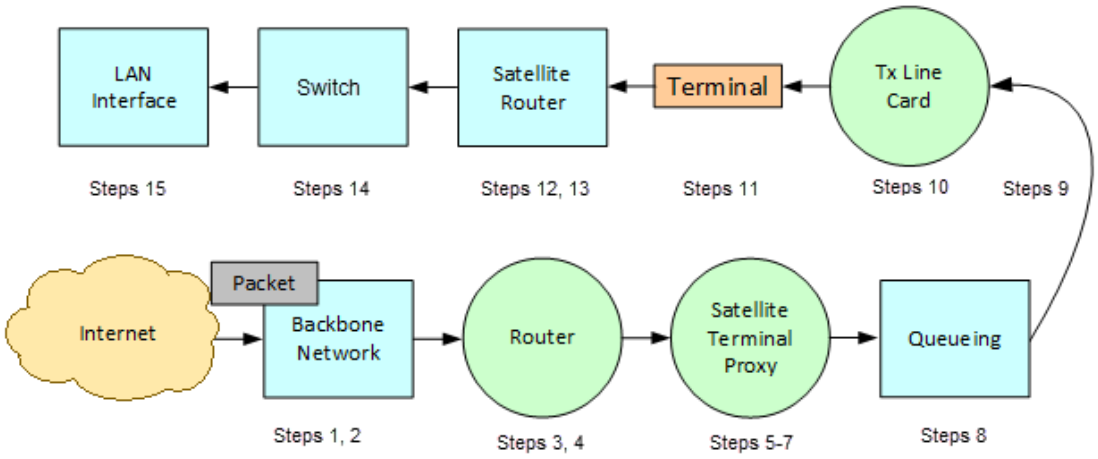


Figure 7-7. Operational Sequence for Traffic from Internet to Satellite Terminal

The normal operational sequence for a packet traveling from the Internet to the satellite terminal is:

1. Packet enters backbone network from the Internet with the satellite terminal's IP address in its destination IP address.
2. Packet is routed across the backbone network to the access router of the appropriate SAS.
3. Packet is routed to the gateway router.
4. Packet is routed by the gateway router within the SVN's VLAN to the appropriate routing process.
5. Packet is routed to the appropriate Satellite Terminal Proxy for the satellite terminal.
6. Packet is processed according to type by the Satellite Terminal Proxy.
7. OTA data layers including link encryption and VLAN headers are added. TCP traffic undergoes processing pertaining to acceleration and compression associated with PEP operations.
8. Packet is queued for downstream transmission.
9. Packet is forwarded to the appropriate Tx line card with an indication of the MODCOD for the satellite terminal.

10. Packet is transmitted by the Tx line card.
11. Packet is received by the satellite terminal.
12. Packet is processed by the satellite router according to type, to produce a standard IP packet. TCP traffic undergoes processing pertaining to acceleration and compression associated with PEP operations.
13. Packet is forwarded within the SVN's VLAN to the satellite terminal's switch.
14. Based on configured VLAN mapping, set the VLAN tag in the Ethernet frame.
15. Packet exits the satellite terminal's switch on the appropriate LAN interface. **Note**: The switch may be internal or external depending on the specific terminal.

7.11 Terminal Beam Switching

Once a terminal has acquired into a network, it will remain supported by the initial beam until a decision to switch has been made. It is not uncommon, however, for a terminal to simultaneously lie within a number of beam footprints. This state can occur for instance with a mobile terminal as it moves from one beam to another; and in stationary terminals where beams overlap. In either case, the terminal will acquire into the network using a single beam.

The beam switching operation has two distinct components — first is the mechanics of the switch, which refers to the actual switching itself and the effect on the physical layer and user traffic. The second is the decision logic, which is how the decision to switch is actually made. This first component, the mechanics of switching, is described in the following sections. See , for the second component — the beam switching decision logic.



NOTE: A switch can occur between iNets on the same satellite or between iNets on different satellites.

The modem timestamps and logs key events during a beam switch; the log is stored in a file on the modem. The log file may be processed offline to measure traffic interruption time or to support debugging. The maximum file size is 200 KB (older entries are over- written). Retrieval of the log file is done through SSH to modem. The log file is located in the /var/log/messages folder.

7.11.1 Mechanics of Terminal Switching

Beam switching can be triggered for the following reasons:

- Geo-location changes of the terminal
- Changes in iNet congestion
- A command through the Pulse NMS API

There are three types of beam switching:

- **Make-before-Break (MBB)** — Set up the connections on the new iNet before performing the actual switching. This is done when there is no change in polarization and no change in the LNB band between the initial and final iNets, and both iNets are on the same satellite and land at the same SAS.
- **Break-before-Make (BBM)** — Break the connections on the existing iNet before switching beams. This is more disruptive, but is necessary when there is a change in polarization or a change in the LNB band or for terminal types that support only one receiver. This also requires both iNets to be on the same satellite and land at the same SAS or land at the same or different RF gateway in the CDC architecture.
- **Satellite Switch** — This occurs when switching between satellites. The terminal reacquires into the new SAS or land at the same or different RF gateway in the CDC architecture.

When a terminal lies within a single beam, as depicted in the diagram, user traffic starting at the server will traverse the terrestrial network to the PP cluster. From the PP cluster the traffic will traverse tunnels to the line cards that are used for that iNet. From the line cards the modulated traffic is converted to the appropriate IF domain and sent over the air to the terminal. In this example, Rx1 is appropriately configured to receive the modulated signal.

The inset shows the presence of the carrier, the location of the terminal L1 in the iNet 1 beam, and that the modem is receiving the traffic on Rx1 of a dual-receive modem.

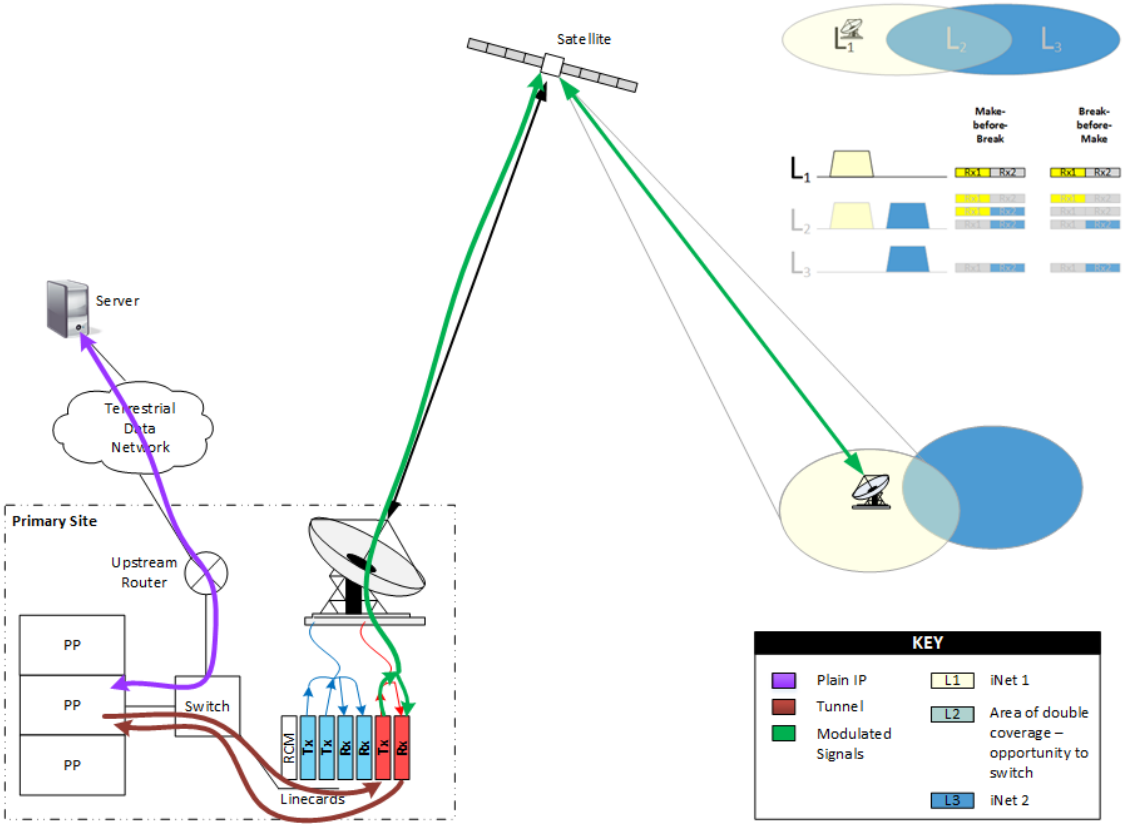


Figure 7-8. Beam Switching, before switch

When the terminal lies within two or more beams and is reporting its location, the PP may decide to switch beams. There are a number of parameters that are considered to enact a beam switch. For details on these parameters, see Beam Switching Decision Logic on page 290.

7.11.2 Terminal Make-Before-Break Beam Switch

The Make-before-Break (MBB) switch, is illustrated in the following diagram. This type of switch occurs when the terminal switches between iNets of the same polarization, but different frequencies.

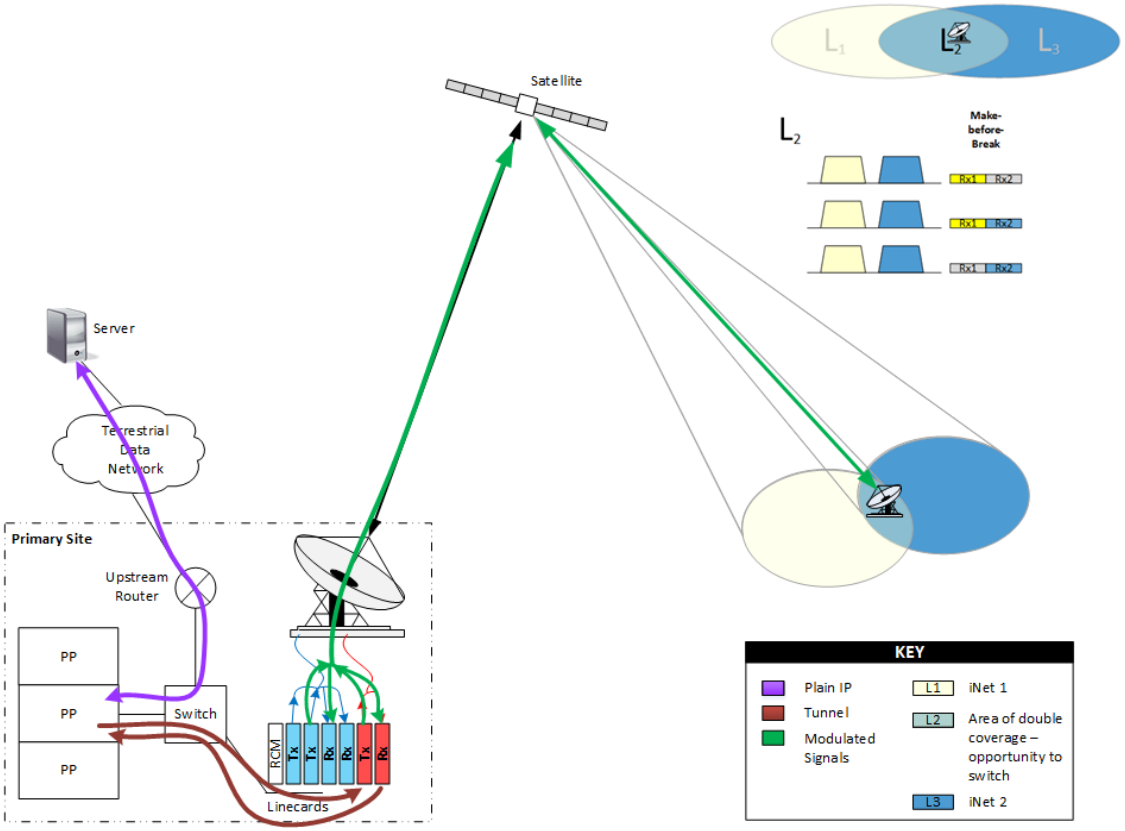


Figure 7-9. Beam Switching, during switch, Make-before-Break

With Make-before-Break, once a decision to switch has been made traffic will flow as follows.

User traffic will traverse the terrestrial network to the PP cluster. (Subsequent actions in the support of a beam switch are transparent to the external network. It is important to note that this terrestrial path is unaffected during a beam switch. The PP will replicate the terminal's downstream traffic onto the new iNet. User traffic will traverse tunnels between the PP cluster and line cards that support the new beam.

The second receiver receives the traffic on the next carrier while the terminal continues to receive the traffic on the current carrier. This allows timing synchronization and transmitter configuration to occur that will eliminate or minimize data loss as the transmitter begins to send traffic in the new beam. If the terminal is not detected in the new beam by the SAS, the beam switch will abort. If the terminal is detected the switch process will continue. The switch process completes by making the first receiver inactive.

It is important to note a few key features of this switch. First, the data flow is through the same PP server, which means there is no change to the external networking. That is, there are no routing updates or MAC table changes needed. There are also no networking changes

on the remote side. This makes the switching transparent from a networking point of view, with no need for the terminal to re-login or re-authenticate.

Because the second connection is fully made before the first one is brought down, the mechanism is robust against failures to synchronize the second channel. While the channel conditions are expected to be similar for both connections (same satellite, same atmospheric fade), differences in footprint EIRP (Effective Isotropic Radiated Power) and interference may cause the second channel to be unusable in some edge cases. When this happens, the switch will abort with no disruption to the user traffic.

After the switch is completed, the terminal may continue moving into the second beam, as is illustrated in the following diagram.

The iQ800 and 3315 modems support MBB.

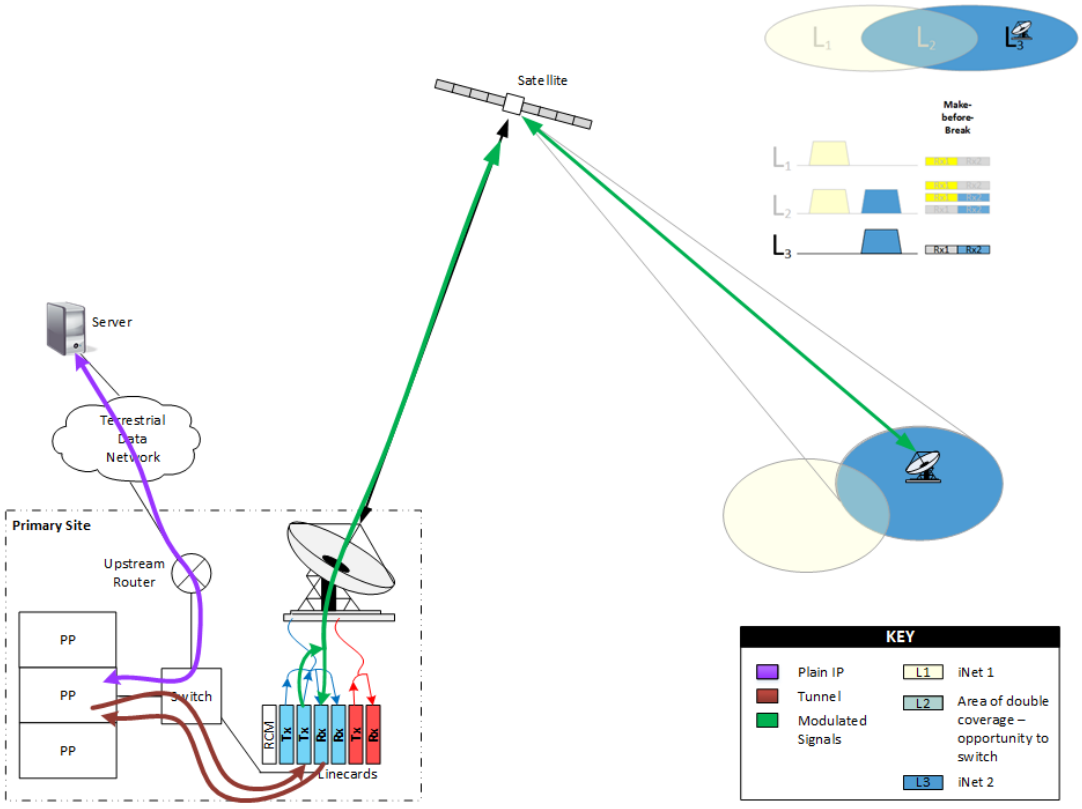


Figure 7-10. Beam Switching, single beam (after switch), Make-before-Break

7.11.3 Terminal Break-Before-Make Beam Switch

The second type of switch, Break-before-Make (BBM), is illustrated in the following diagram. BBM switch does not require dual RF receive paths and is thus more compatible with single-receiver remotes, band or polarization changes, or future operational modes such as multicast overlay. The old connection must be broken before the new one is made.

For iQ200 and the 9-series modems, Break-before-Make is the only supported mode of switch. The connection is broken even when there is no Polarization or Frequency band change. Newer Sabine remotes such as the iQ 800 support Fast Break-before-Make beam switching that significantly reduce traffic disruption time.

	NOTE: The system automatically determines if Make-before-Break or Break-before-Make is needed based on the terminal NMS configuration. No special configuration is needed other than correct configuration of channel polarizations.

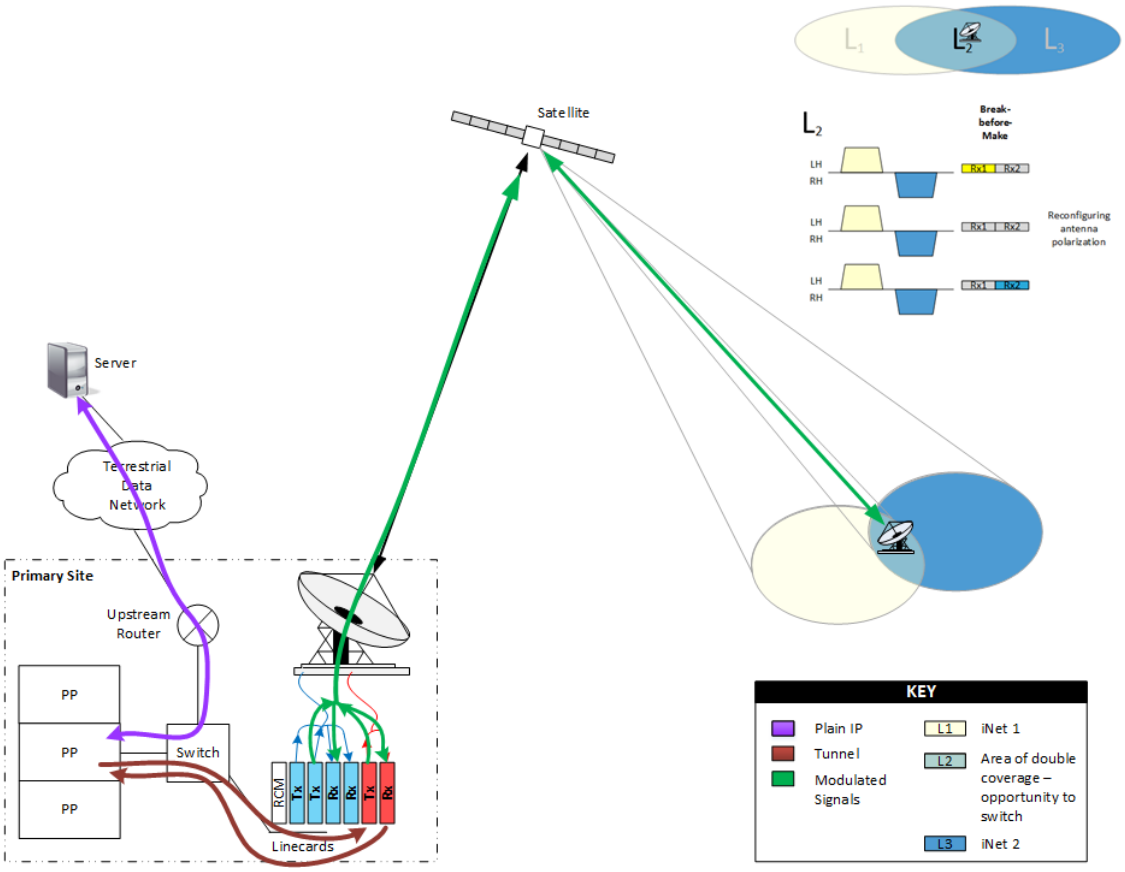


Figure 7-11. Beam Switching, multiple beams (during switch), Break-before-Make

The flow of user data is effectively the same as with Make-before-Break switching; however, the switch process is different. Rather than establishing the connection with the second channel while maintaining the first, the terminal must bring down the initial connection,

command the polarization changes needed for the antenna (or change the LNB band selection), and re-establish the link as quickly as possible. The terminal state is maintained during the beam switch.

There are two differences from a user perspective. The first is that there may be a longer disruption to the traffic (which may depend on how long a terminal takes to change its polarization and how long it takes for its receiver to lock onto the new downstream). The second is that if the second iNet is unusable due to misconfiguration or interference, the terminal may fall out of network. In this case it will automatically log in again. In a well-designed system such occurrences should be rare.

The diagram below shows the state of the terminal after a Break-before-Make switch has completed, and the terminal is subsequently moved out of the overlap area and into the second beam.

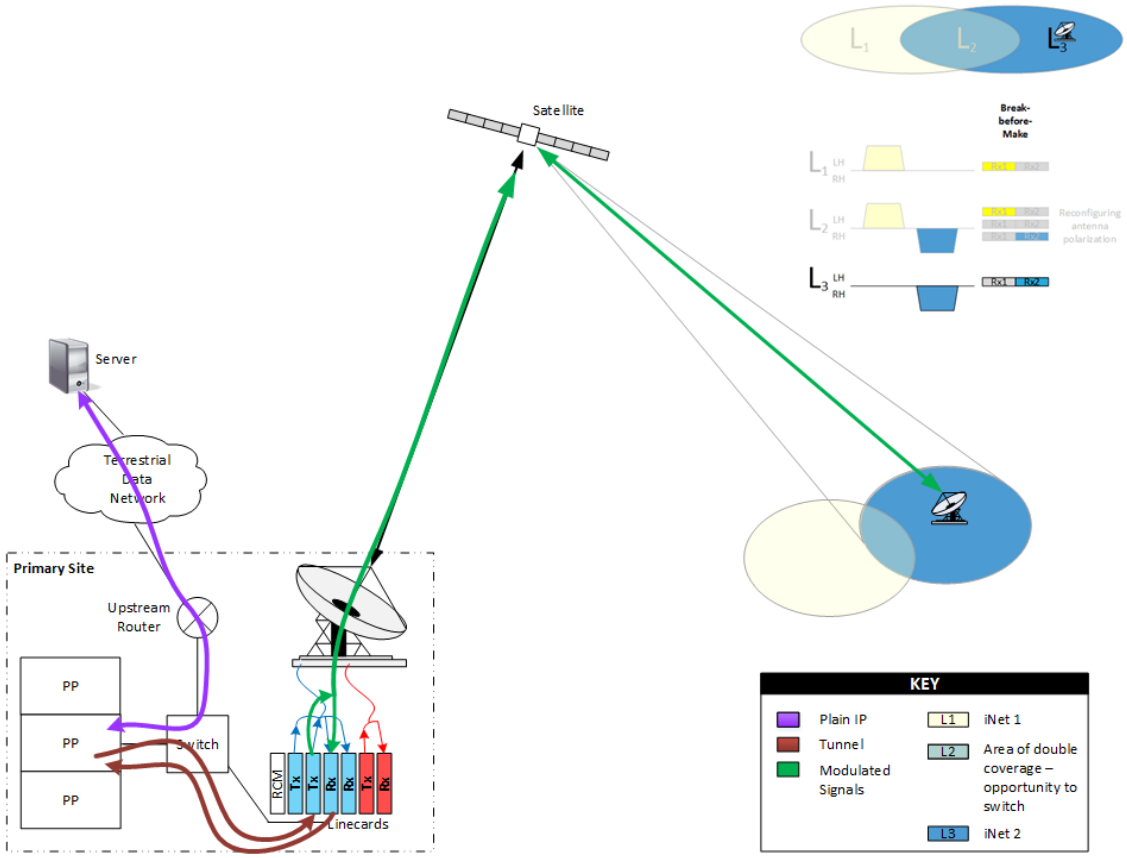


Figure 7-12. Beam Switching, single beam (after switch), Break-before-Make

7.11.4 Terminal Satellite Switch

Satellite switching is more disruptive than the previous two types of switching. In a satellite switch, the terminal logs out of the current SAS site and logs into a new SAS site. This is a complete login, and causes the terminal to land in a new set of PPs at the new SAS site.

In addition to causing sessions to be dropped, re-convergence of the terrestrial network is needed to restore connectivity to the remote. For example, withdrawal of the BGP routes from the original SAS and advertisement of the routes on the new SAS will occur, which will trigger routing updates across the terrestrial network.

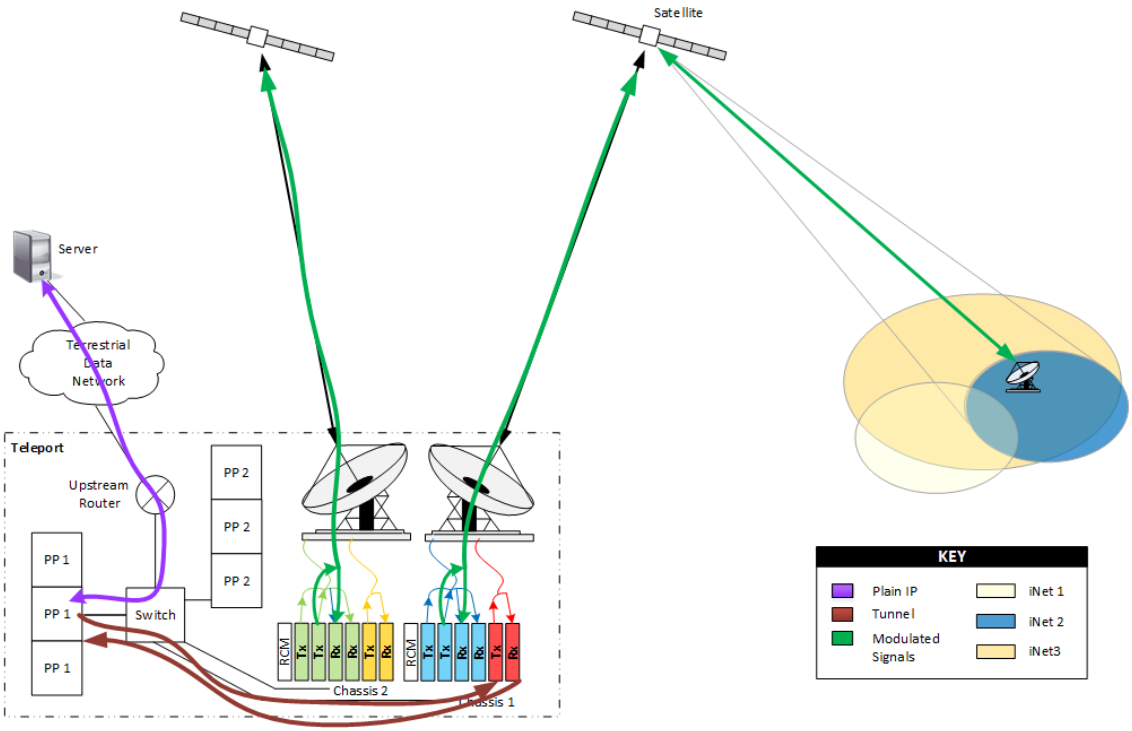


Figure 7-13. Satellite Switching, before switch

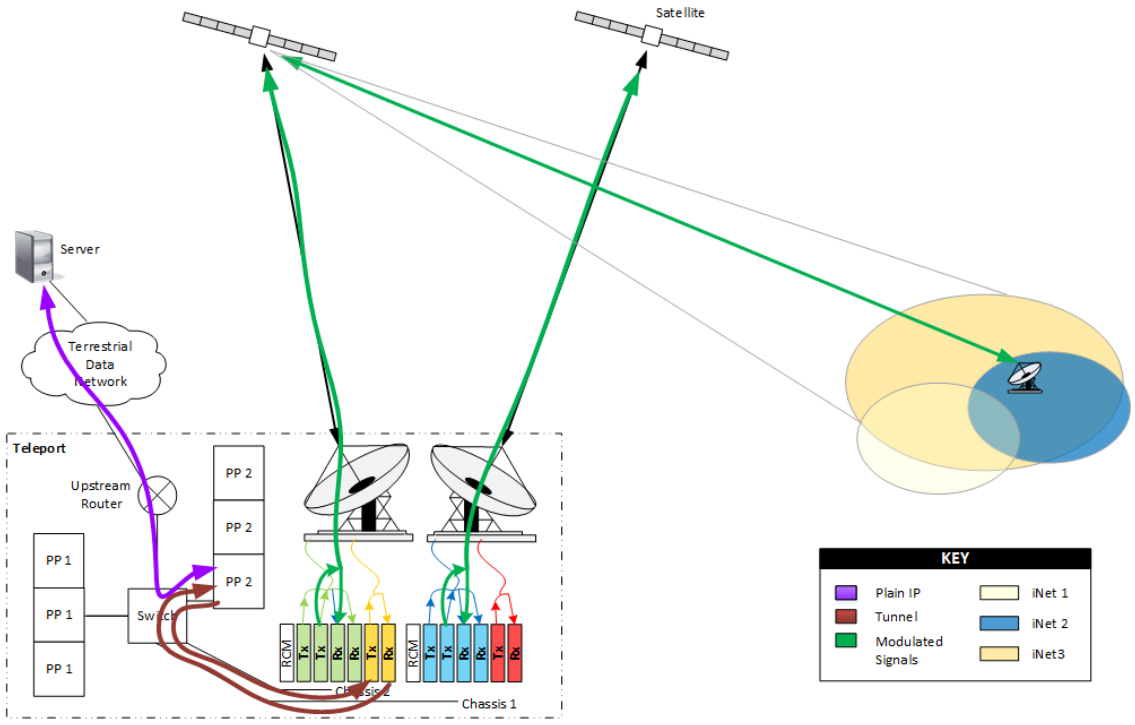


Figure 7-14. Satellite Switching, after switch

7.11.5 Using the Beam Switch API

Beam switches can be triggered through the Web Services API available on Pulse. This allows the use of custom beam-switch logic to replace that provided by iDirect. The proper use of this API requires that the system be configured so as to limit the built-in beam switching to be activated only when necessary to keep terminals from falling out of network.

Use of customer-written switch decision logic using the external beam switch API is desirable under certain conditions:

- The network operator wants to optimize based on data not available to the datapath, such as predicted motion of the terminals, business commitments, or other reasons
- The network operator wants to load balance across satellites
- When the network operator wants to change the behavior of the decision logic for some other reason

7.12 Terminal Congestion Management

The load balancing algorithm runs continuously at each SAS and is largely based on two main components. First, there is a ***decision metric (DM)*** value that is computed for each terminal, for each candidate beam to which the terminal could be switched; and on an actual congestion indication, which is based on a ***congestion metric (CM)*** value that is computed for every iNet. This CM value reflects the total unmet CIR/MIR demand.

Switching decisions between iNets in Velocity are based upon full knowledge of the actual congestion in a candidate iNet/satellite, whether in the same or in a different iNet/satellite.

A summary of the load balancing algorithm is provided below:

- Each SAS bandwidth manager creates a per iNET congestion metric (CM) data table.
- A congestion process resides in both the SAS and NOC PP clusters, and is responsible for communicating congestion metric data between the SASs and the NOC.
- Each SAS reports its CM table, every 30 seconds, to the NOC PP cluster.
- The NOC PP cluster receives and constructs a consolidated CM values table of all iNets in the system (all SASs/all iNets), providing full knowledge of all iNet CM values.
- The NOC then redistributes to all SASs, every 60 seconds, the consolidated iNet CM information.
- Based on the consolidated congestion metric data and decision metric data, the PP then makes appropriate switching decisions for terminals experiencing congestion.

These key functions are described in the remaining topics.

7.12.1 SAS-to-NOC Congestion Metrics Communication

Each SAS in the Velocity network, has full knowledge of the congestion metric (CM) of each iNet in its own network. Based on this knowledge, the SAS bandwidth manager builds a data table structure that contains the SAS ID, iNet ID of each active iNet, and a current iNet CM value. This CM value, is the same value used to report the associated "congestion metric" stat for each iNet. The table also contains a date-time-stamp of when each CM value was sampled and the table constructed.

After assembling its CM value table, each SAS sends the table to the NOC Global Bandwidth Manager (GBWM), based on a polling interval of every 30 seconds. The NOC will always have the latest CM table of each SAS. This SAS-to-NOC communication is illustrated below.

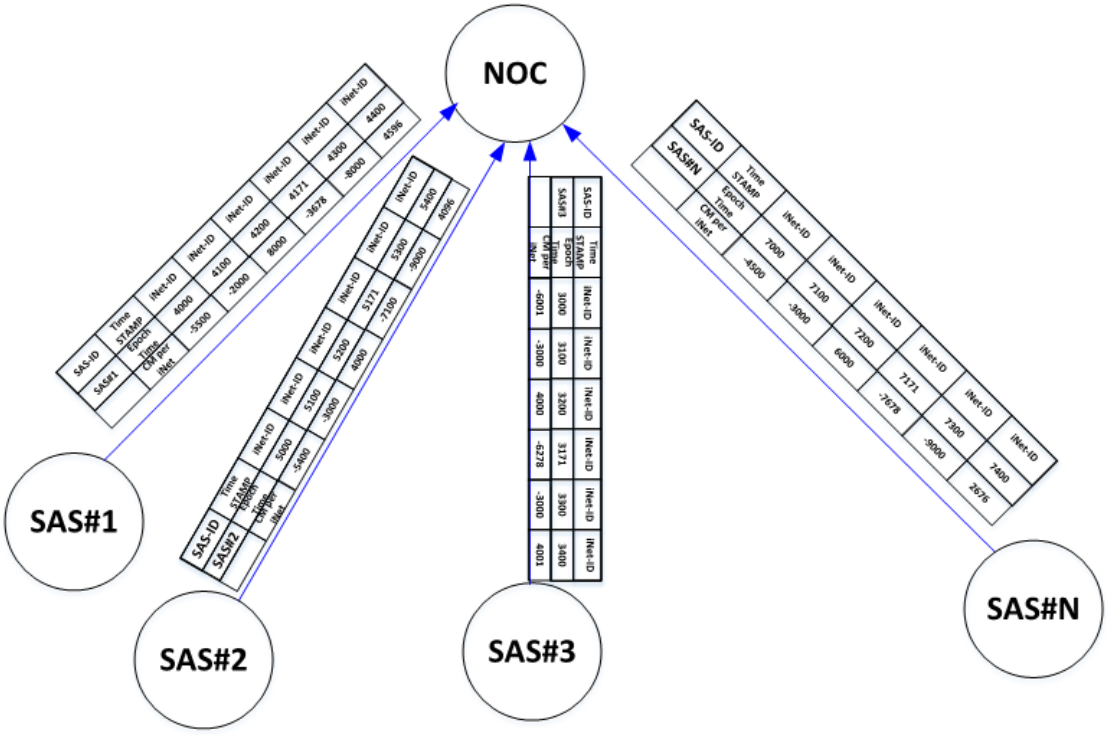


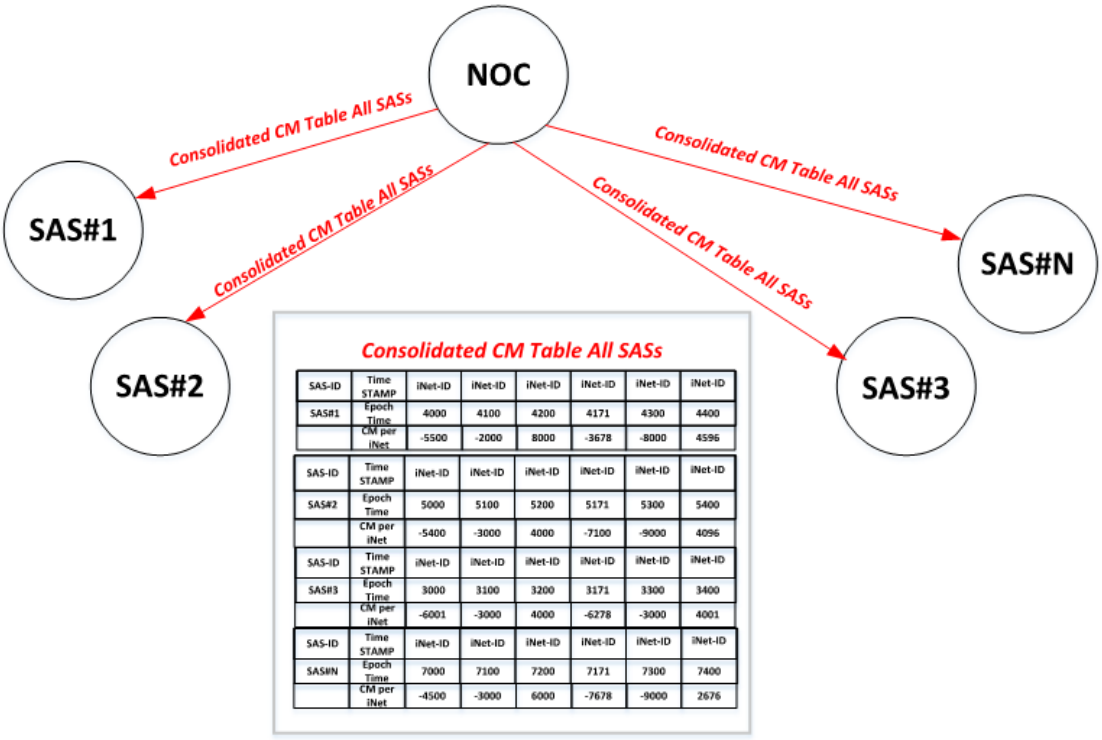
Figure 7-15. SAS-to-NOC iNet CM Values Communication

> ***NOTE:*** The CM values of foreign iNets (not in the same SAS/Satellite) are delayed, on the average, by 30 seconds, compared to the CM values in the local SAS. If during that period, the CM value of the foreign iNet has changed to indicate a congestion condition, then due to the delay in the reporting, the foreign iNet will appear uncongested. The congestion algorithm assumes that the overall system is relatively stable and a large number of such transitions is not expected, and therefore will have minimal impact on the system performance.

7.12.2 NOC-to-SAS Consolidated iNet CM Communication

After receiving a CM value table for all of the iNets, from each SAS, the CNG_Proxy process at the NOC constructs a consolidated table that contains the CM values for all the iNets in the entire system (all SASs and all iNets). The consolidated table includes a version number of the table, and a new date-time stamp generated at the NOC, for the time at which the new table is published. To ensure that each SAS in the network has CM data for all other SASs, the consolidated CM table, is then sent from the NOC to each SAS in the network.

Based on the consolidated congestion information, the PP make appropriate decisions to switch the terminals during congestion. An illustration of how the consolidated CM value table is published, every 60 seconds, from the NOC to each SAS, is shown below.



SAS-ID	Time STAMP	iNet-ID	iNet-ID	iNet-ID	iNet-ID	iNet-ID	iNet-ID
SAS#1	Epoch Time	4000	4100	4200	4171	4300	4400
	CM per iNet	-5500	-2000	8000	-3678	-8000	4596
SAS-ID	Time STAMP	iNet-ID	iNet-ID	iNet-ID	iNet-ID	iNet-ID	iNet-ID
SAS#2	Epoch Time	5000	5100	5200	5171	5300	5400
	CM per iNet	-5400	-3000	4000	-7100	-9000	4096
SAS-ID	Time STAMP	iNet-ID	iNet-ID	iNet-ID	iNet-ID	iNet-ID	iNet-ID
SAS#3	Epoch Time	3000	3100	3200	3171	3300	3400
	CM per iNet	-6001	-3000	4000	-6278	-3000	4001
SAS-ID	Time STAMP	iNet-ID	iNet-ID	iNet-ID	iNet-ID	iNet-ID	iNet-ID
SAS#N	Epoch Time	7000	7100	7200	7171	7300	7400
	CM per iNet	-4500	-3000	6000	-7678	-9000	2676

Figure 7-16. NOC-to-SAS iNet Consolidated CM Values Communication

7.12.3 NOC/SAS Communications

As a part of congestion management implementation, the same PP process runs on both the SAS and NOC PPs. The function of constructing and distributing the congestion metric (CM) tables, sent from each SAS to the NOC, is managed by the per SAS process; while the function of consolidating the CM data from all of the SASs and redistributing it to each of the SASs, is handled by the corresponding process at the NOC. To ensure system-wide awareness of the congestion metrics, a version of this same process runs on each primary and optional backup SAS, each primary and back-up NOCs, and interfaces with the NOC and at each SAS.

The diagram illustrates this intercommunication between all Velocity sites, including primary and optional backups, to facilitate assembly and delivery of system wide congestion values of all iNets in the system to all SASs.

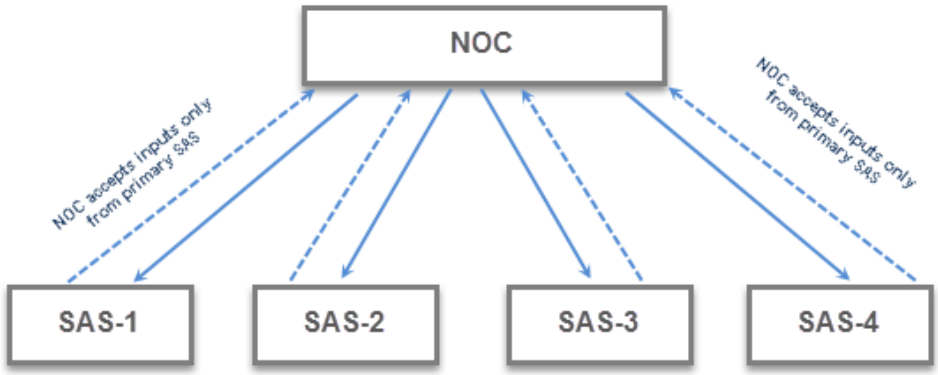


Figure 7-17. SAS-to-NOC iNet CM Values Communication

NOTE: If, for some brief period, there is an interruption of the SAS-to-NOC communication, then for that period the congestion management algorithm will rely on the local SAS CM data alone until the communication is re-established.

7.12.4 Determining Congestion Metric (CM) Values

Calculation of the congestion metric, is every 30 seconds, and is for upstream and downstream. The averaging algorithm, on which the calculation is based, determines a value that best represents the congestion level of a network. Due to the potential bursty nature of user traffic, averaging is done to reduce messaging chatter and to provide a more stable system behavior. An operator at the NOC NMS can configure congestion control parameters on a per-SAS basis.

To understand the congestion metric, it is important to consider the following GSP/SSPC bandwidth allocation properties:

- ***CIR Demand*** — the portion of the bandwidth request (demand) that is less than or equal to the configured CIR

- ***Best Effort Demand*** — the portion of the bandwidth request (demand) above the configured CIR, but less than the configured MIR (capped by the MIR value).

The following diagram show three cases to illustrate examples of how parameters A and B, which are used in the congestion metric calculation, are being calculated. Each of these cases has a **Total Capacity**, a **CIR Demand**, and a **Best Effort** demand. These are all demands at the top level nodes of the QoS allocation tree for the iNet.Consider the three cases as shown in the following diagram.

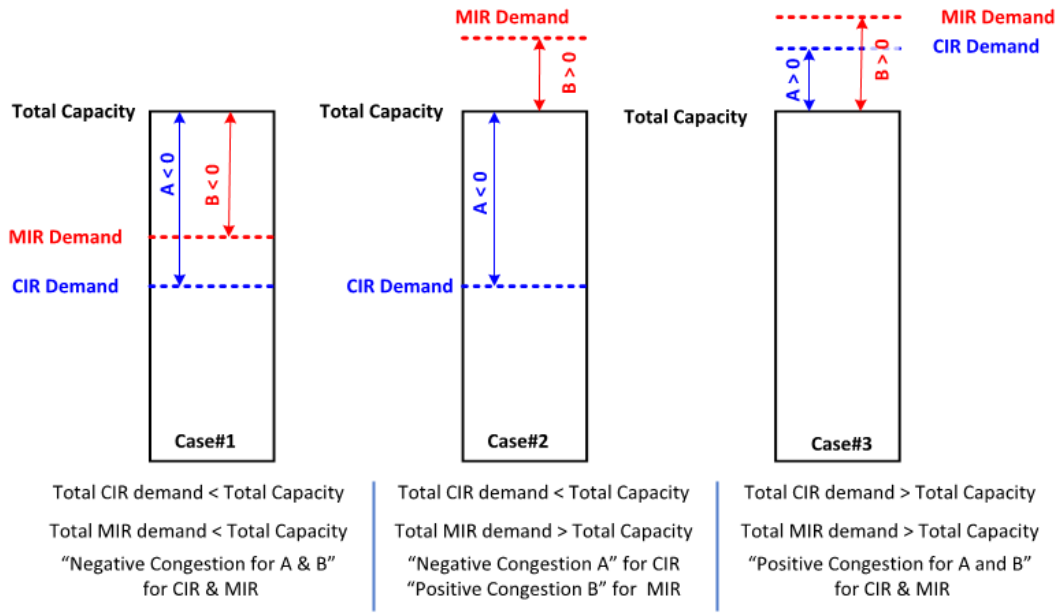


Figure 7-18. Congestion Metric Examples

The CIR and MIR are set, in a dynamic manner, based on the global bandwidth manager. Furthermore, the demand from terminals can vary from one moment to the next. These two influences combined will cause the total CIR Demand and Best Effort Demand to vary up and down as a function of time. Also, the Total Capacity on the inbound can change based on the current inroute group selection, and the capacity on the outbound can change based on configuration changes in the outbound carrier and the current efficiency, in terms of IP bits per-symbol for the outbound DVB-S2 carrier using ACM.

In **Case 1**, the iNet has met both the CIR Demand and the MIR Demand — hence there is "Negative Congestion" for both A & B, and the iNet is uncongested. In **Case 3,** the CIR Demand and the MIR Demand are both obviously greater than the Total Capacity — hence there is "Positive Congestion" for both A & B, and the iNet is congested. Finally, in **Case 2**, the CIR Demand is met, and the MIR Demand is not met — hence there is "Negative Congestion for A" CIR, and "Positive Congestion for B" MIR. Whether or not this is defined as congestion, is a policy decision that is input to the computation as configurable parameters.

7.12.5 Configurable Congestion Parameters

The parameters, described in the following table can be configured to affect the congestion algorithm. These parameters, which were configurable prior to the Velocity 1.7, are all configurable using the Pulse Web UI. Additional configurable parameters are in 7.12.5.

Table 7-2. Configurable Congestion Parameters

Parameter	Pulse UI Name	Description
HYSTERESIS (MBB hysteresis)	Hysteresis Second (**Add Satellite**)	This value, in seconds, is configured per satellite to determine a wait period before a new satellite/beam switch is initiated post the start of a previous switch.
BEAM SWITCH HYSTERESIS	Hysteresis Kbps (**Add Satellite**)	This positive value provides stability with respect to making a beam switch (current beam more preferred). A single value is configured per satellite (SAS).
SAT SWITCH HYSTERESIS	Sat Hysteresis Kbps (**Add Satellite**)	This additional hysteresis, a positive value, affects switching stability between satellites (current satellite more preferred). Higher values discourage the system from switching satellite, as this is a more disruptive switch. A single value is configured per satellite.
SAT SWITCH CONGESTION CEILING	Sat Congestion Ceiling (**Add Satellite**)	This value prevents switching back and forth between iNets on different satellites that are both congested.
CONGESTION FLOOR	Congestion Floor (**Add Satellite**)	This value, in Kbps, is a lower limit to how much negative congestion is considered. At some point all uncongested iNets are treated as equals, even if one is lightly congested compared to others. A single value is configured per satellite.
CONGESTION WEIGHT	Congestion Weight (**Add Satellite**) (**Add Terminal Type**)	This positive value (no units), which specifies the relative weight of the figure-of-merit vs. the congestion metric, can be used to disable beam switching due to congestion. A single default congestion weight is configured for the whole satellite, but can be overwritten by a per **Terminal Type** CW.

Table 7-2. Configurable Congestion Parameters (Cont.)

Parameter	Pulse UI Name	Description
CONGESTION METRIC DOWNSTREAM_DS_ WEIGHT	Congestion Metric Downstream Weight (**Add PP Cluster**)	This positive value (no units), with a range between 0 and 1, specifies a multiplier of the measured downstream congestion, to determine its relative influence on the congestion metric calculation.
CONGESTION METRIC UPSTREAM (US) WEIGHT	Congestion Metric Upstream Weight (**Add PP Cluster**)	This positive value (no units), with a range between 0 and 1, specifies a multiplier of the measured upstream congestion, to determine its relative influence on the congestion metric calculation.
CIR CONGESTION WEIGHT	CIR Congestion Weight (**Add PP Cluster**)	This configurable parameter, which has a '0' to '1' (inclusive) range, determines the influence of CIR/MIR on congestion. **1=CIR only**; **0=MIR only**; values in between indicate a weighted average of CIR and MIR.
CM_BACKOFF	Margin Allows iNETs to be Considered Congested (**Add PP Cluster**)	This parameter, in Kbps , is applied to the load balancing algorithm to allow the iNet to be considered congested before an unsatisfied demand occurs. Due to this factor, terminals are load balanced away from an iNet when the congestion is within **CM_BACKOFF** (margin) of actual congestion.



NOTE: The configurable hysteresis is independent for beam and satellite switch.

7.12.6 Per-iNet Congestion Metrics Calculations

Computation of the congestion metric value, which is computed for each and every iNet in each SAS, is illustrated in the following diagram. On the left side, separate calculations are shown for Upstream and Downstream, where the output of each is averaged every 5 seconds.

The computed CM total value, which is shown on the right side of the diagram, can be based on **CIR** Congestion only, **MIR** Congestion only, or a weighted combination of both the CIR congestion and the MIR congestion. See Configurable Congestion Parameters on page 285.

Configurable parameters that can influence the calculation of the total CM value (CM_{Total}), for example the Upstream/Downstream Congestion Weight values and the CM Backoff value, are listed in 7.12.5.

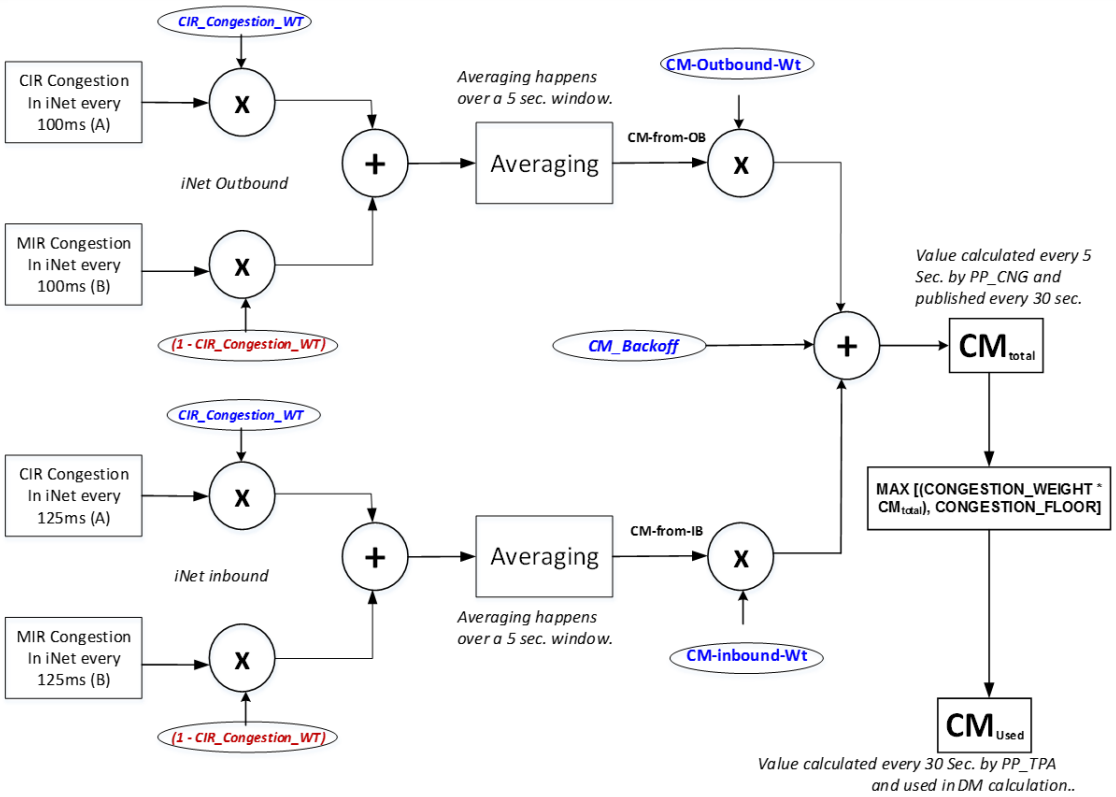


Figure 7-19. Congestion Metric Value Calculation

7.12.6.1 Configurable Congestion Weight Values

Some example guidelines for determining the operator configurable parameter settings for the Downstream and Upstream Congestion Weight values are given below, and following table outlines what the expected behavior will be.

- To consider both downstream/upstream congestion, based on a ratio of the weighted values, configure a non-zero value for US/DS, for example 1/1, 0.5/0.5, 1/0.5 respectively.
- To consider Upstream only, configure DS weight of "0" and Upstream weight of "1".
- To consider CIR only configure **CIR CW =1**; **MIR CW** is automatically set = 0.
- To consider MIR only configure **CIR CW = 0**; **MIR CW** is automatically set = 1.
- To consider a CIR/MIR congestion mix, then configure CIR CW = 0.5 (example), and the MIR CW is automatically set = 0.5.

Table 7-3. Operator Configurable US/DS Congestion Weight (CW) Parameters

DS CW	US CW	Expected Behavior
0	**0**	Switch never occurs since the US/DS weights are both configured as "0"
0	**1**	Switch occurs based on US demand being higher than the configured MIR
1	**0**	Switch occurs based on DS demand being higher than configured MIR/ CIR
1	**1**	Switch will occur since the demand is higher than the configured CIR/MIR

7.12.6.2 Configurable Upstream and Downstream Congestion

The following table illustrate sample calculations for Downstream and Upstream congestion, based on configured CIR/MIR congestion weights. Note that the CIR/MIR congestion values may be different for Downstream (A/B) and Upstream (A'/B'), and therefore the Upstream/Downstream congestion contributions may be different.

Table 7-4. Sample Upstream/Downstream Congestion Calculation

Upstream Congestion (A, B)				Downstream Congestion (A', B')			
CIR CW	**MIR CW**	**CONGESTION**	**Contribution**	**CIR CW**	**MIR CW**	**CONGESTION**	**Contribution**
1	0	A	CIR Only	1	0	A'	CIR Only
0	1	B	MIR Only	0'	1	B'	MIR Only
0.5	0.5	0.5 * (A+B)	CIR/MIR Mix	0.5	0.5	0.5 * (A'+B')	CIR/MIR Mix
0.7	0.3	0.7A + 0.3B	CIR/MIR Mix	0.7	0.3	0.7A' + 0.3B'	CIR/MIR Mix

7.12.7 Congestion Metric and Decision Metric Calculations

Table 7-5 shows congestion metric and decision metric calculations, which are partly based on configurable parameters like SAT_SWITCH_HYSTERESIS. In the following examples, S0 = Same Satellite; S1 = Different Satellite.

Table 7-5. Congestion Metric and Decision Metric Calculations

Parameter	Calculation
Congestion Metric Total	*CMTOTAL = CM_Outbound* CM_Outbound_WT + CM_Inbound * CM_Inbound_WT+ CM_Backoff* CM Outbound WT: CM Weight in Outbound: value =**0**; value=**1**; value **between 0 and 1** CM Inbound WT: CM Weight in Inbound: value =**0**; value=**1**; value **between 0 and 1** CM Backoff: CM backoff allows the iNet to be considered congested before there is unmet demand for the load balancing algorithm
CM used in DM Calculation	CM_{used} *= max [(CONGESTION_WEIGHT (S0) * CM_{total}), CONGESTION_FLOOR(S0)], where **iNet is in same Satellite(S0)**.*
CM used in DM Calculation	CM_{used} *= max [(CONGESTION_WEIGHT(T) * CM_{total}), CONGESTION_FLOOR(S0)], **where iNet is in same Satellite(S0), and where T is the Terminal Congestion Weight based on specific Terminal Type.***
CM used in DM Calculation	CM_{used} *= max[(CONGESTION_WEIGHT(T)* CM_{total}), CONGESTION_FLOOR (S1)],* ***where iNet in different Satellite(S1), and where T is the Terminal Congestion Weight based on specific Terminal Type.***
CM used in DM Calculation	CM_{used} *= max[(CONGESTION_WEIGHT(S1) * CM_{total}), CONGESTION_FLOOR (S1)],* ***where iNet in a different Satellite (S1) and there is no Terminal Specific Congestion Weight based on specific Terminal Type.***
Decision Metric: Switch in Same Satellite	***DMsame-satellite = (CMdestination - CMcurrent) + (FOMdestination - FOMcurrent) + BEAM_SWITCH_HYSTERESIS + (BBM * BBM_HYSTERESIS)*** • ***BBM = 1 if switching mode is break before make*** • ***BBM = 0 if switching mode is make before break*** • ***BBM_HYSTERISIS = 5000 kbps (default)***
Decision Metric: Switch to Different Satellite	***DMdifferent-satellite = (CMdestination - CMcurrent) + (FOMdestination - FOMcurrent) + BEAM_SWITCH_HYSTERESIS + SAT_SWITCH_HYSTERESIS*** ***when CM of the destination iNet is available***
Decision Metric: Switch to Different Satellite	***DMdifferent-satellite = - Min(CMcurrent, SAT_SWITCH_CONGESTION_CEILING) + (FOMdestination - FOMcurrent) + BEAM_SWITCH_HYSTERESIS + SAT_SWITCH_HYSTERESIS*** ***when CM of the destination iNet is not available***

7.12.8 Beam Switching Decision Logic

The beam switching decision logic, which is controlled by a single algorithm, is executed periodically, whenever a geo-location update is received from a terminal. This decision, which can trigger a switch to a beam within the same satellite (S0), or a different satellite (S1), takes into account the following:

- Terminal geo-location, reported from the terminal to the SAS
- Support for the RF Terminal Type
- Congestion of each of the iNets
- A map that describes a figure of merit (FOM) for each beam as a function of location
- In the case of different satellite, satellite visibility from terminal's geo-location, based on the configured look angle parameters

Prior to making a beam switch decision, the PP determines the switch mode of all reachable networks, to determine whether to use a Break-before-Make or Make-before-Break switch. If the current beam and target beam have different upstream/downstream polarizations, or if the target beam downstream frequency falls in a different LNB band from the current beam, then a switch to that target will require a BBM switch. Otherwise, an MBB switch can be used. Finally, the PP also takes into consideration whether the RF terminal type is supported by the target beams. As such, network operators can exclude certain terminal types from being allowed to use a specified Beam/iNet.

NOTE: Certain remote types always use BBM beam switching.

7.12.8.1 Determining the Decision Metric

When the decision logic is executed, by the SAS, a decision metric (DM) value is calculated for each terminal, for every candidate iNet to which that terminal could be switched. An iNet candidate must meet the following criteria:

- The terminal is within the beam footprint of the candidate iNet
- The terminal access class and iNet access class are compatible (**See Note**)
- The iNet polarization/frequencies are compatible with the terminal capabilities
- The beam/iNet can support the RF Terminal Type
- The RF Terminal Type has not been excluded from Beam/iNet by the operator

Computation of the DM takes into account a comparison of the congestion metric (CM) values and the ***Figure-of-Merit (FOM)*** values of both the current iNet and target iNet. The FOM is a relatively static value that is proportional to the beam signal strength. Also taken into account is an additional hysteresis intended to stabilize terminal switching. This hysteresis is slightly biased in favor of the current iNet over any candidate iNets. Hysteresis values can also be operator-configured to control the amount of resistance the system will have to switching to a beam in the same, to a different satellite, or to requiring a BBM switch.

A negative DM value for a candidate iNet indicates that the iNet is potentially a new and better home for the remote. The iNet with the most negative DM value is considered the best candidate, to which the terminal could be immediately switched or scheduled to be switched.

This metric provides a means of comparing the quality of the link obtainable in each candidate beam and weighing that against the level of congestion being experienced in each candidate iNet. Various configuration parameters allow the balance between link quality and network congestion to be adjusted. The configuration of the FOM map allows for the setting of link quality, and various weights can be configured to control the influence of the types of demand that can be included in the calculation of network congestion.



NOTE: The access class is an 8-bit or 40-bit mask configured on individual terminals, as well as on iNets. This is a method by which some iNets can be reserved for use by only certain classes of terminals. For example, exclusive access to an iNet may be given to terminals operated by a government agency. Whether an iNet is usable by a terminal depends on the result of a bitwise logical AND of the iNet access class and the terminal access class. A non-zero result means the iNet is usable.



NOTE: The 40-bit access bitmask option is enabled through a script that is run by iDirect TAC at the request of the customer. Once this feature is enabled by the script, it cannot be disabled.

7.12.8.2 Staggered Terminal Switching

The Velocity enhanced congestion management implementation also leverages an enhanced “staggering algorithm” that ensures switching of terminals to a new iNet/satellite is properly managed to avoid creating congestion in the target iNet.

With the enhancements to the CM calculation and the continuous updates of all SASs, the staggering algorithm is more able to consider the dynamic changes of congestion conditions in both the current iNet and candidate iNets. Furthermore, staggering can be partly controlled as the operator is able, through configuration, to affect the number of terminals that are switched during an active congestion cycle.

Other enhancements to the staggering algorithm are listed below:

- The current congestion state in the target iNet is considered such that terminals will not be switched if the target iNet is congested
- The number of terminals to be switched during an active congestion cycle is controllable via configuration (based on the percentage of active terminals in an iNet).
- Selected terminals are switched in a staggered fashion over a period of 90 seconds, wherein the terminals are switched one-by-one until congestion is relieved.
- During a 90 second duration for staggering, the congestion metric is calculated every 5 seconds — as soon as the CM becomes negative, the remaining terminals (if any) in the queue are immediately released from the staggering queue.

7.12.9 Congestion Switching Scenarios

These congestion switching scenarios represent different cases that involve the same or different SAS/Satellites in which congestion may exist or not. The requirements for handling congestion in these scenarios are all addressed in the congestion management enhancements.

- **Case-1**: Two iNets are in the same or different SAS or Satellite, in which one of the iNets is experiencing congestion, while the other is not.

  In this case, the terminal switches to the iNet that is not experiencing congestion.

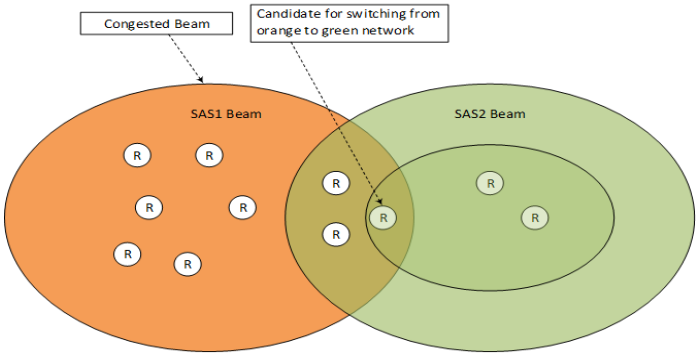


Figure 7-20. Multiple SASs with Different Satellites

- **Case-2**: Two SASs are pointing to the same or different Satellites, in which each satellite has two overlapping beams, each of which is currently experiencing traffic congestion.

  In this case, if the figure-of-merit (FOM) values are identical or close to being equal in the current iNet and the target iNet, then the terminal never switches to the other iNet; and if the two iNets in the two SASs are both congested, then switching the remotes to the other iNet is not a solution.

  A solution to the congestion in this scenario, is either of the following:

  - Increase bandwidth in both iNets
  - Adding iNets to the beam

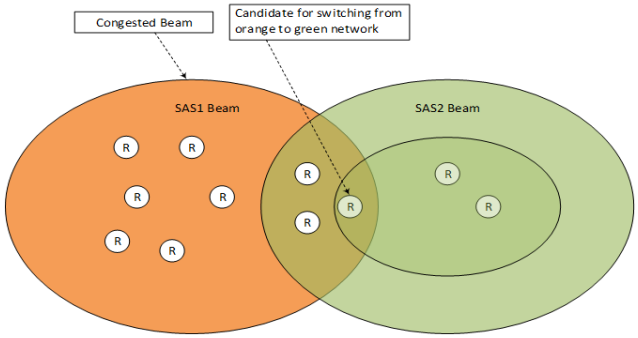


Figure 7-21. Multiple SASs with Different Satellites

- **Case-3**: This case involves multiple remotes in an overlapping area, in the same or different satellite. In such as case, all remotes should not be switched simultaneously to the iNet in the other SAS/satellite. The response should be a staggered switching.

  In this case, where multiple remotes are in an overlapping area, during a period of congestion, the remotes should not be switched simultaneously to a target iNet in the other SAS/satellite — not in the case of the multiple SASs pointed to single satellite; nor in the case of multiple remotes involving SASs pointing to different satellites.

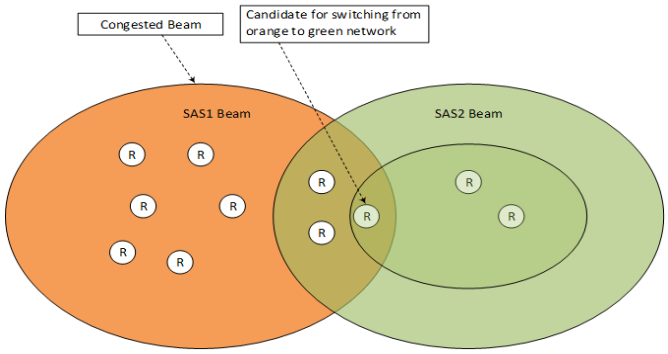


Figure 7-22. Multiple SASs with Different Satellites

	NOTE: The enhanced congestion management does not prevent terminals from acquiring into a congested iNet and staying in that iNet if there are not other candidate iNets available. This is an expected behavior.

7.13 Terminal Global Positioning System Validation

GPS position information is provided to the terminal manually through user input or through connection to an external device. A common external device is an OpenAMIP-enabled antenna. Positional information is an important real-time input for mobile terminals. This information is normally based on precise timing of signals received from GPS satellites.

Occasionally, the reception and timing of these signals is altered by signal path blockage, multi-path reflections, or interference. This can result in some calculations from these altered signals producing erroneous terminal geo-locations. The resulting outliers can induce significant errors into:

- Terminal transmit timing calculations performed during acquisition
- Terminal velocity calculations that can hamper any efforts to predict the apparent shifts in carrier frequencies, which occur due to the Doppler Effect; such predictions that become increasingly critical as a terminal's velocity relative to the satellite increases.

A series of calculations are performed on a set of sequential GPS positional values, and a determination is then made as to whether to reject any of those positional values and which ones to reject. Detection of an outlier is not in itself cause for any action, other than rejection of the corresponding measurements. Only measurements that are not rejected are then used for subsequent calculations and controls.

7.14 Highly-Elliptical Orbit Support in Velocity

The Highly-Elliptical Orbit (HEO) Feature supports HEO Satellites, which operate in a Highly Elliptical-Orbit. HEO satellites provide significantly better coverage to the arctic region where Geo Stationary Orbit (GEO) Satellite access is challenging.

Continuous service over the arctic region is accomplished by the use of multiple HEO Satellites. These satellites are not visible from the service area in the arctic region all the time, but the orbits of these satellites are chosen so that at least one satellite is visible all the time. A switchover between the satellites is required as each satellite transitions between the active and inactive arc. During periods when more than one satellite is visible, switchovers between the satellites (either automatic or through an external management system) ensures that each terminal maintains the best possible service depending on where it falls within the coverage.

7.14.1 HEO Acquisition and Physical Layer Operations Including GSC

The Terminal Acquisition algorithm and mechanism supports remote terminals operating in both Geo Stationary Orbit (GEO) and non-Geo Stationary Orbit (NGSO) satellite networks, and includes Highly Elliptical Orbit (HEO), in addition to Low Earth Orbit (LEO), and Medium Earth Orbit (MEO).

Figure 7-23 shows two (2) HEO Satellites covering the arctic region.



Figure 7-23. HEO Satellites

Before transmitting an acquisition burst to acquire into the network, a remote must accurately estimate its Time Offset (TO) (with reference to a reference terminal located at

the sub-satellite point of the GEO satellite) and Local Frequency Offset (LFO) (derived from the rate of change of the remote's TO).

For a HEO satellite constellation, the Terminal Acquisition algorithm must account for the following:

- The satellite location relative to the terminal or RF GW is no longer fixed
- The satellite is moving at very high speed which can reach up to 9000mph (14484Km/h).

The Terminal Acquisition algorithm applies proper TO and LFO when the user link path length (terminal to satellite) is changing due to a combination of satellite movement and terminal movement.



NOTE: The Antenna Control Unit (ACU) is required to reliably provide accurate location and time from satellite navigation system(s) over the arctic region so that the terminal can find HEO Satellite position and velocity through TLE entries.

7.14.1.1 Acquisition Algorithm

To operate in NGSO satellite networks, the Acquisition Algorithm includes the following features:

- A Two-Line Element (TLE) orbital description can be configured to define the orbit and movement of a satellite. The TLE defines the HEO Satellite orbit and can be used to predict its location given epoch time. The GPS receiver equipped at the remote terminal provides the GPS location and time. The TLE set, which contains multiple HEO Satellites, coupled with GPS time, defines the location and movement of each satellite over time.
- TLE and beam map data are dynamically refreshed. The TLE file accuracy degrades with time; therefore, the operator must update TLE data periodically. After the remote terminal finds the proper satellite and locks with its downstream carrier, it receives the latest TLE info from PP and uses it for timing correction during acquisition or normal in-network operation. Beam maps for NGSO satellites must be dynamic, to account for satellite and beam motion, and the potential for channel reassignments.
- TLE and GPS data allows a remote to dynamically adjust its transmitter frequency and burst timing.
- The algorithm has the ability to select a satellite based on its elevation as seen from the terminal, and its position in the active portion of the orbit. Upon receiving geolocation and time from the ACU, the remote selects an NGSO satellite or GEO satellite. Elevation is computed from its longitude (and assumed GEO orbit) or from its TLE.

  A Sub-Satellite Point (SSP) Contour Map may optionally be used. The operator may configure an SSP Contour to ensure the remote only attempts to use a satellite when the satellite is visible from the RF Gateway location, and is in the active portion of its orbit to support NGSO acquisition.

  This map is installed when commissioning or staging the terminal, alongside TLE information. A contour and TLE are provided for each satellite. The terminal will only

acquire on a satellite if that satellite's SSP is currently within the SSP Contour. The SSP Contour and default TLE are quasi-statically configured in nonvolatile memory in the modem. The SSP Contour can be provided through the Update Manager, or through the Commissioning API/WUI.

The remote will use the TLE to point to any satellite for which it has a TLE (including GEO). It sends the satellite TLE to the ACU via the OpenAMIP 'O' command. The ACU uses the TLE for pointing and tracking the NGSO satellites.

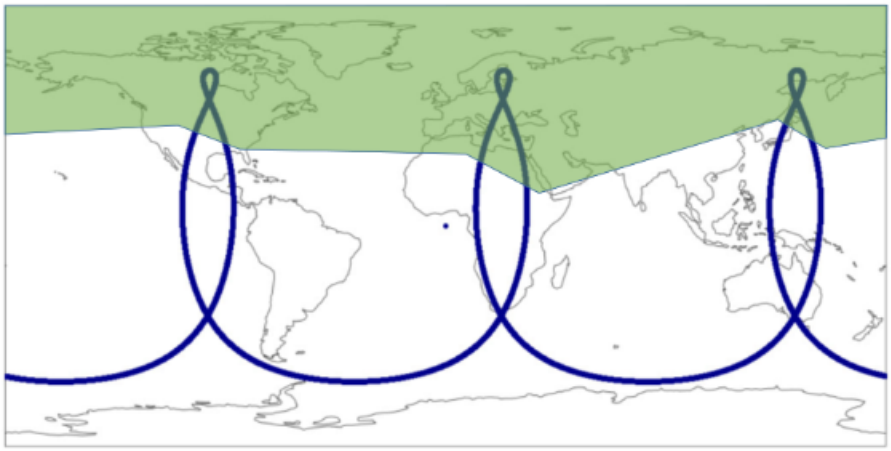


Figure 7-24. SSP Contour Map

7.14.2 Dynamic iNet Creation/Deletion

HEO Satellite beam footprints change as the satellite moves relative to the ground and the terminals move across the beams. Because of this, the capacity required in each beam changes over time. To address the changing capacity required, the operator may choose to activate additional “Floater” iNet(s) when capacity demand increases, and deactivate them when demand decreases. Load Balancing Processes in the iDirect platform perform load balancing when an iNet is added; and terminals are off-loaded in an iNet scheduled to be deactivated to the other active iNet in the beam. A “Floater” iNet can be activated in any beam to for increased demand that is causing congestion in that beam.

Figure 7-25 shows the state of the beam in terms of the number of iNets in the beam at any given time. In State 1, the beam has only a single iNet, which constitutes the base capacity. Adding a “Floater” iNet takes the beam to State 2, where it has a total of two iNets. It can stay in that state for some time and then transition back to State 1 when the “Floater” iNet is deactivated. Assuming there is sufficient hardware, “Floater” iNet(s) can be added to a single beam.

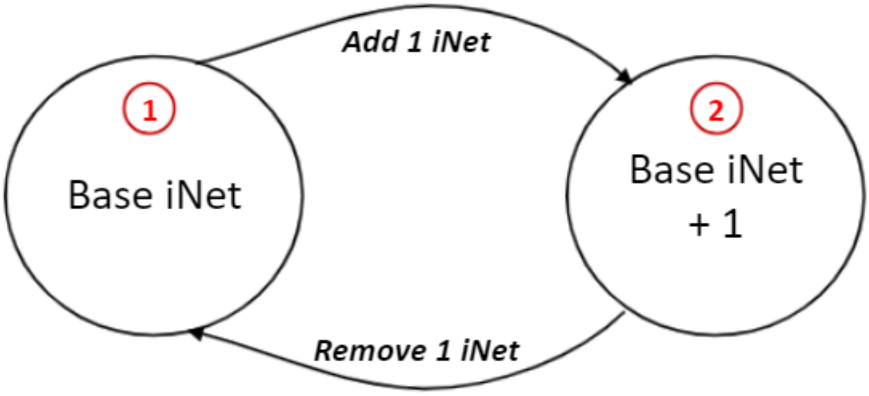


Figure 7-25. Dynamic iNet Creation/Deletion

When the beam is in State 1 it has only the Base iNet. At some point in time, a decision is made (externally) to add the "Floater" iNet to the beam. The system then transitions to State 2, where the beam has two iNets. It can remain in this state for some time. At a certain point in time, an external decision is made to remove the Floater iNet and the beam transitions back to State 1 with the Base iNet.

Figure 7-26 shows the changes in the capacity of a beam as the "Floater" iNet is moved from the blue Beam to the red Beam.

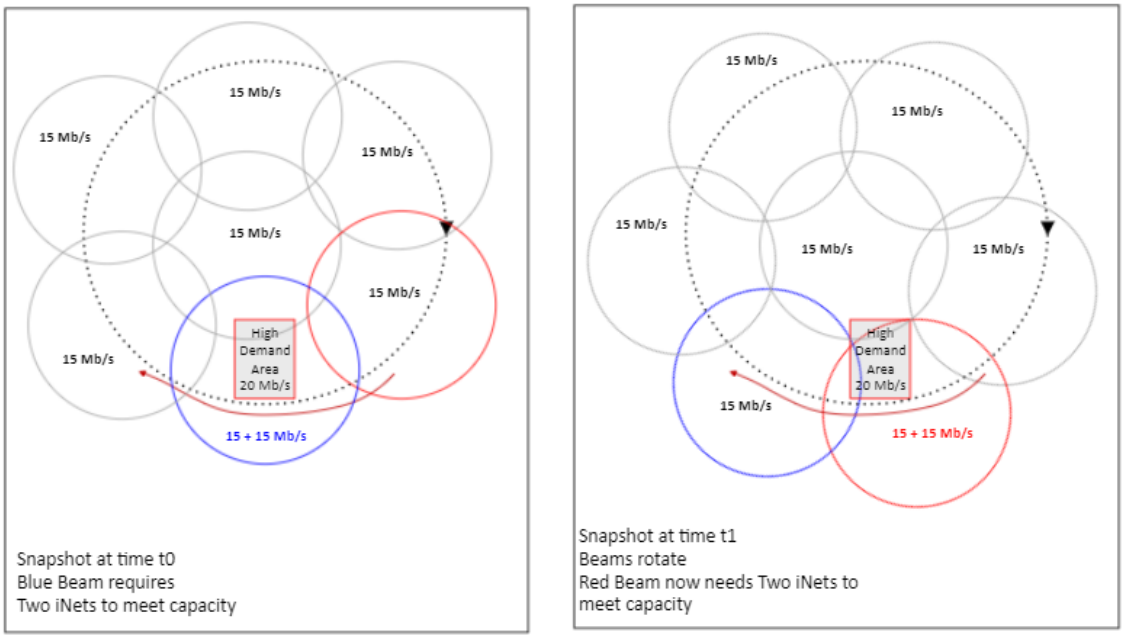


Figure 7-26. Beam Capacity

A value of 15 Mb/s is used as an example for the capacity of each iNet. At t0, the blue beam covers the high-demand area and requires additional capacity. So with the "Floater" iNet activated, the total capacity is 30 Mb/s and can meet demand from the high-demand area. At t1, the beams moved and now the high demand area on the ground falls under the red beam. So the "Floater" iNet is deactivated in the blue beam and reactivated in red beam to meet the demand. This example demonstrates the need to shift the capacity between the beams as

the beams move but the demand remains high in a certain region on the ground (or in the air above the ground).

7.14.3 HEO Satellite and Beam Switching

The Velocity 4.0 system allows the operator to integrate services using a combination of HEO (High Elliptical Orbit) and Geosynchronous Equatorial Orbit (GEO) satellites. This requires the integration of the beam and satellite switching between Geosynchronous Satellites and HEO Satellites. Intra-satellite and inter-satellite switching between the Geosynchronous Satellites is done automatically by the iDirect System. Switching between the Geosynchronous and HEO Satellites can also be done through an external API control performed externally by the customer.

An emergency switchover capability is initiated When the terminal reaches the edge of beam but it has not yet been switched to another iNet (either due to the load balancing algorithm or external command). This emergency switchover is initiated by the iDirect System and is not initiated externally. The purpose of emergency switchover is to prevent a terminal from dropping out of the network. Its purpose is not to optimize or pick the best iNet/beam/ satellite, but to maintain the service. Once the terminal is back in the system, its optimal target iNet/beam/satellite is selected either by the intra-group automated switching or the external API control when the selection of optimum target requires a switch between groups.

7.14.4 Dynamic Beam Changes

Some satellites may project beam footprints that sweep across the ground rather than being stationary, as is common with GEO satellites. This may, for example, be the result of the satellite projecting a fixed pattern from its viewpoint; satellite motion will sweep that fixed pattern across the ground. The exact time interval, number of beams, and number of satellites are operator-configured. The changing beam footprint results in a varying figure of merit on the ground that needs to be constantly updated approximately every 15 minutes.

The Dynamic Beam Feature enables the Velocity System to adapt to a changing beam footprint such that there is minimal impact to the system while ensuring that the system can adapt to the changes within the requisite time requirements.

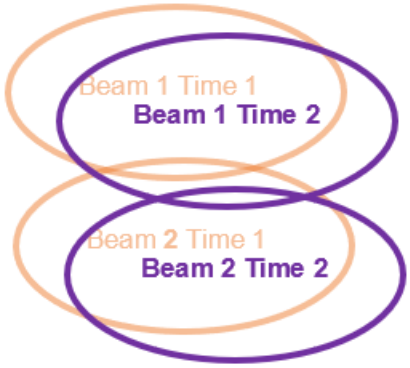


Figure 7-27. Dynamic Beam Changes

The operator generates dynamic beam maps and loads them into Pulse at intervals suitable to the satellite orbits in use. These are then loaded onto the respective data path elements on the CDC/SAS. This process is similar to the existing method of updating configuration.

However, for Dynamic Beam Switching, the configuration loaded has a specific validity time post during which a specific beam and/or channel configuration set can be used. This specific dynamic beam configuration and schedule set are "activated" automatically based on the schedule time in the configuration.

To minimize the impact on the system and also ensure the system can apply the dynamic beam changes quickly (approximately every 15 minutes), the dynamic beam configuration is updated via Pulse API as configuration files. To ensure that the system can switch the currently active beam set in the expected time frame, rather than relying on Pulse to push a new beam update or have an external program trigger the activation via an external command, a pre-determined "dynamic beam configuration" change schedule is provided to the protocol processor cluster via a separate configuration file. However, to provide the operator the capability to manually intervene/change the active beam set configuration, an RT command is also provided.

Note that it is expected that the PP_Cluster is already synchronized to Network Time Protocol (NTP) to ensure the time specified in the schedule is universally aligned. Upon activation (or failure) of the operation, an event is reported to the Pulse NMS for record keeping.

7.15 Enhanced Inclined Orbit Support in Velocity

Velocity can support satellites with up to 15 degrees of inclination without terminal acquisition issues and without sacrificing spectral efficiency. This can help mitigate service degradation on congested beams by enabling satellite operators to access older satellites at an inclined orbit.

	***NOTE:*** iHub and ILC linecards are required to use this feature for an Inclined Orbit Satellite.

To support this, operators can input TLE information and longitude information when configuring a satellite. TLE configuration is available beginning in Pulse 4.0; and beginning in Velocity Release 4.0, Velocity software can utilize TLE information, if available (otherwise it will just use longitude). It is not mandatory to upload TLE information for GEO or GIO Satellites. Existing GEO networks can still be configured in the legacy way; and when adding new GEO satellites, the operator can choose to add TLE information or defer that. If TLE is not configured, as satellites begin to become inclined, the operator can continue configuring longitude and latitude wander for the satellite. However, the terminal's tolerance for orbital inclination will be limited to 2 - 4 degrees depending on the geo-locations of the hub and terminal.

As a satellite begins to drift, operators can update their configurations with TLE information. This ensures that the Protocol Processors (PP) and linecards can compensate for timing and Doppler effects caused by satellite movement. While the hub side can only compensate for about half of the timing and Doppler variation, this adjustment significantly extends the satellite's usable life, even for non-upgraded terminals.

As the satellite's inclination increases, operators can initiate terminal upgrades to Velocity Release 4.0. Upgraded terminals can compensate for residual timing and Doppler errors that the hub alone cannot correct. However, terminals that cannot undergo upgrades will eventually need to be excluded from the GIO iNets.

Eventually, the system determines (based on a configured inclination threshold) that the orbit type is GIO. This will cause the PP to not switch a terminal into the satellite if the terminal does not report GIO capability.

Inclination will have the greatest effect on terminals at high latitudes or near beam edges. The effects are most pronounced at the Northern and Southern extremes of the inclination, when timing offset is most extreme. This mainly affects acquisition; once in the network, UCP can compensate a substantial amount of offset. Terminals can be upgraded gradually. Hub-side GIO support will actually keep many terminals operating much longer as the transition progresses. Eventually, the operator does need to upgrade terminals, especially those operating in high latitudes or near the edges of beams, which are now moving due to the inclination. Some old terminals can probably linger for years if not in high latitudes or near beam edges. Terminals that cannot be upgraded will eventually have to be excluded from the GIO iNets.



NOTE: The full benefit of this feature is realized only when both the hub side and remote side software are at 4.0 version. If only the hub side is running 4.0, the hub side will still provide its half of the compensation, hence providing improvement in incline orbit performance. In this configuration, the actual tolerance depends on the geo-locations of the hub and remotes. Please contact TAC for more details.

7.16 Centroid-Based Initial Beam Selection

The Centroid-Based Initial Beam Selection uses an improved beam selection algorithm on the terminal to make optimal initial beam selection. This new algorithm significantly improves the initial beam selection at a terminal over the previous approach in a constellation with non-circular beams and/or mixed wide/spot beams.

Beam selection logic is used at acquisition by all terminals. For discreet terminals, the beam selection decision is made continuously and not just at acquisition or when a terminal nears a beam edge. Excluding the notion of hysteresis, the previous logic selected the beam in which the modem is furthest from the nearest beam edge. In the case of network operators that use a combination of spot and broad beams, a number of the spot beams might not be selected. For example:

- Spot Beam 1 is selected when in Spot Beam 1 and **no** overlapping coverage with Broad Beam.
- Broad Beam is selected when “distance to Broad Beam edge” is greater than “distance to Spot Beam 1 edge”.
- Broad Beam is selected when in Spot Beam 2, 3, and 4.
- Spot Beam 5 is selected when “distance to Spot Beam 5 beam edge” is greater than “distance to Broad Beam edge”.

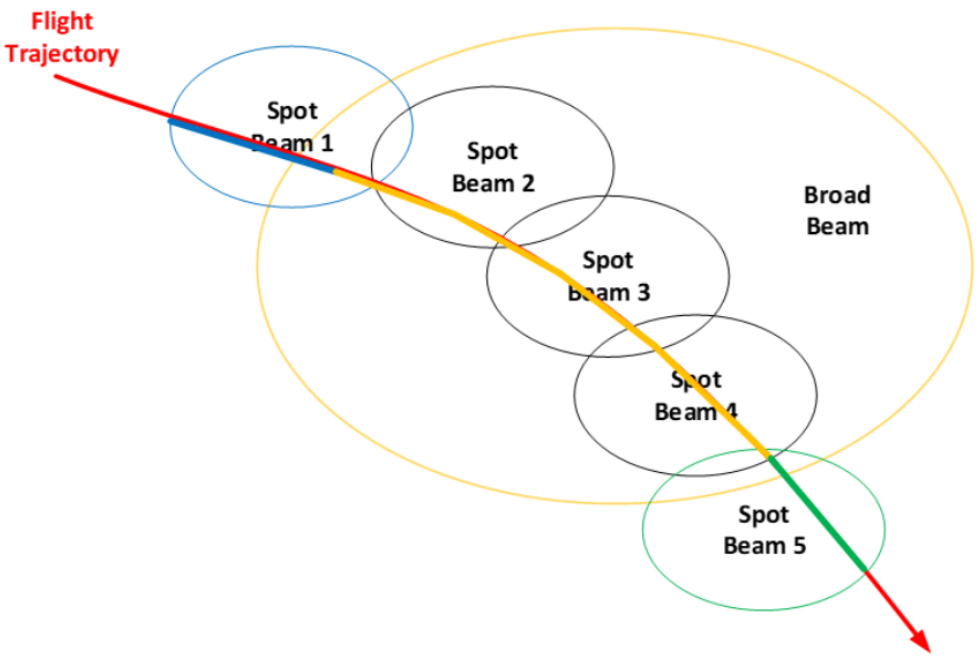


Figure 7-28. Beam Selection

Centroid-Based Beam selection enables the terminal to make better choices in a variety of complex beam scenarios (e.g., split beams, multiple FOM hot spots in a beam, different size beams, hole in beam, exclusions from beam for regulatory purposes).

This Centroid method is based on latitude/longitude proxy for distances. In some cases, the operator can assist by dividing complex beams into simpler polygons.

A terminal selecting a beam using the Centroid method computes a Figure of Merit (FOM) based on beam area (smaller is better) and distance to the centroid of the beam (closer is better). It then selects the beam with the best FOM (after accounting for hysteresis).

8 Satellite Terminal Features

This chapter describes major Satellite Terminal features supported in the Velocity system.

The following sections are included in this chapter:

8.1 Terminal VLAN Mapping 306
8.2 Terminal Component Mismatch 307
8.3 Bonded Modems 308
8.4 Discreet Mode Terminals 310
8.5 Mitigating Skew Impairment in Mobile Terminals 312
8.6 Fast SNR Reporting 320
8.7 External Beam Select Mode 321
8.8 Commissioning Enhancements (API) 322

8.1 Terminal VLAN Mapping

Packets in each segment of the subscriber's virtual network within the Velocity system contain an 802.1q tag (i.e., VLAN ID). Previously, the iDirect system required packets to have the same value in each segment and to be globally unique. In Velocity, the VLAN ID used on the Terminal LAN is locally significant.

VLAN mapping allows the VLAN IDs used on the terminal LAN to be different from the globally unique SVN ID used in the Velocity system. The satellite router software will map between the VLAN ID used in the OTA network and the Terminal LAN VLAN ID and will map a packet's globally unique SVN ID to a locally-significant VLAN ID before sending it on to the terminal LAN. This enables subscribers to connect a Velocity satellite network to an existing satellite network configured to use one or more VLAN IDs.



Figure 8-1. VLAN ID Mapping

NOTE: VLAN mapping on the Velocity Hub-side, is a customer responsibility.

8.2 Terminal Component Mismatch

Satellite terminals can be configured such that upon entering a satellite network, it can send to the SAS, a key/value string that identifies all of its key components — for example, satellite router, BUC, ACU, or Terminal Type. This string is compared to an equivalent string at the SAS, whereby any mismatch is reported to the NMS as an event. The string used for the comparison is generated at the NMS and sent to the SAS as part of the hub-side configuration for the terminal.

If a particular key/value pair from the satellite terminal fails to match an element in the comparison string, the Satellite Terminal Proxy generates an event of the specific Event Type ID, Terminal Component Mismatch and sends it to the NMS. If enabled by the configuration, the terminal will be sent a forced logout command, and is not allowed to enter the network.

The ***terminal component mismatch*** event creates a MAJOR severity event, in the NMS, which immediately triggers a MAJOR severity alarm for the satellite terminal. The result should be that the operator will examine the alarm and makes a case-by-case determination of the correct action to take. Once the corrective action is taken, the NMS operator can manually clear the alarm at the appropriate time.



NOTE: It is likely some key/values pairs not present in the comparison string from the NMS (for example, BUC serial number) will come from the satellite terminal. No comparison is performed on these pairs.

The comparison string from the NMS supports regular expression matching, so more than one component value can match a given comparison string.

8.3 Bonded Modems

The use of ***bonded modems***, in Velocity, allows multiple modems to operate with a single antenna, via the ACU. Third-party solutions such as the Xiplink box can provide link aggregation at higher layers where two or more modems can make use of a single ACU and acquire onto the same satellite/beam. The bonded modems implementation significantly increases throughput — especially on the upstream.

A master/subordinate design is the basis of the bonded modems architecture. In this implementation, one modem is designated to operate in the master mode and controls the ACU, while all other bonded modems are designated to operate in the subordinate mode. If two or more modems are deployed in a bonded configuration, each must be configured to operate in either the master or the subordinate mode. Although most bonded modem deployments will operate with two modems, there is no fixed limit as to the number of modems that may be used.



NOTE: **1)** OpenBMIP capable terminals, configured with a "smart BUC," cannot support bonded modems, as the modems do not share OpenBMIP information.

2) The bonded modems implementation does not bond the IP connections.

See ***Configuring Velocity Networks Using Pulse***, for configuration details.

3) 9-Series remotes do not currently support bonded modems.

8.3.1 Master Mode

In ***master mode*** the designated modem controls the ACU and acts as a relay that forwards geo-location and other status information from the ACU to all connected ***subordinate modems***. In order to support this service, the modem operating in master mode runs an OpenAMIP server that establishes connections with subordinate modem clients to relay any ACU responses. Any incoming OpenAMIP messages from subordinate modems are ignored by the master modem, which listens on the OpenAMIP default TCP port of 5001, unless configured otherwise.

The behavior of the master mode terminal, except for relaying any OpenAMIP messages, is the same as with any other standard mobile terminal. When the master modem drops the ACU connection, all connections with subordinate modems are terminated, which emulates the behavior as if the subordinates were actually connected to the ACU. Once the master re-establishes the ACU connection, it again listens for connections from the subordinate modems. Terminal components such as the BUC/LNB/BIM still rely on the master mode terminal for power, clock or tone/voltage for band selection.

8.3.2 Subordinate Mode

In the ***subordinate mode***, of the bonded modem architecture, modems configured as subordinates receive ACU messages relayed from the master modem. Subordinate modems behave similar to regular modems operating with OpenAMIP enabled, however they are unaware that the connection is not with a physical ACU. Since subordinate modems only

accept messages relayed from the master modem, they can only shadow the master modem as it moves and switches to different beams and satellites.

It should be noted that once the master modem has acquired into the network, subordinate modems, having no ability to choose satellites, will cycle through available satellite/signaling carriers. For each failed attempt to acquire, there is a delay of up to 25 seconds, which causes the slave acquisition to lag the master modem, and to some extent delay the availability of the expected increased capacity.



NOTE: 1) The terminal types configured for the master mode terminal and for the subordinate mode terminals must be configured the same.

8.4 Discreet Mode Terminals

In some Velocity applications, it may be a requirement that the exact location of a terminal remain unknown to all others. iDirect Velocity supports the terminal ***discreet mode***, in which the geo-location of the terminal is never reported to the hub. Since the terminal location is not revealed, mobile terminals operating in the discreet mode, cannot rely on the reported latitude/longitude to support beam switching and advanced skew control. In discreet mode, the terminal determines the next best beam/iNet to switch to, and sends a geo-location message containing the beam ID of the target beam. For fixed terminals, a lat/long of (0,0) is sent, along with the current Beam ID.

8.4.1 Beam Switch Decision

The decision to switch beams is based on the service beam in which the discreet mode terminal is currently operating — initially, only beams in the current satellite configuration are considered as new beam targets. As long as one carrier is supported on a visible beam on the current satellite, the terminal will continue to use that satellite. A satellite switch is only initiated if there are no longer any visible beams/iNETs on the current satellite.

In determining the target beam to switch to, the terminal first sorts through each of the visible beams on the current satellite according to the distance from the terminal to the nearest beam edge (***beam edge distance***). The optimal beam is the beam with the greatest edge distance.

If the current beam is visible, a mapping is created to correlate each beam with its beam ID and beam edge distance. A metric is then assigned to each beam in the map. The metric of the current beam is set to '0', whereas the metric for all other beams is computed against the current beam as follows:

hysteresis_radians - (beam_edge_distance - current_beam_edge_distance)

The optimal beam is then selected as the beam with the lowest metric value. If the selected target is not the current beam, then a beam switch to the optimal beam is initiated by the terminal, which sends a geo-location message that contains the beam ID of the target beam. Subsequently, the PP beam switch process will cause the discreet mode terminal to switch to the new target beam.



NOTE: 1) The default hysteresis value is set to 0.1 degrees. The computed beam edge distance must exceed the current beam edge distance, by this value, in order to initiate a beam switch.

It is important to note that the decision of whether to select a new Beam ID is a continuous process, and not only made as the terminal nears a beam edge. The beam selection logic, excluding the influence of hysteresis, will select the beam that is furthest from beam edge. In the case of network operators that use a combination of spot beams and broad beams, a number of the spot beams may not be selected. This beam selection logic is further illustrated in the following example, which is described in the subsequent steps:

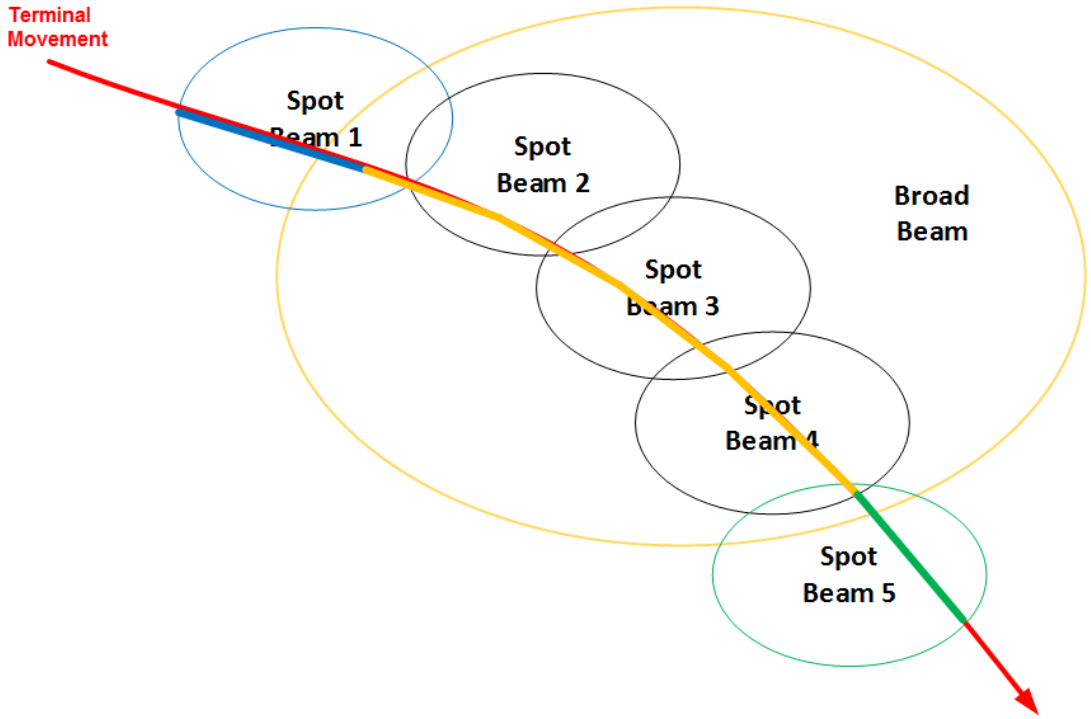


Figure 8-2. Discreet Mode beam selection logic illustration

1. Select Spot Beam 1 when terminal is in Spot Beam 1 and there is No coverage overlap with Broad Beam
2. Select Broad Beam if the “distance to Broad Beam edge” is greater than “distance to Spot Beam 1 edge”
3. Select Broad Beam if the terminal is in Spot Beam 2 and there is coverage overlap with Broad Beam
4. Select Broad Beam if the terminal is in Spot Beam 3 and there is coverage overlap with Broad Beam
5. Select Broad Beam if the terminal is in Spot Beam 4 and there is coverage overlap with Broad Beam
6. Select Spot Beam 5 if the “distance to Spot Beam 5 beam edge” is greater than “distance to Broad Beam edge”

8.4.2 iNet Load Balancing

Compared to other terminals, terminals operating in discreet mode only participate in load balancing to a limited extent, whereby the actual load balancing only occurs among iNets in the same beam. Although the load due to discreet mode terminals is considered in the PP load balancing algorithm, the PP does not move a discreet terminal to a different beam.

8.4.3 Skew Control

Once the terminal is in the network, the hub pushes the PSD-vs-skew curve and the tolerance parameter (maximum acceptable C/N reduction or angular tolerance) to the core module via the DFOE protocol, as part of the session-dependent configuration parameter set. A skew characteristic can be associated with any configured Regulatory Area associated with discreet mode terminals, which can be used in the same manner as regular characteristics. This regulatory area will never change as long as the terminal remains in the same beam/iNet.

Advanced skew, which relies on periodic geo-location updates from the terminal, is not supported for discreet terminals.

8.4.4 Service Areas

The “service area” concept, in the context of both discreet and non-discreet terminals, can be used for the following purposes:

- Accounting
- SLA modification (including barring)
- Regulatory forced routing

The SAS Hub is responsible for identifying the current service area of each remote and for taking appropriate action when a remote moves between service areas. To accomplish these tasks, To accomplish these tasks within the context of discreet mode terminals, the PP uses a per-beam configured service area for the terminal while it is in a given beam. That service area can be used to define service geoscopes and/or to define regulatory areas.

For additional details on maps, RAs, and SAs, see About Maps and Regulatory Areas on page 34.

8.5 Mitigating Skew Impairment in Mobile Terminals

Proper and efficient operation of terminals with radiation patterns that are not circularly symmetric, most often, will require special provisions in order to comply with regulatory off-axis limits. This is particularly the case of antennas mounted in a fixed relation to a vehicle, such as on the fuselage of an aircraft or the roof of some other vehicle.

The orientation of the beam pattern with respect to the geostationary arc, as seen from the terminal location, is known as the “skew angle”. This section first takes a brief look at terminals that operate with non-circular radiation patterns, and then describes the methods used in Velocity systems to mitigate the resulting skew impairment in these terminals.

8.5.1 Non-Circular Aperture Antennas

Low-profile, or flat-panel, antennas commonly used for aeronautical terminals have a non-circular aperture that causes transmissions from these antennas to project a beam that is elliptical rather than circular, with the major axis of the ellipse being horizontal to the platform on which the antenna is mounted. Alignment of the ***minor axis*** of this ellipse along the geostationary arc produces the antenna's widest projection onto this arc, which can easily result in the transmitted signal projecting across multiple satellites.

Conversely, alignment of the ***minor axis*** of this ellipse along the geostationary arc produces the antenna's narrowest projection onto the arc, greatly reducing the projected pattern's reach to neighboring satellites. So, if the antenna is located directly north or south of the satellite and the antenna platform is horizontal to the local horizon, the projected pattern will cross the geostationary arc along the pattern's minor axis, thereby minimizing the transmitted signal's impact on neighboring satellites along that arc (See Figure 8-3 a). If, however, the antenna is located directly east or west of the satellite, the projected pattern's major axis will align with the geostationary arc, thereby maximizing the transmitted signal's impact on neighboring satellites (See Figure 8-3 b).

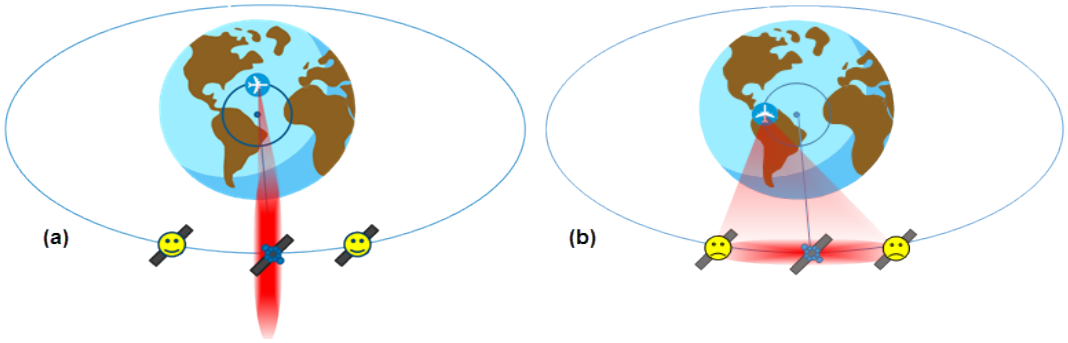


Figure 8-3. Low-Profile Antenna a) Favorable Alignment b) Unfavorable Alignment

Local tilt of the antenna platform can also affect the alignment between the projected pattern and the geostationary arc. If the platform is not locally horizontal, and is tilted about an axis connecting the panel to the satellite, the projected pattern will also be tilted about that axis by the same amount. This local tilt and the tilt related to the geo-location of the antenna together determine the total amount of misalignment of the projected pattern to the geostationary arc.

The amount of misalignment of the projected pattern's minor axis with the geostationary arc is referred to as the skew angle. When the minor axis is aligned with the arc, the skew angle is 0°, and when the minor axis is perpendicular to the arc, the skew angle is 90°. Other possible values of skew angle lie between these extremes. At a skew angle of 0°, the projected pattern is producing the narrowest projection onto the arc, while at a skew angle of 90°, the projected pattern is producing the widest projection onto the arc. So, as the skew angle increases, the impact the transmission has on neighboring satellites also increases.

The satellite router and the antenna controller on a terminal using an antenna prone to generating skew-related interference on adjacent satellites work together to determine the amount of skew and take action to limit the amount of interference. The amount of power spectral density (PSD) allowed at different skew angles can be configured for a terminal, and the terminal will limit its transmit power to ensure the PSD limits are not exceeded at the current amount of skew. Limits on the amount of allowable skew can also be configured, and the terminal will mute its transmissions whenever those limits are exceeded. The PSD and skew limits can be configured separately, where the terminal will either limit PSD or limit skew. The terminal can also be configured to limit PSD within some range of skew, imposing a limit on both. This latter approach allows a wider range of skew to be supported without trying to support operation at an especially low transmit power, allowing an operator to trade average efficiency for link reliability under dynamically varying skew conditions.

8.5.2 PSD Limits Per Beam, Terminal, and Regulatory Area

Velocity PSD (*power spectral density*) Limits Per Beam, Terminal Type, and Regulatory Area (RA) supersedes how skew parameters and PSD limits were configured and applied in earlier releases— Reference Skew and Reference C/N per Channel, for a terminal, were based strictly on the RA page configuration. Previously, regardless of beam switches and the number of points at which the terminal intersected different RAs, in which beam performance could be quite different, the configured worst-case PSD limit of the RA page was always applied.

How RA and skew parameters were configured and applied, introduced operational inefficiencies and limitations. Terminals could not use better available carriers, due to the restricted Max C/N, or at times had to mute due to restricted skew limit or the unavailability of carriers due to limited Max C/N.

With the introduction of this feature, Satellite operators can now define skew versus PSD limits with enough granularity to function accurately in every beam and in every RA; and terminals are not restricted to the worst-case beam in the RA. Instead, the terminal uses parameters configured specifically for the current beam (channel) at the point which the beam intersects the RA. As such, slot assignments for affected terminals can use better carriers that improve inbound efficiency. The reference point (skew, C/N) that ties the relative skew versus PSD limit increase curve to a specific C/N limit at every skew value is now set differently in each beam/channel.

8.5.2.1 Feature Description

The ability to configure a reference skew and PSD limit per regulatory area, beam, and RF Terminal Type is an enhancement of the Real Skew feature. This configurability allows the PSD limit curve to "shift" up or down as a whole in each region, while still maintaining its shape. RA compliance is enhanced greatly and terminal muting, due to skew limits, is largely reduced.

Some key characteristics of PSD Limit per Terminal Type, Channel, and RA configurability are listed below:

- Reference Skew Limits and Reference C/N can be configured manually per RA/Channel/Terminal Type
- Skew Parameters, at acquisition, are unaffected by the new implementation
- Operators have more granular management and control of RA requirements
- PSD limit and Skew curve mapping are based on each beam/channel
- PSD scaling accurately reflects the Maximum C/N of the current beam
- RA page parameters are considered when only the RA page is configured
- Terminal PSD limit no longer restricted to worst-case beam in RA
- Inbound efficiency improved as terminal slot assignments use more efficient carriers
- Selected terminals can be excluded from operating in specific regions, by setting very low maximum C/N limits for certain Terminal Type/Channel/RA intersections

- Terminal Type channel exclusion feature is extended as Terminal Type can be configured non-useable in a more granular area, based on Channel/RA intersection

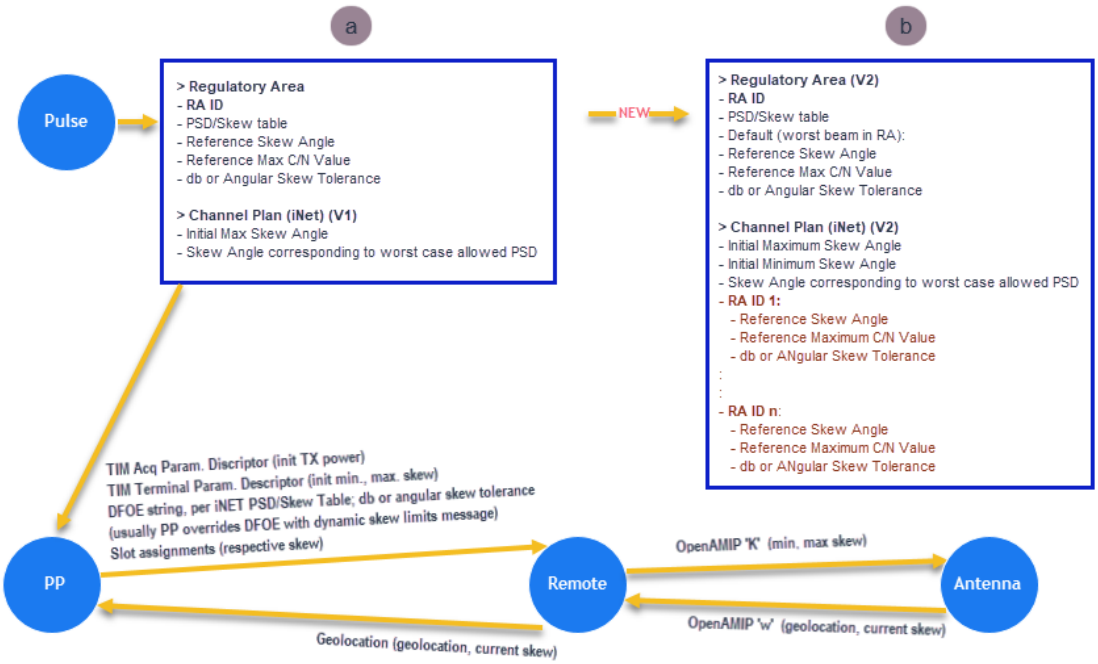


Figure 8-4. Example a) Standard Skew Table b) Pessimized Skew Table.

In the Channel object, which corresponds geographically to the current beam, a Max C/N value is configured for each RA that the channel (beam) intersects. To link the Max C/N configuration in the channel with the correct RA, the Channel object associates each Max C/N with the RA Element ID. Whereas the applied algorithm is the same as with the prior method, the new implementation uses a Max C/N specific to the Channel/RA combination, as opposed to applying one value across the entire RA, as in the prior implementation.

During acquisition, several transactions occur using the default (worst-case) power and PSD limits for slot assignments. For a brief period of less than 1 second, efficiency is affected for a very small amount of management traffic, with no impact on user traffic or risk of regulatory infringement. Also, there is no impact on acquisition, as it applies only after geo-location, when the Regulatory Area is known.

As this new implementation is based on the Regulatory Area, which is known to the PP only after the remote sends a geo-location message—several steps into the acquisition over-the-air process, acquisition behavior is not affected. The PP periodically broadcasts messages tailored to each of the Terminal Types allowed in the iNet. To compute its initial transmit power, and PSD limits, the terminal uses Initial Tx Power from the Acquisition Parameters Descriptor within the TIM. The remote is unaware of the new method of configuration.

As the PP determines slot assignments for a terminal, it now incorporates the max C/N based on the physical location of the terminal (both its RA and its beam). Note that the remote is unaware of the additional information available in the PP; all OTA messaging is unchanged by the new algorithm in the PP. The PP already takes physical layer limits into account for beam selection; setting those limits with more granularity will affect beam switch decisions.

8.5.2.2 Configuration Considerations

Prior to Velocity 3.0, the PSD vs Skew angle curve along with reference values for Skew and Maximum C/N, were configured on the Pulse RA page; however, the Max C/N and Regulatory Offset (RO) maps are beam dependent and not enforced by the RA parameters. The choice, however, of the reference Max C/N is based on the Max C/N link budget per beam. Within an RA, or group of RAs where the same curve is configured, the same reference Skew value should result in many different Max C/N results, as the performance across the beam varies.

As new Terminal Types are deployed, using the Channel configuration, the NMS operator has the option to enter a record for each RA that intersects the channel's beam. Each record contains the RA ID, the Channel, a Skew Reference Angle, and a Skew Reference Max C/N for that RA ID. The table must be entered in its entirety—otherwise if it is omitted, the new implementation is not applied and the Reference Skew Angle and Reference C/N parameters in the RA configuration are used as defaults, to cover the case where per-channel (iNet) configuration have not been entered. RA-level defaults will be applied to any Channel/RF Terminal Type not configured for the current RA.

General Configuration Notes

- PSD Limits per TT per RA is enabled by setting the Pulse Network Version to 3.0 or higher
- There is no inherent limit on how many Regulatory Areas that may intersect a Channel; Channels and RAs are stored as separate objects linked only by coincidence of geography
- When entering each RA ID and Max C/N, the RA to configure per beam must be known
- When computing and configuring existing PSD limiting features, minor work-flow changes are needed to preserve intermediate computation results (PSD for each channel and RA); The value can then be entered under the Channel
- PSD Limits (Max C/N) for specific Terminal Types may be set artificially low in selected regions, to prevent operation of affected terminals on selected channels in those regions.
- When considering capacity planning, more favorable average inbound link budgets may be considered; terminals operating in less-restricted RA portions of a channel are able to use a greater PSD than the worst-case, thereby increasing the aggregate efficiency.
- Channel page configuration of Minimum Skew (deg) and Maximum Skew Angle (deg), used at acquisition, remain as in earlier releases
- Channel/Terminal Type/RA Skew properties configured on Channel Page and are valid within the Channel
- Skew parameters, like Reference Skew Angle, and Reference C/N, can be configured per Terminal Type/Regulatory Area on the Regulatory Area page.
- PSD Curve (PSD versus Skew Angle), provided by manufacturer are still configured on the RA Page as in earlier releases
- This feature supports the configuration and management of PSD/ESD limits required to comply with asymmetric antenna patterns

8.5.3 Hybrid Skew Margin Support

The PSD management implementation required choosing between an ***angular margin*** mode or ***C/N margin*** (***db reduction)*** mode, each of which produced optimal results under different conditions. Large regulatory areas, such as a single RA spanning the Earth's oceans, required an excessive C/N margin (4 to 7 dB) to avoid muting in locations of high skew, which unnecessarily penalized terminals when the skew was favorable.

The ***Hybrid Skew Margin*** implementation combines the best attributes of both the C/N and angular margin modes, by permitting users to configure both parameters, which are simultaneously enforced. In this way, a sufficient operating range of skew angles can be preserved in steep regions of the PSD curve, without penalizing operations in flatter regions. In this straightforward generalization of the existing algorithm, the existing two modes of operation fall out as special cases by either setting the angular margin or dB margin to zero.

By considering angular and C/N margins concurrently, the C/N margin can be greatly reduced. In regions of favorable skew, angular margins are easily achieved such that reduction of the C/N margin permits using better MODCODs. In areas of unfavorable skew, small C/N margins are not effective and the addition of angular margins ensure that resulting skew limits are wide enough to avoid terminal muting.

The hybrid skew margin support extends the PSD management feature to optimize the performance of Aero-terminals across a broader range of geographic regions. Network operators can specify a global minimum C/N margin, to be applied in addition to a configured angular skew margin. When calculating skew limits and maximum C/N, the PP and terminal considers both the C/N and angular margins. A further enhancement limits the application of C/N margins such that the terminal is not penalized beyond its worst-case operating point — the lowest point on the antenna's PSD curve.

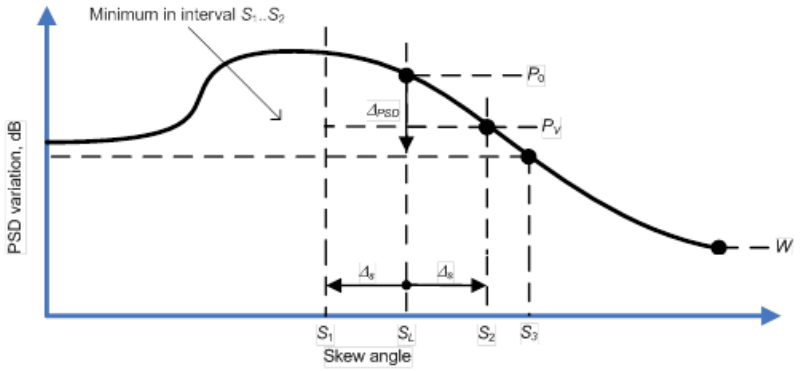


Figure 8-5. Limit determination using combined angular and PSD margins

The following are some key characteristics of Hybrid Skew Margin support:

- Combines C/N (db reduction) method and angular margin method
- Optimizes Aero-terminal performance across a broader range of geographic regions
- Operators can configure a global minimum C/N margin and an angular skew margin
- C/N margin limited so terminals are not penalized beyond worst-case operating point



NOTE: The PSD curve is provided by the antenna manufacturer as a list of data points, where each point is formed by the **Skew Angle** (deg.) and the **Relative PSD Level** (dBm). A PSD curve must be configured for each terminal type.

8.5.4 Real Skew

Satellite routers can now support antennas that report the "real skew." Real skew takes into account the relative positioning of the terminal, and the "local tilt," which results from platform attitude variations (pitch and roll for an aircraft). Using OpenAMIP v1.12 or higher, the antenna control unit (ACU) directly provides a real skew measurement in the OpenAMIP "w" message, which includes pitch, roll, yaw, heading, and skew.

If the skew is provided by the ACU, the PP and modem use this value instead of calculating level-flight skew. The modem will include these values in an updated version of the OTA OOB geo-location reporting message to the PP. If the message is received without the heading, pitch, roll, yaw, and skew parameters, the associated parameter values are all assumed equal to zero, with the exception of the skew. In this case, the system computes a level-flight skew angle value based on the latitude and longitude of the terminal only.

Velocity Real Skew also supports newer antennas with beams in which the shape varies - in this case the interface is the same but the table and the skew value are manipulated by the antenna vendor to account for varying beam shape.

Real skew data should improve regulatory compliance and reduce terminal muting, due to skew limits. The following are some key characteristics of Real Skew support:

- Terminals provide support for antennas that report real skew
- Takes into account the local tilt and relative positioning of terminal
- Real Skew benefits greatly from high update rate of the "w" message - typically 1Hz
- Improves overall performance by allowing the system to more accurately calculate the minimum and maximum skew values thereby reducing terminal muting

8.5.4.1 Real Skew with Flat Plate Antennas

The real skew algorithm in Velocity, without any changes, is also applicable to flat plate antennas. Perhaps the only difference is how the algorithm should be viewed. Real Skew simply asks the antenna to provide a "skew angle," which is an index into a table stored at the Velocity hub (***provided by antenna integrator, in conjunction with network operator***). In turn, the antenna directs the hub to look up a maximum PSD value in the table. The table and the index are managed by the operator and the antenna integrator.

As with a mechanically steered elliptical antenna, the index value for the flat plate antenna does not have to physically represent a classic skew angle. The index, in this case, would be the result of any arbitrary calculation based on the geometry — the platform location and orientation, satellite location, and the instantaneous beam shape.

When the beam shape is variable, depending on beam elevation and azimuth—relative to the plate antenna, the index calculation is likely based on a 3D table resident in the antenna controller. From the table, the controller looks up the skew, elevation, and azimuth (***each***

value calculated from the geometry). The table could potentially provide a PSD limit, or be factored to scale the PSD limit from a value in dB, to a value in the range of 0-90 degrees. The 0-90 value is what the antenna provides as "Real Skew," using OpenAMIP. Next, a lookup of the "Real Skew" from the table configured in Velocity, and then the index is translated back to a PSD limit value (actually C/N limit), which is what the hub actually measures on the ground.

When the beam pattern of the antenna is irregular, and the PSD limit can vary rapidly in some regions of 3D space, then the 3D table will be somewhat irregular as well. Some regions will be smooth, and other regions irregular. These irregular areas are of more concern on a moving platform — in those areas, a bit more margin might be required. This additional margin is added by adjusting the 3D table to "***pessimize***" the PSD limit in those regions.

The concept of "pessimization" can be visualized by simplifying the table from 3D to 2D. Given a raw calculated table as the start, for any cell in the middle, look at the block of 9 cells, which includes the targeted cell in the center of the block. Take the worst PSD for the 9 adjacent cells, and in the new pessimized table, place that worst PSD in the corresponding cell. For edge cells, the concept is similar but fewer adjacent cells to include. The resulting table automatically adjusts skew margin to accommodate regions where skew will vary more rapidly with maneuvers. The following is an example, in which the table on the right is a pessimized version of the table on the left.

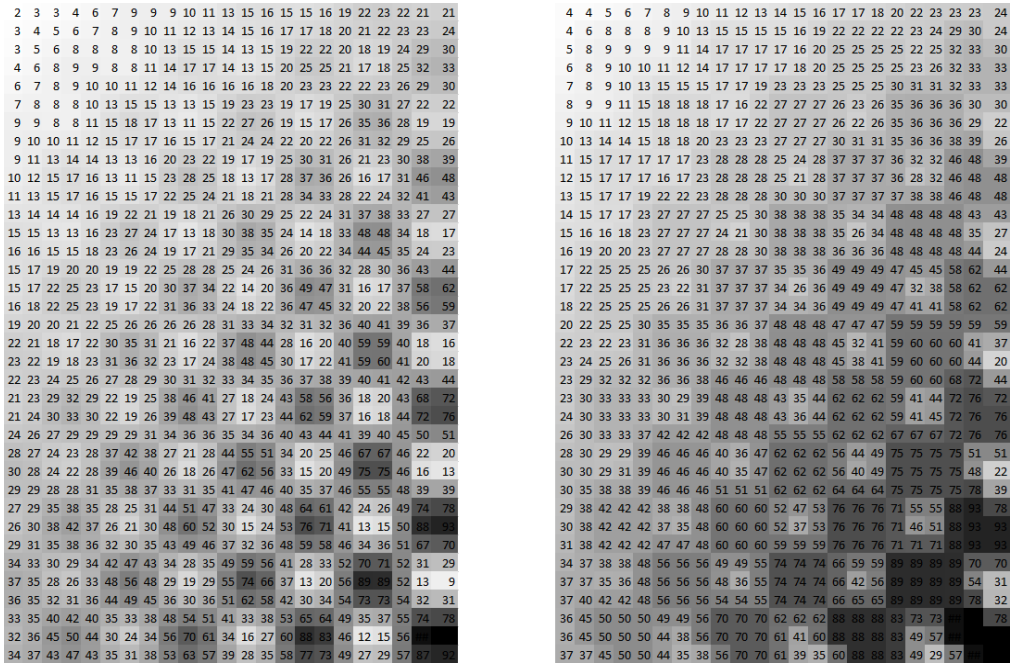


Figure 8-6. Example a) Standard Skew Table b) Pessimized Skew Table.

8.6 Fast SNR Reporting

The fast SNR reporting feature is based on an iDirect modem products feature, referred to as the receive signal strength indicator (RSSI) signal. Fast SNR reporting, in a modem, allows the modem to use OpenAMIP to report its carrier signal-to-noise ratio (SNR) as measured at the demodulator. This information provides an error compensation component, needed to support closed-loop feedback, which allows mobile terminal integrators to dynamically control antenna pointing.

Although these measurements are not calibrated to represent an absolute value of SNR at the satellite router's receive input, differences between consecutive measurements can be considered accurate. Measurements are timestamped and can be streamed to the ACU at rates as high as 60 times per second. This periodic feedback can help the ACU improve dynamic antenna pointing accuracy over the accuracy that can be obtained from the use of platform-mounted navigation sensors, alone. This additional accuracy is especially useful when attempting to perform a conical scan for a satellite with an antenna assembly mounted on a moving platform, such as is the case for maritime terminals.

The following are some key characteristics of fast SNR reporting:

- SNR read/reported at update rate of 0 to 60 Hertz
- Configured entirely on receipt of OpenAMIP “c” message from the ACU
- OpenAMIP “c” message with update rate set to zero (0) disables reporting
- The composite power, although reported in the “c” message at 0-60Hz, is sampled from hardware at 1Hz (samples are duplicated as necessary)

8.7 External Beam Select Mode

External Beam Select Mode is a Remote WebServices API feature that enables VSAT to connect users who move from location to location over time. This Communication On The Pause (COTP) differs from Communication On the Move (COTM) in that the VSAT does not need to transmit or receive while it is moving. Typical use cases of COTP include disaster recovery, military ground comms, news gathering, camping comms, and oil/gas exploration.

External Beam Select Mode enables the user to select a signaling carrier from a satellite of their choice to download the service map (OTA_BEAM_MAP), and select the service beam from a satellite of their choice to acquire into the network. This feature enables the user to completely control the satellite and beam selection, and informs the remote to ignore all satellite switch, beam switch, and emergency beam switch decisions made by the Hub. The remote will, however, still honor requests from the hub to switch to another iNet within the user's selected beam, hence permitting some measure of congestion management and improving the user's experience.

The feature is enabled/configured using the Remote WebServices API by configuring "beam_select" to EXTERNAL, and using the "beam/signaling" and "beam/service" API commands to configure signaling and service carrier selection respectively.

See the *Remote WebServices ICD Reference Guide (Rev T)* for more information.

8.8 Commissioning Enhancements (API)

The commissioning process is carried out to make the terminal come into the network and transmit properly. Commissioning can be performed either by the Terminal Web UI or the Terminal Web APIs.

The commissioning enhancements apply to the commissioning of the terminal through Web APIs, which allow the terminal vendor to select from a list of available satellites and service carriers for the given geolocation where the commissioning has to be performed.

The APIs, beam/signaling and beam/service, introduced as part of the External Beam Select Mode feature, are used for this in the commissioning mode of the terminal.

8.7.1 Prerequisites

The terminal must be staged with the Constellation and Terminal Options Files.

8.7.1.1 First-Time Commissioning of Terminal

In this case, the terminal is newly out of the box and not commissioned. It would be staged with only the Constellation and Terminal Options Files. The OTA Beam Map will not be available. The beam/signaling API can be used to obtain the list of satellites for the given terminal geolocation. The user can select one of the satellites and manually provide the downstream parameters.

8.7.1.2 Previously-Commissioned Terminal

In this case, the terminal already has the OTA Beam Map downloaded from a previous connection to the hub over a satellite. The beam/service API can provide the list of satellites and beams from the OTA Beam Map for the given terminal location, and the user can select one of the beams to configure downstream parameters.

8.7.2 Commissioning Steps

1. Put the terminal into Commissioning Mode via the terminal/config/commission API.
2. Set the terminal geolocation via the terminal/config/location/remote API.
3. For first-time commissioning of the terminal without the OTA Beam Map, use the terminal/beam/signaling API to list the satellites for the given geolocation of the terminal.
4. Choose one of the satellites and manually enter the downstream parameters by contacting NOC.
5. When the OTA Beam Map is already present, choose from the list of valid satellite and service carriers for the given geolocation through the terminal/beam/service API.
6. For downstream configuration and antenna pointing, use the terminal/mode/antenna_ point API by providing the downstream parameters for the chosen satellite and service

carrier (e.g., rf_frequency, symbol rate, roll off, polarization) to lock to the downstream.

7. Exit the commissioning mode via the terminal/config/commission API.

See the *Remote WebServices ICD Reference Guide (Rev T)* for more information on API Commands.

9 Fair Access Policy

A ***Fair Access Policy (FAP)*** is implemented to ensure fair usage of bandwidth and access to the network service by all subscribers. The FAP implementation governs the amount of data that is downloaded or uploaded within a specific period by a subscriber or group. If set limitations are exceeded, the CIR and MIR restrictions take effect during a "***recovery period***," in which service is still usable, although speeds may be reduced or some other penalty may be applied.

This chapter contains the following sections:

9.1 About Fair Access Policy 326

9.2 FAP Volume Allowance 326

9.3 FAP Feature Interaction 333

9.1 About Fair Access Policy

Fair Access Policy (FAP) allows the service provider to cap and control the bytes consumed by a customer during a given time period. It is often used to implement a type of service referred to as “Pay as you go”. FAP provides the service provider with a rich set of tools, including capped overage bytes, account top-up, rollover, and HTTP redirect.

The policies configured using FAP govern the volume of inbound and outbound traffic (in Mbytes) a GSP or subscriber service plan may consume within a given period of time. When that volume of traffic is used up, depending on the FAP rules configured, the service provider may take various actions such as throttling or blocking the customer's data usage, differential billing, or requiring the user to pay for additional traffic.

FAP manages the total amount of data (in Mbytes) used under a service plan within a specific period of time (day, week, or month). FAPs can be configured for both GSPs and SSPCs. Assigning a FAP to a GSP or SSPC is optional, and is not required for the system to function.

Multiple FAP rules may be configured for each service plan, and a violation of the FAP rules may result in a notification to the service provider, bandwidth allocation throttling or blocking, or redirection of the user to an “Out of Data Allowance” website.

9.2 FAP Volume Allowance

Volume allowance is the basic volume of data (in Mbytes) that may be transferred during a FAP allowance period. A FAP allowance can be defined separately for downstream and upstream traffic, or combined for an aggregate allowance — where the same allowance is used for both outbound and inbound traffic. The volume allowance may also be referred to as the regular allowance or periodic allowance.

9.2.1 FAP with Multicast Traffic

When tracking multicast bytes used in a FAP, the traffic transmitted in each beam is counted separately. For example, if a single multicast stream was configured to transmit on five beams, 1 Mbyte of transmitted multicast data should be accounted for in FAP as 5 Mbytes of transmitted data. The goal here is to allow the service provider to track the bandwidth actually occupied by the multicast traffic, and thus not available to other customers. If no multicast-SSPCs have joined the multicast stream in a beam, multicast traffic will not be transmitted in that beam and thus no usage is counted against that FAP.

Use of the FAP is not supported for multicast SSPPs and can be used only on multicast GSPs.

9.2.2 FAP Allowance Period

The allowance period is the applicable period of time in which the transfer volume is being tracked. The allowance period can be a 24-hour, 7-day, or 30-day period. There are two types of allowance period — fixed or rolling.

- **Fixed Allowance** — If the allowance period is configured as fixed, the normal volume allowance is credited to the service plan at the beginning of the allowance period. If the entire allowance is consumed before the end of the allowance period, the service plan will require overage or Top-Up in order to continue consuming data.
- **Rolling Allowance** — If the allowance period is configured as rolling, the service plan's regular allowance is credited in increments over the duration of the allowance period. Rolling allowance periods are intended to avoid situations at the end of the allowance period where the entire allowance has been consumed, and the service plan owner is throttled or forced to consume overage or Top-Up allowances (described below). For rolling allowance periods, the user must also specify the update interval at which the periodic allowance is credited to the service plan.
- Once the entire allowance has been consumed, if there is no overage or Top-Up available, the service plan must wait until a certain number of bytes (referred to as the ***rolling restoration***) are returned to the account via the rolling updates before consumption of data can continue. The time it takes for these bytes to be returned to the allowance is the ***recovery period***.

9.2.3 FAP Restoration Rules

Restoration rules are conditions that need to be fulfilled before service is restored to normal, after being throttled or suspended. These rules cannot be removed by the user, although one of the restoration rules is configurable.

If CIR and MIR for a GSP or subscriber service plan has been throttled or suspended, CIR and MIR will be restored to their regular level if one of the conditions below apply:

1. at the beginning of the next allowance period, if the service plan is configured for fixed allowance.
2. when a (configurable) portion of the allowance for the period has been credited, if configured for rolling allowance
3. if Top-Up bytes become available (See FAP Top-UP Account on page 328)

Top-Up bytes are consumed during an allowance period only if there are no remaining bytes from the allowance granted for the regular allowance period.

9.2.4 FAP Rollover

A service plan may allow the user to roll over unused volume allowance to the next allowance period up to a configurable maximum volume, which is defined as an absolute value or as a percentage of the volume allowance.

Rollover is supported for fixed allowance periods only, and not supported for rolling allowance periods. Separate rollovers for downstream and upstream are applied when separate volume allowances are used and combined when an aggregate volume allowance is used.

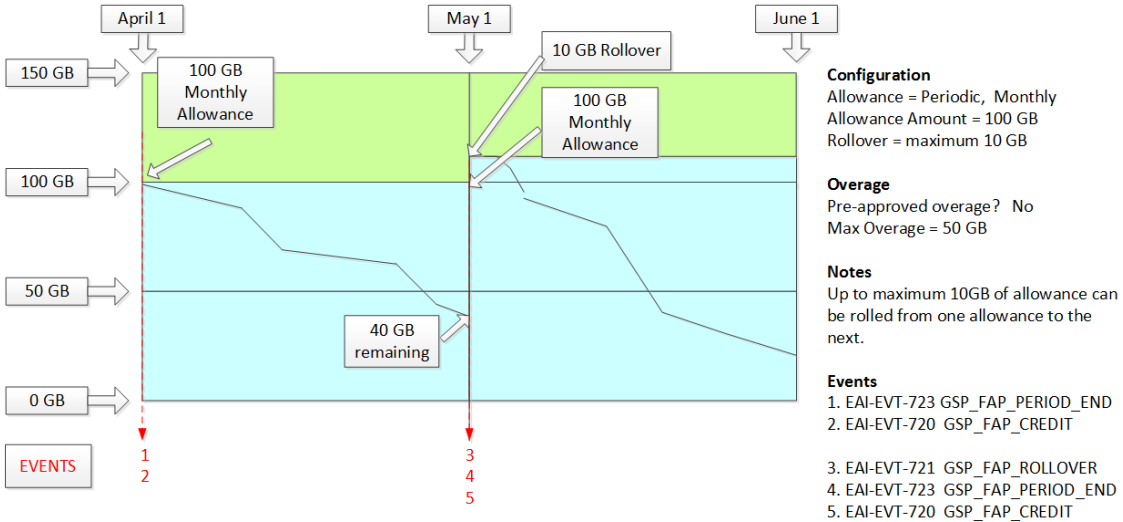


Figure 9-1. Simple Rollover Example

9.2.5 FAP Top-UP Account

A FAP includes a Top-Up account. A Top-Up account is bytes, which are purchased by the owner of the service plan, which do not expire between allowance periods. Top-Up accounts can be configured for both GSPs and SSPCs.

The service provider will be able to add and remove bytes from the Top-Up account for each SSPC and GSP using the Pulse UI (see the FAP Top-Up probe commands in ***Configuring Velocity Networks using Pulse***). The Pulse API may also be used by an OSS/BSS billing application to manage Top-Up bytes. These commands may be issued at any time.



NOTE: The system will not manage the expiry of bytes in the Top-Up account; it is the responsibility of the operator to decrement bytes in the Top-Up account, according to the policies in place to manage credit in this account. Unlike overage, no events are generated when the Top-Up account is empty.

The related GSP or SSPC Top-Up account is maintained regardless of whether the GSP or SSPC is activated or de-activated. The Top-Up account is deleted when the SSPC or GSP is deleted.

Top-Up is not mutually exclusive from overage. Service plans may be configured with both overage and Top-Up, with Top-Up bytes being used before overage.

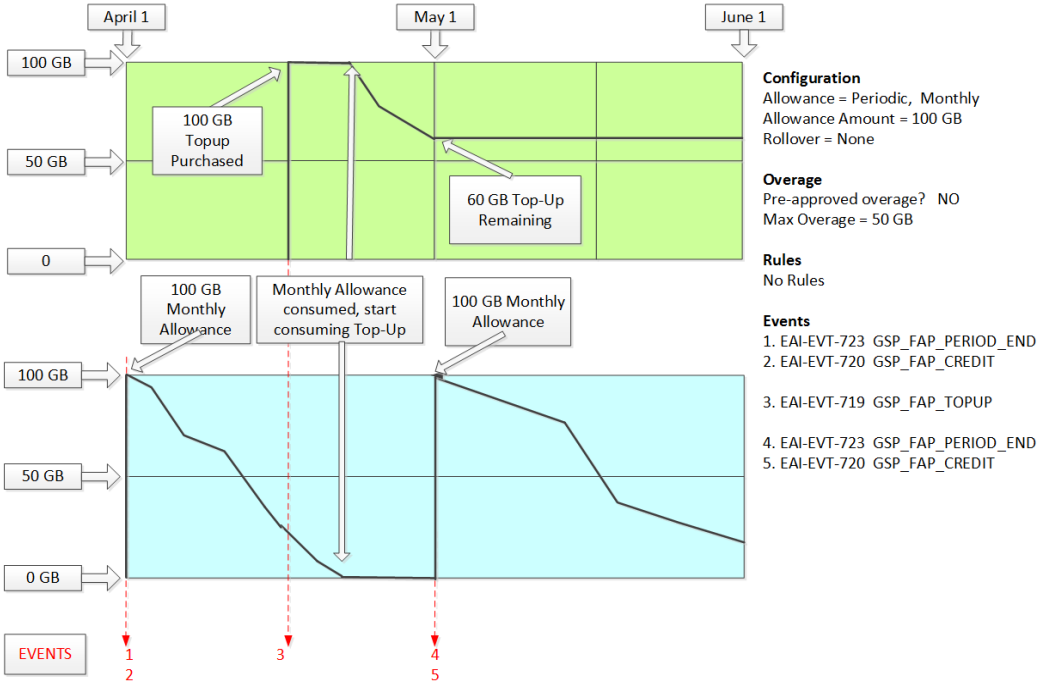


Figure 9-2. Simple Top-Up Example - Fixed Allowance

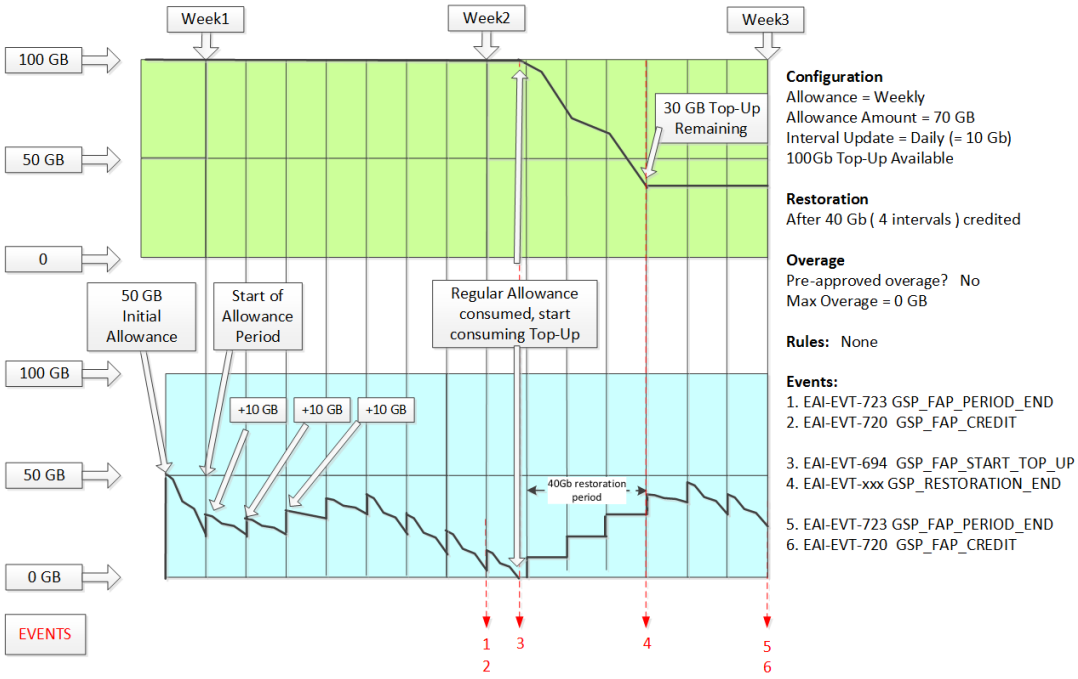


Figure 9-3. Simple Top-Up Example - Rolling Allowance

9.2.6 FAP Overage

The overage state allows:

- The consumption of a limited amount of extra data beyond the normal allowance
- The service provider to charge a different rate for those bytes
- The consumer to choose whether they wish to consume those bytes

When the allowance from the user's periodic allowance has been consumed, and there are no remaining bytes in the user's Top-Up account, the user may be presented with the option to enter an overage state. This overage state is entered via the HTTP redirect feature, described in FAP HTTP Redirect on page 332. At the same time, the system generates an event notification that may trigger an external billing OSS/BSS to query the owner for permission to enter the overage state, or the user may receive the event notification via the EAP.

Using the NMS Web UI, Pulse provides the ability to approve the overage state for a SSPC and GSP using either the Pulse UI (see the FAP Overage Approval probe commands in ***Configuring Velocity Networks using Pulse***) or the associated Pulse API (See ***iDirect DevTool***). Note that the event requesting approval to enter the overage state is only sent once. If no response is received from the user, the assumption is that he does not wish to enter the overage state.

Overage can also be configured to be pre-approved — in which case, the FAP goes directly into the overage state once all of the periodic allowance has been consumed. The system will not request approval from the end user before entering the overage state.

If overage is disabled, when the regular allowance and the Top-Up have been fully consumed, the GSP or SSPC will be suspended.

If approval is received, the object (GSP/SSPC) enters an overage state until one of the conditions for service restoration is met. For details on service restoration, see FAP Restoration Rules on page 327.

An upper limit to the Mbytes consumed while in overage must also be configured. Once these bytes have been consumed, service will be suspended until:

1. The next allowance period is reached
2. The empty Top-Up account is credited with additional allowance

Users can only approve overage one time within a given allowance period. Once consumption has reached the limit for overage, the service will be suspended.

Overage is not mutually exclusive from Top-Up. Service plans may be configured with both overage and Top-Up, with Top-Up bytes being used before overage. If credit is added to the Top-Up account while in overage, the system will immediately drop out of overage, and switch to consuming from the Top-Up account.

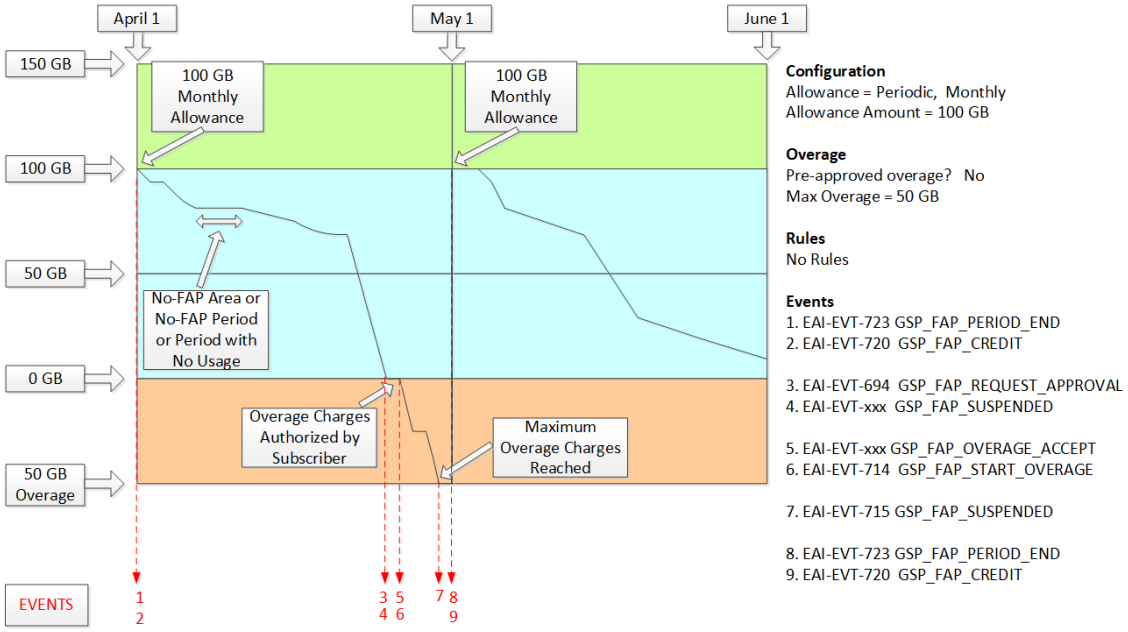


Figure 9-4. Simple Overage Example - Fixed Allowance

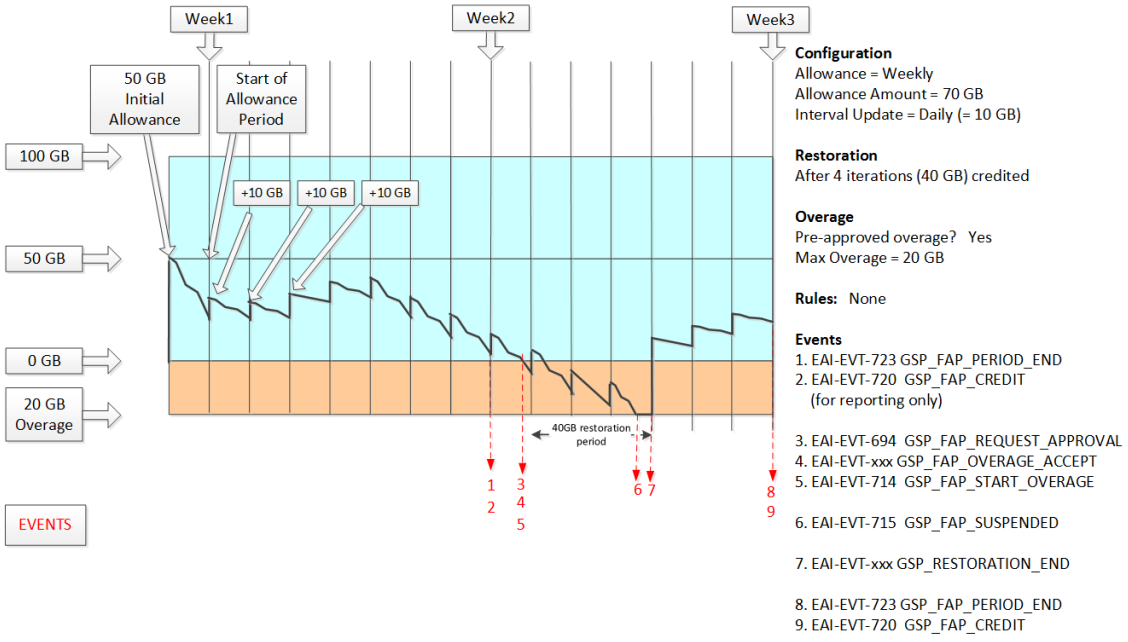


Figure 9-5. Simple Overage Example - Rolling Allowance

9.2.7 FAP HTTP Redirect

HTTP redirect is a means for a service provider to redirect a web-browsing subscriber to a service-provider specified "Out of Data Allowance" web page when their FAP has no remaining allowance available. This web page, which is designed and hosted by the service provider, would inform the subscriber of the situation and ask the subscriber to add and pay for top-up bytes. It is the responsibility of the service provider to ensure that the GBWM configuration ensures a trickle of traffic for HTTP (TCP port 80/443) to allow HTTP redirect.

“No remaining allowance” means:

1. All regular allowance has been consumed
2. There is no Top-Up available
3. Overage has not been pre-approved



NOTE: If Top-Up or overage is available, consumption will continue from those allowances, without recourse to HTTP redirect.

When one of the FAP restoration conditions is met, the subscriber will no longer be redirected to the “Out of Data Allowance” web page. HTTP redirect can be enabled per GSP or subscriber service plan and is supported for HTTP sites. Use of HTTP redirect is optional.

9.3 FAP Feature Interaction

The interaction of FAP features, including 1) Volume Allowance, 2) Overage, 3) Rollover, and 4) Top-Up, is illustrated in the FAP Flowchart, and detailed in the following scenarios.

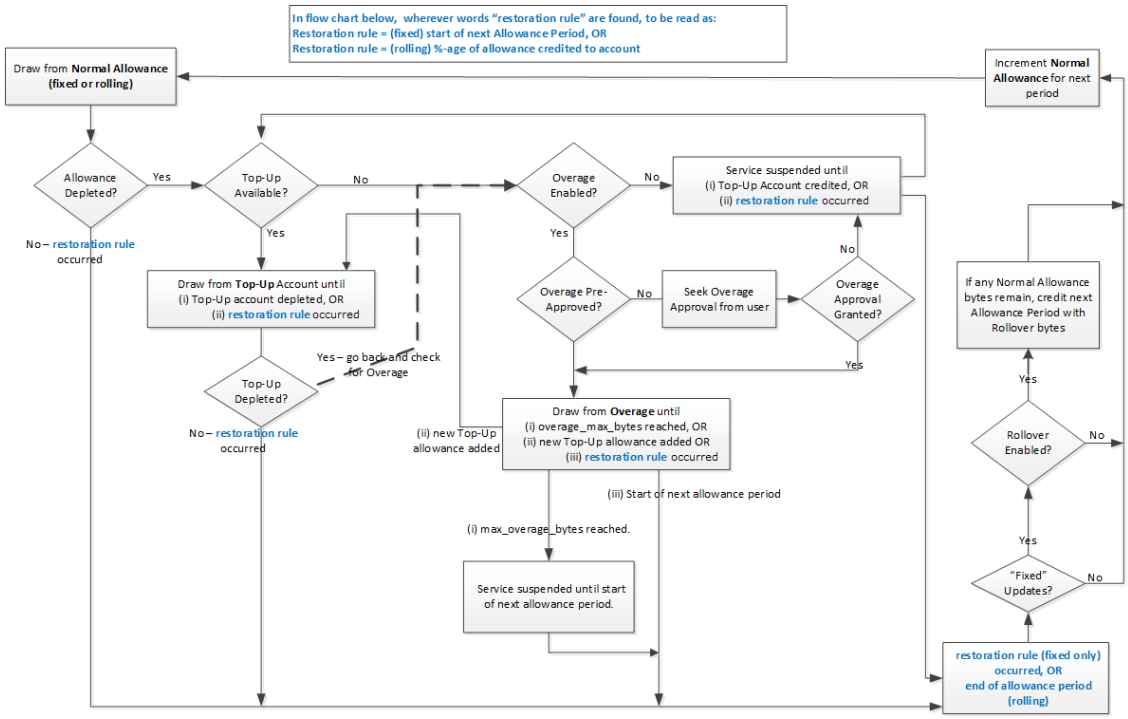


Figure 9-6. FAP Flowchart

9.3.1 Regular Consumption (Fixed or Rolling Updates)

Consumption of bytes is first counted against the regular allowance (plus any rollover allowance) for the allowance period, whether the allowance period is updated on a periodic or rolling basis.

Scenario 1: All regular allowance is consumed, and Top-Up is available

a. **Fixed Updates** — Should the regular allowance (plus any rollover allowance) be consumed before the end of the allowance period, continued consumption is counted against any bytes in the Top-Up account. If there is sufficient credit in the Top-Up account, usage is counted against this Top-Up account **from *the time that the normal allowance (plus any rollover allowance) is consumed*** **until *the beginning of the next allowance period***.

b. **Rolling Updates** — Should the allowance in the regular account be completely consumed, continued consumption is counted against any bytes in the Top-Up account. If there is sufficient credit in the Top-Up account, usage is counted against this Top-Up account **from *the time that there is no normal allowance remaining***,

until *the restoration rule kicks in (percentage of periodic allowance has been restored to account)*.

Scenario 2: All regular allowance is consumed, and no Top-Up balance available — enter overage state

a. If overage is not enabled, service is suspended until the beginning of the next allowance period.

b. If overage is enabled and has been pre-approved, usage continues seamlessly, with continued consumption accounted for as overage.

c. If overage is enabled but not pre-approved, service is suspended, and the user is prompted for consent to enter an overage state. (The user may be warned before the service has been suspended due to consumption of all the allowance. For additional information on this and how the user is prompted, see FAP Top-UP Account on page 328.)

In either of the scenarios above where all regular allowance has been consumed, if there is no Top-Up balance available, the object attempts to enter an overage state. If overage was not pre-approved, the user is prompted for confirmation that he wishes to enter the overage state. Until approval to enter the overage state is received, the user's service will be suspended. The degree of throttling when in suspension is configurable, with 100% throttling corresponding to CIR/MIR = 0.

There is a configurable maximum for the number of bytes that may be consumed as overage, applied equally to a FAP with fixed or rolling update intervals.

a. **Fixed Updates** — The user remains in the overage state until conditions in the restoration rules are met:
   - The next fixed allowance period has started.
   - Balance is added to a Top-Up account, where previously none was available. See **Note**.

b. **Rolling Updates** — The user will remain in the overage state until conditions in the restoration rules are met:
   - A (configurable) percentage of the periodic allowance has been restored to the account through regular rolling interval updates.
   - Balance was added to a Top-Up account, where previously none was available. See **Note**.



NOTE: In both the case of (a) **fixed updates** or (b) **rolling updates**, if a Top-Up balance was previously available, the system would have started consuming those bytes without attempting to enter overage state.

If the SSPC or GSP has consumed all of its allowance, and no bytes remain in the FAP Top-Up account, further consumption is handled as overage. These control flows are shown in the FAP flowchart diagram.

10 Multicast Overlay

With ***Multicast Overlay***, Satellite Network Providers are able to offer broadcast-like services with multicast such as video or television, while at the same time offer high-throughput unicast services such as the Internet. By leveraging the advantage of high-throughput spot beams for unicast traffic, while using wide overlay beams for broadcast services to reach a much larger area, the multicast overlay implementation can offer both services simultaneously in a manner that is economically efficient for both.

This chapter contains the following sections:

10.1 About Multicast Overlay 336
10.2 Multicast Overlay Use Cases 338

10.1 About Multicast Overlay

The Multicast Overlay implementation relies on Velocity dual receiver terminals — one receiver that can be tuned to a 2-way spot beam for unicast traffic and the other tuned to a 1-way broad beam for multicast traffic. In practice the dual receiver remote can be configured to receive both 2-way unicast traffic and 1-way multicast. In these configurations, the second receiver is generally referred to as the multicast overlay receiver.

The multicast overlay functionality incorporates use of the following Velocity capabilities:

- One-Way iNet Support
- Multicast Traffic on Terminal Secondary Receiver
- Persistent Multicast Added to Velocity Multicast
- Overlay Mode on Terminal

10.1.1 One-Way iNet

In Velocity, a beam is generally configured with 2-way iNets, but it can also be configured to support 1-way iNets. The initial Multicast Overlay implementation relies on the configuration of 1-way iNets on which a terminal may receive broadcast traffic only. This 1-way iNet does not require use of a receive line card.

Since current Velocity multicast methods, static multicast and dynamic multicast, both involve the bidirectional IGMP protocol, this 1-way iNet is limited to only supporting Persistent Multicast. See Persistent Multicast on page 337.

10.1.2 Multicast Traffic on Terminal Secondary Receiver

With multicast overlay enabled on the terminal, a Velocity dual receiver remote is able to support 2-way unicast traffic on the primary receiver, while simultaneously receiving 1-way multicast traffic on the secondary receiver. Although there are two receivers, the terminal is constrained to a single antenna, and therefore can only tune to a single satellite.

When multicast overlay is enabled, and using the second receiver to receive multicast traffic, only break-before-make beam switch operation is possible, as the secondary receiver is no longer available for make-before-break operation. In this configuration, both receivers are receiving user data — the primary receives unicast traffic from one beam/2-Way iNet and the secondary receives multicast traffic from another beam/1-Way iNet. It is also possible for the primary and secondary receivers to point to a 2-way iNet and a 1-way iNet in the same beam, thereby enabling more efficient multicast distribution in a beam where terminals that are spread across multiple unicast channels wish to share a single multicast feed.



NOTE: See Terminal Break-Before-Make Beam Switch on page 275.

10.1.3 Persistent Multicast

Persistent Multicast is one of three multicast methods, in Velocity. With persistent multicast, the protocol processor sends the multicast stream on the configured iNets without waiting for an IGMP join. The other methods include the dynamic and static multicast streams, where a 2-way connection is required. The PP sends multicast streams only after a request has been issued via an IGMPv3 connection. Persistent Multicast does not need this 2-way communication to support IGMP, and is therefore able to operate using a 1-way beam/iNet.

Persistent Multicast can be configured either with or without enabling Multicast Overlay. When the Multicast Overlay Mode is enabled on the terminal, the terminal will tune its secondary receiver to lock to the overlay beam. A persistent multicast service configured on the terminal will be received on the multicast receiver.

The control flow for each of the Velocity system multicast types is illustrated in the diagram:

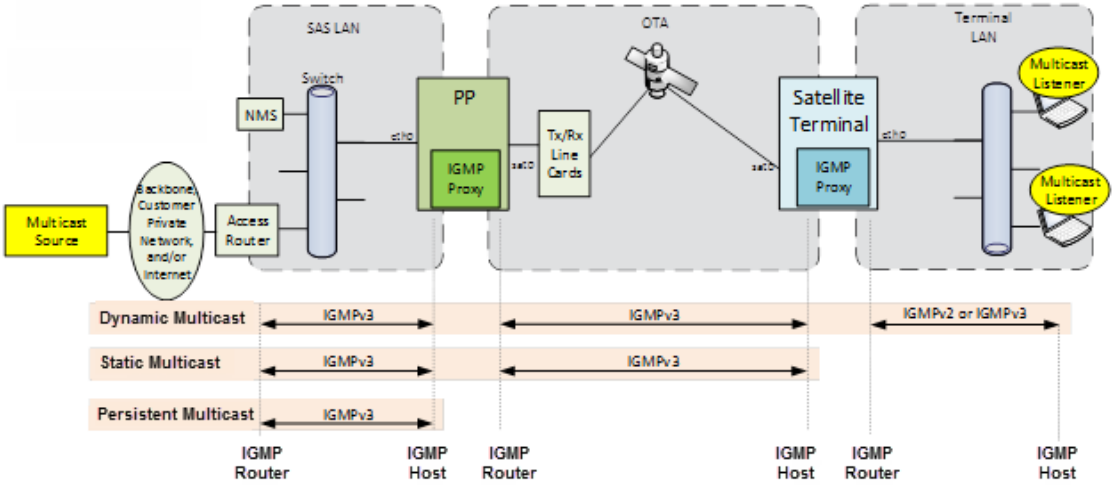


Figure 10-1. IP Multicast in the Velocity System

10.1.3.1 Beam/iNet Switching

Beam/iNet switching, which may require a polarization or band switch, is supported on both the primary and secondary receivers. During a polarization, band or satellite switch, the receivers will switch simultaneously, since the terminal is receiving from only a single receive feed at a time.

When beam switching on the same satellite and same polarization for the unicast traffic, the secondary receiver can maintain connectivity on a wide beam to prevent un-necessary interruption of multicast services. Given this situation, preference is given to beams that do not require a change in polarity or frequency band when the primary receiver needs to beam switch. When changing the antenna to different polarizations, a beam switch will cause an interruption in the multicast traffic on the secondary receiver, of one second or less, in addition to any antenna reconfiguration time to retune and lock.

Although most Service Providers would likely want their terminals to prefer the spot beams for unicast service wherever those spot beams are accessible, and overlay beams for

multicast when they are available, those preferences are not universal, but instead must be configured. To achieve a particular preference model, the operator must configure which beams are preferred — for instance, during persistent multicast service configuration, the beams associated with the service could each be assigned a preference value, with the spot beams all having a lower level of preference than the overlay beams.

10.2 Multicast Overlay Use Cases

The deployment of Multicast Overlay depends largely on the satellite network arrangement — specifically, on the number and position of the available wide beams and spot beams, beam polarizations, and whether the multicast overlay beam contains an inbound channel.

The following are the use cases supported with the multicast overlay feature:

- Persistent Multicast via 1-way Overlay Beam
- Persistent Multicast with Terminal in Receive-Only Mode
- Switching Between Overlay Beams (Same &Different Geographic Regions)
- Unicast on Overlay Beam
- Beam Supporting both 1-way and 2-way iNets

10.2.1 Persistent Multicast via 1-way Overlay

This use case represents a specific deployment scenario in which uninterrupted TV/video services are offered to airline passengers. See the following illustration.

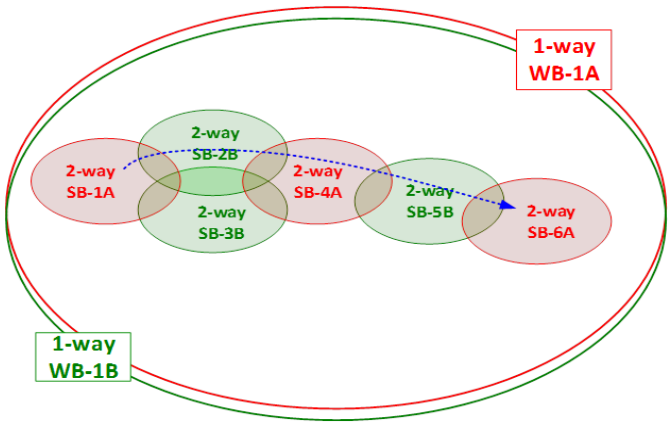


Figure 10-2. Multicast Overlay Aero Services

Key Characteristics:

- Switches between wide beams cover same geographic areas but different polarization
- Primary receiver (2-way) of the terminal locks to the spot beam to receive unicast data
- Secondary receiver locks to wide beam (receive-only mode) to receive multicast traffic
- The 1-way iNet is on the same SAS as the 2-way iNet

Operation:

A Satellite Operator has deployed a satellite that provides a number of overlapping 2-way spot beams, which support continuous coverage across a wide geographic area covering the flight path of aeronautical terminals. These spot beams occupy the same frequency band but may use one of two different polarizations. The same satellite supports two 1-way overlay beams that operate in the same frequency band as the 2-way beams. They each use one of the two available downlink polarizations, and have identical footprints on the ground. See the following illustration.

In this example, as illustrated in the diagram, the mobile terminal starts its service in the area covered by SB-1A and travels to the area served by SB-6A. The terminal exchanges unicast traffic on a 2-way spot beam, using the primary receiver, while simultaneously using its secondary receiver to receive multicast traffic from a 1-way overlay beam. As the aircraft traverses its flight path, the terminal switches between the 2-way spot beam using a break-before-make switch. When a 2-way beam is not available, the terminal continues to receive persistent multicast on its secondary receiver.

10.2.1 shows the use of dual receivers (Rx1/Rx2), as the terminal traverses the flight path.

Table 10-1. Sequential Use of Rx1 and Rx2 as Flight Path is Traversed

#	Rx1	Rx2	Comment
1	SB-1A Vertical Pol	WB-1A Vertical Pol	Unicast on Rx1 Persistent Multicast on Rx2
2	Terminal switches to SB-2B Horizontal Pol	Switches to WB-1B Horizontal Pol	Unicast on Rx1 Persistent Multicast on Rx2
3	Terminal moves & switches to SB-4A Vertical Pol	Switches to WB-1A Vertical Pol	Unicast on Rx1 Persistent Multicast on Rx2
4	Terminal moves & switches to SB-5B Horizontal Pol	Switches to WB-1B Horizontal Pol	Unicast on Rx1 Persistent Multicast on Rx2
5	Terminal moves & switches to SB-6A Vertical Pol	Switches to WB-1A Vertical Pol	Unicast on Rx1 Persistent Multicast on Rx2



NOTE: For configuration requirements for implementing Multicast Overlay, see Multicast Overlay Configuration Requirements.

10.2.2 Persistent Multicast Receive-Only Mode

An important characteristic of Persistent Multicast is that a terminal can continue to receive multicast traffic even when transmitting is prohibited by the system or by mute indications from the terminal — for example, when the terminal is in an aircraft that is on the ground, or

when located in a no-transmit zone. This operation is referred to as “1-way mode” or “receive-only mode.”

In normal operation, Velocity implements no-transmit zones using Regulatory Areas (RA) — a set of service areas in which restrictions such as transmit power levels, transmit allowed/not allowed, and constraints on skew properties are enforced. In such an area, the terminal is still able to receive Persistent Multicast service.

In practical applications, no-transmit conditions may occur in the following cases:

- Aircraft is on the ground, but terminal is not allowed to transmit while at gate
- Aircraft is in-flight, but flying below a certain altitude
- Aircraft enters regulatory area where terminal is not allowed to transmit
- Aircraft moves to area where 2-way beams are not available

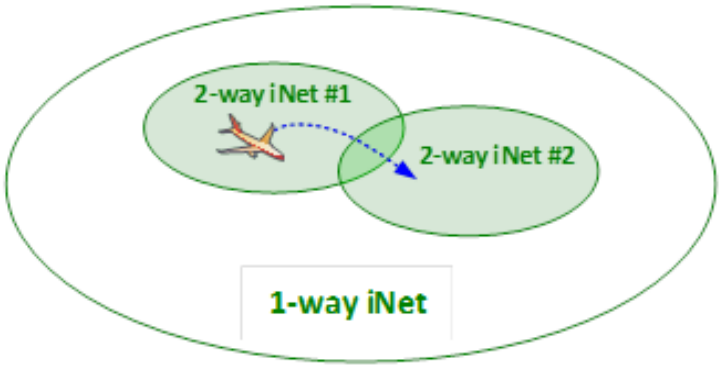


Figure 10-3. Persistent Multicast Receive-Only Mode

10.2.3 Switching Between Overlay Beams

Service Providers can also extend the geographic scope of their persistent multicast service beyond that which is provided by a single wide beam, or completely overlapping wide beams. This extension is accomplished by making the service available across multiple wide beams covering different geographic areas. These beams would likely overlap, but not completely, such that coverage would continue to be available to mobile terminals as they move from one beam to another.

Due to the break-before-make beam switch from one wide-beam to another or from one satellite to another will create a momentary loss of both multicast and unicast services.

Switching between spot beams and switching between overlay beams is independent of each other, with a switch on one receiver not affecting operation on the other receiver. An exception to this, would be when there is a change in polarization during the switch, in which case the polarization controlling the selection of the overlay beam will follow the polarization of the new spot beam, and the overlay beam will be switched with the spot beam (2-way iNet/beam).

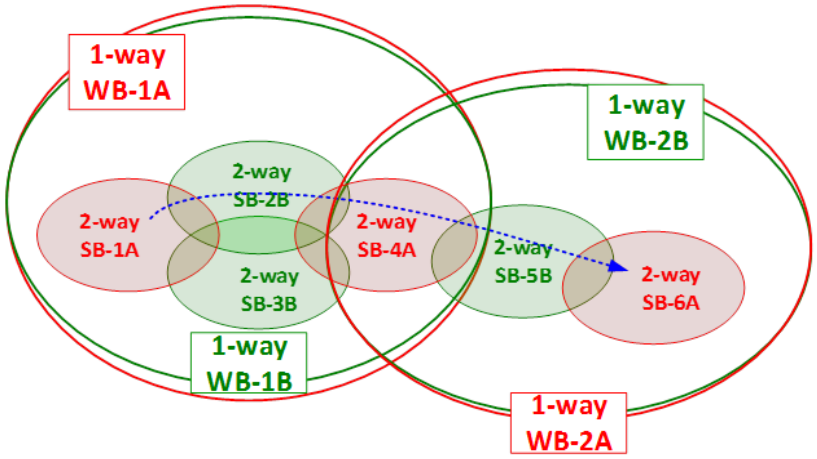


Figure 10-4. Switching Between Overlay Beams

10.2.4 Unicast on Overlay Beams

Service Providers that have gaps in the unicast data coverage provided by their spot beams, can extend the unicast services to cover those gaps that are also covered by their wide beams. The extension of this coverage would be achieved by adding unicast 2-way iNets to those wide beams, to thereby enable unicast coverage in those areas not previously covered by the spot beams. The wide beams would then contain both 2-way iNets and 1-way iNets (broadcast-only), and continue to carry persistent multicast traffic on its own downstream carrier.

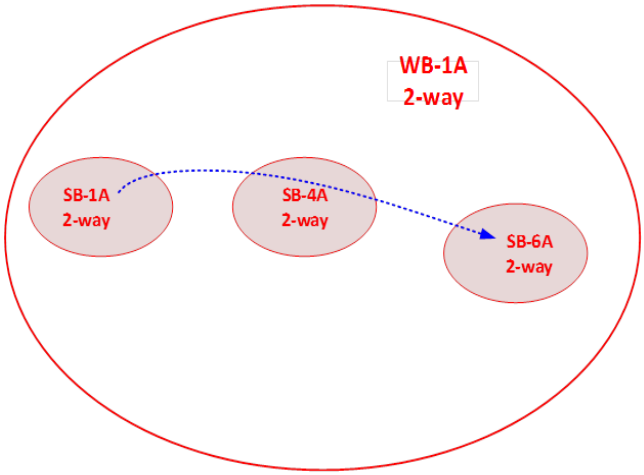


Figure 10-5. Unicast on Overlay Beams

In most situations, providers would likely want terminals to always prefer spot beams for unicast service wherever those spot beams are accessible; and that all terminals have the same preference for spot beams over wide beams. To achieve this preference would require that unicast services be configured such that there is a preference for the spot beams over the wide beams. For instance, in the configuration of the service, the associated beams might each be assigned a level of preference, where all spot beams are assigned one level of preference and all wide beams are given a lower level of preference.

This use case supports the following situations in which the terminal may become situated:

- A spot beam provides unicast while a wide beam provides persistent multicast
- A wide beam provides unicast and persistent multicast where there is no spot beam
- A spot beam provides unicast where there is no wide beam (and no persistent multicast)
- A wide beam provides persistent multicast where there is no spot beam (and no unicast)
- A wide beam provides persistent multicast while the terminal is muted (and no unicast)

Switching from a spot beam to a wide beam and from a wide beam to a spot beams in order to maintain the unicast service would be independent of switching from wide beam to wide beam in order to maintain the persistent multicast service, whereby a switch on one receiver will not impact operation of the other. The only exception to this, is when there is a polarization change during the switch — in which case, the polarization controlling the selection of the wide beam that provides the persistent multicast will need to follow the

polarization of the chosen unicast beam (spot or wide beam), and the wide beam providing the persistent multicast service will need to be switched with the spot beam.

10.2.5 Multicast Overlay Antenna and Terminal Requirements

Multicast overlay operation relies heavily on the associated Antenna and Antenna Control Unit (ACU). The following Terminal, Antenna, and ACU requirements are applicable:

- The terminal has a single antenna
- The terminal has a single Rx input
- The ACU supports OpenAMIP
- The antenna is pointed to a single satellite at a time
- The antenna is tuned to a single polarization at a time (Tx/Rx polarizations may differ)
- Frequencies of the two receivers must be in the same band (on the same LNB)

The following are implications for beams/iNets used by the two independent receivers:

- $Recv_1$/$Recv_2$ must be sourced from the same satellite
- $Recv_1$/$Recv_2$ must be the same polarization
- $Recv_1$/$Recv_2$ must be in the same frequency band

10.2.6 Multicast Overlay Limitations and Constraints

The following items describe specific limitations and constraints of Multicast Overlay with Velocity Release 1.7. Note: These limitations will be documented in the Release Notes.

- ***1-way and 2-way iNets must be configured on same SAS*** — Multicast Overlay can be configured with 2-way and 1-way iNets in the same SAS; however, the configuration of a 1-way iNet on one SAS, and a 2-way iNet on a different SAS is not supported.
- ***1-way iNet Downstream Carrier Change*** — Seamless downstream carrier changes on a 1-way iNet is not supported. Such a change will cause the terminal to lose lock on 1-way carrier until new carrier changes are in sync on Line Card and Remote — resulting in a brief loss of persistent multicast streams.
- ***Terminal Muted When Aircraft on Ground*** — When a terminal is muted it continues its attempt to acquire (searching for available carriers). In cases where the beam polarization is different (compared to the overlay to which it is currently locked), the terminal will try to lock to the corresponding polarization overlay, which would cause a glitch in the multicast. This behavior extends to terminals searching multiple satellites that may or may not be visible — which would also cause loss of multicast connectivity.
- ***Persistent Multicast Limited to 1-way iNets*** — By design, Velocity multicast is streamed to all iNets in a beam. As a result, when 2-way and 1-way iNets are configured in the same beam, persistent multicast would also be streamed to all iNets. In this current design, persistent multicast is limited to 1-way iNets.
- ***Regulatory Area Assumption*** — When in a regulatory area where the terminal is not allowed to transmit, current behavior is to force logout of the terminal and for the terminal to enter the signaling mode; persistent multicast is still received, however the signaling carrier must be of the same polarization of the LNB and as the overlay beam.

- ***Multicast Overlay Only Supported with Terminal in DVB-S2 Mode*** — Multicast overlay only works when terminal is in DVB-S2 mode, including the ASC operation.
- ***Persistent/Dynamic Multicast Streams*** — The same multicast group address must not be configured for both dynamic and persistent multicast for the same iNet. Only the M-SSPP profile loaded first will receive traffic, traffic for all other M-SSPPs is lost.
- ***Multicast Overlay Not Supported with Discreet Terminals*** — Multicast Overlay support is not available for discreet terminals.
- ***Beam-Based Geoscopes Only*** — Multicast overlay is only supported in a regional geoscope, which is based on beams.
- ***Seamless Beam Switch Not Supported*** — The seamless beam switch operation is not supported when the multicast overlay capability is enabled on a terminal, since the terminal secondary receiver is dedicated to the overlay.
- ***Encryption Disabled*** — The encryption feature is disabled since the terminal key exchange requires 2-way communication. This limitation does not impact other traffic when connected in two-way mode.
- ***Use Persistent Multicast is Read-Only on Multicast-SSPP*** — The option “Use Persistent Multicast,” on a Multicast-SSPP cannot be modified once the configuration is saved. The Multicast-SSPP cannot be modified to configure dynamic/static multicast.

11 Statistics Reports and SLA Management

Network operators and subscribers use Service Level Agreement (SLA) management reports to verify that SLAs are in compliance. These reports can be created using various Velocity statistics that are collected and saved are all saved as statistics that can be used for SLA Management Reporting.

This chapter contains the following sections:

11.1 About SLA Management and Statistics Reporting 348
11.2 CIR/MIR Fulfillment Statistics 348
11.3 CIR/MIR Fulfillment Parameters 350
11.4 CIR/MIR Fulfillment Ratio Statistics 352
11.5 CIR/MIR Statistics Calculation Exclusions 359

11.1 About SLA Management and Statistics Reporting

Network operators and subscribers use Service Level Agreement (SLA) management reports to verify that SLAs are in compliance. These reports can be created using various Velocity statistics that are collected and saved — for example, bandwidth management, network availability, GSP and SSPC bandwidth availability, CIR (***Committed Information Rate***) and MIR (***Maximum Information Rate***) *fulfillment statistics* and *CIR/MIR Fulfillment Ratios*.

The SLA compliance report for the Subscriber Service Plan Component (SSPC), for example, shows the subscriber whether the CIR/MIR demand has been satisfied based on the SLA. Velocity statistics used in producing SLA reports are generated in real-time, every 30 seconds at the Velocity SAS NMS, and are derived every 5 minutes at the NOC NMS. The CIR/MIR fulfillment stats can be used externally by network operators or service providers to generate monthly SLA compliance reports.

As an example, using Velocity CIR/MIR Fulfillment Statistics and CIR/MIR Fulfillment Ratios performance metrics, Operators can determine, the percentage of time (for example 97%) that the CIR/MIR demand is satisfied during a period of one month; or, for example, what percentage of the CIR/MIR has been achieved, or by how much the SLA CIR/MIR has been missed.

11.2 CIR/MIR Fulfillment Statistics

The CIR/MIR Fulfillment Stats, as listed below, are used in Satellite Operator and Service Provider SLA reporting. These performance metrics are used in conjunction with the CIR/MIR Fulfillment Ratio metrics, described later in this chapter. Pulse NMS CIR Fulfillment Statistics are listed below:

Table 11-1. CIR/MIR Fulfillment Statistics

CIR/MIR Fulfillment Stat	Description
• sspc_ds_cir_fulfillment	Downstream SSPC CIR Fulfillment
• sspc_us_cir_fulfillment	Upstream SSPC CIR Fulfillment
• sspc_ds_mir_fulfillment	Downstream SSPC MIR Fulfillment
• sspc_us_mir_fulfillment	Upstream SSPC MIR Fulfillment



NOTE: These stats are reported at 30 second intervals at the SAS and 300 second intervals at the NOC.

The frame-by-frame CIR Demand is considered “Fulfilled” or “Pass” when the ***frame-by-frame Allocation*** satisfies the ***CIR Demand*** within the ***Window for Delayed Fulfillment***. If the CIR Demand for any frame is not satisfied within the Window for Delayed Fulfillment, the CIR Fulfillment for the frame is considered “Fail”.

The ***CIR Fulfillment Statistic*** is reported as the percentage of frames, within the reporting interval, with a CIR Fulfillment of "Pass". For example, 228 "Pass" and 12 "Fails" out of 240 upstream frames, during a 30 second interval, result in a 95% CIR Fulfillment for the interval.

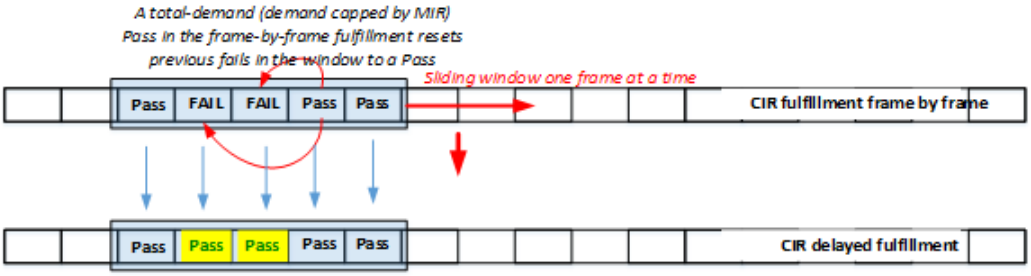


Figure 11-1. Delayed Fulfillment for CIR

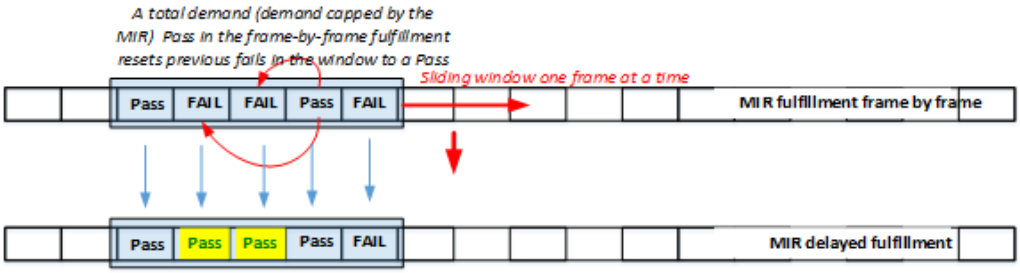


Figure 11-2. Delayed Fulfillment for MIR

The following diagram also demonstrates the new algorithm for the CIR fulfillment.

Frame #	1	2	3	4	5
(Window Size of 5 Frames)					
Current Demand	3	4	8	8	20
CIR Demand (CIR = 10 Units)	3	4	8	8	10
CIR Allocation	1	1	2	3	10
New demand = current demand – (previous demand – previous allocation)	2	2	6	2	5
Total New Demand over 5 frames = 2 + 2 + 6 + 2 + 5 = 17 Total Allocation over 5 frames = 1 + 1 + 2 + 3 + 10 = 17 **Hence total demand has been fulfilled by the total allocation. <u>This is prior to delayed fulfillment.</u>**					
Excess/Shortage	-2	-3	-6	-5	0
Pass/Fail - Frame-by-Frame	F	F	F	F	P
Last Frame in the window met condition of total allocation = total demand Hence reset all previously failed frames to a state of "Pass"	P	P	P	P	P
(Assume that frame # 0 (not shown) had 1 unit of unfulfilled demand.) Total Allocation after the delayed fulfillment = 17					

Figure 11-3. CIR Delayed Fulfillment Algorithm

11.2.1 Delayed Fulfillment Based on CIR/MIR Demand

If any given frame, within a given window, receives an allocation that is equal to the total demand, in the frame capped by MIR, then all of the initially failed frames in the same window are reset to a status of passed. This resetting applies both to the CIR and MIR delayed fulfillment calculations only. The condition of resetting is contingent on the total demand being fulfilled since this demand might have originated from CIR demand that continued accumulating in successive frames till it exceeded the CIR setting. Fulfilling the total demand will reset the (subsequently) failed frames in both the CIR and the MIR delayed fulfillment calculations.



NOTE: See CIR/MIR Fulfillment Parameters on page 350, for additional details of example. Also, see CIR/MIR Statistics Calculation Exclusions on page 359., for periods and cases that are excluded from the CIR/MIR metrics calculations.

11.3 CIR/MIR Fulfillment Parameters

SSPC demand is evaluated ten times a second on the downstream and eight times a second on the upstream based on the total amount of data queued for transmission and arrival rate. Allocation is expressed as the value that is handed out in a single allocation cycle.

The following parameters provide more detail about CIR/MIR demand and allocation:

- ***CIR Demand*** — the frame-by-frame demand from an SSPC, up to but not exceeding the configured CIR. CIR Demand is the lesser of Current Demand and the configured CIR.

  When current demand is lower than the configured CIR, the CIR Demand is equated to the current demand. When the current demand is higher than the CIR, up to the configured MIR, then the CIR Demand is equated to the CIR.

- ***MIR Demand*** — the demand from an SSPC, which is larger than the configured CIR, but is less than or equal to the configured MIR.

  When current demand is higher than the configured CIR, but less than the configured MIR, the MIR Demand is equated to the current demand that exceeds the CIR. When the current demand is higher than the MIR, then the MIR Demand is equated to the MIR-CIR.

- ***Current Demand*** — the frame-by-frame demand from the SSPC. When SSPC demand is lower than the configured CIR, the CIR Demand is equated to the current demand.

- ***Window for Delayed Fulfillment*** — the configurable CIR/MIR fulfillment window is specified independently, with a value of 1-50 for downstream, and a value of 1-40 for the upstream. On the downstream a value of 1 corresponds to 100ms and a value of 50 corresponds to 5 seconds; on the upstream a value of 1 corresponds to 125ms and a value of 40 corresponds to 5 seconds.

  For real-time applications, the Window for Delayed Fulfillment is typically set to a low value on the order of 1 second, and corresponds to a 10-frame window on the downstream and an 8-frame window on the upstream. For non-real-time applications,

the window may be set to a larger value. However one important point to make: See **Note**.



NOTE: The Window for Delayed Fulfillment can be appropriately sized smaller or larger, based on the application. Some applications can tolerate more delay in the allocation of bandwidth than others. For those applications that cannot tolerate any significant delay, the window should be configured smaller; and for those applications that are less sensitive to delay, the window is set larger.

Based on the agreement between the operator and the customer the window size is configured to match the application needs. An SLA with a very small Window for Delayed Fulfillment would likely be more expensive than an SLA with a medium or large window. A larger window allows more time for the system to satisfy the demand, whereas a smaller window will demand more bandwidth to ensure the requirement is met.

A size adjustment in the Window for Delayed Fulfillment does not impact QoS allocation in the downstream or upstream. It does, however, influence SLA computation as described in the respective section ***CIR/MIR Fulfillment Statistics***.

11.4 CIR/MIR Fulfillment Ratio Statistics

The Velocity system has integrated the Fulfillment Ratio Statistics as additional metrics that support unicast SSPCs in the upstream and downstream directions. The Fulfillment Ratio Stats, as listed below, are an enhancement to Satellite Operator and Service Provider SLA reporting.

Table 11-2. CIR/MIR Fulfillment Ratio Statistics

CIR/MIR Fulfillment Ratio Stat	Description
• sspc_ds_cir_fulfillment_ratio	Downstream SSPC CIR Fulfillment Ratio
• sspc_us_cir_fulfillment_ratio	Upstream SSPC CIR Fulfillment Ratio
• sspc_ds_mir_fulfillment_ratio	Downstream SSPC MIR Fulfillment Ratio
• sspc_us_mir_fulfillment_ratio	Upstream SSPC MIR Fulfillment Ratio



NOTE: These stats are reported at 30 second intervals at the SAS/CDC level and 300 second intervals at the NOC level.

Fulfillment Ratio stats provide additional SLA performance metrics that are both complementary and an extension of existing CIR and MIR Fulfillment Stats. Their implementation relies on the results of the CIR and MIR fulfillment stats, which are also applicable only to unicast SSPCs. These stats can be enabled per SSPC, on the upstream, downstream, or both.

Whereas CIR/MIR Fulfillment Stats report the percentage of time that the CIR and MIR are respectively achieved, for a given SSPC during a given period, the Fulfillment Ratio Stats can be used by Service Providers in SLA bill settlement, to determine by how much a CIR or MIR SLA has been missed, in terms of bytes, and apply appropriate credits accordingly. The remainder of this section describes CIR/MIR Fulfillment Ratio stats and the calculation at the frame, SAS and NOC levels.

11.4.1 CIR/MIR Fulfillment Ratio Implementation

CIR/MIR Fulfillment Ratio stats are directly derived from CIR/MIR Fulfillment stats and are similarly calculated on a per frame basis, as described earlier in this chapter. Like the CIR/MIR Fulfillment Stats, the Fulfillment Ratio stats also use the delayed fulfillment and the fulfillment window size mechanisms for measurement and reporting. Since both CIR/MIR Fulfillment Stats and CIR/MIR Fulfillment Ratio Stats use the same fulfillment window size, the data of these combined metrics represents a coherent set of reports for a given SSPC in a given direction, at any time.

The CIR/MIR Fulfillment stats (*cir_fulfillment*, *mir_fulfillment*) are reported as a percentage, every 5 minutes at the NOC. The operator can use these stats externally (to Velocity/Pulse), to derive performance reports for an extended period, for example 30 days, of how the CIR and MIR was respectively met for a given SSPC in the downstream or upstream direction.

These stats, on a per frame basis, are represented in a binary form, where a value of TRUE (1) = CIR/MIR Achieved; and a value of FALSE (0) = CIR/MIR Not Achieved.

These stats, although important and are used in SLAs with end customers, they provide no information regarding what percentage of the CIR/MIR was met in terms of bandwidth (bytes). An iNet that is relatively congested, can be used to illustrate this concept. For a relatively large percentage of the time, the SSPCs for the terminals in this iNet are barely getting the contracted CIR bandwidth. Let us assume the cir_fullfillment is 50%. However the amount of bandwidth they (the SSPCs) are getting is very close to the CIR but is below it for 50% of the time.

The SSPC usually represents an application, used by the end customer and that will have certain bandwidth demands. Assuming that the application receives 99% of its CIR for 99 % of the time, then this would likely be sufficient and is not likely to introduce any noticeable degradation in service. This performance of 99% can be reported using the CIR/MIR Fulfillment Ratio stat. If the SLAs are based on these stats, the first stat would be dismal at 50% while the second stat would be very good at 99%. The Service Provider use the Fulfillment Ratio Stat to differentiate the SLA.

11.4.1.1 Configuring the Fulfillment Window Size

The fulfillment window size is configurable for each terminal SSPP, independently for downstream and upstream directions of the SSPC. The units are in frames, translated to seconds, where for downstream, one frame unit represents 100 ms; and for upstream, one frame unit represents 125 ms. The fulfillment window size that is configured for the CIR/MIR Fulfillment Stats is also used for the corresponding Fulfillment Ratio Stat.

In a real application, having relatively relaxed fulfillment requirements, there would be no noticeable degradation in application performance if the allocation of the demand is delayed by a certain amount— for example, with file transfer or directory replication. These applications are expected to be tolerant of delays in allocation and are not critical with respect to variations in bandwidth over time. In this case we might use a large fulfillment window, for example 5 seconds. In the downstream we would configure 50 (50*0.1 = 5) and in the upstream we configure 40 (40*0.125 = 5).

The customer might also have a low-latency, low-jitter application such as VoIP that requires timely responses and has more stringent timing constraints. Assuming the operator determines a fulfillment window of 0.5 seconds is needed in this case—then the window settings for downstream and upstream would be 5 and 4 respectively. The assumption is that the operator will use different SSPCs for the different applications.

If we continue with this example, and the operator puts both applications in the same SSPC then the window size must be set for the more stringent of applications, to meet the requirements of the transactional application. This will likely be more expensive as we are guaranteeing a level of service that is not required for the more relaxed application of file transfer/directory replication.

11.4.1.2 Enable/Disable Fulfillment Ratio Statistics

When configuring SSPPs, you may enable or disable the generation of the new stats for all SSPCs using that SSPP. Configuration of the SSPP is possible using the NMS Web UI and through an API call to the NMS. The generation of the stats cannot be selectively enabled or disable for a given SSPC, but instead, when applied, has effect on all the SSPCs associated with the SSPP.

The enabling and disabling of the CIR and MIR fulfillment ratio stat on an SSPP can be configure for any one of the following configurations:

a. Enable the downstream and disable the upstream

b. Enable the upstream and disable the downstream

c. Enable both the downstream and the upstream

d. Disable both the downstream and the upstream directions (default configuration).

> ***NOTE:*** Fulfillment Ratio Stats are supported for unicast traffic only. Operators should ensure that only the required SSPPs (SSPCs) have this feature enabled.

11.4.1.3 Deriving CIR/MIR Fulfillment Ratio Statistics

Derivation of the CIR/MIR Ratio Stats can be described in terms of the CIR, as both are derived in the same fashion, except that the MIR fulfillment ratio stats are based on MIR related parameters. Assuming the operator has configured the desired window size for the CIR fulfillment stat, for both the downstream and upstream, the stat is derived as follows.

The CIR fulfillment stat is derived based on a per frame calculation and uses the configured window for delayed fulfillment. After applying the algorithm the system determines which frames met CIR demand and which failed. The number of frames that passed is then divided by the number of intervals—for example, 300 intervals in the downstream and 240 for upstream— each of 30 second intervals. The output of the CIR/MIR fulfillment stats calculation with additional information of how much the demand was satisfied, will be used to derive the stats. The algorithm is best explained through the following diagram.

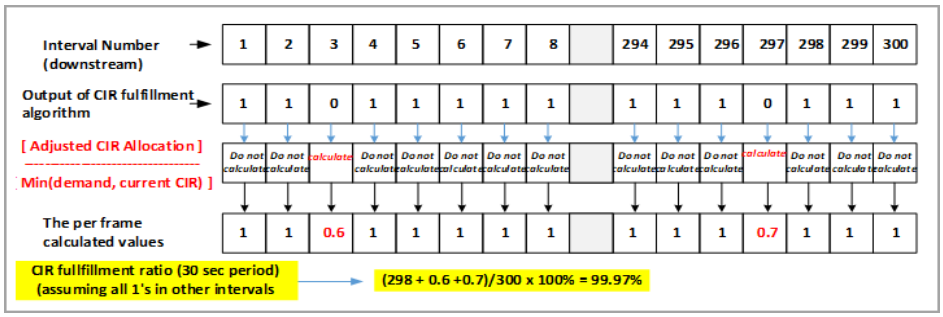


Figure 11-4. Algorithm for Calculating the CIR (MIR) fulfillment Ratio



NOTE: The MIR Fulfillment Ratio Stats are derived in the same fashion as are the CIR Fulfillment Ratio Stats, except that the configured MIR parameters are used.

11.4.2 CIR/MIR Performance Metrics Compared

In some cases, the MIR fulfillment stats will report a better performance than the CIR fulfillment stats. Although the case may seem counter intuitive, this occurrence is possible with both CIR/MIR Fulfillment Stats and CIR/MIR Fulfillment Ratio Stats. This occurrence illustrated with with the reported metrics for an SSPC.

If during operation there is no demand, then there is no allocation. Consider the CIR first—if there is no CIR demand in a frame and consequently no allocation in the same frame, then the calculation will show that the CIR fulfillment stat and fulfillment ratio stat have a value of 1—or 100% fulfillment for both parameters. Now consider the MIR—it is defined as the demand that exceeds the CIR, but is below the configured MIR. MIR is therefore calculated as "Demand - CIR". If negative, then the demand did not exceed CIR and therefore the MIR demand was '0'. The allocation, would also be '0'. In this scenario, the MIR fulfillment Stat and Fulfillment Ratio are '1' —which is 100% fulfillment, as nothing was asked for and nothing was allocated.

There may also be cases where the demand is below CIR, and there is no MIR demand as a result of congestion; the CIR fulfillment is '0' and the CIR fulfillment ratio is less than '1'. So MIR fulfillment and fulfillment ratio will be 1 (100%). This scenario may also seem counter-intuitive for Satellite Operators and Service Providers, as a common belief is that the MIR fulfillment is generally worse than CIR fulfillment. This relationship, however, is reversed whenever the demand is below the CIR and the CIR fulfillment is low. This is a noteworthy point to be considered by operators with regard to SLA agreements.

Demand Level: D	CIR Demand Fulfilled	CIR Fulfillment Reported	MIR Demand Fulfilled	MIR Fulfillment Reported	Comments
D ≤ CIR	Yes	1	Yes	1	No MIR demand & no Allocation MIR considered fulfilled.
D ≤ CIR	No	0	1	1	No MIR demand & no Allocation MIR considered fulfilled.
CIR ≤ D ≤ MIR	Yes	1	Yes	1	
CIR ≤ D ≤ MIR	Yes	1	No	0	
CIR ≤ D ≤ MIR	No	0	No	0	MIR demand made but no allocation.
D > MIR	Yes	1	Yes	1	Both CIR & MIR demand are met.
D > MIR	Yes	1	No	0	CIR demand met, MIR demand not met.
D > MIR	No	0	No	0	Neither CIR nor MIR demands are met.

From an SLA perspective, these scenarios are readily adapted into the billing arrangement, as illustrated in the following examples:

Case 1: Assume that SLA for the CIR fulfillment is for 96% of the time. If the measured performance is below 96% the satellite operator has some predefined penalties depending on by how much the SLA is missed.

Case 2: Assume that the SLA for the MIR Fulfillment is 50%. In the special case where demand is below CIR (all the time) the performance is at 100%. In such as case, there would be no penalties or rebates against the MIR. This result is a reasonable expectation since the service provider demand was below CIR and never crossed into MIR— the Satellite Operator should not be penalized for something that was not used (MIR demand above CIR).

Case 3: It is more likely to have values other than 100% fulfillment for MIR. However that fulfillment can be larger than the CIR fulfillment. This scenario is explained in the same way as for the case where for part of the time the demand is below CIR so the MIR fulfillment is 100% for that portion of time and then below 100 % for other portions. So the overall performance, in this case, will be below 100% but above the CIR fulfillment.

11.4.3 Fulfillment Ratio Stats Calculation and Reporting

This section describes CIR/MIR Fulfillment Ratio stats and the calculation at the frame, SAS and NOC levels.

11.4.3.1 Fulfillment Ratio Calculation at Frame Level

The CIR/MIR Fulfillment Ratio Stats, when enabled, are calculated for each frame in both the downstream and upstream directions.

For CIR: *If the CIR fulfillment value for a given frame has been determined to = 1*

then: *CIR-Fulfilment-Ratio-frame = 1 (for that frame)*

else: *If the CIR fulfillment value for a given frame has been determined to = 0*

then: *CIR-Fulfilment-Ratio-frame = (Adjusted CIR allocation)/[minimum (demand, current CIR)]*

The adjusted CIR allocation is the remaining CIR allocation after the application of the window and the delayed fulfillment. Assuming that the stat is enabled for the SSPC in both the downstream and upstream directions then for every frame and for each direction (one calculation is done for downstream every 100 ms and one for upstream every 125 ms).

For MIR:

MIR demand is defined as the amount of demand above CIR.

If demand <= CIR, then the MIR = 0

(Demand – CIR) = 0 if demand ≤ CIR.

If demand > CIR, then MIR demand will be minimum((Demand – CIR), (MIR-CIR))

If the MIR fulfillment value for a given frame has been determined to be = 1

then: *MIR-Fulfilment-Ratio-frame = 1 (for that frame)*

else: *If the MIR fulfillment value for a given frame has been determined to be = 0*

then: *MIR-Fulfilment-Ratio-frame = (Adjusted MIR allocation)/ [minimum ((demand-CIR), (current (MIR – CIR)))]*

The adjusted MIR allocation is the remaining MIR allocation after the application of the window and the delayed allocation. Assuming that the stat is enabled for the SSPC in both the downstream and upstream directions then for every frame and for each direction (one calculation is done for downstream every 100 ms and one for upstream every 125 ms).

11.4.3.2 Example Calculations at Frame Level

Example 1:

Current CIR = 1 Mbps

Current MIR=2 Mbps

Demand = 0.8 Mbps

CIR Allocation=0.4 Mbps,

MIR Demand = (0.8 – 1) which is equivalent to 0. If demand is zero then allocation is also 0

CIR Fulfilment Ratio-Frame = 0.5 (0.4/0.8)

MIR Fulfilment Ratio-Frame = 1 (Demand is less than CIR, hence MIR demand is 0 and allocation is 0)

Example 2:

Current CIR = 1 Mbps,

Current MIR = 2 Mbps

Demand = 1.5 Mbps

Total Allocation = 0.3 Mb/s

CIR Allocation=0.3 Mbps

CIR Fulfilment Ratio-PCT frame = 0.3 (CIR demand is 1, allocation is 0.3 , hence CIR fulfillment is 0.3/1)

MIR Demand = 1.5 – 1 = 0.5 Mb/s

MIR Allocation = (total allocation – CIR) = 0 Mb/s

MIR Fulfillment Ratio-Frame = 0 / (0.5) = 0

(MIR Demand is 0.5, allocation is 0.0 , hence MIR Fulfillment is 0.0/0.5)

Example 3:

Current CIR = 1 Mbps

Current MIR = 2 Mbps

Demand = 2.5 Mbps

Total Allocation 1.5 Mbps

CIR Allocation=1 Mbps

MIR Allocation = 0.5 Mbps

CIR Fulfilment Ratio frame = 1

MIR Demand = Minimum ((Demand – CIR), (MIR-CIR)) = minimum ((2.5 – 1), (2 – 1)) = 1

MIR Allocation = (total allocation – CIR) = (1.5 – 1) = 0.5

MIR Fulfillment Ratio-Frame = 0.5/1 = 0.5

11.4.3.3 Fulfillment Ratio Reporting at SAS Level

The following stats are reported once every 30 seconds. Note that the "*Number of frames*" depends on whether the stat is for upstream or for downstream. If the terminal was in network and the SSPC was active for all of the reporting period—then, the Number of frames for downstream = 300 and Number of frames for upstream = 240.

sspc_ds_cir_fulfillment_ratio =

*(Σ (downstream CIR Fulfilment Ratio/frame))/(Number of frames) * 100%*

sspc_us_cir_fulfillment_ratio =

*(Σ (upstream CIR Fulfilment Ratio/frame))/(Number of frames) * 100%*

sspc_ds_mir_fulfillment_ratio =

*(Σ (downstream MIR Fulfilment Ratio/frame))/(Number of frames) * 100%*

sspc_us_mir_fulfillment_ratio =

*(Σ (upstream MIR Fulfilment Ratio/frame))/(Number of frames) * 100%*

Also, see CIR/MIR Fulfillment Ratio Statistics on page 352, regarding values excluded from the calculation. If a remote was in the network for only 15 sec of a 30 sec reporting interval, then the downstream number of frames = 300/2 = 150; for the upstream, it is 240/2 = 120.

11.4.3.4 Fulfillment Ratio Reporting at NOC Level

There are ten 30-second SAS reporting intervals, for each 300-second NOC reporting interval. Hence the following equations are used to calculate the CIR and MIR fulfillment ratio stats at the NOC level. The summation used in the following equations is for a 300-second period.

sspc_ds_cir_fulfillment_ratio =

(Σ sspc_ds_cir_fulfillment_ratio at SAS level)/(# of reported intervals)

sspc_us_cir_fulfillment_ratio =

(Σ sspc_us_cir_fulfillment_ratio at SAS level)/(# of reported intervals)

sspc_ds_mir_fulfillment_ratio =

(Σ sspc_ds_mir_fulfillment_ratio at SAS level)/(# of reported intervals)

sspc_us_mir_fulfillment_ratio =

(Σ sspc_us_mir_fulfillment_ratio at SAS level)/(# of reported intervals)

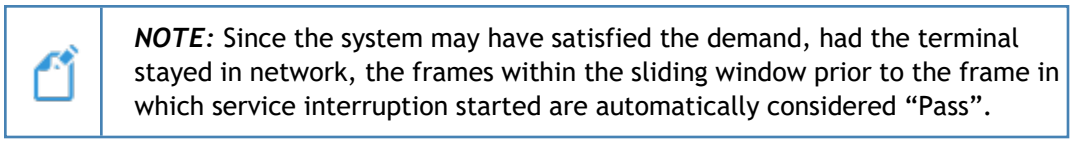
The number of reported intervals is the number of intervals with valid data over the period of for which the data is aggregated. The typical value used is expected to be a value of 10.

11.5 CIR/MIR Statistics Calculation Exclusions

The CIR/MIR Fulfillment Statistics, and the CIR/MIR Fulfillment Ratio Stats both only apply when the associated SSPC is active and the associated terminal is in network—otherwise, the CIR/MIR Fulfillment metrics are invalid, and such periods are not included in the CIR/MIR Fulfillment calculations.

The following are examples of situations and periods that are excluded from the fulfillment calculations:

- When the terminal is muted due to skew, the terminal is out of the network, then the fulfillment metrics are not included in the calculations.
- If a terminal is out of network for 10 frames out of 40, and the other 30 frames were determined to be 20 “Pass” and 10 “Fail,” then the CIR/MIR Fulfillment for the 40 frames is calculated at 66.7%.
- When an SSPC is not active (upstream/downstream) or the terminal is out of the network for an entire reporting interval, the CIR/MIR Fulfillment metrics are invalid and are not reported. Those periods for which the SSPC is not active or the terminal is out of network are not included in the determination of CIR/MIR Fulfillment metrics.
- When a terminal goes out of network, while switching satellites, or for any other reason, queued demand does not have an opportunity to be satisfied in subsequent frames within the Window for Delayed Fulfillment.
- When a terminal is muted due to skew, and is out of network, then the fulfillment metrics are included in the calculations. When the terminal is muted for a short period (< 13 seconds), although the PP does not receive any demand, the terminal is considered to be in network. Therefore, in the upstream the CIR and MIR will be considered as fulfilled for that period. In the downstream the demand/fulfillment calculations will continue while the terminal is in that state. When the 13 seconds expire, the terminal is considered, by the PP, to have dropped out of network and the outage period is excluded from the fulfillment calculations.



NOTE: Since the system may have satisfied the demand, had the terminal stayed in network, the frames within the sliding window prior to the frame in which service interruption started are automatically considered “Pass”.

12 Layer 2 Over Satellite

This chapter contains the following sections:

12.1 About Layer 2 Over Satellite (L2oS) 362
12.2 L2oS Reference Model 364
12.3 SVN and Service Delimiting Tags 366
12.4 Layer 2 Services -SVN 367
12.5 L2oS Service Mode 369
12.6 Layer 2/Layer 3 Hybrid Mode 370
12.7 TCP Acceleration 370
12.8 Advanced Header Compression 370
12.9 MAC Address Learning 373
12.10 L2oS Failover and Resiliency 373
12.11 Forwarding Rules 374
12.12 MEF Carrier Ethernet 2.0 Services 375

12.1 About Layer 2 Over Satellite (L2oS)

Layer 2 over Satellite (L2oS) enables the satellite link to emulate an Ethernet connection. When an iDirect network is configured for L2oS, Ethernet frames, rather than IP packets, are transported across the satellite link. The Velocity model of Transparent Layer 2 Emulation, is shown in Figure 12-1.

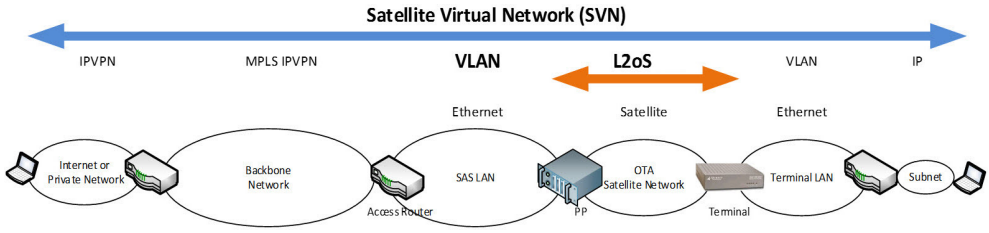


Figure 12-1. Velocity Transparent Layer 2 Emulation

NOTE: 1) L2oS is only supported on iQ Desktop, iQ 200, and iQ LTE Satellite Routers.

2) L2oS supports a Maximum Transmission Unit (MTU) of 1532 bytes in Velocity.

L2oS makes the satellite link appear to be a transparent Ethernet connection between the satellite hub and the satellite router. Any Ethernet protocol can be executed over the L2oS link. In terms of data networking, this is "moving down the stack" from the traditional iDirect (and VSAT) model of IP-connectivity over the satellite link. The iDirect L2oS implementation is based on concepts from various IETF L2 VPN RFCs and Metro Ethernet Forum specifications, but optimized for satellite links and iDirect network topology. In Velocity, the initial implementation is restricted to point-to-point connections.

NOTE: Individual IP protocol stats (TCP, HTTP, UDP, ICMP, and IGMP bytes) are not supported on L2 SVNs. Values reported are inaccurate and must not be used.

The following L2oS features are supported in the iQ200 and IQ desktop only:

- Ethernet frame forwarding based on MAC learning
- Virtual Private Wire Service (point-to-point) - L2VPN topology ONLY
- ARP broadcast support for learning unknown MAC
- IP-based service only (due to reliance on ARP)
- VLAN ONLY support
- L2oS events and statistics reporting on L2oS
- TCP acceleration - IPv4 only

- ROHCv2 Header compression
- QoS

12.1.1 L2oS Benefits

In many VSAT applications, Layer 2 connectivity between the hub and remotes is preferred over the Layer 3 IP connections traditionally offered by iDirect. The benefits of using L2oS include:

- **Layer 3 Transparency**— Since an iDirect L2oS VSAT network behaves as a switched Ethernet network, the satellite network is invisible to higher-layer protocols. Therefore, any Layer 3 protocol, such as IPv6, OSPF, and BGP are transported transparently across the satellite network.
- **Simplified Operation**—In a traditional iDirect Layer 3 network, services such as Virtual Private Networks typically require allocation and configuration of a large number of IP addresses on the iDirect protocol processor and remote satellite routers. With Layer 3 transparency, this IP address allocation is no longer required. The only VLAN IP addresses required on an L2oS network are for the default VLAN used for iDirect network management and in Hybrid mode IP addresses for L3 Data VLANs only.
- **Efficient Physical Layer**—As in traditional iDirect IP networks, L2oS networks carry traffic over iDirect's highly-optimized DVB-S2 outbound and TDMA inbound carriers. All the advantages of features such as DVB-S2 ACM and TDMA 2D 16-State coding are preserved when using L2oS.
- **QoS Support**—L2oS supports all of the QoS features that exist for iDirect IP networks. In addition, for L2oS networks, Layer 2 classifiers, such as source and destination MAC address, have been added to the Service Level and Filter Rules in Pulse NMS. IPV6 source and destination IP address classifiers have also been added.

12.2 L2oS Reference Model

A basic L2oS reference model is shown below, with the associated terms following.

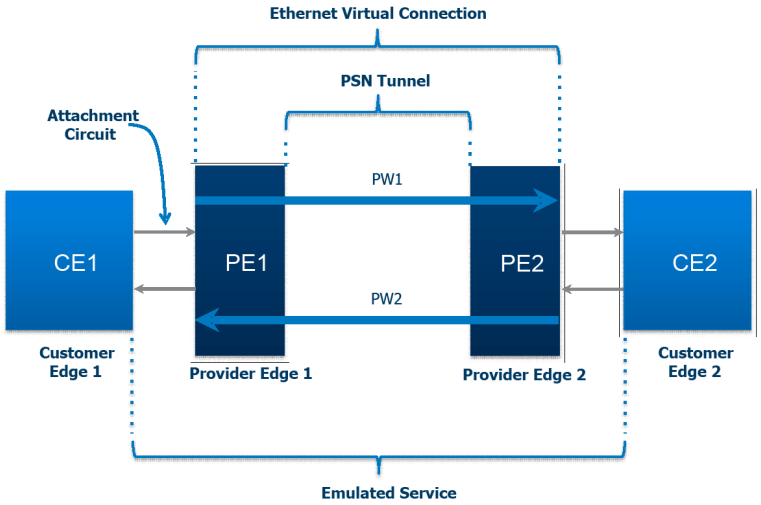


Figure 12-2. L2oS Reference Model

- **Customer Edge** (CE): A Customer Edge is a device such as a switch or router that provides an entry point into the customer network. In the case of the iDirect L2oS feature, a CE is directly connected to iDirect hub equipment or an iDirect satellite router.
- **Provider Edge** (PE): A Provider Edge is a device such as a switch or router that provides an entry point into the provider network. In the case of the iDirect L2oS feature, a PE is iDirect hub equipment or an iDirect satellite router.
- **Attachment Circuit**: An Attachment Circuit is a connection between CE and PE devices. In the case of the iDirect L2oS feature, an Attachment Circuit is an Ethernet connection between a CE and iDirect hub equipment or between a CE and an iDirect satellite router.
- **Ethernet Virtual Connection** (EVC): An Ethernet Virtual Connection is a logical connection between two PEs. In an iDirect satellite network, one end of the EVC is always the hub; the other end of the EVC is always one or more remote modems.

The L2oS reference model, as applied to iDirect Velocity, is shown in the following diagram.

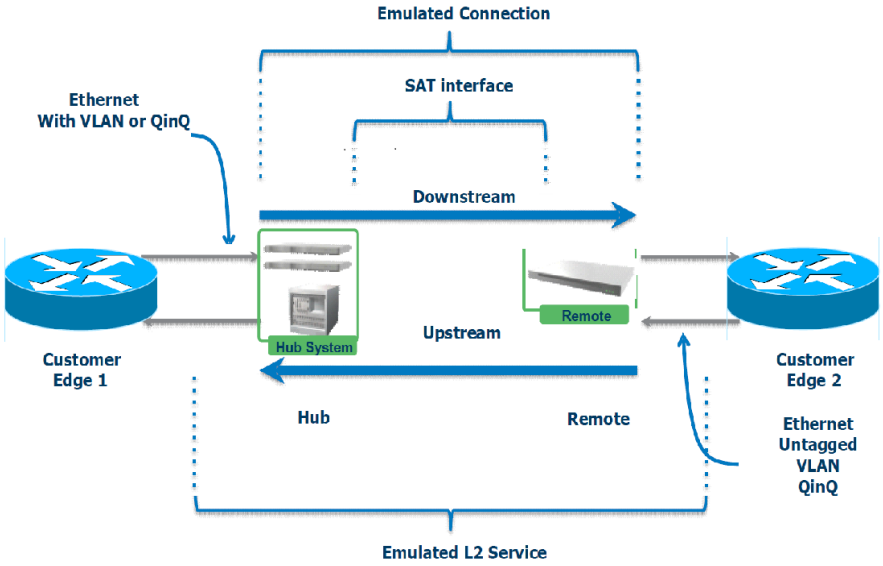


Figure 12-3. L2oS Reference Model Applied to iDirect

Traffic on an L2oS connection may be segregated into Satellite Virtual Networks (SVNs) by the use of VLAN in the Ethernet frames.

12.3 SVN and Service Delimiting Tags

An L2oS satellite network is partitioned into one or more Satellite Virtual Networks (SVNs). SVNs provide segregation of Layer 2 traffic into streams that behave as if user data is being transmitted over separate networks. Each remote in the satellite network belongs to one or more SVNs.

Service Delimiting Tags (SDTs) associate Ethernet frames with specific SVNs. The Velocity iDirect system supports IEEE 802.1Q VLAN tags type of SDT.

VLAN tags allow the Layer 2 emulation to provide traditional VLAN VPN functionality. When VLAN tags are used, only the outermost tag of an Ethernet frame determines the SVN to which the frame belongs. The SVN number is the VLAN ID of the first VLAN tag in the frame. Any additional inner VLAN tags are passed through transparently.

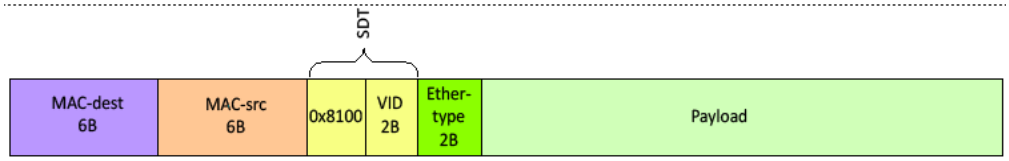


Figure 12-4. SDT Mode = VLAN

SDTs are always used on the hub side. It is the responsibility of the Gateway CE to add the appropriate SDT for downstream traffic; it is the responsibility of the iDirect hub to add the appropriate SDT for upstream traffic. The hub SDT Mode is a global setting for VLAN. The hub and its associated gateway CE should be configured in the same mode.

SDTs are optional on the remote side. When used, it is the responsibility of the CE attached to the remote modem to add the appropriate SDT for upstream traffic; it is the responsibility of the remote to add the appropriate SDT for downstream traffic.

SDTs can be omitted on the remote side by configuring a remote LAN port as an Access Port and associating the port with a single SVN. All Ethernet frames received on the port are transmitted on the configured SVN. The hub tags the frame based on the SVN ID and hub SDT Mode before sending the frame to the hub CE.

12.3 displays the Service Delineating Tag (SDT) support matrix.

Table 12-1. SDT Support Matrix

	Supported on Hub	Supported on iQ Desktop	Supported on iQ 200
Access (untagged)	No	No	No
VLAN	Yes	Yes	Yes

12.4 Layer 2 Services -SVN

SVNs allow the network operator to implement service provider, customer, and subscriber business models by extending the customer private network across the shared Velocity satellite network, allowing the customer remote devices to send and receive data as though they were directly connected to the customer private network.

An SVN is analogous to a terrestrial Virtual Private Network (VPN). Both provide connectivity across a common network infrastructure while segregating each private network's traffic and providing each with a separate, independent IP address space.

The Velocity L2oS implementation allows network operator to design their networks considering the typical Velocity network (terminal, satellite OTA segment, and SAS components/links) as a single Layer 2 broadcast domain. The Velocity system can be viewed as a switch that happens to have interfaces in geographically separated locations.

This allows a network designer to think of the Velocity segment of their end-to-end L2 VPN as shown in Figure 12-5.

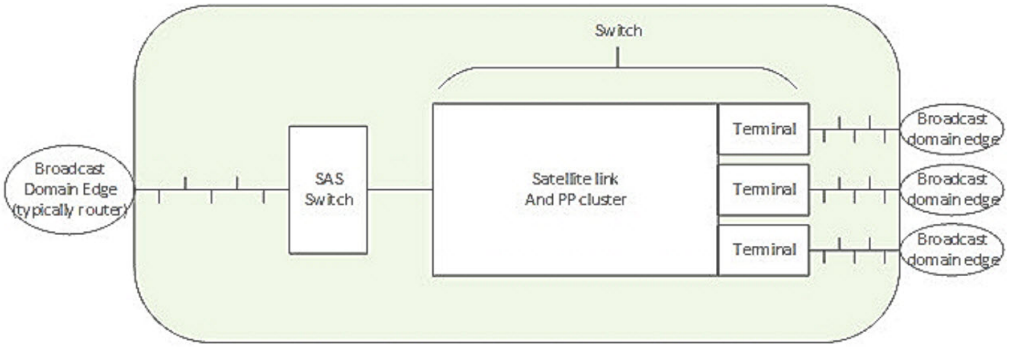


Figure 12-5. Velocity L2oS Broadcast domain

The edge of a broadcast domain is typically an IP router. The term is used here to also include a server, host or laptop located on the LAN.

The implementation of the section of the diagram labeled "switch" relies on L2oS Pseudo-Wires (PW) connecting the terminal and PP. The terminal and PP each implement MAC learning and MAC-based forwarding. See Persistent MAC FIB Tablefor more information.

The following subsections describe end-user services that can be used when is enabled.

- L2 SVN for IP VPN
- Layer 2 Network

12.4.1 L2 SVN for IP VPN

Since Velocity supports L2oS, a network operator who is offering an IPVPN service will be able to choose to implement that IPVPN using L2 SVNs or L3 SVNs.

> ***NOTE:*** In the Pulse NMS, an L3 SVN is called a DataVLan, and an L2 SVN is called DataVLan-L2.

L3 SVN depicts one possible scenario of how an end-user IP VPN could be implemented by a satellite network operator using an iDirect L3/IP functionality.

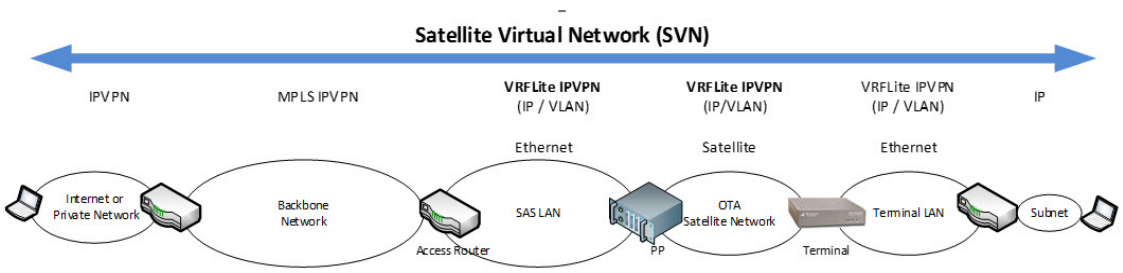


Figure 12-6. L3 SVN

The same SVN could be implemented using L2oS as seen in Figure 12-7.

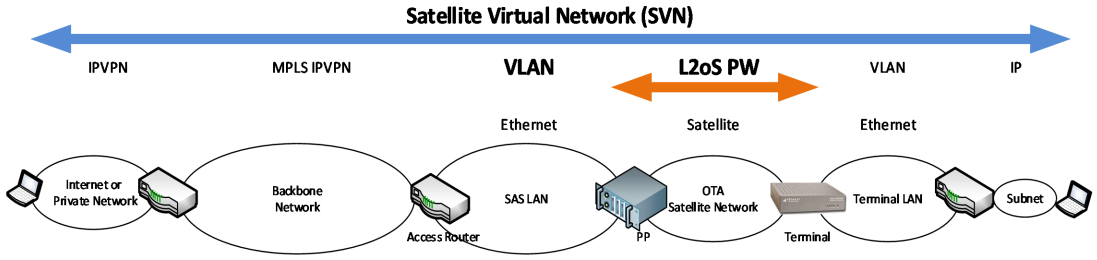


Figure 12-7. L2 SVN

Flooding frames are expensive over the satellite link. When an L2 SVN is used for IPVPN, the Velocity L2oS implementation will prevent flooding by relying on the flow of ARPs to learn MAC addresses and populate the terminal and PP FIBs.

12.4.2 Layer 2 Network

In the Velocity L2oS implementation, the PP does not participate in the L2 network; it simply terminates the L2oS PW instead of functioning as a VLAN switch. Thus from a purely networking perspective, the L2 network is as shown below.

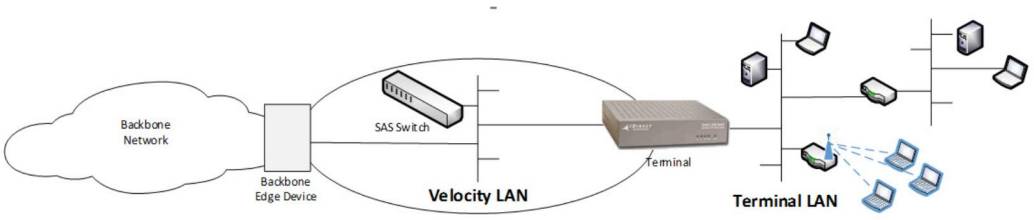


Figure 12-8. Velocity L2 Network

The type of device used for the Backbone Edge Device shown in Figure 12-8 depends on the technology used to construct the backbone network. The network may be MPLS/IP WAN, metro Ethernet or leased lines, and the device may be any one of multi-service router, Carrier Ethernet Switch, or Optical switch.

12.5 L2oS Service Mode

Velocity networks support the Virtual Private Wire Service L2oS service mode. When using Virtual Private Wire Service (VPWS), the satellite network provides logical point-to-point connections between the Customer Edge equipment at the hub and the Customer Edge equipment at the remote sites.

The remote modem and satellite hub are responsible for mapping user Ethernet frames to the appropriate VPWS connection, and for forwarding the frames to the correct destinations. (See Forwarding Rules on page 374.) A VPWS can contain a single VLAN or multiple VLANs.

Although the VPWS model provides multiple point-to-point Ethernet connections in the satellite network, these connections are not bridged or switched together at the hub. Switching is the responsibility of the external Customer Edge equipment or upstream router.

In VPWS mode, when the Hub receives an upstream Ethernet frame destined for a known MAC address behind another remote, it does not forward the frame on the downstream carrier. It does, however, forward the frame to the hub CE to allow the Layer 3 router to determine whether or not to route the frame back to the iDirect hub for downstream transmission.

	NOTE: A frame from a remote on one blade to a remote on another blade might not be delivered to the upstream router. This is because the switch can replicate upstream frames on the downstream carrier before a Layer 3 decision can be made.

12.6 Layer 2/Layer 3 Hybrid Mode

iDirect introduces the Layer 2/Layer 3 Hybrid Mode feature to allow both Layer 2 and Layer 3 traffic to traverse a single VSAT network at the same time, sharing the same NMS, PP stack, line cards, and satellite bandwidth.

When transported in the same network, Layer 2 and Layer 3 traffic is separated by an individual Satellite Virtual Networks (SVN), see SVN and Service Delimiting Tags on page 366. In other words, SVNs will be marked as either Layer 2 or Layer 3 during network configuration.

The most common use cases for this feature include:

- Introducing Layer 2 in Existing Layer 3 Network—In this use case, for a network operator currently running a Layer 3 iDirect network, a new Layer 2 connection to new sites can be introduced, leveraging its protocol transparency and other benefits, without having to set up a brand new network.
- Running Persistent Multicast along with Layer 2 — In this use case, a network operator can take advantage of Layer 2 network architecture while providing multicast service such as video distribution using the tried-and-tested persistent multicast feature in Layer 3.



In Velocity Release 2.0, Layer 2/Layer 3 Hybrid Mode is only supported on the PP. Traffic on ADMIN SVN will operate in L3 Mode.

12.7 TCP Acceleration

TCP acceleration can be enabled for IPv4 TCP packets. TCP acceleration only occurs if the protocol stack in the Ethernet frame is Ethernet, VLAN, IPv4, TCP. Tunneled IPv4 TCP packets or IPv4 TCP packets nested in the payload are not accelerated. IPv6 TCP packets are never accelerated.

12.8 Advanced Header Compression

Robust Header Compression (ROHC) is a standardized method to compress the IP, UDP, UDP-Lite, RTP, and TCP headers of Internet packets. Currently, ROHCv2 compression only supports Layer 2 traffic.

The iDirect ROHCv2 implementation extended the standard compression profiles to include the compression of Ethernet headers with stacked VLAN tags. Enabling the compression of Ethernet headers will almost completely remove the additional header overhead associated with L2oS in comparison to Layer 3. Compression of additional headers will allow L2oS to surpass the overhead efficiency of the traditional Layer 3 networks. The overall efficiency gain depends on the size of the Ethernet frames. With ROHCv2, the smaller the average frame size, the better the efficiency gain.

The ROHC v2 header compression can be enabled by specific protocol profiles. These selections are done at the remote level. The following profiles are supported in Velocity:

- IP

- IP/ESP
- IP/UDP
- IP/UDP/RTP
- IP/UDP/GTP
- IP/GRE
- IP-in-IP (IPv4-in-IPv4, IPv4-in-IPv6)
- Aggressive Compression

For IP/GRE and IP-in-IP, the headers inside the tunnel wrapper will also be compressed according to the profile(s) chosen. There is also an Aggressive setting and when this setting is enabled, the padding bytes in an Ethernet frame will be removed before transmission.

The following tables shows the comparison between uncompressed headers and ROHC v2-compressed headers for various protocol profiles based on IPv4. The actual compression efficiency will be slightly lower from the table below because of the periodic Initialization and Reset (IR) frames that are used to setup the initial compression context and re-sync the information. The IR frames are necessarily less efficient.

The following tables shows the IPv4 header compression.

Table 12-2. IPv4 Header Compression

Effective Header (Bytes)	Compression Profile	ROHCv2 Effective Header (Bytes)
58	RTP+UDP+IP+ETH	6
54	GTP+UDP+IP+ETH	4
46	UDP+IP+ETH	4
46	ESP+IP+ETH	2
38	IP+ETH	2
18	Ethernet with VLAN tag	1



NOTE: On IPv4, when UDP checksum is not enabled, the ROHCv2 effective header sizes for all UDP related compression will decrease by two bytes. The results in 12.8 and 12.8 are obtained with UDP checksum enabled.

Table 12-3. IPv4 Header Compression with IP Extensions

Protocol (IP Extensions)	Effective Header (Bytes)	Compression Profile	ROHC v2 Effective Header (Bytes)
GRE with UDP	70	IP+ETH	6
GRE with UDP	70	IP+ETH+UDP	8
IPv6-in-IPv4 with UDP	86	IP+ETH	6
IPv6-in-IPv4 with UDP	86	IP+ETH+UDP	8

12.9 MAC Address Learning

A generic Layer 2 switch builds the contents of its MAC FIB by learning MACs – that is, inspecting frames received and storing the source MAC address and the interface on which it was received in the MAC FIB. When the switch needs to forward a frame; it can ‘look up’ the MAC destination address of the frame in the MAC FIB to determine how to forward it.

The PP and terminal operate the same way; they learn MAC addresses from the frames they receive and use the resulting MAC FIB to determine where to send the frames. The difference is that the destination for the frames is a PW that carries the frame over the satellite link. It also forwards the frame over-the-air, only when the frame is addressed to a known MAC address on a known SVN.

In both cases, a generic L2 switch and the components of the Velocity L2oS solution, and the learned MAC addresses expire if the remote devices with the MAC addresses have not transmitted for a specific duration. This duration is called the MAC Aging Timeout. For the Velocity PP, the default aging timeout is 6hrs, which is based on the industry standards. The Velocity terminal default timeout is 5mins.

12.9.1 Persistent MAC FIB Table

When a terminal goes out-of-network, the MAC address is kept until 'age out' timer expires - as if the terminal is still in network.

If the terminal comes back into the network, before MAC address ‘age out,’ the (new) PP will recover the MAC addresses while preserving the current address age.

12.10 L2oS Failover and Resiliency

Similar to existing failover systems in Velocity, L2oS on Velocity supports fail over mechanisms that maintain process and data resiliency:

- If an L2oS PP process fails or a blade crashes, these processes are automatically restarted.
- Configured L2 SVNs are preserved when PP processes are restarted or when a PP blade fails.
- FIB records (MAC tables) are preserved when PP processes are restarted or when a PP blade fails.

L2oS failover can support recovery of 160 MAC addresses for up to 250 remotes (40,000 MAC addresses total).

12.11 Forwarding Rules

This section contains the rules for forwarding Ethernet frames received by the iDirect hub equipment and remote modems. iDirect management traffic and Ethernet control frames are not discussed here. Refer to Figure 12-9.

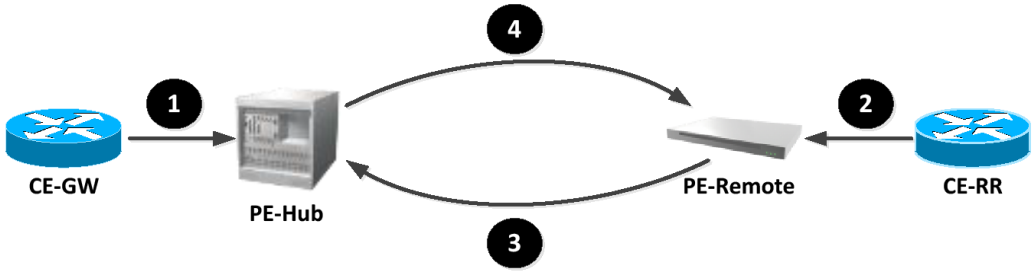


Figure 12-9. L2oS Forwarding Rules

1. When Ethernet frames are received by the hub from the Gateway CE:
   - The hub discards frames that do not map to any known SVN.
   - The hub sends over-the-air all unicast frames that map to a known SVN and destination MAC address.
   - The hub broadcasts over-the-air all multicast and broadcast frames that map to a known SVN.
2. When Ethernet frames are received by the remote from the Remote CE:
   - The remote discards all frames with a local destination MAC Address.
   - If the remote is configured for VLAN tagging:
     - The remote sends over-the-air all frames that map to a configured SVN.
3. When Ethernet frames are received over-the-air by the hub from a remote:
   - If the service mode of the SVN is VPWS:
     - The hub directs the frame back to its local LAN and lets the Gateway CE to make the decision of whether or not to route the frame back for downstream transmission (Required coordination with customer switch configuration—for example: protected ports, etc.)
4. When Ethernet frames are received over-the-air by a remote from the hub:
   - The remote discards all frames that do not match any configured SVN.
   - The remote discards all frames with a local source MAC Address.
   - The remote forwards all remaining frames to the local LAN.

12.11.1 Flooding Avoidance

The term 'Flooding' describes how an Ethernet switch handles frames with unknown destination MAC addresses. In VPWS mode, [[[Undefined variable SW_ Variables.ProductName]]] applies a solution to flooding within a star-topology VSAT network where frames with unknown destination MAC addresses received by the:

- PP:
  - From the attachment circuit (AC) are discarded
  - From a terminal are forwarded to the hub AC
- Terminal:
  - From the Terminal AC are always forwarded to the hub.
  - From the hub are forwarded to the terminal AC

The goal is to protect satellite bandwidth, unlike a "regular" switch and to not flood packets when an entry is not in the FIB. The FIB is updated whenever data arrives on a PW from one of the terminals.

- An FIB entry is added (or refreshed) to show an ***<SVN, MAC-src>*** is accessible on a given PW.
- An FIB update always occurs using the MAC source address of the L2 frame.

The FIB is consulted whenever data arrives on the Hub AC. When an ***f <SVN, MAC-dest>*** is found in the FIB, then the target PW is known, and the target ***<HDLC, PW-id>*** is ready and available.

This deployment assumes one FIB per SVN containing MAC-addresses, however, design implementation does allow a deployment case defined by one large FIB containing ***<SVN, MAC-address>*** pairs.

12.11.2 Ethernet Control Protocols

In general, Ethernet control protocols (for example, Spanning Tree and variants) are not forwarded over satellite link. Ethernet frames containing this type of data will be filtered and discarded by the Forwarder.

12.12 MEF Carrier Ethernet 2.0 Services

Although the Velocity is not an Metro Ethernet Forum (MEF) certified solution, iDirect has attempted to align our terminology and use case definitions with MEF.

MEF 2.0 Carrier Ethernet Services are defined in terms of the behavior seen by the user at the User Network Interface (UNI). MEF services do not specify implementation details within the service provider network, and so a service provider is free to implement these services in a way that best fits the service provider's infrastructure. [MEF-6.3] defines various service types.

Table 12-4. MEF Service Summary

Service Type	Port Based (All-to-1 Bundling)	VLAN-Based (EVC Identified by VLAN ID)
E-Line (Point-to-Point EVC)	Ethernet Private Line (EPL)	Ethernet Virtual Private Line (EPL)
E-LAN	Ethernet Private LAN (EP-LAN)	Ethernet Virtual Private LAN (EVP_LAN)
E-Tree (Rooted-Multipoint EVC)	Ethernet Private Tree (EP-Tree)	Ethernet Virtual Private Tree (EVP-Tree)

- **E-Line**—an E-line service connects two endpoints, which can communicate as if connected by a cable;
- **E-Tree**—an E-tree service connects Leaf endpoints with one or more Root endpoints. Leaf endpoints can only communicate with the Root endpoints. Root endpoints can communicate with Leaf endpoints, and also other Root endpoints. The network learns customer MAC addresses to switch the traffic appropriately, using data plane learning;
- **E-LAN**—an E-LAN service consists of a number of endpoints. Each endpoint can communicate with all the other endpoints. The network learns customer MAC addresses to switch the traffic appropriately, using data plane learning.

The implementation of the service provider network is not specified. Thus, while the service types describe specify data plane learning for MAC addresses, it is perfectly acceptable for a service provider to use an SDN type infrastructure, and move the MAC address learning function into the internal control plane of service provider network. This is transparent from a customer perspective-there is no control plane messaging for MAC addresses across the UNI.

All iDirect use cases are an implementation of MEF like **E-Line** (EPL or EVPL) service type as defined for connecting exactly two User Network Interfaces (UNI) where those two UNIs can communicate only with one another. E-Lines are defined in MEF 6.2 and MEF 10.3.

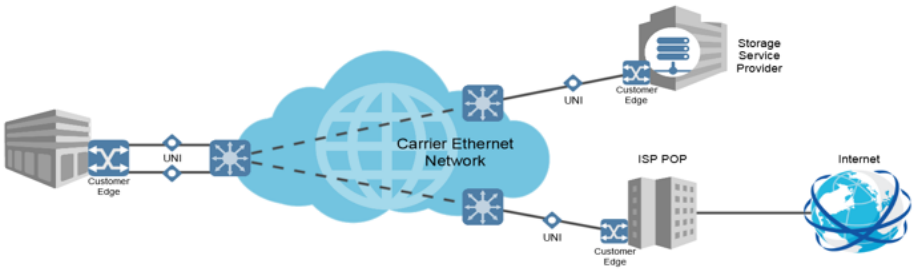


Figure 12-10. MEF E-Line Overview

Any Ethernet Service that is based on a Point-to-Point Ethernet Virtual Connection (EVC) is designated as an Ethernet Line (E-Line) Service Type. The E-Line Service Type is the basis for a broad range of Point-to-Point Services. In its simplest form, an E-Line Service Type can

provide symmetrical bandwidth for data sent in either direction with no performance assurances, for example , best effort Service between two 10Mbps UNIs.

In more sophisticated forms, an E-Line Service Type can be between two UNIs with different line rates and can be defined with performance objectives such as delay, inter-frame delay variation, loss, and availability for a given Class of Service Name (CoS Name).

Service Multiplexing might occur at one or both UNIs in the EVC. For example, more than one Point to-Point EVC might be offered on the same physical port at one or both of the UNIs. One or more CoS Names might be associated with the Service.

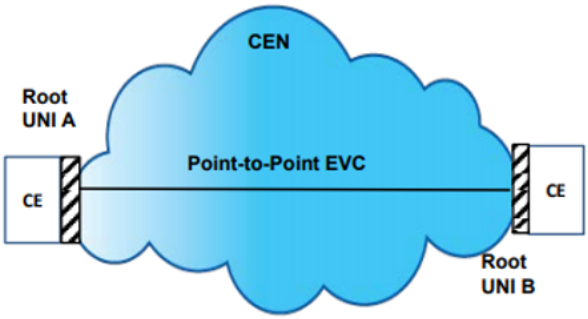


Figure 12-11. MEF Ethernet Virtual Connection

MEF UNI defines the interface between the User and the Network as the User Network Interface, or UNI. For some service types, multiple service endpoints can exist on the same UNI. The UNI is defined as an Ethernet port

MEF E-Line Services are by far the most commonly deployed by global carriers. Understanding this, L2oS enables common E-Line carrier services to operate over iDirect Satellite networks.

A fully equipped MEF certified Network Interface Device, Ethernet Access Device, or Layer 3 functions such as a router, can be deployed at each customer location (behind our remote) and communicate freely with its upstream peer to establish an uninhibited control, data and management plane over the iDirect network to provide each of the following use cases:

- Layer 2 Ethernet Virtual Private Line (EVPL) Service Offering
- Layer 3 Overlay Service Offering

For specific use cases, contact iDirect.

Appendix A: Acronyms & Abbreviations

This list is generic within this document and may contain entries not found in the main body of the document. Some entries may not be defined based on industry standards.

0…9

16APSK

Sixteen Amplitude and Phase Shift Keying

8PSK

Eight Phase Shift Keying

-A-

A-TDMA

Adaptive Time Division Multiple Access

ABS

Automatic Beam Switching

AC

Alternating Current

ACM

Adaptive Coding and Modulation

ACS

Antenna Control System

AES

Advanced Encryption Standard

API

Application Program Interface

APSK

Amplitude and Phase-Shift Keying

ASC

Acquisition Signaling Carrier

AZ

Azimuth

-B-

BB

BaseBand

BGP

Border Gateway Protocol

BIM

Below-Decks Interface Module

BIM

Broadband Interface Module

BIST

Built-In Self-Test

BITE

Built-In Test Equipment

BPN

BUC Part Number

BPSK

Binary Phase Shift Keying

BSN

BUC Serial Number

BSS

Broadcasting Satellite Service

BTP

Burst Time Plan

BUC

Block Up Converter

BW

Bandwidth

-C-

C/N

Carrier-to-Noise Ratio

C/N_0

Carrier-Power-to-Noise Spectral Density Ratio

CA

Certificate Authority

CBIT

Continuous Built In Test

CCM

Constant Coding and Modulation

CDC

Central Data Center

CDR

Critical Design Review

CIR

Committed Information Rate

CMS

Carrier Measurement System

COTM

Comms-On-The-Move

CPE

Customer Premise Equipment

CPU

Central Processing Unit

CRC

Cyclic Redundancy Check

CSA

Canadian Space Agency

CSIR

Communications Signal Interference Remover

-D-

DAC

Digital-to-Analog Converter

dB

deciBel

dBi

deciBel isotropic

dBm

deciBel milli-Watt

dBW

deciBel Watt

DC

Direct Current

DDR

Double Data Rate

DHCP

Dynamic Host Configuration Protocol

DID

Derived ID

DNS

Domain Name Service

DSCP

Differentiated Services Code Point

D-TDMA

Deterministic TDMA

DVB-S2

Digital Video Broadcasting over Satellite, Second Generation

-E-

EAP

External Access Portal

Eb/N0

Bit Energy to Noise Power Spectral Density Ratio

EIR

Enhanced Information Rate

EIRP

Effective Isotropic Radiated Power

EL

Elevation

EMC

ElectroMagnetic Compatibility

EMI

ElectroMagnetic Interference

ER

Embedded Router

ETSI

European Telecommunications Standards Institute

-F-

FAP

Fair Access Policy

FCC

Federal Communication Commission

FEC

Forward Error Correction

FID

Functional ID

FMECA

Failure Mode Effects Criticality Analysis

FOM

Figure of Merit

FPGA

Field Programmable Gate Array

FS

Functional Specification

-G-

G/T

Gain over Temperature

GBWM

Global Bandwidth Manager/Management

GDR

Gateway Diversity Relationship

GHz

GigaHertz

GPIO

General-Purpose Input/Output

GPS

Global Positioning System

GQoS

Global QoS (Quality of Service)

GSP

Group Service Plan

GUI

Graphical User Interface

GW

Gateway

-H-

HCP

High-Capacity Payload Beam

HLC

Hub Line Card

HNO

Host Network Operator

HTS

High Throughput Satellite

HTTP

Hypertext Transfer Protocol

HTTPS

Hypertext Transfer Protocol Secured

-I-

IBIT

Initiated Built In Test

ICD

Interface Control Document

iDX

iDirect Evolution Software Platform

IEC

International Electrotechnical Commission

IF

Intermediate-Frequency

IFL

Inter-Facility Link

IGC

Inroute Group Composition

IGMP
Internet Group Management Protocol
iGW
Intelligent Gateway
iHub
Intelligent Hub
iLC
Intelligent Line Card
IP
Ingress Protection
IP
Internet Protocol
IR
Infrared
I/O
Input/Output
IM
Interference Mitigation

-K-

kbps
kilobits per second
kHz
kilohertz
KRFU
Ku/Ka-band Radio Frequency Unit
ksps
kilo symbols per second

-L-

LAN
Local Area Network
LDPC
Low-Density Parity Check
LED
Light Emitting Diode

LHCP

Left Hand Circular Polarization

LNA

Low Noise Amplifier

LNB

Low Noise Block Converter

LO

Local Oscillator

LOS

Loss of Signal

LRU

Line-Replaceable Unit

-M-

Mbps

Megabits per second

Mcps

Megachips per second

MES

Mobile Earth Station

MF-TDMA

Multi-Frequency TDMA

M-GSP

Multicast Group Service Plan

MHz

Megahertz

MID

Manufacturer ID

MIL-STD

US Military Standard

MIR

Maximum Information Rate

MODCOD

MODulation and CODing

MPLS

Multi-Protocol Label Switching

Msps

Mega Symbols per Second

M-SSPP

Multicast SSPP

MTBF

Mean Time Between Failures

MTBUR

Mean Time Between Unscheduled Removals

-N-

NF

Noise Figure

NMS

Network Management System

NTP

Network Time Protocol

-O-

OAE

Outside Antenna Equipment

ODU

Outdoor Unit

OEM

Original Equipment Manufacturer

OMT

Orthogonal-Mode Transducer

OpenAMIP

Open Antenna-Modem Interface Protocol

OpenBMIP

Open BUC Management Interface Protocol

OSPF

Open Shortest Path First

OTA

Over-The-Air

OTP

One Time Programmable

-P-

PA

Power Amplifier

PAST

Person Activated Self-Test

PCB

Printed Circuit Board

PCBA

Printed Circuit Board Assembly

PDR

Preliminary Design Review

PLL

Phased Locked Loop

PP

Protocol Processor

PPS

Packets Per Second

PPS

Pulses Per Second

PSK

Phase Shift Keying

PSU

Power Supply Unit

PW

Pseudo-Wire

-Q-

QAM

Quadrature Amplitude Modulation

QEF

Quasi Error Free

QoS

Quality of Service

QPSK

Quadrature Phase Shift Keying

-R-

RA

Regulatory Area

RADIUS

Remote Authentication Dial-In User Service

RCM

Reference Clock Module

RDS

Rain Diversity Switchover

RF

Radio Frequency

RFS

Radio Frequency System

RGMII

Reduced Gigabit Media Independent Interface

RHCP

Right Hand Circular Polarization

ROHC

Robust Header Compression

RoHS

Restriction of Hazardous Substances

ROM

Read Only Memory

RSSI

Receive Signal Strength Indication

RTP

Real-Time Protocol

Rx or RX

Receiver

-S-

SA

Service Area

SAS

Satellite Access Station

SCPC

Single Channel Per Carrier (Satellite)

SGMII

Serial Gigabit Media Independent Interface

SIM

Subscriber Identity Module

SLA

Service Level Agreement

SNR

Signal-to-Noise Ratio

SP

Service Provider

SRS

Systems Requirement Specification

SRU

Shop Replaceable Unit

SSB

Single Side Band

SSPC

Subscriber Service Plan Component

SSPP

Subscriber Service Plan Profile

STP

Spanning Tree Protocol

SVN

Satellite Virtual Network

-T-

TAC

Technical Assistance Center

TBD

To Be Determined

TBS

To Be Supplied

TDM

Time Division Multiplexing

TDMA

Time Division Multiple Access

TFI

Terminal Functional ID

TMI

Terminal Manufacturer ID

TOS

Type Of Service

TPCFEC

Turbo Product Code FEC

TPN

Terminal Part Number

TSN

Terminal Serial Number

TSP

Terminal Service Plan

TTC

Terminal Transmit Control

Tx or TX

Transmit

-U-

UDP

User Datagram Protocol

UI

User Interface

UL

Underwriters Laboratories

UMD

Update Manager Daemon

UTC

Universal Time, Coordinated

-V-

VAC

Volts Alternating Current

VCM

Variable Coding and Modulation

VDC

Volts Direct Current

VNO

Virtual Network Operator

VSAT

Very Small Aperture Terminal

-W-

WAN

Wide Area Network

WFQ

Weighted Fair Queuing

WGS

Wideband Global SATCOM

Appendix B: Glossary

This list is of glossary terms, although it may contain generic VSAT introductory terms, is based on the Velocity and Pulse product platforms, and topics covered in this document.

2D 16-State

Type of Forward Error Correction coding available on iDirect inbound carriers in DVB-S2 networks. 2D 16-State coding can provide better link margins, improved IP throughput and faster acquisition than Turbo Product Coding.

ABS

See Automatic Beam Selection (ABS).

ACM

See Adaptive Coding and Modulation (ACM).

Access Class Value

On the channel side, a bit mask value that is logically combined, using the AND operation, with the terminal access class value when the terminal attempts to join the channel. If the logical result is non-zero, the terminal is allowed into the sub-channel — otherwise it is not.

ACM Gain

The ACM Gain represents the increase in performance achieved on a DVB-S2 outbound carrier when the MODCOD used to transmit data is higher than the minimum MODCOD configured for the carrier.

Acquisition Signaling Carrier (ASC)

Used to broadcast the current satellite beam configuration information to the terminals within a network.

Adaptive Coding and Modulation (ACM)

Adaptive Coding and Modulation. A method of applying coding to a data stream in DVB-S2 networks in which every BBFrame can be transmitted on a different MODCOD.

Action Plan

A list of update commands to be applied by the update manager, against a configuration item.

Active Configuration

The element configuration, resident on the network element.

Activity Log

A record of date/time and details of each activity, whether performed by a human or machine user.

Acquisition

The process whereby the satellite router synchronizes its bursts with the upstream TDMA frame timing and joins an iDirect network.

Adaptive Coding and Modulation Gain

The increase in performance achieved on a Digital Video Broadcasting - Satellite - Second Generation (DVB-S2) outbound carrier when the Modulation and Coding (MODCOD) used to transmit data is higher than the minimum Modulation and Coding (MODCOD) configured for the carrier.

Admin VLAN

The Velocity Admin SVN transports internal administrative traffic between Velocity equipment, wherever it is located: within a SAS, between SASs, between the SAS and the NOC, and between the NOC or SAS and satellite routers.

Allocation Fairness Relative to CIR

1) A QoS configuration that when enabled causes satellite bandwidth to be allocated in proportion to the configured Committed Information Rate (CIR) of the Group QoS node or virtual remote. **2)** When not enabled, bandwidth is allocated equally to competing nodes until available bandwidth is exhausted.

Allocation Fairness Relative to MODCOD

Applies only to DVB-S2 outbound carriers using Adaptive Coding and Modulation (ACM). A QoS configuration option that hen enabled, causes satellite bandwidth allocation to be based on information rate rather than raw satellite bandwidth. This method favors remotes at lower MODCODs, since their satellite bandwidth allocations must increase to achieve the same information rate as remotes at higher MODCODs.

Alternate Downstream Carrier

An iDirect feature that allows definition of downstream carrier alternates to be associated with an iDirect network in order to facilitate moving the network to a new downstream carrier.

Antenna Control Unit

An intelligent control device that both monitors and controls the positioning of a VSAT antenna. The device connects to the satellite router over OpenAMIP; provides geolocation information to the satellite router, points the antenna at the satellite, and performs related functions such as providing status and, in some terminals, muting the transmitter when off-pointed or blocked.

Application

The definition of a specific type of service, such as Voice over IP or TCP. An Application is created from an Application Profile. An instance of an Application running on a remote is called a Virtual Remote.

Automatic Beam Selection (ABS)

A process by which roaming remotes select which network to join and automatically lock on to the associated outbound carrier. Also known as Automatic Beam Switching.

Automatic Rain Diversity Swithover

The Velocity Automatic Rain Diversity Feature minimizes the impact of weather events on over the air performance by deploying a second (diversity) site site within the beam footprint of the feeder link. When Automatic Rain Diversity is enabled, and rain fade is detected on the primary site, Velocity automatically switches over to the secondary/backup site. Two (2) rain diversity configurations are supported: SAS Automatic Rain Diversity and RF Gateway Automatic Rain Diversity.

Available Tokens

The number of tokens still open for use by a user group or other associated Elements.

Backup Site

The backup (or secondary) operational Site of any network Site pair. **1)** May assume operations IF failure occurs at the primary Site. **2)** May operate as primary OR backup (secondary) Site.

Beam

The physical footprint from a satellite antenna onto the ground.

BGP

See Border Gateway Protocol (BGP).

BGP Peers

Two or more routers that maintain a TCP connection, through BGP, for the purpose of exchanging BGP route table information.

BGP Peer Group

A group composed of member BGP peers that share common update policies, in order to simplify routing configuration and management.

BGP IP Prefix List

A filter list, which may be applied to a specific route map.

Binary Large Object File (BLOB)

A collection of binary data files stored as a single entity.

Blade

A short name for the Protocol Processor server machine.

Block Up Converter

A device used for uplink satellite transmission and amplification, converting a band from a lower frequency to a higher frequency.

Board Support Package

Also known as Cumulative Update Package. A support package downloaded to remotes before loading remote image files.

Border Gateway Protocol (BGP)

A standardized exterior gateway protocol designed to exchange routing and reachability information among autonomous systems (AS) on the Internet. BGP is often classified as a path vector protocol but is sometimes classed as a distance-vector routing protocol. BGP makes routing decisions based on paths, network policies, or rule-sets configured by a network administrator and is involved in making core routing decisions.

BTP

See Burst Time Plan (BTP).

Burst Time Plan (BTP)

Slot allocation message sent to remote modems to indicate when each remote can transmit on the TDMA upstream carriers.

CA

See Certificate Authority (CA).

CA Foundry

The Certificate Authority (CA) utility that issues the X.509 public key certificates that allow “hosts” to join a secured network.

Carrier

A carrier is a single modulated RF signal or waveform carrying encoded data in one direction. It is referred to as an inbound or outbound carrier with respect to the antenna sending the signal (outbound) or receiving it (inbound).

Central Data Center

A site supporting a single satellite, through one or more RF Gateways, that contains only packet processing components.

Certificate

A certificate, created by the Certificate Authority (CA) Foundry Service, is required by all in Network elements in the Public Key Infrastructure (PKI) Data File system, in order to communicate with one another.

Certificate Authority (CA)

An authority in a network that issues and manages security credentials and public keys for message encryption.

Certificate Authority Foundry Service

The Certificate Authority (CA) utility that issues the X.509 public key certificates that allow “hosts” to join a TRANSEC network.

Channel

A channel is a fixed region of bandwidth on a satellite feeder link that is dynamically mapped onto a beam.

CIR

See Committed Information Rate (CIR).

Classification Rules

Rules and actions that determine how packets are filtered and prioritized.

Committed Information Rate (CIR)

A specified amount of bandwidth that is allocated to a node before additional (non-CIR) bandwidth is allocated to that node for traffic with the same priority.

Comms-on-the-MOVE (COTM)

Communications On-the-Move — an iDirect mobile remote feature.

COTM

See Comms-on-the-MOVE (COTM).

Customer SVN

A Global Satellite Virtual Network (SVN) that connects customer equipment, located in the terminal network, behind a terminal, through the SAS and across the DCN to a customer network. Also called Data SVN.

Derived ID (DID)

The unique identifier of an iDirect remote satellite router derived from the model type and serial number.

Deterministic TDMA (DTDMA)

A technique used to prevent collisions of remotes transmitting simultaneously in which synchronized burst time plan provides the network timing.

Dedicated Acquisition Signaling Carrier (ASC)

A DVB-S2 signaling carrier in which the hub broadcasts current satellite network configuration information, to terminals, on a single uplink and a single downlink.

Dedicated Acquisition Signaling Carrier (ASC) with fan-out

A single DVB-S2 signaling carrier, in a Velocity network, by which the hub broadcasts the current satellite configuration, on a single up-link that is replicated on multiple downlinks in different beams.

DID

See Derived ID (DID).

Digital Video Broadcasting - Satellite - Second Generation (DVB-S2)

A set of open standards for satellite digital broadcasting. DVB-S2 is an extension to the widely-used DVB-S standard and was introduced in March 2005.

Downstream Carrier

Synonymous with Outbound Carrier. The satellite carrier transmitted from the hub to the remote.

DTDMA

See Deterministic TDMA (DTDMA)

DVB-S2

A set of open standards for satellite digital broadcasting. DVB-S2 is an extension to the widely-used DVB-S standard and was introduced in March 2005.

Dynamic Configuration

The terminal configuration information that is passed over the RF interface from the SAS to the remote, using an iDirect over-the-air dynamic configuration exchange protocol.

Dynamic Features and Options Exchange (DFOE)

A protocol used between the PP and the Terminal to allow some remote-side configuration changes to be dynamically applied.

Dynamic Multicast Stream

When this Velocity multicast option is enabled, the terminal software accepts dynamic configuration of the static streams and configures IGMPv3 or MLD on a per terminal, per SVN basis. Dynamic multicast is the default option on all SVNs, but can be disabled.

EAP

See External Access Portal (EAP).

EIR

See Enhanced Information Rate (EIR).

Element

A physical or logical network object, which must also be a unique object known to the iDirect Network data path.

Element Domain

A hierarchical object model in which configurable network Elements are grouped into logically related collections.

Engineering Debug Console

A tool that provides direct access to a selected Element, using a shell window for debugging purposes.

Enhanced Information Rate (EIR)

In iDirect's Group QoS (GQoS), the EIR option allows you to configure the system to maintain CIR or MIR during rain fade for the physical remote (Remote-Based Group QoS) or critical applications (Application-Based Group QoS). EIR only applies to networks that use DVB-S2 with Adaptive Coding and Modulation (ACM).

External Access Portal (EAP)

The EAP is an external instantiation of the Pulse NMS that provides limited access to NMS configuration, statistics, and reporting functionality for any users external to the Satellite Network Provider. The EAP also supports configuration capabilities that allow an EAP user with appropriate permissions to create, modify, and delete network elements.

Fast Fade Margin

For iDirect DVB-S2 outbound carriers, the additional margin added to the SNR thresholds measured at hardware qualification to arrive at the operational threshold during a "fast fade" condition.

Feature License

A license purchased from iDirect allowing NMS operators to configure a license-controlled feature.

Filter Profile

A traffic profile configurable in the NMS and assigned to remotes to filter out unwanted packets.

Free Slots

Slots left in the dynamic sub-frame after all stream, guaranteed (CIR) and preemptive bandwidth requests are satisfied. Free slots are allocated to all VSATs (up or down), except the master, in a round-robin fashion.

Frequency Hopping

The ability of iDirect remotes to switch between TDMA carriers within an inroute group when transmitting to the hub.

Full-Trigger CIR

Committed Information Rate (CIR) (CIR) that is always fully-allocated even if demand is less than the configured CIR.

Gateway Diversity Relationship

The Rain Diversity Feature requires a Gateway Diversity Relationship (GDR) object for each CDC pair. The object pairs a primary RF Gateway with a backup RF Gateway to enable automatic or manual Rain Diversity Switchover. Note that a Carrier Measurement System (CMS) is required at each RF Gateway for this feature.

Geographic Region

A geographic region that is defined as a combination of satellites/beams, and/or service areas that constitutes an area of coverage.

Global SVN

An SVN type that is at the top of the Velocity SVN object structure and represents a VPN that extends across the Velocity core network (DCN) to another Site.

GQoS

See Group QoS (GQoS).

Group QoS (GQoS)

iDirect's Quality of Service (QoS) solution based on a hierarchical tree structure by which satellite bandwidth allocation flows down the tree from the root node to the leaf nodes. GQoS allows advanced network operators a high degree of flexibility in creating subnetworks and groups of remotes with various levels of service tailored to the characteristics of the user applications being supported.

Group Service Plan (GSP)

A plan of service that represents a pool of satellite bandwidth capacity on the Velocity Network. A GSP can be defined by a Satellite Operator/Service provider.

Header Compression

A Velocity network optimization option that enables the compression of the IP header of each data packet prior to transmission. Two types of header compression are supported in Velocity — RTP header compression (used for RTP packets) and TCP header compression (used for TCP packets).

HLC

See Hub Line Card (HLC).

Hub Line Card (HLC)

An iDirect modem deployed at the hub to transmit and/or receive outroutes and inroutes.

Hub Network Operator (HNO)

An NMS operator with privilege to act as an administrator to Service Providers. An HNO can configure Service Provider users and networks and set Service Provider permissions such as visibility and read/write access.

iDirect Tunnel SVN

A Global SVN that connects Protocol Processors and Line Cards in a Velocity network.

IF Domain

An IF domain represents a specific group of line cards and IF frequency ranges at which these line cards operate.

Inbound Carrier

Synonymous with Upstream Carrier. The carrier transmitted from the remote satellite router to the hub.

Indoor Unit (IDU)

The satellite modem and indoor devices (in contrast to Outdoor Unit or ODU).

Information Rate

The rate of transmission of user data over an upstream or downstream carrier including IP headers and iDirect overhead.

Inroute

A TDMA Upstream Carrier.

Inroute Group

An inroute group (IG) represents a collection of upstream carriers. Characteristic of an inroute group, is that member carriers are assigned consistent transmission attributes – for example MODCOD and FEC rate.

Inroute Group Composition (IGC)

A set of defined carriers assigned to an inroute group.

LDCP

Low Density Parity Coding. The error correction coding scheme used in DVB-S2 networks.

LEGS

Lightweight Encapsulation for Generic Streams. An iDirect proprietary protocol for encapsulating data in DVB-S2 networks which maximizes the efficiency of data packing into BBFrames.

Low-Noise Block (LNB) Converter

A receiving device that is integrated in the VSAT terminal to convert the signal gathered by the antenna feed circuit.

Maximum Information Rate (MIR)

A QoS configuration specification of the maximum bandwidth that will be allocated to a node, regardless of demand generated by the node. A node with MIR set will never be granted more bandwidth than the configured MIR bit rate.

Manual Rain Diversity Switchover

As in the automatic rain diversity swithover, a second (diversity) site site within the beam footprint of the feeder link is required. When rain fade is detected on the primary site, an operator can manually switchover to the secondary/backup site. Two (2) rain diversity configurations are supported: SAS Automatic Rain Diversity and RF Gateway Automatic Rain Diversity.

Maximum MODCOD

The highest Modulation and Coding value used in DVB-S2 networks.

MIDAS Controller Board

A type of controller board used on newer iDirect chassis.

Minimum Information Rate (MIN)

A QoS configuration specification of the minimum bandwidth that will be allocated to a node, regardless of demand generated by the node.

MIR

See Maximum Information Rate (MIR).

MODCOD

The combinations of Modulation Types and Error Coding schemes supported on a satellite channel. The higher the modulation the greater the number of bits per symbol (or bits per Hz).

Multicast Group Service Plan (M-GSP)

A service plan that provides a data transmission service in which the data stream is simultaneously transmitted to multiple recipients, rather than sending separately to each recipient.

Multicast Subscriber Service Plan Profile (M-SSPP)

Like the unicast SSPP, the M-SSPP is configured in Pulse by authorized VNOs and used in creating a Terminal Service Plan - particularly, a multicast component of a TSP, to which individual terminals or a group of terminals may subscribe.

When an M-SSPP is selected as part of a TSP, an instance of the M-SSPP is created in the NMS. The TSP configuration may consist of multiple components, where each component is an instance of a Unicast SSPP or Multicast SSPP.

Network Domain

In the Pulse hierarchy of network elements, the network domain that is immediately above the transport element domain — it consists of logical infrastructure elements such as iNet Profiles, Inroute Group Profiles, Upstream Carriers, Downstream Carriers, and Inroute Composition Groups.

Network Address/Port Translation (NAT/PAT or NAPT)

A process of modifying IP addresses as well as TCP/UDP port numbers so that nodes residing on a private network can share a public IP address for communication with the outside world.

Network Management System (NMS)

Software used by network operators to configure, control and manage networks.

Network Operations Center (NOC)

One or more locations from which network monitoring and control, or network management, is exercised over a computer, telecommunication or satellite network.

NMS

See Network Management System (NMS).

NMS Server Cluster

A group of servers acting as a single system to provide the resources required by the Pulse NMS for managing network configuration and control, software version management and updates, as well as for monitoring and reporting on network events and alarms. The NMS server cluster is implemented at both the primary SAS, and optional backup SAS sites where applicable, and at the NOC site.

Nominal MODCOD

In the iDirect DVB-S2 implementation, the Reference Operating Point (ROP) for a remote receiving a downstream DVB-S2 carrier with ACM.

Options File

An iDirect configuration file generated by Pulse. Options files are used to download configuration settings to protocol processors, hub line cards and remote satellite routers.

Outbound Carrier

See Downstream Carrier.

Outroute

See Downstream Carrier.

Package

A package, in the NMS, represents various software files and images required to upgrade a Protocol Processor Server, Line Card, or Satellite Terminal to a particular software version.

Payload Compression

A mechanism whereby a datagram is compressed with the intent of reducing the size of data transmitted over congested or slow network connections, thereby increasing the speed of such networks without losing data.

Pending Configuration

The last saved OR current configuration of NMS information, that is not yet applied to an Element.

Physical Domain

In the Pulse hierarchy of network elements, the physical domain is at the lowest level of the hierarchy, — it consists of hardware infrastructure network elements such as the NOC and SAS sites, VLANs, hub chassis and line cards, NMS and Protocol Processor (PP) servers, and other ancillary servers.

Primary Site

The main operational site of any one of the Velocity network site pairs. Each site and its optional backup/diversity site can operate as the primary or backup (secondary) site.

Protocol Processor

in a Velocity network, the SAS servers (primary/backup), which are responsible for processing functions, traffic routing, load balancing, automatic fail-over and automatic redistribution of load across the remaining PP servers. A PP cluster also resides at the NOC for handling global bandwidth management operations.

Public Key Infrastructure (PKI) Data File

The chain of trust required by network Elements to facilitate inter-communication.

Quality of Service (QoS)

System traffic flow, classification, filter and priority: The classification and prioritization of IP traffic in order to optimize the delivery of packets as it flows through the network. (Attributes of a connection that affect QoS include throughput, latency, jitter and packet loss and others).

RCM

The Reference Clock Module in an iDirect line card chassis.

Real-Time Transport Protocol (RTP)

Real-Time Transport Protocol. A protocol designed to provide end-to-end network transport functions for applications that transmit real-time data, such as audio, video, or simulation data, over unicast or multicast network services. RTP provides such services as payload type identification, sequence numbering, time-stamping, and delivery monitoring to real-time applications.

Remote Locking

An iDirect feature that allows individual remotes to be locked to a particular network. Once a remote is locked with a key, it only functions in a network with the same key.

Revoke Certificate

Retraction of a security certificate that was previously assigned to a network Element.

RF Gateway

A site supporting a single feederlink that contains only RF components.

Roaming Remote

Mobile remotes that use the global NMS feature to "roam" from network to network around the globe. Roaming remotes are not constrained to a single location or limited to any geographic region.

Roll Vault

A Pulse operation used to increase security of the network by requesting that the vault table cease using the existing keys and generate a new set of keys for data encryption.

Route map

A map that defines the routing policies that are considered before a router examines its forwarding table.

Satellite Access Station (SAS)

A site supporting a single satellite that contains both packet processing and RF components. The Velocity network terrestrial segment, also referred to as a hub, provides communications between the Service Provider Data communications Network (DCN) and the remote Satellite Terminals. A SAS site contains the SAS LAN, Protocol Processor (PP) servers, NMS servers, Chassis and Line Cards, the Radio Frequency Subsystem, and other ancillary equipment.

Satellite Operator

A satellite owner/operator who offers satellite bandwidth in MHz or Mbps (Megabits per second) to other service providers. In general, the satellite operator sells bandwidth to service providers such as virtual network operators (VNO).

Satellite Operator/Service Provider

A satellite operator who also provides services directly to subscribers. The satellite operator may simultaneously provide subscriber services and bandwidth to other service providers.

Satellite Terminal Domain

The Element Domain that provides IP connectivity between each remote LAN and the Teleport equipment.

Satellite terminal Identification Message

A terminal identification feature whereby, on each attempt to acquire the network, a terminal is screened, by the Satellite Access Station (SAS), as to whether it should be allowed to acquire the network.

Satellite Virtual Network (SVN)

In a Velocity network, an IP VPN that contains a satellite network segment.

Satellite Terminal Identification Message

A terminal identification feature whereby, on each attempt to acquire the network, a terminal is screened, by the SAS, as to whether it should be allowed to acquire the network.

Server (Blade) SVN

Each Server SVN is generated automatically as a result of creating a Site SVN and its parent Global SVN.

Service Area Group

In Velocity, an SA Group maps to a single active map in the NMS, and represents an identifier pool from which service areas (SAs) can be designated when defining geoscopes or regulatory areas (RAs).

Service Provider

A company that provides services over satellite links to terminals. Such a service may involve simple network connectivity, or a higher-level service like disaster recovery. Service Providers who are not also satellite operators may also be referred to as Virtual Network Operators (VNO).

Signal-to-Noise Ratio (SNR)

The ratio of signal power to noise power, often expressed in decibels. SNR is a measure of sensitivity performance of a receiver. Abbreviated SNR or S/N.

Site

A Site, within an iDirect Velocity™ Network, is a collection of processes and equipment at a designated location — for example a *Satellite Access Station (SAS)* or a *Network Operations Center (NOC)*, or *External Access Portal (EAP)*.

Site SVN

The Site SVN, just beneath the Global SVN in the Velocity SVN hierarchy, represents the segment of an SVN (VPN) located within a Site — for example within a NOC or SAS.

Sleep Mode

An iDirect feature that allows remote modems to conserve power consumption during periods of network inactivity.

Spread Spectrum

A transmission technique in which a pseudo-noise (PN) code is employed as a modulation waveform to “spread” the signal energy over a bandwidth much greater than the signal information bandwidth.

Static Configuration

The terminal configuration information that is stored in the configuration file, and is the minimum required by the terminal software for the terminal to acquire into a specific iNet.

Static Multicast Stream

When this Velocity multicast option is enabled, the NMS provides the terminal with the static multicast configuration.

Steady State Margin

In DVB-S2 networks, the margin added to the SNR thresholds measured at hardware qualification to arrive at the operational SNR threshold during steady state operation.

Subscriber

An end user who receives service through a satellite terminal.

Subscriber Service Plan Component (SSPC)

When a Subscriber Service Plan Profile (SSPP) is selected as part of a Terminal Service Plan (TSP), an instance of the SSPP is created in the NMS. That instance of the SSPP is called a Subscriber Service Plan Component, as it could be one of several components of the Terminal Service Plan (TSP).

Subscriber Service Plan Profile (SSPP)

A set of service plan parameters configured, in Pulse, for use in creating a Terminal Service Plan.

Subscriber Organization

An organization owning many satellite terminals. Examples of subscriber organizations might include ocean-going shipping companies, airlines, companies that manage oil rigs, emergency relief organizations, infrastructure distribution networks, cellular network providers, and store chains.

Subscriber SVN (Data SVN)

A Global SVN that connects customer equipment, located in the terminal network, behind a terminal, through the SAS and across the DCN to a customer network.

Symbol Rate

The number of symbols that are transmitted in one second. From the symbol rate, calculate the bandwidth (total number of bits per second) by multiplying the bits per symbol by the symbol rate.

TAC

See Technical Assistance Center (TAC).

TDMA

See Time Division Multiple Access (TDMA).

Technical Assistance Center (TAC)

iDirect's customer service and technical support center, at http://tac.idirect.net or 703-648-8151. iDirect Government customer service and technical support center, at

http://tac.idirectgov.com.

Teleport

A Teleport, for example, contains elements hub equipment such as the Protocol Processor servers, line cards, and chassis; and the NMS and associated servers, and connecting SVNs. Also called SAS.

Terminal Domain

In the Pulse hierarchy of network element domains, the element domain that comprises Terminals and Terminal Components.

Terminal SVN

That segment of an SVN that is located in the remote network behind a Satellite Terminal.

Terminal Type

A terminal type is a unique element type in the NMS that has a unique Name, is composed of specific terminal components (LNB, BUC, and ACU), a specific Satellite Router Type, and specific RF parameters. A terminal type is used in creating terminals that would include the terminal type.

Terminal Service Plan (TSP)

It is the configured service plan of an individual terminal.

Time Division Multiple Access (TDMA)

A type of over-the-air multiplexing by which two or more channels of information are transmitted simultaneously over the same link by allocating different time slots within TDMA frames for the transmission of each channel.

Token

In a network system, it is a way of managing the number of times a certain operation can be performed or an object may be created.

Token Number

The number of tokens owned by a user group for a specific operation.

Traffic Filters

Traffic filters are created to classify and manage packets presented on the Upstream or Downstream.

Transmission Security (TRANSEC)

The component of communications security that includes the application of measures designed to protect transmissions from interception and exploitation by means other than encryption.

Transmission Rate

A measure of the speed of all over-the-air data. This includes the user data (Information Rate), iDirect overhead, and FEC encoding bits.

Transport Domain

In the Pulse hierarchy of network element domains, the element domain that represents a set of logical elements that are associated with the space segment of an iDirect Velocity™ Network. For example, satellites, beams, and channels.

UDP

See User Datagram Protocol.

Upstream Carrier

Synonymous with Inbound Carrier. The carrier transmitted from the remote satellite router to the hub.

Used Tokens

The number of tokens that are already used or given away.

User

An individual or an internal process or external system that has authorized access to the NMS, but only as a member of one or more user groups and with assigned user roles.

User Datagram Protocol

A connectionless transport layer protocol in the TCP/IP protocol stack. UDP is a simple protocol that exchanges datagrams without acknowledgments or guaranteed delivery, requiring that error processing and retransmission be handled by other protocols. UDP is defined in RFC 768.

User Group

A collection of users with a common set of permissions and visibility.

User Role

A specific set of defined permissions that can be assigned to a user.

Variable Coding and Modulation (VCM)

A method of applying coding to a DVB-S2 data stream in which MODCODs are assigned according to service type. iDirect does not support VCM.

Vault Table

A vault table stores a set of AES keys, used to encrypt all Certificate Authority (CA) private keys that are associated with the same chain of trust branch.

Very Small Aperture Terminal (VSAT)

A two-way satellite ground station with a less-than three meter dish antenna.

VCM

See Variable Coding and Modulation (VCM).

Virtual Network Operator (VNO)

A member of a VNO User Group. A VNO User Group restricts visibility and access rights of group members based on the permissions granted to the group by the Hub Network Operator (HNO).

Virtual Remote

In iDirect Group QoS, an instance of a Group QoS Application running on a remote modem. In Application Based QoS mode, a remote has one Virtual Remote for each Application assigned to the remote. In Remote Based QoS mode, all Applications are combined into a single Virtual Remote.

VLAN

Virtual LAN. Any broadcast domain that is partitioned and isolated in a computer network at the data link layer (OSI layer 2). LAN is an abbreviation of local area network. To subdivide a network into virtual LANs, one configures a network switch or router.

VNO

See Virtual Network Operator (VNO)

Appendix C: Debugging Terminal Events

Overview

When a remote fails to acquire into the network or drops out-of-the network, a common first place to investigate has generally been the terminal. In reality, the following are the most common causes and should be the first place to look:

- BUC/ACU/LNB malfunction
- Remote is in a location not covered by any beams
- iNet misconfiguration (for example, the remote's RF Terminal Type is not configured to use the channel it is attempting to acquire)
- A Force-Log out from the Hub has occurred

In early Velocity releases, troubleshooting an out-of-network remote was not a straight-forward process and root causing the issue was manual and expensive. To speed up the debug process, iDirect has implemented the following tools on the remote to help rule out the most common problems:

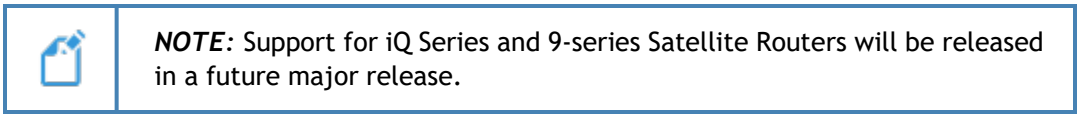
- Events Report Tool
- Events Snapshot Tool

These two tools are intended to streamline the troubleshooting process, and enable users to independently debug and resolve network outages.

Supported Remotes

The Events Report and Snapshot tools are currently only supported on the following remotes:

- X7/CX700/CX751 Satellite Routers
- CX780 Satellite Router
- 980 Satellite Router



Events Report

To expose key events to the end-user in a useful format, the remote now generates a concise report of events. The **Events Report** tool captures and exposes what the remote has been doing over-time. This history can be leveraged to identify why a remote fell out-of-network. For example, what happened immediately before the outage? Did the remote get a force logout or did the ACU instruct the remote to mute, and so on.

The report remains on the disk across reboots/restarts, and can hold up to ~3000 events (which corresponds to up to approximately 2 weeks worth of data). The reports maybe viewed in two ways:

- On the Terminal console, using the events viewer tool, type the command **evtview**
- Using the Terminal Web UI, from the dashboard

Using Event Viewer

The Event Viewer "evtview", is a lightweight package pre-installed on the remote. It loads the machine-readable events report file and converts it into a human-readable table of events that is easy to understand.

The evtview tool can be executed on any linux machine using the following command:

```
evtview -help
```

```
root@iDirectCM:p5 sat_router # evtview -s -n 14
Last 14 events from May 30 18:33:31 to May 31 14:43:24:

Timestamp           Event                        Detail
================================================================================
May 30 18:33:31     MODEM_STARTING_UP            This terminal's modem software is starting up.

May 30 18:33:44     MODEM_STARTING_SEARCH_CYCLE  This terminal's modem is beginning a new carrier search cycle.

May 30 18:33:51     MODEM_TRYING_SERVICE_CHANNEL This terminal's modem is trying to come into network on SAT_1:
                                                 Satellite longitude: 1.0 degrees
                                                 Satellite preference: 7
                                                 Carrier frequency: 19925000 kHz
                                                 Symbol rate: 10000 KSps
                                                 polarization (rx,tx): LR
                                                 Modem location (lat,lon): (-10.1, 350.0)

May 30 18:34:54     MODEM_IN_NETWORK             This terminal is in network and authenticated on iNet 1471.

May 30 19:13:23     MODEM_NETKEY_CHANGE          This terminal is adjusting system time by 2238 seconds
                                                 Old: Thu May 30 18:36:05 2019
                                                 New: Thu May 30 19:13:23 2019
```

Figure C-1. Sample evtview Table

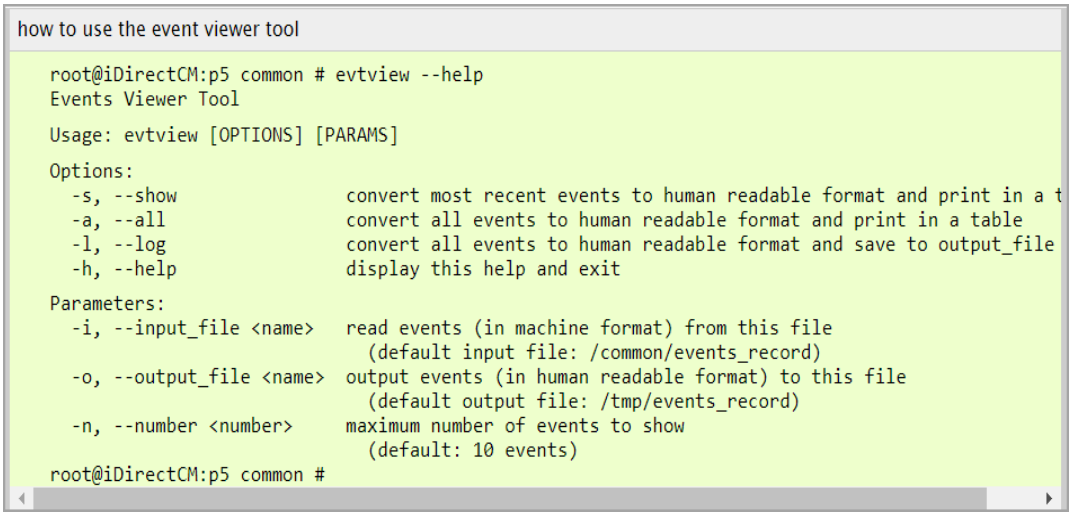


Figure C-2. evtview Commands

NOTE: In order to speed up the debugging process (and avoid the need to manually filter through verbose log files as it is done today), the remote only stores key events on disk. The remote also takes great care to ensure that duplicate events are *not* reported, even across process restarts/reboots. For example, if the remote loses its connection to the ACU, a single event is reported (MODEM_ACU_DISCONNECTED). This event will not be reported again until the ACU connection is restored, even when the remote is rebooted or restarted. This way, the user is not bombarded or distracted with duplicate events, and space is not wasted saving old information on disk.

Using Terminal Web UI Dashboard

On the Terminal WUI dashboard, bottom right-hand corner, the **Terminal Events** table displays the most recent events, to simplify the debug process.

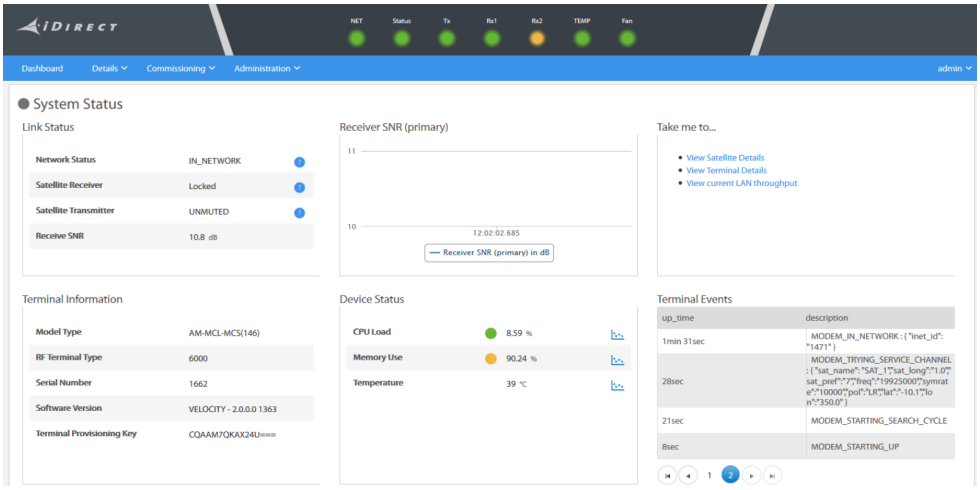


Figure C-3. Terminal Events Displayed on the Terminal WUI

Events Snapshot

To understand the exact reason on why the remote is not coming into the network, execute the **`events snapshot`** command in the terminal console. The remote's current status is displayed which helps in narrowing down the problem.

The events report and events snapshot tool can be used in tandem for debugging; where the events report provides historical context and the events snapshot displays the current status of the remote in real-time.

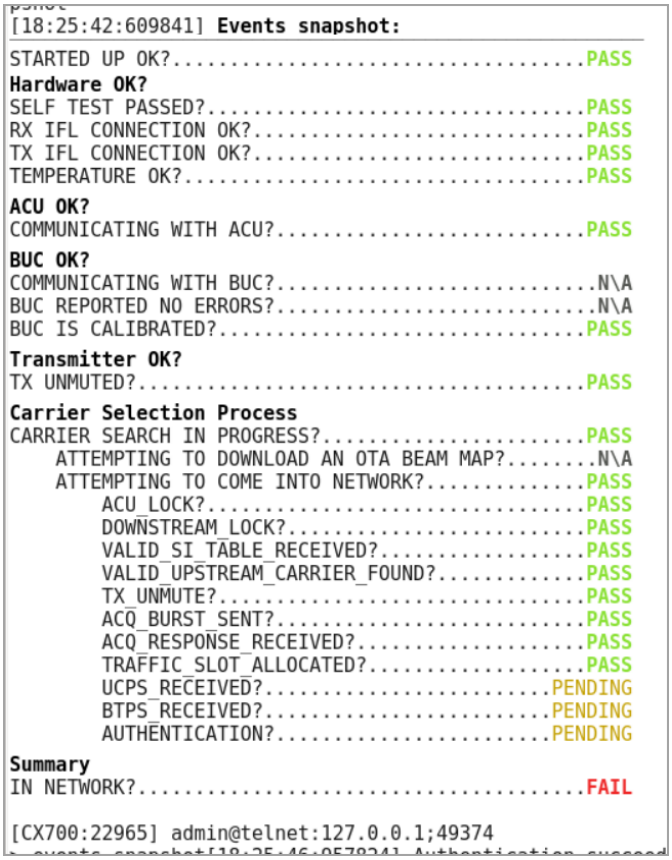


Figure C-4. Event Snapshot

```
Events snapshot:
STARTED UP OK?.....................................PASS
Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS
ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled
BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS
Transmitter OK?
TX UNMUTED?........................................PASS
Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?........N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...............PASS   # Tuned to a service carrier
        ACU_LOCK?..................................PASS
        DOWNSTREAM_LOCK?...........................PASS
        VALID_SI_TABLE_RECEIVED?...................PENDING   # Waiting to receive valid SI tables
        VALID_UPSTREAM_CARRIER_FOUND?..............PENDING
        TX_UNMUTE?.................................PENDING
        ACQ_BURST_SENT?............................PENDING
        ACQ_RESPONSE_RECEIVED?.....................PENDING
        TRAFFIC_SLOT_ALLOCATED?....................PENDING
        UCPS_RECEIVED?.............................PENDING
        BTPS_RECEIVED?.............................PENDING
        AUTHENTICATION?............................PENDING
Summary
IN NETWORK?........................................FAIL
```

Figure C-5. Event Verbose Snapshot

Event Snapshot Example:

This section displays how the event snapshot tool works when there are no errors and the remote is able to:

- Lock to a signaling carrier and download a valid OTA beam map file
- Transition to service mode and acquire into the network

```
Events snapshot:
_____________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................FAIL   # May be starting up, or re-acquiring on same carrier, etc

Summary
IN NETWORK?........................................FAIL
```

Figure C-6. On Startup Before the Search Cycle Begins

```
Events snapshot:
_____________________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?.......FAIL
        FOUND A SIGNALING CARRIER?.................FAIL
        RECEIVED A VALID OTA BEAM MAP?.............FAIL
    ATTEMPTING TO COME INTO NETWORK?...............FAIL

Summary
IN NETWORK?........................................FAIL
```

Figure C-7. Search Cycle Initiated and the Remote Tunes to the Signaling Carrier

```
Events snapshot:
_____________________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?.......PASS
        FOUND A SIGNALING CARRIER?.................PASS   # Tuned to a signaling carrier
        RECEIVED A VALID OTA_BEAM_MAP?.........PENDING   # Waiting to receive an OTA beam map
    ATTEMPTING TO COME INTO NETWORK?..............FAIL

Summary
IN NETWORK?........................................FAIL
```

Figure C-8. Remote Tries to Download the OTA Beam Map from the Signaling Carrier

```
Events snapshot:

STARTED UP OK?.....PASS

Hardware OK?
SELF TEST PASSED?.....PASS
RX IFL CONNECTION OK?.....PASS
TX IFL CONNECTION OK?.....PASS
TEMPERATURE OK?.....PASS

ACU OK?
COMMUNICATING WITH ACU?.....N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?.....N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?.....N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.....PASS

Transmitter OK?
TX UNMUTED?.....PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?.....PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?.....N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?.....PASS   # Tuned to a service carrier
        ACU_LOCK?.....PENDING   # Waiting for ACU to lock to satellite
        DOWNSTREAM_LOCK?.....PENDING
        VALID_SI_TABLE_RECEIVED?.....PENDING
        VALID_UPSTREAM_CARRIER_FOUND?.....PENDING
        TX_UNMUTE?.....PENDING
        ACQ_BURST_SENT?.....PENDING
        ACQ_RESPONSE_RECEIVED?.....PENDING
        TRAFFIC_SLOT_ALLOCATED?.....PENDING
        UCPS_RECEIVED?.....PENDING
        BTPS_RECEIVED?.....PENDING
        AUTHENTICATION?.....PENDING

Summary
IN NETWORK?.....FAIL
```

Figure C-9. Waiting for ACU to Lock to the Satellite

```
Events snapshot:
_____________________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?........N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...............PASS   # Tuned to a service carrier
        ACU_LOCK?..................................PASS
        DOWNSTREAM_LOCK?........................PENDING   # Waiting to get downstream lock
        VALID_SI_TABLE_RECEIVED?................PENDING
        VALID_UPSTREAM_CARRIER_FOUND?...........PENDING
        TX_UNMUTE?..............................PENDING
        ACQ_BURST_SENT?.........................PENDING
        ACQ_RESPONSE_RECEIVED?..................PENDING
        TRAFFIC_SLOT_ALLOCATED?.................PENDING
        UCPS_RECEIVED?..........................PENDING
        BTPS_RECEIVED?..........................PENDING
        AUTHENTICATION?.........................PENDING

Summary
IN NETWORK?........................................FAIL
```

Figure C-10. Waiting for Downstream Lock

```
Events snapshot:

STARTED UP OK?.....PASS

Hardware OK?
SELF TEST PASSED?.....PASS
RX IFL CONNECTION OK?.....PASS
TX IFL CONNECTION OK?.....PASS
TEMPERATURE OK?.....PASS

ACU OK?
COMMUNICATING WITH ACU?.....N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?.....N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?.....N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.....PASS

Transmitter OK?
TX UNMUTED?.....PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?.....PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?.....N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?.....PASS   # Tuned to a service carrier
        ACU_LOCK?.....PASS
        DOWNSTREAM_LOCK?.....PASS
        VALID_SI_TABLE_RECEIVED?.....PENDING   # Waiting to receive valid SI tables
        VALID_UPSTREAM_CARRIER_FOUND?.....PENDING
        TX_UNMUTE?.....PENDING
        ACQ_BURST_SENT?.....PENDING
        ACQ_RESPONSE_RECEIVED?.....PENDING
        TRAFFIC_SLOT_ALLOCATED?.....PENDING
        UCPS_RECEIVED?.....PENDING
        BTPS_RECEIVED?.....PENDING
        AUTHENTICATION?.....PENDING

Summary
IN NETWORK?.....FAIL
```

Figure C-11. Waiting to Receive Valid SI Tables

```
Events snapshot:

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?........N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...............PASS   # Tuned to a service carrier
        ACU_LOCK?..................................PASS
        DOWNSTREAM_LOCK?...........................PASS
        VALID_SI_TABLE_RECEIVED?...................PASS
        VALID_UPSTREAM_CARRIER_FOUND?..........PENDING   # Waiting to find a valid upstream carrier to transmit on
        TX_UNMUTE?.................................PENDING
        ACQ_BURST_SENT?............................PENDING
        ACQ_RESPONSE_RECEIVED?.....................PENDING
        TRAFFIC_SLOT_ALLOCATED?....................PENDING
        UCPS_RECEIVED?.............................PENDING
        BTPS_RECEIVED?.............................PENDING
        AUTHENTICATION?............................PENDING

Summary
IN NETWORK?........................................FAIL
```

Figure C-12. Waiting for a Valid Upstream Carrier to Transmit

```
Events snapshot:
_____________________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?........N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...............PASS   # Tuned to a service carrier
        ACU_LOCK?..................................PASS
        DOWNSTREAM_LOCK?...........................PASS
        VALID_SI_TABLE_RECEIVED?...................PASS
        VALID_UPSTREAM_CARRIER_FOUND?..............PASS
        TX_UNMUTE?.............................PENDING   # Waiting to unmute the transmitter
        ACQ_BURST_SENT?........................PENDING
        ACQ_RESPONSE_RECEIVED?.................PENDING
        TRAFFIC_SLOT_ALLOCATED?................PENDING
        UCPS_RECEIVED?.........................PENDING
        BTPS_RECEIVED?.........................PENDING
        AUTHENTICATION?........................PENDING

Summary
IN NETWORK?........................................FAIL
```

Figure C-13. Waiting to Unmute the Transmitter

```
Events snapshot:
-----------------------------------------------------
STARTED UP OK?.................................PASS
Hardware OK?
SELF TEST PASSED?..............................PASS
RX IFL CONNECTION OK?..........................PASS
TX IFL CONNECTION OK?..........................PASS
TEMPERATURE OK?................................PASS
ACU OK?
COMMUNICATING WITH ACU?........................N\A   # OpenAMIP is disabled
BUC OK?
COMMUNICATING WITH BUC?........................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?........................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.............................PASS
Transmitter OK?
TX UNMUTED?....................................PASS
Carrier Selection Process
CARRIER SEARCH IN PROGRESS?....................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?....N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...........PASS   # Tuned to a service carrier
        ACU_LOCK?..............................PASS
        DOWNSTREAM_LOCK?.......................PASS
        VALID_SI_TABLE_RECEIVED?...............PASS
        VALID_UPSTREAM_CARRIER_FOUND?..........PASS
        TX_UNMUTE?.............................PASS
        ACQ_BURST_SENT?....................PENDING   # Waiting to send an acquisition burst to the hub
        ACQ_RESPONSE_RECEIVED?.............PENDING
        TRAFFIC_SLOT_ALLOCATED?............PENDING
        UCPS_RECEIVED?.....................PENDING
        BTPS_RECEIVED?.....................PENDING
        AUTHENTICATION?....................PENDING
Summary
IN NETWORK?....................................FAIL
```

Figure C-14. Waiting to Send an Acquisition Burst to the Hub

```
Events snapshot:
STARTED UP OK?.....PASS
Hardware OK?
SELF TEST PASSED?.....PASS
RX IFL CONNECTION OK?.....PASS
TX IFL CONNECTION OK?.....PASS
TEMPERATURE OK?.....PASS
ACU OK?
COMMUNICATING WITH ACU?.....N\A   # OpenAMIP is disabled
BUC OK?
COMMUNICATING WITH BUC?.....N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?.....N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.....PASS
Transmitter OK?
TX UNMUTED?.....PASS
Carrier Selection Process
CARRIER SEARCH IN PROGRESS?.....PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?.....N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?.....PASS   # Tuned to a service carrier
        ACU_LOCK?.....PASS
        DOWNSTREAM_LOCK?.....PASS
        VALID_SI_TABLE_RECEIVED?.....PASS
        VALID_UPSTREAM_CARRIER_FOUND?.....PASS
        TX_UNMUTE?.....PASS
        ACQ_BURST_SENT?.....PASS
        ACQ_RESPONSE_RECEIVED?.....PENDING   # Waiting for the hub to acknowledge the acquisition burst
        TRAFFIC_SLOT_ALLOCATED?.....PENDING
        UCPS_RECEIVED?.....PENDING
        BTPS_RECEIVED?.....PENDING
        AUTHENTICATION?.....PENDING
Summary
IN NETWORK?.....FAIL
```

Figure C-15. Waiting for the Hub to Acknowledge the Acquisition Burst

```
Events snapshot:
_____________________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?........N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...............PASS   # Tuned to a service carrier
        ACU_LOCK?..................................PASS
        DOWNSTREAM_LOCK?...........................PASS
        VALID_SI_TABLE_RECEIVED?...................PASS
        VALID_UPSTREAM_CARRIER_FOUND?..............PASS
        TX_UNMUTE?.................................PASS
        ACQ_BURST_SENT?............................PASS
        ACQ_RESPONSE_RECEIVED?.....................PASS
        TRAFFIC_SLOT_ALLOCATED?....................PENDING   # Waiting to get allocated traffic slots
        UCPS_RECEIVED?.............................PENDING
        BTPS_RECEIVED?.............................PENDING
        AUTHENTICATION?............................PENDING

Summary
IN NETWORK?........................................FAIL
```

Figure C-16. Waiting for Allocated Slots

```
Events snapshot:
_____________________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?.........N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...............PASS   # Tuned to a service carrier
        ACU_LOCK?......................................PASS
        DOWNSTREAM_LOCK?...............................PASS
        VALID_SI_TABLE_RECEIVED?.......................PASS
        VALID_UPSTREAM_CARRIER_FOUND?..................PASS
        TX_UNMUTE?.....................................PASS
        ACQ_BURST_SENT?................................PASS
        ACQ_RESPONSE_RECEIVED?.........................PASS
        TRAFFIC_SLOT_ALLOCATED?........................PASS
        UCPS_RECEIVED?.............................PENDING   # Waiting to receive UCP updates
        BTPS_RECEIVED?.............................PENDING
        AUTHENTICATION?............................PENDING

Summary
IN NETWORK?........................................FAIL
```

Figure C-17. Waiting to Receive UCP Updates

```
Events snapshot:
_____________________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?.........N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...............PASS   # Tuned to a service carrier
        ACU_LOCK?..................................PASS
        DOWNSTREAM_LOCK?...........................PASS
        VALID_SI_TABLE_RECEIVED?...................PASS
        VALID_UPSTREAM_CARRIER_FOUND?..............PASS
        TX_UNMUTE?.................................PASS
        ACQ_BURST_SENT?............................PASS
        ACQ_RESPONSE_RECEIVED?.....................PASS
        TRAFFIC_SLOT_ALLOCATED?....................PASS
        UCPS_RECEIVED?.............................PASS
        BTPS_RECEIVED?.........................PENDING   # Waiting to receive BTP updates
        AUTHENTICATION?........................PENDING

Summary
IN NETWORK?........................................FAIL
```

Figure C-18. Waiting to Receive BTP Updates

```
Events snapshot:
_____________________________________________________

STARTED UP OK?.....................................PASS

Hardware OK?
SELF TEST PASSED?..................................PASS
RX IFL CONNECTION OK?..............................PASS
TX IFL CONNECTION OK?..............................PASS
TEMPERATURE OK?....................................PASS

ACU OK?
COMMUNICATING WITH ACU?............................N\A   # OpenAMIP is disabled

BUC OK?
COMMUNICATING WITH BUC?............................N\A   # OpenBMIP is disabled
BUC REPORTED NO ERRORS?............................N\A   # OpenBMIP is disabled
BUC IS CALIBRATED?.................................PASS

Transmitter OK?
TX UNMUTED?........................................PASS

Carrier Selection Process
CARRIER SEARCH IN PROGRESS?........................PASS
    ATTEMPTING TO DOWNLOAD AN OTA BEAM MAP?........N\A   # Terminal already has a valid OTA beam map
    ATTEMPTING TO COME INTO NETWORK?...............PASS   # Tuned to a service carrier
        ACU_LOCK?..................................PASS
        DOWNSTREAM_LOCK?...........................PASS
        VALID_SI_TABLE_RECEIVED?...................PASS
        VALID_UPSTREAM_CARRIER_FOUND?..............PASS
        TX_UNMUTE?.................................PASS
        ACQ_BURST_SENT?............................PASS
        ACQ_RESPONSE_RECEIVED?.....................PASS
        TRAFFIC_SLOT_ALLOCATED?....................PASS
        UCPS_RECEIVED?.............................PASS
        BTPS_RECEIVED?.............................PASS
        AUTHENTICATION?........................PENDING   # Waiting on authentication to complete

Summary
IN NETWORK?........................................FAIL
```

Figure C-19. Waiting for Authentication to Complete

ST Engineering iDirect

13861 Sunrise Valley Drive, Suite 300
Herndon, VA 20171-6126
+1 703.648.8000
+1 866.345.0983
www.idirect.net